



WAF Administration Guide

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Administering Imperva SecureSphere

What would you like to read about?

Administration Guide Contents

Section Name	Description
Introduction to the SecureSphere Administration Guide	This section describes the SecureSphere Administration Guide.
Topologies and Deployment	This section describes the different ways that SecureSphere can be deployed to protect Web and database applications.
Platforms and Installation	This section describes how to install SecureSphere.
Configuring SecureSphere	This section describes how to configure SecureSphere appliances.
Gateways	This section describes how to configure SecureSphere Gateways.
Licenses	This section describes the SecureSphere licensing process.
Users and Permissions	This section describes how to define SecureSphere administrators, users and their privileges.
Sessions	This section describes how to track SecureSphere sessions.
ADC	This section describes how to configure SecureSphere to download content from Imperva's ADC (Application Defense Center).
Job Status	This section describes how to track jobs initiated by SecureSphere.

Section Name	Description
Maintenance	This section describes how to maintain, backup and restore SecureSphere.
System Definitions	This section describes the SecureSphere system definitions.
System Performance	This section describes the system performance data SecureSphere provides about Management Servers, and Gateways.
Gateway High Availability	This section describes how to configure High Availability for SecureSphere Gateways.
Management Server High Availability	This section describes how to configure High Availability for SecureSphere Management Servers.
Network High Availability	This section describes how to provide High Availability for SecureSphere using the network.
Network Services	This section describes how to configure the network in which SecureSphere appliances are deployed.
Add-Ons	This section describes various SecureSphere add-ons, such as IPMI etc.
Command Line Interface	This section describes the SecureSphere command line tools: <code>impcfg</code> and <code>impctl</code> .
PCI Compliance	This section describes how SecureSphere can help administrators achieve PCI compliance for their sites.

Topologies and Deployment

This chapter describes the different deployments of SecureSphere, and includes:

- Choosing a Deployment
- Sniffing Mode
- Inline Gateways (Bridge Mode)
- Reverse Proxy Deployments
- Features Available in Each Deployment Mode

Choosing a Deployment

A SecureSphere Gateway can be deployed either inline, where traffic passes through the SecureSphere Gateway, or as a sniffing Gateway, where traffic does not pass through the Gateway but is copied to it for analysis. The most important differences between these deployments are:

- An inline Gateway can block malicious traffic so that it never reaches the protected network, but on the other hand, it introduces some latency.
- A sniffing Gateway introduces no latency, and it can prevent malicious traffic from reaching the protected network by issuing TCP resets to the server on its blocking interface.

Inline Gateways can be deployed in a number of configurations:

- **Bridge STP:** The SecureSphere Gateway is deployed in bridge mode in existing high-availability deployments. For more information, see [Gateway High Availability](#).
- **Bridge IMPVHA:** The SecureSphere Gateway is deployed in bridge mode in existing high-availability deployments where STP is not supported by the switches, is too complicated to implement, or is in use with Rapid Spanning Tree Protocol (RSTP). For more information, see [Gateway High Availability](#).
- **Reverse Proxy:** The SecureSphere Gateway is deployed in reverse proxy mode where traffic intended for a Web server first passes through the Reverse Proxy. For more information, see [Reverse Proxy Deployments](#).

In reverse proxy mode, SecureSphere Gateways can be configured as Transparent Reverse Proxy for specific Web services, so a SecureSphere Gateway can act as a proxy for some services but not for others. For more information, see [Transparent Reverse Proxy - Selected Web Servers](#).

SecureSphere Gateways can also be deployed in conjunction with Load Balancers. For more information, see [Reverse Proxy with Load Balancer](#).

This chapter presents some typical examples of these deployments. Not all possible deployments are covered here.

CPU Affinity



Note: CPU affinity does not apply in sniffing mode.

For all SecureSphere Gateways with more than one core, SecureSphere implements IRQ affinity (CPU affinity), meaning that each core handles interrupts from specific interfaces.

SecureSphere version 8.0 allocates cores to interfaces at boot time. Beginning with version 8.5, SecureSphere implements dynamic allocation of cores to 10G interfaces, depending on the load. For other interfaces, the boot time allocation remains unchanged.

Sniffing Mode

This section reviews information regarding sniffing mode and includes the following:

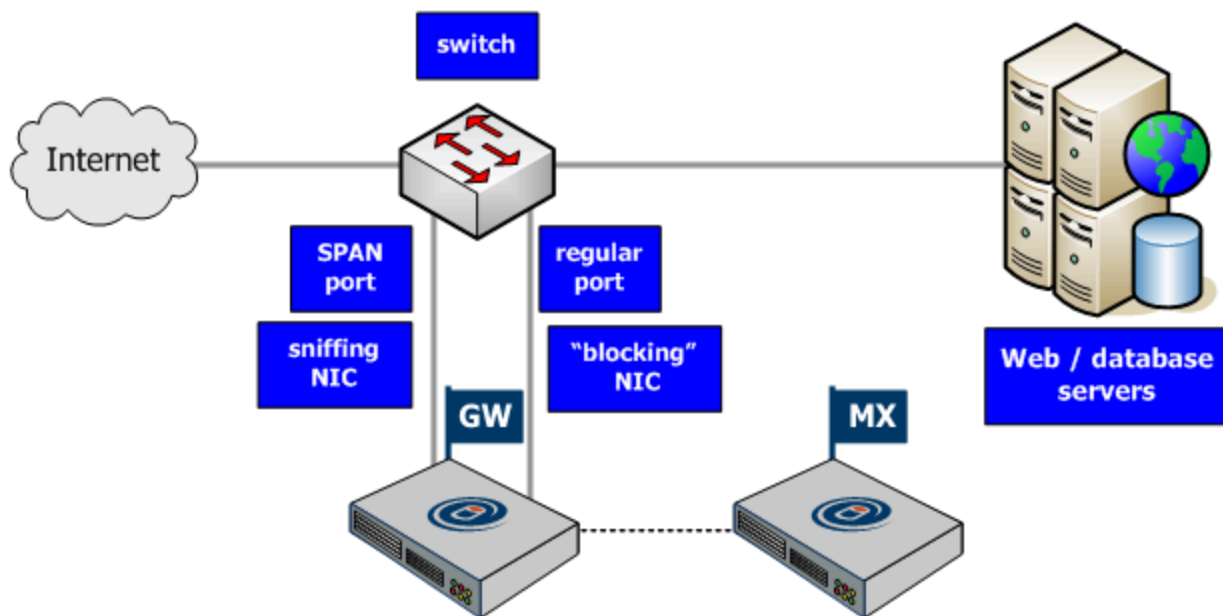
- [Sniffing Mode Topologies](#)

Sniffing Mode Topologies

In a sniffing mode deployment, traffic does not pass through the SecureSphere appliance but is only copied to it. A sniffing appliance blocks malicious Web traffic by issuing TCP resets to the server. A sniffing appliance's advantage is that it introduces no latency in either Web or database application traffic.

The disadvantages of sniffing mode are:

- Issuing TCP resets is not as dependable a method of blocking traffic as is dropping packets.
- Overloaded SPAN ports sometimes drop packets, so SecureSphere will not inspect them.



There are two methods to "sniff" the traffic:

- using a SPAN (mirror) port on the switch, or
- using a dedicated TAP

The SPAN port is less reliable since it depends on the switch resources and has lower priority than forwarding traffic, so it might miss packets. TAP does not have this disadvantage.

Aggregation taps can also be used.

Inline Gateways (Bridge Mode)

Inline gateways operate in Bridge Mode. For more more information on Bridge Mode, see [Bridge Deployments](#).

Bridge Deployments

This section reviews information regarding bridge deployments and includes the following:

- [Bridge Mode Topologies](#)

Bridge Mode Topologies

In a bridge deployment, all traffic passes through the SecureSphere Gateway, which monitors the traffic and

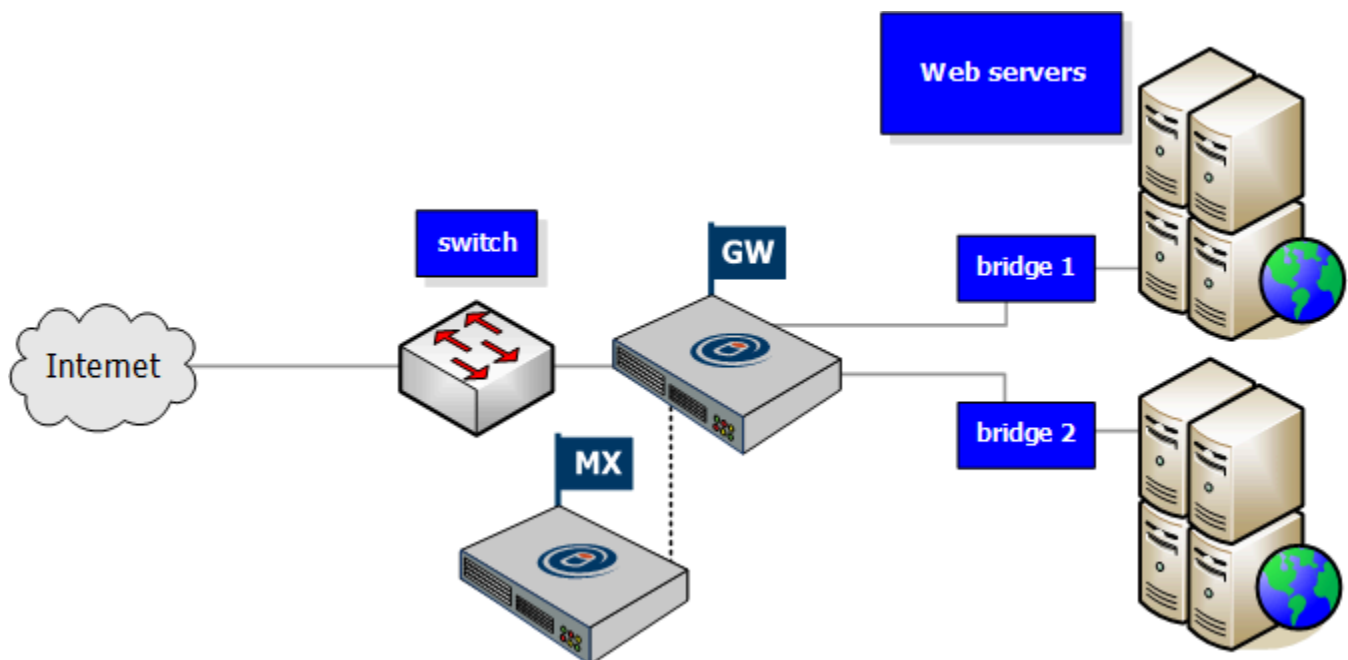
blocks malicious connections by dropping packets.



Note: By default the SecureSphere OVF is installed with 2 NICs for data to form a single bridge. If you decide to install a second bridge, it should be done in a way that prevents the same traffic from being seen on both bridges as SecureSphere only tracks sessions.

SecureSphere supports two bridge topologies

- Transparent bridge
- Non-transparent bridge using RP (transparent and non transparent RP)



The bridge can be invisible (transparent) to both clients and servers, which would then see each others' real IP addresses.

Fail Safe and Fail Open Modes

A transparent bridge can be configured to maintain connectivity in the event of failure. In Fail Open mode, traffic will continue to pass through the bridge, but SecureSphere will not inspect it.

Reverse Proxy Deployments

For Web traffic only, an inline Gateway can be configured as a Reverse Proxy. In a Reverse Proxy, traffic intended for a Web server first passes through the Reverse Proxy.

This section contains the following:

- [Reverse Proxy Overview](#)
- [Transparent Reverse Proxy Overview](#)
- [Non-Transparent Reverse Proxy Overview](#)
- [Reverse Proxy with Load Balancer](#)

Reverse Proxy Overview

Reverse Proxies can be configured as Transparent Reverse Proxies (TRP) or Non-Transparent Reverse Proxies.

Reverse Proxy Mode Advantages

Some of the advantages of Reverse Proxy mode are:

- A Reverse Proxy assembles packets into a complete message, inspects the message and confirms that it complies with the relevant policies, and only then transmits the packets to the server.
- Reverse Proxy mode supports features that require changing data, for example, cookie signing, SSL decryption, *etc.*
- Reverse Proxy mode supports features that require changing the destination IP address.
- Traffic is decrypted only once, by the SecureSphere. The server sees only unencrypted traffic, improving server performance.
- In the same way, SSL decryption is offloaded from the server to the SecureSphere Gateway.
- Reverse Proxy mode enables SecureSphere to be implemented in legacy deployments without the need to reconfigure networks (drop-in replacement).
- Improved proxy functionality (connection termination, including SSL termination)
- Improved proxy performance

Transparent Reverse Proxy Overview

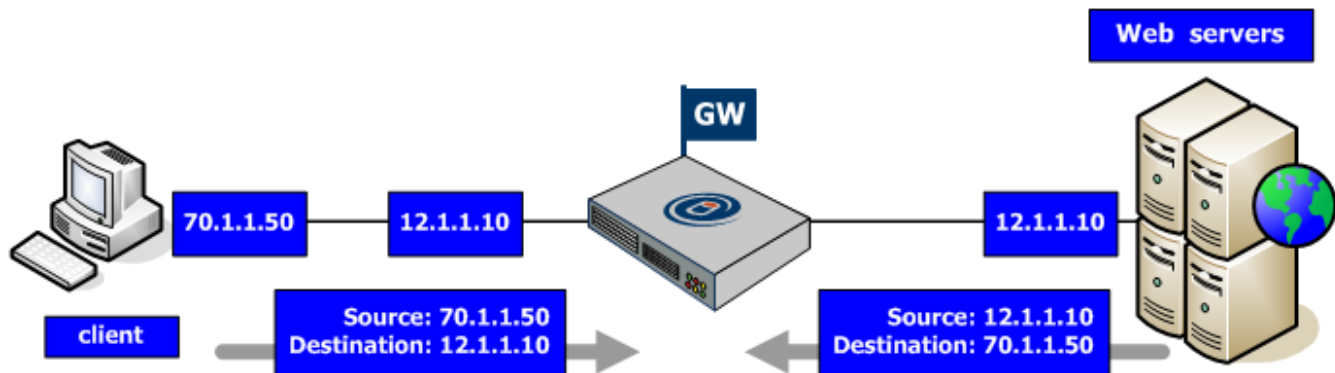
This section reviews the different types of Transparent Proxies and includes the following:

- [Transparent Reverse Proxy](#)
- [Transparent Reverse Proxy - Selected Web Servers](#)

Transparent Reverse Proxy

In Transparent Reverse Proxy mode, the client is unaware of the Reverse Proxy, and connects directly to the server. The SecureSphere Gateway is not visible to the client or to the server, which see each others' real IP addresses.

The SecureSphere Gateway is in bridge mode, but additionally functions as a Reverse Proxy for the servers specified in the SecureSphere GUI (in the **Reverse Proxy Addresses** section of the **Definitions** tab of the server group (**Main > Setup > Sites**). All other traffic, including non-Web traffic, is bridged through. In this way, the same SecureSphere Gateway protecting many Web and database server groups can act as a Reverse Proxy for some or all of the Web server groups it protects, and as a bridge for the other server groups.



The client begins the session by sending packets to the server's IP address (12.1.1.10). The SecureSphere Gateway intercepts connections destined for the server, inspects them and forwards them on to the server.

The server sees the connections as coming from the client's IP address (70.1.1.50) and replies to that address. The return connection passes through the SecureSphere Gateway, which inspects them and forwards them on to the client.

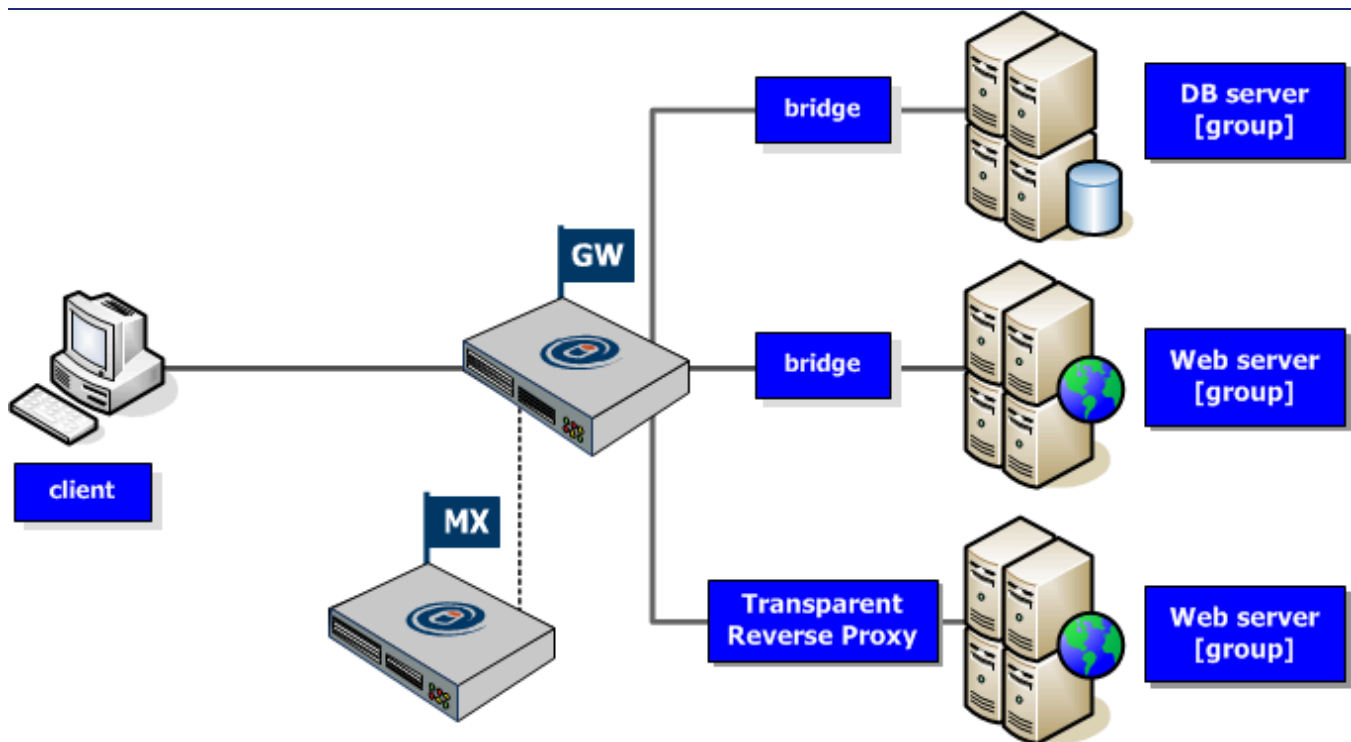


Note: When using the SecureSphere gateway in Transparent Reverse Proxy mode, a default gateway must be configured as well.

For information on configuring a SecureSphere Gateway in Transparent Reverse Proxy, see [Configuring Transparent Reverse Proxy](#).

Transparent Reverse Proxy - Selected Web Servers

In the deployment depicted below, the SecureSphere Gateway acts as a bridge, protecting all the servers (the two Web servers and the database server), but for one of the Web servers, the SecureSphere Gateway also acts as a Transparent Reverse Proxy.



Non-Transparent Reverse Proxy Overview

In non-Transparent Reverse Proxy mode, clients connect to the IP address of the SecureSphere Gateway, which acts as a Reverse Proxy for the Web servers it protects. In contrast to transparent deployments, in non-transparent deployments the SecureSphere Gateway is visible to both client and server, while the client and server are not directly visible to each other.

The Web server groups are specified in the SecureSphere GUI (in the **Reverse Proxy** section of the **Definitions** tab of the server group (**Main > Setup > Sites**). The SecureSphere Gateway does not act as a bridge. The Non-Transparent Reverse Proxy listens on the specified ports and ignores all other traffic.

Among the advantages of non-Transparent Reverse Proxies are:

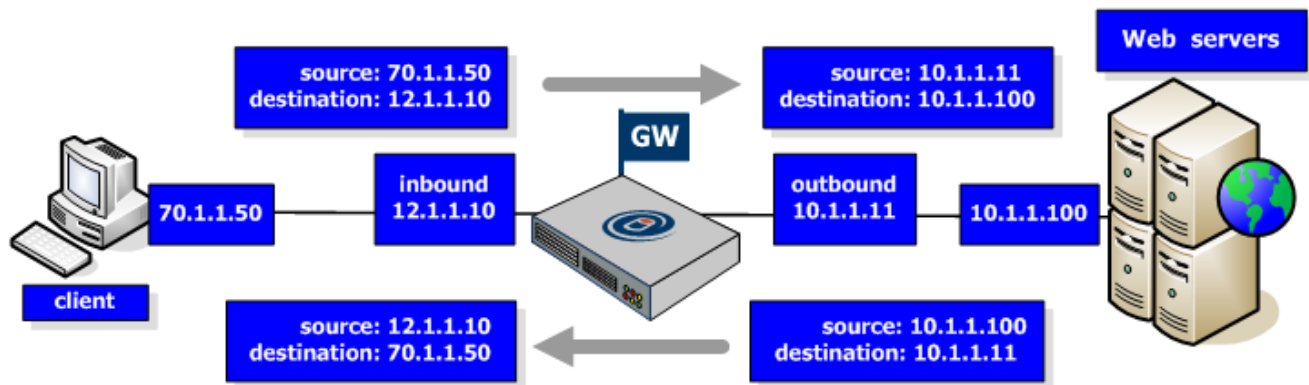
- SSL off-loading: the ability to terminate SSL sessions, reducing the load on the server
- The ability to sign and encrypt cookies
- The ability to protect web servers located in multiple subnets, not necessarily close to each other or to the Gateway
- Improved proxy functionality (connection termination, including SSL termination)
- Improved proxy performance



Note: In a Non-Transparent Reverse Proxy deployment, if the SecureSphere Gateway fails, the Web server becomes unreachable, so High Availability is usually implemented to maintain connectivity. For more information, see [Gateway High Availability](#).

Non-Transparent Reverse Proxy

In a Reverse Proxy deployment, the client begins the session by sending packets to the SecureSphere Gateway's inbound IP address. The SecureSphere Gateway inspects the packets and, based on the rules, opens a second connection to the server, changing the packet's source IP address to the SecureSphere Gateway's outbound IP address and the destination IP address to the real IP address of the server.



The server sends the return packets to the SecureSphere Gateway, which inspects the packets and forwards them to the client, changing the source IP address to the SecureSphere Gateway's inbound IP addresses, and the destination IP address to that of the client.

From the client's point of view, the server is the SecureSphere Gateway's inbound IP address. From the server's point-of-view, the client is the Gateway's outbound IP address.

Notes:

- In Non-Transparent reverse proxy mode, the maximum length of a POST request without multi-parts is 2MB.
- When there is a request sent to the non existing page or the protected server is down, it returns blocking response and no alert is generated, as it is not an attack. For example, when the protected server is down, an error page is sent, but it does not indicate that the user attacked the server or performed a violation to the server.

The SecureSphere Gateway does not act as a bridge. The Reverse Proxy listens on the specified ports and ignores all other traffic.

The following sections describe a number of possible non-Transparent Reverse Proxy deployments.

Non-Transparent Reverse Proxy (One Arm)

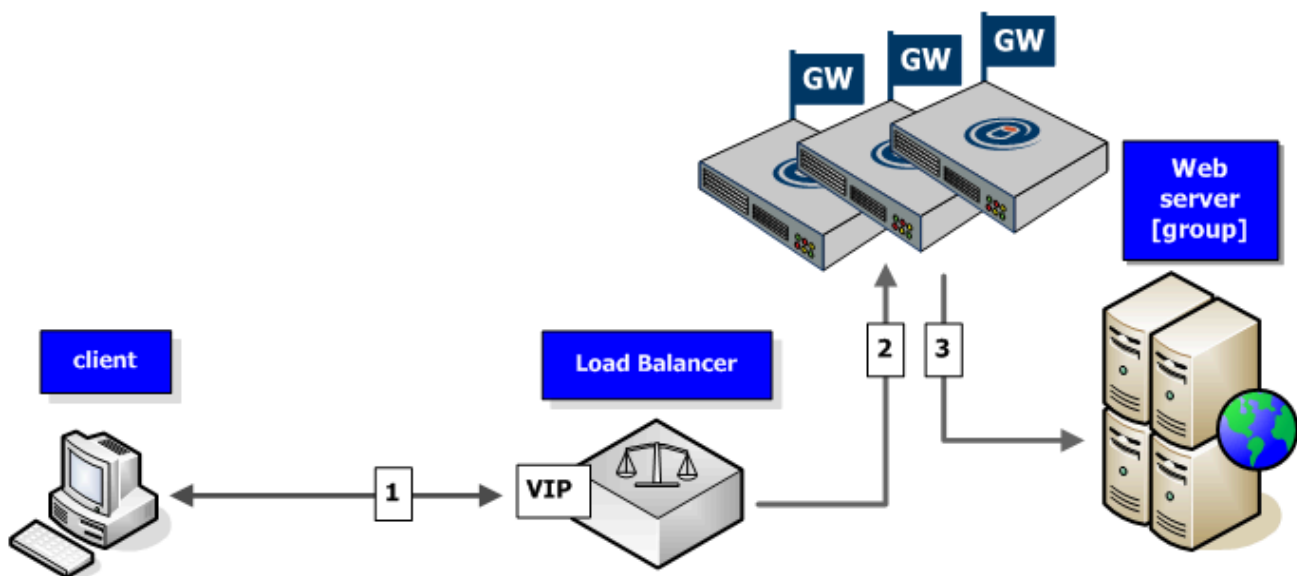
A "one arm" Non-Transparent Reverse Proxy deployment is one where the inbound and outbound interfaces are the same. This configuration simplifies routing.

Reverse Proxy with Load Balancer

Load Balancers provide a number of advantages:

- **High Availability** — If one of the SecureSphere Gateways fails, the Load Balancer will direct traffic to one of the other Gateways.
- **Increased Bandwidth** — Since the processing load is shared among SecureSphere Gateways, more total bandwidth is available.

Reverse Proxy can be implemented with Load Balancers.



1. The client connects to the external VIP (Virtual IP) address of the Load Balancer.
2. The Load Balancer opens a connection to a Reverse Proxy on one of the SecureSphere Gateways.
3. The Reverse Proxy inspects the packets and, based on the rules, sends the connection to one of the Web servers.

Return traffic follows the opposite path.

Features Available in Each Deployment Mode

The table below lists some of the SecureSphere features for which the deployment mode affects the feature's availability.

Some SecureSphere Features for which the Deployment Mode Affects the Feature's Availability

SecureSphere feature	Transparent Bridge	Sniffing	Reverse Proxy Modes	
			Transparent Reverse Proxy	(Non- Transparent)
Inspect TCP handshake	√	√	√	√
Normalize traffic (decode encoded content, such as URL-encoded or UTF-encoded content)	√	√	√	√
Decrypt and inspect SSL (either interface to a FIPS-compliant HSM or store an encrypted copy of the SSL key)	√	√	√	√
HTTP / SQL protocol conformance	√	√	√	√
Session / cookie protection	√	√	√	√
Input validation (allow list)	√	√	√	√
Attack detection (deny list, including application, network, Web server and OS attacks)	√	√	√	√
Correlated attack validation	√	√	√	√
Block attacks	√	√*	√	√
Block data leaks (inspect outgoing traffic for sensitive data such as SSN, credit card numbers, CVV numbers and data matching custom signatures)	√	-*	√	√

SecureSphere feature	Transparent Bridge	Sniffing	Reverse Proxy Modes	
			Transparent Reverse Proxy	(Non- Transparent)
Rewrite content	-	-	√	√
Application level auditing (record complete HTTP request and server response code)	√	√	√	√
Ignore IP Group (Server Groups)	√		√	√
Application user tracking	√	√	√	√
Firewall and Network Protocol Violation Policies enforced	√	-	-	-
Redirect client to an error page	√	-*	√	√

*A SecureSphere Gateway in sniffing mode cannot send an error page or block a connection. Instead, it sends a TCP reset to the server. The client sees a timeout message.

Platforms and Installation

This section describes the installation and configuration for SecureSphere appliances, and includes:

- [Forward Compatibility](#)
- [Preparing the Appliance](#)
- [Downloading and Verifying Software](#)
- [Installing the Software on Physical Machines](#)
- [Installing the Software on Virtual Machines](#)
- [Working with Management Servers After the Installation](#)
- [Replacing Gateways](#)

Forward Compatibility

Forward compatibility means that gateways of version X are able to register and operate with MXs of version Y where $X \geq Y$. For example, a version 14.x Gateway can be managed by a SecureSphere version 13.x MX.

Forward compatibility is intended for upgrades only and is supported for up to two major versions.

When upgrading from SecureSphere version 13.x or higher, you should upgrade the Gateways first and then upgrade the MX. This results in significantly reduced downtime during the upgrade process.



Note: If the MX and the Gateway are different SecureSphere versions, features added to SecureSphere after the version of the MX are not available. For example, a version 14.x Gateway feature cannot be configured on a version 13.x MX. The feature cannot be used until the MX is also upgraded.

Preparing the Appliance

The steps for installing and configuring the SecureSphere appliance are as follows:

1. Connect the appliance to the network as described in the appliance's **Quick Start Guide**, which is packaged with the appliance.
2. If necessary, install the SecureSphere software, as described in [Installing the SecureSphere Software](#).
3. Execute the First Time Login procedure, as described in [Initial Configuration](#). During this procedure, you will define passwords, interfaces and other networking parameters.
4. Review the section [Working with Management Servers After the Installation](#) and perform the procedures relevant to your deployment, as described there.
5. Configure SecureSphere using the GUI, connect to the Management Server by pointing your browser to `https://<IP address>:8083`, where <IP address> is the IP address of the management port on the MX.

Before configuring the SecureSphere software, you should open the required ports on firewalls in the network. For more information, see [Configuring Firewall Ports](#).

Additionally, it is recommended that you provide an uninterrupted power source (UPS) for the appliance.

Downloading and Verifying Software

You can download versions and patches from the FTP. To do so you require an FTP account. You receive an FTP account as part of your software purchase. For more details on obtaining an FTP account please contact Imperva Support.

You can use one of the following Imperva FTP sites according to your geographical location:

- USA: <ftp://ftp-us.imperva.com>
- Europe: <ftp://ftp-eu.imperva.com>

Upgrading and patching is now performed using a single Imperva Update file. You can download the file from **Downloads** section of the [Imperva Customer Portal](#), under **Downloads/Imperva_WAF/Updates/v15/<patch number>**.

For AWS deployments, use the Cloud Template Tool at the following link: <https://cloud-template-tool.imperva.com/>

Once downloaded, you need to verify the software package, before installing it. Imperva supplies MD5 and SHA-256 files to be used for verification with the corresponding binary installation files.

To verify the software package:

1. From the Imperva FTP, download the binary installation file and corresponding MD5 or SHA-256 file of the version you want to install.
2. Under a Windows environment you need to install a third-party verification program and use it to perform the verification.
3. Under a Linux environment, put the downloaded files in a directory and run one of the following commands:

```
sha256sum -c <binary file name>.sha256
```

```
md5sum -c <binary file name>.md5
```

Where <binary file name> is the full name (including extension) of the binary installation file. For example, `sha256sum -c usb-disk-14.1.0.20_0. 32053-x86_64.img.sha256`

You should receive the answer: <binary file name>: OK. For example, `usb-disk-14.1.0.20_0. 32053-x86_64.img: OK.`

Installing the Software on Physical Machines

The appliance is shipped from the factory with the On-Premises software already installed on it. You can install a different version on the appliance, or from a USB device, by following these instructions.

This section contains the following information:

- [X and M Series Appliances](#)
- [Creating a Bootable USB Device Containing the Installation Image](#)

X and M Series Appliances

This section reviews information regarding X and M series appliances and includes the following:

- [Connecting to a X or M Series Appliance Using a Serial Console](#)
- [Installing X and M Series Appliances From USB](#)

Connecting to a X or M Series Appliance Using a Serial Console

To connect to the appliance using a serial console:

Connect a computer or a terminal to the serial port on the appliance using a serial cable. If you are using a program such as Hyperterminal™, configure the serial console settings as follows:

Serial Console Settings

Setting	Value for models: X1010, X2010, X1020, X2020, X2510, X4510, X6510, X8510, X10K, M110, M120, M160
baud rate	9600
data bits	8
parity	none
stop bits	1
flow control	none
terminal emulation	VT-100 or VT-UTF8 (supports colors)

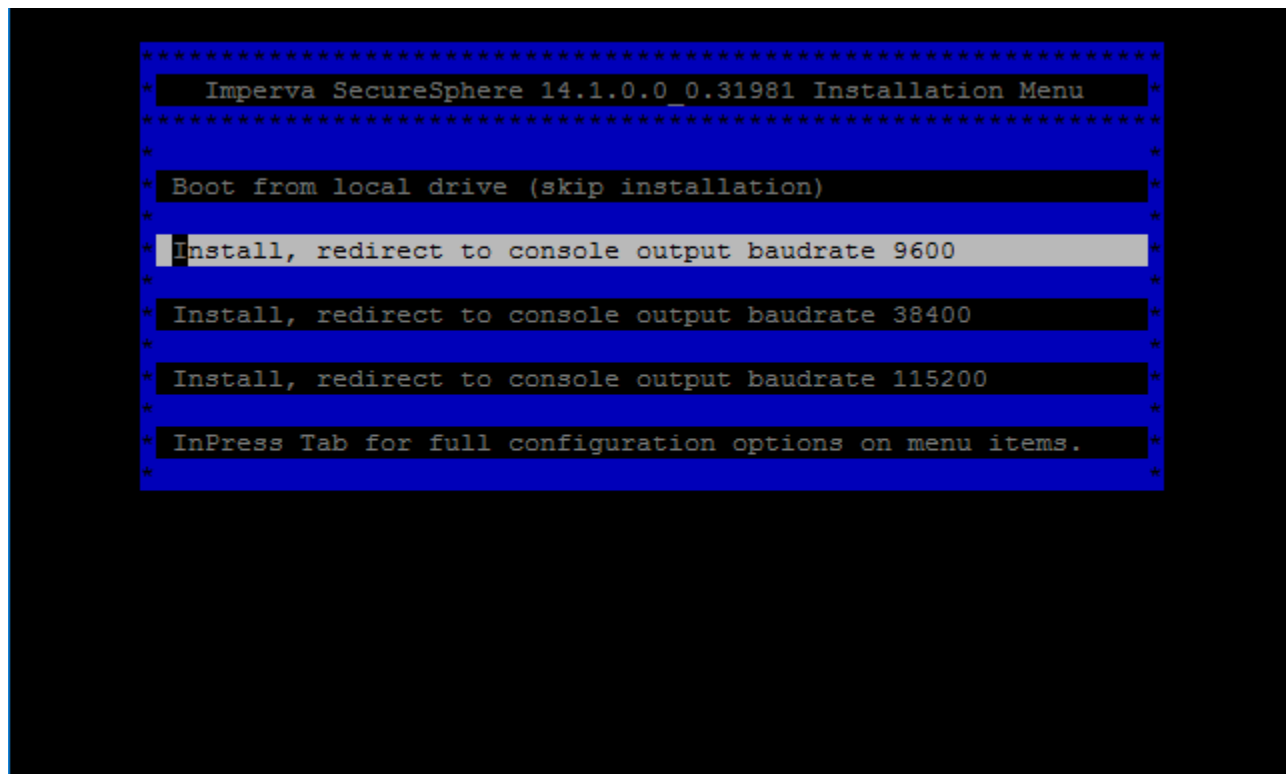
Note: Some terminal emulators correctly interpret the Backspace key to delete the previous character from the stream sent to the application as well as from the displayed text, while others send a control sequence, so that the stream the application sees is different from the displayed text. You should determine the behavior of your terminal emulator before using the Backspace key.

In Hypterterminal, you can avoid this problem by using the DEL option or Ctrl-H (delete character).

Installing X and M Series Appliances From USB

To install from a USB device:

1. Turn off the appliance.
2. Plug a USB device with the software image into the appliance USB port.
3. Connect to the appliance using a serial console (see [Connecting to a X or M Series Appliance Using a Serial Console](#)).
4. Power on the appliance.
5. After the appliance boots, select the **Install, redirect to console output baudrate 9600** option.



6. After installation completes, login using the credentials user: **admin** and password: **admin**.
7. Change the admin password and Execute the First Time Login procedure, as described in [Initial Configuration](#).

Note: Make sure your password has the following characteristics:

- It must have no fewer than 7 characters and no more than 14 characters.
- It must have at least one number, one capital letter, and one special character from:

* + = # % ^ : / ~ . , [] _ \ () | ; @ \$ & - ? { } < >

- It cannot have more than two characters repeated in succession.

Creating a Bootable USB Device Containing the Installation Image

To install the software from a USB device, you must first create an image of the installation file on the device. You will need the following:

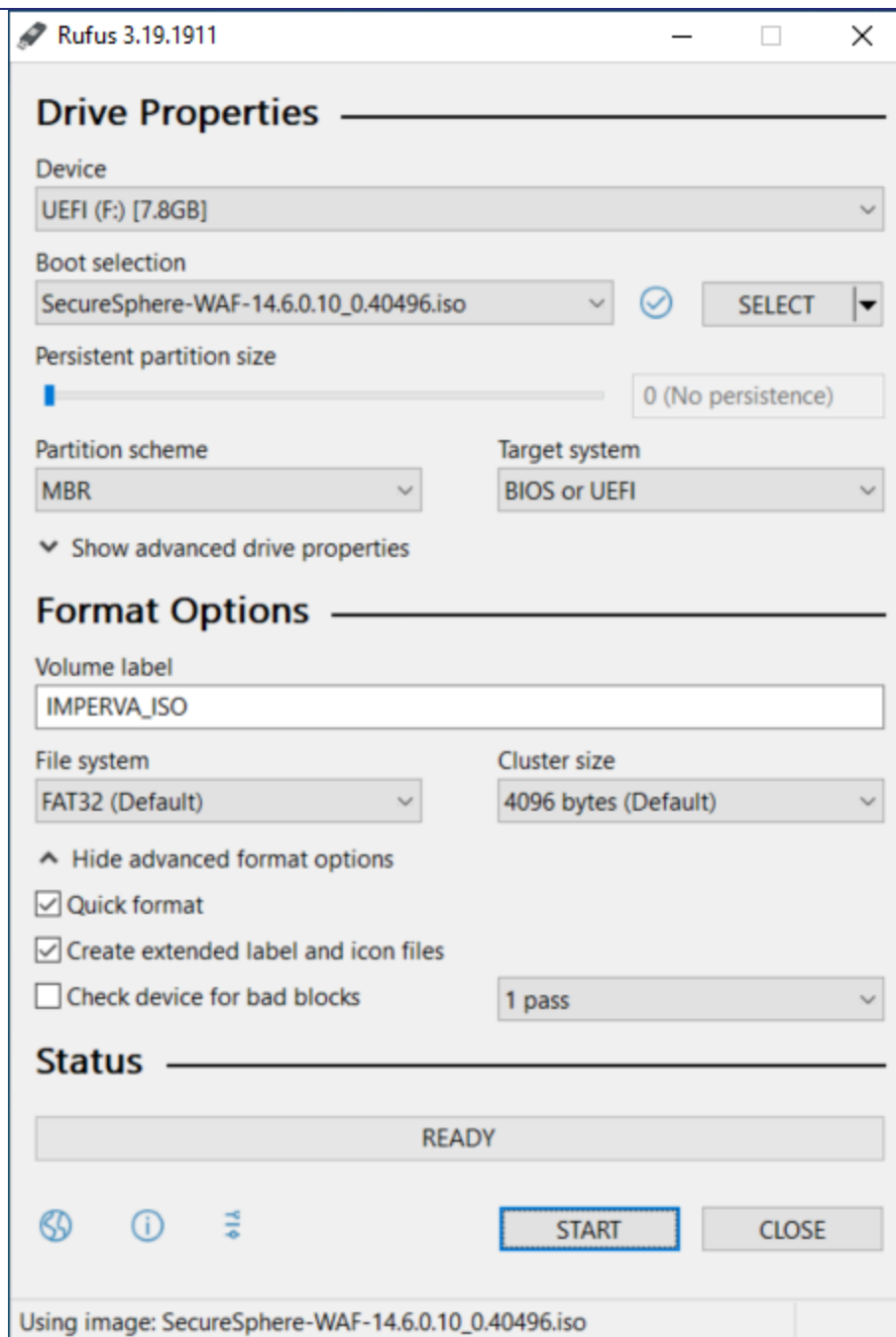
- A USB device (for example, a disk on key). The USB must have sufficient capacity to contain the image. You can check image size for this version in step 1 below.

Note: All data on the device will be erased, so back up the data before starting this procedure.

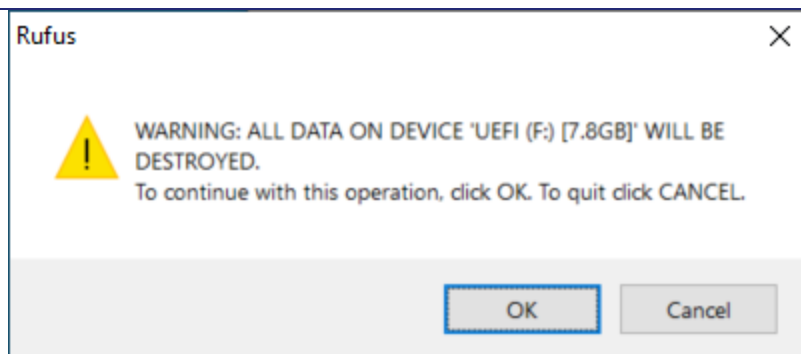
- Access to the Internet

To create a bootable USB device:

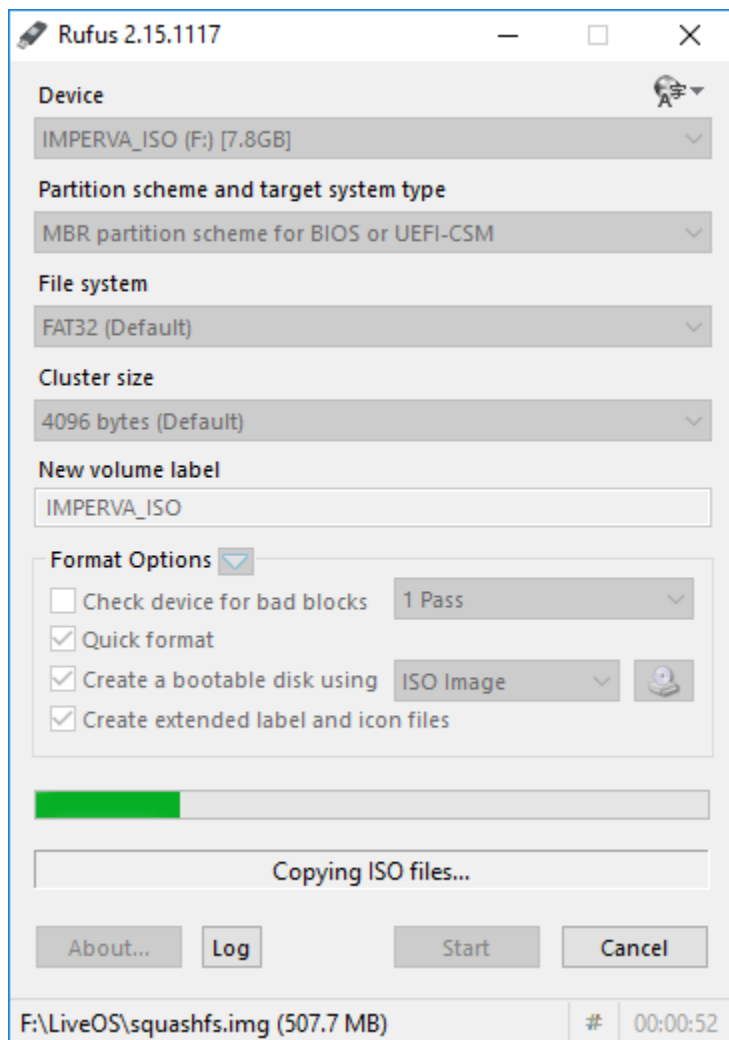
1. Download the image of the USB file: from the [Imperva Customer Portal](#), click **Downloads**, then navigate to `/Downloads/Imperva_WAF/Setup/v15/v15.3/Patch#/USB`
2. Download and install the **Rufus** executable from the Rufus website.
3. Attach your USB device to your computer.
4. Start the Rufus application. The **Rufus** window is displayed.



5. Configure the application as shown in the capture above.
6. Drag the image file you downloaded in step 1 into the Rufus window. The bottom row in the window should read **Using image: <image file name>**.
7. Click **Start**. A confirmation window is displayed.



8. Click **OK**. The bootable USB device creation begins.



9. When the bootable USB device creation finishes, click **Close** and remove the USB device. You can now use it to install the software.

Installing the Software on Virtual Machines

To install the software on virtual machines, you need to refer to the guide that is specific to your machines' environment.

Virtual machines

See:

- **KVM:** [WAF KVM Installation Guide](#)
- **VMWare:** [WAF VMWare Installation Guide](#)

Nutanix

If you are using Nutanix based on KVM, refer to the [WAF KVM Installation Guide](#).

If you are using Nutanix based on VMWare, refer to the [WAF VMWare Installation Guide](#).

Public cloud

See:

- **AWS:** [WAF on Amazon AWS Installation Guide](#)
- **GCP:** [WAF on Google Cloud Platform Installation Guide](#)
- **Azure:** [WAF on Microsoft Azure Installation Guide](#)

Working with Management Servers After the Installation

To work with Management Servers after installation, you need to connect and configure it using the GUI.

- [Connecting to the Management Server](#)

Connecting to the Management Server

To connect to the Management Server, on your browser go to <https://<IP address>:8083>, where <IP address> is the IP address of the management port on the MX.

Note: For more information, see the Quick Start Guide for your MX appliance.

- **M180:** [Performing First-Time Login](#)
- **M170:** [Performing First-Time Login](#)

Replacing Gateways

This section reviews information regarding replacing gateways and includes the following:

- Replacing a Faulty Gateway or Upgrading to a New Gateway Model

Replacing a Faulty Gateway or Upgrading to a New Gateway Model

You can replace a faulty gateway with a new gateway or upgrade to a new Gateway model without losing its configuration. You must take care to configure the new gateway so that it retains the original IP and listener. In this way, MX is unaware of the hardware change.

Notes:

- Make sure you obtain a new license that supports new gateway model. Otherwise, you will not be able to upgrade the model.
- If your MX is in delayed activation mode, please apply any pending changes in the MX GUI in **Main view > Action > Activate** before you proceed. For more information, see Enabling Activate Settings.

To replace a gateway:

1. Ensure the gateway you want to replace is disconnected from the MX and powered off.

Do not delete the gateway from the MX GUI.

2. Power on and connect the replacement gateway.
3. Once the replacement gateway is booted, log into the console using these credentials: user `admin`, password `admin`.
4. Change the default `admin` password as instructed.
5. Type `ftl` to begin the First-Time Login procedure (also referred to as FTL).
6. During First-Time Login, answer the following questions as shown:
 - Is this gateway deployed in order to replace another gateway? `[y/n]: enter y.`
 - Name of gateway to be replaced [Case sensitive]: enter the name of the old gateway to be replaced exactly as it appears in the WAF Gateway GUI, in the **Setup > Gateways** tab.
7. When prompted for the machine's management IP address, enter the same IP address as that of the gateway you are replacing.
8. Continue with login and provide the rest of the appliance parameters.

We recommend that the same values as the previous appliance are used where applicable.

On successful completion of the procedure, the new gateway's status is Connected.

Replacing a Faulty Gateway or Upgrading to a New Gateway Using an Already Registered Gateway

In the event where a new gateway was installed and registered, but has different parameters (hostname, IP) from the gateway it is intended to replace, carry out the following procedure.



Warning: Only use this procedure if you are using a gateway **that was already installed and registered**, to replace another gateway. If you use it on a new gateway, you will corrupt the new gateway. For instructions on replacing a faulty gateway or upgrading to a new model using a new gateway, see [Replacing a Faulty Gateway or Upgrading to a New Gateway Model](#).

To replace a faulty gateway using an already registered gateway:

1. Use SSH to connect to the new gateway.
2. Stop the gateway by entering `impctl gateway stop`.
3. Unregister the gateway by entering `impctl gateway unregister`.
4. In the WAF Gateway GUI, select **Setup > Gateways**.
5. Click on the gateway that you intend on removing and when it shows a Disconnected status, delete it by clicking **x**.

A confirmation dialog appears.

6. To confirm deletion, click **Yes**.
7. Re-register the gateway. Type `impctl gateway register --name-of-gateway-to-replace=<gw_name>`
8. Restart the gateway. Type `impctl gateway start`
9. Configure the IP address using `impcfg`. See [Configuring a Gateway](#). On successful completion of the procedure, the new gateway's status is Connected.

Configuring SecureSphere

This section describes the initial installation procedure for SecureSphere, and also the additional configuration steps required before SecureSphere can become operational, including:

- Initial Configuration
- Automating First Time Login
- Single Sign On (SSO)
- IPv6
- SecureSphere Users
- Appliance OS
- Gateway/Management Server Appliance Initial Configuration
- Configuring SecureSphere Using the Command Line Tools
- Configuring a Management Server
- Configuring a Gateway
- Configuring the Platform
- Configuring SecureSphere Routes
- Configuring Transparent Reverse Proxy
- Configuring the Management Server for High Availability
- Switching Deployments
- Configuring the MX to Communicate with SOM Across Borders
- Using QinQ

Initial Configuration

For instructions for the physical configuration of the SecureSphere appliance, see the **Getting Started Guide**. For instructions on automating first time login, see [Automating First Time Login](#).

Notes:



- For instructions on installing SecureSphere on VMWare or Public Clouds such as AWS or Azure, see the relevant Configuration Guide
- For instruction on installing and configuring a SecureSphere appliance as a SOM, see the **First Time Login** topic in the Quickstart guide for your appliance and the SOM User Guide
- For instruction on installing and configuring a SecureSphere appliance as a Management Server or a Gateway, see the **First Time Login** topic in the Quickstart guide for your appliance, the Admin Guide and the relevant User Guide

During the First Time Login, the administrator is required to define the following:

- The SecureSphere component type (Management Server only, SOM Server only, or Gateway only).
- Password for the appliance's admin user.
- For Management Servers, for the `system` user (the password for the internal SecureSphere database).
- Additional OS users (CLI users), who are able to run a restricted set of OS commands (see [CLI Users](#)).
- Appliance's management interface IP address (in CIDR format).

This interface is usually used for managing SecureSphere Gateways, so that the interface used to manage the Management Server is distinct from the interface the Management Server uses to communicate with the Gateways.

- Appliance's management interface (NIC) for out-of-band management.
- Appliance's hostname.
- Appliance's default Gateway.
- Appliance's operation mode (sniffing, bridge, *etc.*).
- For virtual appliances, the appliance model number.
- Time zone.

Once these parameters have been defined, the administrator defines the additional configuration parameters required to make the SecureSphere operational. The additional parameters which must be defined depend on the appliance's operational mode (Management Server, sniffing Gateway, bridge, *etc.*).



Note: A SecureSphere Management Server is also known as an **MX**.

Automating First Time Login

First time login is typically run using the CLI and is guided with a set of prompts. You can alternatively configure a script to run, thereby enabling you to automate the process. This may be useful when provisioning a number of SecureSphere gateways at once.

Notes:

- This procedure cannot be conducted to configure SecureSphere in bridge mode, it can only be used to configure the SecureSphere gateway in sniffing mode
- This procedure can only be conducted when running first time login on a gateway. It cannot be conducted to install a management server.

To run first time login:

- Use the following command:

```
impctl auto-ftl --management-interface=<value> --management-ip-with-
cidr=<value> --default-gateway=<value> --root-password=<value> --user-
name=<value> --user-password=<value> --host-name=<value> --set-sniffing --
```

```
secure-password=<value> --imperva-password=<value> --grub-password=<value> --
management-server-ip=<value> --appliance-type=<value>
```

Note the arguments listed above are mandatory arguments, whose descriptions are listed below.

You additionally have arguments for the following:

- [Mandatory Arguments](#)
- [General Optional Arguments](#)
- [Optional Listener Arguments](#)
- [Optional Sniffing Arguments](#)
- [Common Arguments](#)

Mandatory Arguments

This section lists mandatory arguments that need to be used when conducting first time login.

Argument	Description
--management-interface=<value>	Sets the management interface for this appliance. For example, eth0.
--management-ip-with-cidr=<value>	Sets the IP address of the management interface. IP address should be IPv4.
--default-gateway=<value>	Sets the IP address of the default gateway. IP address should be IPv4.
--root-password=<value>	Creates a password for the Linux root user.
--user-name=<value>	Creates an individual named account. This account will be used to access the CLI on the appliance. This is done to achieve a higher level of security by introducing an additional layer which provides improved auditing of the appliance users.
--user-password=<value>	Creates a password for the individual named account.
--host-name=<value>	Sets the host name. Host name will also be used as the Gateway name. If you enter a Fully Qualified Domain Name (FQDN) the leftmost part of the name will be used for hostname. For

Argument	Description
	example, if you enter mycomputer.mydomain.com the host name will be mycomputer.
--set-sniffing	Sets the SecureSphere gateway operation mode to sniffing.
--imperva-password=<value>	Configures an imperva password. This password is primarily used to register remote agents to the gateway.
--grub-password=<value>	Changes the bootloader password: Protecting the bootloader with a password is essential to denying unauthorized persons physical access to the console.
--management-server-ip=<value>	Sets the IP Address (IPv4) of the Management Server. The Management Server must be accessible over the management interface of this Gateway.
--appliance-type=<value>	Sets the SecureSphere gateway model. For example x2500, x4510, v2510, v4500, etc.

General Optional Arguments

This section lists general arguments that can to be used when conducting first time login (optional).

Argument	Description
--root-allowed-ips=<value>	Sets IP addresses from which users can login as root user.
--gateway-group=<value>	Sets the gateway group name.
--timezone=<value>	Sets the time zone. You must use Posix TZ format. For example, GST-10 is a zone named GST that is 10 hours ahead (east) of UTC.

Argument	Description
--time-servers=<value>	Sets the IP address of NTP servers. Must be an IPv4 address. To add more than one NTP server IP address, separate IP addresses with a space.
--dns-servers=<value>	Sets the IP address of the DNS name servers. Must be an IPv4 address. To add more than one name server, separate name server IP addresses with a space.
--dns-domain=<value>	Sets the DNS domain name.

Optional Listener Arguments

This section lists arguments specifically related to configuring a listener that can to be used when conducting first time login (optional).

Argument	Description
--listener-ip-with-cidr=<value>	Sets the IP Address of the listener. Use a [IPv4 address/CIDR].
--listener-port=<value>	Sets the port of the listener.
--listener-interface=<value>	Sets the network interface of the listener. For example, eth1.
--listener-ssl=<value>	Enables SSL. The SecureSphere gateway can be configured to utilize SSL to encrypt the remote agent listener. Note this option may increase CPU consumption on the Agent host. Choose true if you would like to enable SSL (otherwise - choose false).

Optional Sniffing Arguments

This section lists arguments specifically related to configuring sniffing or blocking options that can to be used when conducting first time login (optional).

Argument	Description
--sniffing-interface=<value>	Sets the network interface for sniffing. For example, eth1.
--blocking-interface=<value>	Sets the network interface for blocking. For example, eth1.
--blocking-network=<value>	Sets/modifies the subnet used for blocking.

Common Arguments

This section lists options that can be used to set the location of the command log or display help when conducting first time login (optional).

Argument	Description
--log-file=<VALUE>	Changes the command's log file location (default- /var/log/auto-ftl.log).
--help	Displays help for first time login.

Single Sign On (SSO)

This topic explains how to configure and use Single Sign-On (SSO) functionality in the Management Server (MX). The SSO implementation uses SAML 2.0 protocol to enable secure authentication between your Identity Provider (IdP) and the MX, which acts as the Service Provider (SP).

Benefits

- Enhanced security through centralized authentication
- Simplified user experience with single sign-on capability
- Reduced password management overhead
- Standardized authentication process using SAML 2.0
- Improved audit and compliance capabilities

Key Components

- **Management Console (MX):** Acts as the Service Provider (SP) in the SAML authentication flow

-
- **Identity Provider (IdP):** Your organization's chosen authentication system (e.g., Okta, Azure AD, OneLogin)
 - **SAML 2.0:** The underlying protocol that enables secure exchange of authentication and authorization data

Authentication Flow

1. User attempts to access the MX
2. MX redirects the user to the configured IdP
3. User authenticates with the IdP
4. IdP validates credentials and sends SAML response to MX
5. MX validates the SAML response and grants access

Current Implementation Scope

In this phase of the SSO implementation:

- Users must be provisioned in both systems:
 - The Management Console (MX)
 - Your Identity Provider (IdP)
- User attributes and access rights are managed independently in both systems
- The MX supports configuration of one IdP through the MX configuration interface
- Authentication is handled by the IdP while authorization is still managed within MX

Prerequisites

Before implementing SSO, ensure you have:

1. Administrative access to your Identity Provider
2. Administrative access to the Management Console
3. Connectivity from the MX Host to the IdP
4. User information ready for both systems
5. SAML configuration details from your IdP:
 - IdP Entity ID
 - SSO URL
 - X.509 Certificate

To enable SSO login:

In the MX:

1. In the Admin workspace, go to **System Definitions**.
 2. From the **System Definitions** pane, select **SSO Settings**. The SSO Settings pane appears on the right.
 3. Select the **SSO Enabled** check box. If not enabled, sign in via SSO attempts will fail with failure page that will mention that SSO is disabled.
 4. Click . The Create Identity Provider dialog box is displayed.
 5. Fill in the fields:
 - **Name** – Enter any name that will appear in the MX
 - **Registration ID** – Enter an Arbitrary value the MX (Service Provider) will be identified at the IdP, this value will be needed to configure the IdP.
 - **Remote IdP entity ID** – Provided by the IdP
 - **SSO URL** – Provided by the IDP
-

6. After creating the IDP, user needs to register the IDP in order to make SSO login operational.
7. Click **Create**.
8. Click **Register Identity Provider**. The Upload Verification Certificate dialog box is displayed.
9. Click **Choose File** and navigate to your IDP certificate file and select it.
10. Click **Upload & Register**. The Status field changes to **Registered**.

In the IDP:

1. Set **Assertion consumer URL**, which is the endpoint of the Assertion we get from IDP and send to the SP (MX) to: **https://[MX_IP]:8083/SecureSphere/login/saml2/sso/[registration id]**
2. Set **Audience Restriction** (sometimes maybe referred to as SP Entity ID, Service Provider Entity ID, Audience, etc), which is the value sent by the IDP and the SP (MX) and checks that the assertion sends the value that it expects (hence it is the right audience) to **https://[MX_IP]:8083/SecureSphere/saml2/service-provider-metadata/[registration id]**

Notes:

- You can unregister the IDP without deleting it in case we want to disable it with the **Unregister Identity Provider** button. It causes the IDP not to be operational, and to the “SIGN IN WITH SSO” button to appear disabled.
- IDP can be registered again, and even if IDP is already registered, it can be registered again (in order to maybe upload a new IDP verification certificate).

IPv6

This section describes IPv6 support, which is implemented in SecureSphere.

Notes:



- IPV6 configuration is performed using the `impcfg` and `impcli` commands only.
- Additional information regarding IPv6 support is given in [IPv6 Support](#).

SecureSphere Features Which Support IPv6

SecureSphere supports IPv6 in the following SecureSphere features:

- WAF
- MX-Gateway communications
- VRRP

- communications with external systems which support IPv6 (see [External Systems](#))

Overview

The section provides an overview of IPv6 support. It reviews the following:

- [IPv6 Address Formats](#)
- [CIDR Notation](#)

IPv6 Address Formats

IPv6 addresses are 128 bits long and are written in eight groups of four hexadecimal digits separated by colons, for example, **2001:0db8:85a3:0000:0000:8a2e:0370:7334**. The hexadecimal digits are case-insensitive.

An IPv6 address can be abbreviated according to following rules:

- Leading zeroes within a 16-bit value may be omitted. For example, the address **fe80:0000:0000:0202:b3ff:fe1e:8329** may be written as **fe80:0:0:202:b3ff:fe1e:8329**.
- A single occurrence of any number of consecutive groups of zeroes within an address may be replaced by a double colon. For example, **fe80:0:0:202:b3ff:fe1e:8329** becomes **fe80::202:b3ff:fe1e:8329**.

So the same IPv6 address can be represented in several different ways. For example, the following are all equivalent:

2001:db8:0:0:1:0:0:1	2001:0db8::1:0:0:1
2001:0db8:0:0:1:0:0:1	2001:db8:0:0:1::1
2001:db8::1:0:0:1	2001:db8:0000:0:1::1
2001:db8::0:1:0:0:1	2001:DB8:0:0:1::1

Whenever an IPv6 address is entered into SecureSphere, after it is saved it is displayed in the "short" format.



Note: SecureSphere is able to recognize the equivalence of these different forms of the same IPv6 address, except in certain cases where the IPv6 address is treated as a character string, for example, in lookup datasets and in ThreatRadat.

CIDR Notation

Classless Inter-Domain Routing (CIDR) notation explicitly specifies the number of bits in an IP address denoting the network, in contrast to the netmask notation which indirectly specifies the number of bits. For example, a netmask of **255.255.255.0** indicates (for IPv4 addresses) a network in which the first 24 bits of an IP address denote the network. In CIDR notation, one simply writes "/24" after the IP address (for example, **123.124.125.126/24**). The advantages of CIDR notation over the netmask notation are simplicity and scalability from IPv4 to IPv6.

The IP address/netmask notation is no longer permitted for both IPv4 and IPv6 addresses. Instead, CIDR notation is used both in the SecureSphere GUI and `impcfg`.



Note: CIDR notation is used only when specifying a network or when configuring a local IP address where it is necessary to specify the subnet. CIDR is not to be used when specifying a remote IP address, for example, that of an NTP server or a DNS server, or when defining the Management Server of a SecureSphere Gateway.

When `impcfg` requires that CIDR notation be used, it specifies so in the prompt, for example, "Enter IP address/CIDR". If "CIDR" is not specified in the prompt, it must not be entered.

IPv4 / IPv6 Dual Stack

The source and destination IP addresses of an IP packet must both be either IPv4 or IPv6. It is not possible for one of the addresses to be IPv4 and the other to be IPv6. The solution is to assign to a device both an IPv4 and an IPv6 address, so that the device can communicate on both protocols.

So, for example, if an MX manages both IPv4 and IPv6 Gateways, the MX's management or LAN interface (whichever one the Gateways communicate with) must have both an IPv4 and an IPv6 address.

At various points in the SecureSphere GUI and in `impcfg`, users are prevented from trying to define an IPv4-IPv6 connection.

SecureSphere MX and Gateway

SecureSphere Management Servers and Gateways can have IPv6 addresses. An MX can simultaneously manage Gateways with IPv4 addresses together with Gateways with IPv6 addresses, if the MX's management interface or its LAN interface has both an IPv4 and an IPv6 address.

As a SecureSphere GUI user, you can point your browser to an IPv6 SecureSphere MX.

IPv6 is supported for Gateway syslog, MX syslog and MX-Gateway communications.

Hostnames

When entering a hostname, the ":" is not allowed.

For example:

[Short or FQDN (separated by .) hostname, where each part starts with a letter and consists of the following: letters, digits, _ or -]:

[Limitation of hostname]

The first letter is an English letter (not digit) = Yes

Consists of letters, digits, "_" and "-". = Yes

Short hostname example = host_name-123

IPv6 in `impcfg` and `impctl`

The following points are relevant specifically to `impcfg` and `impctl`.

1. `impcfg` and `impctl` accept both IPv4 and IPv6 addresses, for example for DNS and NTP servers. In all cases where IPv6 is supported (note the exceptions below) the user can assign to the same entity:
 - both IPv4 and IPv6 addresses
 - an IPv4 address only
 - an IPv6 address only
 - An IPv6 address can be specified for external servers (for example, DNS or NTP), only if either the appliance's management or LAN interface has an IPv6 address.
2. When `impcfg` asks the user to enter an IP address for a management or LAN interface, it allows the user to enter either an IPv4 or IPv6 address, and then asks the user if they want to enter an address of the other type (so, for example, if they enter an IPv4 address, they will be asked if they want to enter an IPv6 address, and they can either enter one or not enter one).
3. As noted above, network masks have been replaced by CIDR notations for both IPv4 and IPv6 addresses. While in the past, users were asked to enter an IP address/netmask pair (for local IP addresses and network addresses - for example, when specifying a static route - and for sniffing mode blocking masks), users are now asked (for both IPv4 and IPv6 addresses) to enter an IP address in CIDR notation.

For example, instead of entering an IP address of **144.234.123.0** and a netmask of **255.255.255.0**, the user enters only one field: **144.234.123.0/24**.



Note: CIDR notation is used only when specifying a network or when configuring a local IP address where it is necessary to specify the subnet. CIDR is not to be used when specifying a remote IP address, for example, that of an NTP server or a DNS server, or when defining the Management Server of a SecureSphere Gateway.

When `impcfg` requires that CIDR notation be used, it specifies so in the prompt, for example, "Enter IP address/CIDR". If "CIDR" is not specified in the prompt, it must not be entered.

4. Link-local addresses (**fe80::/10**) are not supported and cannot be entered anywhere in `impcfg`.
5. Default Gateway - this is unchanged (for both IPv4 and IPv6), that is, the user is not required to specify the device (interface).
6. VRRP routes (Reverse Proxy, - `impcfg` tries to deduce the device (interface) from the aliases, and if it cannot, it asks the user.
7. `portguard` supports IPv6. Access to the appliance's link-local addresses is blocked.
8. When configuring a SecureSphere Gateway, the user will not be able to specify an IPv6 management interface if the MX management and LAN interfaces both have only IPv4 addresses. If the Gateway's management or LAN interface has both IPv4 and IPv6 addresses, then the communication will depend on whether the user specifies the IPv4 or IPv6 address as the management interface on the Gateway.
9. `ping6` is not blocked for SecureSphere appliances (as `ping` is).

Exceptions

The following exceptions apply:

1. IPv6 is not supported in MX-HA.
2. IPv6 VRRP static routes are not supported.
3. The following commands have been deprecated:
 - `impctl vrrp gateway show`
 - `impctl vrrp gateway config`

External Systems

SecureSphere MX can communicate over IPv6 with the external systems listed below:

External System	
HTTP Proxy	Admin > System Definitions > External HTTP Settings
SMTP	SMTP servers with IPv6 addresses.

External System	
SNMP	SNMP nodes with IPv6 addresses.
LDAP endpoint access	LDAP servers with IPv6 addresses.
LDAP authentication	
FTP archive	Archiving audit files to an IPv6 archive via FTP.
SCP archive	Archiving audit files to an IPv6 archive via SCP.
Remote Access to Infra to IPv6	Enables remote access to servers, to run shell scripts remotely.

SecureSphere MX **cannot** communicate over IPv6 with the external systems listed below:

External System		Reason
Oracle Server	Admin > System Definitions > External Systems	This product is not yet IPv6 compliant.
MSSQL Server		
MySQL Server		
SQL		
HTTP/HTTPS archive	Admin > System Definitions > Action Interfaces	
BMC Remedy		This product is not yet IPv6 compliant.

SecureSphere Users

Several users are created for a SecureSphere appliance, each for a different purpose. Each of these users has its own password.

SecureSphere Users and Passwords

SecureSphere User	Description
CLI Users	
root	This is the all-powerful superuser of the appliance's Linux-based OS. This user is created automatically by SecureSphere when it is installed. Note: You cannot login to the appliance as root over SSH. To administer the appliance remotely, login as a CLI user other than <code>root</code> and <code>secure</code> over SSH and enter the <code>admin</code> command. For more information, see CLI Users. For information about changing this user's password, see Configuring the Platform.
Other CLI users	These are OS users, at least one of whom must be defined during the First Time Login, who are allowed to run a restricted number of OS commands in order to administer the appliance. The sessions of other CLI users (but not of <code>root</code> and <code>secure</code>) are logged to the file <code>/var/log/secure</code> to enforce accountability. For information on logging these sessions to <code>syslog</code>, see the Imperva Knowledge Base. For more information, see CLI Users.
admin	Default user in the <code>impcli</code> shell environment. The default password for this user is admin. Users are required to change the password when logging in for the first time (not FTL).
SecureSphere GUI users	
These are the users of the SecureSphere GUI, some of whom are defined within SecureSphere as administrators. One user ("admin") is pre-defined and has complete read-write privileges, including that of defining other SecureSphere GUI administrators and users.	

SecureSphere User	Description
For more information, see Users and Permissions .	
SecureSphere Database users	
system	This is the user of the internal SecureSphere database, and is required only on a SecureSphere Management Server. This user is created automatically by SecureSphere when it is installed. For information about changing this user's password, see Configuring a Management Server.

Appliance OS

The appliance OS is based on a "hardened" version of Linux.

CLI Users

At least one other CLI user (in addition to the CLI users created automatically by SecureSphere: `root`, and `system`) must be defined after the First Time Login. You can define any number of additional CLI users using `impcfg`.

Non-root and non-admin CLI users are allowed to run only the following OS commands:

- `impcfg` - This command opens a command line utility that enables you to configure the machine (i.e. GW/MX/SOM) you installed.
- `passwd` - This command enables changing the password of the user that is logged in.
- `admin` - This command enables you to login as the admin user.

All commands issued by CLI users (except `root`) are logged to the file `/var/log/secure`. For information on logging these sessions to `syslog`, see the Imperva Knowledge Base.

Note:



- By default, you cannot connect to the appliance as `root` or `secure` over SSH. To login as `root`, you must first connect as a CLI user and use the `admin` command. However, you can specify an IP address from which user `root` is allowed to login over SSH using the following command:

```
impctl hardening config --root-source-ip-exception=<IP
address>
```

- A CLI user name is limited to a maximum length of 14 characters.
- Only lowercase letters can be used to create CLI user names. Uppercase is not supported.

Authentication Failure

CLI users other than `root` and `secure` who fail authentication 6 times are locked out for 30 minutes. An administrator can release the lock before this time using `impcfg` (see [Manage User](#)).

Passwords

Make sure your password has the following characteristics:

- It must have no fewer than 7 characters and no more than 14 characters.
- It must have at least one number, one capital letter, and one special character from:

```
* + = # % ^ : / ~ . , [ ] _ \ ( ) | ; @ $ & - ? { } < >
```

- It cannot have more than two characters repeated in succession.

Warning: It is of the utmost importance that all these passwords be "strong" (that is, difficult to guess), and that they be treated with discretion.

Passwords for other CLI users (not those of `root`) expire after 90 days. After the 90 day period, users trying to log in are requested to change the password immediately before they can continue.

Both GUI and CLI passwords are hashed using SHA-2.



Note: When changing user or system passwords, it is highly recommend to only use `impctl/impcli` commands and not the `passwd` command, as this may result in undesirable issues at a later time.

Timeout

Shell sessions for all users are timed out after 15 minutes.

Date Changes

Whenever the system date is changed, a log event is sent to syslog, so that administrators are notified.

Locally, the events are stored in `/var/log/secure`.

Gateway/Management Server Appliance Initial Configuration

When configuring an appliance for the first time (the first-time login), the `impcfg` wizard guides you through a question and answer procedure. After confirming that all the information is correct, this initial configuration is applied to the system components using the `impctl` application. This process takes between 45 minutes and 1 hour for a Management Server, where a database must be created and initialized. On a Gateway the process takes between 10-15 minutes.



Note: The instructions for the initial configuration (first-time login) are given in the appliance's Quick Start Guide, which is packaged with the appliance.

- If you configure a DNS client during the first-time login, make sure you specify the IP address of a real DNS server.

Additional configuration is performed using the menu-driven `impcfg` command line application. For more information, see [Configuring SecureSphere Using the Command Line Tools](#).

If you run the First-Time configuration script after the appliance has already been configured, the previous configuration is overwritten.

First-Time Login (Configuration)

The First-Time Login is usually via the console port. Once the appliance has a management IP address, further interactions can be via the console, or SSH. Make sure that SSH communication is possible for remote appliance configuration (port 22 on the firewall must be open).



Note: You cannot login to the appliance as `root`, `secure`, or `system` over SSH.

For more information on which firewall ports must be open, see [Configuring Firewall Ports](#).

If you are configuring a Management Server and a Gateway, configure the Management Server first. The reason is that when a Gateway is configured, the Management Server must be running and reachable. If not, the Gateway must be restarted when the Management Server is alive and reachable. Check the Gateway status from the Gateway Management screen (see [Configuring a Gateway](#)).

After the First-Time Login, you can configure SecureSphere using the GUI by pointing the browser to `https://<IP address of MX>:8083`.



Note: During the First Time Login procedure for a Gateway, you must specify a password for the "imperva" user when asked to do so. This step is NOT optional.

Registering a Gateway

In order to establish the relationship between a SecureSphere Gateway and its Management Server, you must register the Gateway to the Management Server. Registration is part of the First Time Login procedure for the Gateway, during which you will be asked to specify the IP address of the Management Server. For more information, see [Initial Configuration](#) and [Configuring a Gateway](#).



Note: You cannot specify a CIDR address for a SecureSphere Gateway's Management Server. For more information about when CIDR can be used, see [CIDR Notation](#).

You can unregister a Gateway from one Management Server and register it to a different Management Server using the `impcfg` application. For more information, see [Configuring a Gateway](#).

Configuring SecureSphere Using the Command Line Tools

There are two command line interface (CLI) tools with which you can configure SecureSphere on the Management Server and Gateways:

-
- `impcfg` is a menu-based configuration tool that can be used to configure both the Management Server and the Gateway.

`impcfg` is described in [Command Line Interface](#).

- `impctl` is a lower level configuration tool. `impcfg` runs `impctl` on the SecureSphere appliance to deploy the configuration which the administrator defines in `impcfg`.

`impctl` is described in [Command Line Interface](#).

Command Line Tools and MX-HA

This section describes command line tools and how to use them in an MX-HA environment. It includes the following:

- [Changing MX Properties in an MX-HA Environment](#)
- [Stopping and Starting the MX in an MX-HA Environment](#)

Changing MX Properties in an MX-HA Environment

In an MX-HA environment, `impcfg` is not supported, and `impctl` supports the following commands:

- `server ha change-ip`
- `server ha change-vip`
- `server ha config`
- `server ha install`
- `server ha password`
- `server ha preparerpm`
- `server ha show`
- `server ha start`
- `server ha startall`
- `server ha status`
- `server ha stop`
- `server ha stopall`
- `server ha uninstall`

However, you can perform the following changes in an MX-HA environment:

- change the MX's network properties, such as its IP addresses and interfaces (devices).
- change the MX's hostname

To change the MX properties in an MX-HA environment:

1. Uninstall MX-HA on both the primary and secondary MXs. For more information, see [Register the Gateway](#).

2. Make the necessary changes.
3. Reinstall MX-HA on both the primary and secondary MXs. For more information, see [Installing Management Server High Availability \(MX-HA\)](#).

Stopping and Starting the MX in an MX-HA Environment

The correct way start and stop the MX in an MX-HA environment is by using the following commands on the Primary and Secondary servers:

```
impctl server ha start
```

```
impctl server ha stop
```

Warning: Do not use `impcfg` or `impctl` to start and stop the MX as you would in a non-MX-HA environment (as described in [Configuring a Management Server](#)). Use only the above commands.

Configuring a Management Server

You can:

- Activate/deactivate the Management Server
- Start/stop/reactivate the Management Server
- Change management and database passwords

Notes:



- In the procedure below, you use **impcfg**. For more information, see [impcfg](#).
- The management and database passwords cannot be changed from the OS command line.

To configure the Management Server:

1. In the **Top** Screen, select option **1) Manage SecureSphere Management Server**.

The **Server Management** screen is displayed.

```

-----
SecureSphere 14.7.1.80_0 - Server Management Screen
-----

Configuration target:      local (appliance, VM150, reachable)
Management Server status: running,Ready
Setting markers:          C: changed, I: invalid, P: pending (saved but not applied)
Navigation:               Top → Management Server

1) Restart Management Server (immediate).
2) Stop Management Server (immediate).

3) Manage 'secure' user password.
4) Manage 'system' user (database) password.

e) End this level. j) Jump to a previous level. t) Top level.
q) Quit (discarding not-saved changes).

Your choice: █

```

2. Select one of the following options, as listed below.

Appliance Management Configuration Options

Option	Name	Description
1	Restart Management Server (immediate)	Stop and then start the Management Server.
2	Stop Management Server (immediate)	Stop the Management Server.
3	Manage 'secure' user password	Change the "secure" SecureSphere administrative user used to register the SecureSphere Gateway with a SecureSphere Management Server. Note: Make sure your password has the following characteristics: - It must have no fewer than 10 characters and no more than 14 characters. - It must have at least one number, one capital letter, and one special character from: <pre>* + = # % ^ : / ~ . , [] _</pre> - It cannot have more than two characters repeated in succession.

Option	Name	Description
		This option restarts the Management Server and is not supported in Management Server High Availability deployments. For more information about Management Server High Availability, see Management Server High Availability (MX-HA). For more information about passwords, see SecureSphere Users.
4	Manage 'system' user (database) password	Change the password of the internal SecureSphere database. Note: Make sure your password has the following characteristics: ▪ It must have no fewer than 10 characters and no more than 14 characters. ▪ It must have at least one number, one capital letter, and one special character from: <code>* + = # % ^ : / ~ . , [] _</code> ▪ It cannot have more than two characters repeated in succession. This option restarts the Management Server and is not supported in Management Server High Availability deployments. For more information about Management Server High Availability, see Management Server High Availability (MX-HA). For more information about passwords, see SecureSphere Users.

Configuring a Gateway

You can:

- Activate/deactivate, register/unregister, start/stop/reactivate the Gateway
- Change the Gateway name, server address/password, and operation mode
- Manage HSM, high availability, SecureSphere Agents, and interfaces



Note: When a Gateway is registered to an MX, the Gateway sends the MX information about its network interfaces, and users may configure the Gateway on the basis of that information. If any of the Gateway network interfaces is deleted or removed, the MX's Gateway configuration based on the deleted interface becomes invalid. The user must then either restore the network interface on the Gateway or modify the Gateway configuration on the MX accordingly.

To configure the Gateway:

1. In the **Top** Screen, select the option **Manage SecureSphere Gateway**. The **Gateway Management** Screen is displayed.

```
-----
SecureSphere 14.4.0.16_0 - Gateway Management Screen
-----

Configuration target:      local (appliance, V2500, reachable)
Gateway status:           registered,running
Setting markers:          C: changed, I: invalid, P: pending (saved but not applied)
Navigation:               Top -> Gateway

Gateway settings
Gateway name:             TW_GW
Gateway mode:             Bridge STP
Advanced Bridge:          Enabled
Server address:           10.100.67.66
'secure' user password:   <is-set>
Cluster Configuration:    Disable

1) Perform actions (start, stop, etc.).
2) Change gateway name.
3) Change Management Server address/password.
4) Change operation mode.
5) Manage hardware security modules (HSM).
6) Manage remote agents.
7) Manage interfaces and routes.
8) Change virtual gateway model.

e) End this level. j) Jump to a previous level. t) Top level.
q) Quit (discarding not-saved changes).

Your choice: █
```

2. Select one of the following options, as listed below.

Gateway Management Configuration Options

Option	Name	Description
1	Perform actions (start, stop, etc.)	Start, stop or unregister the Gateway.
2	Change gateway name	Change the Gateway's name.

Option	Name	Description
		This option is used, for example, when a Gateway is moved to another network.
3	Change Management Server address/password	Specify a different Management Server for the Gateway. This option is used, for example, when a Gateway is moved to another network, and will be managed by a different Management Server. Note: You cannot specify a CIDR address for a Gateway's Management Server. For more information about when CIDR can be used, see CIDR Notation. When you apply this change, the Gateway registers with the Management Server.
4	Change operation mode	Change the mode of the Gateway to one of the following: ▪ sniffing: For more information, see Configuring a Gateway as a Sniffing Gateway. ▪ bridge STP: For more information, see Configuring a Gateway as a Bridge. ▪ bridge IMPVHA: For more information, see Configuring a Gateway as a Bridge. ▪ Reverse Proxy: For more information, see Configuring a Reverse Proxy. See also: Topologies and Deployment.

Gateway Actions

You can:

- Register/unregister the Gateway
- Restart the Gateway (soft or hard restart)
- Stop the Gateway
- Refresh the Gateway's status

To perform Gateway actions:

1. In the Top Screen, select option 2) Manage SecureSphere Gateway.
2. In the Gateway Management screen, select 1) Perform actions (start, stop, etc.).

```

-----
SecureSphere 13.3.0.0_0 - Gateway Management Screen
-----

Configuration target:      local (appliance, V2500, reachable)
Gateway status:           registered, running
Setting markers:          C: changed, I: invalid, P: pending (saved but not applied)
Navigation:               Top -> Gateway -> Actions

Gateway settings
Gateway name:             TW_GW_13
Gateway mode:             Reverse Proxy
Server address:           10.100.67.66
'secure' user password:   <is-set>
Cluster Configuration:    Disable

1) Unregister gateway (immediate).
2) Soft gateway restart (immediate).
3) Hard gateway restart (teardown/prepare resources) (immediate).
4) Stop gateway (immediate).
5) Refresh gateway status (immediate).

e) End this level. j) Jump to a previous level. t) Top level.
q) Quit (discarding not-saved changes).

Your choice: █

```

3. Select one of the following options, as listed below.

Gateway Actions Options

Option	Name	Description
1	Register / Unregister gateway (immediate)	Register or unregister the Gateway from the Management Server.
2	Soft gateway restart (immediate)	Performs a controlled restart of the gateway's software services without rebooting the underlying appliance.
3	Hard gateway restart (immediate)	Forces the gateway to shut down and turn back on without waiting for active tasks to finish.
4	Stop gateway (immediate)	Forces the gateway to terminate instantly, bypassing normal shutdown procedures.

Option	Name	Description
5	Refresh gateway status (immediate)	Forces the gateway to instantly perform a re-check of its connectivity, and configuration.

Configuring a Gateway as a Bridge

To configure a Gateway as a bridge:

1. In the **Gateway Management** screen (as described in [Configuring a Gateway](#)), select option **Change operation mode**.
2. From the list of operation modes (shown below), select one of the bridge modes:
 - Bridge STP or
 - Bridge IMPVHA.

```

The SecureSphere gateway operation modes are:

1) Sniffing.
2) Bridge STP.
3) Bridge IMPVHA.
4) Reverse Proxy.

Operation mode: █

```



Note: Bridge STP mode does not correctly pass Cisco PVST PDU packets. Use Bridge IMPVHA mode if you need to pass these packets.

An option to enable Advanced Bridge is presented.

```

Would you like to enable Advanced Bridge?

1) Yes.
2) No.

Your choice: █

```

3. Select whether to enable Advanced Bridge or not. The **Gateway Management** screen appears.

```

-----
SecureSphere 14.4.0.16_0 - Gateway Management Screen
-----

Configuration target:      local (appliance, V2500, reachable)
Gateway status:           registered,running
Setting markers:          C: changed, I: invalid, P: pending (saved but not applied)
Navigation:               Top -> Gateway

Gateway settings
Gateway name:             TW_GW
Gateway mode:             Bridge STP
Advanced Bridge:          Enabled
Server address:           10.100.67.66
'secure' user password:   <is-set>
Cluster Configuration:    Disable

1) Perform actions (start, stop, etc.).
2) Change gateway name.
3) Change Management Server address/password.
4) Change operation mode.
5) Manage hardware security modules (HSM).
6) Manage remote agents.
7) Manage interfaces and routes.
8) Change virtual gateway model.

e) End this level. j) Jump to a previous level. t) Top level.
q) Quit (discarding not-saved changes).

Your choice: █

```

4. In the **Gateway Management** screen (as described in [Configuring a Gateway](#)), select option **Manage interfaces and routes**.

```

-----
SecureSphere 14.4.0.16_0 - Bridges Configuration Screen
-----

Configuration target:      local (appliance, V2500, reachable)
Setting markers:          C: changed, I: invalid, P: pending (saved but not applied)
Navigation:               Top -> Gateway -> Interfaces

Gateway name:             TW_GW
Gateway mode:             Bridge STP
Server address:           10.100.67.66

Data interfaces:          eth1
Used interfaces:           eth2 eth3
Free interfaces:          eth1

STP Bridges:              br0: eth2 <-> eth3, mtu=1500, high-availability=off, stp=on

1) Create bridge.
2) Delete bridge.
3) Set bridge MTU.
4) Toggle STP on a bridge (Disabling STP will disable HA as well).
5) Toggle high-availability on a bridge (Enabling HA will enable STP as well).

e) End this level. j) Jump to a previous level. t) Top level.
q) Quit (discarding not-saved changes).

Your choice: █

```

5. In the **Bridges Configuration** screen, select one of the following options, as listed below.

Bridges Configuration Options

Option	Name	Description
1	Create bridge	Create a bridge on available interfaces.
2	Delete bridge	Delete an existing bridge.
3	Set bridge MTU	Define the Maximum Transmission Unit (MTU) in octets for a specified bridge.
4	Toggle STP on a bridge	Select an existing bridge and toggle STP on it. For more information, see Topologies and Deployment. Note: When disabling STP, High Availability is disabled as well). This option is only available when the gateway is in STP mode.
5	Toggle high availability on a bridge	Select an existing bridge and toggle High Availability on it. For more information, see Gateway High Availability.

Notes:

- In IMPVHA Bridge mode, SecureSphere gateways allow LACP messages to pass through the bridge interfaces, though it doesn't otherwise support the protocol
- If the Gateway is not in a High Availability deployment, it is recommended that the Gateway be configured as Bridge IMPVHA with HA disabled.
- SecureSphere supports jumbo frames up to 9600 bytes.

Configuring a Gateway as a Sniffing Gateway

Warning: The Management Server must be running when you execute this command, otherwise the Gateway will switch to bypass mode.

To configure a sniffing Gateway:

1. In the **Gateway Management** screen (shown in [Configuring a Gateway](#)), select option **4) Change operation mode**.
2. From the list of operation modes (as described in [Configuring a Gateway as a Bridge](#)), select **1) Sniffing**.
3. Enter **j** and hit **enter** to jump to the previous level. The **Gateway Management** screen appears.
4. In the **Gateway Management** screen (shown in [Configuring a Gateway](#)), select option **7) Manage interfaces and routes**.

```

-----
SecureSphere 13.3.0.0_0 - Sniffing Configuration Screen
-----

Configuration target:      local (appliance, V2500, reachable)
Setting markers:          C: changed, I: invalid, P: pending (saved but not applied)
Navigation:               Top -> Gateway -> Interfaces

Allowed devices:          eth1
Interfaces:               Sniffing:      <not-set>
                          Blocking:      <not-set>
                          Blocking net:  <not-set>

1) Set sniffing interface(s).
2) Set blocking interface.
3) Add/modify blocking net.

e) End this level. j) Jump to a previous level. t) Top level.
q) Quit (discarding not-saved changes).

Your choice: █

```

5. Select **1) Set sniffing interface(s)**.
6. In the **Sniffing Configuration** screen, select one of the following options, as listed below.

Sniffing Configuration Options

Option	Name	Description
1	Set sniffing interface(s)	Specify the interface(s) to be used for sniffing. Use a space to separate multiple interfaces, e.g. eth0 eth1
2	Set blocking interface	Specify the interface to be used for blocking, that is, the interface from which SecureSphere sends RST packets.

Option	Name	Description
3	Add/modify blocking net	Specify the following for the blocking interface: - the blocking device (one of the interfaces) - the blocking interface IP address - the blocking interface IP net

Configuring the Platform

You can:

- Manage network and time settings, and users
- Reboot/shutdown the appliance

To configure the platform:

- In the **Top** Screen, select **Manage platform**. The **Platform Management** screen is displayed.

```

-----
SecureSphere 13.3.0.0_0 - Platform Management Screen
-----

Configuration target:      local (appliance, V2500, reachable)
Setting markers:          C: changed, I: invalid, P: pending (saved but not applied)
Navigation:               Top -> Platform

Network settings
Management Interface:     device=eth0,IPv4=10.100.67.65/24,IPv6=<not-set>
LAN Interface:            device=eth1,IPv4=1.1.1.1/24,IPv6=<not-set>
Default gateway:          IPv4: 10.100.67.254,device=eth0
                           IPv6: <not-set>

Time settings
Time zone:                Asia/Jerusalem
Time servers:              10.2.16.66
                           10.2.16.58

1) Manage network.
2) Manage time.
3) Manage users.
4) Manage host name.
5) Manage bootloader password.
6) Manage SSH keys.

7) Reboot the appliance (immediate).
8) Shutdown the appliance (immediate).

e) End this level. j) Jump to a previous level. t) Top level.
q) Quit (discarding not-saved changes).

Your choice: █

```


2. Select one of the following options, as listed below.

Platform Management Screen Configuration Options

Option	Name	Description
1	Manage network	Change the management and LAN interfaces, the default Gateway, <i>etc.</i> For more information, see Network Management .
2	Manage time	Change the time zone, date and/or time, or configure NTP on the appliance. For more information, see Time Management .
3	Manage users	Manage CLI users. For more information, see Users Management .
4	Manage host name	Manage the appliance host name. For more information, see Hostname Management .
5	Manage bootloader password	Manage the bootloader (GRUB - Grand Unified Bootloader) password. For more information, see Bootloader Password Management .
6	Reboot the appliance (immediate)	Reboot the appliance.
7	Shutdown the appliance (immediate)	Shutdown the appliance.

Network Management

In the **Network Management** screen, you can:

- Specify the management interfaces, and default Gateway
- Configure the DNS client

- Locate network devices

To manage network settings:

1. In the **Platform Management** Screen (see [Configuring the Platform](#)), select **1) Manage network**. The **Network Management** screen is displayed.

```
-----
SecureSphere 14.7.0.170_0 - Network Management Screen
-----

Configuration target:      local (appliance, VM150, reachable)
Setting markers:          C: changed, I: invalid, P: pending (saved but not applied)
Navigation:               Top -> Platform -> Network

Management Interface:     device=eth0,IPv4=10.100.67.66/24,IPv6=<not-set>
LAN Interface:            IPv4: 10.100.67.254,device=eth0
Default gateway:          IPv6: <not-set>
                          IPv4: <not-set>
Static routes:            <not-set>
DNS Client:               enabled=true,domain=imperva.local,search-list=imperva.local,servers=10.2.16.66 10.2.16.58

1) Management interface.
2) LAN interface.
3) Change IPv4 default gateway.
4) Remove IPv4 default gateway.
5) Static routes.
6) Name resolution (DNS client).
7) Locate network devices.

e) End this level. j) Jump to a previous level. t) Top level.
q) Quit (discarding not-saved changes).

Your choice: █
```

2. Select one of the following options, as listed below.

Network Management Screen Configuration Options

Option	Name	Description
1	Management interface	Change the network interface and/or IP address (including CIDR) of the management interface. You can specify: ▪ an IPv4 address only ▪ and IPv6 address only ▪ both an IPv4 address and an IPv6 address Physically locate the network interface used as the management interface. For more information, see Management Interface.

Option	Name	Description
2	LAN interface	Change the network interface and/or IP address (including CIDR) of the LAN interface. You can specify: ▪ an IPv4 address only ▪ and IPv6 address only ▪ both an IPv4 address and an IPv6 address Physically locate the network interface used as the LAN interface. For more information, see LAN Interface.
3	Change IPv4 default gateway	Change the appliance's default Gateway. Whether IPv4 or IPv6 default Gateway options (or both) are available depends on whether the management interface and/or LAN interface has only an IPv4 or only an IPv6 address, or both. In the screen shown above, the management interface has both an IPv4 and an IPv6 address, so both "IPv4 default gateway" and "IPv6 default gateway" options are listed in the menu. For more information, see Default Gateway.
4	Remove IPv4 default gateway	Removes the appliance's default Gateway.
5	Static routes	Add and delete static routes. For more information, see Static Routes.
6	Name Resolution (DNS client)	Enable or disable the DNS client. You must enable the DNS client if the Management Server will be using LDAP to authenticate SecureSphere administrators (see Authentication and Authorization Configuration). Define the domain name and search list, and specify a list of name servers. For more information, see Name Resolution (DNS client).
7	Locate network devices	Physically locate an network interface.

Option	Name	Description
		For more information, see Locate Network Devices5 .

Management Interface

The management interface is used for all management communication between the SOM, Management Server and Gateway, and access to the MX GUI.

In the **Management Interface** screen, you can:

- Change the network interface and/or IP address (including CIDR)
- Physically locate the network interface used as the management interface

To manage the management interface:

1. In the **Network Management** screen (as described in [Network Management](#)), type **1) Management interface**. The **Management Interface** screen is displayed.

```
-----
SecureSphere 13.3.0.0_0 - Management Interface Screen
-----

Configuration target:      local (appliance, V2500, reachable)
Setting markers:          C: changed, I: invalid, P: pending (saved but not applied)
Navigation:               Top -> Platform -> Network -> Management

Management Interface:
1:                         device=eth0,IPv4=10.100.67.65/24,IPv6=<not-set>

LAN Interface:
1:                         device=eth1,IPv4=1.1.1.1/24,IPv6=<not-set>

1) Change device.
2) Change IPv4 address.
3) Set IPv6 address.
4) Remove IPv4 address.
5) Locate device (immediate).

e) End this level. j) Jump to a previous level. t) Top level.
q) Quit (discarding not-saved changes).

Your choice: █
```

2. Select one of the options and perform the desired change. For more information, see the table below.
3. Repeat step 2 for all desired options.
4. After performing the change(s), go to the **Top Screen** (type **t** at the "Your choice:" prompt).
5. Type **S** (capital letter) and press **Enter**.

6. Type **A** (capital letter) and press **Enter**.
7. Type **C** (capital letter) and press **Enter**. The gateway applies the changes and the **Top Screen** is displayed when done.
8. Type **3** and press **Enter**.
9. Type **7** and press **Enter**. The gateway is rebooted.

Management Interface Screen Configuration Options

Option	Name	Description
1	Set/Change device	Set or change the network interface to be used as the management interface. Specify the name, for example "eth4", of one of the available network interfaces.
2	Set/Change IPv4 address	Set or change the IP address of the network interface used as the management interface. Whether IPv4 or IPv6 address options (or both) are available depends on whether the management interface has only an IPv4 or only an IPv6 address, or both. In the screen shown above, the management interface has only an IPv4 address, so only "Change IPv4 address" option is listed in the menu. Important: When changing the IP address, make sure you perform a gateway reboot after applying the change.
3	Set/Change IPv6 address	Set or change the IP address of the network interface used as the management interface. Whether IPv4 or IPv6 address options (or both) are available depends on whether the management interface has only an IPv4 or only an IPv6 address, or both. In the screen shown above, the management interface has only an IPv4 address, so only "Set IPv6 address" option is listed in the menu.
4	Remove IPv4 address	Remove the IP address of the network interface used as the management interface. Whether IPv4 or IPv6 address options (or both) are available depends on whether the management interface has only an IPv4 or only an IPv6 address, or both. In the screen shown above, the management interface has only an IPv4 address, so only "Remove IPv4 address" option is listed in the menu.

Option	Name	Description
5	Remove IPv6 address	Remove the IP address of the network interface used as the management interface. Whether IPv4 or IPv6 address options (or both) are available depends on whether the management interface has only an IPv4 or only an IPv6 address, or both. In the screen shown above, the management interface has only an IPv4 address, so this option is not listed in the menu.
6	Locate device (immediate)	This option will help you physically locate a network interface by attempting to blink one of the LEDs associated with the management network interface. Note: Before activating this option, disconnect, if possible, all the network cables attached to the device.

LAN Interface

The LAN interface is used for communication between Gateways and databases. Optionally, you can, as a SecureSphere administrator, also communicate with the MX GUI over the LAN interface.

In the **LAN Interface** screen, you can:

- Change the network interface, IP address and netmask of the LAN interface
- Physically locate the network interface used as the LAN interface

To manage the LAN interface:

1. In the **Network Management** screen (as described in [Network Management](#)), select **2) LAN interlace**.

The **LAN Interface** screen is displayed. (See the example screenshot below.)

```

-----
SecureSphere 15.5.3.15.3 - LAN Interface Screen
-----

Configuration target:      local (appliance, VM150, reachable)
Setting markers:          C: changed, I: invalid, P: pending (saved but not applied)
Navigation:               Top → Platform → Network → Lan

Management Interface:
1:                         device=eth0,IPv4=10.10.123.77/23,IPv6=<not-set>

LAN Interface:

1) Set device.
2) Set IPv4 address.
3) Set IPv6 address.
4) Locate device (immediate).
5) Delete interface.

e) End this level. j) Jump to a previous level. t) Top level.
q) Quit (discarding not-saved changes).

Your choice: █

```

2. Select one of the following options, as listed below.

LAN Interface Screen Configuration Options

Option	Name	Description
1	Set/Change device	Set or change the network interface to be used as the LAN interface. Specify the name, for example "eth4", of one of the available network interfaces.
2	Set/Change IPv4 address	Set or change the IPv4 address of the network interface used as the LAN interface. Whether IPv4 or IPv6 address options (or both) are available depends on whether the management interface has only an IPv4 or only an IPv6 address, or both. In the screen shown above, the management interface has only an IPv4 address, so only "Change IPv4 address" option is listed in the menu.
3	Set/Change IPv6 address	Set or change the IPv6 address of the network interface used as the LAN interface.

Option	Name	Description
		Whether IPv4 or IPv6 address options (or both) are available depends on whether the management interface has only an IPv4 or only an IPv6 address, or both. In the screen shown above, the management interface has only an IPv4 address, so only "Set IPv6 address" option is listed in the menu.
4	Remove IPv4 address	Set or change the IPv4 address of the network interface used as the LAN interface. Whether IPv4 or IPv6 address options (or both) are available depends on whether the management interface has only an IPv4 or only an IPv6 address, or both. In the screen shown above, the management interface has only an IPv4 address, so only "Remove IPv4 address" option is listed in the menu.
5	Remove IPv6 address	Set or change the IPv4 address of the network interface used as the LAN interface. Whether IPv4 or IPv6 address options (or both) are available depends on whether the management interface has only an IPv4 or only an IPv6 address, or both. In the screen shown above, the management interface has only an IPv4 address, so this option is not listed in the menu.
6	Locate device (immediate)	This option will help you physically locate a network interface by attempting to blink one of the LEDs associated with the LAN network interface. You will be asked to specify an Activation time. Note: Before activating this option, disconnect, if possible, all the network cables attached to the device.
7	Delete interface	Deletes the specified LAN interface.

Default Gateway

You can:

- Change the appliance's default Gateway

To change the appliance's default Gateway:

1. In the **Network Management** screen (as described in [Network Management](#)), select **IPv4 Default gateway** or **IPv6 Default gateway**.
2. You will be asked to specify an IP address and, optionally, a device (network interface) for the default Gateway.

```

Your choice: 4

You will be asked for the IPv6 default gateway's Address and an optional device.

If you specify a device:
  It should be one of the onboard Ethernet interfaces (eth2 eth3).

If you don't specify a device:
  The default gateway must be reachable through one of the onboard interfaces.

The route will become available when the appliance boots and remain available until it shuts down.

Gateway [IPv6 Address]: █

```

Static Routes

In the **Static Routes** screen, you can:

- Add and delete static routes



Note: Changing or adding static routes may cause loss of connectivity for a few seconds.

To configure static routes:

1. In the **Network Management** screen (as described in [Network Management](#)), select **Static routes**. The Static Routes screen is displayed.

```

-----
SecureSphere 13.3.0.0_0 - Static Routes Screen
-----

Configuration target:      local (appliance, VM150, reachable)
Setting markers:          C: changed, I: invalid, P: pending (saved but not applied)
Navigation:               Top -> Platform -> Network ->

Static routes:            <not-set>

1) Add route.
2) Delete route.

e) End this level. j) Jump to a previous level. t) Top level.
q) Quit (discarding not-saved changes).

Your choice: █

```

2. Select one of the following options, as listed below.

Static Routes Screen Configuration Options

Option	Name	Description
1	Add route	Add a static route. You will be asked to specify the type (host or net), the IP address, the Gateway and optionally, a network interface. When defining static routes, the user must define the device (interface). <code>impcfg</code> presents the user with a list of relevant devices (interfaces) as a hint, and the user should select one of these.
2	Delete route	Delete an existing static route.

Name Resolution (DNS client)

In the **DNS Client Configuration** screen, you can:

- Enable or disable the DNS client
- Define the domain name and search list
- Specify a list of name servers

To configure DNS:

1. In the **Network Management** screen (as described in [Network Management](#)), select **Name resolution (DNS client)**. The **DNS Client Configuration** screen is displayed.

```

=====
SecureSphere 13.3.0.0_0 - DNS Client Configuration Screen
=====

Configuration target:      local (appliance, VM150, reachable)
Setting markers:          C: changed, I: invalid, P: pending (saved but not applied)
Navigation:               Top -> Platform -> Network -> DNS

DNS Client:               enabled=true, domain=imperva.local, search-list=imperva.local, servers=10.2.16.66 10.2.16.58

1) Disable DNS client.
2) Set domain.
3) Set search list.
4) Set name servers.

e) End this level. j) Jump to a previous level. t) Top level.
q) Quit (discarding not-saved changes).

Your choice: █

```

2. Select one of the following options, as listed below.

DNS Client Configuration Screen Configuration Options

Option	Name	Description
1	Enable/Disable DNS client	Enable or disable the DNS client. You must enable the DNS client if the Management Server will be using LDAP to authenticate SecureSphere administrators (see Authentication and Authorization Configuration).
2	Set domain	Define the domain name.
3	Set search list	Define the domain search list.
4	Set name servers	Specify a comma-separated list of name servers. Make sure these are real DNS servers, not "dummy" IP addresses.

Locate Network Devices

This option will help you physically locate a network interface by attempting to blink one of the LEDs associated with the network interface.



Note: Before activating this option, disconnect, if possible, all the network cables attached to the device.

To physically locate a network interface:

1. In the **Network Management** screen (as shown in [Network Management](#)), select **Locate network devices**.
2. Select an interface from the list of available interfaces.
3. Specify an Activation time.

Time Management

You can:

- Change the date, time, and time zone
- Configure the NTP server.

To manage time settings:

1. In the **Platform Management** screen (as described in [Configuring the Platform](#)), select **Manage time**. The **Time Management** screen is displayed.

```

-----
SecureSphere 13.3.0.0_0 - Time Management Screen
-----

Configuration target:      local (appliance, V2500, reachable)
Setting markers:          C: changed, I: invalid, P: pending (saved but not applied)
Navigation:               Top -> Platform -> Time

Time zone:                Asia/Jerusalem
Time servers:              10.2.16.66
                           10.2.16.58

1) Change time zone.
2) Change date and time (immediate).
3) Time servers configuration (NTP).

e) End this level. j) Jump to a previous level. t) Top level.
q) Quit (discarding not-saved changes).

Your choice: █

```

2. Select one of the following options, as listed below.

Time Management Screen Configuration Options

Option	Name	Description
1	Change time zone	Change the appliance's time zone. You will be asked to select a time zone and then a country, or to specify the time zone using the Posix TZ format.
2	Change date and time (immediate)	
3	Time servers configuration (NTP)	For more information, see Time Servers .

Time Servers

You can:

- Add and delete NTP servers

To manage NTP servers:

1. In the **Time Management** screen (as described in Time Management), select **Manage time**. The **Time Management** screen is displayed.
2. Select **Time servers configuration (NTP)**. A list of previously-added time servers is displayed under **Time Server(s)**.

```

-----
SecureSphere 13.3.0.0_0 - Time Servers Screen
-----

Configuration target:      local (appliance, V2500, reachable)
Setting markers:          C: changed, I: invalid, P: pending (saved but not applied)
Navigation:               Top -> Platform -> Time

Time server(s):
                           10.2.16.66
                           10.2.16.58

1) Add time server.
2) Delete time server.

e) End this level. j) Jump to a previous level. t) Top level.
q) Quit (discarding not-saved changes).

Your choice: █

```

3. Select one of the following options, as listed below.

Time Servers Screen Configuration Options

Option	Name	Description
1	Add time server	Add an NTP server. You will be asked to enter the IP address of the NTP server to be added. You can add an IPv4 address or an IPv6 address, depending on whether the appliance's LAN interface has an IPv4 or IPv6 address, or both. Note: Synchronization will not take place if the time difference between the appliance and the NTP server is too great. You should manually set the time on the appliance to be correct (within 2-3 minutes) before the initial synchronization attempt.
2	Delete time server	Delete a previously-added NTP server. You will be asked to enter the IP address of the NTP

Option	Name	Description
		server to be deleted. This must be one of the IP addresses displayed under Time Server(s) .

Users Management

You can:

- Change the root user's password
- Create a new user
- Manage an existing user
- Configure external authentication

To manage users:

1. In the **Platform Management** screen (as shown in [Configuring the Platform](#)), select **Manage users**. The **Users Management** screen is displayed.

```

-----
SecureSphere 13.3.0.0_0 - Users Management Screen
-----

Configuration target:      local (appliance, V2500, reachable)
Setting markers:          C: changed, I: invalid, P: pending (saved but not applied)
Navigation:               Top -> Platform -> Users

Root password:            <is-set>

External authentication:   None

1) Change user 'root' password.
2) Create new user.
3) Manage user.
4) Configure RADIUS authentication.

e) End this level. j) Jump to a previous level. t) Top level.
q) Quit (discarding not-saved changes).

Your choice: █

```

2. Select one of the following options, as listed below.

User Management Screen Configuration Options

Option	Name	Description
1	Change user 'root' password.	For more information, see Change Root User Password .
2	Create new user.	For more information, see Create New User .
3	Manage user.	For more information, see Manage User .
4	Configure RADIUS authentication.	For more information, see Configure RADIUS Authentication .
5	Enable / disable RADIUS.	For more information, see Configuring RADIUS Authentication . This option appears only if RADIUS authentication was configured.
6	Edit external authentication configuration.	For more information, see Edit External Authentication Configuration . This option appears only if RADIUS authentication was configured.

Change Root User Password

To change the 'root' user password:

1. In the **Platform Management** screen (as shown in [Configuring the Platform](#)), select **Manage users**. The **Users Management** screen is displayed.
2. Select **Change user 'root' password**.
3. Enter the root user's existing password.
4. Enter a new password for the root user.

You can also enter the string "nochange", in which case you will be returned to the previous menu.

5. Enter the new password again.

Note:

Make sure your password has the following characteristics:



- It must have no fewer than 10 characters and no more than 14 characters.
- It must have at least one number, one capital letter, and one special character from:

* + = # % ^ : / ~ . , [] _

- It cannot have more than two characters repeated in succession.

Create New User

To create a new user:

1. In the **Platform Management** screen (as shown in [Configuring the Platform](#)), select **Manage users**. The **Users Management** screen is displayed.
2. Select **Create new user**.
3. Enter a user name.

Note:

- The user name must begin with a letter followed by letters, digits, dash or underscore characters.
- You cannot create a user whose name is one of the existing system users, for example, root, secure, *etc.*
- Only lowercase letters can be used to create user names. Upper case is not supported.

4. If the authentication method (see [Configure RADIUS Authentication](#)) is "user-defined" , you will be asked to specify whether authentication for this user is local or external.
 - If you choose external, then you have finished creating the new user. All the information about the user will be obtained from the external authentication server (RADIUS) when the user logs in.

Note:

- Only users added after RADIUS has been configured can be configured with RADIUS authentication.
- When configuring RADIUS users, you must configure them with the same username as they appear in the RADIUS server.
- When logging into SecureSphere with a RADIUS authenticated user, you must use the password configured for them in the RADIUS server.

- If you choose "local", then continue with the next step.

5. Enter a full name for the user (see the figure above).
6. Enter a password for the user. The password is a one-time password, and the user will have to change it when he or she logs in for the first time.
7. Enter the password again. The new user will be saved in the SecureSphere database when you apply the pending changes (**Apply changes** in the **Top** screen).

Manage User

To create a new user:

1. In the **Platform Management** screen (as shown in [Configuring the Platform](#)), select **Manage users**. The **Users Management** screen is displayed.
2. Select **Manage user**.

```
-----
SecureSphere 13.3.0.0_0 - 'j123456' Account Details Screen
-----

Configuration target:      local (appliance, V2500, reachable)
Setting markers:          C: changed, I: invalid, P: pending (saved but not applied)
Navigation:               Top -> Platform -> Users -> User 'j123456'

Full name:                John Doe
Locked?:                  false

1) Reset password (immediate).
2) Lock account (immediate).
3) Delete account (immediate).

e) End this level. j) Jump to a previous level. t) Top level.
q) Quit (discarding not-saved changes).

Your choice: █
```

- In the **User Account Details** screen (above), select one of the following options, as listed below.

User Account Details Options

Option	Name	Description
1	Reset password (immediate).	Reset (change) the user's password. You will be asked to enter a new password, and then to re-enter the same password. The password will be changed immediately. The new password is a one-time password, and the user will be asked to change it the next time he or she logs in. Note: Make sure your password has the following characteristics: ▪ It must have no fewer than 10 characters and no more than 14 characters. ▪ It must have at least one number, one capital letter, and one special character from: * + = # % ^ : / ~ . , [] ▪ It cannot have more than two characters repeated in succession. You can also enter the string "nochange", in which case you will be returned to the previous menu.
2	Lock account (immediate).	If the user's account is not locked, you can lock it by selecting this option.
3	Unlock account (immediate).	If the user's account is locked, you can unlock it by selecting this option.
4	Delete account (immediate).	Delete this user's account. You will be asked to confirm the deletion.

Configure RADIUS Authentication

This topic describes how to configure RADIUS Authentication on SecureSphere for CLI users. This includes establishing trusted communication between the SecureSphere appliance and the Radius server using a Shared Secret (internal password). For information on how to disable RADIUS that is already running or re-enable it after it being disabled, see [Enable / Disable RADIUS](#).

Notes:

- CLI RADIUS authentication, supports only PAP. SecureSphere MX RADIUS authentication supports PAP, CHAP, and MS-CHAP.
- Configuring RADIUS authentication is not applicable in MX-HA environments

To Configure RADIUS:

1. Establish an SSH connection to the Management Server or the Gateway via CLI.
2. Log into SecureSphere, then enter **impcfg**.
3. Select **Manage Platform**.
4. Select **Manage Users**.

```
-----
SecureSphere 13.3.0.0_0 - Users Management Screen
-----

Configuration target:      local (appliance, V2500, reachable)
Setting markers:          C: changed, I: invalid, P: pending (saved but not applied)
Navigation:               Top -> Platform -> Users

Root password:            <is-set>

External authentication:   None

1) Change user 'root' password.
2) Create new user.
3) Manage user.
4) Configure RADIUS authentication.

e) End this level. j) Jump to a previous level. t) Top level.
q) Quit (discarding not-saved changes).

Your choice: █
```

5. Select **Configure RADIUS Authentication**.
6. Type the **IP Address** of your RADIUS server, then press **Enter**.
7. Type the **Shared Secret** (an internal password used to authenticate between SecureSphere and the RADIUS server).
8. Define the **Authentication Method**. You have three options:

- **local:** By default, all new users that are created will be authenticated by a password, locally on the SecureSphere appliance (not RADIUS). You can then later go in and change the authentication method per user
 - **radius:** By default, will use RADIUS to authenticate all new users that are created
 - **user-defined:** When a new user is created, will prompt if to use local or RADIUS authentication for the new user
9. Type **T** to go to the Top level.
 10. Type **A** to apply the new settings, then when prompted review the settings and if correct, type **C** to confirm the settings.

The settings are applied. From this point on all new users will be capable of RADIUS authentication. For information on how to create new users, see [Create New User](#).

Once you have configured RADIUS authentication, two new options are added to the User Management Screen, **Enable/Disable RADIUS**, and **Edit external authentication configuration**.

Notes:

- Only users added after RADIUS has been configured can be configured with RADIUS authentication.
- When configuring RADIUS users, you must configure them with the same username as they appear in the RADIUS server.
- When logging into SecureSphere with a RADIUS authenticated user, you must use the password configured for them in the RADIUS server.

Enable / Disable RADIUS

The Enable / Disable RADIUS authentication allows you to disable RADIUS authentication once it has already been enabled. It additionally allows you to then re-enable RADIUS authentication if you previously disabled it. It does not describe how to initially configure and enable RADIUS authentication. For instructions on enabling and configuring RADIUS authentication the first time in SecureSphere, see [Configure RADIUS Authentication](#).

To disable RADIUS authentication, or enable it after it was previously disabled:

1. Establish an SSH connection to the Management Server or the Gateway via CLI.
2. Log into SecureSphere.
3. Select **Manage Platform**.
4. Select **Manage Users**.

```

-----
SecureSphere 13.3.0.0_0 - Users Management Screen
-----

Configuration target:      local (appliance, V2500, reachable)
Setting markers:          C: changed, I: invalid, P: pending (saved but not applied)
Navigation:               Top -> Platform -> Users

Root password:            <is-set>

External authentication:   None

1) Change user 'root' password.
2) Create new user.
3) Manage user.
4) Configure RADIUS authentication.

e) End this level. j) Jump to a previous level. t) Top level.
q) Quit (discarding not-saved changes).

Your choice: █

```

5. Select **Enable/Disable radius**.
6. Type one of the following:
 - **false**: Disables RADIUS Authentication for an existing configuration
 - **true**: Re-enables authentication if it was previously disabled.
7. Press **Enter**. RADIUS Authentication is enabled or disabled as configured.

Edit External Authentication Configuration

The Edit External Authentication Configuration option enables you to directly turn off RADIUS Authentication.

To edit external authentication configuration:

1. Establish an SSH connection to the Management Server or the Gateway via CLI.
2. Log into SecureSphere.
3. Select **Manage Platform**.
4. Select **Manage Users**.

```

-----
SecureSphere 13.3.0.0_0 - Users Management Screen
-----

Configuration target:      local (appliance, V2500, reachable)
Setting markers:          C: changed, I: invalid, P: pending (saved but not applied)
Navigation:               Top -> Platform -> Users

Root password:            <is-set>

External authentication:   None

1) Change user 'root' password.
2) Create new user.
3) Manage user.
4) Configure RADIUS authentication.

e) End this level. j) Jump to a previous level. t) Top level.
q) Quit (discarding not-saved changes).

Your choice: █

```

5. Select **Edit external authentication configuration**.
6. Type one of the following:
 - **none**: No authentication will be used
 - **radius**: Will use RADIUS to authenticate all new users that are created
7. Press **Enter**. RADIUS Authentication is enabled or disabled as configured.

Hostname Management

You can:

- Change the hostname.

To manage the hostname:

1. In the **Platform Management** screen (as described in [Configuring the Platform](#)), select **Manage host name**. The **Hostname Configuration** screen is displayed.

```

-----
SecureSphere 13.3.0.0_0 - Hostname Configuration Screen
-----

Configuration target:      local (appliance, V2500, reachable)
Setting markers:          C: changed, I: invalid, P: pending (saved but not applied)
Navigation:               Top -> Platform -> Hostname

Host name:                TW_GW_13

1) Change Hostname.

e) End this level. j) Jump to a previous level. t) Top level.
q) Quit (discarding not-saved changes).

Your choice: █

```

2. Select one of the following options, as listed below.

Hostname Configuration Screen Options

Option	Name	Description
1	Change Hostname	Change the appliance's hostname.

Bootloader Password Management

To manage the bootloader password:

1. In the **Platform Management** screen (as described in [Configuring the Platform](#)), select **Manage bootloader password**.
2. Enter the bootloader user's existing password.
3. Enter a new password for the bootloader user

You can also enter the string "nochange", in which case you will be returned to the previous menu.

4. Enter the new password again.

Note:

Make sure your password has the following characteristics:



- It must have no fewer than 10 characters and no more than 14 characters.
- It must have at least one number, one capital letter, and one special character from:

* + = # % ^ : / ~ . , [] _

- It cannot have more than two characters repeated in succession.

Configuring SecureSphere Routes

You must define the SecureSphere routes listed in the table below using `impcfg`.

Static Routes Screen Configuration Options

Routes	Description	Data stored in
Platform Routes	These routes are related to the appliance, Management Server or Gateway, management interfaces. For more information, see Static Routes.	▪ <code>/etc/sysconfig/static-routes</code> (executed at boot time) ▪ <code>bootstrap.xml</code>
Reverse Proxy	These routes are executed and added automatically according to the Gateway mode. They are deleted when the Gateway is stopped with the "teardown" option, and added when the Gateway is started with the "prepare" option.	<code>bootstrap.xml</code>
VRRP Reverse Proxy	These routes are executed with relation to the VRRP status, either master or backup. For more information, see Configuring a Reverse Proxy	<code>/etc/keepalived.conf</code>

Configuring Transparent Reverse Proxy

To configure Transparent Reverse Proxy, perform the following tasks after the First Time Login is complete, as listed below.

Transparent Reverse Proxy Configuration Task Overview

	Task Overview	Step-By-Step
1	Verify there is and if not, configure a default gateway.	To verify: Execute the command <code>impcfg</code> and go to Manage platform . Under Network settings > Default gateway , see if an IPv4 address is shown. If an address is shown, continue to step 2. If not, go to Manage network > Change IPv4 default gateway and set the IP address.
2	In <code>impcfg</code> , configure the appliance as a bridge.	Manage SecureSphere Gateway > Change operation mode and select Bridge STP or Bridge IMPVHA .
3	In the SecureSphere GUI, configure the server group.	In the Transparent Reverse Proxy section of the Definitions tab of the server group (Main > Setup > Sites), select Enable Transparent Reverse Proxy .
4	For more information, see the SecureSphere Web Security User Guide .	Define the server IP addresses and ports for which the Gateway will act as Transparent Reverse Proxy.



Note: In the event of a conflict between a Transparent Reverse Proxy rule (**Main > Setup > Sites**) and another service's port configuration, the Transparent Reverse Proxy rule takes precedence.

Configuring the Management Server for High Availability

For information on configuring High Availability for Management Servers, see [Management Server High Availability \(MX-HA\)](#).

Switching Deployments

This section describes how to switch a SecureSphere appliance from one deployment mode to another, for example, from an inline gateway to a network monitor (sniffing) gateway, and includes:

- Converting a Gateway to an MX
- Changing the MX of a Gateway

Converting a Gateway to an MX

It is not possible to convert a gateway appliance to an MX appliance, because of their different hardware configurations.

Changing the MX of a Gateway

To change the MX of a gateway:

1. Start `impcfg`.
2. Select **Manage SecureSphere Gateway > Perform Actions > Stop gateway**.
3. Select **Manage SecureSphere Gateway > Perform Actions > Unregister gateway**.
4. On the gateway, start `impcfg`.
5. Select **Manage SecureSphere Gateway > Change server address / password** and specify the new MX.

Configuring the MX to Communicate with SOM Across Borders

Some user deployments have regulatory restrictions regarding data crossing borders. These restrictions prohibit the transfer of user information across national boundaries. When MXs are in one location and the SOM in a location in a different country, if your organization is governed by these restrictions, you cannot have the MXs connected to the SOM unless they are configured so that no user data is transferred from the MXs to the SOM. This user data includes:

- Audit data
- Security alerts.

You can configure the MX so that it does not send this data when it is connected to the SOM.

To configure the MX to communicate with the SOM across borders:

1. Open the `/opt/SecureSphere/server/SecureSphere/jakarta-tomcat-secsph/webapps/SecureSphere/WEB-INF/properties/common.properties` file.

2. Change the value of the parameter **allowed.to.share.user.data.with.som** to `false`.
3. Save the file.
4. Restart the MX using the **impctl server restart** command.

Using QinQ

QinQ or double VLAN tag is supported in Bridge, TRP and Advanced Bridge (in both IMPVHA and STP) gateway modes.

QinQ is supported with physical appliances only when working with 10GB cards. Our 1GB cards do not support this network configuration.

When using QinQ with 6G appliances, no additional configuration is needed.

When using QinQ with 5G appliances, please add the following according to your network ether-type in `/opt/SecureSphere/etc/bootstrap.xml` under `<current-mode>bridge-impvha</current-mode>`

```
<vlan>

<interface name="ethX" ether-type="0x8100" enabled="true"/>

<interface name="ethX" ether-type="0x8100" enabled="true"/>

</vlan>
```

Gateways

This section describes configuration of SecureSphere Gateways, and includes the following:

- [Configuring Gateways and Gateway Groups](#)
- [Exporting Technical Information from Gateway](#)

Configuring Gateways and Gateway Groups

SecureSphere Gateways monitor traffic, enforce policies and send events to the SecureSphere MX.

A Gateway's MX is defined during the Gateway's initial configuration, using `impcfg`.

Notes:



- Gateways support up to 5 SSH connections
- When a Gateway is registered to an MX, the Gateway sends the MX information about its network interfaces, and users may configure the Gateway on the basis of that information. If any of the Gateway network interfaces is deleted or removed, the MX's Gateway configuration based on the deleted interface becomes invalid. The user must then either restore the network interface on the Gateway or modify the Gateway configuration on the MX accordingly

For more information, see [Configuring a Gateway](#).

You can configure Gateways and Gateway groups as described in the following topics:

- [Gateways](#)
- [Gateway Groups](#)

Gateways

Creating a Gateway means creating a MX-Gateway relationship, that is, specifying that a Gateway is managed by this MX (the MX that the SecureSphere GUI is running on). This is done using `impcfg`.

For more information, see [Configuring a Gateways and Gateway Groups](#).

To view a Gateway:

1. In the **Main** workspace, select **Setup > Gateways**.
2. In the **Filter** pane, select **By Mode**, **By Platform** or **By Server Group**.

3. Select a Gateway. The Gateway's details are displayed in the **Gateway Details** tab.

Details

General Info

Management Interface IP: 10.100.67.65
 Installed Version: 14.7.0.170_0
 Up Since: 11/25/25 2:20:03 PM
 License Level: Enterprise Edition
 Performance Report (CSV): [Download](#)
 Tech Info (ZIP): [Download](#)

Group

Gateway Group: TW_GW

Errors

Errors
 No data found

Server Groups

Name	Status	Errors
No data found		

CPU Load

Kernel: 0
 User: 0

Gateway Details

The following tables detail the configuration sections of the gateway's Details tab.

Details Tab - General Info

Field	Description
Management Interface IP	The IP address of the Gateway's management interface.
Installed Version	The version of SecureSphere currently installed on this Gateway.
Up Since	The date and time since which the Gateway has been continuously running.
License Level	The Imperva license level (Enterprise).
Performance Report CSV	Click Download to export the Gateway's statistics to a CSV file.

Field	Description
Tech Info (ZIP)	Click Download to export Gateway or one-box get-tech-info to a zip file. For more information see Exporting Technical Information from Gateway .

Details Tab - Group

Field	Description
Gateway Group	The name of the Gateway group to which this Gateway belongs. To move the Gateway to another group, select the group from the dropdown list.

Detail Tab - Errors

The following section displays a read-only list of the errors reported for this Gateway.

Field	Description
Errors	A description of the error, for example, "Minimum disk space reached, data is lost."

Detail Tab - Server Groups

The following table displays a read-only list of the server groups which this Gateway protects.

Field	Description
Name	The name of the server group.
Status	The status of the server group (for example, "Running").
Errors	The number of errors for the server group. Expand the server group to display a list of the errors.

Details Tab - CPU Load

The following table the CPU load on the server.

Field	Description
kernel	The CPU load of the kernel mode module.
user	The CPU load of the user mode module.

Gateway Groups

You can view, create and edit Gateway groups.

To view a Gateway group:

1. In the **Main** workspace, select Setup > Gateways.
2. In the **Gateways** pane, click .
3. In the **Create New Gateway Group** window, enter the data in the table below.
4. Click **Create**.
5. Click **Save**.

Parameter	Options
Name	The name of the Gateway group.
Platform	Select a platform from the menu: ▪ Imperva ▪ Crossbeam ▪ Amazon ▪ Other
Gateway Mode	Select a mode from the menu: ▪ STP Bridge ▪ IMPVHA Bridge ▪ Sniffing ▪ Reverse Proxy.

Parameter	Options
	For more information about these modes, see Configuring a Gateway.
Fail Mode	Select a Fail Mode: ▪ Fail Open: In the event of hardware failure, power loss, or system outage, traffic will continue to pass between ports without monitoring. ▪ Fail Close: This mode maintains physical separation between the two NIC ports during normal operation, allowing traffic to be inspected when the software is operational. In the event of hardware failure, power loss, or system outage, traffic between ports will not pass through the gateway. ▪ High Availability: This gateway group is set to High Availability. This option is available only if Gateway Mode is STP Bridge or IMPVHA Bridge. You must also toggle High Availability for each of the gateways in the group using <code>impcfg</code> (Top > Gateway > Interfaces > Toggle HA on a bridge). Note: These bypass modes utilize specialized hardware NICs with physical bypass capabilities. For virtual machine deployments, only the Fail Close mode is supported due to the absence of the required hardware components.
Overload Policy	Select Block on Overload to specify that if the Gateway's throughput capacity is exceeded, all connections will be blocked until the throughput falls. A Gateway is considered overloaded when any of the following occurs: ▪ CPU or memory utilization reach 100%. ▪ Throughput exceeds the defined limit.

Parameter	Options
	Resources are exhausted.

To view a Gateway group:

1. In the **Main** workspace, select Setup > Gateways.
2. In the **Filter** pane, select **By Mode**, **By Platform** or **By Server Group**.
3. Select a Gateway group.

The Gateway group's details are displayed in the **Gateway Group Details** pane.

To edit a Gateway group:

1. In the **Main** workspace, select **Setup > Gateways**.
2. In the **Filter** pane, select **By Mode**, **By Platform** or **By Server Group**.
3. Select a Gateway group.
4. In the **Gateway Group Details** pane, enter the following data:

Parameter	Description
General Info	
Name	The Gateway group's Name is displayed. You can change Name here.
Platform	The Gateway group's Platform is displayed.
Topology Configuration	
Mode	The Gateway group's Mode is displayed.
Fail Mode	Select a Fail Mode : Fail Open: In the event the Gateway fails, it will continue to pass traffic without monitoring it. Fail Close: In the event the Gateway

Parameter	Description
	fails, it will block all traffic.
Overload Policy	Select Block on Overload to specify that if the Gateway's throughput capacity is exceeded, all connections will be blocked until the throughput falls.
Reset untraceable user connections	FAM only - Select Reset untraceable user connections to specify that the SecureSphere Gateway should reset connections that were opened before the Gateway was started, so that the actual user will be identified in the audit file.
Allow Ambiguous Packets	
Allow IP bad checksum	Allow a packet through the Gateway even if it has a bad IP checksum.
Allow TCP bad checksum	Allow a packet through the Gateway even if it has a bad TCP checksum.
Allow TCP reset out-of-window	
Allow TCP sequence before ISN	Allow a packet through the Gateway even if its sequence number is less than the TCP session's ISN (Initial Sequence Number).
Allow TCP ack beyond sequence	
Allow TCP sequence out-of-window	
Allow UDP bad checksum	Allow a packet through the Gateway even if it has a bad UDP checksum.

Parameter	Description
External Logger (not available in WAF-only configurations)	
Syslog transport	Select a transport protocol for issuing logs to syslog: ▪ UDP: The connectionless UDP protocol is used. UDP carries less overhead but has the disadvantage that delivery is not guaranteed. ▪ TCP: The TCP protocol is used. TCP carries a higher overhead than UDP but delivery is guaranteed. Some additional configuration may be necessary. ▪ Note: In audit policies, you can specify (in the External Logger tab) that the configuration defined here be used for the syslog action.
Primary Host	The IP address of the primary host to which syslogs will be sent.
Port	The port on the primary host to which syslogs will be sent.
Secondary Host	The IP address of the secondary host to which syslogs will be sent.
Port	The port on the secondary host to which syslogs will be sent.
Performance Profiling	
Enable	Performance Profiling is a debug tool used for identifying performance issues with SecureSphere. Important Notes on Performance

Parameter	Description
	Profiling ▪ Only enable this feature if requested by Imperva personnel ▪ Enabling this feature can cause severe degradation when enabled for a long period of time ▪ This feature is enabled on the level of the Gateway Group. Subsequently all members of the gateway group may experience an impact on performance as long as the feature is enabled ▪ This feature is not available with SecureSphere on VMware. When enabling Performance Profiling in an environment that runs both VMware and Physical SecureSphere appliances, Performance Profiling will still be disabled on the VMWare environment. For more information see System Performance.
Gateways	
Gateway Name	A read-only list of the Gateways which are members of this group is displayed.
HTTP Connection	Expand this field to configure the Gateway group's connection to outside networks. Allow Gateway to Connect with Outside Systems - Allow the Gateway to connect to outside systems. HTTP Proxy - Specify the HTTP proxy the Gateway should use. Select from the list one of the HTTP proxies defined under Main > Setup > Global Objects > Gateway HTTP Proxy. You can specify an External HTTP proxy, which is designed for use with third party

Parameter	Description
	tools such as AA, ABP & THR. It is used for a single point of exit/entry from your network to the outside world. Validate Certificate - Check this if you want the Gateway to validate the HTTP proxy's certificate.

If you have made any changes in the **Gateway Group Details** pane, click **Save**.

Configuring a Reverse Proxy

Configuring a Reverse Proxy consists of the following:

- Configuring the Gateway appliance as a Reverse Proxy using `impccfg` (see [Configuring a Gateway](#))
- Configuring aliases and routing on the Reverse Proxy, as described in this section
- Configuring Reverse Proxy rules in the HTTP service's **Reverse Proxy** tab (see [Reverse Proxies in the Web Application Firewall User Guide](#))



Note: If a server group's operation mode is set to **Disabled**, a Reverse Proxy Gateway protecting that server group will block all traffic.

Aliases

An alias associates an inbound IP address and an outbound IP address for the Reverse Proxy, enabling the SecureSphere administrator to "route" traffic from a specific inbound interface to a specific Web server. Since a network tends to be dynamic, IP address allocation may change from time to time. Aliases eliminate the need to configure changes in two places (on the Gateway and on the MX).

Beginning with SecureSphere version 10.0, aliases are defined in the SecureSphere GUI, so that administrators can centrally view and manage all Reverse Proxy features: routing, interfaces and virtual routers (VRRP). In addition, configuration can also be accomplished importing a pre-prepared CSV file, simplifying the task of configuring multiple Gateways.



Note: When a Gateway is registered to an MX, the Gateway sends the MX information about its network interfaces, and users may configure the Gateway on the basis of that information. If any of the Gateway network interfaces is deleted or removed, the MX's Gateway configuration based on the deleted interface becomes invalid. The user must then either restore the network interface on the Gateway or modify the Gateway configuration on the MX accordingly.

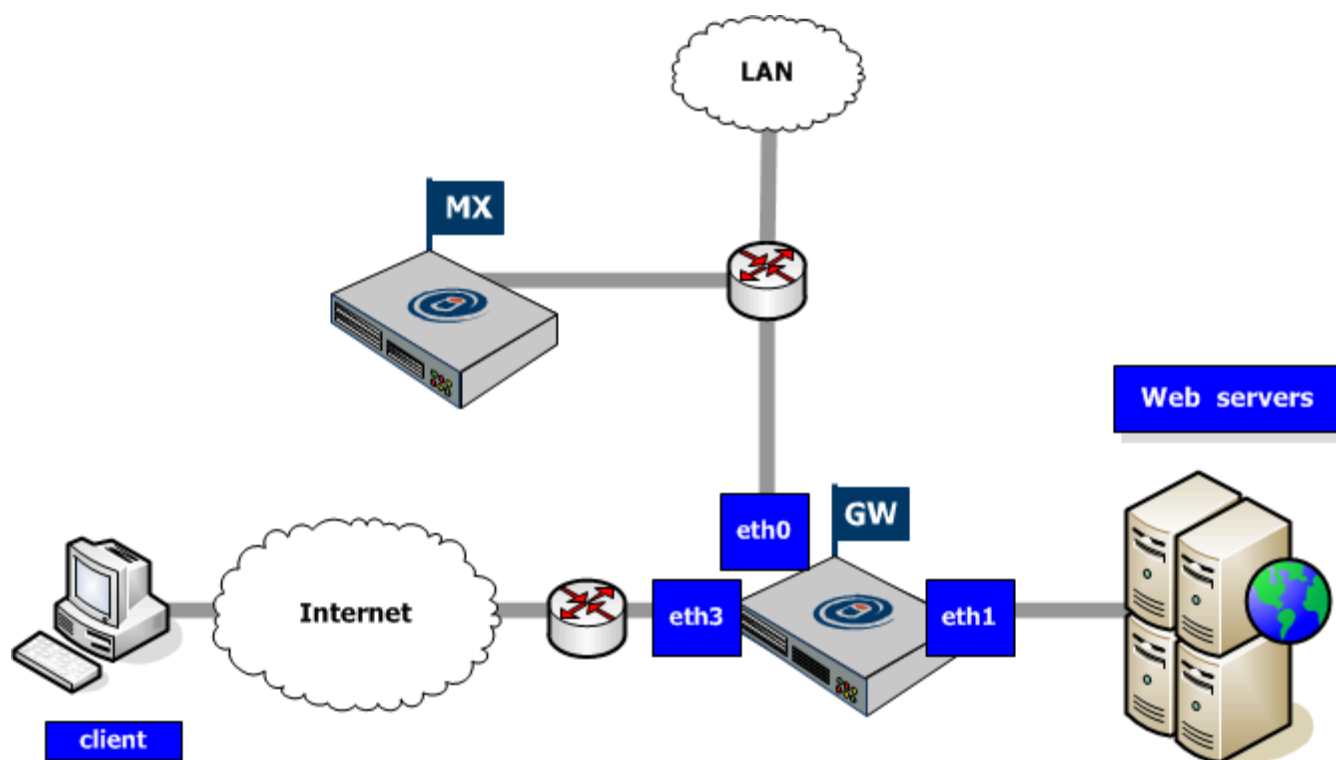
Routing

If the Reverse Proxy's IP addresses (usually outbound addresses) are not on the same network as the destination IP addresses, routing must be configured to enable connectivity. If the default route was changed, it is important to configure new routes (for example, SSH to the Gateway).



Note: If the backend server is not on the same network as the SecureSphere Gateway, make sure communication is allowed through any intervening firewalls.

A typical configuration is shown in the following figure.



There are two routes to be configured:

- **eth0:** to the SecureSphere MX
- **eth1:** to the Web server

All other traffic can be routed out through the Internet. The challenges here are to ensure that:

- The MX traffic is properly routed out through eth0
- Return traffic from the Web servers is routed out through eth3

Configuring Aliases

An alias associates an inbound IP address and an outbound IP address for the Reverse Proxy, enabling the SecureSphere administrator to "route" traffic from a specific inbound interface to a specific Web server. To configure an alias, you must:

1. If you are using VRRP for High Availability, define an IP address for the Gateway.
2. If required, configure a virtual router.
3. Associate a pair of alias IP addresses with the network interfaces.

You can associate multiple IP addresses with each network interface.

Defining an IP Address for the Gateway for VRRP

To define an IP address for the Gateway:

1. In the **Main** workspace, select Setup > Gateways.
2. In the **Filter** pane, select **By Mode**, **By Platform** or **By Server Group**.
3. Select a Gateway.

The Gateway's details are displayed in the **Gateway Details** tab.

The screenshot displays the 'Details' tab for a gateway. The 'General Info' section includes fields for Management Interface IP (10.100.67.65), Installed Version (14.7.0.170_0), Up Since (11/25/25 2:20:03 PM), License Level (Enterprise Edition), and buttons to download the Performance Report (CSV) and Tech Info (ZIP). The 'Group' section shows the Gateway Group set to 'TW_GW'. The 'Errors' section indicates 'No data found'. The 'Server Groups' section is a table with columns for Name, Status, and Errors, also showing 'No data found'. The 'CPU Load' section shows Kernel: 0 and User: 0.

General Info		
Management Interface IP:	10.100.67.65	
Installed Version:	14.7.0.170_0	
Up Since:	11/25/25 2:20:03 PM	
License Level:	Enterprise Edition	
Performance Report (CSV):	Download	
Tech Info (ZIP):	Download	

Group	
Gateway Group:	TW_GW

Errors		
No data found		

Server Groups		
Name	Status	Errors
No data found		

CPU Load	
Kernel:	0
User:	0

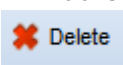
If you are using VRRP to provide High Availability, you must define at least one IP address for the Reverse Proxy Gateway, on which the two primary and secondary Gateways will communicate. To do this:

4. Under **Networking**, click **IP Addresses**.
5. In the **IP Addresses** window, click .
6. In the **IP Address** window, enter the data in the table **IP Address Window Parameters** below.
7. Click **OK** to save.
8. The IP address will be defined on the Gateway, exactly as if you had defined it in the Gateway's CLI. The required CLI commands are executed "behind the scenes."
9. Define additional IP addresses as required, or click **Close** in the **IP Addresses** window.

IP Address Window Parameters

Field	Description
Network Interface	From the dropdown menu, select the network interface to which the IP address will be assigned.
IP Address/Mask (CIDR)	Enter the IP address in CIDR notation.
Virtual	Check this box if the IP address is virtual, that is, an IP address that moves between the primary and secondary Gateways upon failover in the VRRP environment.
VR ID	If Virtual is checked, select the VR ID from the dropdown menu.
Next Hop	Specify the IP address to which to send outgoing traffic from this IP address. This field is used to route return traffic from the Web servers out from the Reverse Proxy, for example, to ensure that traffic from a specific ISP is routed back to the same ISP.
Comment	Optional explanatory text.

To delete an IP address for the Gateway:

1. In the **Main** workspace, select Setup > Gateways.
2. In the **Filter** pane, select **By Mode**, **By Platform** or **By Server Group**.
3. Select a Gateway.
4. Under **Networking**, click **IP Addresses**.
5. In the **IP Addresses** window, select an IP address.
6. Click .

7. Confirm the deletion.

Configuring a Virtual Router

This procedure describes how to configure a virtual router.



Note: When creating Virtual Routers for a High Availability (HA) environment, you must first create the router for the Backup Gateway (default state in the procedure below), and only then the router Master Gateway. This rule should also be followed with any configuration changes.

To define a virtual router:

1. In the **Main** workspace, select **Setup > Gateways**.
2. In the **Filter** pane, select **By Mode**, **By Platform** or **By Server Group**.
3. Select a Gateway.
4. Under **Networking**, click **Virtual Router**.
5. In the **Virtual Routers** window, do one of the following:
 - Click **Upload from CSV** and browse to the CSV file containing the information (see [Importing Reverse Proxy Configuration From a CSV File](#)), or
 - Click **+ New**.
 - In the **Virtual Router** window, enter data in the table **Virtual Router Configuration Options** below.
 - Click **OK**.
 - Define additional virtual routers as required, or click **Close** in the **Virtual Routers** window.

Virtual Router Configuration Options

Field	Description
Network Interface	From the dropdown menu, select the network interface to which the virtual router will be assigned.
VR ID	Enter an identifier for the virtual router.
Primary IP Address	Enter the primary IP address for this virtual router. If you leave this empty, the VRRP protocol will assign an IP address. Optional: Enter the IP address to be used by the Virtual Router for multicast VRRP packets. If empty, VRRP will use the interface's primary IP address.

Field	Description
Default State	Choose Master or Backup from the dropdown menu.

To delete a virtual router:

1. In the **Main** workspace, select Setup > Gateways.
2. In the **Filter** pane, select **By Mode**, **By Platform** or **By Server Group**.
3. Select a Gateway.
4. Under **Networking**, click **Virtual Routers**.
5. In the **Virtual Routers** window, select a virtual router.
6. Click  **Delete**.
7. Confirm the deletion.

Configuring an Alias

This procedure describes how to configure an alias.

To configure an alias:

1. In the **Main** workspace, select Setup > Gateways.
2. In the **Filter** pane, select **By Mode**, **By Platform** or **By Server Group**.
3. Select a Gateway.
4. Under **Networking**, click **Aliases**.
5. In the **Aliases** window, do one of the following:
 - Click  **Upload from CSV** and browse to the CSV file contain the information (see [Importing Reverse Proxy Configuration From a CSV File](#)), or
 - Click  **New**.
 - In the **Alias** window, enter the data in the table **Alias Configuration Options** below.
 - Click **OK**.
 - Define additional aliases as required, or click **Close** in the **Aliases** window.

Alias Configuration Options

Field	Description
Alias Name	Enter an alias name, or select one from the dropdown menu.
Enabled	Specify whether the alias is enabled.

Field	Description
External Address	Select an IP address from the dropdown menu. Note: In a One Arm deployment, this IP address should be the same as the one selected for the Internal Address.
Internal Address	Select an IP address from the dropdown menu. Note: In a One Arm deployment, this IP address should be the same as the one selected for the External Address.
Comment	Optional explanatory text.

To delete an alias:

1. In the **Main** workspace, select Setup > Gateways.
2. In the **Filter** pane, select **By Mode**, **By Platform** or **By Server Group**.
3. Select a Gateway.
4. Under **Networking**, click **Aliases**.
5. In the **Aliases** window, select an alias.
6. Click  **Delete**.
7. Confirm the deletion.

Configuring a Route

This procedure describes how to configure a route.

To configure a route:

1. In the **Main** workspace, select Setup > Gateways.
2. In the **Filter** pane, select **By Mode**, **By Platform** or **By Server Group**.
3. Select a Gateway.
4. Under **Networking**, click **Routes**.
5. In the **Routes** window, click  **New**.
6. In the **Route** window, enter the data in the table **Route Configuration Options** below.
7. Click **OK**.
8. Define additional routes as required, or click **Close** in the **Routes** window.

Route Configuration Options

Field	Description
Type	From the dropdown menu, select Network or Host , specifying whether the route is per network or per a specific host.
Destination Address / Mask	If Type is Network, enter the IP address of the destination in CIDR notation. If Type is Host, enter the IP address of the destination (without CIDR).
Next Hop	Specify the IP address to which to all packets sent to Destination Address will be forwarded.
Virtual	Check this box if the IP address is virtual.
Network Interface	Optional: Select the network interface to use or leave blank. If left blank, the best available interface is selected when routing.
Virtual	If Virtual is checked, select the VR ID from the dropdown menu.
Metric	Optional: The metric to be used by the routing protocol to determine the route to be chosen when more than one route is available.

To delete a route:

1. In the **Main** workspace, select Setup > Gateways.
2. In the **Filter** pane, select **By Mode**, **By Platform** or **By Server Group**.
3. Select a Gateway.
4. Under **Networking**, click **Routes**.
5. In the **Routes** window, select a route.
6. Click  **Delete**.
7. Confirm the deletion.

Importing Reverse Proxy Configuration From a CSV File

When configuring Virtual Routers (see [Configuring a Virtual Router](#)) or Aliases, (see [Configuring an Alias](#)), you have the option of importing the data from a CSV file you prepared in advance (possibly originally in the form of a

spreadsheet).

Virtual Routers

The format of a CSV file for virtual routers is as follows:

Field Number	Field	Description
1	VR ID	The virtual router identifier.
2	Network Interface	for example, eth0, eth1, <i>etc.</i>
3	Primary IP Address in CIDR Format	
4	Default State	One of the following strings: "Master" "Backup"

Aliases

The format of a CSV file for aliases is as follows:

Field Number	Field	Description
1	Alias Name	
2	Enabled	0 or 1
3	External IP Address in CIDR	

Field Number	Field	Description
	format	
4	External IP Address Interface	for example, eth0, eth1, <i>etc.</i>
5	External IP Address Next Hop	
6	External IP Address Virtual Flag	0 or 1
7	External IP Address VR ID	
8	Internal IP Address in CIDR format	
9	Internal IP Address Interface	for example, eth0, eth1, <i>etc.</i>
10	Internal IP Address Next Hop	
11	Comment	



Note: When there is a request sent to the non existing page or the protected server is down, it returns blocking response and no alert is generated, as it is not an attack. For example, when the protected server is down, an error page is sent, but it does not indicate that the server is under attack.

Exporting Technical Information from Gateway

On occasion, when encountering trouble with SecureSphere operation, Imperva support may request that you provide them with technical information that is automatically generated by SecureSphere, so they can analyze logs and other information. You can export this information from the SecureSphere Gateway using the GUI. This procedure describes how to export tech info for a SecureSphere Gateway from the GUI. For information on exporting tech info for SecureSphere MXs, see [Exporting Technical Information from Management Servers](#).

To export technical information for a Gateway:

1. In the **Main** workspace, select **Setup > Gateways**.
2. In the **Gateways** pane, select the **Gateway** whose get tech info you want to export.



Note: The Gateway must be in the status Running.

3. In the **Details** pane, next to **Tech Info (zip)**, click **Download**.

SecureSphere prepares the information for download. Once complete a dialog box appears with a link. Click the link and download the zip file to the desired location. You can then mail it to Imperva support for analysis.

Licenses

Imperva SecureSphere requires a license. This chapter describes the SecureSphere licensing process and reviews the following subjects:

- [License Overview](#)
- [Uploading Licenses](#)
- [Viewing Licenses](#)

License Overview

A license contains the following information:

- Product information - a list of the products enabled by the license, for example, Web, DB, File, etc.
- Expiration date - for evaluation licenses only; non-evaluation licenses are not time-limited
- Maintenance period - specifying the period during which ADC content can be downloaded
- Gateway modules - the WAF gateways enabled by the license and counters for each module (specifying the number of modules enabled)
- Assessment counter - the number of scanned servers enabled

When a license is renewed, it is replaced by a new license.

Licenses for a SecureSphere Management Server and all the gateways it manages are installed on the Management Server.

If you are using a FlexProtect license, it contains the following information:

- License mode - a list of the products enabled by the license, for example, Web
- SecureSphere challenge - a unique identifier that correlates to a specific appliance
- Expiration date - the date when the uploaded license expires

Uploading Licenses

Uploading a license is performed in one of the following scenarios:

- After performing a first time login procedure and accessing the MX UI for the first time
- To update an existing license



Note: You cannot upload a DAM-only license or a DAM-only FlexProtect code.

To upload a license for the first time:

1. Connect to your MX via **https://<IP address of MX>:8083**. The **EULA** window is displayed.
2. Click **Accept**. The **First Time Login** window is displayed.
3. Enter a password for the admin user, retype it and select the UI display language.
4. Click **Login**. The **Upload License** window is displayed.

5. If you have a license file click **Choose File**, browse to the location of the file and select it.
6. If you do not have a license file, you can generate one by clicking the link called **here** which is located alongside the text **To generate a License Activation File**. The SecureSphere License Activation Portal web page opens.
 - a. Fill in the required fields and click **Activate** in order to request a license key be sent to your email by Imperva Support (requires an internet connection). The SecureSphere Challenge string for this MX appears in the **Upload License** window and is pre-populated if you click through to Imperva support.
 - b. Once you have a license file available, click **Choose File**, browse to the location of the file and select it.
7. If you are using a FlexProtect license and the MX is:
 - a. Not located behind a proxy:

Select **FlexProtect activation code** and enter the activation code you received in the email from Imperva.

Click **Upload**. The upload begins and the **Update License Information** progress bar is displayed.

When the status is reported as 100%, click **Continue**. The **Main** workspace is displayed.

- b. Located behind a proxy:

Connect to the MX via SSH.

Run the command `impctl server proxy --set --mx_user [MX User Name] --mx_password [MX Password] --enable --host [Host Name or IP address] --port [Port Number] --proxy_user [Proxy User Name] --proxy_password [Proxy Password] --auth_policy [Authentication Policy Type] --domain [Domain]`

Select **FlexProtect activation code** and enter the activation code you received in the email from Imperva.

Click **Upload**. The upload begins and the **Update License Information** progress bar is displayed.

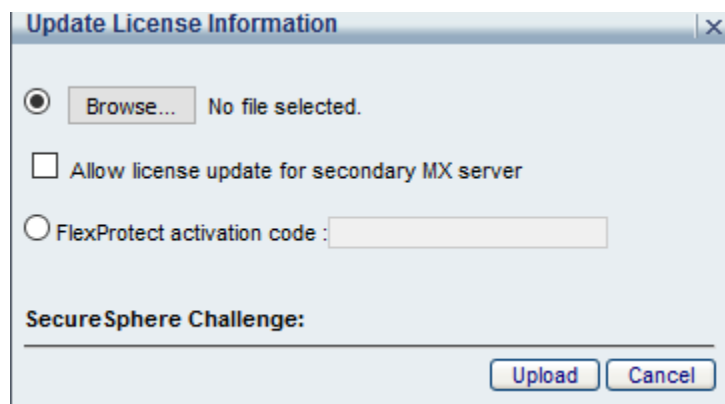
When the status is reported as 100%, click **Continue**. The **Main** workspace is displayed.

To update an existing license:



Important: If you are updating your license in order to change from License File to Flex Protect, you need to first follow the steps in the [How To: Precautions for going from License Files to Flex Protect](#) article to make sure that your system is in the proper state and will not cause any unnecessary outages.

1. In the **Admin** workspace, click **Licensing**.
2. Click **Actions**.
3. From the menu, select **Update License Information**.
4. In the **Update License Information** window, click **Browse** and navigate to the license file.



5. If this is an MX-HA deployment, enable **Allow license upload for secondary MX server**.
6. If you are using a FlexProtect license select **FlexProtect activation code** (instead of Browse) and enter the activation code you received in the email from Imperva.
7. Click **Upload**. The upload begins and the **Update License Information** progress bar is displayed.

When the status is reported as 100%, click **Continue**. The **Main** workspace is displayed.

Viewing Licenses

This procedure describes how to view licenses installed on SecureSphere.

To view all the installed licenses:

1. In the Admin workspace, click **Licensing**.

A summary of the installed licenses is displayed in the **License <license file name>** pane.

In the Licenses pane under **License Summary**, the following information is displayed:

- **Level:** Enterprise Edition.
- **Products:** A list of licensed products, for example, Web.

In the Licenses pane under **License Summary (GW modules)**, the following information is displayed for each gateway license:

- **Module:** The Imperva product, for example WAF.
- **Quantity:** The number of modules included in the license.
- **HW:** The gateway hardware model, for example, GV1000 or V1000.
- **File:** The name of the license file which contains this gateway license.
- **Expiration Date:** The date this gateway license expires.

In the Licenses pane under **License Summary (Cross System modules)**, the following information is displayed about the Imperva Customer Service support license, including ADC content:

- **Module:** For example, ADC Content, Maintenance, etc.
- **License Type:** There are two license types available:
 - Permanent (Perpetual or Commercial) license: This license is only given to a customer that bought the product. When this license expires, the system stays fully operational with the exception of not being able to load ADC content and not being able to upgrade until a valid license is loaded. The UI is not locked, and you can make configuration changes.
 - Evaluation (Time-based or Subscription) license: This license is for for evaluation purposes and is given to prospects, partners and existing customers. When this license expires, the system stops working (no alerts, events, blocking, access to GUI, configuration changes, new gateway registration, etc.).
- **Start Date:** The date from which the license is valid.
- **End Date:** The date the license expires.

In the Licenses pane under **License files**, the following information is displayed for each license file:

- **Expiry:** The license expiration date.
- **Challenge Key:** Since a license is valid for a single appliance, it needs to be verified. Challenge Key is the way to achieve this. Here you can see if the key is valid or not.
- **File Name:** The name of the license file.
- **License Type:** The type of license, for example, evaluation.
- **Issuing Date:** The date the license was issued.
- **Comments:** Other information.

2. Select a license file from among those displayed under License files.

The details of the licenses contained in the selected file are displayed in the right pane, under Details.

The screenshot shows the Imperva WAF Administration console. The 'Licenses' tab is selected in the top navigation bar. The left pane displays a table of licenses with columns for License ID, License Type, and Expiration Date. The right pane shows the details of a selected license, including License Name, Organization Name, License Type, License Version, License Start, License End, License Product, and License ID.

If you are using a FlexProtect license the information displayed is as shown in the figure below.

The screenshot shows the Imperva WAF Administration console. The 'Licenses' tab is selected in the top navigation bar. The left pane displays a table of licenses with columns for License ID, License Type, and Expiration Date. The right pane shows the details of a selected license, including License Name, Organization Name, License Type, License Version, License Start, License End, License Product, and License ID.

Users and Permissions

SecureSphere provides a robust permissions model that enables you to create new users, assign them one or more roles that grant permissions to objects based on those roles, then grant users additional permissions to specific objects in SecureSphere.

This section describes how to manage SecureSphere users and contains the following sections:

- [Understanding Permissions](#)
- [Permissions Workflow](#)
- [Working with Roles](#)
- [Working with Users](#)

Understanding Permissions

This section provides information about the mechanism for managing users and permissions and the impact of granting or removing permissions. It reviews the following subjects:

- [Permission Models in SecureSphere](#)
- [Roles and Users](#)
- [Permission Types](#)
- [Understanding the Permissions Window](#)
- [Adding and Removing Permissions](#)
- [Notes on Permission Behavior](#)

Permission Models in SecureSphere

It is recommended that before you get started in creating roles and users you first determine how to best match your user model with your workflow. There are three primary models that are recommended to understand how to match SecureSphere users with your organization.

Based on:	
Role Based	In this model, users are assigned a specific role and receive permissions for fulfilling this role. For example, perhaps a user is designated specifically to monitor a certain type of traffic. You would assign them the role of Web or DB Administrator based on what kind of traffic they are responsible for.

Based on:	
Product Line Based	In this model, a user may be responsible for all aspects of a specific product line, either Web based or database based products. For example, they may be responsible for the configuration of security policies for that product, the monitoring of traffic, and the creation of reports. You would then create a user and assign them the various role for that product, or create a new role for this purpose and configure it as required.
Specific Access Based	In some cases you may want to give targeted permissions to an outside user. For example, if you are an ISP and want to give your customers the ability to only view the specific web service and monitor that service, you would create a user for this purpose, and only grant him permissions to view the specific objects related to the service. In some cases you may want to give targeted permissions to a user. In that case, you need to create a user for this purpose, and grant him only specific permissions.

Roles and Users

The permissions model in SecureSphere is based on the following primary types of permissions:

- **Roles:** Represents a role in an organization, such as a Web Security Administrator and includes a set of permissions specified for that role. SecureSphere comes with a set of pre-defined roles.
For information on roles see [Working with Roles](#)
- For a list of predefined roles and their descriptions see [PreDefined Roles](#)
- **External Roles:** Represents a group of users defined in an external authorization system (currently Active Directory only). In SecureSphere these roles are mapped to the roles in external systems and can receive SecureSphere roles and permissions, similarly to SecureSphere users. For more information on external roles see [Configuring Integration with Active Directory for User Management](#).
- **Users:** Represents an actual user in SecureSphere to whom permissions are granted. When users are created, they are assigned one or more roles, granting them the permissions associated with those roles, and can additionally be assigned permissions to specific users. For more information on users see [Working with Users](#)
- **External Users:** A user that is defined in the external authorization system (currently Active Directory only) and in SecureSphere this user is created automatically during user's first login. Such users cannot be deleted or assigned roles in SecureSphere, only in the external system. External users can receive SecureSphere permissions, granted explicitly by administrator, or automatically when it creates objects. For more information on external users see [Configuring Integration with Active Directory for User Management](#).

Permission Types

There are three primary types of permissions that users can be granted to various SecureSphere components. These include the following:

Based on:	Description
View	Enables users to only view the specific object. Though users cannot edit their parameters nor create new objects. If a user doesn't have "view" permission to an object, they can still see the object's name.
Edit	Enables users to view a specific object, edit the details of an object (asset configuration, policy rules etc.), but not create new objects.
Create	Enables users to create objects (per category). These users then automatically receive edit and view permissions for this object. A category Subsystem on SOM controls all the permissions that SOM users are granted to work with MXs on that SOM.

In addition to the permissions mentioned above, all users can use all objects available in SecureSphere such as policies, global objects, action sets, etc., on assets they have permissions to. For example, if a user has edit permissions to a specific web service, they can apply any existing policy to that web server, even if they don't have view permissions on the policy.

Working with the Subsystem category objects on SOM:

- When you have **View** permissions, you can perform the following with MXs registered to that SOM:
 - Viewing MXs in the **Registered MXs** window.
 - Viewing MXs in the **Apply to MXs** pane that presents the items that can be downloaded to MXs.
 - Viewing the statistical data from that MX in the **Dashboard**.
 - Drilldown to that MX.
- When you have **Create** permission for MXs, you can create MXs.
- When you have **Edit** permissions, you can perform the following with MXs registered to that SOM:
 - Editing MX properties.
 - Registering an MX, deleting an MX and test connection to that MX.
 - Downloading items to that MX.

Understanding the Permissions Window

This topic provides an overview of the Permissions window.

To access permissions:

1. From the **Admin** workspace, click **Users & Permissions**.
2. Select a user or group. For information on how to create users, see [Creating a SecureSphere User](#).

The screenshot shows the 'Users & Roles' window with the 'admin' user selected. The 'Permissions' tab is active, displaying a list of roles and a table of permissions.

Roles

Available

- DB Security Admin
- Directory Service Security Admin
- File Security Admin
- SharePoint Security Admin
- Web Security Admin

Assigned

- Administrator

Permissions

Read Only: ☐

Category				Effective		
	View	Edit	Create	View	Edit	Create
Site Tree Object Types	☐	☐	☐	✓	✓	✓
Policy Types	☐	☐	☐	✓	✓	✓
Scan Types	☐	☐	☐	✓	✓	✓
Subsystems	☐	☐	☐	✓	✓	—
Global Objects	☐	☐	☐	✓	✓	✓

Adding and Removing Permissions

Adding and removing permissions to objects in the permissions window is accomplished by selecting or deselecting the relevant checkbox then clicking Save. Permissions are granted to the specific object or category selected.



Note: When permissions are granted to categories, they automatically cascade to all objects in that category. The category object is selected, though the its objects appear disabled.

Notes on Permission Behavior

Granting or revoking permissions in SecureSphere affects various permission objects in different ways. The below provides some important information regarding permission behavior to assist in better understanding and managing permissions.

- The default user is "administrator". The Administrator user has create, view and edit permissions to the entire SecureSphere system
- Only Administrators or users assigned the **Administrator** role, have access to the **Admin** workspace
- A regular user has edit permission for any objects they create, except if an Administrator later revokes those permissions
- When enabling Edit permissions, users are automatically granted view permissions

Permissions Workflow

The basic process of creating users and assigning permissions includes the following primary steps:

#	Action	Description	For more information, see...
1	Create or customize roles as desired	Customize existing roles or create new roles to match your organization's needs.	Working with Roles
2	Create a new user	Create a new user to represent a SecureSphere user that needs access.	Creating a SecureSphere User
3	Assign roles to the user	Assign the desired roles to the user that match that user's responsibilities.	Assigning Roles to Users
4	Configure object specific permissions	Configure any additional permissions for specific objects the user or role might require, if applicable.	Configuring Object or Category Level Permissions
5	Configure navigation and activate settings permissions	Configure permissions for access to views and activating settings.	Configuring Navigation Permissions and Activate Settings

#	Action	Description	For more information, see...
6	Configure user information	Configure User information including phone, e-mail, department and more.	Configuring User Information



Note: In addition to the regular SecureSphere users, you can set permissions for the Active Directory users. For more information about integration with Active Directory, see **Configuring Integration with Active Directory** in the **SecureSphere User Guide**.

Working with Roles

Roles in SecureSphere reflect a role in your organization and contain a specific set of permissions required to perform that role. SecureSphere also contains a set of pre-defined roles which can be used.

Additionally, you can create new roles and customize them to match your company's workflow. The following subjects review various aspects of working with Roles:

- [PreDefined Roles](#)
- [Custom Role Examples](#)
- [Creating a Role](#)

PreDefined Roles

SecureSphere comes with a set of pre-defined roles to assist you in easily configuring users with the level of permissions required to fulfill their roles.



Note: Predefined roles cannot be modified.

The following is a list of these pre-defined roles and a short description of each:

Pre-Defined Roles

Administrator	Main administrator user for SecureSphere. Assigned when first logging into SecureSphere. This user is responsible for managing SecureSphere and has permissions to all features and functionality.
Web/DB/File/SharePoint Security Admin	Security Admins are responsible for all the security aspects that deals with a specific product. For example, the Web Security Admin can create and edit services and applications, security policies and all global objects relevant to web products.

Custom Role Examples

The following are examples of custom roles that can be created to match a specific set of requirements.

Example 1: Managed Security Service Provider (MSSP)

MSSPs often offer third party solutions to their customers, and may want to provide some level of access to these products. In this case, you would want to provide view permissions to your customers for only those SecureSphere components that represent their specific services and applications, and provides under special permissions to the alerts screen. They would not obtain permissions to edit assets, policies, or global objects.

This can be done in different ways. For example, if you have 10 customers, each with only 1 user, you may want to create the user separately for each customer. Alternatively, you may want to create a role that would give navigation permissions to items such as monitoring, create individual users and include this role, then customize the user with permissions to view their specific assets.

Example 2: Web Services Engineer

If your organization utilizes a compartmentalized model such as a bank in which specific web services engineers should only have access to view details about specific services, you may want to:

- Create a new Role based the existing Web Security Admin
- Remove the Web Services and Applications that appear under Site Tree Object Type category
- Access each individual user, and under Web and Application Services (under site tree object types) click the edit button for these items, and select only those web services and applications you want the specific user to have access to.

Example 3: Audit Expert

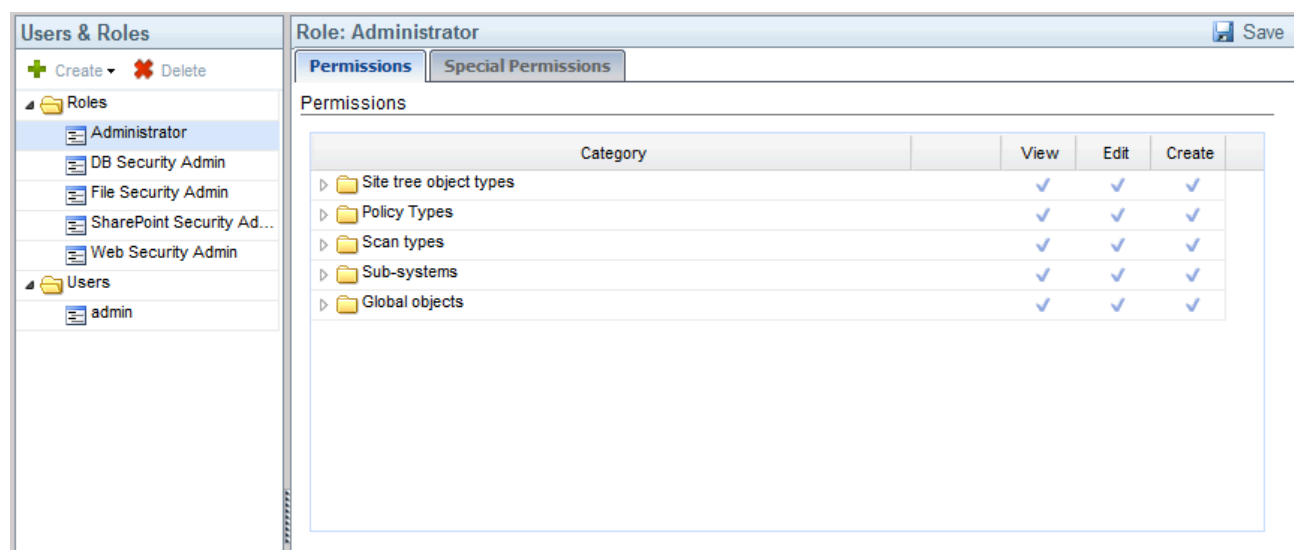
In some organizations there is a designated audit expert that creates audit policies, defines what data is audited, and how this data is handled. You may want to create a special Role for users who are responsible for creating these policies, but which prevent them from seeing actual audit data. In this case you could create a user (or role if there is more than one user with these responsibilities) which would only have create and edit permissions on audit policies. You may also want to provide them view permissions on a test database, which would enable them to test their policy as they are working on it, without exposing sensitive data.

Creating a Role

SecureSphere comes with a number of pre-defined roles. You can additionally create custom roles to better match your company's environment.

To create a Role:

1. In the **Admin** workspace, select Users and Permissions.
2. In the **Users & Roles** pane, click , then **Create New Role**. The **Create New Role** window appears.
3. Type a **Name** for the role, then click **Create**. The new role is created and appears under **Roles** in the **Users and Roles** pane to the left-hand side of the window.



4. Configure this role with the desired permissions by selecting the role and defining permissions. When defining permissions in the **Permissions** pane, the same permissions are added to all the objects under that category.
5. To define different permissions for each object under the same category, click . The **Object Level Permissions** window appears.
6. Define the permissions that you want for each item in the **Object Level Permissions** list.

Working with Users

There are a number of aspects that are involved with working with users. They include the following:

- Creating a SecureSphere User
- Assigning Roles to Users
- Configuring Object or Category Level Permissions
- Configuring Navigation Permissions
- Configuring the Authorization to Activate Settings
- Authorizing Roles and Users to Display File Classification Matched Text
- Configuring User Information
- Locking a User Account

- Resetting a User Password
- Read-Only User

Creating a SecureSphere User

Users in SecureSphere define a number of parameters that include roles to which a specific user is assigned, permissions to various SecureSphere objects ranging from site tree objects to global object, special permissions that provide access to views, and user information that define their full name, e-mail address, geographic location in the world, and more.



Note: This procedure does not apply if the **User Authentication** is set to **External** and if **External Authorization** is selected in the **Authentication & Authorization Configuration**. For more information, see [Authentication and Authorization Configuration](#).

To add a user if the **User Authentication** is set to **External** and if **External Authorization** is selected, see [Using External Roles to Add External Users](#).

To create a SecureSphere User:

1. In the **Admin** workspace, select Users and Permissions.
2. In the **Users & Roles** pane, click , then **Create New User**. The Create New User window appears.
3. Type a **Name** for the user.
4. In **Authenticator**, select **SecureSphere** or **External** as appropriate.
 - **SecureSphere:** The user is authenticated by SecureSphere using a password (entered in **Password** below).
 - **External:** The user is authenticated using an external service, such as LDAP.

In this case, the **Enter Password** and **Re-Enter Password fields** are ignored by SecureSphere, and the user is required to provide the password defined in the LDAP server.

The user's name in SecureSphere must be identical to the user's CN in LDAP. Case sensitivity is enforced, that is, "Mary" is not identical to "mary".

For information about configuring SecureSphere to work with LDAP, see [External Systems - LDAP](#).

If **Authenticator** is **SecureSphere**, enter the user's password in **Enter Password** and again in **Re-Enter Password**.



Note:

Make sure your password has the following characteristics:

- It must have no fewer than 10 characters and no more than 14 characters.
- It must have at least one number, one capital letter, and one special character from:

* + = # % ^ : / ~ . , [] _

- It cannot have more than two characters repeated in succession.

5. If **Authenticator** is **LDAP**, you can lookup the user in the LDAP database by clicking **Lookup User**. However, see the Note in the introductory section.

If **External Authorization** is not selected in the **Authentication & Authorization Configuration** window (see [Authentication and Authorization Configuration](#)), the user's SecureSphere password is used. If **External Authorization** is selected, you are prompted for the LDAP password.

6. Add one or more roles you want to assign the user by moving the desired roles to the Assigned pane. A role is a set of permissions to different objects. For more information on roles see [Working with Roles](#)
7. Click **Create**. The new user is created and appears under **Users** in the **Users and Roles** pane to the left-hand side of the window.

You may now configure additional permissions for that user in addition to any permissions they are granted through roles. For more information see [Configuring Object or Category Level Permissions](#).

Assigning Roles to Users

SecureSphere enables you to configure new users with roles and assign roles to existing users. By assigning roles to users you grant them the permissions that have been configured for the assigned roles.



Note: This procedure describes how to assign roles to existing users. Assigning roles to new users is part of the process of creating a new user. For information on assigning roles to new users, see [Creating a SecureSphere User](#).

To assign a role to existing users:

1. In the **Admin** workspace, select Users and Permissions.
2. In the **Users & Roles** pane, expand the **Users** category, then select the user you want to assign a new role. That user's details are displayed in the details pane.
3. On the permissions tab, move the role you want to assign the user from the **Selected** pane to the **Assigned** pane.



Note: For information on creating new roles, see [Creating a Role](#).

4. Click **Save**. Settings are saved and the user has now been granted the permissions available in the assigned role.

Configuring Object or Category Level Permissions

In addition to assigning users to a role, you have the option to assign them permissions to specific objects or categories of objects which may be relevant for their role or access model. For example, perhaps you are an ISP that provides SecureSphere as a paid service to your customers. And perhaps you'd like to give your customers only view access and only to their own web server. You could create a new user for each customer, and assign them additional (specific) view permissions for their web server.



Note: While you can add permissions on top of permissions provided through a predefined roles, you cannot revoke permissions that are part of these roles by default.

To configure additional permissions:

1. In the **Admin** workspace, select Users and Permissions.
2. In the **Users & Roles** pane, expand the **Users** category, then select the user you want to assign additional permissions. That users details are displayed in the details pane.
3. In the **Permissions** pane, navigate to the category and object you want to grant permissions to. For example, if you want to assign additional permissions to a specific policy, next to the policy category, click **Edit** .
4. Select the desired object level permissions. The item is selected, and the item representing permissions is displayed. For more information on how permissions are represented in the **Permissions** tab, see [Understanding How Permissions are Displayed](#).
5. In the **Object Level Permissions** window click **Save**.
6. In the **Users and Roles** window from the upper right-hand corner of the screen. Click **Save**.
7. Your changes have been saved and the additional permissions have been assigned.

- [Understanding How Permissions are Displayed](#)

Understanding How Permissions are Displayed

The way that permissions are displayed for a user depends on what permissions have been configured on a specific object. Permissions have three possible levels as displayed in the permission window.

Permission Levels

	Type	Description
☐	None	None of that category's object permissions have been assigned to the current user.
—	Partial	Some of the category's object permissions have been assigned to the current users.
✓	Full	All of the category's object permissions have been assigned to the current users.

Object View vs. Sum View

Permissions are displayed in SecureSphere on two levels:

- **Object View Permissions:** Displays permissions for the specific object only. For example, if a user has permissions to all Cross System Related Global objects as seen in the example below. Checkmarks are displayed for that category. Yet for the Global Objects category, you can see that hyphen symbols (partial) are displayed because this user does not have permissions to the other objects in that category.
- **Sum View Permissions:** Displays the sum of all permissions, both Object Level permissions, as well as permissions provided to users by roles they have been granted. So in the example below you can see that many of the objects the users have full permissions in Sum View, while under Object Permissions for the same items they only have partial permissions. This is due to the fact they have been granted additional permissions through their role which result in the user having all permissions for that object.

Permissions

		Object Permissions			Sum View Permissions		
Category		View	Edit	Create	E-View	E-Edit	E-Create
▶ Site tree object types		—	☐	☐	✓	✓	✓
▶ Policy Types		—	☐	☐	✓	✓	✓
▶ Scan types		—	☐	—	✓	✓	✓
▶ Sub-systems		☐	☐	☐	✓	✓	☐
▲ Global objects		—	—	—	✓	✓	✓
Cross System Related Objects		✓	✓	✓	✓	✓	✓
DB Related Objects		☐	☐	☐	✓	✓	✓
File Related Objects		☐	☐	☐	✓	✓	✓
Web Related Objects		☐	☐	☐	✓	✓	✓

Configuring Navigation Permissions

In the Navigation tab you can define the navigation permissions to the enabled GUI windows.

For example, when you give a user permissions to access an agent, that user has access permissions to all data and all GUI windows pertinent to that agent - assuming no limitations have been placed on that user's navigation permissions. There are actions that the user can do on the agent's screen, like alter its configuration. However, the same agent appears in the software update screen thereby enabling the user to install the agent on a remote database or file server. This is an undesirable situation - you do not want the user with that permission to see the associated software update GUI window, and have access to all its functionality. By configuring the navigation permissions, you can define precisely which GUI windows can be seen by any user or role.



Note: An example of navigation permission might be that you want to allow a user to edit a certain database service but not see audit results. You would then give edit permissions on the service, but you wouldn't give navigation permissions on the audit screen.

A new user receives no navigation permissions by default. You must set navigation permissions either directly, or by assigning the user to a role that has navigation permissions.

To configure navigation permissions:

1. In the **Admin** workspace, select **Users and Permissions**.
2. In the **Users & Roles** pane, expand the **Users** category, then select the user or role to whom you want to assign navigation permissions. The users permissions of the selected user are displayed in the details pane. In the case of roles, only the **Access** column appears.
3. Click the **Navigation** tab. The relevant GUI windows categories and screens, together with their options, appear.

Role: DB Security Admin	
Permissions	Navigation
Screen	Access
▶ Discovery & Classification	—
▶ Setup	✓
▶ Profile	✓
▶ Risk Management	—
▶ Policy	✓
▶ Audit	—
▶ Reports	✓
▶ View Results	✓
▶ Reports	✓
▶ Monitor	—
▶ Dashboard	☐
▶ Alerts	✓
▶ Violations	✓
▶ System Events	✓
▶ Blocked Sources	✓
▶ Gateway Management	☐
▶ ThreatRadar	✓
▶ Admin	☐

You can expand one or more categories to view the precise GUI windows of those categories.

Notes:



- The **Access** column is for adding permissions. The **Effective** column displays the default permissions, based on the role.
- A check mark signifies that the permission exists and cannot be removed. An empty check box signifies a non-existent permission that can be added. A dash mark appears only on categories and means that the windows in that category are a mixture of those with existing permissions, and those without.

4. To enable or disable permissions for any category or GUI window, check the appropriate options.
5. Click **Save**. Your settings are saved.

Configuring the Authorization to Activate Settings

A user for whom permission is given to use Activate Settings can modify the SecureSphere database by making the changes and clicking the Activate button. For any user, you can define under which conditions the Activate button will function, using the Activate Settings window.

To configure activate settings:

1. In the **Admin** workspace, select **Users & Permissions**.
2. In the **Users & Roles** pane, expand the **Users** category or the **Roles** category, then select the user or role for whom you want to configure activate settings.
3. In the **Global Settings** tab, select one of the **Activate Settings** options in the following table.
4. Click **Save**. Your settings are saved.

Activate Settings Options

Parameter	Description
The user is not authorized to perform Activate Settings.	The user does not have permission to use Activate Settings.
The user may only perform Activate Settings if authorized to view all changes that will be activated.	The user is allowed to perform Activate Settings only if he has permission to view all the parameters whose settings will be changed when Activate Settings is performed.
The user is authorized to perform Activate Settings unconditionally.	There are no restrictions on the user in regard to the use of Activate Settings.

Authorizing Roles and Users to Display File Classification Matched Text

When a data type is applied to an object based on a file content classification rule, the **File Explorer** can display the text in the object that matches the rule criteria. Since the matched text may include sensitive data, the administrator may want to restrict the ability to display matched text to specific SecureSphere roles or users.

To authorize SecureSphere roles and users to display matched text from file content classification in the File Explorer:

1. In the **Admin** workspace, select **Users & Permissions**.
2. In the **Users & Roles** pane on the left, expand the **Users** category or the **Roles** category, and select a user or role.
3. In the **User** or **Role** pane on the right, on the **Global Settings** tab, select the checkbox for **The user is authorized to view File Classification matched text**.
4. Click **Save**.

Your settings are saved.

Configuring User Information

When creating users it is recommended that you configure all the relevant details to identify that user. This can assist in communicating directly with a user, identifying to what department they belong, knowing their geographic location, and more.

To configure User Information:

1. In the **Admin** workspace, select Users and Permissions.
2. In the **Users & Roles** pane, expand the **Users** category, then select the user you whose information you want to edit. That users permissions are displayed in the details pane.
3. Click the **User Information** tab. User Information is displayed.

The screenshot shows a web interface for configuring user information. At the top, there's a header "User: db". Below it are four tabs: "Permissions", "Navigation", "Global Settings", and "User Information", with "User Information" being the active tab. The form contains several fields: "Username:" with a text box containing "db"; "Full Name:", "Email:", "Phone:", "Title:", and "Location:" each with empty text boxes; "Department:" with an empty text box; "Last Login:" with a text box containing "11/21/13 12:05 PM"; "Password Updated:" with a text box containing "11/21/13 12:05 PM"; "Account Enabled:" with a checked checkbox; and "Authenticator:" with two radio buttons, "SecureSphere" (selected) and "External". At the bottom right, there are two buttons: "Reset Password" and "User Lookup".

The **Username**, **Last login**, **Password Updated** fields are informative and cannot be changed in this window. You can enter data into the other fields without restriction.

The following fields are displayed only when a user is not **admin**: **Account Enabled**, **Authenticator** (with **SecureSphere** and **External** options), and **User Lookup** (available when **Authenticator** is **External**). For more information on **Authenticator** and **User Lookup**, see [Creating a SecureSphere User](#).

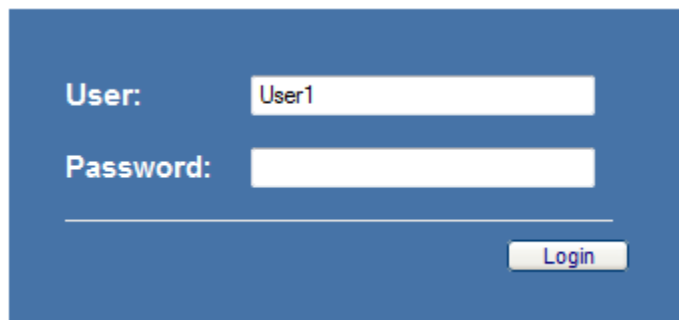


Note: Enable or disable a user's account by selecting or deselecting **Account Enabled**. This field is useful for temporarily disabling a user without having to delete and later re-create the user.

Locking a User Account

A user must type a user name and password to log into the SecureSphere GUI.

If a user types an incorrect password when trying to log in, the user name continues to be displayed and the password field is cleared. The user can try to type a password again.

A screenshot of the SecureSphere login interface. It features a blue background with white text. The 'User:' label is followed by a text input field containing 'User1'. Below it, the 'Password:' label is followed by an empty text input field. A 'Login' button is positioned at the bottom right of the form.

If a user types an incorrect password 3 times within five minutes, SecureSphere locks the user account for 30 minutes. When the user account is locked, the user cannot log into SecureSphere even with the correct password.



Note: To modify the default values related to locking a user account, see [User Lockout Settings](#).

To access a user account after it is locked:

- Thirty minutes after the user account was locked, the user must type the correct password.
- If the user forgot the password, the user must:
 - Ask the SecureSphere administrator to reset the password. See [Resetting a User Password](#)
 - Thirty minutes after the user account was locked, type the new password received from the SecureSphere administrator. See [Using a Reset User Password](#)

Resetting a User Password

An administrator can reset the password for a user account. This can assist users who forget their password.

To reset a user password:

1. In the **Admin** workspace, select Users and Permissions.
2. In the **Users & Roles** pane, expand the **Users** category, and then select the user whose password you want to reset.
3. In the details pane, click the **User Information** tab and then click **Reset Password**.

User: User1

Permissions	Navigation	Activate Settings	User Information
-------------	------------	-------------------	------------------

Username:

Full Name:

Email:

Phone:

Title:

Location:

Department:

Last Login:

Password Updated:

Account Enabled: ☒

Authenticator: ☒ SecureSphere ☐ External

The **Reset Password** window appears.

Reset Password [X]

Your Password:

User's New Password:

Confirm New Password:

4. In the **Reset Password** window:

- In the **Your Password** text box, type your SecureSphere administrator password.
- In the **User's New Password** text box, type a new password for the user. The password must conform to the restrictions defined in [Password Settings](#) regarding its length and allowed characters.
- In the **Confirm New Password** text box, type the new user password again.
- Click **Change** to save the new user password.



Note: When the user logs in using the reset password, the user must define a new password. See [Using a Reset User Password](#).

Using a Reset User Password

The first time that a user logs into SecureSphere using a password that was reset by a SecureSphere Administrator, the user must replace the reset password with a new password.

To use a reset user password:

1. In the SecureSphere GUI login window, type the new password from the SecureSphere administrator and click **Login**. A window appears for defining a new password.

The image shows a login window with a blue background. It contains four text input fields stacked vertically, each with a label to its left: 'User:', 'Old Password:', 'Password:', and 'Retype Password:'. The 'User' field contains the text 'User1'. Below the fields is a 'Login' button. A horizontal line separates the 'Retype Password' field from the 'Login' button.

2. In the **Old Password** text box, type the password from the SecureSphere administrator.
3. In the **Password** text box, type a new password. See the Note below.
4. In the **Retype Password** text box, type the new password again.
5. Click **Login** to log into SecureSphere.

Note:

Make sure your password has the following characteristics:



- It must have no fewer than 10 characters and no more than 14 characters.
- It must have at least one number, one capital letter, and one special character from:

* + = # % ^ : / ~ . , [] _

- It cannot have more than two characters repeated in succession.

Read-Only User

Read-only user can use SecureSphere for viewing items only. This user cannot add or change anything in the system. You can manage read-only user's permissions by defining which items in the system this user can view.

Admin users cannot be defined as read-only.

You can define a SOM user as read-only. Such user cannot navigate to the Admin tab or drill down to MXs from any location in the SOM GUI. When a SOM read-only user enters MX, this user acts on MX as a read-only user as well.

Additional aspects of working with the read-only user:

- **Roles:** When assigning roles to read only users, these users inherit only the View permissions from these roles. You cannot set a role as read-only, only external roles, see [Working with Roles](#).
- **Tasks:** Read-only user can open and edit tasks that this user is assigned to.
- **Preferences:** Read-only user can edit the preferences.
- **Save as:** A read-only user can save data as PDF or as CSV.
- **Reports:** A read-only user can run existing reports.



Note: The default system admin cannot be defined as read-only user. You can define additional admin users in the system and set this user to read-only. By that way you are able to easily add a new user with all the possible permissions and then by setting this user as read-only, you get a user that can view everything and cannot change anything.

SOM-MX Issues for Read-Only Users

You can define a SOM user as read-only. Such user cannot navigate to the Admin tab or drill down to any options under Admin. When a SOM read-only user enters MX, this user acts on MX as a read-only user as well.

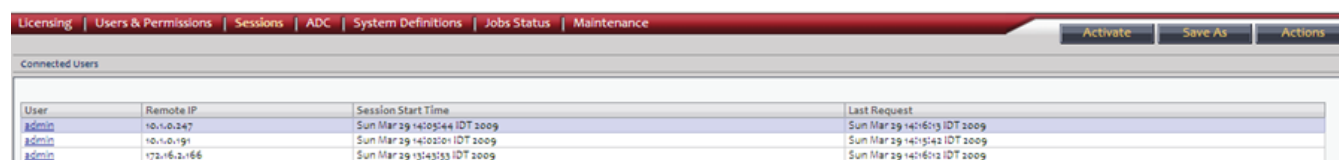
Sessions

This chapter describes the **Connected Users** window, which displays information about the users currently connected to the SecureSphere GUI, and includes the following:

- **Connected Users**

Connected Users

The **Connected Users** pane displays information about the users currently connected to the SecureSphere GUI (have open sessions).



User	Remote IP	Session Start Time	Last Request
admin	10.1.0.247	Sun Mar 29 14:05:44 IDT 2009	Sun Mar 29 14:16:19 IDT 2009
admin	10.1.0.191	Sun Mar 29 14:02:01 IDT 2009	Sun Mar 29 14:15:42 IDT 2009
admin	172.16.2.166	Sun Mar 29 13:43:33 IDT 2009	Sun Mar 29 14:16:12 IDT 2009

For each user, the following information is displayed:

- **User:** The username.
- **Remote IP:** The IP address from which the user is connected.
- **Session Start Time:** The date and time the session started.
- **Last Request:** The date and time of the last action performed by the user.

The last action may be something as simple as displaying another tab in the GUI.

- **Sessions**

Sessions

Administrators can terminate (kill) sessions and disable users that are connected to the MX.

To kill a session:

1. In the **Admin** workspace, click **Sessions**.
2. In the **Connected Users** pane, select a user.
3. Click **Actions**.
4. From the menu, select **Kill User Session(s)**.

All the user's sessions will be killed.



Note: As an administrator, you cannot kill your own sessions or disable yourself.

To disable a user and kill all of the user's sessions:

1. In the **Admin** workspace, click **Sessions**.
2. In the **Connected Users** pane, select a user.
3. Click **Actions**.
4. From the menu, select **Kill User Session(s) and Disable User**.

All the user's sessions will be killed and the user will be logged off. The user will not be able to login again until they are enabled (see [Working with Users](#)).

ADC

This section describes Imperva's Application Defense Center (ADC).

The Imperva Application Defense Center (ADC) is a premier research organization for security analysis, vulnerability discovery and compliance expertise. ADC research combines extensive lab work with hands-on testing in real world environments to ensure that Imperva's products, through advanced data security technology, deliver up-to-date threat protection and unparalleled compliance automation.

Having discovered dozens of commercial application vulnerabilities and having issued numerous security advisories, the ADC offers exceptional insight into both published and unpublished security threats.

ADC research not only provides an indispensable service to commercial application and database vendors and to security professionals, it also delivers the foundation for many Imperva SecureSphere features and services, including attack signature updates, database vulnerability assessments, and pre-defined compliance reports.

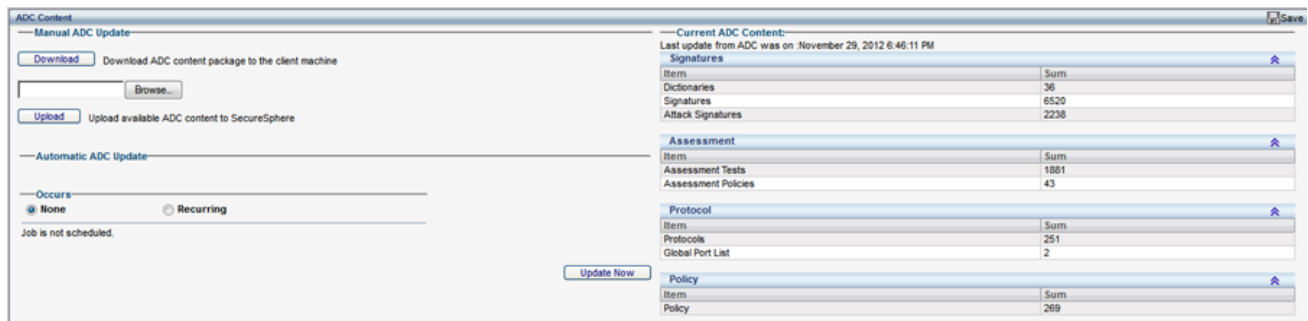
You can view and update ADC content, and view a history of ADC system events.

- [Viewing ADC Content and Status](#)
- [Updating ADC Content](#)
- [Viewing ADC History](#)

Viewing ADC Content and Status

To view the status of ADC updates:

1. In the **Admin** workspace, click **ADC**.



2. In the **ADC Content** window, information about the ADC content last downloaded is displayed in the right half of the window, under **Current ADC Content**.

The number of ADC Content items in each category is given under **Sum**.

To hide the list of items in each category, click  (at the right end).

To expand a hidden list, click .

You can view the ADC content in the SecureSphere GUI, **as listed below**.

ADC Content

Category	Item	Viewing the ADC Content for this category
Signatures	Dictionaries	In the Main workspace, select Setup > Signatures . The ADC dictionaries are listed under Predefined Dictionaries in the Filter pane.
	Signatures	In the Main workspace, select Setup > Signatures . Select a dictionary from those listed under the Predefined Dictionaries folder in the Dictionaries pane. The signatures belonging to the selected dictionary are displayed in the middle pane.
	Attack Signatures	In the Main workspace, select Setup > Signatures . Select a dictionary from those listed under the Predefined Dictionaries folder in the Dictionaries pane. Select a signature in the middle pane and in the Details pane, the attack signature associated with this signature is shown in the Class field in the Matching Target tab.
Assessment	Assessment Policies	In the Main workspace, click Risk Management > DB Assessment Policies . Assessment Policies are displayed.
	Assessment Tests	In the Main workspace, click Risk Management > DB Assessment Policies . Then select a policy from the Assessment Policies pane. The tests belonging to that policy are displayed in the Test List tab of the rightmost pane.
Protocol	Protocols	In the Main workspace, select Setup > Signatures . Select a dictionary from those listed under the Predefined Dictionaries folder in the Dictionaries pane. Select a signature in the middle pane and the protocols associated with this signature are shown in the Signature tab in the Details pane.
	Global Port List	In the Main workspace, select Setup > Global Objects . Under Scope Selection , select Port Names . Note: The number given under Sum for Global Port List

Category	Item	Viewing the ADC Content for this category
		is the number of lists, not the total number of ports in the lists.
Policy	Policies	The various ADC policies can be viewed in several different GUI locations. For example, to view the SAP-related ADC security policies, select Main > Policies and click SAP in the Filter pane (under By ADC Keywords). The SAP-related ADC security policies are displayed in the Policies pane.
Report	Report	In the Main workspace, select Reports > Manage Reports and, in the Filter pane, select ADC under Defined By .
Stored Procedures	Stored Procedures	In the Main workspace, select Setup > Global Objects . Under Scope Selection , select DB Related Objects > Stored Procedure Groups > Built In .
Sql Table Groups	Table Groups	In the Main workspace, select Setup > Global Objects . Under Scope Selection , select DB Related Objects > Database Table Groups .
Sql Default Users	Default Users	In the Main workspace, select Setup > Global Objects . Under Scope Selection , select SQL User Tracking Groups . The ADC defined SQL User Tracking Groups are shown under the Built In folder in the Globals Tree pane.

Updating ADC Content

ADC content can be updated manually or automatically, on a scheduled basis.

Notes:



- For information about using a proxy for ADC updates, see [External HTTP Settings](#).
- For instructions on how to download the adc content package, see the article [ADC--Where can I download ADC content when the MX server is not directly](#)

connected to Internet in the Knowledge Base (requires login).

To manually update ADC content:

1. In the **Admin** workspace, click **ADC**.
2. Under **Manual ADC Update**, click **Download**.
3. Save the MPRV file that will be downloaded.
4. Browse to the downloaded file using the **Browse** button.
5. Click **Upload**.

SecureSphere will be updated with the downloaded content.



Note: Some of the updates will not take effect until you Apply Settings.

To schedule automatic updating of ADC content:

1. In the **Admin** workspace, click **ADC**.
2. Under **Automatic ADC Update**, select **Recurring**.
3. Define the schedule for automatic ADC updates.
4. To update ADC content immediately, click **Update Now**.



Note: For ADC update to succeed, the Default Gateway and the DNS server must be set using `impcfg`.

Viewing ADC History

To view the history of ADC-related events:

1. In the **Main** workspace, select **Monitor > System Events**.
2. In the **Basic Filter** pane, under **By SubSystem**, select **ADC**.

ADC History is displayed.

imperva

Discovery & Classification Setup Profile Risk Management Policies Audit Reports Monitor

Dashboard Alerts Violations System Events Blocked Sources Monitor System

Basic Filter Saved Filters

Filter

By Severity

By SubSystem

ADC

Agent

Alerts

Appliance

Archive

Assessment

Audit

Classifier

Config

Discovery

Entitlement

File Classification

File User Rights

Gateway

Info

System Events

Event Type

User

Event Time

Subsystem

Severity

Message

14

Page

1

of 60

Event Type

User

Event Time

Subsystem

Severity

Message

Subsystem: adc (10)

Old ADC content

System

Jul 23, 2023 1:00 AM

ADC

Medium

The ADC content has not been updated for the last 31 d

Active modules

admin

Jun 21, 2023 9:05 AM

ADC

Low

Install of New Web Vulnerability Management has succe

ADC download succeeded

admin

Jun 21, 2023 9:05 AM

ADC

Medium

ADC content upload has succeeded.

Active modules

admin

Jun 21, 2023 8:55 AM

ADC

Low

Install of New Web Vulnerability Management started.

Old ADC content

System

Apr 14, 2023 1:00 AM

ADC

Medium

The ADC content has not been updated for the last 31 d

Active modules

admin

Mar 13, 2023 2:23 PM

ADC

Low

Install of Track Value Changes has succeeded

Active modules

admin

Mar 13, 2023 2:23 PM

ADC

Low

Install of Change Tracking - External Objects has succee

ADC download succeeded

admin

Mar 13, 2023 2:23 PM

ADC

Medium

ADC content upload has succeeded.

Active modules

admin

Mar 13, 2023 2:13 PM

ADC

Low

Install of Track Value Changes started.

Active modules

admin

Mar 13, 2023 2:13 PM

ADC

Low

Install of Change Tracking - External Objects started.

Subsystem: appliance (2)

Hardware monitor failure

System

Mar 15, 2023 4:21 PM

Appliance

High

Failed to retrieve hardware monitoring data for MX

Appliance became disconnected

System

Mar 15, 2023 4:18 PM

Appliance

High

Gateway ShiraSetup412_gw1 became disconnected.

Subsystem: audit (38)

Audit fast viewing ended successfully

System

Sep 5, 2023 2:00 AM

Audit

Low

Audit fast viewing data was generated

Audit fast viewing ended successfully

System

Sep 4, 2023 2:00 AM

Audit

Low

Audit fast viewing data was generated

Audit fast viewing ended successfully

System

Sep 3, 2023 2:00 AM

Audit

Low

Audit fast viewing data was generated

Audit fast viewing ended successfully

System

Sep 2, 2023 2:00 AM

Audit

Low

Audit fast viewing data was generated

Audit fast viewing ended successfully

System

Sep 1, 2023 2:00 AM

Audit

Low

Audit fast viewing data was generated

Event Details

Event Type:

Old ADC content

Event Time:

2023-07-23 01:00:00.0

Sub System:

ADC

Severity:

Medium

Message:

The ADC content has not been updated for the last 31 days.

User:

System

Reporting Component Name:

MX

Reporting Component IP:

Thales Cybersecurity Docs Hub © 2019–2026, Thales Group. All rights reserved.
WAF Administration Guide | Last Updated: August 31, 2026 | Version: WAF v15.3

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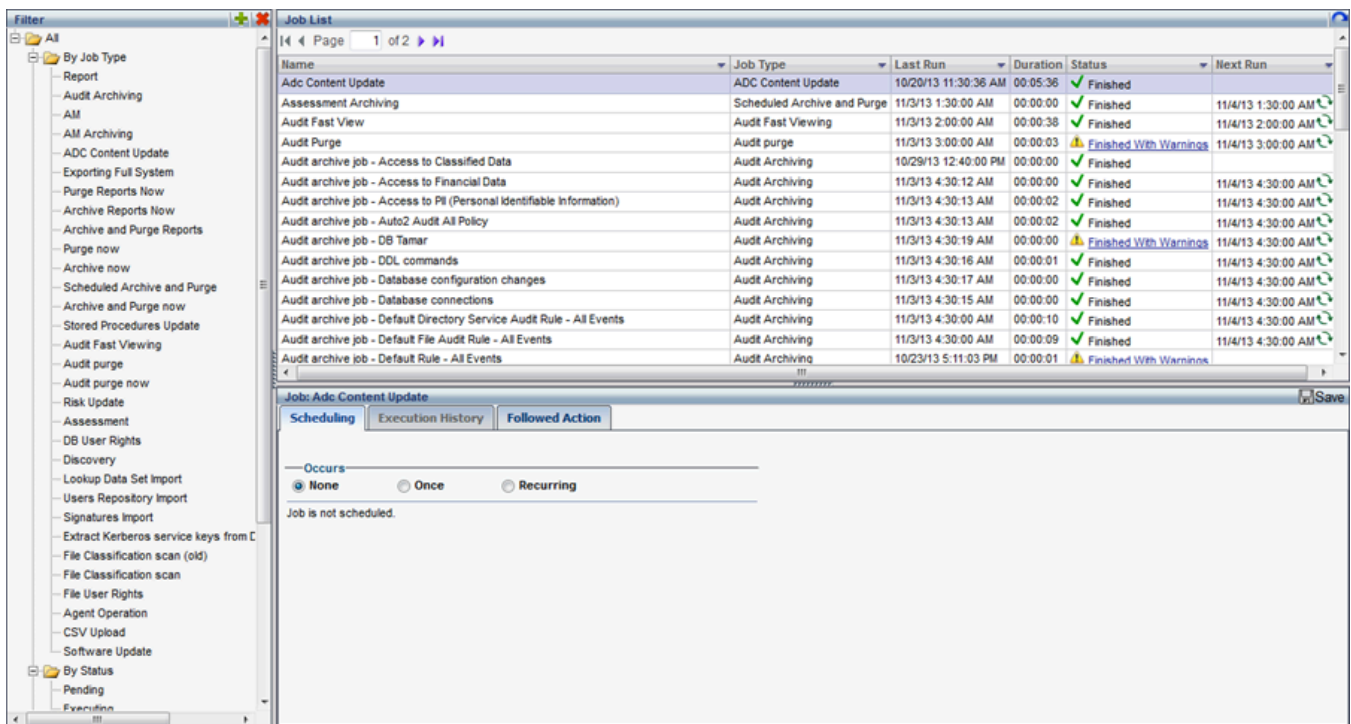
Job Status

This chapter describes the **Jobs Status** window, and includes the following:

- Jobs Status
- Displaying the Jobs Status Window
- Editing a Job
- Aborting a Job
- Using the Log to Analyze Jobs
- Updating Features Configurations from the Cloud

Jobs Status

The **Jobs Status** window displays information about jobs which were run, are currently running, or are scheduled to run.



For each job, the following information is displayed:

- Name:** The job's name.
- Job Type:** The job's type.
- Last Run:** The date and time the job was last executed.
- Duration:** If the job is complete, how long it took. If the job is running, how long it has taken thus far.
- Status:** The status of the job's last execution. If the job's status is not "Finished", you can click on the status to display an explanation of the status.

- **Finished:** The job was completed.
- **Executing:** The job is running. The proportion of the job that has been completed (%) is displayed. You can abort an executing job, see [Aborting a Job](#).
- **Finished with Warnings:** The job was completed but errors occurred. Click on the link to view a window with details of the warnings.
- **Failed:** The job failed to complete.
- **Next Run:** The date and time of the next scheduled run.

The next run  icon indicates that the job is recurring. Hover the mouse over the icon to display the schedule.

In the lower pane, there are three tabs which display information about the job currently selected in the **Job List** pane:

- **Scheduling:** This tab displays the schedule for running the job. You can reschedule the job from this tab. See [Editing a Job](#).
- **Execution History:** A chronologically-ordered list of all executions of the job, with the last execution at the bottom of the list.

Scheduling	Execution History	Followed Action	
Start Time	End Time	Duration	Status
10/20/13 11:30:36 AM	10/20/13 11:36:12 AM	00:05:36	Finished
10/20/13 11:30:19 AM	10/20/13 11:30:19 AM	00:00:00	Finished
10/20/13 11:30:13 AM	10/20/13 11:30:13 AM	00:00:00	Finished
10/20/13 9:54:01 AM			Failed
10/20/13 9:52:01 AM	10/20/13 9:52:21 AM	00:00:20	Finished
10/20/13 9:42:55 AM	10/20/13 9:43:15 AM	00:00:20	Finished

- **Followed Action:** This tab displays the job's followed action.

Displaying the Jobs Status Window

To display the **Jobs Status** window:

1. In the **Admin** workspace, click **Jobs Status**.
2. In the **Filter** pane, you can define a filter on the jobs to be displayed in the **Job List** pane.
3. In the **Job List** pane, the last execution of jobs which have already run are displayed. You can refresh the list by clicking  in the upper-right corner of the **Job List** pane.

Editing a Job

You cannot change jobs which have already run, or have already started running, but you can change the scheduling and Followed Actions for future executions of a job.

Changing a job's scheduling enables you to ease system congestion if a lot of jobs are being run at the same time and causing load problems.



Note: SecureSphere saves a table of your jobs internally, so that in the event of excessive load or other problems, SecureSphere support can access your job schedules and records as a diagnostic aid.

To change a job's scheduling:

1. In the **Admin** workspace, select **Jobs Status**.
2. In the **Filter** pane, you can define a filter on the jobs to be displayed in the **Job List** pane.
3. In the **Job List** pane, select a job.
4. Click the **Scheduling** tab.
5. Under **Occurs**, select one of the following:
 - **None:** The job is never run.
 - **One Time:** The job is run on the date and time you specify.
 - **Recurring:** The job is run periodically, according to the schedule you define.
6. Configure the date and time you want the job to run, and in the case of a recurring job, the run schedule.

7. Click **Save**.

To change a job's followed action:

1. In the **Admin** workspace, click **Jobs Status**.
2. In the **Filter** pane, you can define a filter on the jobs to be displayed in the **Job List** pane.
3. In the **Job List** pane, select a job.
4. Click the **Followed Action** tab.
5. Select a **Followed Action**.

For information on creating followed actions, see [Working with Action Sets and Followed Actions](#).

6. Click **Save**.

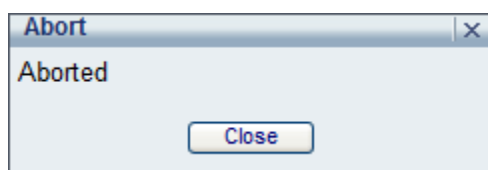
Aborting a Job

To abort a currently running job:

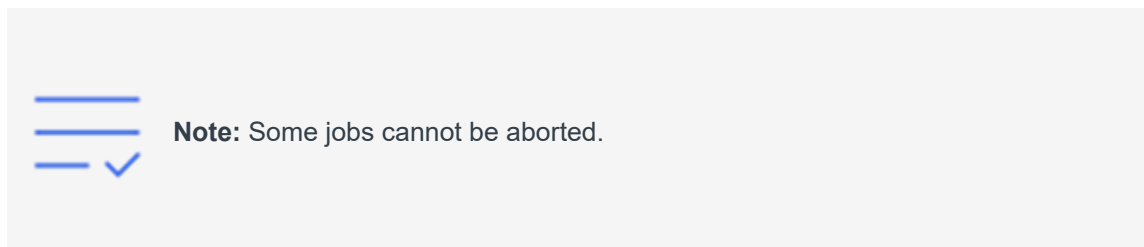
1. In the **Admin** workspace, click **Jobs Status**.
2. In the Job List pane, right-click a job that is executing and select **Abort** from the menu.

Job List				
Page 1 of 1				
Name	Job Type	Last Run	Duration	Status
Report Name: Daily Top 10 All Violations	Report	10/23/13 3:51:16 PM	00:00:05	Executing (5%)
Report Name: Daily Top 10 DB Violations	Report	10/23/13 4:00:24 PM	00:00:	Finished
Report Name: List Of Alerts	Report	10/23/13 3:53:28 PM	00:00:	Failed
Report Name: SharePoint - Changes to Permission Level Report	Report	10/23/13 3:56:33 PM	00:00:	Finished

3. The **Abort** message appears.



4. Click **Close** to confirm. The job is aborted.



Using the Log to Analyze Jobs

Whenever there is a change to the job status, a new line is added to a log file. The file is a .csv file with the following fields:

- IDType
- Status
- Name
- Date
- Duration

The image below shows when lines are added to the log file.

ID	Type	Status	Name	Date	Duration
6693909166359770000	SynchronizePackage	Failed	Software Updates Synchronization	21-05-13 1:07	0:00:06
2249388024808310000	ArchivePurgeJob	Started	Assessment Archiving	21-05-13 1:30	
-837301415778153000	EndpointImport	Started	Update item - Lookup Data Set ThreatRadar - Phishing URLs	21-05-13 1:33	
-837301415778153000	EndpointImport	Failed	Update item - Lookup Data Set ThreatRadar - Phishing URLs	21-05-13 1:33	0:00:01
-837302652728734000	EndpointImport	Started	Update item - Lookup Data Set ThreatRadar - Anonymous Pro	21-05-13 1:34	
-837308837481640000	EndpointImport	Started	Update item - Lookup Data Set ThreatRadar - TOR IPs	21-05-13 1:34	
-837308837481640000	EndpointImport	Failed	Update item - Lookup Data Set ThreatRadar - TOR IPs	21-05-13 1:34	0:00:01
-837302652728734000	EndpointImport	Failed	Update item - Lookup Data Set ThreatRadar - Anonymous Pro	21-05-13 1:34	0:00:03
2249388024808310000	ArchivePurgeJob	Finished	Assessment Archiving	21-05-13 1:35	0:05:50
-5683273311917220000	EagerAudit	Started	Audit Fast View	21-05-13 2:00	
-1936196467902340000	ArchivePurgeJob	Started	Vulnerabilities Archiving	21-05-13 2:00	

An example of the log appears below:

1	2	3	4	5	6
91815;	Report;	Started;	Daily Top 10 All Violations;	07:45:17	08-12-2013
91815;	Report;	Finished;	Daily Top 10 All Violations;	07:45:23	08-12-2013; 00:00:06

Key:

- Execution ID
- Job type
- State
- Name
- Timestamp
- Duration

The log can be found at `/opt/SecureSphere/server/SecureSphereWork/logs/jobs/job_status.csv`.

Updating Features Configurations from the Cloud

You can update features configurations from the cloud, using the Jobs Status page.

You can update the following features configurations:

- FPS Provider Definition
- Geo IP Configuration
- ThreatRadar Bot Mitigation

For each of the above, there is a predefined instance of a job that accesses the cloud, gets the configuration and downloads it to the MX.

You can also perform a manual download of one of these configurations, say, to a test environment, where you can examine it before manually uploading it to a production environment.

To edit/abort a cloud configuration update job:

1. In the **Admin** workspace, select **Jobs Status**.
2. In the **Filter** pane, under **By Job Type**, select **Cloud Configuration Update**. In the **Job List** pane, the Cloud Configuration jobs appear.
3. Select a job.
4. Configure the job. For more information see [Editing a Job](#) and [Aborting a Job](#).

To perform a manual download of a feature configuration file from the cloud:

1. In the **Admin** workspace, select **Jobs Status**.
2. In the **Filter** pane, under **By Job Type**, select **Cloud Configuration Update**. In the **Job List** pane, the **Cloud Configuration** jobs appear.
3. Select a job.
4. Select the **Settings** tab.
5. Click **Download**. Select the location in which you want to save the file.

To perform a manual upload of a feature configuration file:

1. In the **Admin** workspace, select **Jobs Status**.
2. In the **Filter** pane, under **By Job Type**, select **Cloud Configuration Update**. In the **Job List** pane, the Cloud Configuration jobs appear.
3. Select a job.
4. Select the **Settings** tab.
5. Click **Upload**. Navigate to the location of the configuration file and select it. It is loaded into your management server.

Notes:



- If you select **Main > ThreatRadar > Dashboard** to view ThreatRadar services, and you have not yet downloaded a configuration from the cloud, you see a warning to that effect.
- If the ThreatRadar service is not licensed, you see a warning to that effect and the job is not run.
- If you try to use the **Source Geo Location** match criterion in a policy definition, and you have not yet downloaded a configuration from the cloud, you see a warning to that effect.

Maintenance

This section describes various system maintenance tasks, and includes:

- Discovery Results Archive
- Exporting and Importing the Management Server
- File Explorer Maintenance
- Reports Archive
- System Events Archive
- Vulnerabilities Purge

Discovery Results Archive

The **Discovery Results Archive** window specifies the parameters for archiving assessment results.

Archive and purge work together to ensure, on the one hand, that the audit data is preserved and, on the other hand, that the Gateway's available disk space is not exceeded so that audit data can still be recorded. The former is achieved by archiving, and the latter is achieved by purging. These actions must be carefully timed so that a purge does not destroy unarchived data.

Scheduled archives do not archive already archived data or data recorded from last night at midnight. Scheduled purges do not purge data less than seven days old (this is the default value, it can be configured), nor does it purge unarchived data.

You can do **Archive and Purge Now**, or **Purge Now** but note the following. **Archive and Purge Now** archives everything, including already archived data and current data, and then purges it. **Purge Now** purges everything including already archived and current data. You are advised to exercise great care in using either of these buttons.

To define the parameters for archiving of Discovery results:

1. In the **Admin** workspace, click **Maintenance**.
2. In the **Maintenance** pane, click **Discovery Results Archive**.

3. In the **Archiving Definitions** section, set the parameters as described in the table **Discovery Results Archiving Definitions Parameters** below.



Note: The Archive Setting menu item **Default Archive Settings** does not by default encrypt the archive file. To change these settings, in the **Main** workspace, select **Setup > Settings** and then select **Archive Settings** from the **Settings** pane, select **Default Archive Settings** and enter the encryption keys. Alternatively, you can define another archive setting in **Main > Setup > Settings** and use it here.

4. In the **Purge Definitions** section, select an option as described in the table **Discovery Results Purge Definition Parameters** below.



Note: The default is **By Size - Purge Oldest Records When there are more than 5 Job Runs**, meaning that by default, no more than the results of five (5) job runs are available - this default value does not include the baseline Job Run or the latest Job Run which are never purged. (Thus with the default value of 5, there will be 7 Job Runs before a purge occurs - and those 7 will not be purged).

5. In the **Scheduling** section, under **Occurs**, select a an option as described in the table **Discovery Results Scheduling Options** below.
6. Select **Perform Archive** to archive the data before purging.
7. Click **Archive and Purge Now** to archive the data immediately then purge it from SecureSphere.
8. Click **Purge Now** to purge results data immediately.
9. Click **Save**. For information on how to extract assessment or discovery result archives for viewing in its source format, see [Extracting Archives for Viewing](#).

Discovery Results Archiving Definitions Parameters

Parameter	Description
Archiving Action	From the menu, select the Action Set to be performed after archiving assessment results.
Archive Settings	From the menu, select the archive settings.

Discovery Results Purge Definition Parameters

Parameter	Description
By Time - Purge Records Older Than	Purge results older than the specified age.
By Size - Purge Oldest Records When there are more than ... Job Runs	When the number of job runs exceeds the specified number, purge the oldest records. This number does not include the baseline Job Run or the latest Job Run which are never purged. Example: if this value is set to 3, there will be 5 Job Runs before a purge occurs - the baseline, the latest and the three next latest that will not be purged.

Discovery Results Scheduling Options

Parameter	Description
None	Results data are not purged automatically. If this option is selected, the only way to purge results data is to do so manually, by clicking Run Now .
One Time	Results data are purged automatically on the date and time you specify in At and At Time .
Recurring	Results data are purged automatically according to the schedule you define.

Exporting and Importing the Management Server

You can export and import the SecureSphere data from the Management Server, including configuration files, the SecureSphere database, and (optionally) alerts data. Any data that arrives at the Management Server after the export operation begins is not included in the export file.

Exporting and importing are useful for the following purposes:

- Copying a database to another machine
- Backing up a database

You can perform these functions in any of the following ways:

- Export using the management interface
- Import and/or export using the interactive command line interface (CLI)
- Import and/or export using the non-interactive command line interface (CLI)



Note: You cannot use the SecureSphere GUI to import a SecureSphere database, because you must stop the SecureSphere Management Server during the import process.

Export Using the SecureSphere GUI

To export a SecureSphere database:

1. In the **Admin** workspace, click **Maintenance**.
2. In the **Maintenance** pane, click **Export System**.

Export Definitions

Database Dump Encryption password (optional)

Verify Password

Archiving Action

☒ Archive audit data together with system export

Scheduling

Schedule **Occurs**

☒ None ☐ Once ☐ Recurring

Job is not scheduled.

Immediate Actions

This password is used to encrypt the database dump. It will be required to import the dump back into SecureSphere. If a password is not configured before export, SecureSphere uses the database System user's password.

3. In the **Export Settings** pane, in the **Export Definition** section, enter data as shown in the table **Export Definition Options** below.

4. In the **Scheduling** section, under **Occurs**, select an option as described in the table **Export Scheduling Options** below. Alternatively, you can perform the export immediately by clicking **Export Now**.
5. Click **Save**.

The name of the export file is of the form: `/tmp/SecureSphere_<date>.dmp.gz`, where `<date>` is the current date in the format `YYYYMMDD`.

Export Definition Options

Parameter	Description
Database Dump Encryption Password	The dump file is always encrypted. You can use the system password (the password of the DBA of the SecureSphere database) or specify another password.
Verify Password	To specify another password, enter the password in Database Dump Encryption Password and re-enter the same password in Verify Password. Make sure your password has the following characteristics: ▪ It must have no fewer than 10 characters and no more than 14 characters. ▪ It must have at least one number, one capital letter, and one special character from: * + = # % ^ : / ~ . , [] _ \ () ; @ \$ & - ? { } < > ▪ It cannot have more than two characters repeated in succession. If you do not enter an encryption password, the export file will be encrypted with the system password (the password of the DBA of the SecureSphere database). Note: You will not be able to import the encrypted dump file without the password. There is no workaround.
Archiving Action	Select an Action Set from the dropdown list. Note: By default, only the Default Archive Action Set is available. To use a different Action Set: 1. Define an Action Interface of type NFS Archive (Admin > System Definitions > Action Interfaces).

Parameter	Description
	- Define an Action Set of type Archiving which uses the Action Interface you defined in the previous step (Main > Policies > Action Sets). - In the Action Interface's Destination Directory, specify the Destination Directory for the archive. - If the destination directory is not the default directory (/var/tmp), then you must also ensure that SecureSphere has the required permissions in that directory by executing the following OS commands: chmod 777 chown mxserver:mxserver Select the Action Set you have just configured for Archiving Action.
Archive audit data together with system export	Select this option to also export the archive audit data temporarily stored on the MX. This does not export the audit data stored on the Gateways.

Export Scheduling Options

Parameter	Description
None	The export is never scheduled, but you can run it at any time by clicking Export Now .
One Time	The export is run on the date and time you specify in At and At Time .
Recurring	The export is run periodically according to the schedule you define.

Export / Import Manually Using the Interactive CLI

Note: The following users are defined on a Management Server:



- **root:** the OS superuser
- **admin:** a SecureSphere administrator with access to the SecureSphere GUI; any number of SecureSphere administrators and users can be defined
- **[configured]:** SecureSphere users that are created by the SecureSphere admin



Note: Do not export while the ADC database is being updated, otherwise the exported file will be corrupted.

To export SecureSphere data using the interactive CLI:

1. Login to the SecureSphere Management Server as `root`.
2. Execute the following command:

```
cd /var/tmp
```

3. Execute the following command:

```
full_expimp.sh
```

4. Enter the following data when requested:
 - **operation:** Select `1` (export).
 - **system password:** Enter the system password (the password of the DBA of the SecureSphere database).
 - **export type:** Select `1` to export the alert data as well.
 - **export failed archive files:** Select `y` to include failed archive data (default is not to export).
 - **encryption password:** This password is used to encrypt the database dump. It will be required in order to import the dump back into SecureSphere.

If a password is not configured before export, SecureSphere will use the database System user password.

- **export file:** Enter the name of the export file. The full name of the export file will be `<file_name>.gz`, where `<file_name>` is the name you enter here.
5. Enter `y` to confirm the export.

Note:

Make sure your password has the following characteristics:



- It must have no fewer than 10 characters and no more than 14 characters.
- It must have at least one number, one capital letter, and one special character from:
 * + = # % ^ : / ~ . , [] _
- It cannot have more than two characters repeated in succession.

To import SecureSphere data using the interactive CLI:

1. Login to the SecureSphere Management Server as `root`.
2. Execute the following command:

```
cd /var/tmp
```

3. Stop the SecureSphere Management Server by executing the following command:

```
impctl server stop
```

4. Execute the following command:

```
full_expimp.sh
```

5. Enter the following data when requested:
 - **operation:** Select 2 (import).
 - **system password:** Enter the system password (the password of the DBA of the SecureSphere database).
 - **import type:** Select 1 to drop the target schema, that is, before starting the import operation, delete the schema on the machine to which the export file is being imported.

This is the recommended option, since it ensures that the resulting data will consist only of the imported data.

- **copy option:** Select 1 to copy the configuration files, that is, files not part of the SecureSphere database, for example:
`SecureSphereWork/firsttime/firsttimeIds,`
`SecureSphereWork/firsttime/mutableFirsttimeIds,`
`SecureSphereWork/activated-configuration`
- **import file:** Enter the fullpath name of the import file, that is, the previously exported file.

6. You will be asked whether the file is encrypted. If you answer `Y` (the default), then you will be asked to enter the encryption password.
7. Enter `Y` to confirm the import.
8. Start the SecureSphere Management Server by executing the following command:

```
impctl server start
```

- Review the import log for errors or warnings.

Export / Import Manually Using the Non-Interactive CLI

You can export and/or import the SecureSphere data using the `full_expimp.sh` command while specifying the options in the command line.

To export SecureSphere data using the non-interactive CLI:



Note: Do not export while the ADC database is being updated, otherwise the exported file will be corrupted.

- Login to the SecureSphere Management Server as `root`.
- Execute the following command, adding some or all the additional parameters listed in the following table:

```
full_expimp.sh --operation=1 --pwd=<system_password> --  
copyfiles=<copy_option> --logfile=<file_name>" --schema=<schema>
```

Note:



- The exported file is always encrypted. The default password is the system password, that is, the password of the DBA of the SecureSphere database), although you can specify a different password using the `pwd` argument.
- The table below lists only the most frequently used import and export options. For a complete list of these options, enter the following command:
`full_expimp.sh ?`

full_expimp.sh Export Parameters

Parameter	Description
<code>--dmpfile=<file></code>	The destination file for the exported data. If not specified, the default is <code>/tmp/SecureSphere_<date></code> where <code><date></code> is the current date in the format <code>YYYYMMDD</code> .

Parameter	Description
<code>--exptype=<n></code>	Specifies whether the export will include alert data: 1 - include the alert data 2 - do not include alert data default: 1
<code>--schema=<name></code>	Export only the <code><name></code> schema from the database. If not specified, the script will export the following database schemas: - SECURE - SECURE_DA - all ODM schemas
<code>--copyfiles=<copy_option></code>	Specify 1 to copy the configuration files, that is, files not part of the SecureSphere database, for example: - SecureSphereWork/firsttime/firsttimeIds, - SecureSphereWork/firsttime/mutableFirsttimeIds, - SecureSphereWork/activated-configuration
<code>--logfile=<file_name></code>	Specifies the location of the log file.

Note: The SecureSphere database comprises the following schemas:



- SECURE: The SecureSphere configuration and data
- SECURE_DA: Changes made to the SecureSphere configuration which have not yet been applied
- ODM: Active Module schemas

To import SecureSphere data using the non-interactive CLI:

1. Login to the Management Server as **root**.
2. Stop the Management Server by executing the following command:

```
impctl server stop
```

3. Execute the following command, adding some or all the additional parameters listed in the following table:


```
full_expimp.sh --operation=2 --pwd=<system_password> --dmpfile=<file_name> --
encryptpwd=<encryptpwd> --secure pwd=<secure_password>
```

Note:



- The exported file is always encrypted. The default password is the system password, that is, the password of the DBA of the SecureSphere database, although you can specify a different password using the `pwd` argument.
Make sure your password has the following characteristics:
 - It must have no fewer than 7 characters and no more than 14 characters.
 - It must have at least one number, one capital letter, and one special character from: `* + = # % ^ : / ~ . , [ ] _ \ ( ) | ; @ $ & - ? { } < >`
 - It cannot have more than two characters repeated in succession.
- The table below lists only the most frequently used import and export options. For a complete list of these options, enter the following command:
`full_expimp.sh ?`
 - a. Start the Management Server by executing the following command:

```
impctl server start
```
 - b. Review the import log for errors or warnings.

full_expimp.sh Import Parameters

Parameter	Description
<code>--imptype=<import_type></code>	Specify whether the import will first delete imported schemas from the database: ▪ 1 - delete imported schemas from the database ▪ 2 - do not delete imported schemas from the database default: 1 If you specify 1 (delete imported schemas), then if the import file includes only one schema, only that schema is deleted from the database. If the import file includes all schemas, the following database schemas are deleted from the database: ▪ SECURE ▪ SECURE_DA ▪ all ODM schemas

Parameter	Description
<code>--copyfiles=<copy_option></code>	Specify whether to import configuration files (files not part of the SecureSphere database), for example: SecureSphereWork/firsttime/firsttimeIds, SecureSphereWork/firsttime/mutableFirsttimeIds, SecureSphereWork/activated-configuration. 1 - import configuration files 2 - do not import configuration files
<code>--secure pwd=<secure_password></code>	Specify the secure user password that is needed when importing the secure user and the target database that already contains this user.
<code>--encryptpwd</code>	Specify the password for entire dumpfile encryption and decryption.
<code>--src=<source_schema></code>	Specify the schema in the dump file to import.
<code>--dest=<destination_schema></code>	If you have specified the <code>--src</code> parameter, the <code>--dest</code> parameter enables you to rename the imported schema in the target database. For example, if the exported file contains the SECURE schema, you can import it to SECURE_OLD by specifying: <pre>--src=secure --dest=secure_old</pre>
<code>--destpwd=<destination_password></code>	Specify the password for the newly created target schema (must be specified when destination schema is specified) if source and destination schemas are not defined, the process imports the entire dump file.
<code>--tbsname=<name></code>	Import all the imported objects to the <code><name></code> tablespace instead of the tablespace they were in the source database.
<code>--tbsloc=<path></code>	This option is set if you want to import the entire data of the imported objects to a specific tablespace instead of the tablespaces they were in the source database.

Parameter	Description
<code>--nostatistics</code>	Use this option if you don't want to gather Oracle statistics after the import.

Listing the Contents of an Exported File

The procedure describes how to list the contents of an exported file.

To list the contents of an exported file:

1. Login to the SecureSphere Management Server as `root`.
2. Execute the following command:

```
full_expimp.sh
```

3. Enter the following data when requested:
 - **operation:** Select `3` (List dump file content).
 - **encryption password:** If the file is encrypted, enter a password (the default password is system password, that is, the password of the DBA of the SecureSphere database).
 - **Enter a file name for the operation:** Enter the fullpath name of a previously-exported file.
4. The output will be similar to the following:

```
full_expimp (version 2.0) started on Sun May 3 15:02:56 IDT 2009
```

```
Decompressing dump file
```

```
Dump file contains the following schemas:
```

```
SECURE
```

```
SECURE_DA
```

```
Dump contains SecureSphere version: 7.0.0.7014.Release
```

```
Dump was created using full_expimp version: 2.0
```

```
full_expimp completed successfully on Sun May 3 15:02:57 IDT 2009
```

Miscellaneous Maintenance Tasks

To list the contents of an exported file:

1. Login to the SecureSphere Management Server as `root`.
2. Execute the following command:

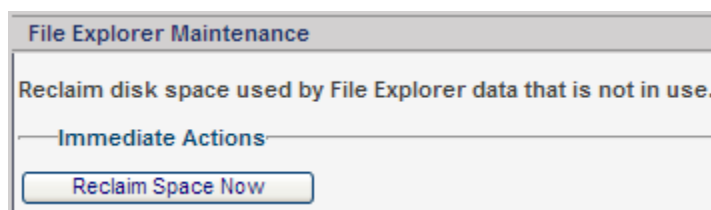
```
full_expimp.sh --operation=3 --dmpfile=<file>
```

File Explorer Maintenance

When you delete data types and data owners in the **File Explorer**, the disk space used by the data is reclaimed on a daily basis. You can improve **File Explorer** performance by reclaiming the disk space manually.

To immediately reclaim disk space from unused File Explorer data:

1. In the **Admin** workspace, click **Maintenance**.
2. In the **Maintenance** pane, click **File Explorer Maintenance**.



3. Click **Reclaim Space Now**.

Reports Archive

The **Reports Archive** window specifies the parameters for archiving reports.

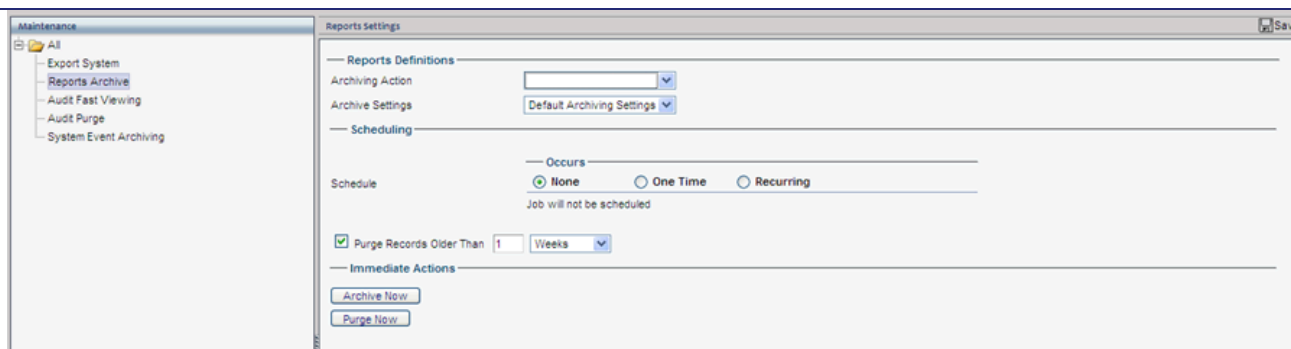
Archive and purge work together to ensure, on the one hand, that the audit data is preserved and, on the other hand, that the Gateway's available disk space is not exceeded so that audit data can still be recorded. The former is achieved by archiving, and the latter is achieved by purging. These actions must be carefully timed so that a purge does not destroy unarchived data.

Scheduled archives do not archive already archived data or data recorded from last night at midnight. Scheduled purges do not purge data less than seven days old (this is the default value, it can be configured), nor does it purge unarchived data.

You can do **Archive Now**, or **Purge Now** but note the following. **Archive Now** archives everything, including already archived data and current data. **Purge Now** purges everything including already archived and current data. You are advised to exercise great care in using either of these buttons.

To define the parameters for archiving reports:

1. In the **Admin** workspace, click **Maintenance**.
2. In the **Maintenance** pane, click **Reports Archive**.



3. In the **Reports Definition** section, set the parameters as described in the table **Report Archive Settings** below.



Note: The Archive Setting menu item Default Archive Settings does not by default encrypt the archive file. To change these settings, in the Main workspace, select Setup > Settings and then select Archive Settings from the Settings pane, select Default Archive Settings and enter the encryption keys. Alternatively, you can define another archive setting in Main > Setup > Settings and use it here.

4. In the **Scheduling** section, under **Occurs**, select one of the options as describes in the table **Report Archive Scheduling** below.
5. Click **Archive Now** to archive reports.
6. Click **Purge Now** to purge report data.

Before purging, make sure the reports have been successfully archived.

7. Click **Save**. Report archiving has been configured. For information on how to extract report archives for viewing in its source format, see Extracting Archives for Viewing.

Report Archive Settings

Parameter	Description
Archiving Action	From the menu, select the Action Set to be performed after archiving reports.
Archive Settings	From the menu, select the archive settings.

Report Archive Scheduling

Parameter	Description
None	Reports are not archived automatically. If this option is selected, the only way to archive reports is to do so manually, by clicking Archive Now .
One Time	Reports are archived automatically on the date and time you specify in At and At Time .
Recurring	Reports are archived automatically according to the schedule you define.

System Events Archive

The **System Event Archive** window specifies the parameters for archiving system event data.

Archive and purge work together to ensure, on the one hand, that the audit data is preserved and, on the other hand, that the Gateway's available disk space is not exceeded so that audit data can still be recorded. The former is achieved by archiving, and the latter is achieved by purging. These actions must be carefully timed so that a purge does not destroy unarchived data.

Scheduled archives do not archive already archived data or data recorded from last night at midnight. Scheduled purges do not purge data less than seven days old (this is the default value, it can be configured), nor does it purge unarchived data.

You can do **Archive and Purge Now**, or **Purge Now** but note the following. **Archive and Purge Now** archives everything, including already archived data and current data, and then purges it. **Purge Now** purges everything including already archived and current data. You are advised to exercise great care in using either of these buttons.

To define the parameters for archiving system event data:

1. In the **Admin** workspace, click **Maintenance**.
2. In the **Maintenance** pane, click **System Events Archive**.

- In the **Archiving Definitions** section, set the parameters described in the table **System Event Archive Settings** below.



Note: The Archive Setting menu item Default Archive Settings does not by default encrypt the archive file. To change these settings, in the Main workspace, select Setup > Settings and then select Archive Settings from the Settings pane, select Default Archive Settings and enter the encryption keys. Alternatively, you can define another archive setting in Main > Setup > Settings and use it here.

- In the **Purge Definitions** section, select an option as described in the table **System Event Archive Purge Definition Parameters** below.



Note: The default is **By Size - Purge Oldest Records When there are more than 100,000 Records**, meaning that by default, no more than 100,000 system events are available.

- In the **Scheduling** section, under **Occurs**, select an option as described in the table **System Event Archive Scheduling Options** below.
- Select **Perform Archive** to archive the system event data before purging.
- Click **Archive and Purge Now** to archive the data immediately then purge it from SecureSphere.
- Click **Purge Now** to purge the system event data immediately.
- Click **Save**. For information on how to extract system event archives for viewing in its source format, see [Extracting Archives for Viewing](#).

System Event Archive Settings

Parameter	Description
Archiving Action	From the menu, select the Action Set to be performed after archiving system events.
Archive Settings	From the menu, select the archive settings.

System Event Archive Purge Definition Parameters

Parameter	Description
By Time - Purge Records Older Than	Purge system events older than the specified age.
By Size - Purge Oldest Records When there are more than Than ... Records	When the number of system events exceeds the specified number, purge the oldest records.

System Event Archive Scheduling Options

Parameter	Description
None	System event data are not purged automatically. If this option is selected, the only way to purge system event data is to do so manually, by clicking Run Now .
One Time	System event data are purged automatically on the date and time you specify in At and At Time .
Recurring	System event data are purged automatically according to the schedule you define.

Vulnerabilities Purge

The **Vulnerabilities Purge** window specifies the parameters for purging vulnerability data for all Gateways being

managed by SecureSphere.

To define the parameters for purging vulnerability data:

1. In the **Admin** workspace, click **Maintenance**.
2. In the **Maintenance** pane, click **Vulnerability Purge**.

3. In the Purge Definitions section, configure the table **Vulnerability Purge Definitions** below.
4. In the **Scheduling** section, under **Occurs**, select an option as described in the table **Vulnerability Purge Schedule Settings** below.
5. Click **Run Now** to purge audit data immediately.

Before purging, make sure the audit data have been successfully archived.

6. Click **Save**.

Vulnerability Purge Definitions

Parameter	Description
Purge Observations Older Than	Determines the period of time for which vulnerabilities should be purged from SecureSphere. You can enter a number, then a time period that includes days, months weeks and months. For example, you can configure SecureSphere to purge vulnerabilities that have been observed for the past 12 weeks.
Closed Vulnerabilities	When selected, purges all vulnerabilities in the SecureSphere server that have been closed.

Vulnerability Purge Schedule Settings

Parameter	Description
None	Audit data are not purged automatically. If this option is selected, the only way to retrieve purge data is to do so manually, by clicking Run Now .
One Time	Audit data are purged automatically on the date and time you specify in At and At Time .
Recurring	Audit data are purged automatically according to the schedule you define.

System Definitions

This section describes the parameters defined under **System Definitions** in the **Admin** workspace, and includes the items listed below:

- [Dynamic Profiling](#)
- [Gateway Settings](#)
- [Security and Authentication](#)
- [Management Server Settings](#)
- [User Interface Settings](#)
- [Cloud Settings](#)

Dynamic Profiling

This section describes the windows related to profiling, and includes the following items:

- [Learning Exceptions](#)
- [Profile Optimization](#)
- [Profile Size Limits](#)
- [Switching to Protect Mode Thresholds \(Web\)](#)

Learning Exceptions

These parameters define exceptions to the Web profile learning process.

If you change any of these parameters, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see [Activating Settings](#).

Learning Exceptions Window Parameters

Parameter	Description
Don't learn requests that have no referer, unless the response code appears in this list:	As a rule, HTTP requests with no referer information are not learned, unless the target URL was already observed at least once. However, if the request's response codes appears in this comma-separated list, the request will be learned. Specify "none" to indicate no exclusions. Default: 200, 304

Parameter	Description
Don't learn requests with an unknown external host, unless the response code appears in this list:	As a rule, if an HTTP request has a host name which was not yet learned, and the request's referer is external (that is, it does not appear in the list of learned hosts), the request will not be learned. However, if the request's response codes is this comma-separated list, the request will be learned. Specify "none" to indicate no exclusions. Default: 200, 304

Profile Optimization

These parameters define exceptions to the profile optimization process.

If you change any of these parameters, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see [Activating Settings](#).

Profile Optimization Window Parameters

Parameter	Description
Alert Occurrences	The number of times alerts need to take place within Alert Timeframe in Hours to generate an optimization issue.
Alert Timeframe in Hours	The period of time in which the number of alerts need to occur to generate an optimization issue.
Size Limit (%)	The percentage of the Profile Size Limit (see Profile Size Limits) which, when reached, generates an optimization issue.

Profile Size Limits

When one of these limits is exceeded, SecureSphere stops learning the application and issues a profile size violation system event.

You should not increase any of these values unless there is a compelling reason to believe that the limits will be exceeded, because a profile's size can impact both Gateway and MX performance. If you find yourself regularly exceeding these limits, then you should attempt to address the issue by using URL patterns, plugins or other methods before concluding that the limits must be increased.

The following table lists the parameters in the **Profile Size Limit** pane.

If you change any of these parameters, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see [Activating Settings](#).

Profile Size Limits Window Parameters

Parameter	Description
Maximum allowed source applications per user	For DB profiles, the maximum number of applications allowed in a user's profile. When this number is exceeded, no additional applications are added to the relevant profile and a warning message is written to the system log. If the profile reaches this limit, consider allowing any application for the relevant user. Default: 1000
Maximum allowed databases per user	For DB profiles, the maximum number of databases allowed in a user's profile. When this number is exceeded, no additional databases are added to the relevant profile and a warning message is written to the system log. If the profile reaches this limit, consider allowing any database for the relevant user. Default: 1000
Maximum allowed source host names per user	For DB profiles, the maximum number of source host names allowed in a user's profile. When SecureSphere reaches this number, no additional host names are added to the relevant profile and a warning message is written to the system log. If the profile reaches this limit, consider allowing any host for the relevant user. Default: 1000
Maximum allowed source IP addresses per user	For DB profiles, the maximum number of source IP addresses allowed in a user's profile. When SecureSphere reaches this number, no additional source IP addresses are added to the relevant profile and a warning message is written to the system log. If the profile reaches this limit, consider allowing any IP address for the relevant user. Default: 1000

Parameter	Description
Maximum allowed source OS users per user	For DB profiles, the maximum number of source operating-system user names allowed in a user's profile. When SecureSphere reaches this number, no additional OS user are added to the relevant profile and a warning message are written to the system log. If the profile reaches this limit, consider allowing any user for the relevant user. Default: 1000
Maximum allowed hosts per application	For DB profiles, the maximum number of hosts in a profile. When SecureSphere reaches this number, no additional hosts are added to the profile, and they also cannot be learned. Default: 4000
Maximum allowed URLs per directory	For Web profiles, the maximum number of URLs profiled in each directory. Note: The limit defined by this parameter is not enforced when URLs are added manually, only when they are learned. Default: 500
Maximum allowed queries per application	For DB profiles, the maximum number of queries allowed per profile. When SecureSphere reaches this number, no additional queries are added to the profile. Default: 3000
Maximum allowed parameters per URL	For Web profiles, the maximum number of different parameter names allowed per URL in the profile. When SecureSphere reaches this number, a certain percentage of the URL's parameters are deleted and a warning message is written to the system log. If these deletions occur repeatedly you should check whether some parameter names are dynamically generated by the application. In this case the profile will never stop growing and you should add a parameter prefix that matches all dynamic parameters. SecureSphere automatically generates parameter prefixes for parameters which start with letters and end with numbers.

Parameter	Description
	Note: The limit defined by this parameter is not enforced when parameters are added manually, only when they are learned. Default: 200
Maximum allowed parameters per application	For Web profiles, the maximum number of different parameter names allowed for each application in the profile. If these deletions occur repeatedly you should check whether some parameter names are dynamically generated by the application. In this case the profile will never stop growing and you should add a parameter prefix that matches all dynamic parameters. Note that SecureSphere automatically generates parameter prefixes for parameters which start with letters and end with numbers. Note: The limit defined by this parameter is not enforced when parameters are added manually, only when they are learned. Default: 3000
Maximum allowed URLs per application	For Web profiles, the maximum number of URLs in each profile. Note: The limit defined by this parameter is not enforced when URLs are added manually, only when they are learned. Default: 5000
Maximum allowed sub-directories per directory	For Web profiles, the maximum number of sub-directories allowed in a single directory in a server group's profile. When SecureSphere reaches this number, no additional sub-directories are added to the relevant directory and a warning message is written to the system log. Note that by default SecureSphere does not profile static URLs (such as image files) so only directories that include dynamic URL such as ASP and JSP files are counted. If the profile reaches this limit you should first check whether some directory names are dynamically generated by the application. In this case the profile will never stop growing and you should consider adding a URL pattern (prefix or suffix) which matches all dynamic directories. Default: 500

Parameter	Description
Maximum allowed parameters per SOAP action	For Web profiles, the maximum number of different parameter names allowed for each SOAP Action in the profile. When this limit is exceeded, no additional elements will be added to the relevant SOAP action and a warning message is written to the system log. Default: 100
Maximum allowed database users per application	For DB profiles, the maximum number of database users allowed per profile. Default: 100
Maximum database user groups per application	For DB profiles, the maximum number of database user groups allowed per profile. Default: 100
Maximum allowed queries per Query Group	For DB profiles, the maximum number of queries allowed per query group in a server group's profile. When SecureSphere reaches this number, no additional queries are added to the relevant query group and a warning message is written to the system log. By default SecureSphere switches query groups with many queries to Dynamic Protect and stops learning additional queries in these query groups. When the profile reaches this limit, check the recommended mode of this query group. If the recommended mode is Dynamic Protect and the query is in either Static Protect or Learning mode, consider manually switching it into Dynamic Protect. Default: 2000
Maximum allowed Query Groups per User	For DB profiles, the maximum number of query groups allowed in a user's profile. When SecureSphere reaches this number, no additional query groups are added to the relevant database user and a warning message is written to the system log. If the profile reaches this limit, check whether some query groups are dynamically generated by the application. In this case the profile will never stop growing and you should add an SQL Text Replacement rule which matches all dynamic SQLs. To define a Text Replacement rule, click Main > Setup > Sites, select a DB service, and click the Operation tab. You

Parameter	Description
	can define a new rule under Text Replacement. Default: 500
Maximum allowed SOAP actions per URL	For Web profiles, the maximum number of SOAP actions allowed for each URL in the profile. When this limit is exceeded, no additional SOAP actions are added to the relevant URL's profile and a warning message is written to the system log. Default: 100
Maximum allowed cookies per application	For Web profiles, the maximum number of cookies allowed for each profile. When this limit is exceeded, no additional cookies are added. If this limit is repeatedly exceeded, check whether some cookie names are dynamically generated by the application. In this case, the profile will never stop growing and you should consider adding a cookie prefix which matches all dynamic cookies. Default: 200
Maximum allowed action URLs per application	For Web profiles, the maximum number of action URLs allowed for each profile. Default: 200
Maximum allowed login URLs per application	For Web profiles, the maximum number of login URLs allowed for each profile. Default: 200

Switching to Protect Mode Thresholds (Web)

These thresholds determine when a Web profile's components are switched from Learning mode to Protect mode.

The table below describes the Web-related parameters in this window.

If you change any of these parameters, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For further information, see the **SecureSphere Web Security User Guide**.

Protect Mode Thresholds (Web) Window Parameters

Parameter	Description
Traceability Percentage required to switch a cookie into partial Protect mode	These four parameters taken together determine when a cookie is switched from Learning mode. If the parameters Duration (in hours) required to switch all Cookies to protect or ignored mode and Minimum occurrences required to switch a cookie into Protect mode are both exceeded, then: - If the cookie was not changed during at least the number of sessions defined by Traceability Percentage required to switch a cookie into partial Protect mode, the cookie is switched to Protect mode without protection against injection. - If SecureSphere saw the SET cookie command for at least the number of sessions defined by Traceability Percentage required to switch a cookie into full Protect mode, and the cookie was not changed during at least the number of sessions defined by Traceability Percentage required to switch a cookie into full Protect mode, the cookie is switched to Protect mode with protection against injection. Note that Traceability Percentage required to switch a cookie into partial Protect mode and Traceability Percentage required to switch a cookie into full Protect mode are percentages, not absolute numbers, and relate to a percentage of the total sessions seen by SecureSphere. The minimum value of Duration (in hours) required to switch all Cookies to protect or ignored mode is 24 hours. If it is set to a smaller value, SecureSphere considers it as if it were set to 24 hours. Defaults: Traceability Percentage required to switch a cookie into partial Protect mode: 95.0 Traceability Percentage required to switch a cookie into full Protect mode: 95.0 Duration (in hours) required to switch all Cookies to protect or ignored mode: 240.0 Minimum occurrences required to switch a cookie into Protect mode: 10
Traceability Percentage required to switch a cookie into full Protect mode	
Duration (in hours) required to switch all Cookies to protect or ignored mode	
Minimum occurrences required to switch a cookie into Protect mode	

Parameter	Description
Duration (in hours) required to switch all URLs to Protect mode	A URL is automatically switched to Protect mode based on the following factors: - the number of occurrences in the traffic - whether the URL has parameters - the time passed since it was first seen by SecureSphere The more frequently the URL occurs in traffic, the sooner it is switched to Protect mode. This parameter specifies the number of hours since the first time this URL was seen after which the URL is automatically switched to Protect mode, without regard to the number of its occurrences in the traffic or whether it has parameters. Default: 240.0
Success rate required to set a specific parameter as read-only	If a parameter has been seen Number of occurrences required to determine if a parameter is read-only times, it is set to read-only, if in Success rate required to set a specific parameter as read-only percent of the time it was successfully identified as read-only. If you are testing the system in a lab with a low amount of traffic, consider changing this parameter to 50, in order to be able to test the system and the way it sets parameters as read-only. The lower this number, the greater the chances of false positives and false negatives. Default: 99.9
Number of occurrences required to determine if a parameter is read-only	See the previous parameter. If you are testing the system in a lab with a low amount of traffic, consider changing this parameter to 6 in order to be able to test the system and the way it sets parameters as read-only. The lower this number, the greater the chances of false positives and false negatives. Default: 10

Gateway Settings

This section describes the windows related to Gateways, and the items below:

- [Response Page Display in Alerts Settings](#)
- [Large Scale Configuration](#)

Response Page Display in Alerts Settings

These parameters relate to the display of response pages.

If you change any of these parameters, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see [Activating Settings](#).

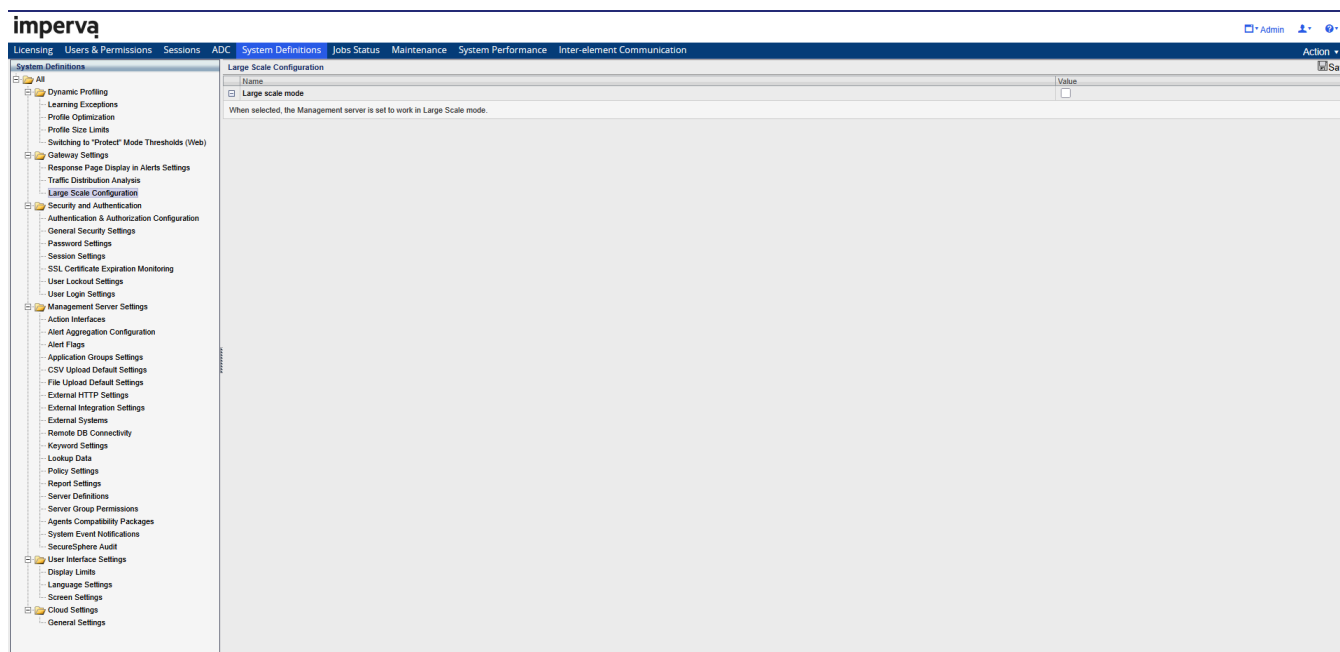
Response Page Display in Alerts Settings Window Parameters

Parameter	Description
Maximal disk space on gateway for responses	Maximum disk space on Gateway that will be allocated for responses (in MB). When this disk space fills up, new response pages will overwrite old ones. Default: 5000 MB
Maximal response page size	The maximum response page size (in KB.) Default: 1000 KB

Large Scale Configuration

This feature is where you configure to the management server to work in Large Scale mode.

Select the check box under the Value column to enable Large Scale mode or clear the check box to disable Large Scale mode.



Security and Authentication

This section describes the windows related to security and authentication, and includes the following:

- Authentication and Authorization Configuration
- FIPS Settings
- General Security Settings
- Password Settings
- Session Settings
- SSL Certificate Expiration Monitoring
- User Lockout Settings
- User Login Settings

Authentication and Authorization Configuration

The **Authentication and Authorization** configuration includes the following two parts:

- In the **Authentication** section, add username and password to identify the user.
- In the **Authorization** section, define the process of granting roles and permissions for SecureSphere/external system users.

To configure user authentication:

1. In the **Admin** workspace, select **System Definitions > Authentication & Authorization Configuration**. The **Authentication & Authorization Configuration** pane appears.
2. For **User Authentication**, select one of the following options:

- **SecureSphere:** Allows authenticating using the internal SecureSphere database only.
- **External:** Allows authenticating using external authentication systems. If you select this option, you must select an external system you previously configured from the **External Authentication System** dropdown list. The external systems that you can choose from are defined in the **External Systems** pane, see [External Systems](#).



Note: When changing an external system to an X.509 certificate the user logged into SecureSphere must be connected using CAC. If not, a message is displayed asking you to first configure an Administrator as the External User, then log in as this administration. Only then to change User Authentication to External.

- **User Specific:** Allows authenticating using SecureSphere database or external authentication systems. If you select this option, you must select the external system from the **External Authentication System** dropdown list. The external systems that you can choose from are defined in the **External Systems** pane, [External Systems](#).
3. To enable managing SecureSphere users in LDAP, select the **External Authorization** checkbox. The LDAP group permissions are reflected in the SecureSphere external role permissions. For **External Authorization System**, select an external system from the dropdown list.

For information about external systems, see [External Systems](#).

4. Click **Save**.

Notes:



- When de-selecting the External Authorization option, all the external users are moved to the regular users section in the **Users & Permissions** window. They maintain their external system authentication, but lose their external role assignment.
- When selecting the **External Authorization** option, all Users with external system authentication become externally authorized: they are moved to the External Users section in the Users & Permissions window, and their permissions are based on their External Roles.

FIPS Settings

This parameter relates to FIPS (Federal Information Processing Standard) 140-2.

For more information about SecureSphere FIPS 140-2 compliance, see [FIPS 140 Compliance](#).

If the **FIPS Mode** parameter is enabled, the Management Server and the Gateways will communicate in a manner compliant with the FIPS 140-2 standard. By default, this parameter is not enabled.

If you change this parameter, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see [Activating Settings](#).

General Security Settings

These parameters define SecureSphere security settings.

The **Enable Iframe integration** parameter enables embedding SecureSphere within an Iframe. The change will be effective after server restart.

If you change this parameter, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see [Activating Settings](#).

Password Settings

These parameters define requirements for passwords of SecureSphere users and administrators.

These settings apply only to internal SecureSphere users and administrators, that is, the ones defined in the SecureSphere GUI. For more information about user permissions, see [Understanding Permissions](#).

If you change any of these parameters, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see the relevant product's documentation.

The longer a password and the greater the variety of its required characters, the more difficult it is to guess and the greater the security it confers.

Password Settings Window Parameters

Parameter	Description
Password validity period (in days): Passwords expire at the end of this period	The number of days after which a password expires. At the end of this period, the user or administrator is required to change the password. The new password must be different from all of the four (4) previous passwords. Default: 100 days
Password length: Minimum required number of characters in a password	The minimum length of a password (in characters). Default: 7 characters
A password must be significantly different from last password used	Specifies whether a new password must be significantly different from the password it replaces. (If a new password is significantly different from the password it replaces, knowledge of the previous password does not confer an advantage on attackers.) Default: not selected

Parameter	Description
A password must include capital letters	Specifies whether a password must include at least one upper-case character. Default: not selected
A password must include numbers	Specifies whether a password must include at least one numeric character. Default: selected
A password must include lower case letters	Specifies whether a password must include at least one lower-case character. Default: selected
A password must include non alpha-numeric characters	Specifies whether a password must include one of the following characters: ! @ # \$ % ^ & * () < > ? . _ + - [] { } \ : ; , / ` ~ = Default: not selected
Number of previous passwords a password must be different from	New Password must be different from the set number of old passwords Default: 1

Make sure your password has the following characteristics:

- It must have no fewer than 10 characters and no more than 14 characters.
- It must have at least one number, one capital letter, and one special character from:

* + = # % ^ : / ~ . , [] _ \ () | ; @ \$ & - ? { } < >

- It cannot have more than two characters repeated in succession.

Session Settings

In the **Session Settings** window, you can do the following:

- Specify the maximum number of concurrent sessions per user. When the user reaches this number of sessions, an error message is displayed.

- Enable or disable the **Force re-authentication following authorization change** option. If this option is enabled, the user session is reset after you make changes to a user in SecureSphere under **Users & Permissions**. If this option is disabled, the user session is not reset. The changes in the user settings are applied only after re-authentication, so you need to enable this option if you want to apply changes immediately.

SSL Certificate Expiration Monitoring

Every day at 01:00 SecureSphere scans the SSL keys and identifies those keys about to expire and those which have already expired (including an optional grace period). The parameters below define the deletion process and the issue system event messages process.

In the **SSL Settings** pane, configure the options.

SSL Settings Pane Parameters

Parameter	Description
Number of days before SSL key expiration to issue system event warning	The number of days before an SSL key's expiration date on which to issue a low-severity system event warning. If Number of days before SSL key expiration to issue system event warning is not empty: ▪ A system event is issued for expired soon keys ▪ A system event is issued for keys that expire on the date of the scan (today) If Number of days before SSL key expiration to issue system event warning is empty, no warnings are issued including for the expired today certificates Default: None
Expired SSL key deletion grace period	The number of days after an SSL key's expiration date on which to delete the key. If Expired SSL key deletion grace period is not empty, expired keys are deleted after the grace period has passed. If Expired SSL key deletion grace period is empty, no keys are deleted. Note: The MX deletes only those expired certificates which are not configured to a web-service. If a certificate is still configured to a web-service, it cannot be deleted manually, and the MX process cannot delete it as well, unless it is removed from the web-service.

Parameter	Description
	Default: None

If both fields are empty, the scan is not run.

User Lockout Settings

These parameters define the values related to a locked user account. For more information, see [Locking a User Account](#).

If you change any of these parameters, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see [Activating Settings](#).

Lockout Settings Window Parameters

Parameter	Description
Login failed attempts period	The period (in minutes) in which entering an incorrect password multiple times (defined by the Number of failed login attempts parameter) locks an account. Default: 5 minutes
Number of failed login attempts	The number of times that entering an incorrect password (within the period defined by the Login failed attempts period parameter) locks an account. Default: 3 times
Lock duration	The period (in minutes) that an account remains locked. Default: 30 minutes

User Login Settings

These parameters relate to the number of days of user inactivity that will cause the user to be locked out, and enable users to see their recent login and permissions activity.

The Admin user can grant access to a user who was locked out due to inactivity.

Lockout Settings Window Parameters

Parameter	Description
Lock inactive user after specified number of days	The period (in days) after which an inactive user is locked out of their account. Default: 35 days
Lock user after a period of inactivity	If enabled, the user will be locked out of SecureSphere after the defined number of days of user inactivity has passed. Default: Disabled.
Show user login info	If enabled, when a user logs in, a pop-up informs them of their recent login attempts.
Show user permission changes	If enabled, when a user logs in, a pop-up informs them of permission changes granted for them by the Admin user.

Management Server Settings

This section describes the windows related to Management Servers, and includes:

- Action Interfaces
- Alert Aggregation Configuration
- Alert Flags
- Application Groups Settings
- CSV Upload Default Settings
- External HTTP Settings
- External Systems
- Keywords Settings
- Lookup Data
- Policy Settings
- Report Settings
- Server Definitions
- System Events Notifications
- SecureSphere Audit

Action Interfaces

Parameters of the following Action Interfaces are defined in the **Action Interfaces** window:

- Archive to an FTP Location
- Archive to an SCP Location
- Assign as Task
- Block a User
- Block an HTTP Session
- Block an IP (Address)
- Create a Review Task
- Log DAM audit events to System Log (Gateway syslog) using the CEF standard
- Log FAM audit events to System Log (Gateway syslog) using the CEF standard
- Log SharePoint audit events to System Log (Gateway syslog) using the CEF standard
- Log audit events to System Log (Gateway syslog)
- Log custom security event to System Log (syslog) using the CEF standard
- Log network security event to System Log (syslog) using the CEF standard
- Log security event to RSA enVision
- Log security event to System Log (syslog) using the CEF standard
- Log system event to RSA enVision (RAW format)
- Log system event to System Log (syslog) using the CEF standard
- Log to System Log (syslog)
- Monitor a User
- Monitor an HTTP Session
- Monitor an IP (Address)
- Remedy Create Incident
- Run a Shell Command
- SNMP Trap
- Send an Email
- Terminate Session

The parameters in all these windows are explained in Working with Action Sets and Followed Actions **in the SecureSphere User Guide**.

The windows here (**Admin > System Definitions > Action Interfaces**) enable an administrator to:

- Define Action Interfaces
- Pre-define a value for an Action Interfaces parameter used in Action Sets by deselecting **User-Configured** for that parameter, so that when a user includes the Action Interface in an Action Set (**Main > Policies > Action Sets**), the values of these parameters are pre-defined in the Action Interface and the user cannot change them.

For example, if the Action Interface **Send an Email** is defined, then when the user defines an Action Set which includes it, the **SMTP Server Address** is pre-defined and the user cannot change its value.

This is illustrated in the following figures.

Parameter	User-Configured	Value
SMTP Server Address	☐	mail
Source Email Address	☒	User-Configured
Destination Address	☒	User-Configured
CC Address	☒	User-Configured
Email Format	☒	User-Configured
Run on Every Event	☒	User-Configured

Parameter	Value
SMTP Server Address	mail
Source Email Address	
Destination Address	
CC Address	
Email Format	Html
Run on Every Event	☐

In the Interfaces pane:

- The **Type** column specifies in which types of objects the Action Interface can be used.
- The **Usage Counter** column specifies the number of Action Sets in which the Action Interface is used.



Note: For most Types, there are pre-defined Action Interfaces. For the Types HTTP Archive, HTTPS Archive and NFS Archive, no Action Interfaces are pre-defined, but administrators can create new Action Interfaces of these Types.

Creating and Configuring Action Interfaces

You can create a new Action Interface, or edit and delete an existing one.

Creating an Action Interface

To create a new Action Interface:

1. In the **Admin** workspace, click **System Definitions**.
2. In the **System Definitions** pane, select **Action Interfaces**.
3. In the **Interfaces** pane, click .
4. Enter the Action Interface's **Name**.

Name should be meaningful, because the Action Interface will be used in Action Sets.

5. Select the Action Interface's **Type** from the drop-down list.

An Action Interface's **Type** determines the Action Set event types in which it can be used.

For more information, see [Working with Action Sets and Followed Actions](#).

6. Click **Create**.
7. When the new Action Interface is displayed in the list, expand it by clicking on the plus-sign to the left of its name.
8. Edit the Action Interface's parameters.

To pre-define a value for one of the Action Interface's parameters, deselect **User-Defined** for that parameter and enter a value. If you select **User-Defined**, the user will have to specify a value for the parameter when defining an Action Set that uses this Action Interface.

9. Click **Save**. Settings are saved. If you are in delayed activation mode, you need to activate these settings. For more information, see [Activating Settings](#).

Configuring an Existing Action Interface

To edit an existing Action Interface:

1. In the **Admin** workspace, click **System Definitions**.
2. In the **System Definitions** pane, select **Action Interfaces**.
3. In the **Interfaces** pane, select an Action Interface from the list.
4. Expand the Action Interface by clicking on the plus-sign to the left of its name.
5. Edit the Action Interface's parameters.

To pre-define a value for one of the Action Interface's parameters, deselect **User-Defined** for that parameter and enter a value. If you select **User-Defined**, the user will have to specify a value for the parameter when defining an Action Set that uses this Action Interface.

For more information, see [Working with Action Sets and Followed Actions](#).

6. Click **Save**. Settings are saved. If you are in delayed activation mode, you need to activate these settings. For more information, see [Activating Settings](#).

Deleting an Existing Action Interface

To delete an existing Action Interface:

1. In the **Admin** workspace, click **System Definitions**.
2. In the **System Definitions** pane, select **Action Interfaces**.
3. In the **Interfaces** pane, select an Action Interface from the list.
4. Click .
5. Click **Save**. Settings are saved. If you are in delayed activation mode, you need to activate these settings. For more information, see [Activating Settings](#).

Logging System Events for Auditing

To log system events for Auditing, SecureSphere offers two pre-configured Action Interfaces. These action interfaces can then be associated with policies and used to send out messages to your SIEM. You can additionally configure custom Action Interfaces. For more information on using Action Sets, Action Interfaces and Followed Actions, see the topic titled **Working with Action Sets and Followed Actions** in the **Advanced Configuration** chapter of your product's SecureSphere User Guide.

For more information on the placeholders used by these messages, see the Appendix titled **Placeholders** in your product's SecureSphere User Guide.

By default, the data included in these system events is as listed below. You also have the option of configuring custom action sets:

Log system event to RSA enVision

The syntax of the default action set to RSA enVision is as follows:

```
%IMPERVA-
Imperva,event#=$!{Event.dn},createTime=$!{Event.createTime},eventType=$!{Event.eventType}
```

Log system event to System Log (syslog) using the CEF Standard

The syntax of the default action set to Arcsight CEF is as follows:

```
CEF:0|Imperva
Inc.|SecureSphere|${SecureSphereVersion}|${Event.eventType}|#cefEscapeMessage(${Event.message})
rt=#arcsightDate(${Event.createTime}) cat=SystemEvent
```

Alert Aggregation Configuration

These parameters define how alerts are aggregated.

If you change any of these parameters, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see [Activating Settings](#).

Alert Aggregation Configuration Window Parameters

Parameter	Description
Alert aggregation mechanism is active	Enable this parameter in order to activate the alert aggregation mechanism. Warning: Disabling this parameter will cause SecureSphere to produce large number of alerts of the same type. Default: enabled
Maximal no. of regular alerts	When an aggregated regular (non-monitoring) alert reaches

Parameter	Description
that can be aggregated into a single alert	this limit, no new alerts will be aggregated to it. Default: -1 (meaning there is no limit)
Maximal no. of monitoring alerts that can be aggregated into a single alert	When an aggregated monitoring alert reaches this limit, no new alerts will be aggregated to it. Default: -1 (meaning there is no limit)
Maximal no. of detailed alerts kept per aggregated alert	When alerts are aggregated, SecureSphere stores, in addition to the aggregated alert information, a certain number of unaggregated alerts, which you can see in the alert viewer when browsing the aggregated alert. This parameter limits the number of raw alerts kept per aggregated alerts. Increase this value with caution, as large values may result in a very quick growth of the database, and will cause the system to discard old alerts more frequently in order to make room for new alerts. Default: 30

Alert Flags

These parameters define alert custom flags that you can then use in the Alerts screen to flag desired alerts for sorting and reporting purposes.

If you change any of these parameters, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see [Activating Settings](#).

Alert Flags Parameters

Parameter	Description
Enable Custom alert flags	Enable or disable Custom flags for alerts.
Custom flag 1 to	Custom flags 1-20 for security alerts. You can give a name to each custom flag that will then appear in the menu when you the right click on an alert in the Alerts

Parameter	Description
Custom flag 20	screen.

Application Groups Settings

This parameter relates to whether the Sites hierarchy or the Application Group hierarchy is used to apply policies.

A site can be organized either based on a server group hierarchy (**Main > Setup > Sites**) or on an application hierarchy (**Main > Setup > Applications**), but not both. Once the decision has been made, the organization cannot be changed, that is, the site cannot be migrated from one hierarchy to the other.

If you enable the **Apply application-level policy objects using application group hierarchy**, the Application Group hierarchy will be used rather than the Sites hierarchy.

By default, this parameter is not enabled.

If you change this parameter, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see [Activating Settings](#).

CSV Upload Default Settings

This parameter relates to the CSV Upload Settings when uploading IP Groups and Protected IP Addresses.

If you change this parameter, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see [Activating Settings](#).

Assessments Window Parameters

Parameter	Description
Case Sensitive	Specifies whether the data in the CSV file are case sensitive, so if existing data match the data in the CSV file, they are not considered duplicates.
Encoding	Select either ASCII or UTF8 .
Format	Select UNIX or Windows .

Parameter	Description
Use First Line As Header	Specifies whether the first line of the CSV file contains data to be imported or column headings.
Override Existing Entries	If the data in the CSV file matches existing data, then the data in the CSV file will overwrite the exiting data.

External HTTP Settings

To perform various updates SecureSphere needs to establish an Internet connection. The External HTTP Settings capability is used to establish the Internet connection using a proxy. The following updates can be performed:

- ADC content updates
- Web scanners updates
- ThreatRadars updates
- Software Updates
- Community Defense updates
- Key Exchange updates

To configure an Internet connection using HTTP proxy:

1. In the **Admin** workspace, click **System Definitions**.
2. From the **System Definitions** pane, select **External HTTP Settings**. The **External HTTP Settings** pane appears on the right.
3. Enter the parameters as described below.
4. Click **Save**.

Parameter	Description
Ignore HTTP Certificate	Enabling this checkbox instructs SecureSphere to not check the SSL certificate Common Name. This can be used if a proxy terminates the SSL session and submits its own certificate.
HTTP Proxy Configuration	Select to enable the following parameters.
IP Address or Host Name	Enter the proxy's IP address or host name.

Parameter	Description
Port	Enter the proxy's port, for example, 80 for HTTP or 443 for HTTPS.
Authentication Policy	Select an authentication protocol from the dropdown list.
User/Password/Verify Password	Enter the username and the password defined for SecureSphere user account on the proxy server.
Domain	When the NTLM is selected as the Authentication policy, you must specify the domain.

External Systems

These parameters specify how to define external systems that can be used in the SecureSphere login process. To use an external system, it must be enabled in the External Systems pane.

For information about the authentication configuration see [Authentication and Authorization Configuration](#).

If you change these parameters, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see [Activating Settings](#).

Defining External Systems

You can create a new External System, or edit and delete an existing one.

To create a new External System:

1. In the **Admin** workspace, click **System Definitions**.
2. In the **System Definitions** pane, select **External Systems**.
3. In the **External System** pane, click . The **External Systems** window appears.
4. In the **Name** text box, enter the name for the external system.
5. From the **Type** drop-down list, select one of the options described in the table below.
6. If you change any of these parameters, click **Save**.

Option	Description
Active Directory Collection	Describes how to configure Domain Forests, known as Active Directory Collections in SecureSphere. For more information see External Systems - Active Directory Collection (Forest) .
Kerberos Authentication	Describes how to configure SecureSphere to connect to the Kerberos domain account and authenticate users so users do not need to login with a user and password. For more information see External Systems - Kerberos Authentication .
LDAP	For more information, see External Systems - LDAP .
LDAP Authentication & Authorization	For more information, see External System - LDAP Authentication and Authorization .
RADIUS Authentication	For more information, see External System - RADIUS Authentication .
SQL	External System - SQL .
SSL Proxy	For more information, see External System - SQL .
X.509 Authentication	Describes how to configure support for X.509 Authentication. For more information see External System - X.509 Authentication .

To edit an existing External System:

1. In the **Admin** workspace, click **System Definitions**.
2. In the **System Definitions** pane, select **External Systems**.
3. In the **External System** pane, select an External System from the list.
4. Expand the External System by clicking on the plus-sign to the left of its name.
5. Edit the External System's parameters. See [External Systems - LDAP](#) for an explanation of External System parameters.

If you change any of these parameters, click **Save**.

To delete an existing External System:

1. In the **Admin** workspace, click **System Definitions**.
2. In the **System Definitions** pane, select **External Systems**.
3. In the **External System** pane, select an External System from the list.
4. Click .
5. Click **Save**.

If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see [Activating Settings](#).

External Systems - Active Directory Collection (Forest)

SecureSphere supports the use of Domain Forests. In SecureSphere, this feature is known as Active Directory Collections. By configuring an Active Directory Collection you enable the system to integrate user information across various Active Directory domains. Members of the collection can be from different forests as long as they have two way trust.

For information on creating and managing an Active Directory Collection external system, see [Defining External Systems](#).

Active Directory Collection Configuration Parameters

Parameter	Description
Available LDAP System	Lists the various LDAP systems that have been configured in SecureSphere.
Collection Members	Lists those LDAP systems you have selected to participate in the LDAP Collection (members of the forest). Note: Using the Test Connection button, you can test connections to all enabled members after saving.

External Systems - Kerberos Authentication

SecureSphere enables you to configure connectivity with your Kerberos Domain account to enable Windows users to be automatically logged into SecureSphere through Kerberos and subsequently eliminate the need for them to enter a username and password.

This procedure describes how to configure the required components in order to support this form of authentication. Kerberos involves the following primary steps:

Configuring Automatic Kerberos Authentication

	Action	Description	For more information, see...
1	Setup the Domain Controller	Configure the Service Principal on the domain controller and generate a keytab file for import into SecureSphere.	See your Windows server documentation for assistance.
2	Add Hostnames and IP addresses to Domain Controller	Configure your Domain Controller with host names and IP addresses. Note: Hostnames must be compliant with RFC 7230, meaning they must start with a letter, end with a letter or digit and have only letters, digits, or hyphen as interior characters.	See your Domain Controller documentation for assistance.
3	Configure Browsers	Configure the browser you want to use to access SecureSphere.	Configuring the Browser for Kerberos Authentication
4	Configure SecureSphere	Configure SecureSphere Admin definitions required to support authentication.	Configuring SecureSphere for Automatic Kerberos Authentication

Supported Components

Kerberos Authentication is supported for the following Microsoft Windows versions: Window XP, Windows 7, Windows Vista, and Windows 2000. The following browsers are supported:

Browser Support

Browser	Version(s) Supported
Microsoft Internet Explorer	10 - 11

Browser	Version(s) Supported
Mozilla Firefox	Most recent stable version.
Google Chrome	Most recent stable version.

Configuring the Browser for Kerberos Authentication

The first step in configuring kerberos authentication and enabling users to access SecureSphere and be automatically authenticated using Kerberos (and not need to enter a username and password), involves making some minor changes in the browser.

Notes:



- When logging in, users must navigate to the host name of the SecureSphere server. Navigating to the IP address will not enable them to automatically log in and they will be asked for credentials.
- Clients must be in the same domain of the Domain Controller, or in a domain with a trust or sub-domain.
- When users are in a domain with trust, they need to navigate using the SecureSphere server hostname + full domain. For example, `https://MX_hostname.domain.com:8083/`

Currently, Kerberos authentication requires configuring the browser:

- **Windows IE:** see [Configuring Windows Internet Explorer \(IE\) for Kerberos Authentication](#)
- **Firefox:** see [Configuring Firefox for Kerberos Authentication](#)

Configuring Windows Internet Explorer (IE) for Kerberos Authentication

This procedure describes how to configure Windows IE to support Kerberos authentication to SecureSphere. For instructions on how to configure Firefox, see [Configuring Firefox for Kerberos Authentication](#).



Note: Windows Internet Explorer is no longer supported by Microsoft. For more information, refer to Microsoft's web site.

To configure Windows IE:

1. In **Windows IE**, from the **Tools** menu, click **Internet Options**. The **Internet Options** window opens.
2. Click the **Security** tab.
3. Select the **Local Intranet** zone, then click the **Sites** button. The **Local Intranet** window opens.
4. Click **Advanced**, advanced settings are displayed.
5. Type the fully qualified URL for the website that is to be authenticated using Kerberos.

For example: `http://mywebsite.mycompany.com`

6. Click **OK**.
7. Configure SecureSphere settings as described in [Configuring SecureSphere for Automatic Kerberos Authentication](#).

Configuring Firefox for Kerberos Authentication

This procedure describes how to configure Firefox to support Kerberos authentication to SecureSphere. For instructions on how to configure Windows IE, see [Configuring Windows Internet Explorer \(IE\) for Kerberos Authentication](#).

To configure Firefox:

1. Open **Firefox**.
2. In the **Address** bar, type **About:Config**. A warning message is displayed. Click **I'll be careful, I promise**. Configuration settings are displayed.
3. Find the Preference Name **network.negotiate-auth.trusted-uris** and double-click it. The **Enter String Value** window opens.
4. Type the Domain for the website that is to be authenticated using Kerberos.

For example: `il.imperva.com`

5. Click **OK**. Settings have been changed.
6. Configure SecureSphere settings as described in [Configuring SecureSphere for Automatic Kerberos Authentication](#).

Configuring SecureSphere for Automatic Kerberos Authentication

To support Kerberos authentication of SecureSphere users, you need to configure settings in the Admin workspace of SecureSphere.



Note: Time needs to be synced between clients, the SecureSphere Server, and the Domain Controller using the same timeserver.

To configure SecureSphere Kerberos Admin settings:

1. In the **Admin** workspace, click **System Definitions > External Systems**. External Systems are displayed.
2. In the **External System** pane, click **New**. The **New External System** window opens.
3. Type a **Name** for the External System, select **Kerberos Authentication** from the dropdown menu, and click **Create**. A new **External System** is added to the **Details** pane.
4. Expand the **External System** by clicking the plus sign to the left of its name.
5. Click **Upload Kerberos Details**. Then do the following:
 - In the **Service Principal** field, type the **Service Principal Key** you configured in the Domain Controller. The syntax of a Service Principal is username@mycompany.com.
 - Under **File**, click **Browse** and navigate to the location of your **Keytab file**.
 - Click **Upload**.



Note: For instructions on obtaining a Service Principal name or generating a Keytab file, see your Domain Controller documentation or refer to the Imperva Support Knowledge Base.

6. Under **Username Type**, select a **Username Type**. This option represents how user names are formatted in SecureSphere. Select one of the following:
 - **Username with domain:** Select this option when SecureSphere usernames are required to have domain names attached. For example, if your user name is JamesT and your company name is Company, users must enter: JamesT@Company.com
 - **Username without domain:** Select this option when using internally when no domain is required as part of the username. Using the above example, your user would type in JamesT.



Note: If there is the possibility that you have two users from different domains with the same username, be sure your system is configured to use Usernames with domains.

7. Click **Save**.
8. Enable the external system by selecting the **Enabled** checkbox.
9. Click **Save**.
10. In the **System Definitions** tree in the left-hand side of the window, click **Authentication Configuration**.
11. In the **Details** pane, click one of the following:

- **External:** Authenticating with an external system such as Radius, LDAP and Kerberos
- **User Specific:** Takes the settings as configured in the Authenticator field of the User Information Tab under **Users and Permissions**.

12. Select the **External System** you configured in step above.

13. Click **Save**.

You can now create users in the Users and Permissions section of the Admin workspace. These users need to match the ones you created on your Domain Controller.

External Systems - LDAP

You can define the LDAP external system.

The connection can be established once this external system is enabled.



Note: This facility is not used for authentication and authorization of SecureSphere users into the MX or SOM. For authentication and authorization into the MX or SOM, use LDAP Authentication and Authorization. For more information, see [External System - LDAP Authentication and Authorization](#).

LDAP configuration Parameters

Parameter	Description
Name	The external system's name.
Primary Server	The IPv4 or IPv6 address or the host name of the primary external system.
Secondary Server	The IPv4 or IPv6 address or the host name of the secondary external system.
Use SSL	Specifies whether to connect to the external system using SSL.
Port	The port number on the external system.

Parameter	Description
Account Name DN	The account name or DN (Distinguished Name) under which to connect to the external system.
Password	The password to use when connecting to the external system.
Verify Password	Re-enter the password.
Base DN	The Base Distinguished Name is the domain account name in the LDAP server's navigation tree.
Follow Referrals	Select this option to tell/direct the LDAP system to follow referrals.
LDAP Custom Fields	In the Custom Field text box, type the LDAP name of the custom field, to receive data from a specific Active Directory field. For example, "physicalDeliveryOfficeName" instead of "Office".

Click **Test Connection** to verify that the connection to the external system is functioning properly.

External System - LDAP Authentication and Authorization

You can set the parameters related to LDAP authentication and authorization of SecureSphere users.

SecureSphere supports LDAP Authentication and Authorization with Windows Server 2022.

The connection can be established once this external system is enabled.

If you change any of these parameters, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see the relevant product's **SecureSphere User Guide**.

Note:



- The admin user cannot be authenticated using LDAP, but is always authenticated by SecureSphere.
- When configuring the external system using a hostname, and not an IP address: In order to use LDAP to authenticate SecureSphere users, a DNS

client must be enabled on the Management Server using the `impcfg` command. For more information, see Name Resolution (DNS client).

LDAP Authentication Window Parameters

Parameter	Description
Connection Settings	
Enabled	Whether the connection is enabled. Make sure this checkbox is checked, otherwise, the other settings are not available.
Name	The external system's name.
Primary Server	The IP address or the host name of the primary LDAP server.
Port	The port number of the LDAP service on the primary LDAP server. Default: 389.
Secondary Server	The IP address or the host name of the secondary LDAP server.
Port	The port number of the LDAP service on the secondary LDAP server.
Use SSL	Select to specify that communication with LDAP servers is encrypted. If this option is selected, ensure that the port number is changed accordingly.
Validate Certificate	You can authenticate the LDAP server using its certificate. Validate Certificate is enabled only if Use SSL is checked. If you check Validate Certificate , then you must upload the LDAP server's certificate to the MX by clicking Upload Certificate . For more information, see Setting Up SSL with Certificate Authentication on MX in the User Guide for your product.

Parameter	Description
Upload Certificate	Upload the LDAP server's certificate to the MX. Click Upload Certificate and in the Upload Certificate window, browse to a .cer certificate file and click Upload . For more information, see Setting Up SSL with Certificate Authentication on MX in the User Guide for your product.
Access Mode These parameters specify whether access to the LDAP server is anonymous or conducted under a specific user. Select one of the following:	
Anonymous	Use the "guest" account on the LDAP server, if it is configured.
User Account	If User Account is selected, then the following parameters must be defined: ▪ Account DN: The Distinguished Name of the domain account under which the LDAP server is to be accessed. ▪ Password: The domain account's password. ▪ Verify Password: Re-enter the password.
Search Settings The following parameters define how to locate a user in LDAP.	
Identifier Field	The unique identifier that SecureSphere uses to identify the external LDAP users. If Identifier Field is empty, the user name in SecureSphere is the same as on the LDAP server. Note: Case-Sensitivity: Depending on the LDAP server, the login process (authentication by sam-name/email, etc.) may or may not be case-sensitive. However regarding Identifier Field vs. name, the comparison in SecureSphere is case-sensitive (this is a cross-system behavior). For example if the user CN in LDAP is "John Smith", you must set the SecureSphere user name to "John Smith" not "john smith". Default: CN.
Base Path for Users	The domain path (branch) in the authentication server's navigation tree where the LDAP users are stored., for example, "ou=research,dc=sf,dc=la".

Parameter	Description
	Note: You can set this parameter only when the External Authentication parameter is enabled, see step
Search Filter	The search criteria for the unique identifier. You can specify the conditions that identify a user in LDAP Base Path for Users. For example, in Active Directory you would specify that the search should compare the name with which the user logged in to the field sAMAccountName as follows: "sAMAccountName={0}". In OpenLDAP, you would specify the same thing as follows: "uid={0}". Additionally, you might specify a compound condition like: "(&(objectCategory=person)(objectClass=user)(sAMAccountName={0}))". This would restrict the comparisons to only persons/users and exclude other types of entities. Note: If you change the search filter in your authentication server's account (for example, from username to email) users will then have to use a different login name.
Users Search Scope	Select one of the following: ▪ One Level: Limits the search to the base path only. ▪ Subtree: Allow searching subtree branches.
Authorization Settings	
Enable	Enables reflecting LDAP permission group structure in SecureSphere, where LDAP groups are mapped to SecureSphere external roles.
Base Path for LDAP Groups	The domain path in the authentication server's navigation tree where the LDAP groups are stored. Note: You can set this parameter only when the External Authentication parameter is enabled, see step 3 in Authentication and Authorization Configuration.
Groups Search Scope	Select one of the following: ▪ One Level: Limits the search to the base path only. ▪ Subtree: Allow searching subtree branches.

Click **Test Connection** to verify that the connection to the LDAP servers is functioning properly.

Notes:

- To point SecureSphere to the Active Directory Global Catalog, set **Base Path** to the domain root (for example, DC=corpmain,DC=com) and **Port** to 3268.
- If you are having difficulty configuring or logging into the MX using an Active Directory external user, do the following:



1. In a text editor, open the **authentication-service.properties** file located in **/opt/SecureSphere/server/SecureSphere/jakarta-tomcat-secsph/webapps/SecureSphere/WEB-INF/properties**

2. Update/add the values for the following properties:

```
authenticationService.ldap.referral = ignore;
```

```
authenticationService.ldap.direct.groups = true;
```

3. Save the file.

4. Restart the MX.

To define, view, or modify the LDAP authentication configuration for individual users:

- For a new user, see [Creating a SecureSphere User](#).
- For an existing user, see [Configuring User Information](#).

Third-Party Authentication Task Flow:

1. Obtain the enterprise account details (server's IP addresses, domain account name to be used for authentication, its password, etc.) from your Domain Administrator.
2. Configure the system for third-party authentication. See [Authentication and Authorization Configuration](#).
3. Configure the users who are to be authenticated via third-party authentication.
 - For a new user, see [Creating a SecureSphere User](#).
 - For an existing user, see [Configuring User Information](#).

External System - RADIUS Authentication

You can define the RADIUS external system to connect to the RADIUS server.

The connection can be established once this external system is enabled.

RADIUS Authentication Parameters

Parameter	Description
Name	The external system's name.
Primary Server	The IP address or the host name of the primary external system.
Secondary Server	The IP address or the host name of the secondary external system.
Primary/Secondary Server Port	The port number on the external system used by primary/secondary server. Default: 1812.
Shared Secret	The shared password to use when connecting to the RADIUS server.
Verify Shared Secret	Re-enter the password.
Authentication Protocol	The authentication method that is used to establish the connection with the RADIUS server.
Additional Attributes	Provide the RADIUS attributes which should be specified to communicate with the RADIUS server. For each attribute, provide the following: ▪ Id: The attribute Id refers to the numerical value of the "RADIUS attribute Type" as defined by RADIUS RFC 286, section 5 (http://tools.ietf.org/html/rfc2865). For example, the Id of Attribute "NAS-IP-Address" is 4, the id of attribute "Service-Type" is 6 and so on. ▪ Value: When providing values for the attributes which have a fixed set of predefined values (for example: "Service Type"), please provide the numeric ID of the value (as defined in the RFC286 for each attribute). For example: If you wish to set Attribute "Service-Type" with value of "Framed", you would provide this attribute with the

Parameter	Description
	value of 6 (see: http://tools.ietf.org/html/rfc2865#section-5.6) Note: You can use ID 27 (session-timeout) to adjust the RADIUS session timeout which can be helpful if you are using a MFA with your RADIUS server (see: https://datatracker.ietf.org/doc/html/rfc2865#section-5.27).

Click **Test Connection** to verify that the connection to the external system is functioning properly.

External System - SQL

You can define the SQL external system to connect to an SQL server.

The connection can be established once this external system is enabled.

SQL External System Parameters

Parameter	Description
Name	The external system's name.
Database Type	The database type.
Server IP	The IP address of the SQL server. This field is optional, if you enter the host name: then the IP address is resolved automatically.
Host	The host name of the SQL server. This field is optional: you can enter the IP address instead. However, if you enter the host name, the IP address is resolved automatically.
Port	The port number of the SQL service on the SQL server.
Server Name/Domain	Enables you to specify a specific domain name.

Parameter	Description
Database Name	Name of the database.
User Name	The name of the user as it appears on the SQL server.
Password	The password to use when connecting to the SQL server.
Verify Password	Re-enter the password.
Certificate	Only relevant for Snowflake (see section Working with Snowflake in DAM User Guide), when you use a certificate instead of credentials.

Click **Test Connection** to verify that the connection to the external system is functioning properly.

External System - SSH Proxy

If the network topology is such that the MX does not have access to the servers which it needs to scan, you can define an SSH Proxy through which the scans will be run. The MX will open an SSH session to the SSH proxy and access the scanned servers through the proxy.

The SSH proxy can be any machine, including a SecureSphere Gateway, which:

- has access to the servers to be scanned, *and*
- which the MX can access via SSH.

Once you have defined an SSH proxy, it will appear as a choice in the **Scan Proxy** dropdown menu (**Servers** tab in **Main > Sites > Server Group**).

If you want to use a SecureSphere Gateway as an SSH proxy, you still must define it here, otherwise it will not appear as one of the choices under **Scan Proxy**.

SSH Proxy External System Parameters

Parameter	Description
Enabled	Check Enabled if you will be running scans using this proxy.

Parameter	Description
Name	The SSH proxy's name. This name will appear in the Scan Proxy dropdown menu. It does not have to be the same as the proxy's hostname.
Server IP	The IP address of the SSH proxy.
User Name	The name of the user under whose account the SSH session on the proxy will be opened.
Password	The user's password.
Verify Password	Re-enter the password.

After entering the above parameters and clicking **Save**, you can test the connection by clicking **Test Connection**.

External System - X.509 Authentication and Authorization

X.509 Client Certificate Authentication and Authorization (X.509 Authentication and Authorization) is a standard identification used to enable access to buildings and controlled spaces, as well as computer networks and systems. X.509 authentication involves an X.509 certificate which identifies people authorized to use the system and authenticates their identity. X.509 authorization is similar, but does not require configuring users in SecureSphere as part of the process since it works with the LDAP authentication and authorization external system.



Note: By combining SecureSphere X.509 support with the proper client side software, SecureSphere supports the CAC standard.

The tables below lists the actions you must take to enable X.509 Authentication and Authorization of SecureSphere administrators. For an overview of the X.509 Authentication and authorization process, see [X.509 Authentication Process - Overview](#).

Configuring X.509 Authentication

	Action	For more information, see
1	Create an External System of type X.509 Authentication.	Defining External Systems
2	Configure the X.509 Authentication External System you created in step 1.	External System - X.509 Configuration
3	Under System Definitions , configure Authentication and Authorization Configuration by selecting External , then choosing the X.509 External System you configured in step 1, so it can use X.509 Certificates.	Authentication and Authorization Configuration
4	Configure users in SecureSphere Admin workspace. To use SecureSphere with a X.509 certificate, they need to be defined in SecureSphere. Note: Names of SecureSphere users must be identical to how they appear in the X.509 certificate.	Creating a SecureSphere User

Configuring X.509 Authentication and Authorization

	Action	For more information, see
1	Create an External System of type LDAP Authentication & Authorization.	Defining External Systems
2	Configure it, including authorization mode and enable it.	External System - LDAP Authentication and Authorization
3	Create an External System of type X.509 Authentication & Authorization.	Defining External Systems
4	Configure it, including the authorization mode (select your created LDAP external system) and enable it.	External System - X.509 Configuration
5	Under System Definitions , configure Authentication and Authorization Configuration by selecting	Authentication and Authorization Configuration

	Action	For more information, see
	External , then choosing the X.509 External System you configured in step 3, so it can use X.509 Certificates.	

X.509 Authentication and Authorization Process - Overview

When a user who is defined as being authenticated by X.509 directs their browser towards the SecureSphere Management Server, this is what happens:

1. In some systems, the user may be asked to enter a PIN.
2. If the PIN is correct, the certificate is passed to the Management Server.
3. The Management Server confirms that it trusts the issuer of the client certificate.
4. The Management Server confirms that the certificate has not expired.
5. The Management Server confirms that the certificate has not been revoked.
6. The Management Server logs the user in.



Note: If a user certificate has been revoked, that user will not be able to login. The one user that can always log in is the SecureSphere Administrator (named "admin") who can login by navigating to the SecureSphere login page and using their password.

External System - X.509 Configuration

The following are parameters that need to be configured to support X.509 Authentication and Authorization.

X.509 Authentication and Authorization System Parameters

Parameter	Description
Enabled	Check Enabled to make the external system available in SecureSphere.
Name	Type a name for the X.509 External System. This name appears when trying to associate an external system with Authentication and Authorization Configuration.

Parameter	Description
Allow Login Screen	Determines if and when to display a login screen when authentication fails, enabling users to manually login. ▪ Never: Never display a login screen no matter what the source of authentication failure. ▪ Only on OCSP Communication Failure: Displays the login screen only if authentication failure is due to a failure to communicate with the OCSP server configured under Trusted Certificate Authorities below. ▪ Always: Displays the login screen no matter the reason authentication with the X.509 certificate failed.
Username Identified Field	Used to identify users for authentication. Options include: ▪ Subject: Authenticates users by matching the user name that appears in the subject field of the CA. If the user being authenticated doesn't match the name in that field, authentication fails. ▪ Email Address: Authenticates users based on their full email address, including the domain name. For example: John.Smith@Imperva.com. ▪ Email Address no Domain: Authenticates users based on the user portion of their email address without the domain. For example, John.Smith. ▪ Subject Alternate Name - User Principal Name: Authenticates users based on the UPN field.
Trusted Certificate Authorities	Enables uploading CA and OCSP certificates and configure supporting parameters. Click Choose File and select the file(s) to upload CA and OCSP certificates as required. Uploading a CA certificate is mandatory, uploading an OCSP certificate is optional. Additional Parameters: ▪ OCSP URL: URL to the OCSP server used to validate that the client certificate is valid. Format should be http://10.0.0.0:8989. This field is mandatory. ▪ Consider OCSP Response of "Unknown" As Revoked: If the response received from the OCSP server is unknown, it will consider the user certificate as invalid and user will not be provided access. ▪ Overwrite Existing Certificates with Same Subject: When uploading

Parameter	Description
	new certificates, overwrites existing certificates that include an identical subject .
Authorization Settings	Enables X.509 authorization and sets the external LDAP authorization system defined. ▪ Enable Authorization: Select this check box to enable integration with LDAP authorization. ▪ External Authorization System: Select the LDAP authentication & authorization enabled external system, with authorization mode enabled from the dropdown.

Keywords Settings

These parameters allow administrators to define keywords to be used in filtering reports.

The following procedures can be used to manage keywords:

- [Defining a New Keyword](#)
- [Deleting a Keyword](#)
- [Associating a User-defined Keyword with a Report](#)

The ADC keywords in the following table are ADC pre-defined.

Keywords Settings Window Parameters

Key Name	Description
Basel II	Indicates that a report is a Basel II report. Defaults: Filter Enabled and ADC: selected
EBS	Indicates that a report is an EBS report. Defaults: Filter Enabled and ADC: selected

Key Name	Description
GLBA	Indicates that a report is a GLBA report. Defaults: Filter Enabled and ADC : selected
HIPAA	Indicates that a report is a HIPAA report. Defaults: Filter Enabled and ADC : selected
ISO 27001	Indicates that a report is an ISO 27001 report. Defaults: Filter Enabled and ADC : selected
PCI	Indicates that a report is a PCI report. Defaults: Filter Enabled and ADC : selected
PeopleSoft	Indicates that a report is a PeopleSoft report. Defaults: Filter Enabled and ADC : selected
Privacy	Indicates that a report is a Privacy report. Defaults: Filter Enabled and ADC : selected
SAP	Indicates that a report is a SAP report. Defaults: Filter Enabled and ADC : selected
SOX	Indicates that a report is a SOX report. Defaults: Filter Enabled and ADC : selected
SharePoint	Indicates that a report is a SharePoint report. Defaults: Filter Enabled and ADC : selected

Defining a New Keyword

This procedure describes how to define a new keyword. For more information on Keyword settings, see [Keyword Settings](#).

To define a new keyword:

1. Click .
2. Enter the keyword's **Name**.
3. If you want to be able to filter the list of reports based on this keyword, select **Filter Enabled**.

If **Filter Enabled** is not selected, then you cannot filter the list of reports by this keyword, but you can still associate the keyword with a report.

For user-defined keywords, **ADC** is by default not selected, and this cannot be changed.

4. Click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see the relevant product's **SecureSphere User Guide**.

Deleting a Keyword

This procedure describes how to delete an existing keyword. For more information on Keyword settings, see [Keyword Settings](#).

To delete a keyword:

1. Select the keyword.
2. Click . Click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see the relevant product's **SecureSphere User Guide**.

Associating a User-defined Keyword with a Report

This procedure describes how to associate a user-defined keyword with a report. For more information on Keyword settings, see [Keyword Settings](#).

To associate a user-defined keyword with a report:

1. In the **Main** workspace, click **Reports > Manage Reports**.
2. Select a report.
3. In the **General Details** tab, move the user-defined keyword from **User Defined Keywords** to **Report Keywords**.
4. Click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see the relevant product's **SecureSphere User Guide**.

Lookup Data

These parameters are related to lookup data.

If you change this parameter, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see the relevant product's **SecureSphere User Guide**.

Lookup Data Window Parameters

Parameter	Description
Query Timeout	Determines the period of time after which no response is received that the query shall time out. Default: 15 minutes
Maximal Lookup Data Set Size	Maximum Lookup Dataset size, that is, the maximum number of results returned from the query. Lookup Dataset size is defined in Main > Setup > Global Objects. Default: 10000

Policy Settings

These parameters relate to Security Policies.

If you change any of these parameters, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see the **SecureSphere User Guide**.

Policy Settings Window Parameters

Parameter	Description
Enable comments for policy changes	If enabled, users are able to add comments when modifying a Security Policy. Default: Disabled

Report Settings

These parameters are related to reports.

If you change any of these parameters, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see the **SecureSphere User Guide**.

Report Settings Window Parameters

Parameter	Description
Maximal number of rows in a report in CSV format	The maximum number of rows in a CSV format report. Additional data beyond this number of rows will not be included in the report. Default: 10000
Require user to install East Asian fonts before displaying PDF reports	If checked, SecureSphere assumes that all PDF reports include East Asian fonts, and requires that the user install these fonts before viewing PDF reports. If not checked, SecureSphere assumes that PDF reports do not include East Asian fonts, so if a report does include these fonts it will be incorrectly displayed.
Archive report without metadata	If checked, SecureSphere archives only the reports themselves without the metadata defining how the reports were produced. Reports are accessible via the file system, not via SecureSphere.
Maximal number of pages in a report in PDF format	The maximum number of pages in a PDF format report. Additional data beyond this number of pages will not be included in the report. Default: 500
Template for results files	A sequence of placeholders which defines the report's file name. Default: \${reportName}_\${username}_\${date}
Include report sent by e-mail followed action as link instead of attachment	Include a direct link to a report when an e-mail followed action is used. When disabled, the report is sent as an attachment. Default: Disabled.

Parameter	Description
URL for reports and tasks links	The URL of SecureSphere which hosts reports, for example https://10.3.250.103:8083. SecureSphere then appends the URL so that it directs users to the specific report.

Server Definitions

These parameters relate to the MX server.

Server Definition Window Parameters

Parameter	Description
Server Name	You can assign a name to the Management Server. This name will be displayed in the title of the SecureSphere GUI window. This name can be different from the MX appliance's hostname. Default: none

System Events Notifications

These parameters relate to system event notifications.

If you change any of these parameters, click **Save**.

System Events Notifications Settings Window Parameters

Parameter	Description
Gaps in minutes between consecutive gateway overload messages	When a Gateway cannot process all the traffic, a system event is generated as a warning. Another system event is created if the Gateway is still overloaded after this interval. Default: 15 minutes
Gap in minutes between	Generate an ambiguous packet message only if this interval

Parameter	Description
consecutive ambiguous packets messages	has passed since the last such message. Default: 5 minutes
Issue a system event when a gateway blocks more than the following number of ambiguous packets per minute	Generate a system event if the Gateway blocks more than this number of packets in one minute. Default: 60 packets
Issue a system event when the following CPU utilization percentage is exceeded	Generate a system event if CPU utilization exceeds this percentage. Default: 90 percent
Gap in minutes between consecutive CPU utilization messages	Generate a CPU utilization system event only if this interval has passed since the last CPU utilization system event was generated. Default: 60 minutes
Gap in minutes between consecutive gateway throughput messages	Generate a Gateway throughput system event only if this interval has passed since the last Gateway throughput system event was generated. Default: 60 minutes
Issue a system event when the following throughput is exceeded	Generate a system event if the Gateway throughput exceeds this rate. Default: 90 Mbps
Issue a system event when the ADC content has not been updated for the following number of days	Generate a system event if this interval has passed since ADC content was updated. Default: 30 days

SecureSphere Audit

These parameters configure the syslog messages sent when there are user changes to the SecureSphere system or SecureSphere security events.

Check the box to enable the sending of SecureSphere audit and security events to syslog.

SecureSphere Audit Parameters

Parameter	Description
Syslog Host	Type the IP address or host name of the syslog server. Note: The port used to send to the syslog server is the default port 514. If you want to use a different port, add :<port number> after the IP address or host name.
Syslog Log Level	Select the desired syslog log level from the dropdown list (info, warn, debug or error).
Facility	Select the required facility (type of authorization required by your installation of the SIEM).
Message	Type a message with placeholder information to be used by syslog to create a message readable by the SIEM.

User Interface Settings

This section describes the windows related to the user interface (GUI), and includes the following:

- Display Limits
- Language Settings
- Screen Settings

Display Limits

These parameters relate to the display of items in the SecureSphere GUI.

If you change any of these parameters, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see the **SecureSphere User Guide**.

Display Limits Window Parameters

Parameter	Description
Maximum options in pushdown menu	The maximum number of items displayed in drop-down menus. If the list includes more items, they are not displayed and are not available for selection. Default: 500 items
Maximum rows in table	The maximum number of items displayed in tables. If the list includes more items, they are not displayed and are not available for editing. Default: 500 item

Language Settings

These parameters define the language used in the SecureSphere GUI.

If you change this parameter, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see the **SecureSphere User Guide**.

Language Settings Window Parameters

Parameter	Description
Select Display Language	The language in which GUI elements will be displayed. Default: English
Change language without affecting the database	If enabled, texts in the SecureSphere database, for example, policy names, will not be translated into the Display Language when displayed in the GUI. Default: Enabled

Screen Settings

These parameters define the screen settings used in the SecureSphere GUI.

If you change any of these parameters, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see the **SecureSphere User Guide**.

Screen Settings Window Parameters

Parameter	Description
Auto refresh screen	Enables an automatic refresh of the Dashboard screen. Clearing the check box disables the action and the screen loads initial data only once. Default: Enabled
Display Response Headers in Alerts Screen	Show the response headers in the violation when expanding a specific violation in the alerts screen. Default: Disabled

Cloud Settings

This section describes the windows related to Imperva cloud applications, and includes the following:

General Settings

General Settings

These parameters allow you to enable or disable Imperva cloud application settings.

If you change any of these parameters, click **Save**. If you are in delayed activation mode, you must activate these settings before they take effect. For more information, see the **SecureSphere Web Security User Guide**.

General Settings Window Parameters

Parameter	Description
ThreatRadar is Enabled	If enabled, SecureSphere ThreatRadar is enabled. Default: selected
Attack Analytics is Enabled	If enabled, SecureSphere Attack Analytics is enabled. Default: selected

System Performance

This section describes the **System Performance** window, which displays performance data about Management Servers, and Gateways. It includes the following:

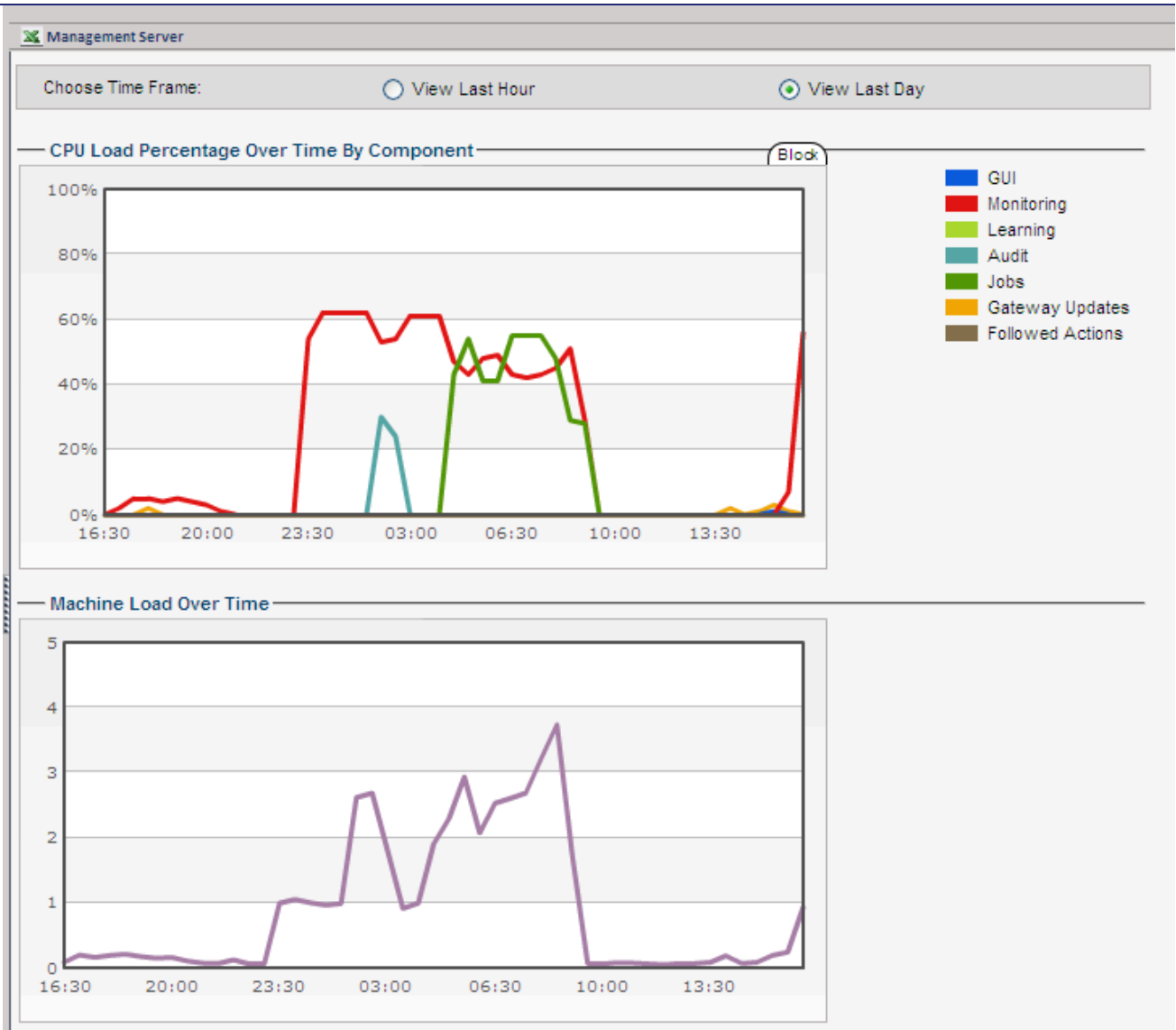
- Management Server
- Exporting Technical Information from Management Servers
- Troubleshooting Performance

Management Server

The **Management Server** window displays real-time information about the Management Server.

To display real-time information about the Management Server:

1. In the **Admin** workspace, click **System Performance**.
2. In the **System Performance** pane on the left, click **Management Server**.



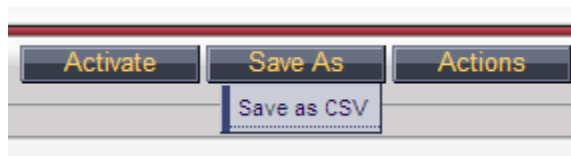
If the Management Server is in an MX-HA configuration, the IP address of the active server is displayed at the top of the window.

The graphs display the following information:

Graph	Displays
CPU Load Percentage Over Time By Component	The CPU load attributable to each of the Management Server's functions. Each line is differently colored and represents a different function, according to the color-coded list of functions alongside the graph. The total for all the functions at any given time is the approximate total CPU load at that time. For example, in the graph shown above, at around 5AM the Monitoring function consumed about 50% of the CPU and the Audit function consumed about 40%, for a total of around 90%.

Graph	Displays
	Note: The Jobs graph displays only jobs whose CPU usage is greater than 10%.
Machine Load Over Time	The overall machine load over time, as returned by the Linux "top" command. This represents the number of processes divided by the processor's capacity to process them. A value of 1.00 means that the available capacity exactly matches the number of processes. This is not ideal as there is no available space should the number of processes increase. A value of 0.70 or less is considered optimum. For multi-core/processor systems, the load is relative to the number of cores/processors, So for a system with two cores, a machine load of about 1.5 is considered optimum. Occasional spikes above the optimum value are acceptable, but if they are continuous there will be lag.

- You can change the time period displayed in the graphs by selecting **View Last Hour** or **View Last Day**.
- To output the data in the graphs to a CSV file:
 - Click  (to the left of the words **Management Server** at the top of the window).
 - Alternatively, you can click **Save As** (in the upper right corner) and select **Save as CSV**.



- To move the generation of the CSV file to a background process, click **Move to Background** in the **Export CSV window**.

The data in the CSV file is the same data displayed in the graphs.



Note: The exported data relate to the previous 24 hours.

Exporting Technical Information from Management Servers

On occasion, when encountering trouble with SecureSphere operation, Imperva support may request that you provide them with technical information that is automatically generated by SecureSphere, so they can analyze logs and other information. You can export this information from the SecureSphere Management Server using the GUI. This procedure describes how to export technical information for a SecureSphere Management Server from the GUI. For information on exporting technical information from SecureSphere Gateways architectures, see [Exporting Technical Information from Gateway](#).

To export technical information from Management Servers:

1. In the **Admin** workspace, select **System Performance > Management Server**.
2. From the top of the **Details** pane, click **MX Tech Info**.

SecureSphere prepares the information for download. Once complete a dialog box appears with a link. Click the link and download the zip file to the desired location. You can then mail it to Imperva support for analysis.

Troubleshooting Performance

By periodically reviewing the **System Performance** data, you can identify bottlenecks and other problems in your SecureSphere deployment. This section provides some guidelines to assist in troubleshooting, and includes the following:

- **Management Server**
- **Gateways**

Management Server

If your Management Server's performance seems excessive over a long time frame, identify the components that use the most CPU resources. Peaks in CPU usage are common and are to be expected, so you should investigate only consistent and prolonged high CPU usage.

Bear in mind that high CPU usage indicate nothing more than high traffic, so review the CPU usage at irregular intervals to determine whether there really is a problem or not.

Component showing consistently high CPU usage	
GUI	It may be that too many users are connected to the GUI at the same time.

Component showing consistently high CPU usage	
Monitoring	You should tune alerts.
Learning	Implement gradual learning.
Audit	It may be that you are trying to view too much audit data at the same time. Consider creating custom time frames that implement the fast view capabilities.
Jobs	These include reports, assessments, scans, archive, Active Modules, Lookup Data sets imports and purges. Consider scheduling these jobs to distribute the load evenly during the day/week. Alternatively, review the jobs to confirm that they are running on relevant data.
Gateway Updates	It may be that one or more of the Gateways is repeatedly requesting configuration. Look for errors in Setup > Gateways .
Followed Actions	These include emails, syslog actions, archive storage, SNMP and OS commands. Some of these may be CPU intensive because of file size or errors. Review the followed actions and identify these problems.

The values on the vertical scale of the **Machine Load Over Time** graph are the "processes" values returned by the Linux "top" command, defined as follows:

The total number of processes running at the time of the last update. This is also broken down into the number of tasks which are running, sleeping, stopped, or undead. The processes and states display may be toggled by the t interactive command.

Gateways

This Gateways window summarizes the resource usage of various SecureSphere components on a specific Gateway. This section review subjects regarding the Gateways window and includes the following:

- [System Performance - CPU Load](#)
- [System Performance - Policies](#)

- System Performance - Signature and Dictionaries
- System Performance - SSL

System Performance - CPU Load

If the CPU usage for a specific Gateway seems excessive over a long time frame, identify peak periods of CPU usage, narrowing the time frame.

In the **CPU Load By Service** graph, you can identify the services that incur the highest CPU loads. For each of these services, review the policies, signatures and dictionaries associated with the service to determine which of these might be responsible.

You may find, after investigation, that you need an additional Gateway, or that the Gateway needs to be upgraded to a more powerful model.

System Performance - Policies

For each service with a high CPU load, review the following:

- Identify the alerts that match the policy and rule of this service. If these are false positives, fine tune the policy or unapply it from the application(s), service(s) or server group(s) where the false positives occurred.

Consider also the possibility that the policy is not required.

- Alternatively, edit the policy by changing some of the criteria or unapply it from some services.

System Performance - Signature and Dictionaries

For each service with a high CPU load, review the following:

Component	Description
Security Signatures	Review the signature (Main > Setup > Signatures) and determine whether it can be safely disabled (ADC signatures) or modified (custom signatures).
SQL Dictionaries (including Stored Procedures & Privileged Operations)	In Main > Setup > Global Objects , review stored procedures and privileged operations and disable those which are not required.

Component	Description
Generic Dictionaries (including Sensitive Data Dictionaries)	Identify the policies using the dictionary, and edit or unapply them accordingly.
SQL injection / XSS	In Monitor > Alerts , review the most frequently-occurring SSL injection and XSS alerts, and create exceptions for those which are false positives.

System Performance - SSL

If SSL operations are consuming significant CPU resources, consider installing an SSL accelerator.

Alternatively, configure the Web server to reuse SSL sessions.

Inter-element Communication

This section describes the **Inter-element Communication** window, which enables SecureSphere to authenticate its elements using certificates generated by the management server. It includes the following:

- [Inter-element Communication Overview](#)
- [Activating Certificate-based Communication](#)
- [Working with Certificate-based Communication](#)
- [Registration Flows](#)
- [Maintaining Inter-element Communication](#)
- [Migration](#)

Inter-element Communication Overview

Inter-element communication between SecureSphere elements (SOM, MX, Gateways and Agents) is performed by default in password-based communication mode (i.e. using a username and password).

Inter-element communication can also be setup in a certificate-based communication mode, which allows SecureSphere elements such as SOM, MX, Gateways and Agents to communicate with each other using signed certificates for higher security. The certificates are signed by the Management server (SOM or MX) acting as a local Certificate Authority. Agents that cannot communicate using certificates continue to use password-based communication mode, and certificate-based communication mode is completely disabled for them.

In certificate-based communication, when two elements establish a communication channel, each of them sends its certificate to the other. Certificates of all trusted Certificate Authorities are preinstalled on the receiving element, which verifies the received certificate before allowing data exchange to proceed. Certificate-based communication mode is activated using a simple three-step wizard.

Activating Certificate-based Communication

To activate certificate-based communication perform the following tasks:

Enabling Certificate-based Communication Task Overview

	Task Overview	Description	For more information, see
1	Before you begin	Prerequisites required before enabling certificate-based communication.	Before You Begin

	Task Overview	Description	For more information, see
2	Run the Certificate-based Communication wizard	Enabling certificate-based communication.	Certificate-Based Communication Wizard

- [Before You Begin](#)
- [Certificate-Based Communication Wizard](#)

Before You Begin

Prior to enabling certificate-based communications you need to ensure that port 8085 is open between all MX and SOM machines, since SecureSphere uses this port for certificate-based communication.

Certificate-Based Communication Wizard

Changing the Inter-element communication mode to certificate-based communication is done using a simple three-step wizard as follows.



Note: Once activated, certificate-based communication mode cannot be changed back to password-based communication mode.

To activate certificate-based communication mode:

1. In MX, select the **Admin** workspace.
2. In SOM, from the Landing Page, click **System Setup**, or from Classic View, select the **Admin** workspace.
3. Click the **Inter-Element Communication** tab.
4. Click the **Let's start** button. The Welcome screen is displayed.
5. Click **Continue**. The Define Certificate Authority screen is displayed.
6. Type a name for your Certificate Authority.
7. Select the amount of years the signed certificates will be valid for.
8. Click **Continue**. The Review Configuration screen is displayed.
9. Verify the settings are correct. If not, click **Change Settings** to go back to the Define Certificate Authority screen and make your changes. Otherwise, click **Finish**.

Working with Certificate-based Communication

Once you start working in certificate-based communication mode, you can re-register your Gateways and Agents to the MX (and your MX to SOM, if using SOM) or register new elements.

Registration Flows

The following topics describe the registration flows between the Gateway and MX and the MX and SOM (if SOM is used) when certificate-based communication mode is active.

- [MX to SOM Registration Flow](#)

MX to SOM Registration Flow

To use SOM to manage certificate-based communication on MX, perform the following:

1. Activate certificate-based communication mode on SOM.
2. Activate certificate-based communication mode on MX.
3. Register the MX to SOM.

The following process describes the MX registration to SOM:

1. The MX and SOM synchronize their trusted CAs list and certificate Certificate Revocation List (CRL).
2. The MX generates and sends a CSR to SOM for an intermediate CA, which is then signed by SOM and returned to the MX.
3. The MX and SOM start communicating over port 8085. Communication is done securely by using SSL and bi-directional certificate enforcement.

Notes:



- SOM can work with an MX that is in certificate-based communication mode and with an MX that is in password-based communication mode
- SOM does not enforce a certificate-based communication mode connection on an MX that is in password-based communication mode
- An MX that is in certificate-based communication mode can be registered to a SOM that is in password-based communication mode. The communication between them is password-based communication mode (over port 8084). The MX keeps working in certificate-based communication mode with the elements registered to it
- In case of a failure, an "MX failed to establish certificate-based communication with SOM" alarm on the SOM is raised. For more information, see the **Working with Alarms** section in the **SecureSphere User Guide**.
- Once the MX and SOM are working in certificate-based communication mode, all the elements that are under the MX are added to the SOM UI and the MX UI is

changed to read-only

- The MX signs Gateways and Agents CSRs during registration using the intermediate CA. The purpose is not to spend time on forwarding requests to SOM so the MX will be able to sign in fast response
- When removing an MX from SOM, the MX generates a new CA called **defaultCA**

Maintaining Inter-element Communication

Views

Once you have certificate-based communication activated, you can continue to maintain your registered elements using the Inter-element Communication tab. Maintenance operations are divided into the following views:

- **Certificate Authorities** - Here you can view the Signing Certificate Authority and change the signing validity period, and any additional trusted certificate authorities. You can also export all certificate authorities to a file, by clicking the Export all Certificate Authorities button, in order to import to another MX or SOM.
- **Elements** - Here you can view, renew and revoke certificates. Additionally, you can filter and sort this view to show only desired information and in your preferred order.

Notes:



- If you are using SOM as part of your deployment, the Parent MX Name and Certificate Installation Status columns are available only in the SOM UI.
- If you are using SOM as part of your deployment, all maintenance operations are managed by your SOM and the MX is used for viewing only.
 - **Communication Dashboard** - This view is available in SOM only. Here you can see a graphical overview of your elements, which also includes a **View all** link that send you to the **Elements** view. In addition, you can see an overview of the certificate authorities, which also includes a **View all** link that send you to the **Certificate Authorities** view.

In addition, the following information is useful to know for the ongoing maintenance of Inter-element communication.

Certificate Validity

- The certificates signed by the signing CA are valid for the amount of years selected by the user.
- The certificates signed by the intermediate CA are valid for 30 days.
- An auto-renew job runs once a day and is responsible for renewing certificates that have a validity of less than

60 days.

- The validity check sends alarms in the following severity:
 - Validity less than 27 days - minor alarm
 - Less than 14 days - major alarm
 - When expired - the element is disconnected
- Users can renew a certificate from the MX UI. When using SOM, the renewal process is done from the SOM UI and the MX UI becomes read-only.
- In the renewal process MX sends the elements a request to send it a new CSR. When the renewal process is issued from SOM, the CSR "climbs" up to SOM.

Registration Status

- When the registration process is in progress and not completed, the certificate installation status indication in the UI is **Being installed**.
- When the registration process is completed successfully, the element status shows **Valid Certificate**.
- When an MX that is not in certificate-based communication mode is registered to a SOM, it receives a certificate generated by the SOM and starts using certificate-based communication.

Agents

- In certificate-based communication mode, agents can be authenticated based on certificates.
- There is no need to unregister and re-register the legacy agents.
- If an agent disconnects due to a certificate expiration, you must re-register the agent, otherwise, it will not connect again.

Migration

You can migrate existing environments to using certificates, and allow setting a mode in which unauthenticated communication is not be accepted by SecureSphere elements at all.

Notes:



- Migrating existing environments to certificate-based communication is possible for older SecureSphere elements that work in password-based communication only.
- Existing environments can start working in certificate-based communication mode without the need to recreate the entire environment.
- After starting Inter-element communication, there is no way back, but it will be possible to register elements from versions earlier than 13.0 to the system when in Non Enforce mode.

- [Enabling Migration](#)
- [Enforce and Non Enforce Modes](#)

- Working with SOM
- Upgrade

Enabling Migration

To enable migration:

1. On the MX UI, go to **Admin > Inter-element Communication** and click '**LET'S START!**'
2. On the SOM landing page, click **System Setup > Inter-element Communication**, or from Classic View, select the **Admin** workspace and then click **Inter-element Communication**.
3. Provide Certificate Authority name and certificate signing validity period.
4. Click **Finish**. The setup moves to **Trust Non-Enforced** mode that allows usage of self-signed certificates and passwords when establishing connections.

In this mode, if some elements failed to establish certificate-based communication (element is down, port is closed, etc.), the element's relevant status in the Trust Elements table is **No Valid Certificate**. Communication with this element continues to be password-based (as before activating Inter-element communication).



Note: In case an element had a valid Trust certificate that is no longer valid (expired/revoked), communication with this element will not be allowed and it will not be possible to switch it back to password-based communication.

When allowing unauthenticated channels, SecureSphere accepts gateways that are in FIPS and non-FIPS modes, and SSL/non-SSL communication between agents and gateways.

This mode is allowed in order to accommodate customers who want to avoid authentication by username and passwords, but also want to keep the same performance as without trust.

There is a banner at the top of the screen that shows the current Inter-element communication mode (Non-Enforced or Enforced) and a link enabling you to change it.

When clicking on the **Change** link, a dialog appears with an explanation of the current and new state. For example, for switching from Non-Enforced to Enforced.

Enforce and Non Enforce Modes

Inter-element communication works in one of two modes:

- **Enforce:** In this mode, all elements use certificates in all communications. All elements must have valid certificates. This also means that all channels are SSL. It is not mandatory for the gateway to be in FIPS mode for Inter-element communication to be in Enforce mode. This means that when the gateway is configured to work in non FIPS mode, the data channel and data sync communicate is in password-based mode.
- **Non Enforce:** In this mode, registering gateways is possible even when the certificate distribution to the

gateways fails.

Working with SOM

When working with SOM and it is in certificate-based communication mode, all Inter-element communication related activities (e.g. CA creation, certificate renew and revoke actions) are managed from SOM. All Inter-element communication related automatic jobs (certificate renewal, retry, etc.) run on SOM and not on MX.



Note: If an MX in certificate-based communication mode registers to a SOM not in certificate-based communication mode, control of the Inter-element communication status of this MX is done from the MX screen and not from SOM. This means it behaves like a standalone MX until SOM starts working in certificate-based communication mode.

Once Inter-element communication is activated on SOM, periodic SOM-MX sync takes place. You need to wait approximately 10 minutes for the first sync after starting Inter-element communication on SOM, in order for SOM to receive the information on all relevant elements from its MXs.

Upgrade

When upgrading a SecureSphere from version 12 and its sub versions or version 13.0 that has been setup to work in Inter-element communication to v13.1, note the following:

- When upgrading SOM:
 - Nothing changes regarding communication. The MXs that used to communicate with SOM using certificates go on doing so. The ones that did not, continue not to.
 - SOM moves to non-enforce mode
 - SOM screen shows the elements that have valid certificates
- When upgrading a gateway that is connected to an MX with version prior to 12.0 to version 13.1, nothing changes. The MX is not trust-capable and therefore so are its elements.
- When upgrading a gateway that is connected to an MX with version 12.0 to version 13.1, nothing changes regarding communication. The gateway communicates with the MX in certificate-based mode as it did before the upgrade.
- When upgrading a gateway and the MX it is connected to from a version prior to 12.0 or from version 12.0 to version 13.1:
 - The MX gets CSR requests from SOM and starts using Inter-element communication
 - The MX reports all elements to SOM
 - SOM starts Inter-element communication with the gateways of the MX (they are all at least 13.1)
 - SOM starts Inter-element communication with the agents of the MX that can communicate in certificate-based mode (version 12.0 and later)
 - All elements that can communicate in certificate-based mode receive certificates. Internally or when it's a standalone MX, MX changes to Non-Enforce. The screen shows all elements that communicate in certificate-based mode and have certificates.

- In order to start enforcing, make sure all the elements can work in certificate-based mode and click on the **Change** link.

Notes:

- After upgrade from version 12.0 that is in certificate-based mode, SecureSphere is in Non-Enforce mode even though it may be thought of as being in enforced mode prior to the upgrade. This is done for simplicity. You need to change to enforce after all elements support it.
- After upgrade from version 13.0 that is in certificate-based mode and Enforced mode, SecureSphere is in Enforce mode.

Gateway High Availability

The term "High Availability" refers to SecureSphere's ability to maintain connectivity and monitoring even when a SecureSphere appliance fails.



Note: Two Gateways in a High Availability deployment must be configured identically (same policies, objects, *etc.*)

This section describes the High Availability options for deploying SecureSphere, and includes:

- Layer 2 High Availability Deployments
- Layer 3 (Proxy) High Availability Deployments
- Other High Availability Deployments

Layer 2 High Availability Deployments

It is possible to deploy High Availability at layer 2 as follows:

- At the Network Level
- At the SecureSphere Level

At the Network Level

Examples of layer 2 High Availability deployments at the network level include:

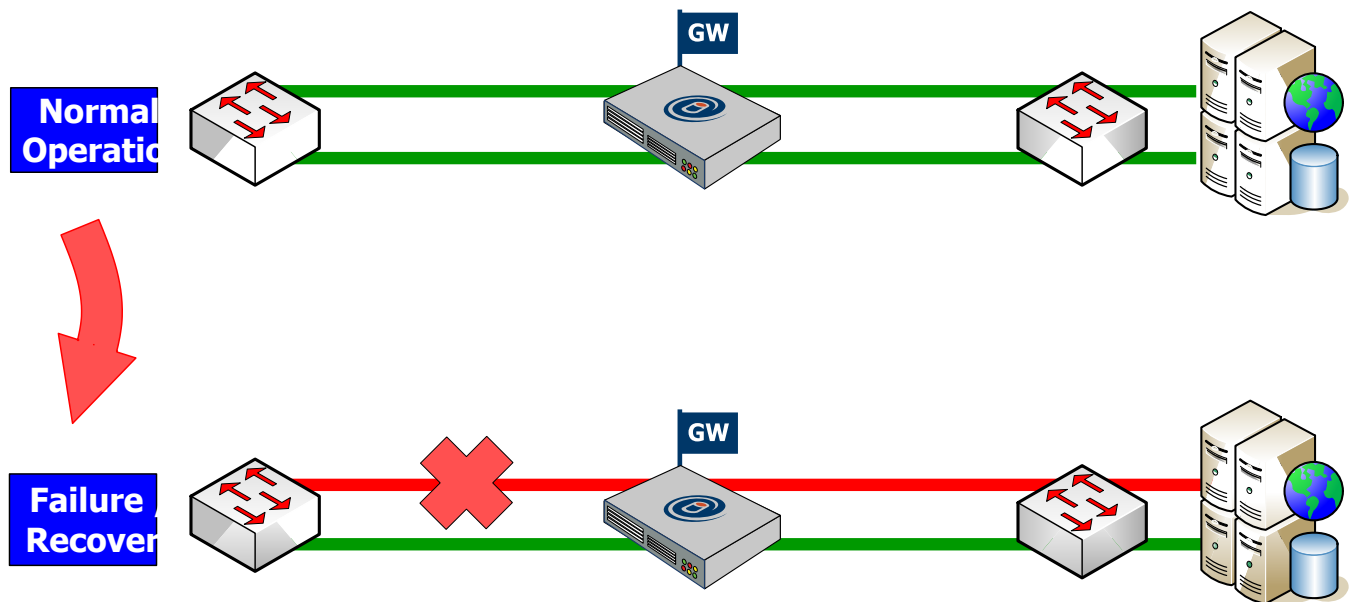
- Link Aggregation (Etherchannel)
- RSTP (Rapid Spanning Tree Protocol)

Link Aggregation (Etherchannel)

One or more SecureSphere Gateways can support link aggregation. For example, two interfaces on a SecureSphere Gateway can be connected to two interfaces on a network load balancer. Traffic can pass through both connections. For example, uplink traffic can travel through the first interface card while downlink traffic goes through the second interface card.

In bridge mode, each SecureSphere Gateway includes four network interfaces dedicated to application security. So it is possible to aggregate traffic through two connections. If the load balancer has 100Mbps Fast Ethernet

interfaces, then aggregating links can also increase total throughput. Even if the traffic is split between the two segments, SecureSphere can accurately inspect all traffic and block attacks.



If an Ethernet cable is disconnected or a network interface fails, then traffic will continue to be forwarded through the active link on the SecureSphere Gateway.

At the SecureSphere Level

Examples of layer 2 High Availability deployments at the SecureSphere level include:

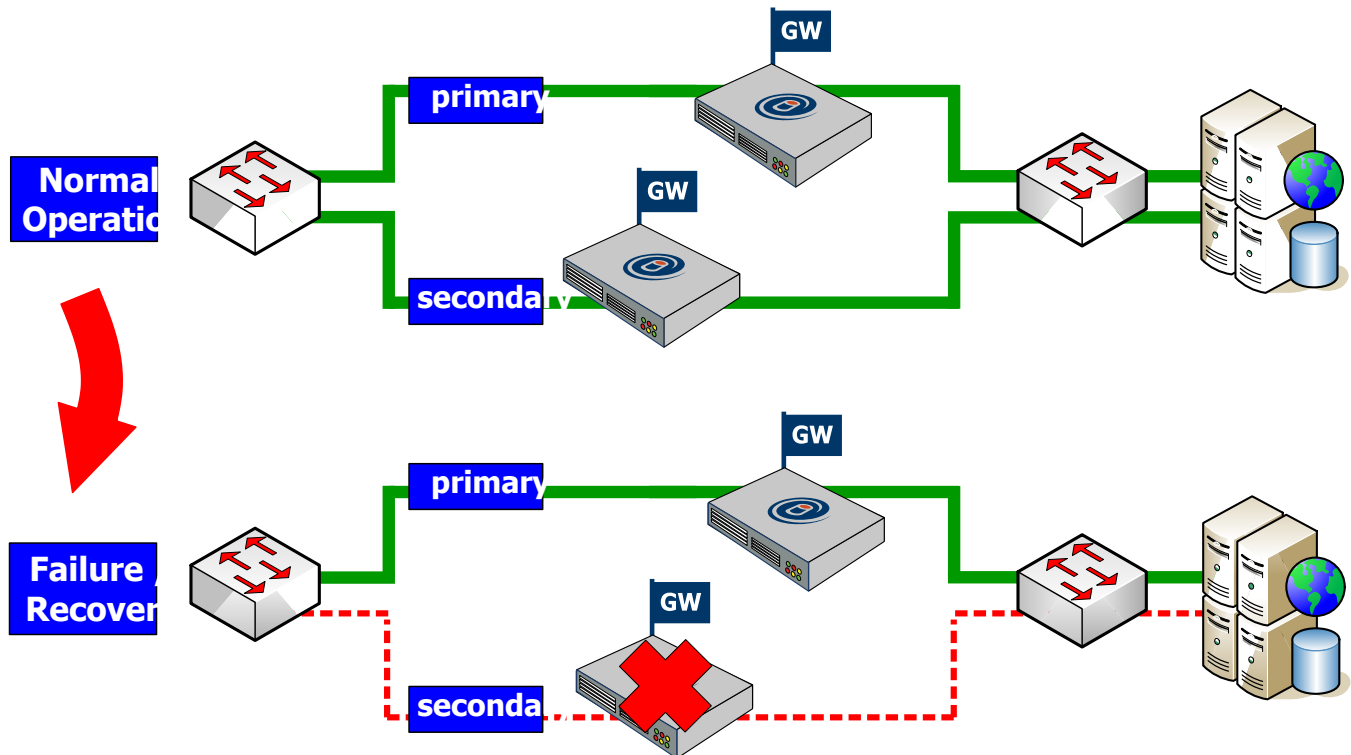
- IMPVHA Protocol — Active-Passive and Active-Active Failover
- Bridge STP

IMPVHA Protocol — Active-Passive and Active-Active Failover

IMPVHA is a proprietary redundancy protocol developed by Imperva for environments which do not have a redundant network architecture and STP cannot be implemented. For some environments, STP convergence times may be unacceptably long or STP may conflict with existing protocols, such as RSTP. In these situations, the Imperva High Availability (IMPVHA) protocol is the ideal solution.

With IMPVHA, two SecureSphere Gateways can be configured as layer 2 bridges in either active-passive or active-active mode. The configuration and failover mechanism is similar to VRRP, except that when a failover occurs, the backup device informs connected devices to redirect traffic via ARP requests rather than assuming the primary device's IP address. When an IMPVHA pair is deployed, the backup Gateway sends periodic loopcheck broadcast messages. If no response is received within the configurable failover time (default is 0.5 seconds), the backup Gateway becomes active.

Since a SecureSphere Gateway supports two bridges through its four network interface cards, it is possible to configure one bridge as the IMPVHA primary for one network and the second bridge as a backup for another network. In this configuration, two SecureSphere Gateways can support an active-active failover configuration.



When both gateways have two IMPVHA bridges, and a different bridge is configured as Primary on each gateway (as seen below), then it will be active - active (Relevant to IMPVHA only).



IMPVHA — Configuration

IMPVHA is configured for each SecureSphere Gateway using the `impcfg` command line utility. Configure each Gateway as Bridge IMPVHA with IMPVHA on.

For more information, see [Bridge IMPVHA Configuration](#).

Bridge STP

SecureSphere can be deployed in bridge mode in existing high-availability deployments, for example, STP. One of

the SecureSphere Gateways is designated as the primary and the other as its backup. Initially, the network directs traffic destined for the server to the primary.

If the primary Gateway fails, the failure appears to the network as a failure of a switch port. Connected devices automatically recognize that the failed Gateway is no longer reachable and then redirect traffic to the backup Gateway, which becomes the new primary Gateway.

If the previous primary is restarted, it becomes the backup.

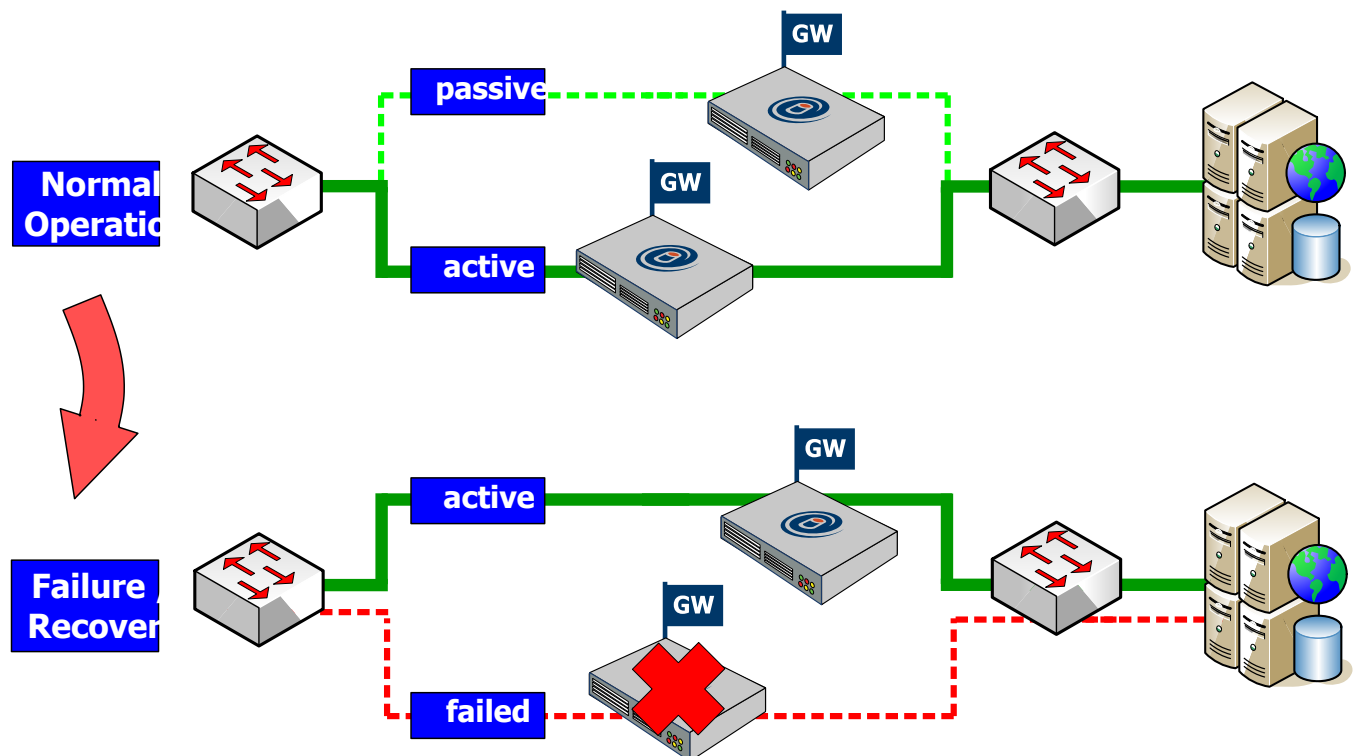
Because the two SecureSphere Gateways are deployed as bridges (that is, they have the same policies and the SecureSphere Management Server sees them as two active Gateways), TCP connections are negotiated directly between the client and the back-end server. A failover event does not disrupt TCP connections or affect user sessions.

When the SecureSphere Gateway works in Bridge-STP mode, it acts like any other network device that is part of the STP. This means that if the STP priority of the Gateway is lower than the priority of the Switch, the Gateway becomes the STP root. By default, the STP priority of the SecureSphere Gateway is 32768. Therefore, in order to make sure that the SecureSphere Gateway does not become an STP root, set it to IMPVHA mode, or configure the switch STP priority to a lower number.

For more information, see [Configuring SecureSphere](#).

Bridge STP — Active-Passive Failover

In this deployment, only the SecureSphere Gateway configured as the primary is active. The backup SecureSphere Gateway is in standby mode, waiting to take over if the first SecureSphere Gateway fails.

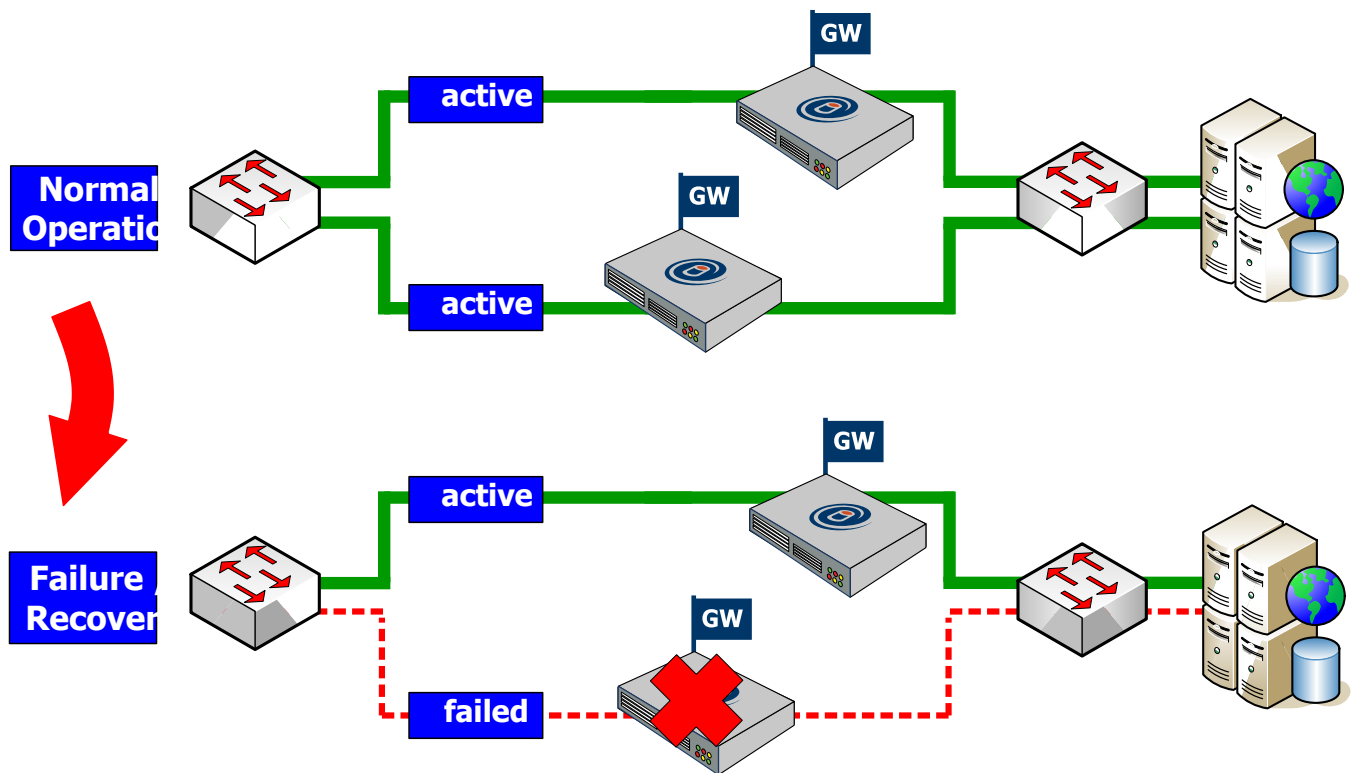


This deployment is somewhat inefficient because at any given time, one of the SecureSphere Gateways is idle — it is functioning properly but is not processing any traffic.

This deployment can also be used with transparent reverse proxies.

Bridge STP — Active-Active Failover

This deployment is the same as the previous one, except that both SecureSphere Gateways are active and monitor traffic simultaneously. If one of them fails, all traffic flows through the other Gateway. Total network capacity is reduced until the failure is corrected.



This deployment supports load sharing if sessions are sticky, that is, the same session always passes through the same Gateway.

When configuring such a solution, ensure that each Gateway is able to handle the bandwidth of both Gateways, should one of them fail.

Redundant Architecture (STP) — Configuration

If STP or a similar capability is available in the network, configure the Gateways (with `impcfg`) as either:

- Bridge IMPVHA with IMPVHA off
- Bridge STP with STP on

For more information, see [Configuring SecureSphere](#).

Layer 3 (Proxy) High Availability Deployments

It is possible to deploy High Availability at layer 3 as follows:

- At the Network Level
- At the SecureSphere Level



Note: Proxy deployments are relevant for Web traffic only. For more information, see [Topologies and Deployment](#).

At the Network Level

SecureSphere is often deployed as a direct replacement for legacy reverse proxy appliances. In many cases, administrators choose to configure SecureSphere as a bridge because it fits into the existing architecture and reduces the operational burden associated with Web proxies. However, in some cases, administrators deploy SecureSphere as in proxy mode to meet architectural requirements or for content modification.

The SecureSphere Web Application Firewall supports both reverse proxy and transparent proxy configurations. Both proxy configurations offer content modification, including cookie signing and URL rewriting. Transparent proxy mode enables transparent deployment without network or DNS changes.

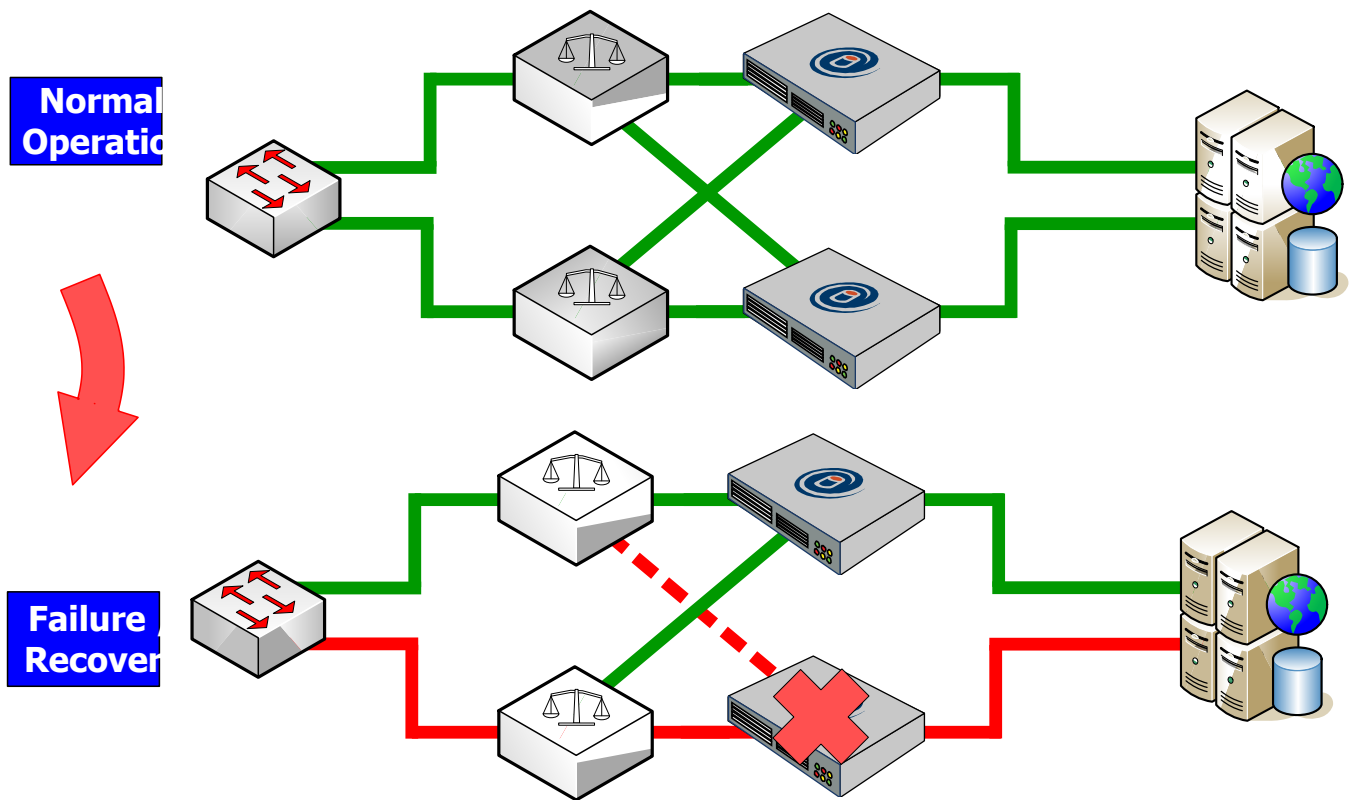
Examples of layer 3 High Availability deployments at the network level include:

[Load Balancer Integration](#)

Load Balancer Integration

In proxy mode, one or more SecureSphere Gateways are deployed as either a Reverse Proxy or a Transparent Reverse Proxy. A load balancer balances the traffic between the Gateways and the Gateways forward the traffic to the Web servers. It is possible to configure load balancer to work in active-active or active-passive modes.

In this mode, the IP address of the Web site resolves to the SecureSphere Gateway's IP address. If SSL is used, then the Gateway should be configured to terminate the SSL connection. Note that in proxy mode, SecureSphere establishes a new connection to the Web server on behalf of the client.



At the SecureSphere Level

Examples of layer 3 High Availability deployments at the SecureSphere level include:

- Redundant Architecture (VRRP) — Configuration

Redundant Architecture (VRRP) — Configuration

VRRP is suitable for a deployment where the SecureSphere Gateway has an IP address, that is, it is not configured as a bridge or as a transparent reverse proxy, but as a non-transparent reverse proxy

To configure a SecureSphere Gateway for VRRP, configure the aliases as virtual. For more information, see [Configuring Aliases](#).



Note: If you encounter an extended period of downtime on VRRP failover, Imperva recommends configuring both the virtual routers as Backup (not Master).

Other High Availability Deployments

Other High Availability deployments include:

- Fail-Open Bridge
- Sniffing Gateway High Availability

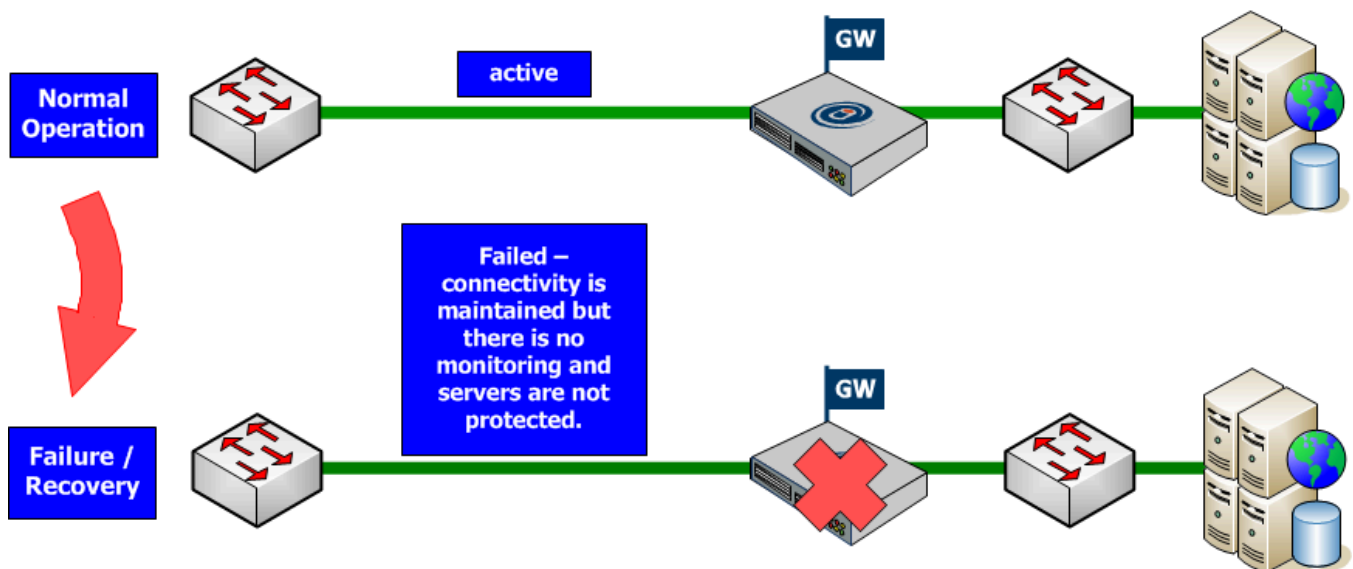
Fail-Open Bridge

Fail-open configuration preserves uptime without the cost or management effort of building and maintaining a redundant architecture. In the event of an appliance failure, SecureSphere's specialized network interface card detects the failure and physically completes the connection through the SecureSphere Gateway. The NIC card will fail open in milliseconds, minimizing network downtime.

Fail open works when :

- Appliance reboots
- During patch installation
- During upgrade
- Gateway is down
- Gateway appliance fails

Though connectivity is maintained, application servers will not be monitored or protected from attacks until the SecureSphere Gateway resumes operations.



Note: A SecureSphere Gateway deployed in sniffing mode provides a similar level of connectivity continuity in the event of failure: traffic continues to flow through the network but it is not monitored.

Fail Open — Configuration

Physical Gateways (not VM Gateways) are configured as fail open by default. When you first connect the Gateway appliance to the network, no sites or server groups are defined, so no policies are implemented, nothing is monitored or blocked, and all traffic is passed through the appliance.

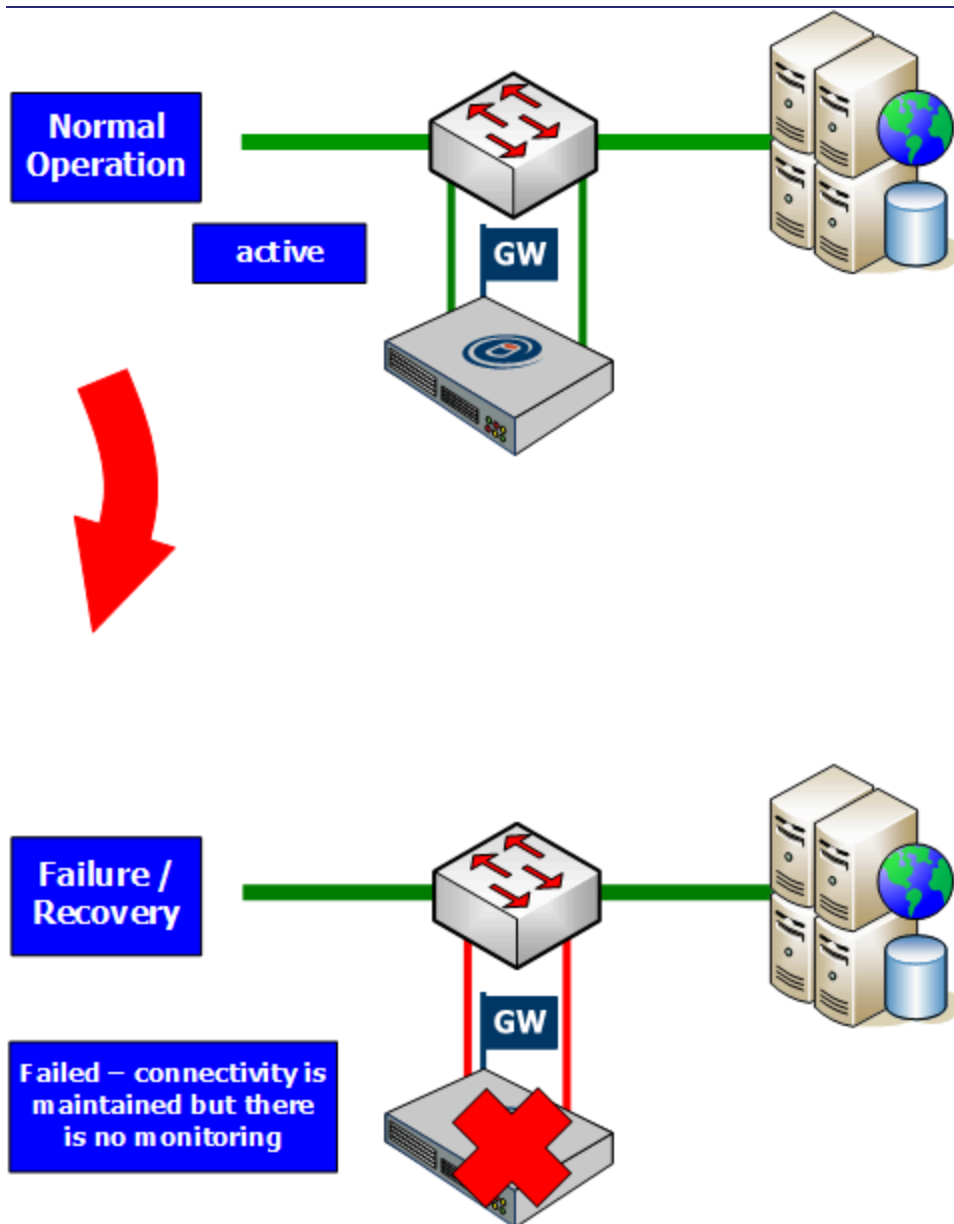
You can change the mode using the `impctl` command line utility. For more information, see [Configuring SecureSphere](#).

Sniffing Gateway High Availability

Configuring SecureSphere as a sniffing Gateway enables organizations to protect critical servers without introducing a new point of failure. It is a lower price alternative to deploying two or more redundant SecureSphere Gateways.

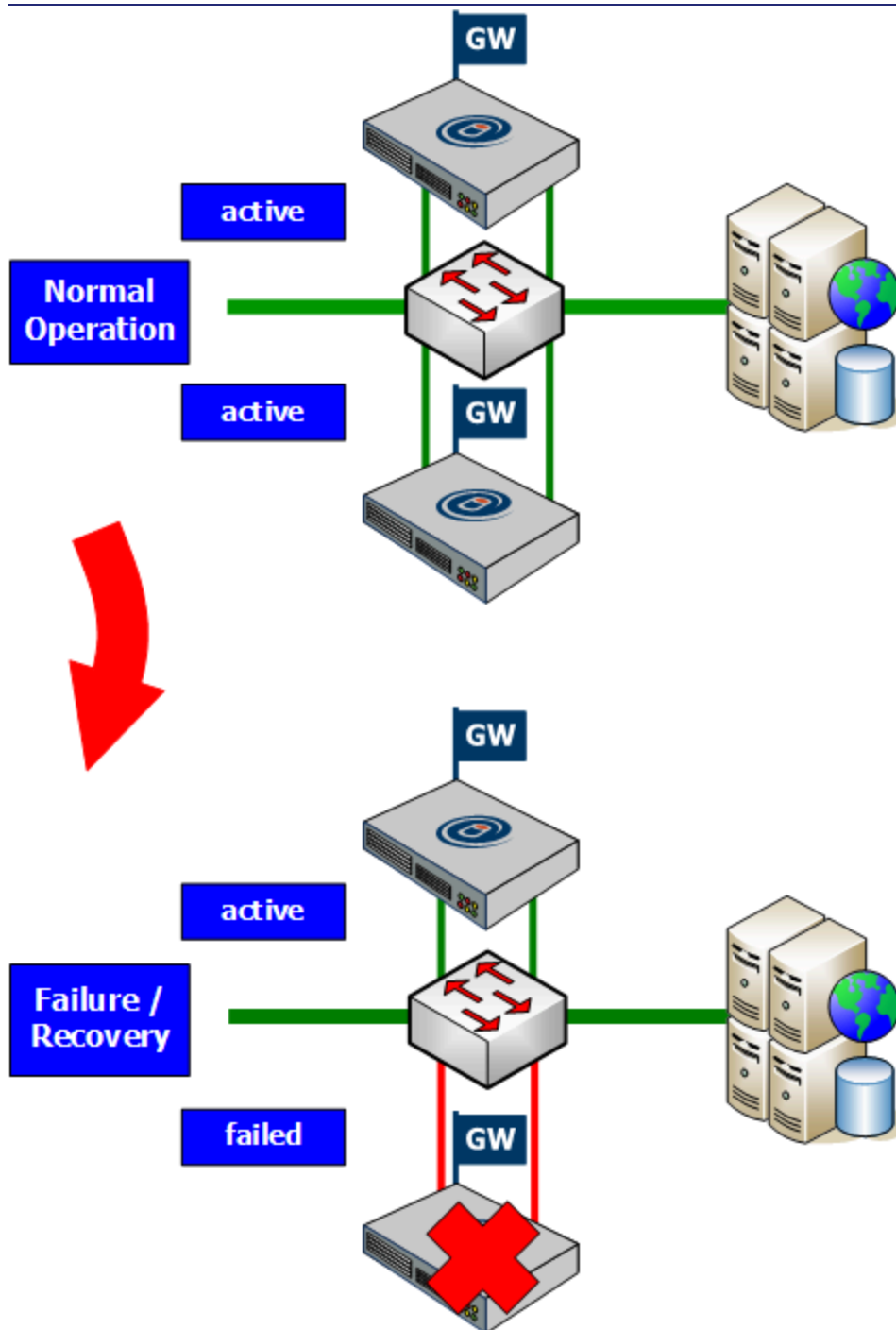
To deploy SecureSphere as a sniffing Gateway, connect it to an open SPAN port or a network TAP to observe traffic. A second connection to an active port on the switch allows SecureSphere to block malicious connections by sending TCP resets to the server.

In the event of a SecureSphere Gateway failure, network traffic is unaffected, but it will not be monitored or protected until the SecureSphere Gateway becomes operational again.



For some organizations, it may be imperative to monitor and audit all Web traffic without disruption. These organizations can deploy multiple SecureSphere Gateways in non-inline mode. If one of the Gateways fail, the other Gateway will continue to monitor and record traffic.

In this configuration, each Gateway should be managed by a separate Management Server.



Bridge IMPVHA Configuration

As stated, gateway high availability is a model of adding a redundant gateway to network protection. High Availability means: one of the gateways is active and performs all gateway tasks, and the other is standby and waits for the order to “jump in”. When SecureSphere gateway operates in a single gateway mode and an exception or system failure occurs, there are two possible scenarios: all internet traffic passes through the gateway un-inspected or there is no connectivity to the servers.

Imperva's Gateway HA is implemented in several ways:

- Bridge STP
- Bridge IMPVHA
- Reverse Proxy (Hades) - VRRP
- Router Mode – VRRP.

This section provides information on Bridge IMPVHA configuration. IMPVHA mode operates similarly to STP mode. It is designed to be used in networks where STP:

- Is not supported by the switches.
- Is too complicated to implement.
- Is in use with Rapid Spanning Tree Protocol (RSTP).

Bridge IMPVHA Overview

After boot, a SecureSphere gateway initializes as a secondary (i.e. each bridge initializes as a secondary). Each secondary bridge is defined with a priority that is calculated based on the bridge devices' MAC addresses. Since there is no primary bridge defined, the bridge defined with the highest priority becomes the primary bridge.



Note: Once the primary bridge is defined, it is not redefined as a secondary bridge due to priority changes. This means that the primary bridge may not always be the bridge with the highest priority.

Under normal conditions, a secondary bridge sends a special ethernet broadcast packet (loop check packet) to the switch, which in turn sends it to the primary bridge. The primary bridge then sends the packet to the second switch, which again forwards it back to the secondary bridge creating the loop. If the secondary bridge gets its packet back, it knows that the primary bridge is alive and it remains as a secondary. If the packet does not reach the secondary bridge, it assumes that there is no primary. The secondary bridge then drops all packets it receives including the loop check packets so that no harmful endless network loop is created between the switches.

When a secondary bridge becomes a primary bridge, it has to tell the switch that all the traffic should be passed to it. This is done by sending ARP requests to both sides of the bridge requesting the IP addresses of all the defined server groups. The ARP response goes back to the primary bridge and pass through it to the switch on the other side that in turn learns the source MAC of the server group on the port to which the new primary bridge is connected.

To configure the gateway in Bridge IMPVHA mode:

1. In the **Gateway Management** screen (as described in [Configuring a Gateway](#)), select the option **Change operation mode**.
2. Select the option **Bridge IMPVHA**. The **Gateway Management** screen appears.
3. Select the option **Manage interfaces and routes**. The **Bridges Configuration** screen appears.
4. Select the option **Toggle high-availability on a bridge**. The **Gateway Management** screen appears.
5. Type **t**. The **impcfg Top** screen appears.
6. Type **a**. The changes are applied.

Notes:

- When gateway HA needs to be configured in Bridge IMPVHA mode, both the individual gateways must be configured as Bridge IMPVHA individually.
- In the MX GUI, a gateway group under **Setup->Gateways** with "HA Fail Mode" needs to be created and the two gateways need to be a member of this group.
- When configuring the gateway using the `impcfg` utility, you need to enable the Bridge IMPVHA option on each gateway and apply it to `br0`, `br1` or both.

Management Server High Availability (MX-HA)

This section describes High Availability options for SecureSphere Management Servers (MXs), and includes:

- [Management Server High Availability \(MX-HA\) Overview](#)
- [Management Server High Availability \(MX-HA\) Implementation](#)
- [Co-Locating Management Servers with MX-HA](#)
- [Management Server High Availability \(MX-HA\) Components](#)
- [Before Installing Management Server High Availability \(MX-HA\)](#)
- [Installing Management Server High Availability \(MX-HA\)](#)
- [After Installing Management Server High Availability \(MX-HA\)](#)
- [Monitoring Management Server High Availability \(MX-HA\)](#)
- [Maintaining Management Server High Availability \(MX-HA\)](#)

Management Server High Availability (MX-HA) Overview

SecureSphere is comprised of two main components, the Gateway and the Management Server (MX). Real time protection is the primary task of the Gateway, whereas the MX's main roles are to configure the system, collect events from the Gateway, analyze them and display resulting alerts.

The SecureSphere Gateway is a mission-critical component — if a Gateway fails, the Web and database servers behind the Gateway are no longer protected. On the other hand, if a SecureSphere MX fails, protection is not interrupted.

An MX failure which continues for an extended period of time impacts the level of protection which the SecureSphere Gateways provides, for the following reasons:

- **Profiling:** Application Profiling is performed by the MX based on data collected from the gateways.
- **Policies and Profiles:** No changes can be made to existing profiles or policies.
- **Alerts:** The gateway passes events to the MX, which stores and analyzes them and displays alerts derived from the events. If connectivity to the MX is down, the gateway saves events locally for later transmission, but the gateway eventually overwrites older events as available disk space is exhausted. When connectivity to the MX is restored and the gateway transmits its store of accumulated events to the MX, the MX will have incomplete data. Also, alerts normally pushed to predefined recipients via email now remain unnoticed.

For these reasons, it is important that there is redundancy in the system so that if your MX fails, its functions can be continued by another machine. This redundancy is known as Management Server High Availability (MX-HA). This configuration means setting up an additional management server to back up the primary management server, ready to take on the primary management server's functions in case of a failure. The backup management server is also known as the MX-HA machine.

Notes:

- In an MX-HA environment, there are many configuration changes which cannot be performed using the command line tools `impcfg` and `impctl`. For more information, see [Command Line Tools and MX-HA](#).
- You cannot configure an MX-HA environment with a Large Scale MX.
- MX HA requires purchasing two license files (one for each server).

Management Server High Availability (MX-HA) Implementation

The MX-High Availability capability uses redundant MXs in a fully-automatic active/standby failover implementation, requiring no user interaction. The system senses the loss of the active MX, and the failover mechanism begins. In the field the typical failover time period (which depends on the size of the system, and time elapsed since the last scheduled database synchronization) has been under measured as under 10 minutes, and the maximum time encountered was 30 minutes.

- The HA configuration comprises two MXs in the active/standby mode. Although only one MX is active (i.e., managing the SecureSphere system), both MXs are running to allow database synchronization (using a standard Oracle process).
- Database synchronization is a continuous asynchronous process. This means that whenever the active MX fails, there may be some data on its disk that are not mirrored on the standby MX.
- A heartbeat function (Linux) enables close monitoring of the active MX. During a server or network failure, the heartbeat function triggers the failover process and dispatches system logs (syslogs) to warn users that the active MX requires attention.

Co-Locating Management Servers with MX-HA

MX-HA is supported both for collocated MXs (same Data Center) or MXs located in different Data Centers, there are however two constraints that needs to be taken into account:

- Both MX use the same IP when becoming active (floating IP), therefore they must reside on the same LAN or Virtual LAN, so in case of remote Data Centers, the VLAN must go across the WAN/Internet.
- A minimum of 50Mbps network capacity should be available for the MX-HA protocols to function and converge reliably and in timely manner.

Note also the prerequisites in [Before Installing Management Server High Availability \(MX-HA\)](#).

Management Server High Availability (MX-HA) Components

The following components are included in the Management Server-High Availability solution:

- Linux Heartbeat
- Oracle Standby Database (Data Guard)
- SecureSphere Server
- HA Health Check
- Imperva Watchdog

Linux Heartbeat

Linux heartbeat is a Linux utility which enables implementing active-passive configurations. The heartbeat is the heart of MX-HA that is responsible for the database and MX and where they run.

The heartbeat configuration includes the following resources:

- Virtual IP address (VIP)
- Database server
- SecureSphere server

Once the heartbeat is started, all the resources are allocated to a specific MX.

The heartbeat checks the status of the resources every minute. If one of the resources fails to run on the MX, all resources are restarted on the same MX. If the restart is not successful, failover is initiated.

Oracle Standby Database (Data Guard)

Database synchronization is based on Oracle standby database (or Data Guard). Once MX-HA is installed, the database on the primary MX is configured to support Oracle standby database. The database on the secondary MX is deleted and copied from the primary server as a standby database.

After the installation, the standby database is constantly synchronized with the primary database using Oracle solution.

If the standby database fails to synchronize with the primary, it is recreated from the primary database.

SecureSphere Server

The SecureSphere Management Server is one of the resources in the heartbeat, but it is not modified during the MX-HA installation.

HA Health Check

In order to verify that everything is up and running, there is a health check mechanism that checks both servers. The heart of this mechanism is the `healthCheck.sh` script.

Imperva Watchdog

The Imperva watchdog script checks the heartbeat status, and if it is invalid, it restarts the heartbeat. The watchdog checks the heartbeat only after a successful start of the MX-HA using `impcctl`. If MX-HA is stopped manually, it is removed from the watchdog's watch list.

Before Installing Management Server High Availability (MX-HA)

This section reviews items that need to be taken before installing the Management Server in High Availability (MX-HA) mode, and includes the following:

- Hardware and Software Requirements
- Ports Required for MX-HA
- Pre-Installation Tasks

Note: The installation of the MX-HA takes a long time. It is possible that due to issues on your workstation or on the network, the SSH connection to the MX will be disconnected.



To overcome the consequences of an SSH connection loss, Imperva recommends running the MX-HA installation commands inside the screen utility, which is supplied on the MX and gateway appliances. This allows the installation scripts to continue working even if the controlling terminal is disconnected.

You can find a short usage tutorial of the screen utility on <https://docs.imperva.com/bundle/z-kb-articles-knowledgebase-support/page/290194466.html>.

Hardware and Software Requirements

In MX-HA, the private network that links the two MX servers is used to constantly copy incremental database and file changes. Additionally, during installation and occasionally at other times, all database data will need to be transferred on that network. While there is no minimum required bandwidth, it is recommended that the network connection between the two be reliable and quick, and fast enough to handle the required amount of traffic for your deployment.

Management Server High Availability (MX-HA) requires a specific hardware and software configuration that includes:

- Two MXs - MX-High Availability cannot be installed on Gateways.
- You must have licenses for both servers.
- Both Management Servers in an MX-HA pair need to be the same appliance model (for example, both MXs in the MX-HA pair need to be M160).
- Physical and virtual hardware models cannot work together in MX-HA. For example, the Virtual Management Server VM150 cannot work together with the Physical Management Server M160 in MX-HA.
- **Both servers must have the same amount of memory (RAM) and disk space.** If this requirement is not met, the installation will fail.
- Both servers must have the same version of SecureSphere installed.
- Each server must have the following two network interfaces:
 - An interface for public network.
 - An interface for interconnect (using an ethernet crossover cable if connecting directly or using another cable as required when connecting through another appliance) to the other server. This interface is used by Linux heartbeat and by Oracle to synchronize the standby database.

Ports to Open for MX-HA

In order for MX-HA to synch between Management Servers part of the MX-HA environment, make sure the following ports are open between the two Management Servers:

- **Ping:** ICMP
- **SSH:** 22 TCP
- **Oracle:** 1521 TCP
- **Heartbeat:** 5405 UDP
- **Conf:** 443 https

Pre-Installation Tasks

This section lists the pre-installation tasks that need to be conducted before installing the Management Server in High Availability Mode and includes the items listed below.



Note: The public IP addresses of both MX servers must be on the same subnet. If the servers are located at different sites, there must be a VLAN between the sites with the same subnet.

- Install the Latest Patch
- Configure the Interconnected Interfaces
- Test the Interconnected Interfaces
- Download the RPMs and Prepare the Primary MX

Install the Latest Patch

Patching a deployment is conducted when you want to install an updated patch of the same major version. If you want to upgrade from one major version to another, please see the [Upgrade Guide](#) for the version you want to upgrade to.

Notes:

- When Patching On-Premises, you should patch in the following sequence.
 - SOM (if relevant)
 - Gateways
 - Management Servers
- For instructions on Patching AWS, see the AWS Installation Guide
- It is recommended to back up Management Servers before applying a patch. For more information, see **Backing up the System** in the Upgrade Guide

1. Download the latest patch (and corresponding MD5 or SHA-256 file):
 - a. From the [Imperva Customer Portal](#), click **Downloads**.
 - b. Navigate to `/Imperva_WAF/Updates/v15/v15.3/Patch#/` for the relevant version.

Note: Starting with v14.1 the same file is used both to patch the On-Premises product and to upgrade it. The file is referred to as **SecureSphere-WAF-Update**.

2. In Linux, run one of the following commands (Windows requires a third-party verification program):
 - `sha256sum -c <binary file name>.sha256`
 - `md5sum -c <binary file name>.md5`
3. Copy the patch to `/var/tmp` of the machine you want to patch
4. Change permissions by typing `chmod +x [filename]`

For example: `chmod +x SecureSphere-WAF-Update-Q3_21-14.4.0.16_0.39028.x`

5. Run the patch by typing `./[patch filename]`

For example: `./SecureSphere-WAF-Update-Q3_21-14.4.0.16_0.39028.x`

6. Reboot the machine only after receiving the message that the patch has been successfully installed.
7. Using the web GUI, download the ADC content file from the Imperva site and upload it to the system (Admin -> ADC).

Configure the Interconnected Interfaces

You must define the interconnected network during the first-time login procedure on the MX by setting up the LAN interfaces. However, if this was not done during the first-time login, you can configure the LAN interfaces later using `impcfg`. (For more information about this command, see `impcfg`.)

Pre-requisites:

- Both interconnected interfaces must be on the same subnet.
- LAN and Management interfaces must not be on the same subnet.

To configure the interconnected interfaces on the MX, perform the following on both MXs:

1. Run `impcfg`.
2. In the **Top** Screen, select **2) Manage platform**.
3. In the **Platform Management** Screen, select **1) Manage network**.
4. In the **Network Management** Screen, select **2) LAN interface**.
5. Configure these options:
 - **Set device**
 - **Set IPv4 address**
6. Return to the **Top** screen by selecting option **t**.
7. Apply the settings by selecting option **A**.
8. Confirm the new settings by selecting option **C**.

The interconnected interfaces on the MXs are configured.

Test the Interconnected Interfaces

To test the interconnected interfaces:

1. Ping from the first MX to the second MX on its internal interface.

The ping will fail, because the MX blocks ICMP by default.

2. Immediately after the ping, execute `arp -a` and locate the arp entry of the other server.

If the arp entry has a valid MAC address for the other server, the interconnect was successfully configured.

Download the RPMs and Prepare the Primary MX

Perform the following:

1. On the primary MX, login as user `root`.
2. Execute the following commands:

```
mkdir /var/tmp/mxha
```

```
cd /var/tmp/mxha
```

3. Run the command `impctl platform show` and verify that **asset-tag** indicates the assigned asset identifier. For example: `asset-tag VM150`
4. Verify you are logged in to the [Imperva Customer Portal](#).
5. Go to [Downloads](#).
6. Navigate to the directory **Imperva_WAF/Setup/v15/v15.3/patch<num>/MX-HA**.
7. Download the **oracle-19c_EE-2.x86_64.rpm** files in the MX-HA directory to the `/var/tmp/mxha` directory.

Installing Management Server High Availability (MX-HA)

MX-HA is installed on an already functioning Management Server (MX). You should have two MXs, one of which you designate as primary and the other as secondary.

To install MX-HA:

1. Perform the pre-installation tasks (see [Pre-Installation Tasks](#)).
2. Upload licenses for both the primary MX and the secondary MX to the primary MX.

Alternatively, you can upload only the primary MX license, install MX-HA on the primary MX and install the secondary license (on the primary MX) at a later time. This alternative is not recommended, because it temporarily leaves the configuration without a license for the secondary MX, but it can be done.

3. Login to the primary MX as user `root`.



Note: The installation is performed only on the primary MX. The secondary MX is installed as part of the installation of the primary MX, and any data on the secondary MX is deleted.

4. On the primary MX, execute the following command, entering parameters as listed in the table **Installation Parameters** below.

```
impctl server ha install
```

Notes:



- After the `impctl server ha install` command finishes, it will take the secondary MX several minutes to synchronize with the primary MX. During that time, the `impctl server ha status` command will produce misleading results.
- You will have to manually perform the hardening on the secondary MX. See below for instructions on how to do this.

Installation Parameters

Parameter	Description	Default Value
Directory for temporary data	The directory in which temporary files will be written during the installation process.	<code>/var/tmp/secsph-ha</code>
Keep alive IP address (pingable server)	This IP address is used to check the network status. You can use any IP address on the management network as long as it remains constant, active and pingable. If the keep alive address does not respond to the ping, a failover occurs.	
Secondary server public IP address	The public address of the secondary MX.	
Virtual server IP address	The IP address used to access the MX-HA servers.	
Heartbeat interface name	The interface which is the direct connection between the two MXs.	
Public interface name	The management interface name.	



Note: You have the option of exporting the database before the installation. You can export the database before installing MX-HA, or optionally, during the MX-HA installation.

Hardening

During MXHA installation, the installation process opens hardening between the MXs. Opening the hardening on the secondary MX must be performed manually by the user, so during the installation process, the commands that must be run on the secondary MX are displayed on the screen.

At this stage, you must do the following:

1. Open an SSH session to the secondary MX.

2. Copy the displayed commands.
3. Paste them into the SSH window and execute them.
4. These commands can be safely executed, even if they were already executed in the past. However, if they were already executed in the past, there is no need to run them again.

After Installing Management Server High Availability (MX-HA)

This section reviews what can be done after Management Server High Availability has been installed and includes the following:

- [Verify the Primary Server is Active](#)
- [Register the Gateway](#)

Verify the Primary Server is Active

Once the installation is completed, verify that the primary server is active and running.

To verify that the primary server is active and running:

1. Execute the following command on both the primary and secondary servers:

```
impctl server ha status
```

For information about the output of this command, see [impctl server ha status Output](#).

Register the Gateway

After installing MX-HA, the Gateway must be (re)registered with MX-HA.

To register the Gateway:

1. Configure the `admin-ips` section in the `bootstrap.xml` file located under `/opt/SecureSphere/etc/` by adding the following xml part at the end just before the `</bootstrap>` entry on each of the gateways:

```
<admin-ips>

<admin-ip ip="server-real-ip1"/>

<admin-ip ip="server-real-ip2"/>

</admin-ips>
```

Substitute the real IP addresses (inside double quotes) of each of the two servers for `server-real-ip1` and `server-real-ip2s`.

2. Re-register the gateway using the virtual IP address (VIP) of the MXs. For more information, see [Changing the Virtual IP Address](#).

To uninstall MX-HA:

To uninstall MX-HA, the primary server must be available, otherwise, there is no guarantee that the uninstall process will successfully complete.

1. Login to the primary MX as user **root**.
2. Execute the following command:

```
impctl server ha uninstall
```

Note: SSH trust and the `/etc/hosts` entries are not deleted by this command.

Once the uninstall process is complete, the primary MX functions normally as a single MX. The secondary MX is unusable, and its database must be recreated using the following command: `impctl db create`

Monitoring Management Server High Availability (MX-HA)

This section includes the following:

- MX-HA Status
- [Confirming that MX-HA is Correctly Configured](#)
- MX-HA Logs
- Heartbeat Behavior
- Known Issues

MX-HA Status

This section reviews MX-HA status and includes the following:

- `impctl server ha status` Command
- `impctl server ha status` Output

impctl server ha status Command

The **impctl** command displays the status of MX-HA. The command syntax is as follows:

```
impctl server ha status
```

This command returns information about whether the MX-HA is started and on which server the resources are running. The output of this command returns the server that holds each resource (IP, database and server) even if only some resources are up.



Note: When you run this command on a secondary MX, it shows the status of the primary MX.

impctl server ha status Output

The output of the `impctl ha status` command will be similar to one of the following:

- If the secondary MX is synchronized:

```
started,ip-on-mxha1,db-on-mxha1(secondary-synchronized),server-on-  
mxha1,running
```

- If the secondary MX database is down or inaccessible:

```
started,ip-on-mxha1,db-on-mxha1(secondary-status-unknown),server-on-  
mxha1,running
```

- If the secondary MX is not synchronized:

```
started,ip-on-mxha1,db-on-mxha1(secondary-synchronization-lag:9m-26s),server-  
on-mxha1,running
```

- If the secondary MX is down:

```
started,ip-on-mxha1,db-on-mxha1(secondary-unavailable),server-on-  
mxha1,running
```

Confirming that MX-HA is Correctly Configured

After configuring MX-HA, you may wish to confirm that the configuration is correct rather than to wait for the primary MX to fail. This section describes the following:

-
- Initiating a Failover in an MX-HA Environment
 - Confirming That the MX-HA Servers are Synchronized

Initiating a Failover in an MX-HA Environment

This section describes how to perform a safe and successful failover in MX-HA. A failover is sometimes required for testing.

To initiate a failover:

1. Execute the following command on both MXs:

```
impctl server ha status
```

If HA is running on both MXs, the output on each MX should resemble the following:

```
started,ip-on-mxha1,db-on-mxha1(secondary-synchronized),server-on-  
mxha1,running
```

2. Confirm that the MXs are synchronized (see [Confirming That the MX-HA Servers are Synchronized](#)).
3. Initiate a failover by performing one of the following:
 - On the primary MX, stop the primary MX by executing the following command:

```
impctl server ha stop
```

or,

- On either MX, execute the following command:

```
impctl server ha failover
```

4. Wait until the secondary MX becomes active (that is, it has become the primary MX). You can determine whether it is active by using the following command:

```
impctl server ha status
```

- If the secondary MX is now the primary MX, that is, if failover has successfully occurred, the output should resemble the following:

```
started,ip-on-mxha2,db-on-mxha2(secondary-synchronized),server-on-  
mxha2,running
```

At this point, the MXs have switched roles:

- The MX that was formerly the secondary MX is now the primary MX, and its GUI is now usable.
 - The MX that was formerly the primary MX is now the secondary MX.
5. If you initiated the failover with the `impctl server ha stop` command, then in order to restore the HA functionality, proceed as follows:

-
- Start the new secondary MX (the MX that was previously the primary one) by executing the following command:

```
impctl server ha start
```

If you initiated the failover with the `impctl server ha failover` command (as in step 3), then the secondary MX will automatically be started and HA functionality restored. There is nothing you need do.

In either case, the secondary MX must perform a full synchronization with the primary MX, and will not be ready for another failover until the synchronization is complete.

To determine when the standby MX has re-synchronized and is ready for failover, follow the instructions in [Confirming That the MX-HA Servers are Synchronized](#) below.

Confirming That the MX-HA Servers are Synchronized

This procedure describes how to verify the status of an MX HA server.

To verify the status of an MX-HA server:

1. Login as root to the secondary MX.
2. Check the last entries in `server_ha_debug.log`, usually located in `/opt/SecureSphere/server-ha/log` and confirm that:
 - the date and time of the entries are recent
 - the status is "Database is standby"
 - The lag is either 0 or the databases are synchronized

For example, in `server_ha.log`, the output might be:

```
[12/11/12 14:58:06 healthCheck] - Secondary database is fully synchronized
```

MX-HA Logs

This section reviews MX-HA logs and includes the following:

- [MX-HA Components Writing to the Logs](#)
- [Logs in the /opt/SecureSphere/server-ha/log Directory](#)
- `alert_secsph.log`
- [Heartbeat Statuses](#)
- [Resource Status](#)
- [Server Status](#)
- `impctl watchdog`

MX-HA Components Writing to the Logs

The following are the main components that write to the logs:

- `ha-dbor` is responsible for managing the database. `Ha-dbor` writes to the log when the heartbeat is requesting to stop, start or get status of the database server. This component runs only on the primary server.
- `ha-secsph` is responsible for managing the SecureSphere server. `Ha-secsph` writes to the log when heartbeat is requesting to stop, start or get status of the SecureSphere MX. This component runs only on the primary server.
- `healthCheck` is responsible for the database health on both servers. This component checks if the server is the primary or secondary, and according to the answer verifies that the database is in the correct status. If needed, this component runs other components to recreate and standby database or changing the standby database to primary.
- `syncDirs` is responsible for synchronization of directories and files from the primary server to the secondary server, this is done using `rsync` utility and according the `rsync.conf` file. This component runs only on the secondary server.
- `watchdog`: every time the watchdog checks the heartbeat status, it writes information to the log file about the current status and whether it is going to restart the heartbeat or not.

Logs in the `/opt/SecureSphere/server-ha/log` Directory

- The `server_ha.log` file contains overview information, mostly changes in status.
- The `server_ha_debug.log` file contains detailed information file about the heartbeat, resources and transitions between the servers.

Information from `ha-dbor`, `ha-secsph` and the `watchdog` is also written to this file.

`alert_secsph.log`

This file is the Oracle database log, and contains information from Oracle database, including information relevant to the standby solution. The log is located under the `/opt/oracle/diag/rdbms` directory, assuming that `/opt/oracle` is the Oracle installation directory.

Heartbeat Statuses

MX-HA uses the Linux heartbeat solution. Part of the heartbeat is the Cluster Resource Manager (CRM), which takes care for the group of resources and handles starting, stopping and moving them between the servers.

MX-HA uses `/usr/lib/heartbeat/crm_resource` to determine where the resources are running and the status of each server.

Resource Status

To determine where a resource is running, MX-HA uses `crm_resource -W -r <resource>`, where `<resource>` is one of the following:

- **PrimaryGroup:ip_resource**, **PrimaryGroup:OracleDB**
- **PrimaryGroup:SecureSphere**

The output of this command is the server name that holds the resource, or a message if the resource is not running or nothing if the heartbeat is down.

The `impctl server ha status` command uses the `crmadmin` to list the owners of the resources.

The `healthCheck` script uses `crmadmin` in order to decide which server is the primary, that is, the server that holds the IP resource.

Server Status

To determine the server status, MX-HA uses the `crmadmin -S <hostname>` command. The output of this command is the current status of the server.

The two most important statuses are `S_IDLE` and `S_NOT_DC`. Both statuses mean that the server is idle, either primary or secondary. Other statuses, such as `S_ELECTION`, `S_TRANSITION_ENGINE`, `S_STARTING` and so on, meaning that the server is performing a task, such as starting, stopping, change status from primary to secondary and so on.

impctl watchdog

The `watchdog` is responsible for health checking of the management components. In a regular management configuration, the `watchdog` verifies that the database and the SecureSphere server are working properly.

Under MX-HA configuration, the `watchdog` command does not check the database or the server. It checks only the heartbeat.

Every time the `watchdog` runs, it checks the heartbeat status and the owner of the IP resource. If the status is not idle or if `crmadmin` returned nothing, it means that everything is functioning properly.

The `watchdog` restarts the heartbeat, if the heartbeat is idle and but the IP address is not running on both nodes.

Heartbeat Behavior

This section reviews heartbeat behavior and includes the following:

- [Failover Sequence](#)
- [Standby Recreation](#)

Failover Sequence

The heartbeat uses the internal scripts `ha-dbor` and `ha-secsph` to verify the status of the IP address, database and SecureSphere server. If one of the resources has an invalid status, the heartbeat stops higher resources. If IP address is down, both database and SecureSphere server are stopped. If the database is down, only the SecureSphere server is stopped.

Then, the heartbeat tries to start the failed resource again. If the start is successful, heartbeat continues to start all other dependent resources. If the start operation fails, heartbeat performs failover and moves all the resources to the other server.

During a failover, the heartbeat stops all the resources on the primary server (server A) and starts them on the other server (server B). At this time, server A is in the `TRANSITION` status. When server B finishes starting all the resources, server A status becomes `S_IDLE`, and since the database on server A is not standby, it is recreated.

When the process completes, server A is still in the `S_IDLE` status, which is invalid for the secondary server. At this time, the watchdog restarts the heartbeat on server A and the entire environment is valid again.

If during the failover from server A to server B, one of the resources could not start on server B, the heartbeat stops everything on both servers. At this point, the `watchdog` command restarts the heartbeat on both servers.

In addition, failover occurs when the primary server reboots or the `server-ha` service is stopped.

Standby Recreation

The standby database is constantly being synchronized with the primary database. There are several situations in which the standby database is recreated. This means that the database is deleted, and is copied from the primary database, as follows:

- The standby database has a lag higher than a certain time. The time is configured in the `bootstrap.xml` file, and its value is 10 minutes.
- The healthcheck process cannot check the synchronization time.
- The database is not a valid standby. For example, after a failover, the new secondary server contains a primary database and not a standby database. In this case, the database on the new secondary server is recreated.

The standby database can be recreated only if the primary database and server are running. Otherwise, the standby writes a message to the `healthCheck.log` and is not recreated.

Known Issues

This section reviews MX-HA known issues and includes the following:

- [MX-HA Installation Failure](#)
- [Failure While Building the Standby Database](#)
- [Management Server \(MX\) Replacement](#)

MX-HA Installation Failure

A failure of the MX-HA installation procedure does not invoke an uninstall process. Before proceeding, you must "clean up" the unsuccessful installation on the primary server and re-initialize the database on the secondary server.

1. On the primary server, execute the following command:

```
impctl server ha uninstall
```

2. On the secondary server, execute the following command:

```
impctl db start
```

Failure While Building the Standby Database

In very rare cases, the standby database needs to be recreated. The creation process can take some time, and during this time the standby database is not operational and cannot become the primary database. If there is an unrecoverable failure on the primary server while the standby is recreated, MX-HA cannot overcome this failure. In some cases, a manual intervention can solve the problem, but in other cases there is no solution.

For example, if the database files are deleted from the primary server during the standby recreation, there is no valid database to use, and MX-HA or manual operations cannot start the system.

If this occurs, contact Imperva support.

Management Server (MX) Replacement

Currently, there is no supported option to install only a secondary server in the MX-HA solution. If the secondary server crashes and needs to be replaced, replace it using the following procedure.

To replace a MX-HA secondary server:

1. Uninstall MX-HA from the primary MX. For more information, see [Register the Gateway](#).
2. Connect and configure the new secondary MX, including all prerequisites. For more information, see [Before Installing Management Server High Availability \(MX-HA\)](#).
3. Reinstall MX-HA from the primary MX. For more information, see [Installing Management Server High Availability \(MX-HA\)](#)

Maintaining Management Server High Availability (MX-HA)

In an MX-HA environment, it is not possible to perform any appliance-level actions on the MX using the command line tools `impcfg` and `impctl`, except for the ones included in this section.



Warning: `impcfg` and `impctl` will let you make changes, and you will not receive an error message, but the MX and the MX-HA environment may become non-operational and you will have to re-install MX-HA.

If you need to perform any changes other than the ones included in this section, you must uninstall MX-HA on both the primary and secondary MXs, make the changes and then reinstall MX-HA.

This section includes the following:

- Stopping the Entire MX-HA Environment
- Starting the Entire MX-HA Environment
- Changing the Physical IP Addresses
- Changing the Virtual IP Address
- Adding a Static Route in an MX-HA Environment

Stopping the Entire MX-HA Environment

The following procedure performs a complete stop of the MX-HA heartbeat function on both the primary and secondary MX units. This procedure can be run from either the primary or secondary MX unit and first stops the secondary and then the primary MX unit.

To stop the entire MX-HA environment:

1. Execute the following command on both MXs:

```
impctl server ha status
```

If HA is running on both MXs, the output on each MX should resemble the following:

```
started,ip-on-mxha1,db-on-mxha1(secondary-synchronized),server-on-
mxha1,running
```

2. Execute the following command on either MX units:

```
impctl server ha stopall
```

Starting the Entire MX-HA Environment

The following procedure performs a complete start of the MX-HA heartbeat function on both the primary and secondary MX units. This procedure sets the MX unit it is run from as the primary MX unit.

To start the entire MX-HA environment:

1. Execute the following command on the MX unit you want set as primary:

```
impctl server ha startall
```

The message `Startall` will start this MX as the primary, continue? [y/n] is displayed.

2. Type `Y` to continue or `N` to abort.

If you want to skip the above confirmation message, execute the command:

```
impctl server ha startall --quiet
```

Changing the Physical IP Addresses

The following procedures change the Public and/or LAN (heartbeat) IP addresses of the primary and/or secondary MX units. The procedures describe how to change the Public and LAN IP addresses of one MX unit.



Note: Whenever you change IP address(es) (Public and/or LAN) on one MX unit, you must perform the change on the other MX unit using the same new IP address(es), but opposite parameters. For example, changing the Public IP on one MX unit is performed using the `local-public-ip` parameter. You must perform this change on the second MX as well using the `remote-public-ip`.

To change the Public IP address:

1. Execute the following command on both MXs:

```
impctl server ha status
```

If HA is running on both MXs, the output on each MX should resemble the following:

```
started,ip-on-mxha1,db-on-mxha1(secondary-synchronized),server-on-mxha1,running
```

2. Execute the following command on either MX units:

```
impctl server ha stopall
```

3. Execute the following command on the MX unit you want to change the IP address of.

```
impctl server ha change-ip --local-public-ip=<local-public-ip>
```

where `<local-public-ip>` is the new IP address in the format IP/cidr.

4. Execute the following command on the second MX unit.

```
impctl server ha change-ip --remote-public-ip=<remote-public-ip>
```

where <remote-public-ip> is the new IP address (the one from step 3) in the format IP/cidr.

5. Execute the following command on the MX unit you designate as primary:

```
impctl server ha startall
```

To change the LAN IP address:

1. Execute the following command on both MXs:

```
impctl server ha status
```

If HA is running on both MXs, the output on each MX should resemble the following:

```
started,ip-on-mxha1,db-on-mxha1(secondary-synchronized),server-on-mxha1,
running
```

2. Execute the following command on either MX units:

```
impctl server ha stopall
```

3. Execute the following command on the MX unit you want to change the IP address of.

```
impctl server ha change-ip --local-heartbeat-ip=<local-heartbeat-ip>
```

where <local-heartbeat-ip> is the new IP address (the one from step 3) in the format IP/cidr.

4. Execute the following command on the second MX unit.

```
impctl server ha change-ip --remote-heartbeat-ip=<remote-heartbeat-ip>
```

where <remote-heartbeat-ip> is the new IP address in the format IP/cidr.

5. Execute the following command on the MX unit you designate as primary:

```
impctl server ha startall
```

Advanced Examples

- If you want to change the Public IP addresses of both MX units, perform the following procedure:

- a. On both MX units, run `impctl server ha status`

If HA is running correctly you should see an output similar to this: `started,ip-on-mxha1,db-on-mxha1(secondary-synchronized),server-on-mxha1,running`

- b. On either MX unit, run `impctl server ha stopall`

- c. On the first MX unit, run `impctl server ha change-ip --local-public-ip=<local-public-ip> --remote-public-ip=<remote-public-ip>` where <local-public-ip> is the

new Public IP address of the first MX unit in the format IP/cidr and `<remote-public-ip>` is the new Public IP address of the second MX unit in the format IP/cidr.

- d. On the second MX unit, run `impctl server ha change-ip --local-public-ip=<local-public-ip> --remote-public-ip=<remote-public-ip>` where `<local-public-ip>` is the new Public IP address of the second MX unit in the format IP/cidr and `<remote-public-ip>` is the new Public IP address of the first MX unit in the format IP/cidr.
- e. On the MX unit you designate as primary, run `impctl server ha startall`

- If you want to change the LAN IP addresses of both MX units, perform the following procedure:

- a. On both MX units, run `impctl server ha status`

If HA is running correctly you should see an output similar to this: `started, ip-on-mxha1, db-on-mxha1 (secondary-synchronized), server-on-mxha1, running`

- b. On either MX unit, run `impctl server ha stopall`
- c. On the first MX unit, run `impctl server ha change-ip --local-heartbeat-ip=<local-heartbeat-ip> --remote-heartbeat-ip=<remote-heartbeat-ip>` where `<local-heartbeat-ip>` is the new LAN IP address of the first MX unit in the format IP/cidr and `<remote-heartbeat-ip>` is the new LAN IP address of the second MX unit in the format IP/cidr.
- d. On the second MX unit, run `impctl server ha change-ip --local-heartbeat-ip=<local-heartbeat-ip> --remote-heartbeat-ip=<remote-heartbeat-ip>` where `<local-heartbeat-ip>` is the new LAN IP address of the second MX unit in the format IP/cidr and `<remote-heartbeat-ip>` is the new LAN IP address of the first MX unit in the format IP/cidr.
- e. On the MX unit you designate as primary, run `impctl server ha startall`

- If you want to change the Public and LAN IP addresses of both MX units, perform the following procedure:

- a. On both MX units, run `impctl server ha status`

If HA is running correctly you should see an output similar to this: `started, ip-on-mxha1, db-on-mxha1 (secondary-synchronized), server-on-mxha1, running`

- b. On either MX unit, run `impctl server ha stopall`
 - c. On the first MX unit, run `impctl server ha change-ip --local-public-ip=<local-public-ip> --remote-public-ip=<remote-public-ip> --local-heartbeat-ip=<local-heartbeat-ip> --remote-heartbeat-ip=<remote-heartbeat-ip>` where `<local-public-ip>` is the new Public IP address of the first MX unit in the format IP/cidr and `<remote-public-ip>` is the new Public IP address of the second MX unit in the format IP/cidr and `<local-heartbeat-ip>` is the new LAN IP address of the first MX unit in the format IP/cidr and `<remote-heartbeat-ip>` is the new LAN IP address of the second MX unit in the format IP/cidr.
 - d. On the second MX unit, run `impctl server ha change-ip --local-public-ip=<local-public-ip> --remote-public-ip=<remote-public-ip> --local-heartbeat-ip=<local-heartbeat-ip> --remote-heartbeat-ip=<remote-heartbeat-ip>` where `<local-public-ip>` is the new Public IP address of the second MX unit in the format IP/cidr and `<remote-public-ip>` is the new Public IP address of the first MX unit in the format IP/cidr and `<local-heartbeat-ip>` is the new LAN IP address of the second MX unit in the format IP/cidr and
-

<remote-heartbeat-ip> is the new LAN IP address of the first MX unit in the format IP/cidr.

- e. On the MX unit you designate as primary, run `impctl server ha startall`

Changing the Virtual IP Address

The following procedure changes the virtual IP address of the primary and secondary MX units. This procedure should be run only once from either the primary or the secondary MX units. A server and database restart is automatically performed at the end of the procedure.

To change the Virtual IP address:

1. Execute the following command on both MXs:

```
impctl server ha status
```

If HA is running on both MXs, the output on each MX should resemble the following:

```
started,ip-on-mxha1,db-on-mxha1(secondary-synchronized),server-on-mxha1,
running
```

2. Execute the following command on either MX units:

```
impctl server ha change-vip --vip=<IP/CIDR>
```

where <IP/CIDR> is the new IP address in the format IP/cidr.

Adding a Static Route in an MX-HA Environment

This procedure describes how to add a static route on the MX in an HA environment.

To add a static route in an MX-HA Environment:

1. Login to the MX via the CLI .
2. Run the following command, replacing the <> with your details:

```
impctl platform network route config --context=<context> --type=<type> --
address=<ipaddress> --gateway=<gatewayipaddress> --device=<interface>
```

For example:

```
impctl platform network route config --context=platform --type=host --
address=10.10.10.0 --gateway=192.168.10.100 --device=eth0
```

Network High Availability

In addition to high availability provided by both the Management Server and the Gateway, the network itself can also be used to provide high availability. This option relies on a mechanism which detects when an opposing network interface is down, and subsequently takes the operating interface on the opposite side of the network link offline to prevent data from being sent to an inoperative interface.

Once the problematic network interface becomes available, the SecureSphere appliance gives the operating interface (which was manually taken down) a period of time to come up and stabilize. If it does not succeed, then the original interface is taken back offline. This is in order to prevent a situation in which the interfaces get involved in an ongoing pattern of going up and down. The amount of time given for the interface to come back up is configurable by editing a supporting file.



Note: This chapter describes how to achieve high availability through the network. It detects link status and enables HA only if it is network related. It does not facilitate HA when an issue is software related. For information on how to provide HA in these cases, see [Management Server High Availability \(MX-HA\)](#) or [Gateway High Availability](#).

This chapter includes the following:

- [Network High Availability Support](#)
- [Network Interface Card \(NIC\) Bypass](#)
- [Configuring Network High Availability](#)

Network High Availability Support

Network high availability is supported in bridge mode (STP / Impvha). It is not available in Reverse Proxy or sniffing modes. The following table lists the interface types and their support for network high availability.

Interface Support for Network High Availability

Interface Type	G-Series	X-Series
Copper	Supported	Supported
1 G	Not Supported	Supported
10 G	Not Supported	Not Supported

Interface Type	G-Series	X-Series
10 G with Bypass Support	Not Supported	Supported

Network Interface Card (NIC) Bypass

The SecureSphere gateway implements a bypass mechanism on bridge ports that are configured in fail-open mode.

During normal system operation, the gateway inspects traffic and the bypass state is OFF. When the gateway malfunctions or the system is rebooted or powered down, the bypass state is turned to ON.

The bypass mechanism is based on a hardware watchdog. When the gateway is up and running, it triggers the hardware watchdog holding the bypass at OFF state. When the gateway malfunctions, it stops sending triggers and the hardware watchdog switches the bypass relays to ON. The gateway triggers the hardware watch dog every one second, and the hardware watch dog timeout is set to six seconds. Once the gateway fails to send six consecutive triggers the bypass is switched to ON.

The bypass switching time between ON and OFF states is up to 500 milliseconds.



Note: The SecureSphere gateway in most deployments is placed between two network switches. When the NIC switches between bypass ON and OFF states (up to 500 milliseconds), the network switches require additional time to gain a stabilized traffic pass through that depend on each network switch configuration.

Configuring Network High Availability

You can configure the amount of retries, and subsequently the period of time that is provided for the operating interface to come back up.

To modify the number of retries for an interface:

1. Navigate to the following location on the SecureSphere appliance:

```
opt/SecureSphere/etc
```

2. Access the following file:

```
hades.cfg.template
```

3. Modify the following parameter:

`bridge_relief_count_[#]`

where [#] is the value for the number retries you want SecureSphere to attempt before it manually takes the interface offline.

Default: 5 retries (10 seconds)

Please note that each retry is made after 2 seconds. So the overall amount of time an interface is given to come back up is the number of retries multiplied by 2.

EXAMPLE:

If you set the parameter to: **bridge_relief_value_1**

It attempts 1 retry after two seconds, or give the interface 2 seconds to come online until the original interface is taken down again.

If you set the parameter to: **bridge_relief_value_6**

It attempts 6 retries two seconds apart, or give the interface 12 seconds to come online the original interface is taken down again.

Network Services

This appendix describes various network configuration issues related to deploying SecureSphere, and includes:

- SecureSphere Component Communication
- Configuring Firewall Ports
- Serial Console Access to SecureSphere

SecureSphere Component Communication

The table **SecureSphere Component Communication** below lists the protocols through which the various SecureSphere components communicate with each other.

SecureSphere Component Communication

Source	Destination	Protocol	Destination Port	Remarks
Gateway	MX	HTTPS	TCP 8083	If the Gateway and MX server have a dedicated network (either with a cross cable or through a dedicated switch) then there is no need to open these ports on the firewall.
MX	Gateway	SSH	TCP 22	
MX	Gateway	HTTPS	TCP 443	
MX	SOM	HTTPS	TCP 8084	Used for SOM / MX Communication.
SOM	MX	HTTPS	TCP 8084	
Gateway / SOM	MX	HTTPS	TCP 8085	Used for certificate communication between the SOM, MX and Gateway as part of Trusted Connections.
SOM	MX	HTTPS	TCP 8085	
MX	www.imperva.com	HTTPS	TCP 443	Allows firewall access for ADC updates, automatic signature updates, and

Source	Destination	Protocol	Destination Port	Remarks
				software updates.
MX	*.impervacloud.com	HTTPS	TCP 443	Allows firewall access for all ThreatRadar updates. For users requiring a higher level of granularity on what domains (destinations) to allow, see the table Granular Network Access for Threat Intelligence Communication below.
Gateway	*.impervacloud.com	HTTPS	TCP 443	Allows firewall access for ThreatRadar features that need to obtain additional threat intelligence required by the feature including Account Takeover (ATO) and Fraud Prevention Services (FPS).
Gateway	syslog server	syslog	UDP 514 TCP 514	Used to send audit if configured. For more information on using action interfaces to send audit, see the section Gateway Syslog > Log ... audit events in Configuring Action Interface Parameters .
MX	syslog server	syslog	UDP 514	
Desktop	MX	HTTPS	TCP 8083	
Desktop	MX	SSH	TCP 22	
MX	DNS server	DNS	TCP 53 UDP 53	

Source	Destination	Protocol	Destination Port	Remarks
Gateway	DNS server	DNS	TCP 53 UDP 53	DNS is required to resolve host names defined in SecureSphere policies.
MX / Gateway	NTP server	NTP	UDP 123	NTP is needed to guarantee accuracy of timestamps.
Desktop	console.imperva.com	HTTPS	TCP 443	Allow client access to Unified Management Console UI (Attack Analytics console)
Gateway	ses-uploader.service.imperva.com	HTTPS	TCP 443	Allow Gateway to upload data to Attack Analytics cloud servers
MX	entitlement.service.imperva.com	HTTPS	TCP 443	Allow MX to get service entitlements (for Attack Analytics & Flex Protect)
MX	auth.service.imperva.com	HTTPS	TCP 443	Allow MX to authenticate with cloud servers (for Attack Analytics & Flex Protect)
AWS ELB/ALB	Any Gateway in the scaling group	HTTP / HTTPS	TCP 541	The port on which the ELB/ALB execute a health check on the Gateway.
MX	thr-auth.service.imperva.com	HTTPS	TCP 443	Allows firewall access for ThreatRadar feature

Granular Network Access for Threat Intelligence Communication

This table lists specific domains which can be used when wanting to configure granular access per domain. While it is recommended to use the asterisk wildcard for access to these domains, users wishing to configure access per domain can use the below in place of the asterisk as listed for the MX ThreatRadar updates listed above.



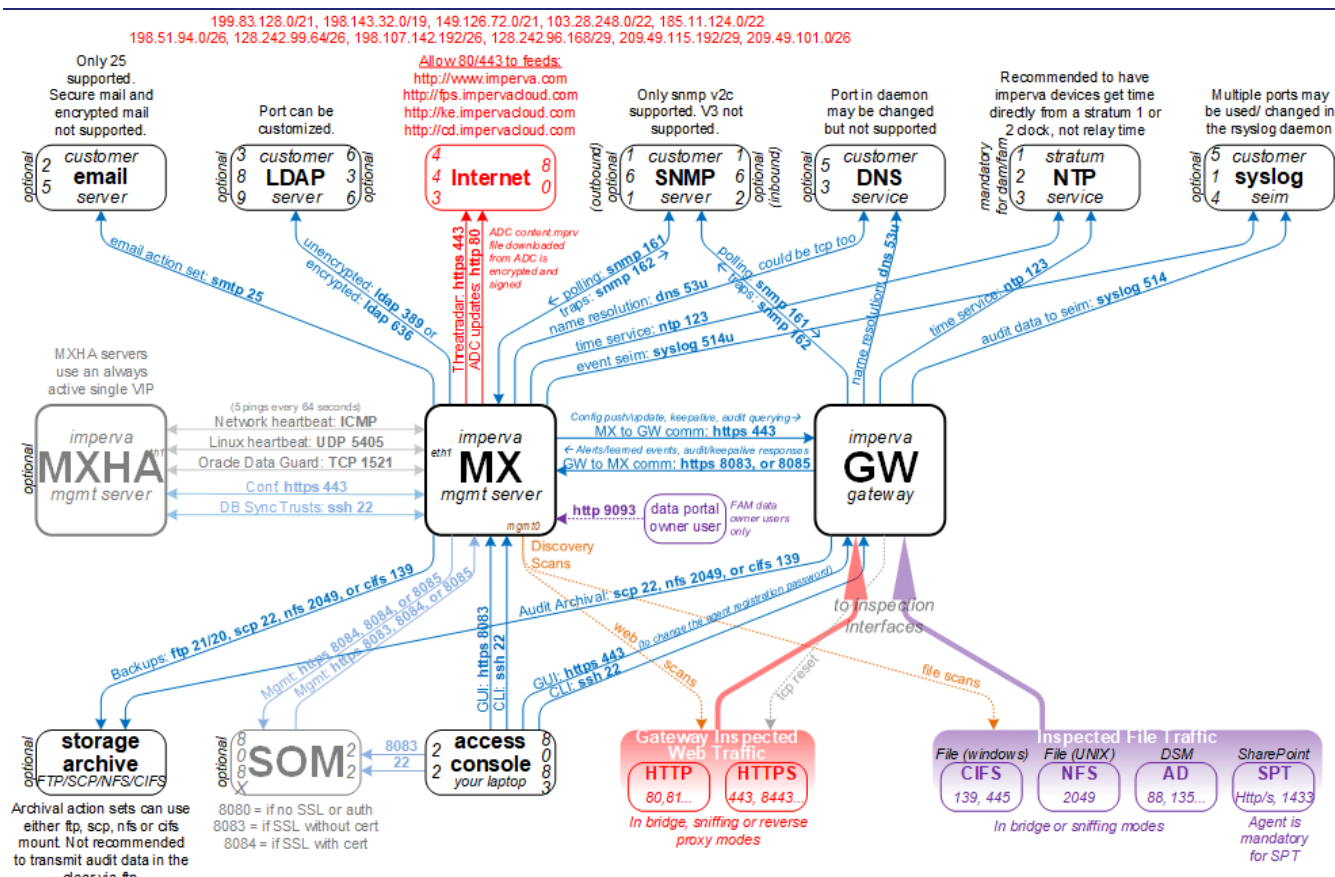
Note: If using the below domains, you **should not** use the general configuration for the MX *.impervacloud.com shown in the table above.

Source	Destination	Protocol	Destination Port	Remarks
MX	ke.impervacloud.com	HTTPS	TCP 443	Allows firewall access Threat Intelligence updates (GeoLocation, Fraud Prevention (FPS), Account Takeover (ATO), and Bot Protection.
MX	cd.impervacloud.com	HTTPS	TCP 443	Allows firewall access to allow sharing for Community Defense.
Gateway	fps.impervacloud.com	HTTPS	TCP 443	Allows firewall access for ThreatRadar features that need to obtain additional threat intelligence required by the feature including Account Takeover (ATO) and Fraud Prevention Services (FPS).
MX	thr.impervacloud.com	HTTPS	TCP 443	Allows firewall access for Reputation Services.

Configuring Firewall Ports

The SecureSphere allows a great deal of flexibility in its deployment modes. Firewalls can be placed wherever they are required.

The figure below shows that ports that need to be open to enable different functions required by SecureSphere to communicate through the deployed firewalls.



Imperva SecureSphere Component Communications

Blue = Mgmt Interface Traffic, Red = Web/Internet Traffic, Orange = Scan Traffic, Purple = File Traffic, Grey = Optional
u = UDP, without u = tcp

Serial Console Access to SecureSphere

This section describes the steps administrators need to take in order interact with SecureSphere's Command Line Interfaces (CLIs) through a serial console, rather than a VGA and PS2 keyboard and review the following:

- Supported Versions and Upgrades
- Supported Serial Console Settings
- Before You Begin: Checking OS Layer Serial Console Access
- Configuring BIOS Settings for Serial Console
- Editing the GRUB Configuration File

Supported Versions and Upgrades

This section describes supported versions and upgrades and reviews the following:

- Hardware

- Software
- Upgrades

Hardware

The steps described in this document are supported only on the FTL versions of G4, G8, G16 and MX appliances, with BIOS versions 74 and 85.

Software

The minimum SecureSphere software version required is 6.0.6.6302. Prior versions are not supported.

Upgrades

The changes described in this document will survive an upgrade from SecureSphere release 6.0.6.6302 to 6.2.0.6442. Other upgrades have not been tested. Furthermore, whenever SecureSphere is reinstalled, administrators must reapply the `grub.conf` changes described in [Editing the GRUB Configuration File](#).

Supported Serial Console Settings

Throughout this document, the serial console settings administrators should use are as follows:

Serial Console Settings

Setting	Value
baud rate	Depending on the network, you can select either 9600, 38400, or 115200. Default values: ▪ XX10 and XX20 machines: 9600 ▪ XX30 machines: 115200
data bits	8

Setting	Value
parity	none
stop bits	1
flow control	none
terminal emulation	VT 100+

VT 100 Escape Sequences

VT 100 key	Escape sequence sent
HOME	<ESC> h
END	<ESC> k
INSERT	<ESC> +
DELETE	<ESC> -
PAGE UP	<ESC> ?
PAGE DOWN	<ESC> /
ALT	<ESC> ^A
CONTROL	<ESC> ^C
F1	<ESC> 1

VT 100 key	Escape sequence sent
F2	<ESC> 2
F3	<ESC> 3
F4	<ESC> 4
F5	<ESC> 5
F6	<ESC> 6
F7	<ESC> 7
F8	<ESC> 8
F9	<ESC> 9
F10	<ESC> 0
F11	<ESC> !
F12	<ESC> @

Before You Begin: Checking OS Layer Serial Console Access

In its default configuration, SecureSphere supports OS layer access via a serial console. It is highly recommended to first confirm that they you such access as follows:

1. Set the connections parameters on the console client (for example, using `putty`) to those defined in the table Serial Console Settings found in [Supported Serial Console Settings](#).
2. Connect to the SecureSphere appliance via its DB9 serial port.
3. Initialize the connection and confirm that you get a Linux prompt and are able to log in, for example, with the

"secure" user credentials.



Note: If serial console access only at the OS layer is sufficient for your needs, then no configuration changes are needed.

Configuring BIOS Settings for Serial Console

After making the changes described below and rebooting the machine, you will be able to interact with the BIOS UI through both the VGA and the serial console.

Use your existing VGA port and PS2 keyboard to make the following changes:

1. Power on the SecureSphere appliance.
2. Press F2 to enter BIOS configuration
3. Set Server Management-> Console Redirection-> Console Redirection to one of the following:
 - [Serial Port A] (DB9)
 - [Serial Port B] (Rj45)
4. Set Flow Control to [None].
5. Set Baud Rate to [38.4K].
6. Set Terminal Type to [VT100+].
7. Set Legacy OS Redirection to [Enable].
8. Save & Exit using "F10".

At this stage the Keyboard and VGA are no longer required.

Editing the GRUB Configuration File

Changing the settings below will allow you to interact with the bootloader and see the messages it prints to the console as modules are loaded.



Warning: Once these settings take effect, boot time messages will only be available through the serial console and not through the VGA console. Only BIOS and OS access will still be available through the VGA and PS2 keyboard.

1. Edit the `/boot/grub/grub.conf` file.
2. Add the following text at the end of each line that start with the word "kernel":

```
onsole=ttyS0,38400
```

3. Save the file and exit.

The `/boot/grub/grub.conf` file should now look like this (the added text is highlighted):

```
# grub.conf generated by anaconda
```

```
#
```

```
# Note that you do not have to rerun grub after making changes to this file
```

```
# NOTICE: You have a /boot partition. This means that
```

```
#         all kernel and initrd paths are relative to /boot/, eg.
```

```
#         root (hd0,0)
```

```
#         kernel /vmlinuz-version ro root=/dev/sysvg/root.vol
```

```
#         initrd /initrd-version.img
```

```
#boot=/dev/sda
```

```
default=0
```

```
timeout=5
```

```
serial --unit=0 --speed=38400
```

```
terminal --timeout=5 serial console
```

```
title CentOS (2.6.18-53.1.4.el5.imp40smp)
```

```
root (hd0,0)
```



```
kernel /vmlinuz-2.6.18-53.1.4.el5.imp40smp ro root=/dev/sysv  
g/root.vol rhgb quiet panic=10  
console=ttyS0,38400
```

```
initrd /initrd-2.6.18-53.1.4.el5.imp40smp.img
```

```
title CentOS (2.6.18-53.el5)
```

```
root (hd0,0)
```

```
kernel /vmlinuz-2.6.18-53.el5 ro root=/dev/sysvg/root.vol rh  
gb quiet panic=10  
console=ttyS0,38400
```

```
initrd /initrd-2.6.18-53.el5.img
```

Hardware Security Modules (HSM)

Hardware Security Module devices (HSMs) store private encryption keys in specialized key storage devices. These devices protect the key information by making it impossible to gain access to the keys, even if the device is stolen.

A network HSM (Hardware Security Module) device is a central key Management Server which stores private keys. The network HSM provides a service to other devices on the network for performing operations with the secret keys.

Imperva SecureSphere supports the use of the following HSMs:

- nCipher card: [nCipher Card](#)
- nCipher netHSM: [nCipher netHSM](#)



Note: Beginning with SecureSphere version 10.0, SafeNet LunaSA is no longer supported.

All these solutions provide:

- Secure storage of up to 256 private keys
- SSL acceleration
- Luna Network HSM
- nCipher Card
- nCipher netHSM

Luna Network HSM

In today's security landscape, protecting cryptographic keys is paramount to ensuring the integrity and confidentiality of SSL/TLS communications. To enhance the security of private keys used for encryption, Imperva WAF Gateway now integrates with Thales Luna Network HSM. This integration allows organizations to securely generate, store, and manage private keys within a FIPS 140-2 Level 3 or FIPS 140-3 Level 3 validated hardware environment.

What the Integration Is About

The integration between Imperva WAF Gateway and Thales Luna Network HSM provides a seamless solution for handling cryptographic operations:

- **Secure Key Management:** Private keys used by the WAF Gateway for SSL/TLS encryption are generated and stored within the Luna Network HSM, ensuring they never leave the secure hardware environment.
- **Cryptographic Offloading:** Offloads intensive cryptographic operations from the WAF Gateway to the Luna

Network HSM.

- **Full Lifecycle Management:** Offers comprehensive management of cryptographic keys throughout their lifecycle, including generation, storage, rotation, and retirement.

Benefits and Value for Users

- **Enhanced Security:** By utilizing hardware-based key storage and management, organizations significantly reduce the risk of key compromise. The Luna Network HSM protects against unauthorized access with tamper-resistant hardware.
- **Seamless Integration:** Designed to work effortlessly with Imperva WAF Gateway without requiring significant changes to existing infrastructure or workflows.
- **Regulatory Compliance:** Achieves compliance with stringent security standards such as FIPS 140-2 Level 3 and FIPS 140-3 Level 3, which are often required in regulated industries like finance and healthcare.
- **Transparency and Accountability:** The integration provides comprehensive audit logs for all cryptographic operations, enhancing transparency and simplifying auditing processes.
- **Scalability and Reliability:** Supports the use of multiple Luna Network HSMs configured in high-availability (HA) groups, ensuring continuous operation and reliability across the network.

How It Works

- **Key Generation and Storage:** Private keys are generated within the Luna Network HSM and stored securely inside the hardware module.
 - **SSL/TLS Handshake Offloading:** During SSL/TLS handshakes, the WAF Gateway delegates cryptographic operations to the Luna Network HSM, which performs the necessary computations.
 - **Secure Key Usage:** The WAF Gateway accesses the keys within the HSM for encryption and decryption processes without exposing the keys to the external environment.
 - **Audit and Monitoring:** All key management activities are logged within the HSM's audit system, providing detailed records for security monitoring and compliance checks.
-
- [Integrate Luna Network HSM with WAF Gateway](#)
 - [Working with Luna Network HSM](#)

Integrate Luna Network HSM with WAF Gateway

To integrate Luna Network HSM with WAF Gateway perform the following tasks:

- [Prerequisites](#)
- [Register Imperva Gateway Luna Client to Luna Network HSM](#)
- [Configure HA group in WAF Gateway Luna Client \(optional\)](#)
- [Configure WAF Gateway](#)

Prerequisites

Before starting configuration of WAF Gateway with Luna Network HSM, complete the following steps:

1. Check connectivity between gateways to the Luna Network HSM
2. Receive the following data from the Luna Network HSM administrator:
 - HSM IP address or hostname
 - Partition name
 - Appliance username
 - Appliance password
 - Crypto Officer password for partition

Register Imperva Gateway Luna Client to Luna Network HSM

Note: The Luna Client is a set of utilities that, starting from version 15.3, is automatically installed on the gateway.

There are 2 options for Luna Client registration:

- One step NTLS with self signed certificate

Note: Client certificate will be generated and exchanged with Luna Network HSM by default.

- Multi step NTLS with custom certificate

One step NTLS

1. SSH to WAF Gateway and open lunaCM

```
/usr/safenet/lunaclient/bin/lunacm
```

2. Register the Luna client to the Luna Network HSM by executing the command: `clientconfig deploy -server [HSM HOSTNAME/IPADDR] -client [WAF GW IPADDR] -partition [PARTITION NAME] -user [USERNAME] -password [PASSWORD]`

Example:

```
clientconfig deploy -server hsm.mynet.com -client 10.10.10.11 -partition my_partition -user my_user
```

Multi step NTLS

The following procedure uses a self-signed certificate. Alternatively, you can use a custom CA certificate, which is also supported.

1. Open SSH to the WAF Gateway.
2. Create a client certificate by executing the command: `/usr/safenet/lunaclient/bin/vtl`

```
createCert -n [WAF GW IPADDR]
```

3. Obtain the HSMs certificate by executing the command: `scp admin@[HSM IPADDR]:server.pem /usr/safenet/lunaclient/`
4. Add the HSM to your local server list by executing the command: `/usr/safenet/lunaclient/bin/vtl addServer -n [HSM IPADDR] -c /usr/safenet/lunaclient/server.pem`
5. Send your client certificate to the HSM by executing the command: `scp /usr/safenet/lunaclient/cert/client/[CLIENT IPADDR].pem admin@[HSM IPADDR]:`
6. Open SSH to Luna Network HSM
7. Register the client to HSM by executing the command: `client register -client [CLIENT NAME] -ip [CLIENT IPADDR]`
8. Assign client to specific partition by executing the command: `client assignPartition -client [CLIENT NAME] -partition [PARTITION NAME]`
9. Verify if client has been registered correctly as follows:
 - a. Open SSH to WAF Gateway
 - b. Execute following command: `/usr/safenet/lunaclient/bin/vtl verify`

Output should be similar to:

```
The following Luna SA Slots/Partitions were found:
```

```
Slot Serial # Label
```

```
==== =====
```

```
0 1624158744741 my_partition
```

Configure HA group in WAF Gateway Luna Client (optional)

Note: Before setting up HA, you are required to create another partition on the 2nd HSM and register the gateway to this partition.

To configure HA group in WAF Gateway Luna client:

1. Open SSH to WAF Gateway and execute the command: `/usr/safenet/lunaclient/bin/lunacm`

At least two partitions should be listed on the Luna Client

2. Create HA group by executing the following commands:

```
hagroup creategroup -slot [first partition slot ID] -label [HA group label]
```

Note: The slot ID is the slot number of your first partition (usually 0)

If objects exist on the partition, you are prompted to either copy these objects or remove them. Select copy to continue.

```
hagroup addmember -group [HA group label] -slot [additional partition slot ID]
```

Note: Repeat “addmember” for all partitions that will be participate in HA group

```
hagroup synchronize -group [HA group label]
```

3. Configure HA group as follows:

- a. Check current HA settings by executing the command: `hagroup listGroups`
- b. Prepare directory for HA log by exiting from lunacm and executing the command: `mkdir /usr/safenet/lunaclient/[HA log directory]`
- c. Select HA log directory, by executing the command `hagroup halog -path "/usr/safenet/lunaclient/[HA log directory]"`
- d. Modify the recovery mode to activeEnhanced to allow sessions to be automatically re-established, by executing the command: `hagroup recoveryMode -mode activeEnhanced`
- e. Check current HA settings by executing the command: `hagroup listGroups`

Configure WAF Gateway

Luna Client is a set of utilities and since version 15.3 has been installed on GW automatically.

To configure WAF Gateway:

1. Add the GemEngine section to the Chrystoki.conf file as follows:
 - a. Open SSH to the WAF Gateway.
 - b. Open the file for edit using the command `vim /etc/Chrystoki.conf`
 - c. Add the following whole section to the end of file:

```
GemEngine = {

LibPath = /usr/lib/libCryptoki2_64.so;

LibPath64 = /usr/lib/libCryptoki2_64.so;

EnableDsaGenKeyPair = 1;

EnableRsaGenKeyPair = 1;

EnableEcGenKeyPair = 1;

DisablePublicCrypto = 1;

EnableRsaSignVerify = 1;
```

```

EnableLoadPubKey = 1;

EnableLoadPrivKey = 1;

DisableCheckFinalize = 1;

DisableEcdsa = 0;

DisableDsa = 0;

DisableRand = 1;

DisableSessionCache=0;

EngineInit = 0:10:11;

}

```

The EngineInit field is composed of the following:

'0:10:11' = <Slot ID>:<Application ID first>:<Application ID second>

Note:

Slot ID could be standalone partition slot or HA group slot.

Application ID is number specified by user in two parts and different for each GW for identification.

It could be any number as 10:11 or 11:12

2. Add a new value "FinalizeOnClose = 1; " to "Misc" section as follows:

```

Misc = {

PE1746Enabled = 0;

ValidateHost = 0;

= /usr/safenet/lunaclient/lock;

ToolsDir = /usr/safenet/lunaclient/bin;

PartitionPolicyTemplatePath = /usr/safenet/lunaclient/data/
partition_policy_templates;

ProtectedAuthenticationPathFlagStatus = 0;

PluginModuleDir = /usr/safenet/lunaclient/plugins;

FinalizeOnClose = 1; # This is the new value to be added

```

```
}
```

3. Provide partition credentials by executing the command: `impctl gateway hsm config --name luna --password '<Crypto Officer password>' --app-id <application id> --slot <slot number>`

Notes:

- Pay attention to the '' that surrounds the --password
- Slot number and application id should be the same as was provided in GemEngine section in Chrystoki file

Example:

```
impctl gateway hsm config --name luna --password 'mycryptoofficerpassword' --
app-id 10:11 --slot 0
```

4. Verify the changes in the bootstrap.xml file by executing the command: `grep 'net-hsm\|acceleration' /opt/SecureSphere/etc/bootstrap.xml`

Output should be like:

```
<net-hsm slot="0" app-id="10:11" user="myuser"
hostname="lunahsm.thaleselab.com" partition="mypartition" login-cmd="/usr/
sbin/sautil -o -s 0 -i 10:11 -p
'U57MmVKdlBICObey5DR8Yf7PuK6H854gozCArPdI5zFoaLhBapGvLwDzWyDY8+M0jiGwwS9R2eJOKfRwsTlJu
32mm3/zO+AfYHUylB87TEKfrVNAn4IerWlZezOFwKnsskhIrXnA8KViLUnJOM9brTc/
m8ws4TjnpBnTCv1SIw7vadqEys4h15WTEdmzBVv3NTtBS7MaRwMppD1AFPXuudWhw/
JOC07V4fzXwz3zpJw5wgn39lOZMlw73hirsmXyyauDQB6XirUFsv7lMMnS3U7PelX4gIA=="
logout-cmd="/usr/sbin/sautil -c -s 0 -i 10:11"/>

. . .

<acceleration-engine>gem</acceleration-engine>

<acceleration-enabled>true</acceleration-enabled>
```

5. Reboot the gateway.

Working with Luna Network HSM

To integrate Luna Network HSM with WAF Gateway perform the following tasks:

- [Create Key Pair in Luna Network HSM Partition with Luna Client](#)
 - [Create Certificate from Key Pair Created in HSM Partition](#)
 - [Upload Key and Certificate to MX](#)
 - [Troubleshooting](#)
-

Create Key Pair in Luna Network HSM Partition with Luna Client

Notes:



- Refer to the Luna Network HSM documentation for more variants and detailed steps on how to create keys in the HSM partition.
- When the gateway mode is ABR or NGRP and an RSA certificate that was generated in a Luna partition where FIPS Mode is enabled is used, only ECDHE and DHE key exchange algorithms are supported.

To create the key pair in the HSM partition with Luna Client:

1. Open SSH to the WAF Gateway.

Possible keys variants:

A) ECDSA

B) DSA

C) RSA

2. For ECDSA:

- a. Create the ecparam file to use the prime256v1 curve by executing the command: `openssl111 ecparam -out ecparam.pem -name prime256v1`

Note: This is a one time operation for all ECDSA keys.

- b. Create the key pair by executing the command: `sautil -v -s <slot> -i <application id> -m OID_:ecparam.pem -f <key_name>.key`

Note: The Key pair is created internally in the HSM and the reference key is exported and saved to root folder automatically.

3. For DSA:

- a. Create the dsaparam file by executing the command: `openssl111 dsaparam -out dsaparam.pem 2048`

Note: This should be re-created for each new DSA key.

- b. Create the key pair by executing the command: `sautil -v -s <slot> -i <application id> -d 2048:dsaparam.pem -f <key_name>.key`

Note: The Key pair is created internally in the HSM and the reference key is exported and saved to root folder automatically.

- For RSA, create the key pair by executing the command: `sautil -v -s <slot_number> -i <app:id> -g <size> -f <key_name>.key`

Note: The key pair is created internally in the HSM and the reference key is exported and saved to root folder automatically.

Create Certificate from Key Pair Created in HSM Partition

To create the certificate from the key pair Created in the HSM partition:

- For Self-signed certificate execute the command: `openssl11 req -engine gem -x509 -new -key <key_name>.key -out <cert_name>.cert`

Note: <key_name> should match with <key_name> provided during key generation.

- For Certificate Sign Request (CSR) for CA signed certificate execute the command: `openssl11 req -engine gem -new -key <key_name>.key -out <csr_name>.csr`

Note: <key_name> should match with <key_name> provided during key generation Certificate can be uploaded to the MX only after has been signed by CA.

Upload Key and Certificate to MX

Note: Before this step the key pair should be already created in HSM partition and reference key should be exported.

To upload key and certificate to the MX:

- Open the MX web UI.
- Go to the definitions tab of the service and add a new key.
- Upload reference key to the private key section and the certificate to the public key section.
- Select the HSM check box and click the upload button.
- Attach the created keys to the protected IP rule.

Troubleshooting

Note: Use debug with caution, debug logs have high impact on the gateway's performance.

If you experience issues with Luna Network HSM integration perform the following step to try and resolve:

1. Execute the below command to check if a session has been established with the HSM

```
netstat -tn | grep <HSM IP/HSM host name>
```

Output should be similar to:

```
tcp 0 0 10.10.73.99:55429 10.10.73.101:1792 ESTABLISHED
```

Note: For NTLS connection the port is 1792. For SCP connection the port is 5656

nCipher Card

The nCipher card is installed inside the SecureSphere Gateway appliance. This section reviews topics about the nCipher Card and includes:

- [Installing the nCipher Card](#)
- [Installing the nCipher Card Driver](#)
- [Security World Initialization](#)
- [Managing Keys](#)
- [Load the Certificate Keys](#)
- [Working in High Availability](#)

Installing the nCipher Card

The nCipher card is installed in an available PCI slot in the appliance.

To install the nCipher Card:

1. Shut down the appliance.
2. Power down the appliance.
3. Remove the appliance cover.
4. Insert the nCipher card into the PCI slot and verify that it is firmly in place.
5. Replace the cover.
6. Power on the appliance.
7. Perform the procedure described in [Installing the nCipher Card Driver](#).

Installing the nCipher Card Driver

Once you have installed the nCipher card in the gateway appliance, you need to install the card driver.

To install the nCipher card driver:

1. On the card's bracket, move the switch to the **O** position (operational).
2. Connect to the gateway via SSH.
3. Execute the following commands:

```
impctl gateway ncipher install
```

```
impctl gateway ncipher enable
```

```
impctl gateway restart
```

4. Verify the driver was installed correctly by executing the command:

```
impctl gateway ncipher show
```

The output resembles the following:

```
Server mode: operational
```

```
Module mode: operational
```

5. Perform the procedure described in [Security World Initialization](#).

Security World Initialization

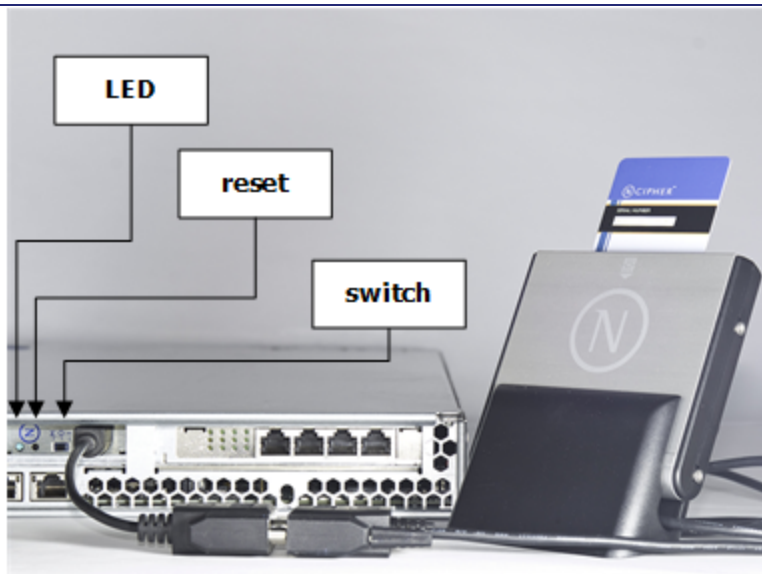
Once you have installed the card and driver properly, you need to initialize the nCipher security world. The nCipher security world must be initialized whenever one or more of the following conditions are true:

- The HSM is newly installed, so it does not yet have a security world.
- An existing security world is discarded.
- The appliance is reinstalled.

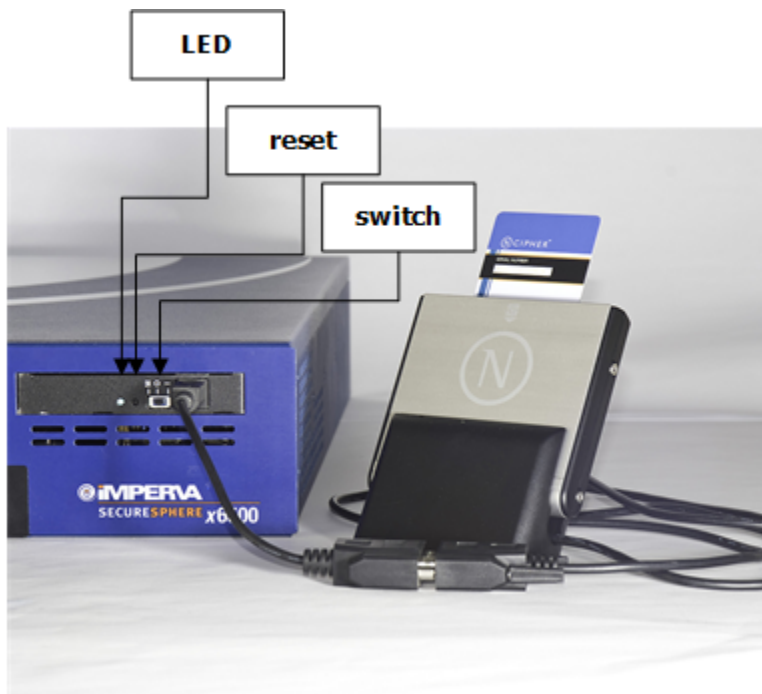
To initialize the security world:

1. On the card's bracket, move the switch to the **I** position (initialization).

On G series appliances, the bracket is accessible from the rear.



On X series appliances, the bracket is accessible from the front.



2. Press the RESET button (use a paper clip).

The LED will blink for a few seconds.

3. When the LED stops blinking, use the nCipher `show` command to confirm that the card is in the pre-initialization mode that is, that "module mode" is "pre-initialization".
4. Execute the following command:

```
/opt/nfast/bin/new-world -i -Q 1/1
```

-
- When you are asked for a passphrase, press <Enter>.

The output should resemble the following:

```
16:46:12 WARNING: Module #1: preemptively erasing module to see its slots!
```

```
Create Security World:
```

```
Module 1: 0 cards of 1 written
```

```
Module 1 slot 0: unknown card
```

```
Module 1 slot 0:- passphrase specified - overwriting card
```

```
Card writing complete.
```

```
security world generated on module #0; hknso =  
746a847b13f443d3a9561960e9071694c4678633
```

- Use the nCipher `show` command to confirm that the card is now in the initialization mode (see the output example in Verifying the Installation), that is, that "module mode" is "initialization".
- On the card's bracket, move the switch to the **O** position (operational).
- Press the RESET button (use a paper clip).

The LED will blink for a few seconds and then will remain mostly on, that is, it will blink only every few seconds.

- At this point, use the nCipher `show` command to confirm that the card is in the operational mode that is, that "module mode" is "operational".
- Execute the following command:

```
/opt/nfast/bin/cryptest
```

The card will run some internal tests.

- Confirm that the internal tests were successful by reviewing the output that resembles the following:

```
KeyType_DES: length 64 bits
```

```
Mech_Any: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384  
65536
```

```
Mech_DESmCBCi64pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992  
8192 16384 65536
```

```
Mech_DESmECBpPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192  
16384 65536
```

Mech_DESmECBpNONE: 0 8 16 64 256 2048 4096 7992 8192 16384 65536

Mech_DESmCBCpNONE: 0 8 16 64 256 2048 4096 7992 8192 16384 65536

Mech_DESmCBCMACi0pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992
8192 16384 65536

Mech_DESmCBCMACi64pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992
8192 16384 65536

KeyType_DES2: length 128 bits

Mech_Any: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384
65536

Mech_DES2mCBCi64pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992
8192 16384 65536

Mech_DES2mECBpPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536

Mech_DES2mECBpNONE: 0 8 16 64 256 2048 4096 7992 8192 16384 65536

Mech_DES2mCBCpNONE: 0 8 16 64 256 2048 4096 7992 8192 16384 65536

Mech_DES2mCBCMACi0pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992
8192 16384 65536

Mech_DES2mCBCMACi64pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096
7992 8192 16384 65536

KeyType_DES3: length 192 bits

Mech_Any: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384
65536

Mech_DES3mCBCi64pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992
8192 16384 65536

Mech_DES3mECBpPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536

Mech_DES3mECBpNONE: 0 8 16 64 256 2048 4096 7992 8192 16384 65536

Mech_DES3mCBCpNONE: 0 8 16 64 256 2048 4096 7992 8192 16384 65536

Mech_DES3mCBCMACi0pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992
8192 16384 65536

```
Mech_DES3mCBCMACi64pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096
7992 8192 16384 65536
```

```
KeyType_CAST:
```

```
KeyType_Void:
```

```
Mech_SHA1Hash: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384
65536
```

```
Mech_MD5Hash: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384
65536
```

```
Mech_RIPEMD160Hash: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536
```

```
Mech_SHA256Hash: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536
```

```
Mech_SHA384Hash: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536
```

```
Mech_SHA512Hash: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536
```

```
Mech_TigerHash: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536
```

```
Mech_HAS160Hash: 0
```

```
KeyType_HMACMD5: length 128 bits
```

```
Mech_HMACMD5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384
65536
```

```
KeyType_HMACSHA1: length 128 bits
```

```
Mech_HMACSHA1: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384
65536
```

```
KeyType_HMACRIPEMD160: length 128 bits
```

```
Mech_HMACRIPEMD160: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536
```

```
KeyType_HMACSHA256: length 128 bits
```

```
Mech_HMACSHA256: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
```

16384 65536

KeyType_HMACSHA384: length 128 bits

Mech_HMACSHA384: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536

KeyType_HMACSHA512: length 128 bits

Mech_HMACSHA512: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536

KeyType_HMACTiger: length 128 bits

Mech_HMACTiger: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536

KeyType_Random: length 128 bits

KeyType_ArcFour: length 128 bits

Mech_Any: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384
65536

Mech_ArcFourpNONE: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536

KeyType_Blowfish:

KeyType_Serpent:

KeyType_Rijndael: length 128 bits

Mech_Any: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384
65536

Mech_RijndaelmCBCi128pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096
7992 8192 16384 65536

Mech_RijndaelmECBpPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992
8192 16384 65536

Mech_RijndaelmECBpNONE: 0 16 64 256 2048 4096 8192 16384 65536

Mech_RijndaelmCBCpNONE: 0 16 64 256 2048 4096 8192 16384 65536

Mech_RijndaelmCBCMACi0pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096
7992 8192 16384 65536

```
Mech_RijndaelmCBCMACi128pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048
4096 7992 8192 16384 65536
```

```
Mech_RijndaelmCMAC: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536
```

```
KeyType_Twofish:
```

```
KeyType_CAST256: length 128 bits
```

```
Mech_Any: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384
65536
```

```
Mech_CAST256mCBCi128pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096
7992 8192 16384 65536
```

```
Mech_CAST256mECBpPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992
8192 16384 65536
```

```
Mech_CAST256mECBpNONE: 0 16 64 256 2048 4096 8192 16384 65536
```

```
Mech_CAST256mCBCpNONE: 0 16 64 256 2048 4096 8192 16384 65536
```

```
Mech_CAST256mCBCMACi0pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096
7992 8192 16384 65536
```

```
Mech_CAST256mCBCMACi128pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096
7992 8192 16384 65536
```

```
KeyType_SEED:
```

```
KeyType_ARIA:
```

```
KeyType_Camellia: length 128 bits
```

```
Mech_CamelliamCBCpNONE: 0 16 64 256 2048 4096 8192 16384 65536
```

```
Mech_CamelliamECBpNONE: 0 16 64 256 2048 4096 8192 16384 65536
```

12. At this point, you can remove the SmartCard from the card reader.
13. Perform the procedure described in [Managing Keys](#).

Managing Keys

This section provides an overview of the process. For detailed information on how to do this, see the nCipher documentation.

Once you have initialized the nCipher security world, you need to manage the keys. The process is composed of:

1. Importing the keys into the gateway. The keys are stored in shared memory and removed on every reboot.
2. Exporting the keys to another machine, from where they can be loaded to the MX.

To import a key file into the gateway:

1. Place the key you want to import on a machine you can connect to via SSH.
2. Execute the following command from the gateway:

```
impctl gateway ncipher import --import-type=pfx --protocol=scp --curl-
user=<ssh-username>:<ssh-password> --insecure --url=<ssh-host ip>/<path-to-
key.pfx> --key-password=<password for the pfx key>
```

The output should resemble the following:

```
INFO: Successfully downloaded 'newkey.enc.pfx'
```

```
INFO: Successfully extracted CN, key and crt from pfx, deleting
'newkey.enc.pfx'
```

```
INFO: Key Successfully imported into HSM, deleting 'www.example.com.key'
```

```
INFO: Output files: fake_key='www.example.com_fake.key',
crt='www.example.com.crt'
```

To export the keys to another machine:

1. Execute the following command:

```
impctl gateway ncipher export --export-type=keys --curl-user=<ssh-
username>:<ssh-password> --url=<ssh-host ip>/<path to keys zip file location>
--protocol=scp --insecure
```

The output should resemble the following:

```
INFO: Testing upload URL...
```

```
INFO: Successfully exported 'keys_export_GW_11_5_2017-03-15_22-48-45.zip' to
'scp://<ssh-host>/var/tmp/'
```

2. Using a disk-on-key or some other method, transfer the zip file to the computer from which you run the SecureSphere UI.
3. Unzip the file to a temporary directory. The output should be two files, a .key file and a .crt file.
4. Perform the procedure described in [Load the Certificate Keys](#).



Warning: These files contain a pointer to the private key's location on the HSM card. You must erase this extremely sensitive information in accordance with your site's procedures.

Important Notes

Note 1

If your .PFX file contain multiple certificates, chain the certificate and intermediate CA certificate(s) into a single .PEM certificate file. This can be done by copying and pasting all intermediate certificates to the file containing your own certificate, with the highest-level certificate **last**, so your cert file looks like this:

```
-----BEGIN CERTIFICATE-----
```

```
ABCDE.....
```

```
(Your certificate)
```

```
....XYZ=
```

```
-----END CERTIFICATE-----
```

```
-----BEGIN CERTIFICATE-----
```

```
ABCDE.....
```

```
(Intermediatecertificate)
```

```
....XYZ=
```

```
-----END CERTIFICATE-----
```

```
-----BEGIN CERTIFICATE-----
```

```
ABCDE.....
```

```
(Root certificate) (i.e. the certificate of you CA, which is itself signed by a  
root CA)
```

```
....XYZ=
```

```
-----END CERTIFICATE-----
```

After saving the file, it can be loaded to SecureSphere as a single public key along with the private key.

Note 2

If there are multiple gateways protecting the same server group, see the **Using-more-than-one-gateway-with-HSM-card-to-protect-the-same-web-server** article in the Imperva knowledge base.

Load the Certificate Keys

This section provides an overview of the process. For detailed information on how to do this, see the **SecureSphere User Guide**.

Once you have transferred and unzipped the zip file, you need to load the certificate keys into the MX.

To load the certificate keys into the MX:

1. Start the SecureSphere UI.
2. Got to **Setup > Sites**, select (or create) an HTTP service.
3. In the HTTP service's **Definitions** tab, under **Encryption Support > TLS Key Name**, click . The Add TLS Keys window is displayed.
4. In the **Name** field, type a name for the TLS Key.
5. Select the **HSM** check box.
6. Select **PEM file format**.
7. In **Private Key**, click **Browse**, navigate to the temporary directory you created and select the .key file.
8. In **Public Key**, click **Browse**, navigate to the temporary directory you created and select the .crt file.
9. Click **Upload**.

Working in High Availability

When you work in a High Availability (HA) environment, you need to set up the gateways with the nCipher cards accordingly. The following process details how to set up an nCipher card in an HA environment. The process is composed of:

1. Installing the nCipher card on the primary gateway. See [Installing the nCipher Card](#) for details.
2. Exporting the nCipher security world from the primary gateway. See procedure below for details.
3. Importing the nCipher security world to the secondary gateway. See procedure below for details.

To export the nCipher security world from the primary gateway:

1. Execute the following command:

```
impctl gateway ncipher export --export-type=kmdata --curl-user=<ssh-username>:<ssh-password> --url=<ssh-host ip>/<path to kmdata zip file location> --protocol=scp --insecure
```

The output should resemble the following:

```
INFO: Testing upload URL (scp://<ssh-host ip>/tmp/keys/)
```

```
INFO: Uploading <path to kmdata zip file location>/kmdata_export_x4500_v12_2017-11-01_14-26-49.zip
```

```
INFO: Successfully exported 'kmdata_export_x4500_v12_2017-11-01_14-26-49.zip' to 'scp://<ssh-host ip>/tmp/keys/'
```

2. Execute the following command to print a list of loaded keys (will be used later for comparison):

```
/opt/nfast/bin/nfkminfo -l
```

The output should resemble the following:

Keys with module protection:

```
key_embed_009a5de15a4264273d154b7bfa1aa45a128347ab `www.example.com'
```

```
key_embed_01d2f6025a785226dceb71a566151ff9e67e83ab `testing'
```

```
key_embed_076cdedd5fee0db67379b60989b457f7b138ffc5 <none>
```

3. Remove the SmartCard from the card reader (will be used in the secondary gateway).

To import the nCipher security world to the secondary gateway:

1. Insert the SmartCard you removed from the primary gateway into the card reader of the secondary gateway.
2. On the secondary nCipher card, move the switch to the **O** position (operational).
3. Execute the following commands:

```
impctl gateway ncipher enable
```

```
impctl gateway stop
```

```
impctl gateway ncipher install
```

4. Verify the driver was installed correctly by executing the command:

```
impctl gateway ncipher show
```

The output resembles the following:

```
Server mode: operational
```

```
Module mode: operational
```

5. On the secondary nCipher card, move the switch to the **I** position (initialization).
6. Execute the following commands:

```
impctl gateway ncipher import --import-type=kmdata --curl-user=<ssh-username>:<ssh-password> --url=<ssh-host ip>/<path to kmdata zip file location> --protocol=scp --insecure
```

```
hsm new-world-1240 -l
```

7. On the secondary nCipher card, move the switch to the **O** position (operational).
8. Execute the following command:

```
impctl gateway ncipher restart
```

```
impctl gateway start
```

9. Execute the following command:

```
impctl gateway ncipher show
```

The output resembles the following:

```
Server mode: operational
```

```
Module mode: operational
```

10. Execute the following command:

```
/opt/nfast/bin/cryptest
```

The output resembles the following:

```
KeyType_DES: length 64 bits
```

```
Mech_Any: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384
65536
```

```
Mech_DESmCBCi64pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992
8192 16384 65536
```

```
Mech_DESmECBpPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536
```

```
Mech_DESmECBpNONE: 0 8 16 64 256 2048 4096 7992 8192 16384 65536
```

```
Mech_DESmCBCpNONE: 0 8 16 64 256 2048 4096 7992 8192 16384 65536
```

```
Mech_DESmCBCMACi0pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992
8192 16384 65536
```

```
Mech_DESmCBCMACi64pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992
8192 16384 65536
```

```
KeyType_DES2: length 128 bits
```

```
Mech_Any: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384
65536
```

```
Mech_DES2mCBCi64pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992
8192 16384 65536
```

```
Mech_DES2mECBpPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536
```

```
Mech_DES2mECBpNONE: 0 8 16 64 256 2048 4096 7992 8192 16384 65536
```

```
Mech_DES2mCBCpNONE: 0 8 16 64 256 2048 4096 7992 8192 16384 65536
```

```
Mech_DES2mCBCMACi0pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992
8192 16384 65536
```

```
Mech_DES2mCBCMACi64pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096
7992 8192 16384 65536
```

```
KeyType_DES3: length 192 bits
```

```
Mech_Any: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384
65536
```

```
Mech_DES3mCBCi64pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992
8192 16384 65536
```

```
Mech_DES3mECBpPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536
```

```
Mech_DES3mECBpNONE: 0 8 16 64 256 2048 4096 7992 8192 16384 65536
```

```
Mech_DES3mCBCpNONE: 0 8 16 64 256 2048 4096 7992 8192 16384 65536
```

```
Mech_DES3mCBCMACi0pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992
8192 16384 65536
```

```
Mech_DES3mCBCMACi64pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096
7992 8192 16384 65536
```

```
KeyType_CAST:
```

```
KeyType_Void:
```

```
Mech_SHA1Hash: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384
65536
```

```
Mech_MD5Hash: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384
65536
```

```
Mech_RIPEMD160Hash: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536
```

```
Mech_SHA256Hash: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
```

16384 65536

Mech_SHA384Hash: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536

Mech_SHA512Hash: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536

Mech_TigerHash: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536

Mech_HAS160Hash: 0

KeyType_HMACMD5: length 128 bits

Mech_HMACMD5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384
65536

KeyType_HMACSHA1: length 128 bits

Mech_HMACSHA1: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384
65536

KeyType_HMACRIPEMD160: length 128 bits

Mech_HMACRIPEMD160: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536

KeyType_HMACSHA256: length 128 bits

Mech_HMACSHA256: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536

KeyType_HMACSHA384: length 128 bits

Mech_HMACSHA384: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536

KeyType_HMACSHA512: length 128 bits

Mech_HMACSHA512: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536

KeyType_HMACTiger: length 128 bits

Mech_HMACTiger: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192
16384 65536

KeyType_Random: length 128 bits

KeyType_ArcFour: length 128 bits

Mech_Any: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384 65536

Mech_ArcFourpNONE: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384 65536

KeyType_Blowfish:

KeyType_Serpent:

KeyType_Rijndael: length 128 bits

Mech_Any: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384 65536

Mech_RijndaelmCBCi128pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384 65536

Mech_RijndaelmECBpPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384 65536

Mech_RijndaelmECBpNONE: 0 16 64 256 2048 4096 8192 16384 65536

Mech_RijndaelmCBCpNONE: 0 16 64 256 2048 4096 8192 16384 65536

Mech_RijndaelmCBCMACi0pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384 65536

Mech_RijndaelmCBCMACi128pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384 65536

Mech_RijndaelmCMAC: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384 65536

KeyType_Twofish:

KeyType_CAST256: length 128 bits

Mech_Any: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384 65536

Mech_CAST256mCBCi128pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384 65536

Mech_CAST256mECBpPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096 7992 8192 16384 65536

```
Mech_CAST256mECBpNONE: 0 16 64 256 2048 4096 8192 16384 65536
```

```
Mech_CAST256mCBCpNONE: 0 16 64 256 2048 4096 8192 16384 65536
```

```
Mech_CAST256mCBCMACi0pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096
7992 8192 16384 65536
```

```
Mech_CAST256mCBCMACi128pPKCS5: 0 1 7 8 9 15 16 17 20 36 64 256 1025 2048 4096
7992 8192 16384 65536
```

```
KeyType_SEED:
```

```
KeyType_ARIA:
```

```
KeyType_Camellia: length 128 bits
```

```
Mech_CamelliamCBCpNONE: 0 16 64 256 2048 4096 8192 16384 65536
```

```
Mech_CamelliamECBpNONE: 0 16 64 256 2048 4096 8192 16384 65536
```

11. Execute the following command to print a list of loaded keys (compare with the list from the primary gateway):

```
/opt/nfast/bin/nfkminfo -l
```

12. Fail over to the secondary gateway and validate SSL traffic is inspected.

nCipher netHSM

The nCipher netHSM is a Network HSM appliance that can be integrated with SecureSphere. This section reviews topics about nCipher HSM and includes:

- [High Level Functional Description](#)
- [Components Taking Part in the Integration](#)
- [High Level Installation Process](#)
- [nCipher HSM Installation](#)
- [Setting Up the Integration](#)
- [Basic netHSM Configuration](#)

High Level Functional Description

This section describes the integration with the network HSM provided by the nCipher company, netHSM.

- SecureSphere supports storing keys on the netHSM and accessing the keys in real-time in order to decrypt SSL web traffic. The following options are supported:
 - Accessing the same key from multiple SecureSphere Gateways, including the case of Gateways that are in
-

high availability mode.

- Working with a mix of keys, where some keys are stored in an HSM device while other keys are stored in SecureSphere.

Components Taking Part in the Integration

The following components are involved in the integration:

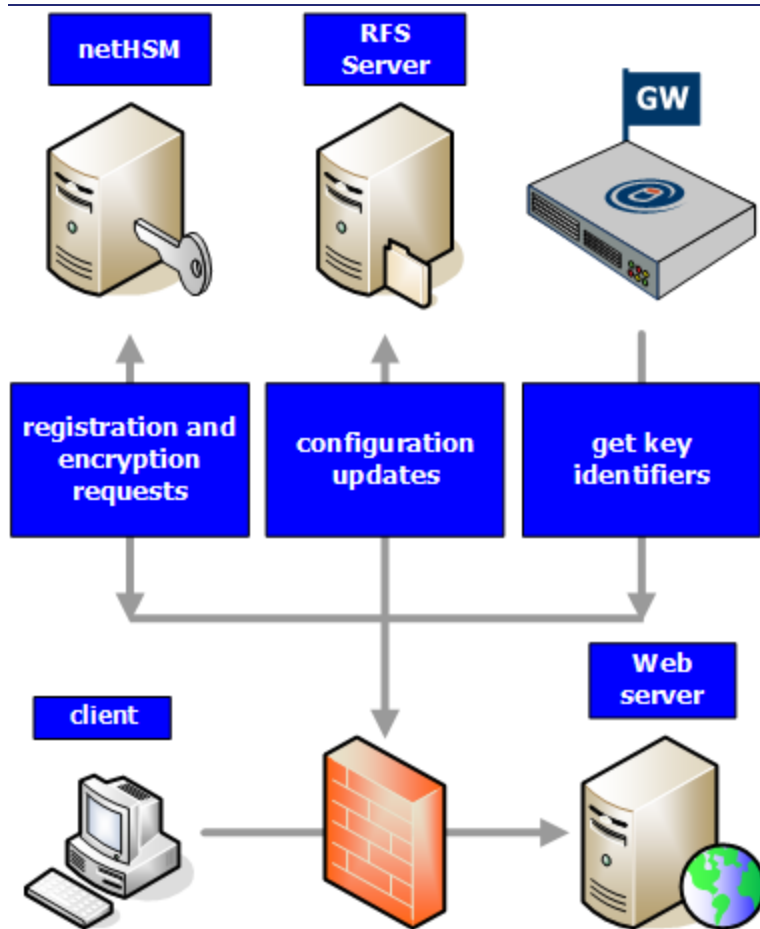
- The nCipher netHSM – the server that holds the secret keys (referred to in this document as the netHSM).
- The nCipher RFS Server – the remote file system server (referred to in this document as the RFS Server).
- The SecureSphere Gateway – performs the traffic decryption and receives key services from the netHSM. The SecureSphere Gateway will be referred to in this document as the Gateway.
- The SecureSphere MX Server – holds the security policy instructing which keys should be applied to the encrypted traffic. The SecureSphere MX Server will be referred to in this document as the MX Server.

High Level Installation Process

To integrate SecureSphere to the netHSM, proceed as follows:

1. Register the Gateway to the netHSM.
2. Register the Gateway to the RFS Server.
3. Enable nCipher support on the SecureSphere Gateway.
4. Synchronize between the Gateway and the RFS server.
5. Define the security policy on the MX Server.

The following figure shows the relationship between the components which comprise an nCipher HSM configuration.



nCipher HSM Installation

This section review aspects of nCipher HSM installation including the following:

- Assumptions
- Installing the nfast Software Package

Assumptions

This section outlines the process of integrating SecureSphere with a netHSM. It is assumed that the netHSM is already functional, so what remains to be done is to integrate SecureSphere to the device.

Consult with your netHSM administrator to confirm that the following prerequisites have been met.

1. netHSM is operational, and that the security world of the HSM device is appropriately initialized.
2. The correct version of SecureSphere is already installed and has successfully gone through the setup process.

netHSM integration is enabled in SecureSphere version 6.0.6 patch 5 and above.

-
3. The RFS server is already installed and you know its IP address.

Any Gateway using netHSM should have network access to the RFS through its management interface.

4. All connections between the Gateway and the netHSM and the RFS server pass through the Gateway's management interface.

It is assumed that you have network connectivity between the Gateway and the RFS Server and the netHSM.

5. Confirm that you have access to the RFS Server and the netHSM.

The RFS Server and the netHSM are separate devices (see the figure in [High Level Installation Process](#)) under the responsibility of the netHSM administrator. Several steps in this procedure are performed on the nCipher machines. If you do not have access to these machines, please contact the nCipher administrator to obtain access.

Installing the nfast Software Package

To install the nCipher application:

1. Connect to the gateway via SSH.
2. Execute the following commands:

```
impctl gateway ncipher enable
```

```
impctl gateway restart
```

```
impctl gateway ncipher install
```

3. Verify the driver was installed correctly by executing the command:

```
impctl gateway ncipher show
```

The output resembles the following:

```
Server mode: operational
```

```
Module mode: N/A
```

Setting Up the Integration

This section reviews setting up the integration of the nCipher HSM card and includes the following:

- [Pre-installation Checklist](#)
- [Terminating the Integration](#)

Pre-installation Checklist

Before setting up the integration procedure between SecureSphere and the netHSM, review the following checklist:

DB Pre-installation checklist

Step	ü	Action
1		netHSM is operational
2		RFS server is operational
3		SecureSphere is operational
4		You have the netHSM IP address
5		You have the RFS Server IP address
6		nfast RPM is installed on the Gateway

To integrate between SecureSphere and the netHSM:

1. Ensure that the SecureSphere Gateway accesses the netHSM and to the RFS Server through the appliance's management interface.

Do this in `impcfg` (**Manage Platform > Manage Networks > Static Routes**).

2. Define the Gateway as a client to netHSM.

On the netHSM, select **Menu > System > System configuration > Client config > New/Edit client** from the menu, and enter the Gateway's management IP address.

3. On the SecureSphere Gateway, get the netHSM's electronic serial number and key-hash (which will be used in subsequent commands) by executing the following command:

```
hsm anonkneti <netHSM-IP-address>
```

You will get the following output: ESN KEY-HASH.

-
4. Register the Gateway as a client to netHSM by executing the following command:

```
hsm nethsmenroll <nethSM-IP-address> <ESN> <KEY-HASH>
```

where <ESN> and <HASH> are the values obtained in the previous step.

5. Synchronize the Gateway with the RFS Server as follows:
 - a. Add the Gateway as a permitted entity on the RFS server by executing the following command on the RFS Server for each Gateway:

```
/opt/nfast/bin/rfs-setup --gang-client --write-noauth <Gateway-IP-Address>
```

- b. Bond and synchronize each Gateway with the RFS Server by executing the following command on the Gateway:

```
hsm rfs-sync --setup --no-authenticate <RFS-IP-address>
```

- c. Update the secure world and certificates files by running the following command:

```
hsm rfs-sync --update
```

6. Configure the keys on the SecureSphere Management Server by editing the web service and adding the appropriate fake keys and matching certificate files.
7. Sign these keys as HSM keys by selecting **HSM** for these keys.

Once the Gateway is updated, you can start decrypting traffic using your netHSM device.

Terminating the Integration

To terminate the integration between the netHSM and SecureSphere:

1. Remove the keys from the MX by deleting all the keys marked as HSM, and replacing them with the real keys as appropriate.
2. Disable the nCipher HSM by disabling HSM (**Impcfg > Manage SecureSphere Gateway > Manage hardware security modules (HSM)**)

Basic netHSM Configuration

This section provides an overview of the process. For detailed information on how to do this, see the nCipher documentation.

Once you have setup the nCipher integration, you need to:

- Import the keys into the gateway. The keys are stored in shared memory and removed on every reboot.
- Export the keys to another machine, from where they can be loaded to the MX.

To import a key file into the gateway:

1. Place the key you want to import on a machine you can connect to via SSH.
2. Execute the following command from the gateway:

```
impctl gateway ncipher import --import-type=pfx --protocol=scp --curl-
user=<ssh-username>:<ssh-password> --insecure --url=<ssh-host ip>/<path-to-
key.pfx> --key-password=<password for the pfx key>
```

The output should resemble the following:

```
INFO: Successfully downloaded 'newkey.enc.pfx'
```

```
INFO: Successfully extracted CN, key and crt from pfx, deleting
'newkey.enc.pfx'
```

```
INFO: Key Successfully imported into HSM, deleting 'www.example.com.key'
```

```
INFO: Output files: fake_key='www.example.com_fake.key',
crt='www.example.com.crt'
```

To export the keys to another machine:

1. Execute the following command:

```
impctl gateway ncipher export --export-type=keys --curl-user=<ssh-
username>:<ssh-password> --url=<ssh-host ip>/<path to keys zip file location>
--protocol=scp --insecure
```

The output should resemble the following:

```
INFO: Testing upload URL...
```

```
INFO: Successfully exported 'keys_export_GW_11_5_2017-03-15_22-48-45.zip' to
'scp://<ssh-host>/var/tmp/'
```

2. Using a disk-on-key or some other method, transfer the zip file to the computer from which you run the SecureSphere UI.
3. Unzip the file to a temporary directory. The output should be two files, a .key file and a .crt file.
4. Perform the procedure described in [Load the Certificate Keys](#).



Warning: These files contain a pointer to the private key's location on the HSM card. You must erase this extremely sensitive information in accordance with your site's procedures.

Add-Ons

This appendix describes various network configuration issues related to configuring add-on hardware in conjunction with SecureSphere, and includes:

- [SSL Accelerator](#)
- [IPMI - 5G2U Appliances](#)
- [IPMI - 6G2U Appliances](#)
- [IPMI - 7G2U Appliances](#)

SSL Accelerator

SSL accelerator cards are used to speed SSL processing in SecureSphere appliances. The cards are available for all SecureSphere appliances.

SecureSphere SSL accelerator cards support keys of length 1024 bits and higher.

For more information on SecureSphere SSL accelerator cards, see the Imperva Customer Support Portal (CSP) at www.imperva.com.

IPMI - 5G2U Appliances

IPMI (Intelligent Platform Management Interface) involves the use of a dedicated management channel for server maintenance. It allows a system administrator to monitor and manage servers by remote control regardless of whether the machine is powered on, or if the Imperva On-Premises system is running or not.

A complete remote management system allows remote reboot, shutdown, powering on; broadcasting of video output to remote terminals and receiving of input from remote keyboard and mouse (KVM over IP). It also can access local media like a DVD drive, or disk images, from the remote machine. If necessary, this allows one to perform remote installation of the operating system.

The remote system is accessed through various web browsers.

- [Preface - 5G2U Appliances](#)
- [IPMI and System Management Overview - 5G2U Appliances](#)
- [Using the IPMI WebGUI - 5G2U Appliances](#)
- [System Configuration from WebGUI - 5G2U Appliances](#)
- [System BMC Management from WebGUI - 5G2U Appliances](#)
- [Utilities - 5G2U Appliances](#)
- [Using the ipmitool Utility - 5G2U Appliances](#)

Preface - 5G2U Appliances

This section provides instructions for managing server using the IPMI. IPMI is included on certain servers. If users have one of these servers, it will include an IPMI Supplement which contains platform-specific information, such as sensors and thresholds, and details about the hardware.

IPMI and System Management Overview - 5G2U Appliances

IPMI is a dedicated system of hardware and supporting software that allows users to manage a server independent of the operating system. The IPMI includes the following components:

- **Service Processor:** This is the hardware. It consists of a dedicated processor board that communicates through the system serial port and a dedicated Ethernet port.
- **WebGUI:** The WebGUI provides a powerful, easy-to-use browser interface that allows users to log in to perform system management, monitoring, and IPMI tasks. Users only need to install the Java client appliance on first use.
- **Remote Console/Java™ Client:** The Java client supports the Remote Console functionality, which allows the user to access the server's console remotely. It redirects the keyboard and video screen, and can redirect input and output from the local machine's storage such as CD/DVD-ROM and hard disk drives.

Using the IPMI WebGUI - 5G2U Appliances

This section describes how to use the IPMI WebGUI.

- [Overview of WebGUI Features - 5G2U Appliances](#)
- [Requirements before using WebGUI - 5G2U Appliances](#)
- [Users and Privileges - 5G2U Appliances](#)
- [Logging In and Out of the WebGUI - 5G2U Appliances](#)

Overview of WebGUI Features - 5G2U Appliances

The UI enables the user to monitor and manage remote systems. The user can quickly activate the WebGUI using a web browser. One of the features of the IPMI is the ability to redirect the server's graphical console to a remote workstation or laptop system. When the user redirects the host console, the user can:

- configure the remote system's keyboard to act as the server's keyboard
- configure the disk drive, or CD/DVD-ROM drive images on the remote system as a device virtually connected to the server
- redirect CD/DVD-ROM images for remote access

Requirements before using WebGUI - 5G2U Appliances

The WebGUI has been tested successfully with recently released popular web browsers, and may be compatible with other web browsers.

Notes:



- You need to install Java™ on the host system. The IPMI remote console currently supports Java versions 1.5.7 through 7. To work with Java version 8, you need to perform an IPMI firmware version upgrade. For information on performing this upgrade, contact Imperva Support
- Java Web Start 1.6 is required to launch the KVM over an IPv6 network

Users and Privileges - 5G2U Appliances

After logging in, users can do the following actions:

- basic software provisioning
- Intelligent Platform Management Interface (IPMI) tasks.

IPMI user accounts include a role which defines what can do.

- **Administrator:** Enables full access to functions and commands.
- **Operator:** Enables limited access to functions and commands.
- **User:** Enables more limited access to functions and commands.



Note: Operators and users cannot change their own assigned roles or privileges.

Logging In and Out of the WebGUI - 5G2U Appliances

When the server is cabled appropriately and connected to an AC supply, the IPMI boots up automatically. Booting up is a very fast process. The default Ethernet configuration is the static IP Address. However, if the management Ethernet is not connected, or if the IPMI's Dynamic Host Configuration Protocol (DHCP) process fails due to the absence of a DHCP server on the management network, the IPMI may take a few minutes to boot. If you are using a browser proxy server, disable it. This may speed up access to the management network.

If you want to refresh information like the sensor reading on the web pages, or you want to logout from the web page, use the **Refresh** or **Log Out** buttons at the top right of the WebGUI window. Do not use the **Refresh** or **Close window** buttons in the browser.

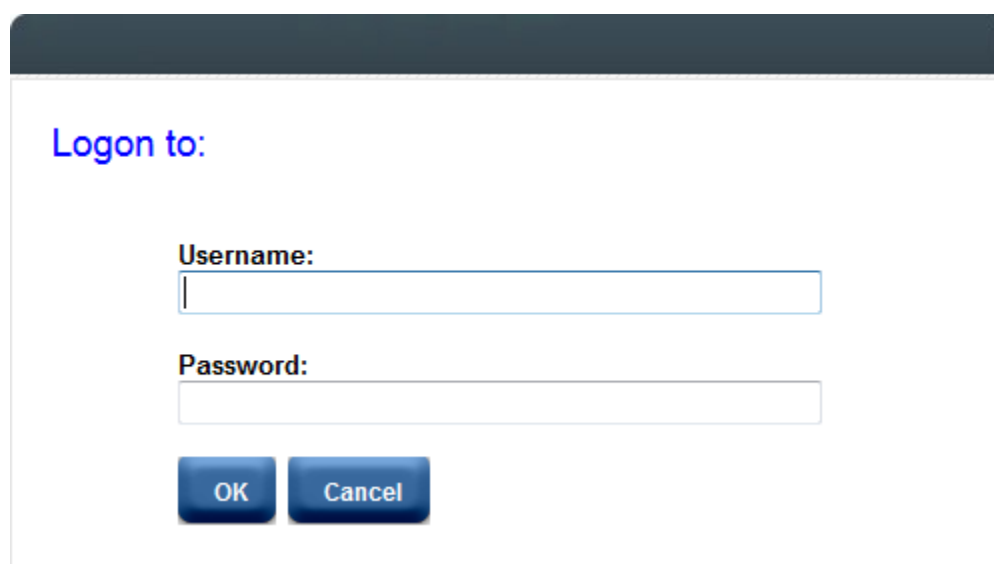
To log in to the WebGUI:

1. Enter the IP address of the IPMI into web browser: `https://192.168.1.1`

The **Security Alert** window appears.



2. Click **Yes**. The WebGUI login screen appears.

A screenshot of the WebGUI login screen. It has a dark blue header bar. Below it, the text 'Logon to:' is in blue. There are two input fields: 'Username:' and 'Password:'. Below the input fields are two buttons: 'OK' and 'Cancel'.

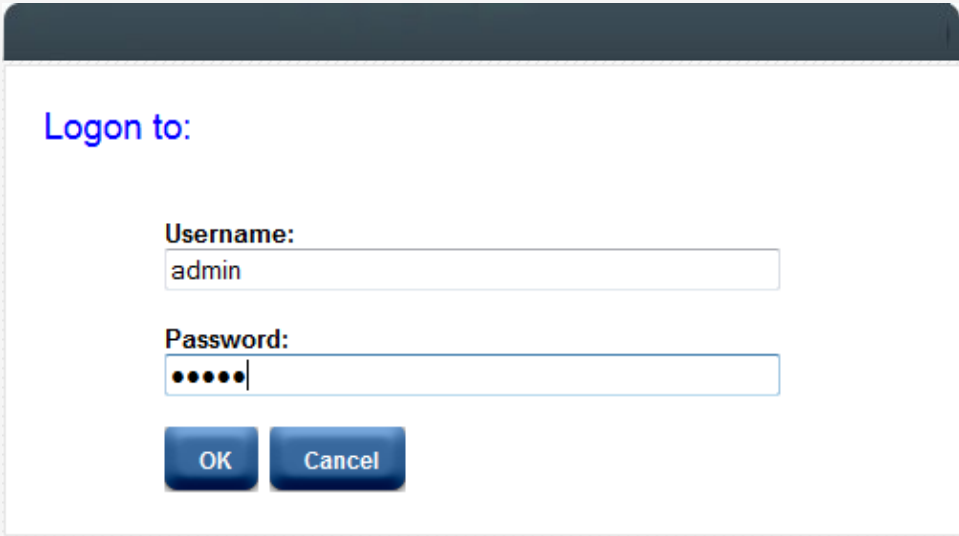
3. Enter your **Username** and **Password**.

The default values for an Administrator are:

- Username: admin
- Password: <machine serial number>

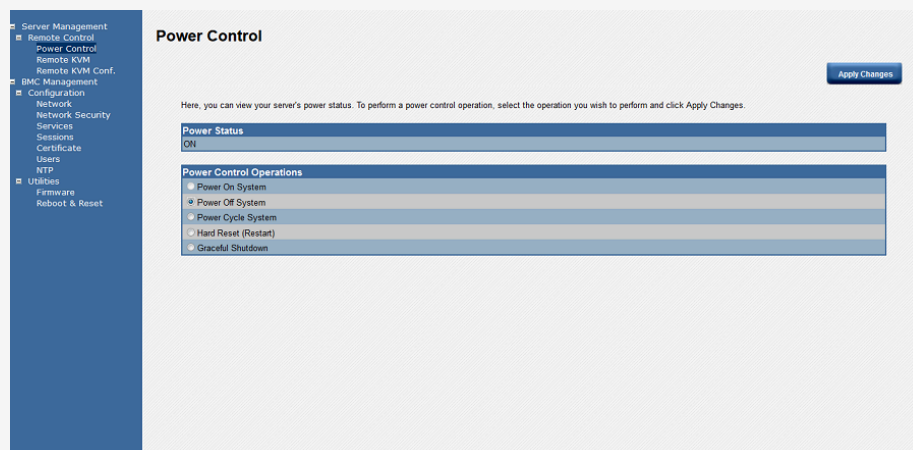
Notes:

- The machine serial number appears either on a sticker on the machine, or on the packaging, or both. Alternatively, SSH to the machine and use the command **ipmtl platform dmi show | grep Serial**.
- It is strongly recommended that you change your password from the default. For more information, see [Users Configuration](#).



A login dialog box with a dark header bar. The text "Login to:" is in blue. Below it, the label "Username:" is followed by a text input field containing "admin". Below that, the label "Password:" is followed by a password input field with six dots. At the bottom are two buttons: "OK" and "Cancel". To the left of the dialog box is a blue icon consisting of three horizontal lines and a checkmark.

- Click **OK**. The WebGUI screen appears.



To log out of the WebGUI:

- Click the **Logout** button at the top right of the WebGUI. The login screen appears.

System Configuration from WebGUI - 5G2U Appliances

This section describes how to configure the system using the WebGUI.

- [Power Control - 5G2U Appliances](#)
- [About the Remote KVM - 5G2U Appliances](#)
- [Starting the Remote Console Application - 5G2U Appliances](#)
- [Remote KVM Application Settings - 5G2U Appliances](#)
- [Starting the Virtual Media \(vMedia\) - 5G2U Appliances](#)

Power Control - 5G2U Appliances

Use the **Power Control** page to control the power on/off and hardware reset functions. See the table below for details.

Power Control

Here, you can view your server's power status. To perform a power control operation, select the operation you wish to perform and click Apply Changes.

Power Status
ON

Power Control Operations

- ☐ Power On System
- ☒ Power Off System
- ☐ Power Cycle System
- ☐ Hard Reset (Restart)
- ☐ Graceful Shutdown

Apply Changes

Field Name	Description
Power Status	Current status of the Power Control (OFF or ON).
Power On System	Turns on the system when it is in the "off" state.
Power Off System	Turns off the system when it is in the "on" state.

Field Name	Description
Power Cycle System	Turns off, then reboots the system (cold boot).
Hard Reset (Restart)	Reboots the system without turning it off (warm boot).
Graceful Shutdown	Shuts down the system without losing data, by shutting down systems in sequence and then powering off.

About the Remote KVM - 5G2U Appliances

The KVM starts up when the WebGUI appears. It allows you to:

- remotely control your server's operating system
- remotely use the server's screen and keyboard
- redirect local CD/DVD-ROM and hard drives as if they were connected directly to the server.

The screen and keyboard functionality allows you to use the operating system and other GUI-based programs, instead of being limited to command-line-based utilities provided by terminals and emulators.

The ability to redirect CD/DVD-ROM and hard drives allows you to download and upload software to and from the server as if accessing its own CD/DVD-ROM and hard drives.

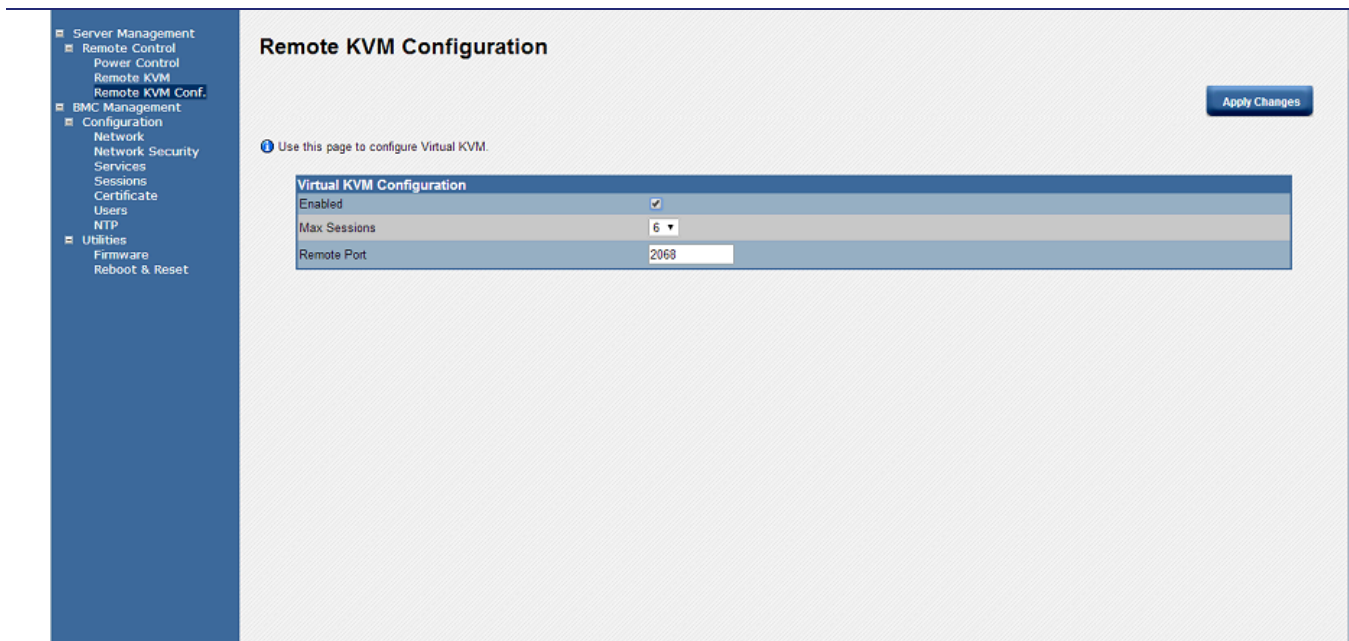
Starting the Remote Console Application - 5G2U Appliances

When you are logged in as an Administrator or Operator, you can modify the configuration settings and also launch the Applet JNLP file.

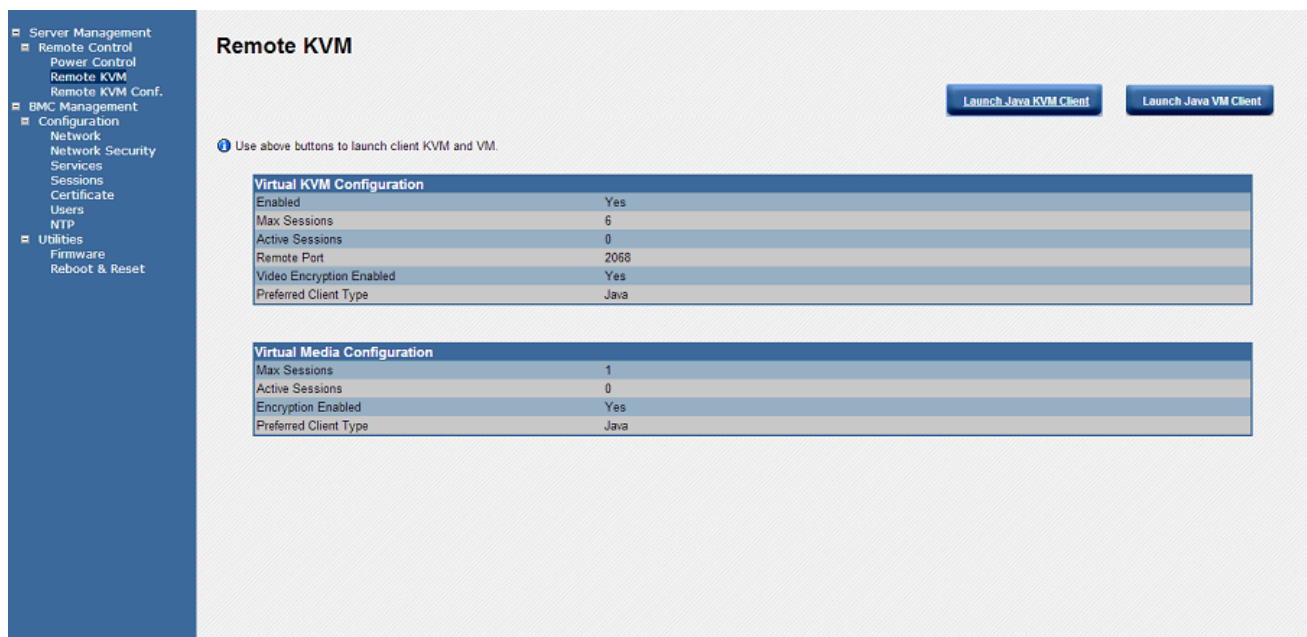
To start the Remote Console Application from the WebGUI:

1. Log in to the IPMI as an Administrator. For more information, see [Logging In and Out of the WebGUI](#).
2. Click **Remote KVM Conf**. The **Remote KVM Configuration** page appears.
3. Check the **Enable** checkbox, then click **Apply Changes**.

See the table at the end of this section for a summary of the Remote KVM Configuration.



4. Click **Remote KVM**. The **Remote KVM** page appears



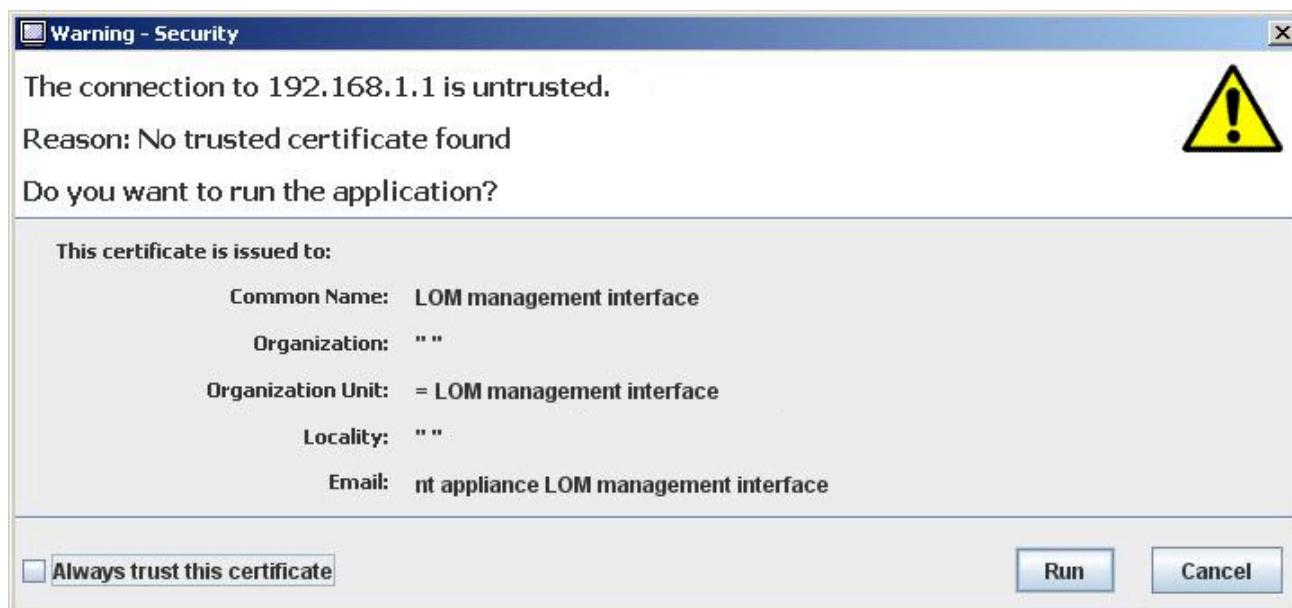
5. Click **Launch Java KVM Client**. The warning message dialog appears.



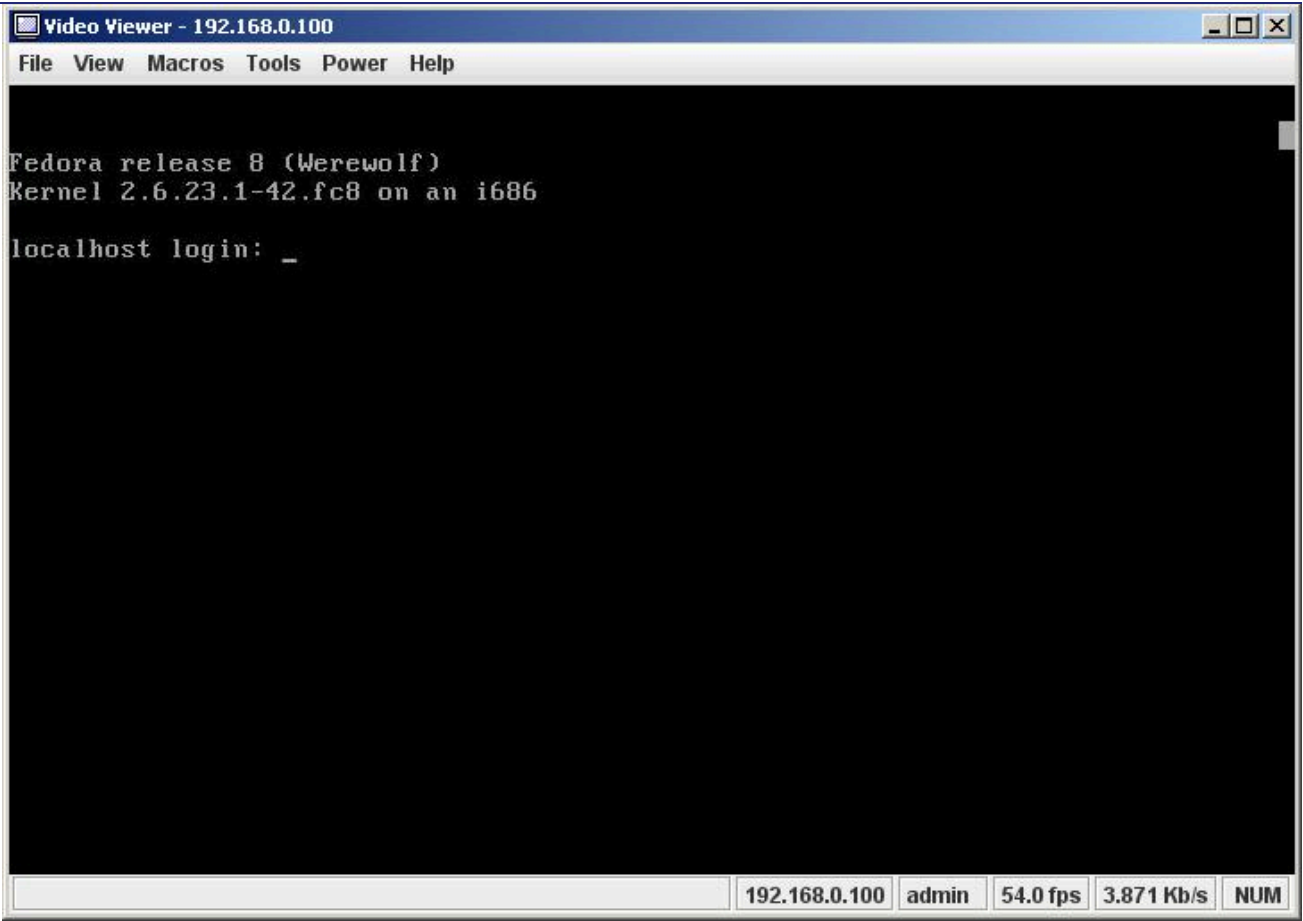
6. Click **Run**. The Java application message appears.



After a few moments, the next Java warning message dialog appears.



7. Click **Run**. The Java application runs, after which the remote console appears.



Field Name	Description
Enabled	Configures access if enabled (checked).
Max Sessions	Configures the max number of sessions allowed.
Remote Port	Configures remote access connection port. Integer range between 1 and 65535. The preset value is 2068. Max length should be 5.

Remote KVM Application Settings - 5G2U Appliances

This section describes the menu options in Remote KVM.

- [File Menu - 5G2U Appliances](#)
- [View Menu - 5G2U Appliances](#)
- [Tools Menu - 5G2U Appliances](#)

File Menu - 5G2U Appliances

- **File > Capture to File:** To capture an image. A dialog box appears that allows user to save the file to a specified location.
- **File > Exit:** The window closes.

View Menu - 5G2U Appliances

- **View > Hide Status Bar:** Toggles the information bar at the bottom of the vKVM application.
- **View > Refresh:** Updates the Video Viewer. The Viewer requests a reference video frame from the server.
- **View > Full Screen/Windowed:** Toggles full screen and window modes.
- **View > Fit:** Resizes the Video Viewer window to the minimum size required to display the server's video. (This menu item is not available in Full Screen mode.)

Tools Menu - 5G2U Appliances

- **Tools > Session Options > General:** You can control the keyboard pass-through mode and select **Pass all keystrokes to target** to pass your management station's keystrokes to the remote system. Some keystrokes are intercepted by the management station OS and will not be passed on.
- **Tools > Session Options > Video Quality:** The compression modes allow for a low quality (420) or high quality (444) compression configuration. The Luminance and Chrominance settings allow for picture color configuration.
- **Tools > Status:** Launches a dialog which displays viewer performance statistics: Frame Rate, Bandwidth, Compression and Packet Rate.

Starting the Virtual Media (vMedia) - 5G2U Appliances

The Virtual Media page enables you to mount remotely a local resource - a hard drive or CD/DVD drive image - from your computer, so that it appears on the managed server. You can then run a remote installation from your computer.

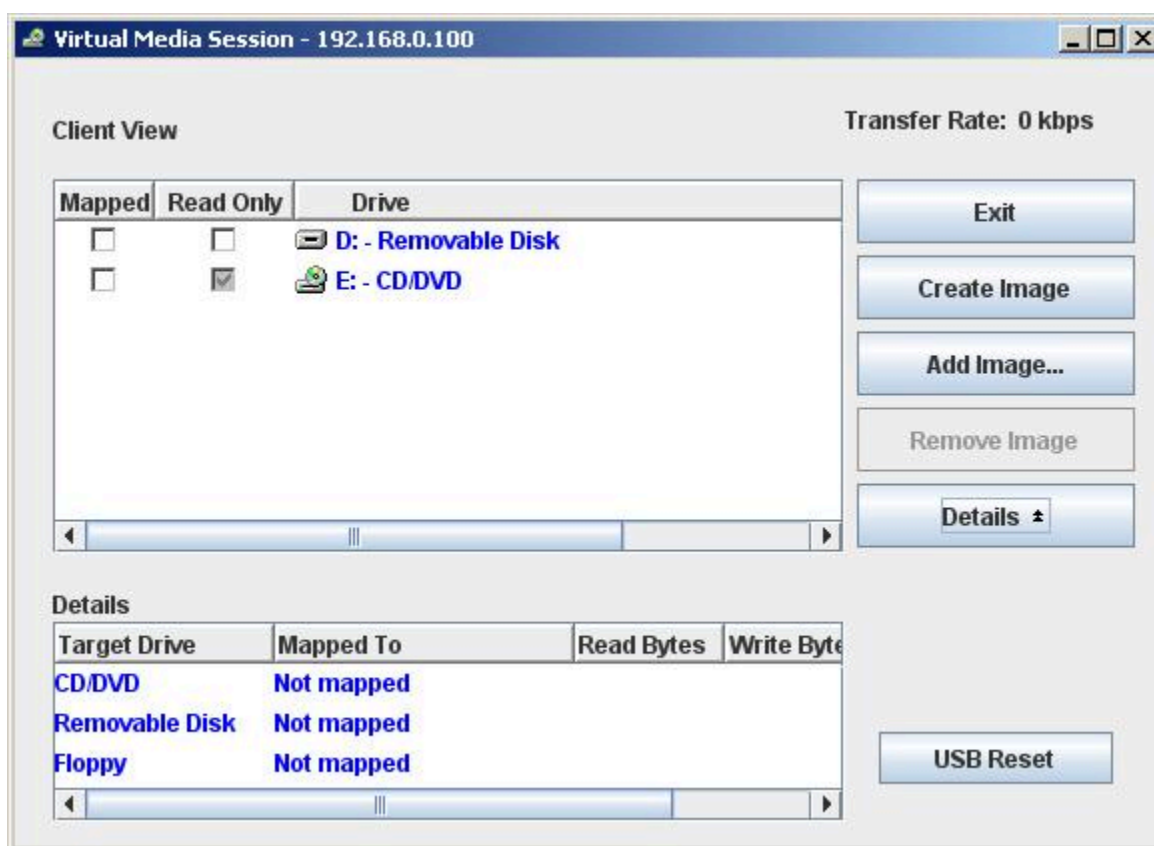
To start Virtual Media from the WebGUI:

1. Log in to the IPMI as an Administrator. For more information, see [Logging In and Out of the WebGUI](#).
2. Click **Remote KVM**. The **Remote KVM** page appears.
3. Click **Launch Java VM Client**.

Warning messages appear as in [Starting the Remote Console Application \(vKVM\)](#). The Virtual Media Session dialog box appears.



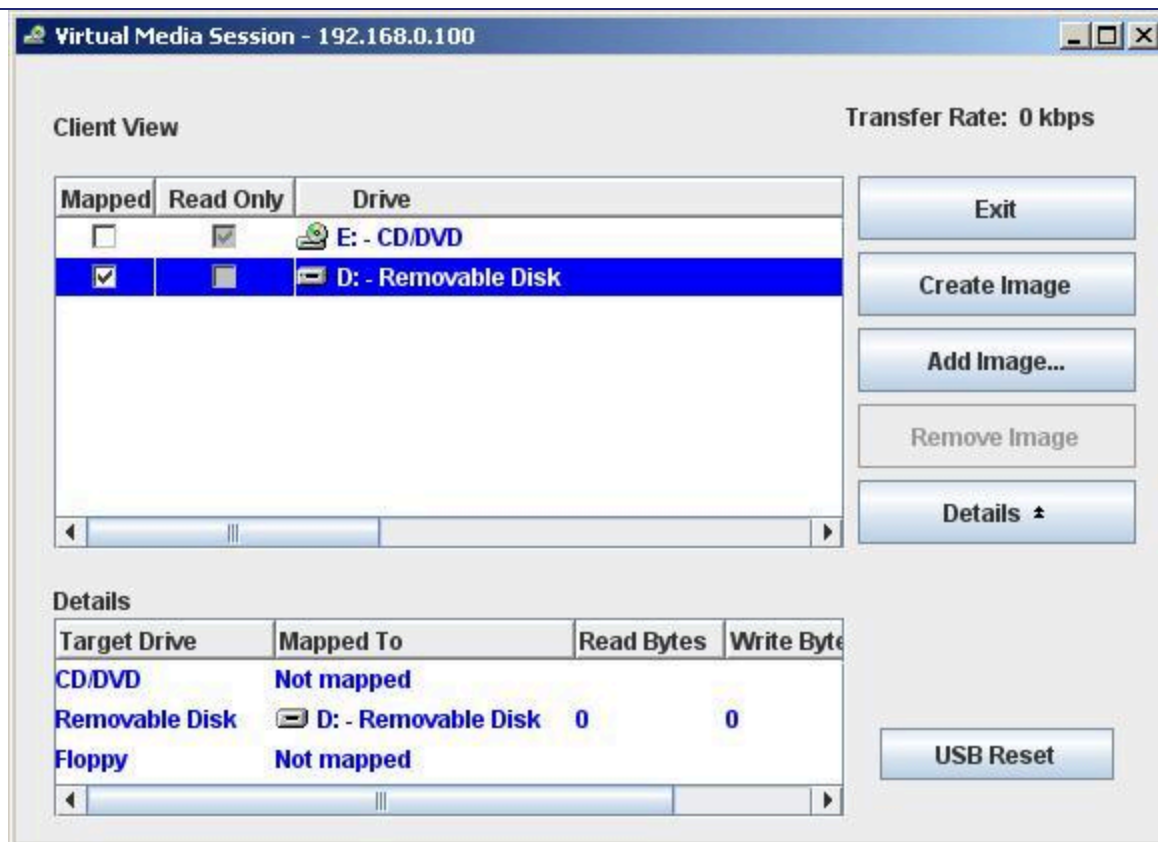
4. Click **Details**. The list of virtual devices, and the read/write activity for each device, appear.



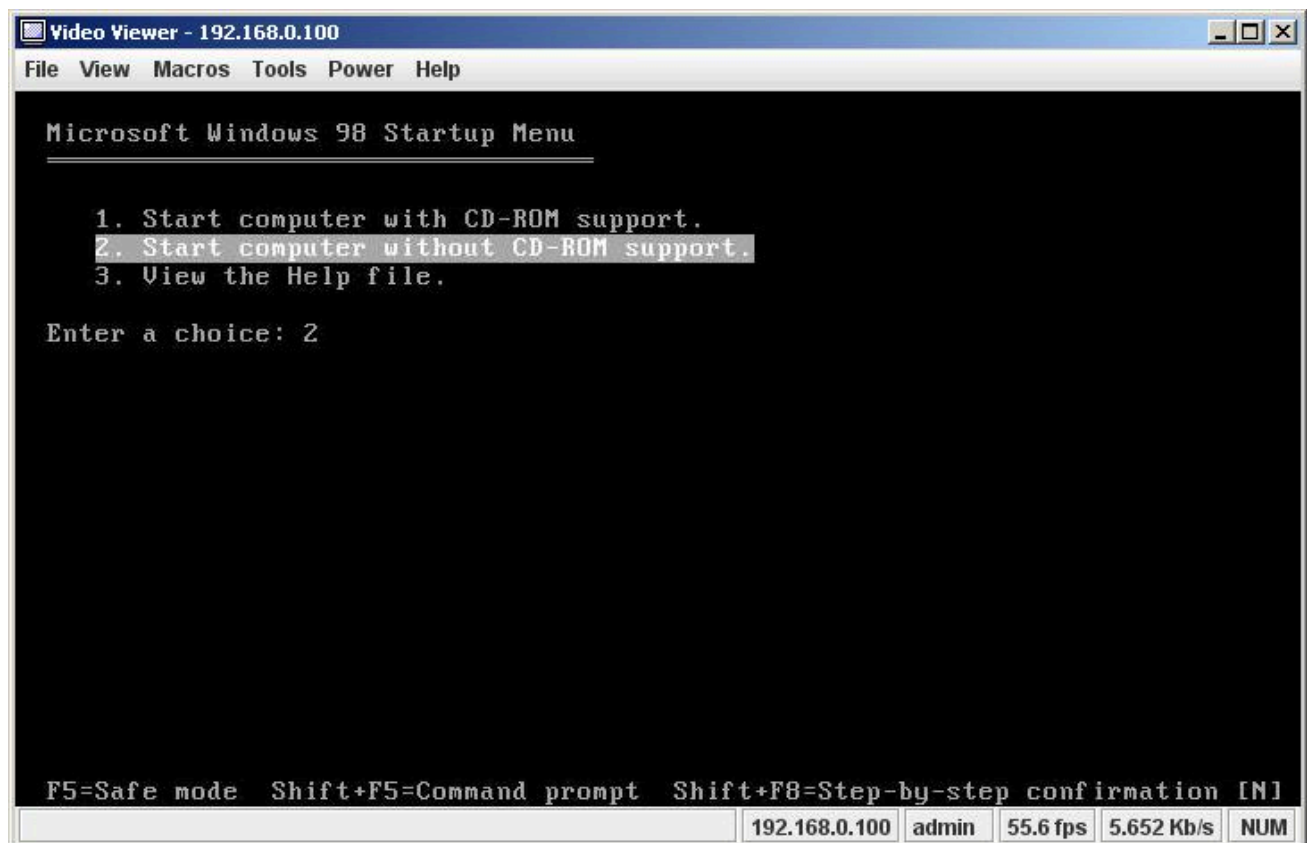
Note: The active session displays if a virtual media session is currently active either from the current GUI session or from any other GUI session.

5. To mount removable storage from the local to the remote server, check the box in the **Mapped** column of the table.

The device is mapped to the server.



The removable storage is now available as a boot source.





Note: The user must have Access Virtual Media permission to virtualize or disconnect a drive.

System BMC Management from WebGUI - 5G2U Appliances

This section describes how to configure the IPMI system from WebGUI.

- [Network Configuration - 5G2U Appliances](#)
- [Network Security Configuration - 5G2U Appliances](#)
- [Services Configuration - 5G2U Appliances](#)
- [Sessions Configuration - 5G2U Appliances](#)
- [Security Configuration - 5G2U Appliances](#)
- [Users Configuration - 5G2U Appliances](#)

Network Configuration - 5G2U Appliances

You can configure the IPMI system network using the WebGUI. The IPMI default IP address is "192.168.1.1".

To configure the IPMI system network using the WebGUI:

1. Log in to the IPMI as an Administrator. For more information, see [Logging In and Out of the WebGUI](#).
2. Select **Configuration > Network**.

The **Network General Settings** page appears.

- Server Management
 - Remote Control
 - Power Control
 - Remote KVM
 - Remote KVM Conf.
- BMC Management
 - Configuration
 - Network
 - Network Security
 - Services
 - Sessions
 - Certificate
 - Users
 - NTP
 - Utilities
 - Firmware
 - Reboot & Reset

Network

Apply Changes

Refresh

General Settings

Changes to the mode settings may change IP address settings. This may cause a brief loss in connectivity. Changes may not take effect immediately, and require a refresh.

Mode

1

Host Name

imperva

DNS Domain Name

Network Interface Configuration

Name	IF Enabled	IPv4 Enabled	IPv4 Address	IPv6 Enabled	IPv6 Address
eth1	Enabled	Enabled	10.1.11.82	Disabled	:::0

You can set the **Host Name** and the **DNS Domain Name**. Set the **DNS Domain Name** before changing any Network Configuration.

- Under **Network Interface Configuration**, click **eth1**.

The Network Interface Configuration page appears.

- Server Management
 - Remote Control
 - Power Control
 - Remote KVM
 - Remote KVM Conf.
- BMC Management
 - Configuration
 - Network
 - Network Security
 - Services
 - Sessions
 - Certificate
 - Users
 - NTP
 - Utilities
 - Firmware
 - Reboot & Reset

Network Interface Configuration eth1

Apply Changes

Refresh

Changes to the NIC IP address settings will close all user sessions and require IP address settings. All other changes will require the NIC to be reset, which may cause a brief loss in connectivity. Changes may not take effect immediately, and require a refresh.

Network Interface Settings

Device Type

Dedicated

MAC Address

08:35:71:00:24:73

Auto Negotiation

On

Network Speed

10 Gb

Duplex Mode

Full

General Settings

Enable Dynamic DNS

Use DHCP for DNS Domain Name

Respond to ARP

IPv4 Settings

Enabled

Use DHCP

IP Address

10.1.11.82

Subnet Mask

255.255.240.0

Gateway

10.1.1.254

Use DHCP to obtain DNS server addresses

Preferred DNS Server

10.2.16.66

Alternate DNS Server

IPv6 Settings

Enabled

Auto Configuration

IP Address 1

:::0

Gateway

Link Local Address

:::0

IP Address 2

:::0

Use DHCP to obtain DNS server addresses

Preferred DNS Server

Alternate DNS Server

VLAN Settings

Enable VLAN ID

VLAN ID

0

Priority

0

- Use this page to configure the IPMI network. See the table below for details. When done, click **Apply**

Changes.

Note: If you have forgotten the default IP address, you can reset it. Enter the BIOS, select the **Server Mgmt** tab, then set **IPMI Reset to def** to **Yes, on next boot**. Then log in again using the following parameter values:



- IP Address: 192.168.1.1
- Username: admin
- Password: <machine serial number>

For more information, see [Logging In and Out of the WebGUI](#).

Field Name	Description
Network Interface Settings	
MAC Address	Displays the MAC address of the interface that uniquely identifies this device on the network. This is read-only information.
Auto Negotiation	Toggles on/off the auto negotiation of the connection speed and duplex mode. This only applies if in dedicated NIC mode. The preset value is On.
Network Speed	Toggles the network speed to 10Mb or 100Mb to match your network environment. This option only applies in dedicated NIC mode and is not available if auto negotiation is set to On.
Duplex Mode	Toggles the duplex mode to Full or Half to match your network environment. This option only applies in dedicated NIC mode and is not available if auto

Field Name	Description
	negotiation is set to On.
General Settings	
Enable Dynamic DNS	Enables user to register this firmware address with the DNS when the checkbox is checked.
Use DHCP for DNS Domain Name	Enables or disables the DHCP to acquire the DNS Domain Name. If checked, the DNS Domain Name field is disabled, and the DNS Domain Name will be acquired from the DHCP server.
Respond to ARP	Enables to open Address Resolution Protocol (ARP) proxy respond.
IPv4 Settings (Enter 0.0.0.0 to clear an IPv4 field. Empty strings are not allowed for IPv4 addresses.)	
Enabled	Enables the availability of the IPv4 protocol when checked, and enables all other fields in this section except Use DHCP to obtain DNS server addresses.
Use DHCP	When enabled (checked), the IP address for the Appliance management NIC is obtained from the DHCP server and the IP Address, Subnet Mask and Gateway fields are deactivated. Use DHCP must be

Field Name	Description
	checked before you enable the Use DHCP to obtain DNS server addresses setting. (The default value is Off.)
IP Address	If Use DHCP is enabled (checked), the IP Address value is automatically supplied from the DHCPv4 server. If disabled, the IP Address value must be entered manually using format 192.168.1.1.
Subnet Mask	If Use DHCP is enabled (checked), the Subnet Mask value is automatically supplied from the DHCPv4 server. If disabled, the Subnet Mask value must be entered manually using format 255.255.255.0.
Gateway	If Use DHCP is enabled (checked), the Gateway value is automatically supplied from the DHCPv4 server. If disabled, the Gateway value must be entered manually using format 192.168.1.254.
Use DHCP to obtain DNS server addresses	If enabled (checked), the DNS server address is obtained from the DHCPv4 server. Use DHCP must be checked before you enable the Use DHCP to obtain DNS server addresses setting. If unchecked, you can specify the address manually in the following Preferred and Alternate DNS Server fields.

Field Name	Description
Preferred DNS Server	IP address of the preferred DNS server. Use format 0.0.0.0 if entering the address manually. To change this setting, you must first uncheck the <i>Use DHCP to obtain DNS Server Addresses</i> option.
Alternate DNS Server	IP address of the alternate DNS server. Use format 0.0.0.0 if entering the address manually. To change this setting, you must first uncheck the <i>Use DHCP to obtain DNS Server Addresses</i> option.
IPv6 Settings (Enter ; to clear IPv6 fields.)	
Enabled	Enables the availability of the IPv6 protocol when checked, and enables all other fields in this section except Use DHCP to obtain DNS server addresses.
Auto Configuration	When enabled (checked), the IPv6 address for the Appliance management NIC is obtained from the DHCPv6 server and the IP Address, Prefix Length and IP Gateway fields are deactivated. Auto Configuration must be checked before you enable the Use DHCP to obtain DNS server addresses setting. (The default value is Off.)
IP Address 1	If Auto Configuration is enabled (checked), the IP Address 1 value is

Field Name	Description
	automatically supplied from the DHCPv6 server. If disabled, the IP Address 1 value must be entered manually.
Prefix Length	If Auto Configuration is enabled (checked), the Prefix Length is automatically supplied from the DHCPv6 server. If disabled, the Prefix Length value must be entered manually (a value from 1 to 128).
Gateway	If Auto Configuration is enabled (checked), the Gateway value is automatically supplied from the DHCPv6 server. If disabled, the Gateway value must be entered manually.
Link Local Address	Specifies the IPv6 address for the Appliance management NIC (read-only).
IP Address 2	Specifies the additional IPv6 address for the Appliance management NIC if one is available (read-only).
Use DHCP to obtain DNS server addresses	If checked, the DNS server address is obtained from the DHCPv6 server. Auto Configuration must be checked before you enable the Use DHCP to obtain DNS server addresses setting. If unchecked, you can specify the address manually in the following Preferred and Alternate DNS Server

Field Name	Description
	fields.
Preferred DNS Server	Displays the IP address of the preferred DNS server. To change this setting, you must first uncheck the <i>Use DHCP to obtain DNS Server Addresses</i> option.
Alternate DNS Server	Displays the IP address of the alternate DNS server. To change this setting, you must first uncheck the <i>Use DHCP to obtain DNS Server Addresses</i> option.
VLAN Settings	
Enable VLAN ID	If enabled, only matched Virtual LAN (VLAN) ID traffic is accepted.
VLAN ID	Displays the VLAN ID of 802.1q fields (must be a number from 1 to 4094).
Priority	Priority field of 802.1q fields. To set the priority of the VLAN ID, enter a number from 0 to 7.

NOTE: For both IPv4 and IPv6 settings, when Use DHCP to obtain DNS server addresses is enabled, the IP address, Subnet Mask and Gateway fields are disabled. The Preferred DNS Server and Alternate DNS Server fields are also disabled.

Network Security Configuration - 5G2U Appliances

You can view the network security values. If you are logged in as an Administrator or Operator, you can modify

these values. The new values are available to the firewall immediately, but may not be utilized until the next security event occurs.

To configure Network Security from the WebGUI:

1. Log in to the IPMI as an Administrator. For more information, see [Logging In and Out of the WebGUI](#).
2. Select **Configuration > Network Security**.

The **Network Security** page appears. You can configure network security parameters. For more information, see the table below.

3. Make the desired changes. When done, click **Apply Changes**.

Field Name	Description
IP Blocking Enabled	Enables/disables the IP address blocking feature, which limits the number of failed login attempts from a specific IP address for a pre-selected time span.
IP Blocking Fail Count	Sets the number of login failures attempted from an IP address before the login attempts are rejected. The range is from 2-16.
IP Blocking Fail Window	Determines the time span (in seconds) that IP Blocking Fail Count failures must occur to trigger the IP Blocking Penalty Time. The range is from 10-65535 seconds.
IP Blocking Penalty Time	Sets the time span (in seconds) that login attempts from an IP address with excessive failures are rejected. The range is from 10-65535 seconds.

Services Configuration - 5G2U Appliances

You can view the services parameters. If you are logged in as an Administrator or Operator, you can modify these values.

To configure Network Security from the WebGUI:

1. Log in to the IPMI as an Administrator. For more information, see [Logging In and Out of the WebGUI](#).
2. Select **Configuration > Services**.

The **Services** configuration page appears. You can configure the service parameters. For more information, see the table below.

Web Server	
HTTP Port Number	80
HTTPS Port Number	443
Timeout	1800 seconds
Max Sessions	10
Active Sessions	2

3. Make the desired changes. When done, click **Apply Changes**.

Field Name	Description
HTTP Port Number	Port used by the embedded software that listens for a server connection. The preset value is 80.
HTTPS Port Number	Port used by the embedded software that listens for a secure server connection. The preset value is 443.
Timeout	Time that a connection is allowed to remain idle (60 to 10800 seconds). The session is canceled when the time-out is reached.
Max Sessions	Maximum number of simultaneous sessions allowed for this system.

Field Name	Description
Active Sessions	Number of current sessions on the system, less than or equal to the setting for Maximum Sessions.

Sessions Configuration - 5G2U Appliances

You can view information about the active sessions.

To view information about the active sessions:

1. Log in to the IPMI. For more information, see [Logging In and Out of the WebGUI](#).
2. Select **Configuration > Sessions**. The **Sessions** page appears.
3. Click **Refresh**.

Sessions

Use this page to view information about the active sessions. Additionally, privileged users can click on the trash can icon to kill an active session.

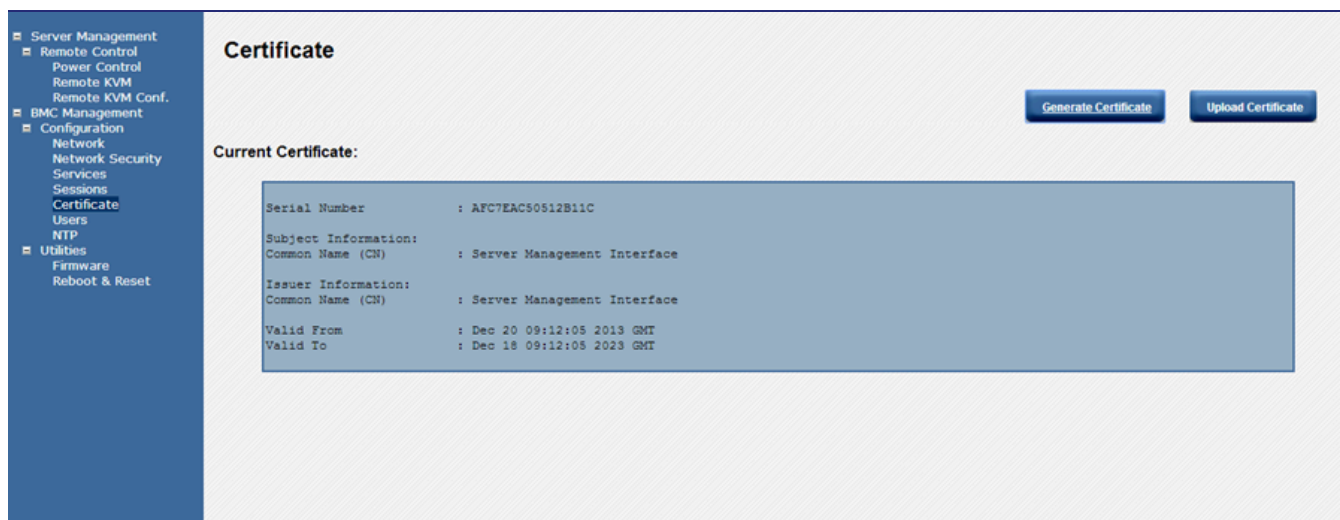
Session ID	User Name	IP Address	Session Type	Kill
1	admin	10.2.140.179	GUI	
2	admin	10.2.140.179	GUI	N/A

Refresh

Security Configuration - 5G2U Appliances

You can view the current server certificate. Secure server certificates ensure the identity of the remote system and ensure that information exchanged with the remote system cannot be viewed or changed by others. The list of SSL ciphers is included in your imported certificate.

If you are logged in as an Administrator or Operator, you can click generate a CSR (Certificate Signing Request). To ensure a secure system, generate a CSR, submit the CSR to a CertAgent and upload the certificate returned from the Certificate Authority (CA).



Certificate

Generate Certificate Upload Certificate

Current Certificate:

Serial Number	: AFC7EAC50512B11C
Subject Information:	
Common Name (CN)	: Server Management Interface
Issuer Information:	
Common Name (CN)	: Server Management Interface
Valid From	: Dec 20 09:12:05 2013 GMT
Valid To	: Dec 18 09:12:05 2023 GMT

To generate a CSR:

1. Log in to the IPMI as an Administrator. For more information, see [Logging In and Out of the WebGUI](#).
2. Select **Configuration > Certificate**. The **Certificate** page appears.
3. Click **Generate Certificate**.



Note: All properties are acquired from a CertAgent and must match the certificate returned from the CA.

Users Configuration - 5G2U Appliances

You can view the users information. If you are logged in as an Administrator or Operator, you can modify these values.

To modify user values:

1. Log in to the IPMI as an Administrator. For more information, see [Logging In and Out of the WebGUI](#).
2. Select **Configuration > Users**. The **Users** page appears. See the **Users** table below for more details.

- Server Management
 - Remote Control
 - Power Control
 - Remote KVM
 - Remote KVM Conf.
- BMC Management
 - Configuration
 - Network
 - Network Security
 - Services
 - Sessions
 - Certificate
 - Users
 - NTP
 - Utilities
 - Firmware
 - Reboot & Reset

Users

To configure a particular user, click the User ID.

User ID	State	User Name	User Role	IPMI LAN Privilege	IPMI Serial Privilege	Serial Over LAN
1	Disabled		None	None	None	Disabled
2	Enabled	admin	Administrator	Administrator	Administrator	Enabled
3	Disabled		None	None	None	Disabled
4	Disabled		None	None	None	Disabled
5	Disabled		None	None	None	Disabled
6	Disabled		None	None	None	Disabled
7	Disabled		None	None	None	Disabled
8	Disabled		None	None	None	Disabled
9	Disabled		None	None	None	Disabled
10	Disabled		None	None	None	Disabled
11	Disabled		None	None	None	Disabled
12	Disabled		None	None	None	Disabled
13	Disabled		None	None	None	Disabled
14	Disabled		None	None	None	Disabled
15	Disabled		None	None	None	Disabled
16	Disabled		None	None	None	Disabled

3. Click the ID of the user whose account you wish to configure. The **User Configuration** page for that user appears. See the **User Configuration** table below for more details.

- Server Management
 - Remote Control
 - Power Control
 - Remote KVM
 - Remote KVM Conf.
- BMC Management
 - Configuration
 - Network
 - Network Security
 - Services
 - Sessions
 - Certificate
 - Users
 - NTP
 - Utilities
 - Firmware
 - Reboot & Reset

User Configuration

[Apply Changes](#) [Delete](#) [Back](#)

General

User ID	2
Enable User	☒
User Name	admin
Change Password	☐
New Password	
Confirm New Password	

User Privileges

User Role	Administrator
IPMI LAN Privilege	Administrator
IPMI Serial Privilege	Administrator
Enable Serial Over LAN	☒

4. Make the changes in the values as desired, then click **Apply Changes**.

Users

Field Name	Description
User ID	Displays a sequential User ID number. If the user has configure privilege, the User ID is a hyperlink that will load a page specific to the selected user where the user settings can be modified.
State	Indicates the status of each User ID, either enabled or disabled (default).

Field Name	Description
User Name	Displays the login name of the user.
User Role	Displays the assigned role of each user. None indicates no role is assigned.
IPMI LAN Privilege	Displays the assigned IPMI LAN privilege of each user (Administrator, Operator, User or None).
IPMI Serial Privilege	Displays the assigned IPMI serial privilege of each user (Administrator, Operator, User or None).
Serial Over LAN	Indicates the status of Serial Over LAN privilege for each user, either enabled or disabled.

User Configuration

Field Name	Description
General	
User ID	One of 16 selected User IDs. Unchecked sets the preset value.
Enable User	Selecting this allows user to access the embedded software.
User Name	Up to 16 characters consisting of alphanumeric (a-z, A-Z and 0-9) characters, underscore (_) and dash (-). Click the Apply Changes button. If validation fails, the GUI displays an error message.
Change Password	A checked box enables user to change the password. New Password and Confirm Password text fields will be enabled also. Unchecked sets the preset value.
New Password	New Password for selected user. A checked box enables user

Field Name	Description
	to enter a new password using up to 16 printable US-ASCII (Code: 33-126) characters. After changes are made, click the Apply Changes button. If validation fails, the GUI displays an error message. New Password for selected user.
Confirm New Password	Re-enter new password to confirm.
User Privileges	
User Role	Role of the user (Administrator, Operator or User) can be selected from the drop-down list.
IPMI Serial Privilege	Role of the IPMI Serial Privilege (Administrator, Operator, User or None).
IPMI LAN Privilege	Role of the IPMI LAN Privilege (Administrator, Operator, User or None).
Enable Serial Over LAN	Enables the user to have Serial Over LAN privilege (uncheck for the preset value).

To change a user's password:

1. Log in to the IPMI as an Administrator. For more information, see [Logging In and Out of the WebGUI](#).
2. Select **Configuration > Users**. The **Users** page appears.
3. Click the ID of the user whose password you wish to change. The **User Configuration** page for that user appears.
4. Check the **Change Password** box.
5. Enter a new password in the **Password** field.
6. Retype the password in the **Confirm Password** field to confirm the password.
7. Click **Apply Changes**.

Note:



Make sure your password has the following characteristics:

- It must have no fewer than 10 characters and no more than 14 characters.
- It must have at least one number, one capital letter, and one special character from:

* + = # % ^ : / ~ . , [] _

- It cannot have more than two characters repeated in succession.

Utilities - 5G2U Appliances

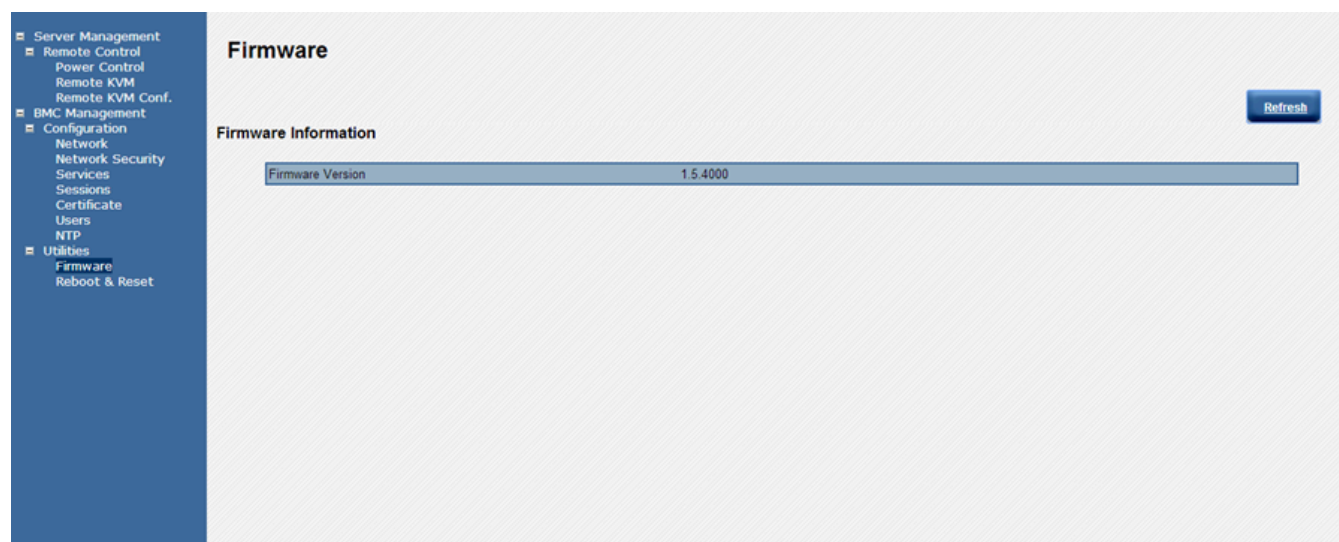
This section presents additional features in the WebGUI.

- Firmware version - 5G2U Appliances
- Reboot and Reset - 5G2U Appliances

Firmware version - 5G2U Appliances

To view the firmware version:

1. Log in to the IPMI. For more information, see [Logging In and Out of the WebGUI](#).
2. Select **Utilities >Firmware**. The current IPMI firmware version is displayed.



Reboot and Reset - 5G2U Appliances

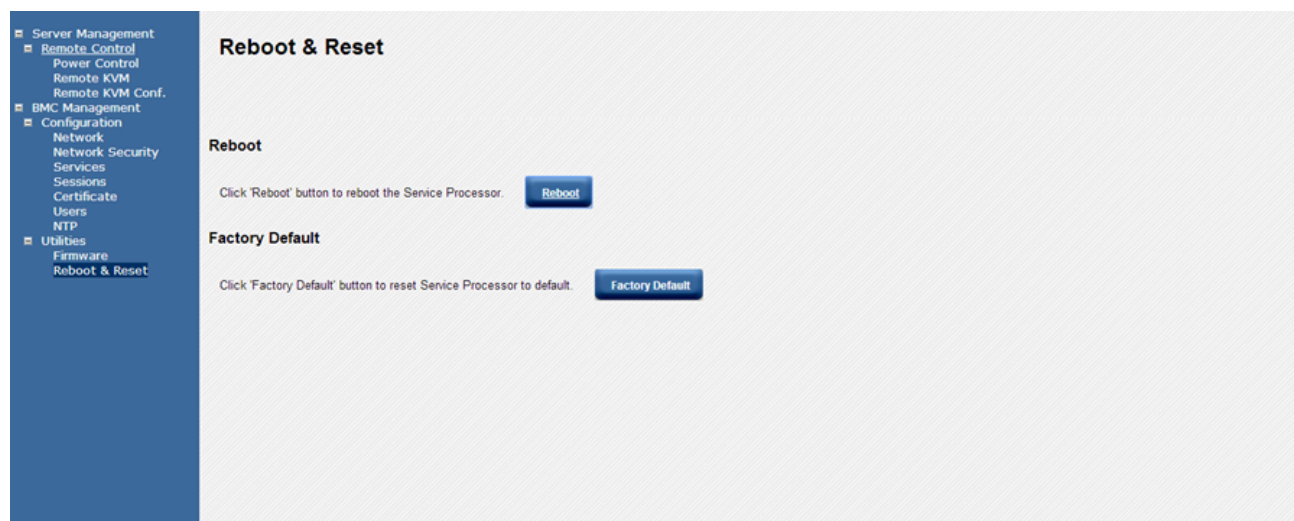
You can reboot the IPMI. You can also restore factory defaults.

To reboot the IPMI Service Processor:

1. Log in to the IPMI. For more information, see [Logging In and Out of the WebGUI](#).
2. Select **Utilities > Reboot & Reset**.
3. Click Reboot.

To restore factory defaults:

1. Log in to the IPMI. For more information, see [Logging In and Out of the WebGUI](#).
2. Select **Utilities > Reboot & Reset**.
3. Click **Factory Default**.



Using the ipmitool Utility - 5G2U Appliances

The ipmitool utility is used for controlling IPMI-enabled devices. This utility lets you manage IPMI functions of either the local system, via a kernel device driver, or a remote system. These functions include printing FRU information, LAN configuration, sensor readings, and remote chassis power control.

To use the ipmitool:

1. Connect to the appliance via SSH and log in.
2. Run the command `ipmctl platform ipmi load-modules`
3. Run the command `ipmitool` with any of its options or commands



Note: You must follow this procedure after every time the appliance is powered down.

IPMI - 6G2U Appliances

IPMI (Intelligent Platform Management Interface) involves the use of a dedicated management channel for server maintenance. It allows a system administrator to monitor and manage servers by remote control regardless of whether the machine is powered on, or if the SecureSphere system is running or not.

A complete remote management system allows remote reboot, shutdown, powering on; broadcasting of video output to remote terminals and receiving of input from remote keyboard and mouse (KVM over IP). It also can access local media like a DVD drive, or disk images, from the remote machine. If necessary, this allows one to perform remote installation of the operating system.

The remote system is accessed through various web browsers.

- [Preface to IPMI for 6G2U Appliances](#)
- [IPMI and System Management Overview - IPMI for 6G2U Appliances](#)
- [Using the IPMI WebGUI - IPMI for 6G2U Appliances](#)
- [Menu Bar - IPMI for 6G2U Appliances](#)
- [Dashboard - IPMI for 6G2U Appliances](#)
- [Configuration - IPMI for 6G2U Appliances](#)
- [Remote Control - IPMI for 6G2U Appliances](#)
- [Maintenance - IPMI for 6G2U Appliances](#)
- [Firmware Update - IPMI for 6G2U Appliances](#)

Preface to IPMI for 6G2U Appliances

This section provides instructions for managing the server using the IPMI. IPMI is included on certain servers. If users have one of these servers, it will include an IPMI Supplement which contains platform-specific information, such as sensors and thresholds, and details about the hardware.

IPMI and System Management Overview - IPMI for 6G2U Appliances

IPMI is a dedicated system of hardware and supporting software that allows users to manage a server independent of the operating system. The IPMI includes the following components:

- **Service Processor:** This is the hardware. It consists of a dedicated processor board that communicates through the system serial port and a dedicated Ethernet port.
- **WebGUI:** The WebGUI provides a powerful, easy-to-use browser interface that allows users to log in to perform system management, monitoring, and IPMI tasks. Users only need to install the Java client appliance on first use.
- **Remote Console/Java™ Client:** The Java client supports the Remote Console functionality, which allows the user to access the server's console remotely. It redirects the keyboard and video screen, and can redirect input and output from the local machine's storage such as CD/DVD-ROM and hard disk drives.

Using the IPMI WebGUI - IPMI for 6G2U Appliances

This section describes how to use the IPMI WebGUI.

- [Overview of WebGUI Features - IPMI for 6G2U Appliances](#)
- [Requirements before using WebGUI - IPMI for 6G2U Appliances](#)
- [Users and Privileges - IPMI for 6G2U Appliances](#)
- [Logging In and Out of the WebGUI - IPMI for 6G2U Appliances](#)

Overview of WebGUI Features - IPMI for 6G2U Appliances

The UI enables the user to monitor and manage remote systems. The user can quickly activate the WebGUI using a web browser. One of the features of the IPMI is the ability to redirect the server's graphical console to a remote workstation or laptop system. When the user redirects the host console, the user can:

- Configure the remote system's keyboard and mouse to act as the server's mouse and keyboard
- Configure the disk drive, or CD/DVD-ROM drive on the remote system as a device virtually connected to the server
- Redirect CD/DVD-ROM images for remote access

Requirements before using WebGUI - IPMI for 6G2U Appliances

The WebGUI has been tested successfully with recently released popular web browsers, and may be compatible with other web browsers.

Notes:



- You need to install Java™ on the host system. The IPMI remote console currently supports Java versions 8. To work with Java version 8, you need to perform an IPMI firmware version upgrade. For information on performing this upgrade, contact Imperva Support
- Java Web Start 1.6 is required to launch the KVM over an IPv6 network

Users and Privileges - IPMI for 6G2U Appliances

After logging in, users can do the following actions:

- Basic software provisioning
- Intelligent Platform Management Interface (IPMI) tasks
- System monitoring

IPMI user accounts include a role which defines what the user can do.

- **Administrator:** Enables full access to functions and commands.
- **Operator:** Enables limited access to functions and commands.
- **User:** Enables more limited access to functions and commands.



Note: Operators and users cannot change their own assigned roles or privileges.

Logging In and Out of the WebGUI - IPMI for 6G2U Appliances

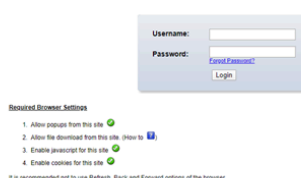
When the server is cabled appropriately and connected to an AC supply, the IPMI boots up automatically. Booting up is a very fast process. The default Ethernet configuration is the static IP Address. However, if the management Ethernet is not connected, or if the IPMI's Dynamic Host Configuration Protocol (DHCP) process fails due to the absence of a DHCP server on the management network, the IPMI may take a few minutes to boot. If you are using a browser proxy server, disable it. This may speed up access to the management network.

If you want to refresh information like the sensor reading on the web pages, or you want to logout from the web page, use the Refresh or Log Out buttons at the top right of the WebGUI window. Do not use the Refresh or Close window buttons in the browser.

To log in to the WebGUI:

1. Enter the IP address of the IPMI into web browser: **https://192.168.1.1**. The WebGUI login screen is displayed.

imperva



2. Enter your Username and Password. The default values for an Administrator are:

- Username: **admin**
- Password: **<appliance's serial number>**



Notes:

- The machine serial number appears either on a sticker on the machine, or on the packaging, or both. Alternatively, SSH to the machine and use the command `impctl platform dmi show | grep Serial`.

- Imperva strongly recommended that you change your password from the default. For more information, see [Users](#).

1. Click **Login**. The Dashboard screen is displayed.



To log out of the WebGUI, click the **Logout** button at the top right of the WebGUI. The login screen is displayed.

Menu Bar - IPMI for 6G2U Appliances

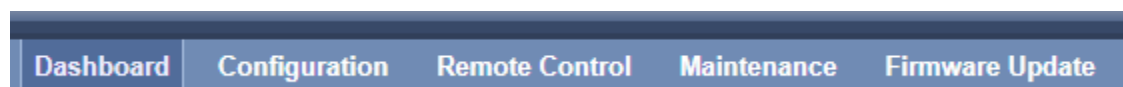
This section describes the Menu Bar of the WebGUI. The Menu Bar consists of:

- Left menu bar
- Right menu bar
- Left Menu Bar - IPMI for 6G2U Appliances
- Right Menu Bar - IPMI for 6G2U Appliances

Left Menu Bar - IPMI for 6G2U Appliances

The left menu bar consists of the following buttons:

- Dashboard
- Configuration
- Remote Control
- Maintenance
- Firmware Update



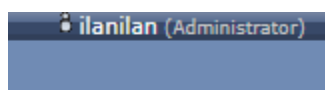
Right Menu Bar - IPMI for 6G2U Appliances

The right menu bar consists of the following:

- Logged-in user information

The logged-in user information shows the logged-in user name and privilege. The following types of privileges are available:

- User - Only valid commands are allowed
- Operator – All commands are allowed except for the configuration commands that can change the behavior of the out-of-hand interfaces
- Administrator – All commands are allowed
- No Access – Login access denied



- Quick buttons

The available quick buttons are:

- Refresh – Reload the current page
- Print – Print out of the current page
- Logout – Log out of the webGUI
- HELP – View the help page



Dashboard - IPMI for 6G2U Appliances

The Dashboard page gives the overall information about the status of a device. This information includes:

- Device Information - The displayed information is:
 - Firmware Revision – The revision number of the firmware.
 - Firmware Build Time – This field shows the date and time on which the firmware is built.
- Network Information - The displayed information is:
 - MAC Address – Read only field shows the MAC address of the device.
 - V4 Network Mode – The v4 network mode of the device could be either disable, static or DHCP.
 - IPv4 Address – The IPv4 address of the device (could be static or DHCP).
 - V6 Network Mode – The v6 network mode of the device could be either disable, static or DHCP.
 - IPv6 Address – The IPv6 address of the device.



Note: You can edit the Network settings by clicking Edit

- Remote Control - To redirect the host remotely, click the Launch button. This downloads the jviewer.jnlp file which after downloaded and launched opens the Java redirection window.



Note: If you want to Launch JViewer from the Dashboard page, the KVM option should be enabled in the Extended Privileges for the logged-in user.



Configuration - IPMI for 6G2U Appliances

This section describes how to configure the system using the WebGUI. The items under this menu allow you to access various configuration settings.

- Active Directory - IPMI for 6G2U Appliances
- DNS - IPMI for 6G2U Appliances
- Images Redirection - IPMI for 6G2U Appliances
- Mouse Mode - IPMI for 6G2U Appliances
- Network - IPMI for 6G2U Appliances
- Network Link - IPMI for 6G2U Appliances
- NTP - IPMI for 6G2U Appliances
- PAM Order - IPMI for 6G2U Appliances
- PEF - IPMI for 6G2U Appliances
- RADIUS - IPMI for 6G2U Appliances
- Remote Session - IPMI for 6G2U Appliances
- Services - IPMI for 6G2U Appliances
- SMTP - IPMI for 6G2U Appliances

- SSL - IPMI for 6G2U Appliances
- System Firewall - IPMI for 6G2U Appliances
- Users - IPMI for 6G2U Appliances
- Virtual Media - IPMI for 6G2U Appliances

Active Directory - IPMI for 6G2U Appliances

An active directory is a directory structure used on Microsoft Windows based computers and servers to store information and data about networks and domains. An active directory (sometimes referred to as AD) does a variety of functions including the ability to provide information on objects. It also helps to organize these objects for easy retrieval and access, allows access by end users and administrators and allows the administrator to set up security for the directory.

Active Directory allows you to configure the Active Directory Server Settings. The displayed table shows any configured Role Groups and the available slots. You can modify, add or delete role groups from here. Group domain can be the AD domain or a trusted domain. Group Name should correspond to the name of an actual AD group.



Note: To view the page, you must be at least a User and to modify or add a group, you must be an Administrator.

To access the Active Directory Settings page:

1. Login to the WebGUI.
2. Click **Configuration > Active Directory**. The Active Directory Settings screen is displayed.

Active Directory Settings Fields

Field Name	Description
Advanced Settings	This option is used to configure Active Directory Advanced Settings. Options are Enable Active Directory Authentication, Secret User Name, Secret Password, User Domain name, Time Out and up to three Domain Controller Server Addresses.

Field Name	Description
Role Group Name	Displays the name that identifies the role group in the Active Directory. Notes: ▪ Role Group Name is a string of 255 alpha-numeric characters. ▪ Special symbols hyphen and underscore are allowed.
Group Name	Displays this name identifies the role group in Active Directory. Notes: ▪ Role Group Name is a string of 255 alpha-numeric characters. ▪ Special symbols hyphen and underscore are allowed.
Group Domain	Displays the domain where the role group is located. Notes: ▪ Role Group Name is a string of 255 alpha-numeric characters. ▪ Special symbols hyphen and underscore are allowed.
Group Privilege	Displays the level of privilege to assign to this role group.
Add Role Group	To add a new role group to the device.
Modify role Group	To modify the existing role group.
Delete Role Group	To delete an existing Role Group.

- [Advanced Active Directory Settings - IPMI for 6G2U Appliances](#)
- [Add New Role Group - IPMI for 6G2U Appliances](#)
- [Modify a Role Group - IPMI for 6G2U Appliances](#)
- [Delete a Role Group - IPMI for 6G2U Appliances](#)

Advanced Active Directory Settings - IPMI for 6G2U Appliances

You can perform advanced settings for the active directory you configured.

To perform advanced settings:

1. In the Active Directory Settings page, click **Advanced Settings**. The Advanced Active Directory Settings window is displayed.

2. In the Active Directory Settings page, select or clear the **Enable** check box to enable or disable Active Directory Authentication respectively.



Note: If you enable Active Directory Authentication, you need to enter the required information to access the Active Directory server.

3. Specify the Secret user name and password in the Secret User Name and Secret Password fields respectively.

Notes:



- Secret username/password for AD is not mandatory. If the AD's secret username/password is not provided, AD should be kept in the last location in PAM order.
- User Name is a string of 1 to 64 alpha-numeric characters. It must start with an alphabetical character and it is case-sensitive.
- Special characters like comma, period, colon, semicolon, slash, backslash, square brackets, angle brackets, pipe, equal, plus, asterisk, question mark, ampersand, double quotes, space are not allowed.
- Password must have no fewer than 7 characters and no more than 14 characters.
- Password must have at least one number, one capital letter, and one special character from:

* + = # % ^ : / ~ . , [] _

- Password cannot have more than two characters repeated in succession.

4. Specify the Domain Name for the user in the User Domain Name field. For example, My-Domain.com.
5. Specify the time (in seconds) to wait for Active Directory queries to complete in the Time Out field.

**Notes:**

- Default Time out value: 120 seconds.
- Range from 15 to 300 allowed.

6. Configure IP addresses in Domain Controller Server Address1, Domain Controller Server Address2 and Domain Controller Server Address3.

**Notes:**

- For the IP address of Active Directory server, at least one Domain Controller Server Address must be configured.
- The IP Address must be made of 4 numbers separated by dots as in "xxx.xxx.xxx.xxx".
- Each number ranges from 0 to 255.
- First number must not be 0.
- Domain Controller Server Addresses will supports IPv4 Address format and IPv6 Address format.

7. Click **Save** to save the entered settings and return to Active Directory Settings Page.
8. Click **Cancel** to cancel the entry and return to Active Directory Settings Page.

Add New Role Group - IPMI for 6G2U Appliances

You can add a new Role Group in the Active Directory Settings page.

To add a new Role Group:

1. In the Active Directory Settings page, select a blank row and click **Add Role Group** or alternatively double click on the blank row to open the Add Role group page.

2. In the Role Group Name field, enter the name that identifies the role group in the Active Directory.

**Notes:**

- Role Group Name is a string of 255 alpha-numeric characters.
- Special symbols hyphen and underscore are allowed.

3. In the Role Group Domain field, enter the domain where the role group is located.

**Notes:**

- Domain Name is a string of 255 alpha-numeric characters.
- Special symbols hyphen, underscore and dot are allowed.

4. In the Role Group Privilege field, enter the level of privilege to assign to this role group.
5. In the Extended Privileges option, select the one of the options: **KVM** or **VMedia**.
6. Click **Add** to save the new role group and return to the Role Group List.
7. Click **Cancel** to cancel the settings and return to the Role Group List.

Modify a Role Group - IPMI for 6G2U Appliances

You can modify a Role Group in the Advanced Directory Settings page.

To modify a Role Group:

1. In the Advanced Directory Settings page, select the row that you wish to modify and click **Modify Role Group** or double click the row that you wish to modify.
2. Make the necessary changes and click **Save**.

Delete a Role Group - IPMI for 6G2U Appliances

You can delete a Role Group in the Advanced Directory Settings page.

To delete a Role Group:

1. In the Advanced Directory Settings page, select the row that you wish to delete.
2. Click **Delete Role Group**.

DNS - IPMI for 6G2U Appliances

The Domain Name System (DNS) is a distributed hierarchical naming system for computers, services, or any resource connected to the Internet or a private network. It associates the information with domain names assigned to each of the participants. Most importantly, it translates domain names meaningful to humans into the numerical (binary) identifiers associated with networking equipment for the purpose of locating and addressing these devices worldwide.

The DNS Server settings page is used to manage the DNS settings of a device.

To open DNS Server Settings page, click **Configuration > DNS** from the menu bar. The DNS Server Settings screen is displayed.

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Dashboard Configuration Remote Control Maintenance Firmware Update

Manage DNS settings of the device.

Domain Name Service Configuration

DNS Service ☒ Enable

Host Configuration

Host Settings

Host Name

Register BMC

eth1 ☒ Register BMC ☐ No update ☐ DHCP Client FQDN ☐ Hostname

Domain Name Configuration

Domain Settings

Domain Name

Domain Name Server Configuration

DNS Server Settings

IP Priority ☒ IPv4 ☐ IPv6

DNS Server1

DNS Server2

DNS Server3

Save Reset

DNS Server Settings Fields

Field Name	Description
Domain Name Service Configuration	
DNS Service	To enable/disable all the DNS Service Configurations.
Multicast DNS Support	
mDNS Settings	To enable/disable the mDNS Support Configurations.
Host configuration	
Host Settings	Choose either Automatic or Manual settings. Note: If you choose Automatic, you need not enter the Host Name and if you choose Manual, you need to enter the Host Name.
Host Name	Displays host name of the device. If the Host setting is chosen as Manual, then specify the host name of the device. Notes: - Value ranges from 1 to 64 alpha-numeric characters. - Special characters '-'(hyphen) and '_'(underscore) are allowed. - It must not start or end with a '-'(hyphen). IE browsers won't work correctly if any part of the host name contain underscore (_)character.

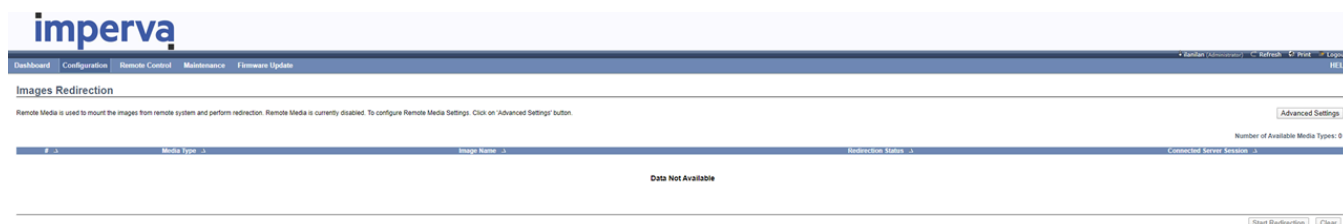
Field Name	Description
Register BMC)	Option to register the BMC either through NS Update or DHCP Client FQDN or Hostname.
Domain Name Configuration	
Domain Settings	Lists the option for domain interface as Manual, v4 or v6 for multiLAN channels. Note: If you choose DHCP, then select v4 or v6 for DHCP servers.
Domain Name	Displays the domain name of the device. If the Domain setting is chosen as Manual, then specify the domain name of the device. If you chose Automatic, the Domain Name cannot be configured as it will be done automatically. The field will be disabled.
Domain Name Server Configuration	
DNS Server Settings	Lists the option for v4 DNS settings for the device, Manual and available LAN interfaces.
IP Priority	If IP Priority is IPv4, it will have 2 IPv4 DNS servers and 1 IPv6 DNS server. If IP Priority is IPv6, it will have 2 IPv6 DNS servers and 1 IPv4 DNS server. Note: This is not applicable for Manual configuration.
DNS Server 1, 2 & 3	To specify the DNS (Domain Name System) server address to be configured for the BMC. Notes: ▪ IPv4 Address made of 4 numbers separated by dots as in "xxx.xxx.xxx.xxx". ▪ Each number ranges from 0 to 255. ▪ First number must not be 0. ▪ IPv4/IPv6 Address format.
Save	To save the entered changes.

Field Name	Description
Reset	To reset the entered changes.

Images Redirection - IPMI for 6G2U Appliances

This page is used to configure for redirection. This can be done by mounting the image from the remote system, Remote Media.

To open the Images Redirection page, click **Configuration > Images Redirection** from the menu bar. The Images Redirection screen is displayed.



- Advanced Images Redirection Settings - IPMI for 6G2U Appliances
- Remote Media Server Image Configuration - IPMI for 6G2U Appliances

Advanced Images Redirection Settings - IPMI for 6G2U Appliances

You can perform advanced settings for the media types you configured.

To perform advanced settings, in the Images Redirection page click **Advanced Settings**. The Advanced Media Settings window is displayed.

Advanced Media Settings Fields

Field Name	Description
Remote Media	

Field Name	Description
Remote Media Support	To enable or disable Remote Media support, select/clear the Enable check box.
Enable Media Types	To enable or disable Media Types support, select/clear the desired media type check box. Note: You can configure different settings for different remote media types by enableing corresponding media types.
All Media Settings	
Server Address	Server address where the remote media images are stored.
Source Path	Source path where the remote media images are stored.
Share Type	Share Type of the remote media server either NFS/Samba(CIFS).
Username, Password and Domain Name	If share Type is selected as Samba(CIFS), then user credentials need to authenticate the server. In that case, enter the valid details in the respective fields. Note: Domain Name field is optional.
Save	Click Save to save the settings.
Cancel	Click Cancel to cancel the modifications and return to Image list.

Remote Media Server Image Configuration - IPMI for 6G2U Appliances

In the Images Redirection page, the displayed table shows configured images on BMC. You can configure images of the remote media server here.

Notes:

- More than one image can be configured for each image type. At maximum 4 images can be configurable.
- To configure the image, You need to enable Remote Media support using 'Advanced Settings'.
- To start/stop redirection and to delete an image, you must have Administrator Privileges.
- Free slots are denoted by “ ”.

To Start/Stop Redirection and configure remote media images:

1. In the Images Redirection page, click **Advanced Settings** and make sure the Remote Media Support check box is selected.



Note: The Start Redirection button is active only for VMedia enabled users.

2. Return to the Images Redirection page and select a configured slot and click **Start Redirection** to start the remote media redirection. A pop-up message appears stating Local Media Redirection is stopped.
3. Click **OK**. If the image is successfully redirected, the button changes to Stop Redirection.
4. Click **Stop Redirection** to stop the remote media redirection. A pop-up message appears stating Local Media Redirection is stopped.
5. Click **OK**.

To clear an image:

1. Click the configured slot.
2. Select an image to be deleted from the drop down box.
3. Click **Clear Image**. A verification pop-up message appears.
4. Click **OK**. The image is cleared.



Note: Redirection needs to be stopped to clear the image.

Mouse Mode - IPMI for 6G2U Appliances

In webGUI, Redirection Console handles mouse emulation from local window to remote screen in either of three

methods. User has to be an Administrator to configure this option.

To open the Mouse Mode page, click **Configuration > Mouse Mode** from the menu bar. The Mouse Mode Settings screen is displayed.

To modify redirection console mouse mode settings:

1. In the Mouse Mode Settings page, select one of the mouse modes:
 - Absolute Mode - The absolute position of the local mouse is sent to the server. This mode is applicable for all Windows versions, versions above RHEL6, and versions above FC14.
 - Relative Mode - This mode is applicable for all Linux versions, versions less than RHEL6, and versions less than FC14.
 - Other Mode - The calculated displacement from the local mouse in the center position is sent to the server. This mode is recommended for SLES-11 OS Installation.
2. Click **Save** button to save the changes made.
3. Click **Reset** to reset the modified changes.

Network - IPMI for 6G2U Appliances

In WebGUI, the Network Settings page is used to configure the network settings for the available LAN channels.

To open the Network Settings page, click **Configuration > Network** from the menu bar. The Network Settings screen is displayed.

Network Settings Fields

Field Name	Description
LAN Interface	Lists the LAN interfaces.
LAN Settings	To enable or disable the LAN Settings.
MAC Address	This field displays the MAC Address of the device. This is a read only field.
IPv4 Configuration	
IPv4 Settings	This option is to enable/disable the IPv4 settings in the device.
Obtain IP Address automatically	This option is to dynamically configure IPv4 address using DHCP (Dynamic Host Configuration Protocol).
IPv4 Address, Subnet Mask, and Default Gateway	These fields are for specifying the static IPv4 address, Subnet Mask and Default Gateway to be configured to the device. Notes: ▪ IP Address made of 4 numbers separated by dots as in xxx.xxx.xxx.xxx". ▪ Each Number ranges from 0 to 255. ▪ First Number must not be 0.
IPv6 Configuration	
IPv6 Settings	This option is to enable/disable the IPv6 settings in the device.
Obtain an IPv6 address automatically	This option is to dynamically configure IPv6 address using DHCP (Dynamic Host Configuration Protocol).
IPv6 Address	To specify a static IPv6 address to be configured to the device. For example, 2004::2010
Subnet Prefix length	To specify the subnet prefix length for the IPv6 settings.

Field Name	Description
	Note: Value ranges from 0 to 128.
Default Gateway	Specify v6 default gateway for the IPv6 settings. Note: If core feature IPV6_COMPLIANCE is enabled, the IPV6 default Gateway field will not be displayed.
VLAN Configuration	
VLAN Settings	To enable/disable the VLAN support for selected interface.

Network Link - IPMI for 6G2U Appliances

In webGUI, this page is used to configure the network link configuration for available network interfaces.

To open the Network Link page, click **Configuration > Network Link** from the menu bar. The Network Link Configuration screen is displayed.



Network Link Configuration Fields

Field Name	Description
LAN Interface	Select the required network interface from the list to which the Link speed and duplex mode to be configured.
Auto Negotiation	This field allows the device to perform automatic configuration to achieve the best possible mode of operation (speed and duplex) over a link. It can be ON/OFF.

Field Name	Description
Link Speed	A list of all the supported capabilities of the network interface. It can be 10/100/1000 Mbps. This field is active only when Auto Negotiation is set to OFF .
Duplex Mode	Duplex Mode could be either Half Duplex or Full Duplex. This field is active only when Auto Negotiation is set to OFF .
Save	To save the settings.
Reset	To reset the modified changes.

NTP - IPMI for 6G2U Appliances

The Network Time Protocol (NTP) is a protocol for synchronizing the clocks of computer systems over packet-switched, variable-latency data networks. It is designed particularly to resist the effects of variable latency by using a jitter buffer.

In WebGUI, this page displays the device current date and time settings. It can be used to configure either Date & Time or NTP server settings for the device.

To open the NTP Settings page, click **Configuration > NTP** from the menu bar. The NTP Settings screen is displayed.

NTP Settings Fields

Field Name	Description
Date	To specify the current date for the device.

Field Name	Description
	Note: This field is enabled only when the Automatically synchronizes Date & Time with NTP Server check box is not selected.
Time	To specify the current time for the device. Notes: - This field is enabled only when the Automatically synchronizes Date & Time with NTP Server check box is not selected. - As Year 2038 Problem exists, Date and Time should be configured within the range.
TimeZone	Lists the UTC offset along with the locations and Manual UTC offset for NTP server, which can be used to display the exact local time.
Primary NTP Server & Secondary NTP Server	Supports IP Address (Both IPv4 and IPv6 format)/FQDN (Fully qualified domain name) format/FQDN Value ranges from 1 to 128 alpha-numeric characters. Note: The Secondary NTP server is an optional field. If the Primary NTP server is not working fine, then the Secondary NTP Server is selected.
Automatically synchronize Date & Time with NTP Server	Select or clear check box to automatically synchronize Date and Time with the NTP Server.
Refresh	To reload the current date and time settings.
Save	To save the settings.
Reset	To reset the modified changes.

PAM Order - IPMI for 6G2U Appliances

This page is used to configure the PAM order for user authentication in to the BMC.

To open the PAM Order page, click **Configuration > PAM Order** from the menu bar. The PAM Order screen is

displayed.



To configure the PAM order for user authentication:

1. Select the required PAM module and click the UP/DOWN arrow button to move the module one step before/after the existing module.

Notes:



- Whenever the configuration is modified, the web server will be restarted automatically. Logged-in session will be logged out.
- If AD Authentication fails, the reason of fail could be invalid User or Invalid Password. So it is always treated as Invalid Password error. For Invalid Password error PAM will not try other Authentication Methods. So it is recommended to keep AD in the last location in PAM order.

2. Click **Save** to save any changes made.
3. Click **Reset** to reset the modified changes.

PEF - IPMI for 6G2U Appliances

Platform Event Filtering (PEF) provides a mechanism for configuring the BMC to take selected actions on event messages that it receives or has internally generated. These actions include operations such as system power-off, system reset, as well as triggering the generation of an alert.

In webGUI, PEF Management is used to configure the following:

- Event Filter
- Alert Policy
- LAN Destination

To open the PEF Management page, click **Configurations > PEF** from the menu bar. The PEF Management screen is displayed.

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Dashboard Configuration Remote Control Maintenance Firmware Update Random Information Refresh Print Logout

PEF Management

Use this page to configure Event Filter, Alert Policy and LAN Destination. To delete or modify a entry, select it in the list and click "Delete" or "Modify". To add a new entry, select an unconfigured slot and click "Add".

Event Filter Alert Policy LAN Destination

Configured Event Filter count: 15

PEF ID	Filter Configuration	Event Filter Action	Event Severity	Sensor Name
1	Enabled	[Alert]	Unspecified	Any
2	Enabled	[Alert]	Unspecified	Any
3	Enabled	[Alert]	Unspecified	Any
4	Enabled	[Alert]	Unspecified	Any
5	Enabled	[Alert]	Unspecified	Any
6	Enabled	[Alert]	Unspecified	Any
7	Enabled	[Alert]	Unspecified	Any
8	Enabled	[Alert]	Unspecified	Any
9	Enabled	[Alert]	Unspecified	Any
10	Enabled	[Alert]	Unspecified	Any
11	Enabled	[Alert]	Unspecified	Any
12	Enabled	[Alert]	Unspecified	Any
13	Enabled	[Alert]	Unspecified	Any
14	Enabled	[Alert]	Unspecified	Any
15	Enabled	[Alert]	Unspecified	Any
16	-	-	-	-
17	-	-	-	-
18	-	-	-	-
19	-	-	-	-
20	-	-	-	-
21	-	-	-	-
22	-	-	-	-
23	-	-	-	-
24	-	-	-	-
25	-	-	-	-
26	-	-	-	-
27	-	-	-	-
28	-	-	-	-
29	-	-	-	-
30	-	-	-	-
31	-	-	-	-
32	-	-	-	-
33	-	-	-	-
34	-	-	-	-
35	-	-	-	-
36	-	-	-	-
37	-	-	-	-
38	-	-	-	-
39	-	-	-	-
40	-	-	-	-

[Add](#) [Modify](#) [Delete](#)

- Event Filter Tab - IPMI for 6G2U Appliances
- Alert Policy Tab - IPMI for 6G2U Appliances
- LAN Destination Tab - IPMI for 6G2U Appliances

Event Filter Tab - IPMI for 6G2U Appliances

A PEF implementation is recommended to provide at least 15 entries in the event filter table. A subset of these entries should be pre-configured for common system failure events, such as over- temperature, power system failure, fan failure events, etc. Remaining entries can be made available for 'OEM' or System Management Software configured events. Note that individual entries can be tagged as being reserved for system use - so this ratio of pre-configured entries to run-time configurable entries can be reallocated if necessary.

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Dashboard Configuration Remote Control Maintenance Firmware Update Random Information Refresh Cr Print Logout HELP

PEF Management

Use this page to configure Event Filter, Alert Policy and LAN Destination. To delete or modify a entry, select it in the list and click "Delete" or "Modify". To add a new entry, select an unconfigured slot and click "Add".

Event Filter Alert Policy LAN Destination

PEF ID	Filter Configuration	Event Filter Action	Event Severity	Sensor Name
1	Enabled	Alerts	Unspecified	Any
2	Enabled	Alerts	Unspecified	Any
3	Enabled	Alerts	Unspecified	Any
4	Enabled	Alerts	Unspecified	Any
5	Enabled	Alerts	Unspecified	Any
6	Enabled	Alerts	Unspecified	Any
7	Enabled	Alerts	Unspecified	Any
8	Enabled	Alerts	Unspecified	Any
9	Enabled	Alerts	Unspecified	Any
10	Enabled	Alerts	Unspecified	Any
11	Enabled	Alerts	Unspecified	Any
12	Enabled	Alerts	Unspecified	Any
13	Enabled	Alerts	Unspecified	Any
14	Enabled	Alerts	Unspecified	Any
15	Enabled	Alerts	Unspecified	Any
16	-	-	-	-
17	-	-	-	-
18	-	-	-	-
19	-	-	-	-
20	-	-	-	-
21	-	-	-	-
22	-	-	-	-
23	-	-	-	-
24	-	-	-	-
25	-	-	-	-
26	-	-	-	-
27	-	-	-	-
28	-	-	-	-
29	-	-	-	-
30	-	-	-	-
31	-	-	-	-
32	-	-	-	-
33	-	-	-	-
34	-	-	-	-
35	-	-	-	-
36	-	-	-	-
37	-	-	-	-
38	-	-	-	-
39	-	-	-	-
40	-	-	-	-

Configured Event Filter count: 15

[Add](#) [Modify](#) [Delete](#)

PEF Management – Event Filter Fields

Field Name	Description
PEF ID	Displays the ID for the newly configured PEF entry (read-only).
Filter configuration	Displays if the Filter Configuration check box is selected.
Event Filter Action	Displays if the Event Filter Action check box is selected.
Event Severity	Displays the configured Event Severity.
Sensor Name	Displays the configured Sensor Name.
Add	To add the new event filter entry and return to Event filter list.
Modify	To modify the existing entries.
Delete	To delete the configured event filter.

To add Event Filter entries:

1. In the Event Filter tab, select a free slot and click **Add** or alternatively double click the empty slot to open the Add Event Filter entry page.

2. In the Event Filter Configuration section:
 - PEF ID - Displays the ID for configured PEF entry (read-only).
 - Filter Configuration - Check box to enable the PEF settings.
 - Event Severity - A list of the available event severities.
3. In the Filter Action Configuration section:
 - Event Filter Action - This check box is a mandatory and selected by default, which enables PEF Alert action (read-only).
 - Power Action - A list of the available power actions (Power down, Power reset Power cycle or None)
 - Alert Policy Number - A list of the available Alert policies.



Note: Alert Policy has to be configured under the Alert Policy tab.

4. In the Generator ID Configuration section:
 - Generator ID Data - Check box to fill the Generator ID with raw data.
 - Generator ID 1 - Used to give raw generator ID1 data value.
 - Generator ID 2 - Used to give raw generator ID2 data value.



Note: In RAW data field, specify hexadecimal value prefix with '0x'.

- Event Generator - Choose the event generator as **Slave type** if the event was generated from IPMB or as **Software type** if the event was generated from system software.
 - Slave Address/Software ID - Specify corresponding I2C Slave Address or System Software ID.
 - Channel Number - A list of channels that the event message was received over. Choose '0' if the event message was received via the system interface, primary IPMB, or internally generated by the BMC.
 - IPMB Device LUN - A list of the corresponding device logical unit number if the event is generated by IPMB.
5. In the Sensor Configuration section:
 - Sensor Type - A list of the types of sensor that trigger the event filter action.
 - Sensor Name - A list of particular sensors
 - Event Options - Can be either All Events or Sensor Events.
 6. In the Event Data Configuration section:
 - Event Trigger - Used to give Event/Reading type value. The value ranges from 1 to 255.
 - Event Data 1 AND Mask - Used to indicate wildcarded or compared bits. The value ranges from 1 to 255.
 - Event Data 1 Compare 1 & Event Data 1 Compare 2 - Used to indicate whether each bit position's comparison is an exact comparison or not. The value ranges from 1 to 255.
 7. In the Event Data 2 Configuration section:
 - Event Data 2 AND Mask - Similar to Event Data 1 AND Mask.
 - Event Data 2 Compare 1 & Event Data 2 Compare 2 - Similar to Event Data 1 Compare 1 and Event Data 1 Compare 2 respectively.
 8. In the Event Data 3 Configuration section:
 - Event Data 3 AND Mask - Similar to Event Data 1 AND Mask.
 - Event Data 3 Compare 1 & Event Data 3 Compare 2 - Similar to Event Data 1 Compare 1 and Event Data 1 Compare 2 respectively.
 9. Click **Add** to save the changes and return to event filter list.
 10. Click **Cancel** to cancel the modification and return to Event filter list.

To modify Event Filter entries:

1. In the Event Filter tab, select the configured slot and click **Modify** or alternatively double click the configured slot.
2. Perform modifications to the existing event filter entry.
3. Click **Modify**.

To delete Event Filter entries:

1. In the Event Filter tab, select the configured slot.
2. Click **Delete**.

Alert Policy Tab - IPMI for 6G2U Appliances

This page is used to configure the Alert Policy for the PEF configuration. You can add, delete or modify an entry in this page.

Use this page to configure Event Filter, Alert Policy and LAN Destination. To delete or modify a entry, select it in the list and click "Delete" or "Modify". To add a new entry, select an unconfigured slot and click "Add".

Event Filter | **Alert Policy** | LAN Destination

Configured Alert Policy count: 15

Policy Entry #	Policy Number	Policy Configuration	Policy Set	Action	Destination
1	1	Disabled	Always send alert to this destination	1	0
2	2	Disabled	Always send alert to this destination	1	0
3	3	Disabled	Always send alert to this destination	1	0
4	4	Disabled	Always send alert to this destination	1	0
5	5	Disabled	Always send alert to this destination	1	0
6	6	Disabled	Always send alert to this destination	1	0
7	7	Disabled	Always send alert to this destination	1	0
8	8	Disabled	Always send alert to this destination	1	0
9	9	Disabled	Always send alert to this destination	1	0
10	10	Disabled	Always send alert to this destination	1	0
11	11	Disabled	Always send alert to this destination	1	0
12	12	Disabled	Always send alert to this destination	1	0
13	13	Disabled	Always send alert to this destination	1	0
14	14	Disabled	Always send alert to this destination	1	0
15	15	Disabled	Always send alert to this destination	1	0
16	-	-	-	-	-
17	-	-	-	-	-
18	-	-	-	-	-
19	-	-	-	-	-
20	-	-	-	-	-
21	-	-	-	-	-
22	-	-	-	-	-
23	-	-	-	-	-
24	-	-	-	-	-
25	-	-	-	-	-
26	-	-	-	-	-
27	-	-	-	-	-
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34	-	-	-	-	-
35	-	-	-	-	-
36	-	-	-	-	-
37	-	-	-	-	-
38	-	-	-	-	-
39	-	-	-	-	-
40	-	-	-	-	-
41	-	-	-	-	-
42	-	-	-	-	-
43	-	-	-	-	-
44	-	-	-	-	-
45	-	-	-	-	-
46	-	-	-	-	-

Add Modify Delete

The table below explains the Alert Policy fields shown in the Alert Policy tab.

PEF Management – Alert Policy Fields

Field Name	Description
Policy Entry #	Displays Policy entry number for the newly configured entry (read-only).
Policy Number	Displays the Policy number of the configuration.
Policy Configuration	Displays if the Policy Configuration check box is selected.
Policy Set	Displays the configured Policy Set. The available values are: 0 - Always send alert to this destination. 1 - If alert to previous destination was successful, do not send alert to this destination. Proceed to next entry in this policy set. 2 - If alert to previous destination was successful, do not send alert to this destination. Do not process any more entries in this policy set.

Field Name	Description
	3 - If alert to previous destination was successful, do not send alert to this destination. Proceed to next entry in this policy set that is to a different channel. 4 - If alert to previous destination was successful, do not send alert to this destination. Proceed to next entry in this policy set that is to a different destination type.
Channel Number	Displays the configured Channel Number.
Destination Selector	Displays the configured Destination Selector. Note: LAN Destination has to be configured in the LAN Destination tab.
Add	To save the new alert policy and return to Alert Policy list.
Modify	To modify the existing entries.
Delete	To delete the configured Alert Policy.

To add Event Filter entries:

1. In the Alert Policy tab, select the slot for which you have to configure the Alert policy. For example, in the Event Filter Entry page, if you have chosen Alert Policy number as 4, you have to configure the 4th slot (the slot with Policy Number 4) in the Alert Policy Tab.
2. Click **Add** or alternatively double click on the empty slot to open the Add Alert Policy Entry page.

3. Policy Entry # - A read only field.
4. Policy Number - Select from the list.
5. Policy Configuration - Select the check box to enable the policy settings.
6. Policy Set - Select from the list. For more information, see table above.
7. Channel Number - Select from the list.

-
8. Destination Selector - Select from the list.



Note: LAN Destination has to be configured under Configuration->PEF->LAN Destination. That is if you select the number 4 for destination selector in Alert Policy Entry page, then you have to configure the 4th slot (LAN Destination Number 4) in the LAN Destination tab.

9. Alert String - Select the check box to set the Alert policy entry as Event Specific.
10. Alert String Key - Select from the list the value used to look up the Alert String to be sent for this Alert Policy entry.
11. Click **Add** to save the new alert policy and return to Alert Policy list.
12. Click **Cancel** to cancel the modification and return to Alert Policy list.

To modify Alert Policy entries:

1. In the Alert Policy tab, select the configured slot and click **Modify** or alternatively double click the configured slot.
2. Perform modifications to the existing alert policy entry.
3. Click **Modify**.

To delete Event Filter entries:

1. In the Alert Policy tab, select the configured slot.
2. Click **Delete**.

LAN Destination Tab - IPMI for 6G2U Appliances

This page is used to configure the LAN destination of PEF configuration.

The table below explains the Alert Policy fields shown in the Alert Policy tab.

PEF Management LAN Destination Fields

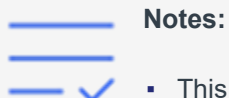
Field Name	Description
LAN Channel	Select the LAN Channel to have its settings displayed.
LAN Destination	Displays the configured LAN Destination.
Destination Type	Displays the configured Destination type. The available values are SNMP Trap or Email Alert. Note: The SMTP server information needs to be added under Configuration->SMTP .
Destination Address	Displays the configured Destination Address.
Send Test Alert	To send sample alert to configured destination. Note: Test alert can sent only with enabled SMTP configuration. SMTP support can be enabled under Configuration->SMTP .
Add	To add a new entry to the device.

Field Name	Description
Modify	To modify that entry.
Delete	To delete the selected configured LAN Destination.

To add LAN Destination entries:

1. In the LAN Destination Tab, choose the slot to be configured. This should be the same slot that you have selected in the Alert Policy Entry- Destination Selector field. For example, if you have chosen the Destination Selector as 4 in the Alert Policy Entry page of Alert Policy Tab, then you have to configure the 4th slot of LAN Destination Page.
2. Click **Add** or alternatively double click on the empty slot. The Add LAN Destination entry window is displayed.

3. LAN Channel Number - Displays the LAN Channel Number for the selected slot. This is a read only field.
4. LAN Destination - Displays the destination for the newly configured entry. This is a read only field.
5. Destination Type - Select from the list. For more information, see table above.
6. Destination Address - Type the destination address of the system that receives the alert.



Notes:

- This field is enabled only when the Destination Type is set to SNMP Trap
- This field supports IPv4/IPv6 address format.

7. Username - Select from the list.



Notes:

- This field is enabled only when the Destination Type is set to Email Alert
- The list indicates the users that were configured in **Configuration->Users**.

8. Subject - Type the subject of the email.



Notes: This field is enabled only when the Destination Type is set to Email Alert and the Email Format field for the user is set to FixedSubject-Format.

9. Message - Type the email body that is sent.



Note: This field is enabled only when the Destination Type is set to Email Alert and the Email Format field for the user is set to FixedSubject-Format.

10. Click **Add** to save the new LAN destination and return to LAN Destination list.
11. Click **Cancel** to cancel the modification and return to LAN Destination list.

To modify LAN Destination entries:

1. In the LAN Destination tab, select the configured slot and click **Modify** or alternatively double click the configured slot.
2. Perform modifications to the existing alert policy entry.
3. Click **Modify**.

To delete Event Filter entries:

1. In the LAN Destination tab, select the configured slot.
2. Click **Delete**.

RADIUS - IPMI for 6G2U Appliances

RADIUS is a modular, high performance and feature-rich RADIUS suite including server, clients, development libraries and numerous additional RADIUS related utilities.

In webGUI GUI, this page is used to set the RADIUS Authentication.

To open the RADIUS Settings page, click **Configuration > RADIUS** from the menu bar. The RADIUS Settings screen is displayed.

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RADIUS Settings

The RADIUS Authentication is currently disabled. To enable RADIUS Authentication and enter the required information to access the RADIUS server. Press the Save button to save your changes. To configure the Advanced settings, RADIUS Server authentication should be enabled. [Advanced Settings](#)

RADIUS Authentication ☒ Enable

Port

Server Address

Secret

Extended privileges ☐ XDM ☐ VMedia

[Save](#) [Reset](#)

RADIUS Settings Fields

Field Name	Description
RADIUS Authentication	To enable/disable RADIUS authentication.
Port	The RADIUS Port number. Notes: ▪ Default Port is 1812. ▪ Port value ranges from 1 to 65535.
Server Address	The IP address of the RADIUS server. Notes: ▪ IP Address made of 4 numbers separated by dots as in “xxx.xxx.xxx.xxx”. ▪ Each Number ranges from 0 to 255. ▪ First Number must not be 0.
Secret	The Authentication Secret for the RADIUS server. Notes: ▪ This field allows up to 31 characters. ▪ Secret must be at least 4 characters long. ▪ White space is not allowed.
Extended privileges	To assign KVM or VMedia privileges for the user.
Save	To save the settings.
Reset	To reset the modified changes.

Once RADIUS Authentication is enabled, you need to configure advanced settings.

To configure advanced settings:

1. In the RADIUS Settings page, click **Advanced Settings**. The RADIUS Authorization window is displayed.

Radius Authorization	
Administrator	H=4
Operator	H=3
User	H=2
OEM Proprietary	H=1
No Access	H=0

Save Cancel

For Authorization Purpose, configure the Radius user with Vendor Specific Attribute in Server side. See examples below.

Example:1

testadmin Auth-Type :=PAP,Cleartext-Password:="admin"

Auth-Type :=PAP, Vendor-Specific="H=4"

Example:2

testoperator Auth-Type := PAP,Cleartext-Password := "operator"

Auth-Type :=PAP, Vendor-Specific="H=3"

Notes:



- If you change the Vendor-Specific value in server then you should change the same values in this page.
- These fields allow up to 127 characters.
- '#' is not allowed.

2. Click **Save** to save the changes made.
3. Click **Cancel** to go back to the previous screen.

Remote Session - IPMI for 6G2U Appliances

Use this page to configure virtual media configuration settings for the next redirection session. "Single Port Application" is enabled by default. While disabling "Single Port Application" KVM and Media Encryption are disabled by default.

To open Remote Session page, click Configuration > Remote Session from the menu bar. A sample screenshot of Remote Session Page is shown below.

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Configure Remote Session

This page is used to configure virtual media configuration settings

Single Port Application ☒ Enable

Keyboard Language

Retry Count

Retry Time Interval Seconds

☒ Local Monitor Off

☐ Automatically Off Local Monitor When Zlifer Launches

Save Reset

Remote Session Fields

Field Name	Description
Single Port Application	To enable/disable single port support by runtime, On changing this configuration, KVM and VMedia Sessions will be restarted. If this support is enabled, KVM session will not use its dedicated port whereas both Web and KVM sessions will be established only via Web Port. If this support is disabled, KVM and Web sessions will use their own dedicated ports respectively.
KVM Encryption	To enable/disable encryption of KVM data for the next redirection session. If KVM Encryption is enabled, the KVM session will use the Secure port which has been configured in Configuration -> Services Page. If KVM Encryption is disabled, the KVM session will use the Non-Secure port which has been configured in Configuration -> Services Page Note: This option is disabled if Single Port is enabled.
Keyboard Languages	This option is used to select the keyboard supported languages.
Retry Count	This option is used to retry the redirection session for certain number of attempts.
Retry Interval	This option is used to give time interval for each attempts.
Local Monitor OFF	To enable/disable Local Monitor OFF. If this option is enabled, You can Lock or Unlock the Local host monitor from the remote KVM window. If this option is disabled, you cannot Lock or Unlock the Local host monitor from the remote KVM window.
Automatically OFF Local Monitor, When JViewer Launches	To enable/disable Automatically OFF Local Monitor, When JViewer Launches.
Save	To save the current changes. Note: This automatically closes the existing remote redirection either KVM or Virtual media sessions, if any.
Reset	To reset the modified changes.

Services - IPMI for 6G2U Appliances

This page displays the basic information about services running in the BMC. Only Administrator can modify the service.

To open Services page, click Configuration > Services from the menu bar. The Services screen is displayed.

Slot	Service Name	Current State	Interfaces	Nonsecure Port	Secure Port	Timeout	Maximum Sessions	Active Sessions
1	web	Active	eth1	80	443	1800	20	0/20
2	kum	Active	eth1	7578	7582	1800	4	0/4
3	cd-media	Active	eth1	5120	5124	N/A	4	0/4
4	fd-media	Active	eth1	5123	5126	N/A	4	0/4
5	ssh	Active	eth1	N/A	22	N/A	4	0/4
6	telnet	Inactive	N/A	23	N/A	600	N/A	0/0

Services Fields

Field Name	Description
Service Name	Displays the service name of the selected slot (read-only).
Current State	Displays the current status of the service, either active or inactive state.
Interfaces	Displays the interface in which service is running.
Nonsecure Port	Displays the non-secure port number for the service. Notes: - KVM default port is 7578. - CD Media default port is 5120. - HD Media default port is 5123. - Telnet default port is 23. - SSH service will not support non secure port. If single port feature is enabled, KVM, CD Media, FD Media and HD Media ports cannot be edited.
Secure Port	Displays the secure port number for the service. Notes: - Web default port is 443. - KVM default port is 7582. - CD Media default port is 5124. - FD Media default port is 5126.

Field Name	Description
	- SSH default port is 23. - Telnet service will not support secure port. If single port feature is enabled, KVM, CD Media, FD Media and HD Media ports cannot be edited.
Timeout	Displays the session timeout value of the service. For web, SSH and telnet service, user can configure the session timeout value. Notes: - Web timeout value ranges from 300 to 1800 seconds. - KVM timeout value ranges from 300 to 1800 seconds. - SSH and Telnet timeout value ranges from 60 to 1800 seconds. - SSH and telnet timeout value should be in multiples of 60 seconds. - SSH default port is 23. - If KVM is launched then the web session timeout will not take effect.
Maximum Sessions	Displays the maximum number of allowed sessions for the service.
Active Sessions	Enables viewing the current active sessions for the service.

To perform actions on active sessions of a service:

1. In the Services page, click **View** for one of the services to view the details about the active sessions for the service. The Active Session - <session type> window is displayed.

#	Session ID	Session Type	IP Address	User ID	User Name	User Privilege
1	37	HTTP	192.140.125	3	Admin	Administrator

Number of Sessions: 1

Terminate Cancel

2. Select a slot and click **Terminate** to terminate the particular session of the service.
3. Click **Cancel** to cancel the modification and return to Services list.

To perform actions on existing services:

1. In the Services page, select a slot and click **Modify** to modify the configuration of the service. Alternatively, double click on the slot. The Modify Service window is displayed.



Note: Whenever the configuration is modified, the service is restarted automatically and you need to close the existing opened session for the service if needed.

2. Service Name - Displays the service name. This is a read only field.
3. Current State - Select/clear the check box to activate/deactivate the service.



Note: Interfaces, Nonsecure port, Secure port, Time out and Maximum Sessions are not active unless the current state is active.

4. Interfaces - Select from the list.
5. Nonsecure Port - Type the port number of the non-secure port.
6. Secure Port - Type the port number of the secure port.
7. Timeout - Type the session timeout value of the service. For more information, see table above.
8. Maximum Sessions - Displays the maximum number of allowed sessions for the service. This is a read only field.
9. Click **Modify** to save the entered changes and return to the Services page.
10. Click **Cancel** to exit.

SMTP - IPMI for 6G2U Appliances

Simple Mail Transfer Protocol (SMTP) is an Internet standard for electronic mail (email) transmission across Internet Protocol (IP) networks.

Using webGUI, you can configure the SMTP settings of the device.

To open SMTP Settings page, click **Configuration > SMTP** from the menu bar. The SMTP Settings screen is displayed.

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Manage SMTP settings of the device.

LAN Channel Number: 1

Sender Address:

Machine Name:

Primary SMTP Server

SMTP Support: ☒ Enable

Port: 25

Server Address:

☐ SMTP Server requires Authentication

User Name:

Password:

☐ Enable STARTTLS Support

Secondary SMTP Server

SMTP Support: ☐ Enable

Port: 25

Server Address:

☐ SMTP Server requires Authentication

User Name:

Password:

☐ Enable STARTTLS Support

Save Reset

SMTP Settings Fields

Field Name	Description
LAN Channel Number	To select the channel number from the list.
Sender Address	Type the sender address valid on the SMTP Server.
Machine Name	Type the machine name of the SMTP Server. Notes: - Machine Name is a string of maximum 15 alpha-numeric characters. - Space, special characters are not allowed.
Primary SMTP Server SMTP Support	Select/clear the check box to enable/disable SMTP support for the BMC.
Port	Type the SMTP Port. Notes: - Default Port is 25. - Port value ranges from 1 to 65535.
Server Address	Type the IP address of the SMTP Server. This is a mandatory field. Notes:

Field Name	Description
	▪ IP Address made of 4 numbers separated by dots as in “xxx.xxx. xxx.xxx”. ▪ Each Number ranges from 0 to 255. ▪ First Number must not be 0. ▪ Supports IPv4 Address format and IPv6 Address format. ▪ This field is enabled only when the Primary SMTP Server SMTP Support check box is selected..
SMTP Server requires Authentication	Select/cleat the check box to enable/disable SMTP Authentication. Notes: ▪ The supported SMTP Server Authentication Types are CRAM-MD5/LOGIN/PLAIN ▪ If the SMTP server does not support any one of the above authentication types, the user will get an error message stating, Authentication type is not supported by SMTP Server. ▪ This check box is enabled only when the SMTP Support check box is selected.
Username	Type the username to access SMTP Accounts. Notes: ▪ The value can be of length 4 to 64 alpha-numeric characters, dot(.), dash(-), and underline(_). ▪ It must start with an alphabet. ▪ Other special characters are not allowed. ▪ This field is enabled only when the SMTP Server requires Authentication check box is selected.
Password	Type the password for the SMTP User Account. Notes: ▪ Password must be at least 4 characters long. ▪ White space is not allowed. ▪ This field allows up to 64 characters. ▪ This field is enabled only when the SMTP Server requires Authentication check box is selected.

Field Name	Description
Enable STARTTLS Support	Select/clear the check box to enable/disable STARTTLS support for the SMTP Client. Notes: - CA Certificate File: File that contains the certificate of the trusted CA certs. - Certificate File: Client certificate filename. - Private Key: Client private key filename. - This field is enabled only when the Primary SMTP Server SMTP Support check box is selected.
Secondary SMTP Server SMTP Support	Lists the Secondary SMTP Server configuration. It is an optional field. If the Primary SMTP server is not working fine, then it tries with Secondary SMTP Server configuration.
Save	To save the new SMTP server configuration.
Reset	To reset the modified changes

SSL - IPMI for 6G2U Appliances

The Secure Socket Layer protocol was created by Netscape to ensure secure transactions between web servers and browsers. The protocol uses a third party, a Certificate Authority (CA), to identify one end or both end of the transactions.

Using webGUI, configure SSL certificate into the BMC. Using this, the device can be accessed in a secured mode.

To open SSL Certificate Configuration page, click **Configuration > SSL** from the menu bar. There are three tabs in this page.

- Upload SSL – Tab is used to upload the certificate and private key file into the BMC.
- Generate SSL – Tab is used to generate the SSL certificate based on configuration details.
- View SSL – Tab is used to view the uploaded SSL certificate in readable format.

- Upload SSL Tab - IPMI for 6G2U Appliances
- Generate SSL Tab - IPMI for 6G2U Appliances
- View SSL Tab - IPMI for 6G2U Appliances

Upload SSL Tab - IPMI for 6G2U Appliances

This page is used to upload the certificate and private key file into the BMC.

SSL Certificate Configuration – Upload SSL Fields

Field Name	Description
Current Certificate	Displays the Current certificate information (read-only).
New Certificate	Browse and navigate to the certificate file. The Certificate file should be of pem type.
Current Privacy Key	Displays the current privacy key information (read-only).
New Privacy Key	Browse and navigate to the private key file. The Private key file should be of the pem type.
Upload	Click Upload to upload the SSL certificate and private key into the BMC. Note: Upon successful upload, the HTTPs service is restarted to use the newly uploaded SSL certificate.

Generate SSL Tab - IPMI for 6G2U Appliances

This page is used to generate the SSL certificate based on configuration details.

This page is used to configure SSL certificate into the BMC. Using this, the device can be accessed in a secured mode. Upload SSL option is used to upload the certificate and private key file into the BMC. Generate SSL option is used to generate the SSL certificate based on configuration details. View SSL option is used to view the uploaded SSL certificate in readable format.

Upload SSL | **Generate SSL** | View SSL

Common Name(CN)

Organization(O)

Organization Unit(OU)

City or Locality(L)

State or Province(ST)

Country(C)

Email Address

Valid for days

Key Length bits

SSL Certificate Configuration – Generate SSL Fields

Field Name	Description
Common Name(CN)	Type the common name for which certificate is to be generated. Notes: - Maximum length of 64 characters. - Special characters '#' and '\$' are not allowed.
Organization(O)	Type the organization name for which the certificate is to be generated. Notes: - Maximum length of 64 characters. - Special characters '#' and '\$' are not allowed.
Organization Unit(OU)	Type the over all organization section unit name for which certificate is to be generated. Notes: - Maximum length of 64 characters. - Special characters '#' and '\$' are not allowed.
City or Locality(L)	Type the city or locality of the organization (mandatory). Notes: - Maximum length of 64 characters. - Special characters '#' and '\$' are not allowed.
State or Province(ST)	Type the state or province of the organization (mandatory).

Field Name	Description
	Notes: - Maximum length of 64 characters. - Special characters '#' and '\$' are not allowed.
Country(C)	Notes: Type the country code of the organization (mandatory). - Only two characters are allowed. - Special characters are not allowed.
Email Address	Type the email address of the organization (mandatory).
Valid for	Type the number of days the certificate is valid for. Note: Value ranges from 1 to 3650 days.
Key Length	Select the key length bit value of the certificate.
Generate	Click to generate the new SSL certificate. Notes: - Upon successful upload, the HTTPs service is restarted to use the newly uploaded SSL certificate - HTTPs session does not work in some browsers for 512 bits RSA Keys

View SSL Tab - IPMI for 6G2U Appliances

This page is used to view the uploaded SSL certificate in readable format.

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SSL Certificate Configuration

This page is used to configure SSL certificate into the BMC. Using this, the device can be accessed in a secured mode. Upload SSL option is used to upload the certificate and private key file into the BMC. Generate SSL option is used to generate the SSL certificate based on configuration details. View SSL option is used to view the uploaded SSL certificate in readable format.

Upload SSL Generate SSL View SSL

Basic Information	
Version	3
Serial Number	ED10C02064F43A8
Signature Algorithm	sha1WithRSAEncryption
Public Key	
Issued From	
Common Name(CN)	AMR
Organization(O)	American Megatrends Inc
Organization Unit(OU)	Service Processors
City or Locality(L)	Atlanta
State or Province(ST)	Georgia
Country(C)	US
Email Address	support@ami.com
Validity Information	
Valid From	Jul 29 06:47:29 2014 GMT
Valid To	Jul 29 06:47:29 2024 GMT
Issued To	
Common Name(CN)	AMR
Organization(O)	American Megatrends Inc
Organization Unit(OU)	Service Processors
City or Locality(L)	Atlanta
State or Province(ST)	Georgia
Country(C)	US
Email Address	support@ami.com

SSL Certificate Configuration – View SSL Fields

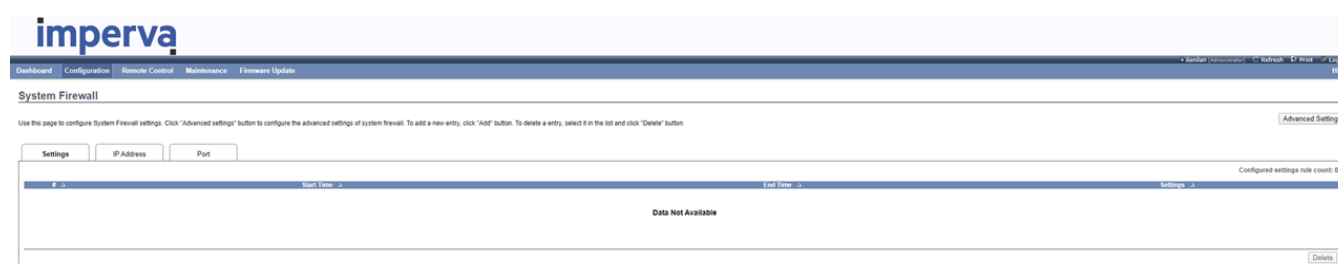
Field Name	Description
Basic Information	This section displays the basic information about the uploaded SSL certificate. It displays the following fields: - Version - Serial Number - Signature Algorithm - Public Key
Issued From	This section displays the Certificate Issuer information. It displays the following fields: - Common Name(CN) - Organization(O) - Organization Unit(OU) - City or Locality(L) - State or Province(ST) - Country(C) - E-mail Address
Validity Information	This section displays the validity period of the uploaded certificate. It displays the following fields: - Valid From - Valid To

Field Name	Description
Issued To	This section display the information about to whom the certificate is issued. It displays the following fields: ▪ Common Name(CN) ▪ Organization(O) ▪ Organization Unit(OU) ▪ City or Locality(L) ▪ State or Province(ST) ▪ Country(C) ▪ E-mail Address

System Firewall - IPMI for 6G2U Appliances

In webGUI, the System Firewall page allows you to configure the firewall settings. The firewall rule can be set for an IP or range of IP Addresses or Port numbers. To view this page, you must at least be an operator. Only administrators can add or delete a firewall.

To open System Firewall page, click **Configuration > System Firewall** from the menu bar.



- Advanced System Firewall Settings - IPMI for 6G2U Appliances
- Settings Tab - IPMI for 6G2U Appliances
- IP Address Tab - IPMI for 6G2U Appliances
- Port Tab - IPMI for 6G2U Appliances

Advanced System Firewall Settings - IPMI for 6G2U Appliances

You can perform advanced settings for the system firewall.

To perform advanced settings:

1. In the System Firewall page, click **Advanced Settings**. The Advanced Firewall Settings window is displayed.

Advanced Firewall Settings

Block All ☐

Flush All ☐

Timeout ☐

Start Time [dd-mm-yy hh-mm] 2 February 2012 18:30

End Time [dd-mm-yy hh-mm] 2 February 2012 18:30

Save Cancel

2. Block All - Select the type of incoming IPs and Ports to block.
3. Flush All - Select/clear check box to enable/disable flushing of all the system firewall rules.
4. Timeout - Select/clear check box to enable/disable firewall rules with timeout.
5. Start Time - Set the start time and date of the respective firewall rule effect.
6. End Time - Set the end time and date of the respective firewall rule effect.
7. Click **Save** to save the changes made.
8. Click **Cancel** to cancel the modification to the existing settings.

Settings Tab - IPMI for 6G2U Appliances

This page displays the configured start time, end time and settings for each entry.

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System Firewall

Use this page to configure System Firewall settings. Click "Advanced settings" button to configure the advanced settings of system firewall. To add a new entry, click "Add" button. To delete a entry, select it in the list and click "Delete" button.

Advanced Settings

Settings IP Address Port

Start Time End Time Settings

Data Not Available

Delete

System Firewall - Settings Fields

Field Name	Description
Start Time	The respective firewall rule effect will start from this time.
End Time	The respective firewall rule effect will end from this time.
Settings	This column indicates the current setting of the firewall rule.

IP Address Tab - IPMI for 6G2U Appliances

This tab enables you to block or allow an IP address or range of IP addresses.

To block or allow an IP address or range of IP addresses:

1. Click **Add** to add a new IP or range of IP address. The Add new rule for IP window is displayed.

2. IP/IP Range - Type the IP address or a range of IP addresses.

Notes:



- IP Address supports IPv4 Address format only.
- IPv4 Address needs to be made of 4 numbers separated by dots as in xxx.xxx.xxx.xxx.
- Each number ranges from 0 to 255.
- First number must not be 0.

3. Timeout - Select check box to enable firewall rules with timeout.
4. Start Time - Set the start time and date of the respective firewall rule effect.
5. End Time - Set the end time and date of the respective firewall rule effect.
6. IP Settings - Select if the rule blocks or accepts.
7. Click **Save** to save the changes made.
8. Click **Cancel** to cancel the modification to the existing settings.
9. Select a slot and click **Delete** to delete an IP address or a range of IP addresses.

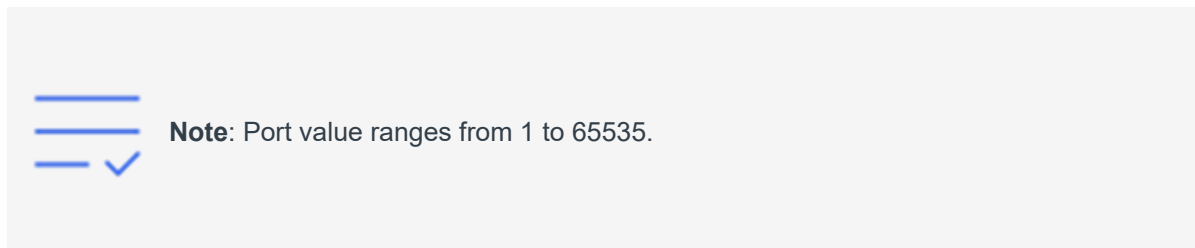
Port Tab - IPMI for 6G2U Appliances

This tab enables you to block or allow a port or range of ports.

To block or allow a port or range of ports:

1. Click **Add** to add a new port or range of ports. The Add new rule for Port window is displayed.

2. Port/Port Range - Type the port or a range of ports.



3. Protocol - Select the protocol type.
4. Timeout - Select check box to enable firewall rules with timeout.
5. Start Time - Set the start time and date of the respective firewall rule effect.
6. End Time - Set the end time and date of the respective firewall rule effect.
7. Port Settings - Select if the rule blocks or accepts.
8. Click **Save** to save the changes made.
9. Click **Cancel** to cancel the modification to the existing settings.
10. Select a slot and click **Delete** to delete a port or a range of ports.

Users - IPMI for 6G2U Appliances

In webGUI, the User Management page allows you to view the current list of user slots for the server. You can add a new user and modify or delete the existing users.

To open User Management page, click **Configuration > Users** from the menu bar. The User Management screen is displayed.

UserID	Username	User Access	Network's Privilege	Email ID
1	admin	Enabled	Administrator	
2	admin	Enabled	Administrator	
3	admin	Enabled	Administrator	
4				
5				
6				
7				
8				
9				
10				

User Management Fields

Field Name	Description
User ID	Displays the ID number of the user. Note: The list contains a maximum of ten users only.
Username	Displays the name of the user.
User Access	Displays the access privilege of the user.
Network Privilege	Displays the network access privilege of the user.
E-mail ID	Displays the email address of the user.
Add User	Click to add a new user.
Modify User	Click to modify an existing user.
Delete User	Click to delete an existing user.

- [Add New User - IPMI for 6G2U Appliances](#)
- [Modify a User - IPMI for 6G2U Appliances](#)
- [Delete a User - IPMI for 6G2U Appliances](#)

Add New User - IPMI for 6G2U Appliances

You can add a new user in the User Management page.

To add a new user:

1. Select an empty slot and click **Add User** or alternatively double click on the empty slot. The Add User window is displayed.

2. Username - Type the name of the user.

Notes:



- User Name is a string of 4 to 16 alpha-numeric characters.
- It must start with an alphabetical character.
- It is case-sensitive.
- Special characters ',', '.' (period), ':' (colon), ';' (semicolon), ' ' (space), '/' (slash), '\' (backslash), '(' (left bracket) and ')' (right bracket) are not allowed.

3. Password Size - Select the size of the password.
4. Password - Type the password for the user. See the Note below.
- Confirm Password - Confirm the new password.
5. User Access - Select check box to enable user access privileges.
6. Network Privilege - Select the network privilege assigned to the user.
7. Extended Privileges - Select the desired option.



Note: Imperva recommends that the Extended privileges support be provided only to the ADMIN user and should not be provided to USER and OPERATOR privilege level users. The Admin user can provide the Extended privilege support to USER and OPERATOR privilege level users at their own risk.

8. Email ID - Type the email ID of the user. If the user forgets the password, the new password will be mailed to the configured email address.



Note: SMTP Server must be configured to send emails.

9. Email Format - Select the types of email format.
 - AMI-Format: The subject of this mail format is 'Alert from (your Host name)'. The mail content shows sensor information. For example, Sensor type and Description.
 - Fixed-Subject Format: This format displays the message according to user's setting. You must set the subject and message for email alert.
10. New SSH Key - Click Choose File to browse and select the SSH key file.



Note: SSH key file should be of pub type.

11. Click **Add** to save the new user and return to the users list.
12. Click **Cancel** to cancel the modification and return to the users list.

Note:

Make sure your password has the following characteristics:



- It must have no fewer than 10 characters and no more than 14 characters.
- It must have at least one number, one capital letter, and one special character from:

* + = # % ^ : / ~ . , [] _

- It cannot have more than two characters repeated in succession.

Modify a User - IPMI for 6G2U Appliances

You can modify a user in the User Management page.

To modify an existing user:

1. Select an existing user from the list and click **Modify User** or alternatively double click on the configured slot. The **Modify User** window is displayed.

2. Edit the required fields.
3. To change the password, enable the **Change Password** option.
4. Click **Modify** to accept changes.
5. Click **Cancel** to close the page without saving.

Note:

Make sure your password has the following characteristics:



- It must have no fewer than 10 characters and no more than 14 characters.
- It must have at least one number, one capital letter, and one special character from:

* + = # % ^ : / ~ . , [] _

- It cannot have more than two characters repeated in succession.

Delete a User - IPMI for 6G2U Appliances

You can delete a user in the User Management page.

To delete a user:

1. Select the user from the list and click **Delete User**.

Notes:

- There is a list of reserved users which cannot be added or modified as BMC users.
- Reserved Users: There are certain reserved users which cannot be added as BMC Users. The list of reserved users are given below.



– sysadmin

– daemon

– sshd

– ntp

– stunnel4

Virtual Media - IPMI for 6G2U Appliances

This page is to configure Virtual Media device settings. If you change the configuration of the virtual media in this page, it will show the appropriate devices in the JViewer Vmedia Wizard. For example, if you select two floppy devices in Configure -> Virtual Media page, then in Jviewer -> VMedia Wizard, you can view two floppy devices available for redirection.

To open the Virtual Media page, click **Configuration > Virtual Media** from the menu bar. The Virtual Media Devices screen is displayed.

Virtual Media Devices Fields

Field Name	Description
Floppy devices	Select the number of floppy devices supported for Virtual Media redirection.
CD/DVD devices	Select the number of CD/DVD devices supported for Virtual Media redirection.
Hard disk devices	Select the number of harddisk devices supported for Virtual Media redirection.
Remote KVM Floppy Devices	Select the number of floppy devices supported for KVM Virtual Media redirection.
Remote KVM CD/DVD Devices	Select the number of CD/DVD devices supported for Virtual Media redirection.
Remote KVM Hard disk Devices	Select the number of Hard disk devices supported for Virtual Media redirection.
Media Encryption	Select check box to enable encryption of media data for the next redirection session. If Media Encryption is enabled, each CD, FD and HD media sessions uses their specific secure ports which have been configured in Configuration -> Services page. Note: This option is disabled if Single Port is enabled in Configuration -> Remote Session page.
Power Save Mode	Select check box to enable the virtual USB devices visibility in the host. If enabled, virtual media devices are connected to the Host machine only at the

Field Name	Description
	instance launching KVM session. If cleared, virtual media devices remain connected to the host machine all the time irrespective of KVM session status.
Save	Click to save the configured settings.
Reset	Click to reset the previously-saved values.

Remote Control - IPMI for 6G2U Appliances

The Remote Control consists of the following menu items.

- Console Redirection
- Server Power Control
- Java SOL
- Console Redirection - IPMI for 6G2U Appliances
- Server Power Control - IPMI for 6G2U Appliances
- Java SOL - IPMI for 6G2U Appliances

Console Redirection - IPMI for 6G2U Appliances

The remote console application, which is started using the WebGUI, allows you to control your server's operating system remotely, using the screen, mouse, and keyboard, and to redirect local CD/DVD, Floppy diskette and Hard disk/USB thumb drives as if they were connected directly to the server.



Note: If you wish to launch JViewer from the Console Redirection Page, the KVM option should be enabled in the Extended Privileges of the logged in user.

- List of Supported Client Operating Systems - IPMI for 6G2U Appliances
- List of Supported Host Operating Systems - IPMI for 6G2U Appliances
- Browser Settings - IPMI for 6G2U Appliances
- Java Console - IPMI for 6G2U Appliances

List of Supported Client Operating Systems - IPMI for 6G2U Appliances

The supported client operating systems are:

- winxp
- Windows Vista
- w2k3 - 32 bit
- w2k3 - 64 bit
- Windows 7 – 32 bit
- Windows 7 – 64 bit
- RHEL 4 - 32 bit
- RHEL 4 - 64 bit
- RHEL 5.4 - 32 bit
- RHEL 5.4 - 64 bit
- RHEL 6.0 - 64 bit
- RHEL 6.0 - 32 bit
- Ubuntu 9.10 LTS - 32
- Ubuntu 9.10 LTS - 64
- Ubuntu 10.04 LTS - 32 bit
- Ubuntu 10.04 LTS - 64 bit
- Ubuntu 8.10 -32
- Ubuntu 8.10 -64
- Ubuntu 11.10 Server - 32 bit
- Ubuntu 11.10 Server - 64 bit
- OpenSuse 11.2 -32
- OpenSuse 11.2 -64
- FC 9 - 32
- FC 9 - 64
- FC 10 - 32
- FC 10 - 64
- FC 12 - 32
- FC 12 - 64
- FC 13 - 32
- FC 13 - 64
- FC 14 - 32
- FC 14 - 64
- FC 15
- FC 16
- MAC -32
- MAC-64

List of Supported Host Operating Systems - IPMI for 6G2U Appliances

The supported host operating systems are:

-
- RHEL 5
 - RHEL 5.3
 - RHEL 5.4
 - RHEL 6
 - w2k3
 - w2k8
 - Win 2012 (64 bit)
 - RHEL 4
 - OpenSuse 11.2
 - OpenSuse 10.x
 - Ubuntu 8.10
 - Ubuntu 9.10
 - Ubuntu 11.04
 - Ubuntu 11.10 Server
 - Ubuntu Server 12.04 (64)
 - SLES 11
 - Debian 6
 - CentOS 6.0

Browser Settings - IPMI for 6G2U Appliances

For Launching the KVM, pop-up block should be disabled. For Internet explorer, enable the download file options from the settings.

Java Console - IPMI for 6G2U Appliances

This is an OS independent plug-in which can be used in Windows as well as Linux with the help of JRE. JRE should be installed in the client's system.

In webGUI, the Java Console can be launched in two ways.

- Open the Dashboard Page and in Remote control section, click **Launch** for Java Console.
- Open **Remote Control>Console Redirection** page and click **Java Console**.

This will download the .jnlp file from BMC. To open the .jnlp file, use the appropriate JRE version (Javaws). When the downloading is done, it opens the Console Redirection window.

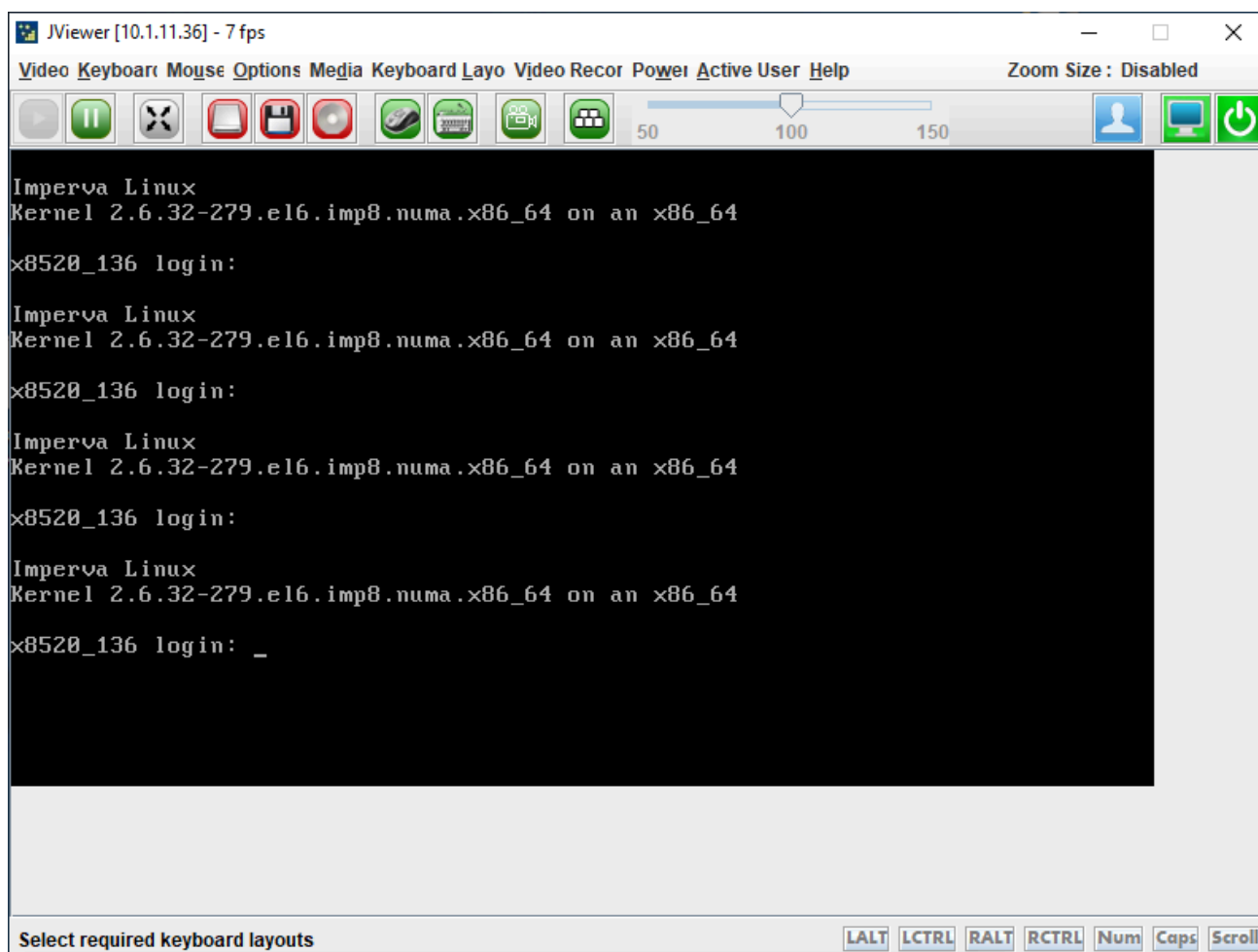
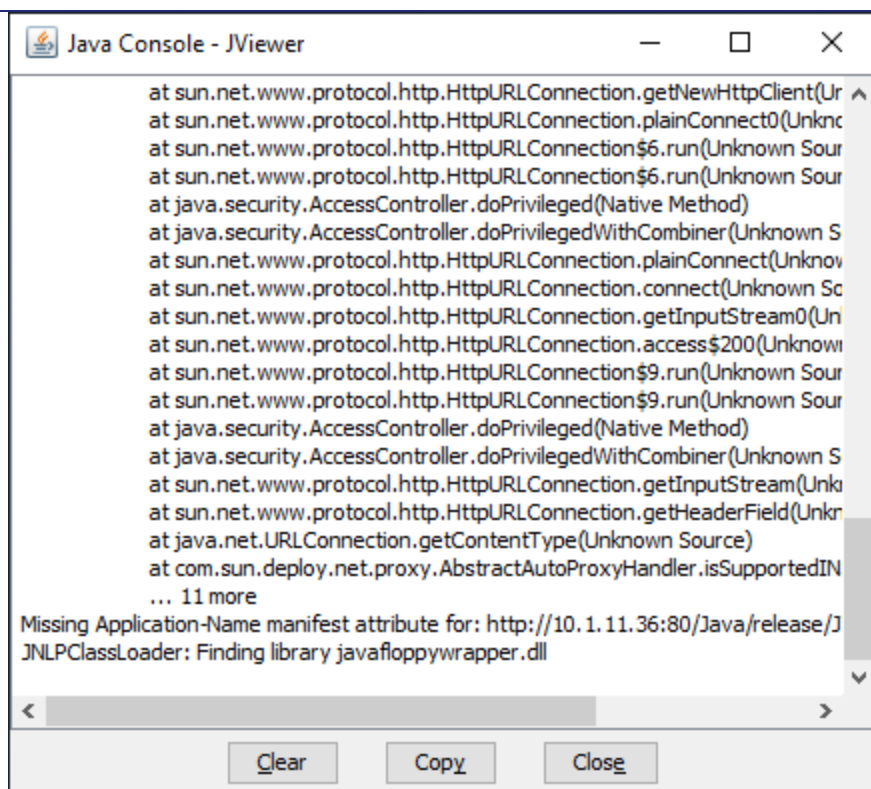
The Console Redirection menu bar consists of the following menu items:

- Video
- Keyboard
- Mouse
- Options
- Media

-
- Keyboard Layout
 - Video Record
 - Power
 - Active Users
 - Help

To start the Java Console:

1. Verify you have Java version 8 installed on the client machine.
2. Open the Java Control Panel application. For example, in Windows OS:
 - a. Open Control Panel.
 - b. In **View by**, select **Small icons** from the drop down menu.
 - c. Click on **Java**. The **Java Control Panel** window is displayed.
3. Go to the **Security** tab and click **Edit Site List**. The **Exception Site List** window is displayed.
4. Click **Add** and type the following http://<IP address of the IPMI machine>
5. Click **Add** and type the following https://<IP address of the IPMI machine>
6. Click **OK**.
7. Click **Apply**.
8. Click **OK**.
9. Log into the WebGUI and go to the **Remote Control -> Console Redirection**.
10. Click **Java Console**. The **jviewer.jnlp** file will be downloaded to your machine (click **Save** or **Keep** if asked).
11. Open a command prompt as administrator.
12. Navigate to the folder where you saved the jviewer.jnlp file.
13. Run the file using the **Java Web Launcher** application. A **Security Warning** pop up is displayed.
14. Select the **I accept...** check box and click **Run**. The **Java Console - JViewer** log and the **JViewer** console open.



To install SecureSphere using the Java Console:

1. In the JViewer console, click **Media**. The **Virtual Media** wizard is displayed.

Virtual Media

Floppy Key Media : I

☒ Floppy Image

CD/DVD Media : I

☒ CD Image

☐ D

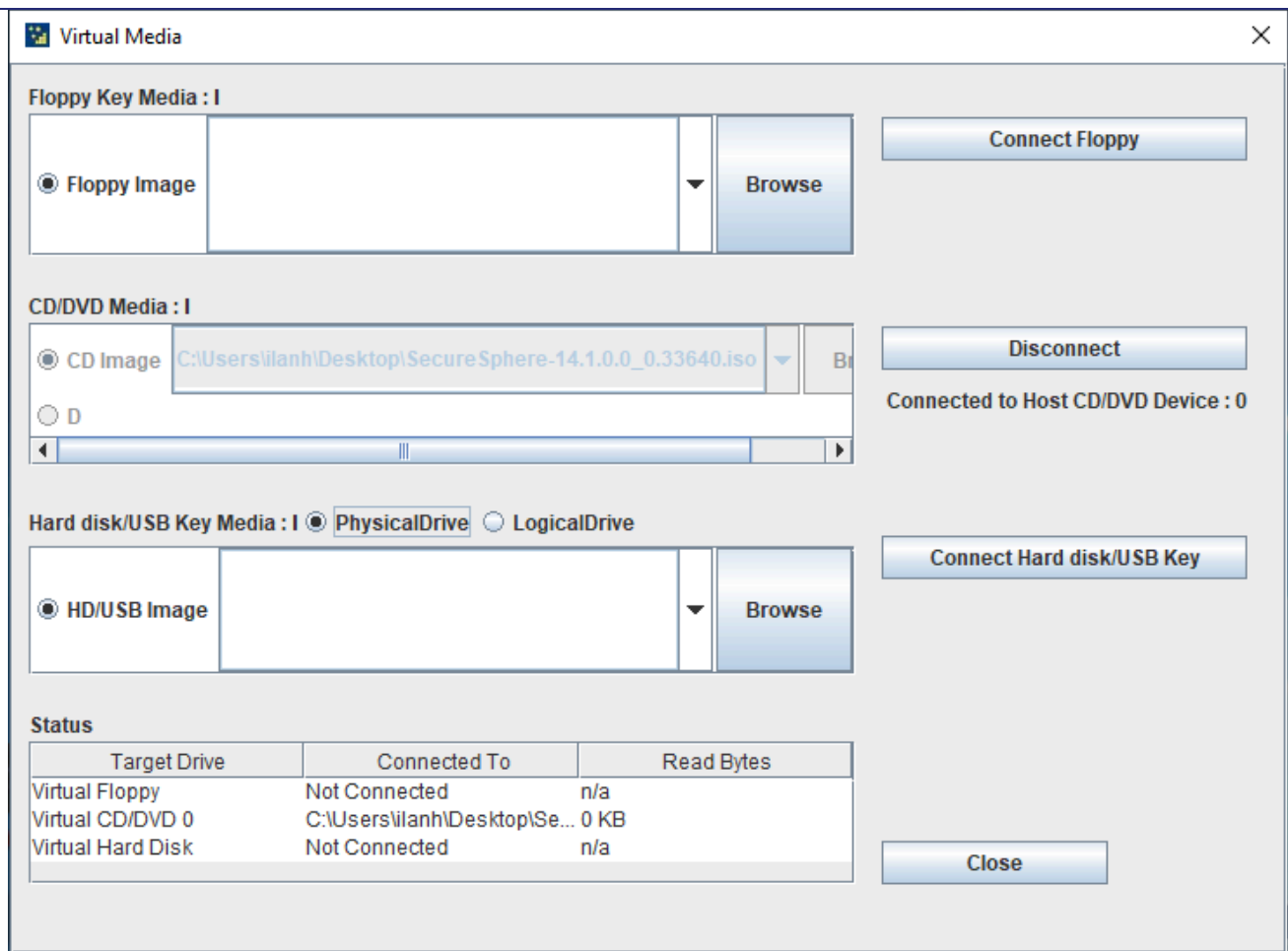
Hard disk/USB Key Media : I ☒ PhysicalDrive ☐ LogicalDrive

☒ HD/USB Image

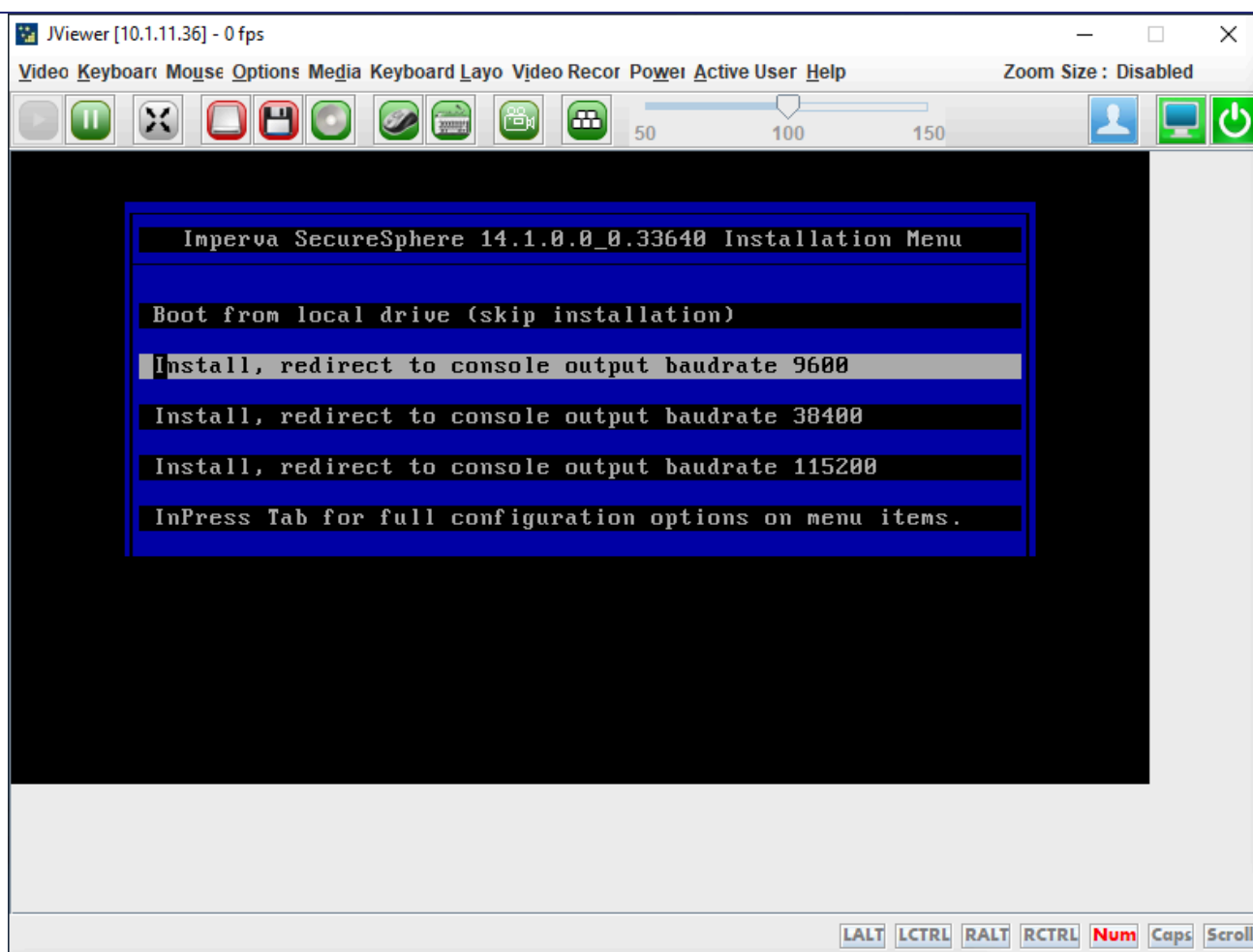
Status

Target Drive	Connected To	Read Bytes
Virtual Floppy	Not Connected	n/a
Virtual CD/DVD	Not Connected	n/a
Virtual Hard Disk	Not Connected	n/a

2. Under CD/DVD Media, select **CD Image** and click **Browse**.
3. Navigate to the location of the SecureSphere ISO file and click **Open**.
4. Click **Connect CD/DVD**. An **Information** pop up is displayed.
5. Click **OK**. The **Virtual Media** wizard is displayed showing the selected file and that a connection to the media was established.



6. Click **Close**.
7. Reboot the SecureSphere machine. The machine boots and the **Imperva SecureSphere Installation Menu** is displayed.



8. Select the 9600 baudrate option and press **Enter**. The installation begins.

- Video - IPMI for 6G2U Appliances
- Keyboard - IPMI for 6G2U Appliances
- Mouse - IPMI for 6G2U Appliances
- Options - IPMI for 6G2U Appliances
- Media - IPMI for 6G2U Appliances
- Keyboard Layout - IPMI for 6G2U Appliances
- Video Record - IPMI for 6G2U Appliances
- Power - IPMI for 6G2U Appliances
- Active Users - IPMI for 6G2U Appliances
- Help - IPMI for 6G2U Appliances

Video - IPMI for 6G2U Appliances

This menu contains the following sub menu items:

- Pause redirection: This option is used for pausing Console Redirection.
- Resume Redirection: This option is used to resume the Console Redirection when the session is paused.

- Refresh Video: This option can be used to update the display shown in the Console Redirection window.
- Capture Screen: This option helps to take the screenshot of the host screen and save it in the client's system
- *Compression Mode : This option helps to compress the Video data transfer to the specific mode.
- *DTC Quantization Table: This option helps to choose the video quality.
- Turn OFF Host Display/ *Host Video Output: If you enable this option, the server display will be blank but you can view the screen in Console Redirection. If you disable this option, the display will be back in the server screen.
- **Low Bandwidth Mode: This option is used to control the video packet dataflow in the network.
- Full Screen: This option is used to view the Console Redirection in full screen mode (Maximize). This menu is enabled only when both the client and host resolution are the same.
- Exit: This option is used to exit the console redirection screen.



Notes:

- * Specific to AST2300
- ** Specific to Hornet

Keyboard - IPMI for 6G2U Appliances

This menu contains the following sub menu items:

- Hold Right Ctrl Key: This menu item can be used to act as the right-side <CTRL> key when in Console Redirection.
- Hold Right Alt Key: This menu item can be used to act as the right-side <ALT> key when in Console Redirection.
- Hold Left Ctrl Key: This menu item can be used to act as the left-side <CTRL> key when in Console Redirection.
- Hold Left Alt Key: This menu item can be used to act as the left-side <ALT> key when in Console Redirection.
- Left Windows Key: This menu item can be used to act as the left-side <WIN> key when in Console Redirection. You can also decide how the key should be pressed: Hold Down or Press and Release.
- Right Windows Key: This menu item can be used to act as the right-side <WIN> key when in Console Redirection. You can also decide how the key should be pressed: Hold Down or Press and Release.
- Ctrl+Alt+Del: This menu item can be used to act as if you depressed the <CTRL>, <ALT> and keys down simultaneously on the server that you are redirecting. Context menu: This menu item can be used to act as the context menu key, when in Console Redirection.
- Hot Keys: This menu is used to add the user configurable shortcut keys to invoke in the host machine. The configured key events are saved in the BMC. Full Keyboard Support: Enable this option to provide full keyboard support. This option is used to trigger the Ctrl and Alt key directly to host from the physical keyboard.

Mouse - IPMI for 6G2U Appliances

- **Show Cursor:** This menu item can be used to show or hide the local mouse cursor on the remote client system.
- **Mouse Calibration:** This menu item can be used only if the mouse mode is relative. In this step, the mouse threshold settings on the remote server will be discovered. The local mouse cursor is displayed in RED color and the remote cursor is part of the remote video screen. Both the cursors will be synchronized in the beginning. Please use '+' or '-' keys to change the threshold settings until both the cursors go out of synch. Please detect the first reading on which cursors go out of synch. Once this is detected, use 'ALT-T' to save the threshold value.
- **Show Host Cursor:** This option is used to enable or disable the visibility of the host cursor.



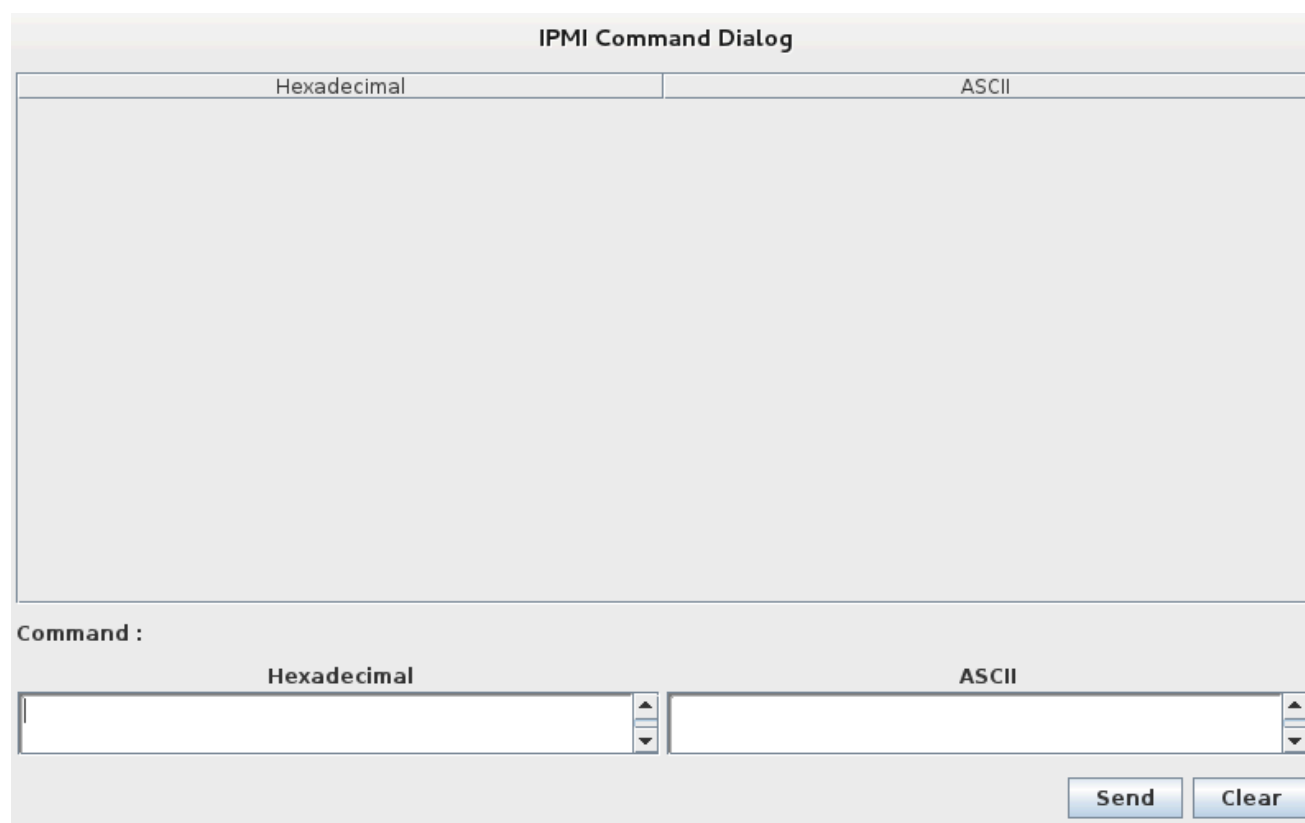
Note: Client cursor will be hidden always. If you want to enable, use Alt + C to access the menu.

- **Mouse Mode:** This option handles mouse emulation from local window to remote screen using either of the two methods. Only 'Administrator' has the right to configure this option.
 - **Absolute mouse mode:** The absolute position of the local mouse is sent to the server if this option is selected.
 - **Relative mouse mode:** The Relative mode sends the calculated relative mouse position displacement to the server if this option is selected.
 - **Other mouse mode:** This mouse mode sets the client cursor in the middle of the client system and will send the deviation to the host. This mouse mode is specific for SUSE Linux installation.

Options - IPMI for 6G2U Appliances

- **Band width (Except Hornet):** The Bandwidth Usage option allows you to adjust the bandwidth. You can select one of the following:
 - Auto Detect - This option is used to detect the network bandwidth usage of the BMC automatically.
 - 256 Kbps
 - 512 Kbps
 - 1 Mbps
 - 10 Mbps
- **Keyboard/Mouse Encryption:** This option allows you to encrypt keyboard inputs and mouse movements sent between the connections.
- **Zoom :** This option is available only when you launch the Java Console.
 - Zoom In – For increasing the screen size. This zoom varies from 100% to 150% with an interval of 10%
 - Zoom Out – For decreasing the screen size. This zoom varies from 100% to 50% with an interval of 10%
 - Actual Size - By default this option is selected
 - Fit to Client Resolution - If the host screen resolution is greater than the client screen resolution, choose this option to fit the host screen to client screen.

- **Fit to Host Resolution** -If the host screen resolution is lesser than the client screen resolution,choose this option to resize the JViewer frame to the host resolution.
- **Send IPMI Command**: This option opens the IPMI Command dialog. Enter the raw IPMI command in Hexadecimal field as Hexadecimal value and click Send.



The image shows a window titled "IPMI Command Dialog". It has a large text area at the top with two tabs: "Hexadecimal" and "ASCII". Below this is a section labeled "Command :" with two input fields, also with "Hexadecimal" and "ASCII" tabs. At the bottom right are "Send" and "Clear" buttons.

- **GUI Languages** : Choose the desired GUI language.
- **Request Full Permission** : Partially Permitted sessions can use this option to request the Full permission from the existing full permitted session.



Note: This menu option is available only for partially privileged session. Full permissions does not have this option in the menu.

Media - IPMI for 6G2U Appliances

- **Virtual Media Wizard**: To add or modify a media, select and click Virtual Media Wizard button, which pops out a box named Virtual Media where you can configure the media.

Virtual Media

Floppy Key Media : I

☒ Floppy Image

CD/DVD Media : I

☒ CD Image

☐ D

Hard disk/USB Key Media : I ☒ PhysicalDrive ☐ LogicalDrive

☒ HD/USB Image

Status

Target Drive	Connected To	Read Bytes
Virtual Floppy	Not Connected	n/a
Virtual CD/DVD	Not Connected	n/a
Virtual Hard Disk	Not Connected	n/a

- Floppy Key Media: This menu item can be used to start or stop the redirection of a physical floppy drive and floppy image types such as img.



Note: Floppy Redirection is not an available feature on all versions of the Imperva SPXs.

- CD/DVD Media: This menu item can be used to start or stop the redirection of a physical DVD/ CD-ROM drive and cd image types such as iso.
- Hard disk/USB Key Media: This menu item can be used to start or stop the redirection of a Hard Disk/USB key image and USB key image such as img.

Notes:



- For redirecting Floppy and Hard disk drives, the user should have administrator privilege (root user in the case of Linux clients).
- For Windows 7 and above, the web browser from which the KVM redirection will be initiated, should be launched using “Run as Administrator” option. If there are

multiple instances of the web browser open simultaneously, ensure that all the instances are launched using the “Run as Administrator” option.

- For Windows client, if the logical drive of the physical drive is dismounted then the logical device is redirected with Read/Write Permission else it is redirected with Read permission only.
- For MAC client, External USB Hard disk redirection is only supported.
- For Linux client, fixed hard drive is redirected only as Read Mode. It is not Write mode supported.
- For USB key image redirection, support FAT 16, FAT 32 and NTFS.
- Media redirection supports only Basic Hard disk Redirection.

Keyboard Layout - IPMI for 6G2U Appliances

- Auto Detect: This option is used to detect keyboard layout automatically. The languages supported automatically are English – US, French – France, Spanish – Spain, German- Germany. If the client and host languages are same, then for all the languages other than English mentioned above, you must select this option to avoid typo errors. If the host and client languages differ, user can choose the host language layout in the menu and thereby can directly use the physical keyboard.
- Soft Keyboard: This option allows you to select the keyboard layout. It will show the dialog as similar to onscreen keyboard. If the client and host languages are different, then for all the languages other than English mentioned above, you must select the appropriate language in the list shown in JViewer and use the soft keyboard to avoid typo errors.
- We have list of the language support in JViewer.

1. English –US
2. English – UK
3. Spanish
4. French
5. Germany (German)
6. Italian
7. Danish
8. Finnish
9. German (Switzerland)
10. Norwegian (Norway)
11. Portuguese (Portugal)

-
- 12. Swedish
 - 13. Hebrew
 - 14. French(Belgium)
 - 15. Dutch(Belgium)
 - 16. Russian
 - 17. Japanese
 - 18. Turkish – F
 - 19. Turkish – Q



Note: Soft keyboard is applicable only for JViewer Application not for other application in the client system.

Video Record - IPMI for 6G2U Appliances

- Start Record: This option is to start recording the screen.
- Stop Record: This option is used to stop the recording.
- Settings: To set the settings for video recording.



Note: Soft keyboard is applicable only for JViewer Application not for other application in the client system.

To perform video recording:

1. Click **Video Record > Settings** to open the settings page

Video Record

Video Length Seconds

Video to be Saved

Browse

OK

☒ **Normalized video resolution to 1024 X 768.**
This might reduce the video quality!

Cancel

2. Video Length - type the video length in seconds.
3. Video to be saved - Click **Browse** and navigate to the location where you want the video to be saved.
4. Normalized video resolution to 1024X768 - Select check box.
5. Click **OK** to save the entries and return to the Console Redirection screen.
6. Click **Cancel** if you don't wish to save the entries.
7. In the Console Redirection window, click **Video Record > Start Record**. The recording starts.
8. To stop the recording, click **Video Record > Stop Record**.

Power - IPMI for 6G2U Appliances

The power option is to perform any power cycle operation. Click on the required option to perform the following operation.

- Reset Server: To reboot the system without powering off (warm boot).
- Immediate Shutdown: To immediately power off the server.
- Orderly Shutdown: To initiate operating system shutdown prior to the shutdown.
- Power On Server: To power on the server.
- Power Cycle Server: To first power off, and then reboot the system (cold boot).

Active Users - IPMI for 6G2U Appliances

Click this option to displays the active users and their system ip address.

Help - IPMI for 6G2U Appliances

Jviewer: Displays the copyright and version information.

Server Power Control - IPMI for 6G2U Appliances

This page allows you to view and control the power of your server.

To open the Power Control and Status page, click **Remote Control > Server Power Control** from the menu bar. The Power Control and Status screen is displayed.



Power Control and Status Fields

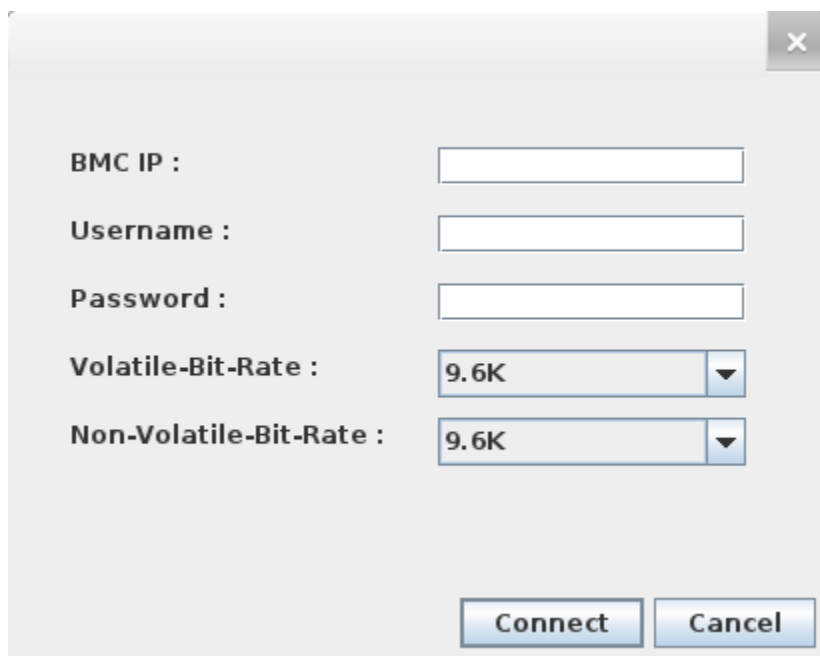
Field Name	Description
Reset Server	Select this option to reboot the system without powering off (warm boot).
Power Off Server – Immediate	Select this option to immediately power off the server.
Power Off Server – Orderly Shutdown	Select this option to initiate operating system shutdown prior to the shutdown.
Power On Server	Select this option to power on the server.
Power Cycle Server	Select this option to first power off, and then reboot the system (cold boot).
Perform Action	Click to perform the selected operation. A confirmation dialog box is displayed. Upon confirmation, the command will be executed and you will be informed of the status.

Java SOL - IPMI for 6G2U Appliances

This page allows you to launch the Java SOL. The Java SOL is used to view the host screen using the SOL Redirection.

To open the Java SOL page:

1. Click **Remote Control > Java SOL** from the menu bar. The Java SOL window is displayed.
2. Click the **Java SOL** button to open the Java SOL window.



The screenshot shows a Java SOL window with the following fields and controls:

- BMC IP :** A text input field.
- Username :** A text input field.
- Password :** A text input field.
- Volatile-Bit-Rate :** A dropdown menu with '9.6K' selected.
- Non-Volatile-Bit-Rate :** A dropdown menu with '9.6K' selected.
- Connect** and **Cancel** buttons at the bottom.

3. BMC IP - Type the BMC IP address, Username and Password values in the respective fields.
4. Volatile-Bit-Rate - Select the appropriate value from the drop down list.
5. Non-Volatile-Bit-Rate - Select the appropriate value from the drop down list.
6. Click **Connect**. The SOL redirection window opens.

Maintenance - IPMI for 6G2U Appliances

This group of pages allows you to do maintenance tasks on the device. The menu contains the following items.

- Restore Configuration
- Backup and Restore Configuration
- [Restore Configuration - IPMI for 6G2U Appliances](#)
- [Backup and Restore Configuration - IPMI for 6G2U Appliances](#)

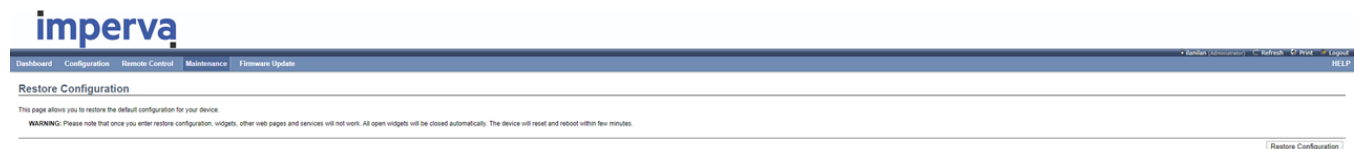
Restore Configuration - IPMI for 6G2U Appliances

In webGUI, this option is used to restore the factory defaults of the device firmware. This section lists the configuration items that will be preserved during restore factory default configuration.



Warning: After entering restore factory widgets, other web pages and services will not work. All open widgets will be closed automatically. The device will reset and reboot within few minutes.

To open the Restore Configuration page, click **Maintenance > Restore Configuration** from the menu bar. The Restore Configuration screen is displayed.

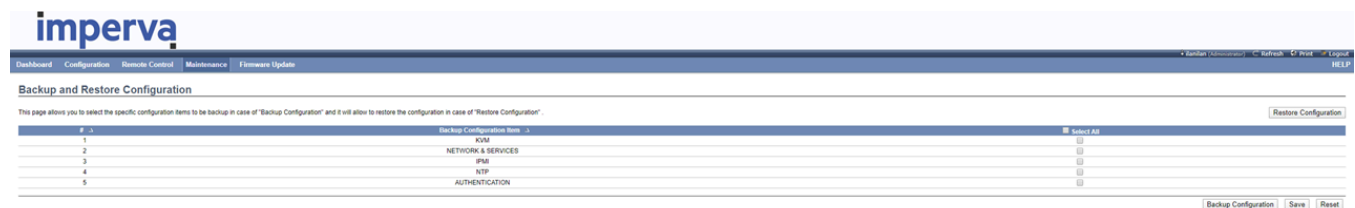


To to restore the factory defaults of the device firmware, click **Restore Configuration**.

Backup and Restore Configuration - IPMI for 6G2U Appliances

This page allows you to select the specific configuration items to be backup in case of “Backup Configuration” and it allows you to restore the configuration in case of “Restore Configuration”.

To open Backup and Restore Configuration page, click Maintenance > Backup and Restore Configuration from the menu bar. A sample screenshot of Backup and Restore Configuration page is shown below.



Backup and Restore Configuration Fields

Field Name	Description
Restore Configuration	Click to restore the configuration items, which were backup earlier. In the Restore Configuration page, click Choose File and browse to select the configuration file.
Backup Configuration Item	Select check box to include the item in the backup from BMC to client system.
Select All	Select check box to select all the configuration list.

Field Name	Description
Backup Configuration	Click to backup configuration items, that were selected.
Save	To save any changes made.
Reset	To reset the modified changes.

Firmware Update - IPMI for 6G2U Appliances

This group of pages allows you to do the following. The menu contains the following items.

- Firmware Update
- Protocol Configuration
- Dual Image Configuration
- [Firmware Update - IPMI for 6G2U Appliances](#)
- [Protocol Configuration - IPMI for 6G2U Appliances](#)
- [Dual Image Configuration - IPMI for 6G2U Appliances](#)

Firmware Update - IPMI for 6G2U Appliances

This wizard takes you through the process of firmware upgrade. A reset of the box will automatically follow if the upgrade is completed or cancelled. An option to preserve configuration will be presented. Enable it, if you wish to preserve configured settings through the upgrade.

	Warning: After entering update mode widgets, other web pages and services will not work. All open widgets will be closed automatically. If upgrade process is canceled in the middle of the wizard, the device will be reset.
---	---

To open the Firmware Update page, click **Firmware Update > Firmware Update** from the menu bar. The Firmware Update screen is displayed.

imperva

Dashboard Configuration Remote Control Maintenance **Firmware Update** Random Information Refresh Print Logout HELP

Firmware Update

Upgrade firmware of the device. Press "Enter Update Mode" to put the device in update mode.

The protocol information to be used for firmware image transfer during this update is as follows. To configure, choose "Protocol Configuration" under Firmware Update menu.
Protocol Type : HTTP/HTTPS

The dual image formation to be used for firmware update is displayed as follows. To configure image to be booted from upon Reset, choose "Dual Image Configuration" under Firmware Update menu.
Current Active Image : Image 1
Image to be updated : Both Images
☒ Reboot the device after update

WARNING: Please note that after entering the update mode, the widgets, other web pages and services will not work. All the open widgets will be automatically closed. If the upgradation is cancelled in the middle of the wizard, the device will be reset.

☐ Preserve All Configuration. This will preserve all the configuration settings during the firmware update.

Enter Update Mode

Firmware Update Fields

Field Name	Description
Current Active Image	Displays the name of current active Image.
Image to be uploaded	Select the image to be uploaded. If required both the images can be chosen.
Reboot the device after update	Select check box to reboot the machine after the update is done.
Preserve All Configurations	Select check box to preserve all the listed configurations.
Enter Update Mode	Click to upgrade the current device firmware. Warning: After entering the update mode, the widgets, other web pages and services will not work. All the open widgets will be automatically closed. If the upgrade is canceled in the middle of the wizard, the device will be reset.

To perform a firmware update:

1. Select the **Preserve All Configuration** check box.
2. From the Image to be updated drop-down list select the image to be updated.
3. Select the **Reboot the device after update** check box if desired.
4. Click **Enter Update Mode**. The Firmware update undergoes the following steps.
 - a. Closing all active client requests
 - b. Preparing Device for Firmware Upgrade
 - c. Uploading Firmware Image.

A file upload pop-up is displayed for http/https. Click **Choose File** and navigate to select the Firmware image to flash and then click **Upload**.

For tftp files, the file is automatically uploaded displaying the status of upload.

- d. Verifying Firmware Image

- In Section Based Firmware Update, you can configure the firmware image for section based flashing. Check the required sections and click Proceed to update the firmware.
- If flashing is required for all images, select the option Full Flash.
- If you select Version Compare Flash option from web, the current and uploaded module versions, FMHlocation, size will be compared.
- If the modules differ in size and location, proceed with force firmware upgrade.
- If all the module versions are same, restart BMC by saying all the module versions are similar.
- If only few module versions are differ, those module will be flashed.



Note: Only selected sections of the firmware will be updated. Other sections are skipped. Before starting flash operation, you are advised to verify the compatibility between image sections.

- Flashing Firmware Image
- Resetting Device

Notes:



- You cannot perform any other tasks until firmware upgrade is complete and the device is rebooted.
- You can now follow the instructions presented in the subsequent pages to successfully update the card's firmware. The device resets if update is canceled or upon successful completion of firmware update.

Protocol Configuration - IPMI for 6G2U Appliances

This page is used to configure the firmware image protocol information.

To open the Protocol Configuration, click **Firmware Update > Protocol Configuration** from the menu bar. The Image Transfer Protocol screen is displayed.

imperva

Dashboard Configuration Remote Control Maintenance Firmware Update

Image Transfer Protocol

The following option allows you to configure firmware image protocol information.

Protocol Type: HTTP/HTTPS

Server Address: [Text Field]

Image Name: [Text Field]

Retry Count: [Text Field]

Save Reset

Transfer Protocol Fields

Field Name	Description
Protocol Type	Select the protocol type used to transfer the firmware image into the BMC.
Server Address	Type the Server IP address where the firmware image is stored. This field is enabled for TFTP protocol type only. Notes: - IP Address made of 4 numbers separated by dots as in “xxx.xxx.xxx.xxx”. - Each number ranges from 0 to 255. - First number must not be 0.
Image Name	Type the name of the image. This field is enabled for TFTP protocol type only.
Retry Count	Type the number of times to be retried when transfer failure occurs. Retry count ranges from 0 to 255. This field is enabled for TFTP protocol type only.
Save	Click to save the configured settings.
Reset	Click to reset the modified changes.

Dual Image Configuration - IPMI for 6G2U Appliances

This page is used to configure the dual image information. Dual Image support feature is helpful to store two firmware images on two 32MB SPI's, and boot any of the image according to users request. The running firmware is responsible for setting the Boot Selector options and Firmware Upload Selector options.

To open the Dual Image Configuration, click **Firmware Update > Dual Image Configuration** from the menu bar. The Dual Image Configuration screen is displayed.

imperva

Dashboard Configuration Remote Control Maintenance Firmware Update

Dual Image Configuration

The following option will allow to configure firmware dual image information.

Image 1		Image 2	
Firmware Version	1.01.7000	Firmware Version	1.01.7000
State	Active	State	Stand-by
☐ Image to be booted from upon reset		☐ Image to be booted from upon reset	

☒ Higher firmware version
☐ Lower firmware version
☐ Most recently updated firmware
☐ Least recently updated firmware

Save Reset

Dual Image Configuration Fields

Field Name	Description
Firmware Version	Displays the firmware version of image 1 and 2.
State	Displays the current state of image 1 and 2.
Image to be booted from upon reset	Select if to boot from image 1 or 2 in the next boot up process.
Higher firmware version	Select to boot a higher firmware version image than image 1 or 2 in the next boot up process.
Lower firmware version	Select to boot a lower firmware version image than image 1 or 2 in the next boot up process.
Most recently updated firmware	Select to boot the most recently updated firmware image than image 1 or 2 in the next boot up process.
Least recently updated firmware	Select to boot the least recently updated firmware image than image 1 or 2 in the next boot up process.
Save	Click to save the configured settings.
Reset	Click to reset the modified changes.

IPMI - 7G2U Appliances

This section provides instructions for managing the server using the IPMI. IPMI is included on all 7G2U servers. If users have one of these servers, it will include an IPMI Supplement which contains platform-specific information, and details about the hardware.

- [IPMI and System Management Overview - IPMI for 7G2U Appliances](#)
- [Using the IPMI WebGUI - IPMI for 7G2U Appliances](#)
- [Dashboard](#)
- [Logs & Reports](#)

-
- [Settings](#)
 - [Remote Control](#)
 - [Image Redirection](#)
 - [Power Control](#)
 - [Maintenance](#)
 - [Sign Out](#)

IPMI and System Management Overview - IPMI for 7G2U Appliances

IPMI is a dedicated system of hardware and supporting software that allows users to manage a server independent of the operating system. The IPMI includes the following components:

- **Service Processor:** This is the hardware. It consists of a dedicated processor board that communicates through the system serial port and a dedicated Ethernet port.
- **WebGUI:** The WebGUI provides a powerful, easy-to-use browser interface that allows users to log in to perform system management, monitoring, and IPMI tasks. Users only need to install the Java client appliance on first use.
- **Remote Console**
 - **H5 Viewer:** Browser-based viewer supporting Chrome latest version, Firefox (with limited support), Microsoft Chromium-based Edge and Safari (On Mac only)
 - **Java™ Client:** The Java client supports the Remote Console functionality, which allows the user to access the server's console remotely. It redirects the keyboard and video screen, and can redirect input and output from the local machine's storage such as CD/DVD-ROM and hard disk drives.

Using the IPMI WebGUI - IPMI for 7G2U Appliances

This section describes how to use the IPMI WebGUI.

- [Overview of WebGUI Features - IPMI for 7G2U Appliances](#)
- [Logging In and Out of the WebGUI - IPMI for 7G2U Appliances](#)
- [Menu Bar - IPMI for 7G2U Appliances](#)
- [Quick Buttons](#)
- [Help](#)

Overview of WebGUI Features - IPMI for 7G2U Appliances

The UI enables the user to monitor and manage remote systems. The user can quickly activate the WebGUI using a web browser. The following list indicates the recommended browsers to launch the WebGUI in.

- [Google Chrome](#)

- Firefox
- Microsoft Chromium-based Edge
- Safari (On Mac only)

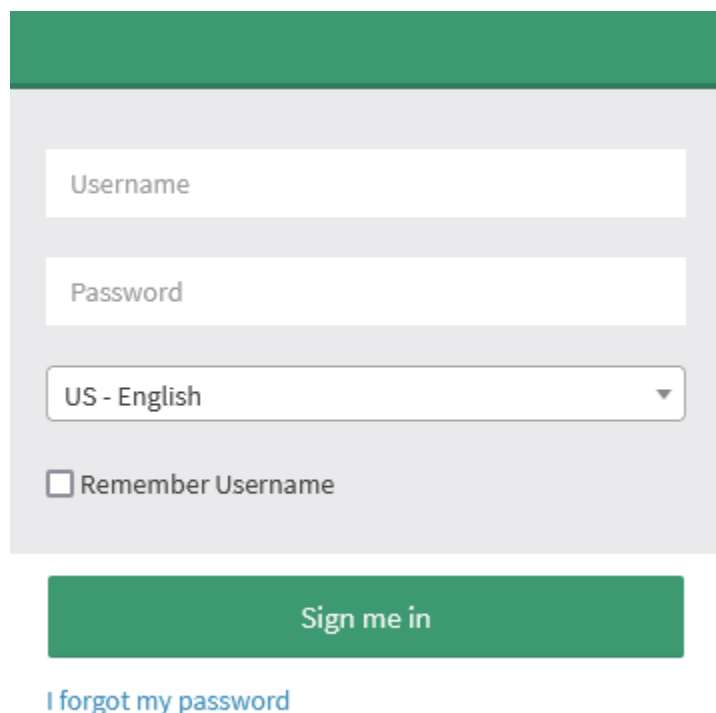
Notes:

- Imperva does not provide debug support for any issues that occur in browsers that are not recommended.
- When using the Chrome browser in private mode, the session is copied to another private tab for WebGUI management purposes. This does not occur when using the other recommended browsers.

Logging In and Out of the WebGUI - IPMI for 7G2U Appliances

To log in to the WebGUI for the first time:

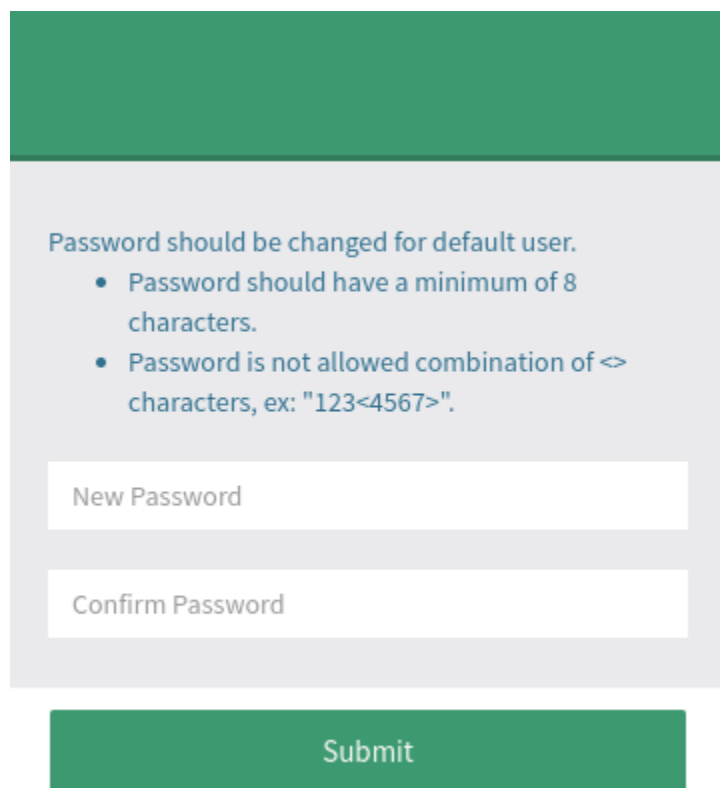
1. Enter the IP address of the IPMI into web browser: **https://192.168.1.1**. The WebGUI login screen is displayed.

The image shows a web browser window displaying the WebGUI login screen. At the top is a green header bar. Below it is a light gray login form. The form contains three input fields: 'Username', 'Password', and a language selection dropdown menu currently showing 'US - English'. Below these fields is a checkbox labeled 'Remember Username'. At the bottom of the form is a large green button labeled 'Sign me in'. Below the button is a blue link that says 'I forgot my password'.

2. In the Username field, enter **admin**.
3. In the Password field, enter **ImpSN#<machine serial number>**. For IPMI versions lower than 1.0.6, enter **<machine serial number>**.
4. In the Language Selection field, select your preferred language from the drop-down. The drop down is

populated based on supported languages in the WebGUI as a part of multi language support feature.

5. Click **Sign me in**. The modify default user's password dialog box appears.



6. Create a new password. The WebGUI login screen is displayed.
7. In the Username field, enter **admin**.
8. In the Password field, enter the new password.
9. In the Language Selection field, select your preferred language from the drop-down. The drop down is populated based on supported languages in the WebGUI as a part of multi language support feature.
10. Select the **Remember Username** check box to remember your login credentials.
11. If you forget your password, click **I Forgot my Password** to generate a new password.
12. Click **Sign me in**. The Dashboard screen is displayed.

To logout of the WebGUI:

Click **admin > Sign Out** at the top right of the WebGUI. The login screen is displayed.

Notes:

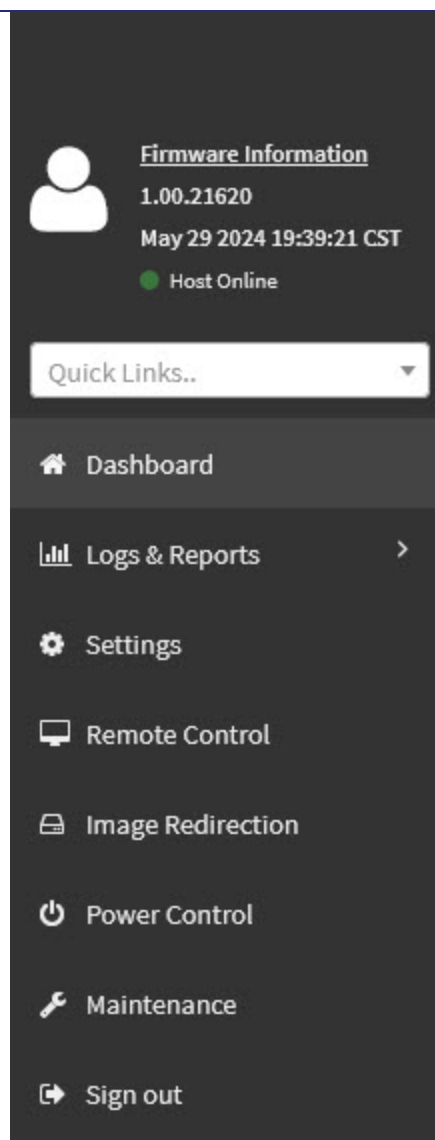
- Duplicate user names shouldn't exist across various authentication methods like AD, LDAP, RADIUS or IPMI since the privilege of one Authentication method is overwritten by another authentication method when login and hence the correct privilege cannot be returned properly.
- Duplicate user names shouldn't exist across different channels in IPMI.
- Once you login to the application, do not use the following options:

- Refresh button of the browser
- Refresh menu of the browser
- Back and Forward options of the browser
- F5 on the keyboard
- Backspace on the keyboard

Menu Bar - IPMI for 7G2U Appliances

The menu bar is located on the left side of the page and displays the following:

- **Firmware Information:** Displays the latest version, built date and time.
- **Power Control Status:** Displayed as Host Online when the host is powered on and vice versa.
- **Quick Links:** A short-cut for the available menu and sub-menu pages. It displays available search queries. As you type search terms in the field, the suggestions will be displayed in a drop-down list below the Quick Links field. On selecting your search term from the drop-down list, it will directly go to the specific page which you have searched.



Quick Buttons

The quick buttons are located at the top right of the WebGUI.



They allow you to perform the following functions:

- **Message:** Click the  icon to view the event log alert messages. On clicking the messages, it will navigate to the Logs and Reports page.
- **Notification:** Click  to view the notification received.
- **Language Selection:** Change the language to view the language strings in different languages.
- **Sync:** Click the  Sync icon to synchronize with Latest Sensor and Event Log updates.
- **Refresh:** Click the  Refresh icon to reload the current page.
- **Logged-in User:** Click the  icon to view the logged-in user name and privileges. There are five kinds of

privileges:

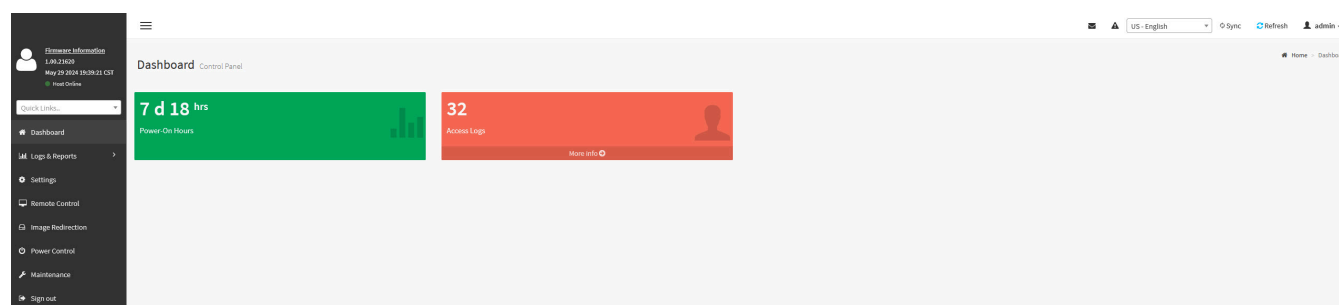
- **User:** Only valid commands are allowed.
- **Operator:** All commands are allowed except for the configuration commands that can change the behavior of the out-of-hand interfaces.
- **Administrator:** All commands are allowed.
- **No Access:** Login access denied.
- **OEM:** All OEM commands are allowed.

Help

The Help icon  is Located at the top right of each page in the WebGUI. Click this help icon to view more detailed field descriptions.

Dashboard

The Dashboard page gives the overall information about the status of a device. To open the Dashboard page, click **Dashboard**.



Power-On Hours

Power-On Hours keeps on accumulating and is reset to zero when you flash a new image.

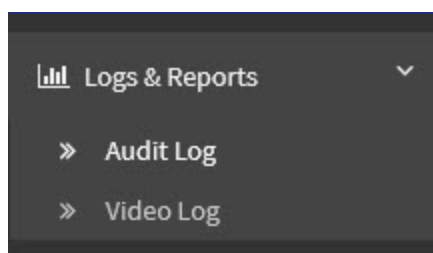
Access Logs

Access Logs displays the number of times the unit was accessed. To know the detailed information, click **More info** to view the Audit Log page.

Logs & Reports

The Logs & Reports contains the following information:

- Audit Log
- Video Log



- Audit Log
- Video Log

Audit Log

Audit Log displays all the audit events for this device that has been already configured. To open the Event Log page, click **Logs & Reports > Audit Log**.

 The screenshot shows the 'Audit Log' page with the breadcrumb 'Home > Audit Log'. Below the header is a 'Filter by Date' section with 'Start Date' and 'End Date' input fields, each with a calendar icon. Below this is a section titled 'Audit Log: 8 out of 8 event entries'. A red tab labeled 'March 2023' is active. A vertical timeline on the left shows five event entries, each with a flag icon:

- ID: 8 March 2nd 2023, 6:30:12 pm AMI0835712178F9 spx_restservice: spx_restservice -- [1546 : 1546 INFO]https Login from IP:192.168.122.131 user:admin -
- ID: 7 March 2nd 2023, 6:30:08 pm AMI0835712178F9 spx_restservice: spx_restservice -- [1546 : 1546 INFO]HTTPS logout from IP:192.168.122.131 user:admin -
- ID: 6 March 2nd 2023, 6:30:04 pm AMI0835712178F9 spx_restservice: spx_restservice -- [1546 : 1546 INFO]https Login from IP:192.168.122.131 user:admin -
- ID: 5 March 2nd 2023, 6:29:59 pm AMI0835712178F9 login[2424]: login 2424 - [2424 : 2424 INFO]SERIAL Login from IP:127.0.0.1 user:sysadmin -
- ID: 4 March 2nd 2023, 6:29:52 pm AMI0835712178F9 login[1651]: login 1651 - [1651 : 1651 WARNING]SERIAL session timeout from IP:127.0.0.1 user:sysadmin -

You can **Filter by Date**, by selecting the Start Date and End Date using the pop up calendar.

Video Log

Video Log displays available recorded video files. To open the Event Log page, click **Logs & Reports > Video Log**.

Note: Video Trigger Settings needs to be enabled in order to display the Video Log page. To enable, go to **Settings -> Video Recording -> Auto Video Settings -> Video Trigger Settings** and select your desired triggers.

Video Log All video event logs Home > Video Log

Filter by Date -

Video Log: 2 out of 2 event entries

March 2023

- video_dump_2023y-03m-02d_18h-32m-36s.dat Post Event Recorded on Thursday, March 2nd 2023, 6:32:37 pm ID: 2 a few seconds ago
- video_dump_2023y-03m-02d_18h-32m-10s.dat Post Event Recorded on Thursday, March 2nd 2023, 6:32:30 pm ID: 1 a few seconds ago

To view the video:

1. In the Video Log page, click on the Video Log entry.
2. You can Download , Play/Pause  and Delete  the video by clicking the respective icons.

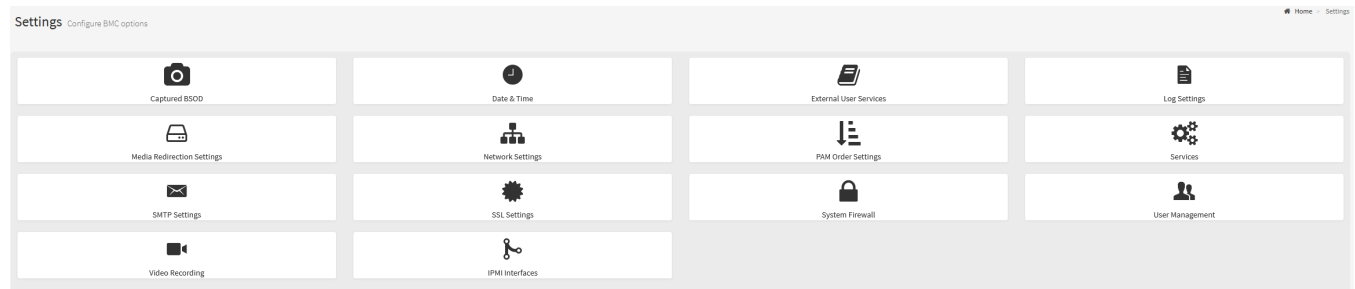
Notes:

- A video can be played or downloaded only if file size is less than 40MB. Browsers have various memory restrictions, due to this browser cannot store and process data greater than 40MB (approximately). If file size is greater than 40MB, you are notified with a message to use a Java player Application.
- The video data may not be proper if the browser zoom in/out settings are changed during video playback.
- The video is downloaded as a .avi file.
- The generated video raw file (.dat file) can only be played using the WebGUI. It cannot be played using other media players.

Note:

Settings

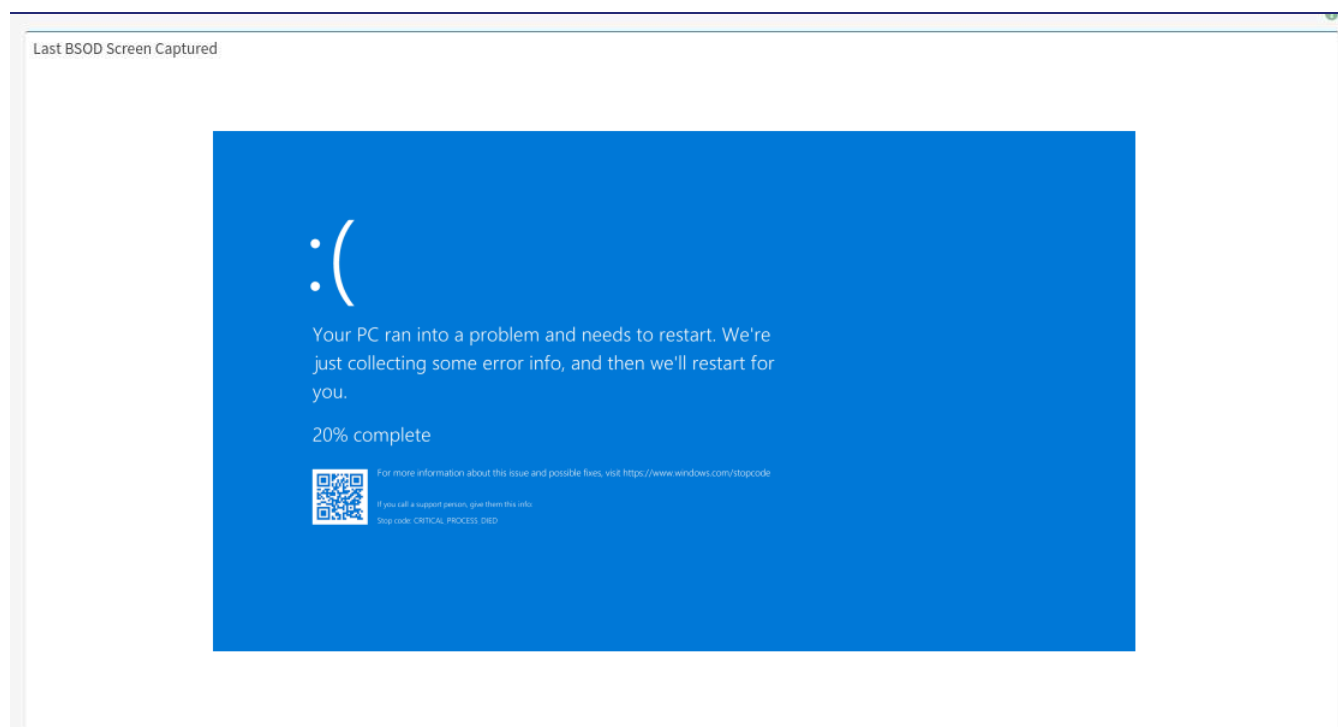
The Settings page allows you to access various configuration settings. To open the Settings page, click **Settings**.



- [Captured BSOD](#)
- [Date & Time](#)
- [External User Services](#)
- [Log Settings](#)
- [Media Redirection Settings](#)
- [Network Settings](#)
- [PAM Order Settings](#)
- [Services](#)
- [SMTP Settings](#)
- [SSL Settings](#)
- [System Firewall](#)
- [User Management](#)
- [Video Recording](#)
- [IPMI Interfaces](#)

Captured BSOD

This page displays a snapshot of the blue screen captured if the host system crashed since last reboot.



Note: A KVM service needs to be enabled in order to display the BSOD screen. To enable, go to **Settings > Services > KVM** and set up the service.

Date & Time

This page is used to set the date and time. To open the Date & Time page, click **Setting > Date & Time**.

[Home](#) > [Settings](#) > [Date & Time](#)

Date & Time

Note:
If the time zone is selected from the group of Manual offset (GMT/ETC time zones), the interactive map selection feature will be disabled.
The new Time Zone settings will be reflected on the page only after being saved.

Configure Date & Time ?



Asia/Taipei

Mar 2, 2023 6:44:35 PM (GMT+08:00 CST) - Asia/Taipei

⌂

☒ Automatic NTP Date & Time

Primary NTP Server

pool.ntp.org

Secondary NTP Server

time.nist.gov

Save

The available configuration options are:

- **Configure Date & Time:** Displays Timezone list containing the UTC offset along with the locations and Navigational line to select the location which can be used to display the exact local time.
- **Select Time Zone:** This field is used to set the date and time.

Note: Based on the manual selection of GMT or ETC/GMT, time zone alone is changed and date & time remains the same.

- **Automatic NTP Date & Time:** To automatically synchronize Date and Time with the NTP Server.
 - **Primary NTP Server:** To configure a primary NTP server to use when automatically setting the date and time.
 - **Secondary NTP Server:** To configure a secondary NTP server to use when automatically setting the date and time.
- **Save:** To save the settings.

Note: If the timezone is selected as Manual Offset, the map selection is disabled. The Time Zone settings will be reflected only after saving the settings.

To configure the date and time:

1. Select the **Timezone** location either using drop down or Map.
2. Enable **Automatic Date & Time** option to enable/disable the use of NTP servers to automatically set the date and time.
3. In the **Primary NTP Server** and **Secondary NTP Server** fields, specify the NTP servers of the device respectively.

Note: Secondary NTP server is optional. If the Primary NTP server is not working fine, then the Secondary NTP Server will be tried.

4. Click **Save** to save the settings.

External User Services

This page is used to set the date and time on the BMC. To open the External User Services page, click **Setting > External User Services**.

External User Services

[Home](#) > [Settings](#) > [External User Services](#)



LDAP/E-Directory Settings



Active Directory Settings



RADIUS Settings

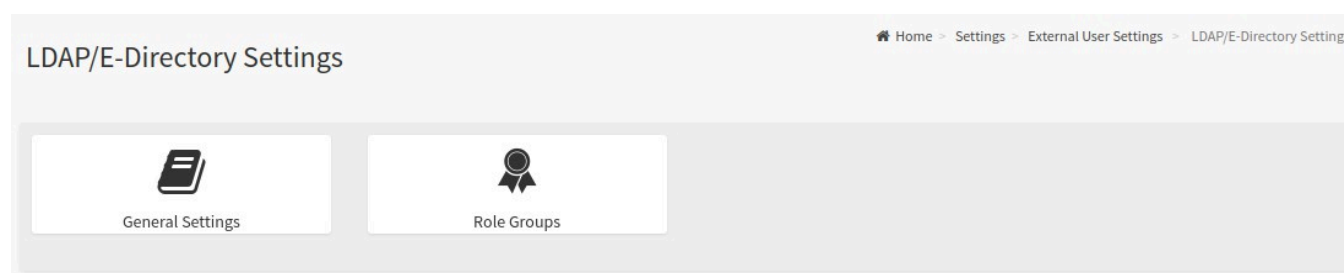
- [LDAP/E-Directory Settings](#)
- [Active Directory Settings](#)
- [RADIUS Settings](#)

LDAP/E-Directory Settings

The Lightweight Directory Access Protocol (LDAP) / E-Directory Settings is an application protocol for querying and modifying data of directory services implemented in Internet Protocol (IP) networks.

In IPMI WebGUI, LDAP is an Internet protocol used to authenticate users. If you have an LDAP server configured on your network, you can use it as an easy way to add, manage and authenticate LOM card users. This is done by passing login requests to your LDAP Server. This means that there is no need to define an additional authentication mechanism. Since your existing LDAP Server keeps an authentication centralized, you will always know who is accessing the network resources and can easily define the user or group-based policies to control access.

To open the LDAP/E-DIRECTORY Settings page, click **Settings > External User Services > LDAP/E- Directory Settings**.



The available setting are:

- **General Settings:** To configure LDAP/E-Directory Settings. Options are Enable LDAP/E-Directory Authentication, IP Address, Port and Search base.
- **Role Groups:** To add a new role group to the device. Alternatively, double click on a free slot to add a role group.

To configure General LDAP/E-Directory Settings:

1. In the LDAP/E-Directory Settings page, click **General Settings**. The General LDAP Settings page is displayed.

General LDAP Settings

[Home](#) > [Settings](#) > [External User Settings](#) > [LDAP/E-Directory Settings](#) > [General LDAP Settings](#)

☒ Enable LDAP/E-Directory Authentication

Encryption Type
☒ No Encryption ☐ SSL ☐ StartTLS

Common Name Type
☒ IP Address ☐ FQDN

Server Address

Port

Bind DN

Password

Search Base

Attribute of User Login

2. Select the **Enable LDAP/E-Directory Authentication** check box to enable LDAP/E-Directory Settings.

Note: During login prompt, use username to login as an LDAP Group member.

3. For Encryption Type select **No Encryption**, **SSL** or **StartTLS**.

Note: Configure proper port number when SSL is enabled.

4. For Common Name Type select **IP Address**.
5. In the Server Address field, type the IP address of the LDAP server.

Notes:

- IP Address made of 4 numbers separated by dots as in 'xxx.xxx.xxx.xxx'.
- Each Number ranges from 0 to 255.
- First Number must not be 0.
- Supports IPv4 Address format and IPv6 Address format.
- Configure FQDN address, when using StartTLS with FQDN.

6. In the Port field, type the LDAP port.

Notes:

- Default Port is 389.
- For SSL connections, the default port is 636. The Port value ranges from 1 to 65535.

7. In the Bind DN field, type the distinguished name that is used during the bind operation, which authenticates the client to the server.

Notes:

- Bind DN is a string of 4 to 253 alpha-numeric characters.
- It must start with an alphabetical character.
- Special Symbols like dot(.), comma(,), hyphen(-), underscore(_), equal-to(=) are allowed.
- Example: cn=manager,ou=login, dc=domain,dc=com

8. In the Password field, type the password used to connect to the LDAP server.

Notes:

- Password must be at least 1 character long.
- White space is not allowed.
- This field does not allow more than 48 characters.

9. In the Search Base field, type the search base that allows the LDAP server to find which part of the external directory tree is to be searched. The search base may be something equivalent to the organization, or group of external directory.

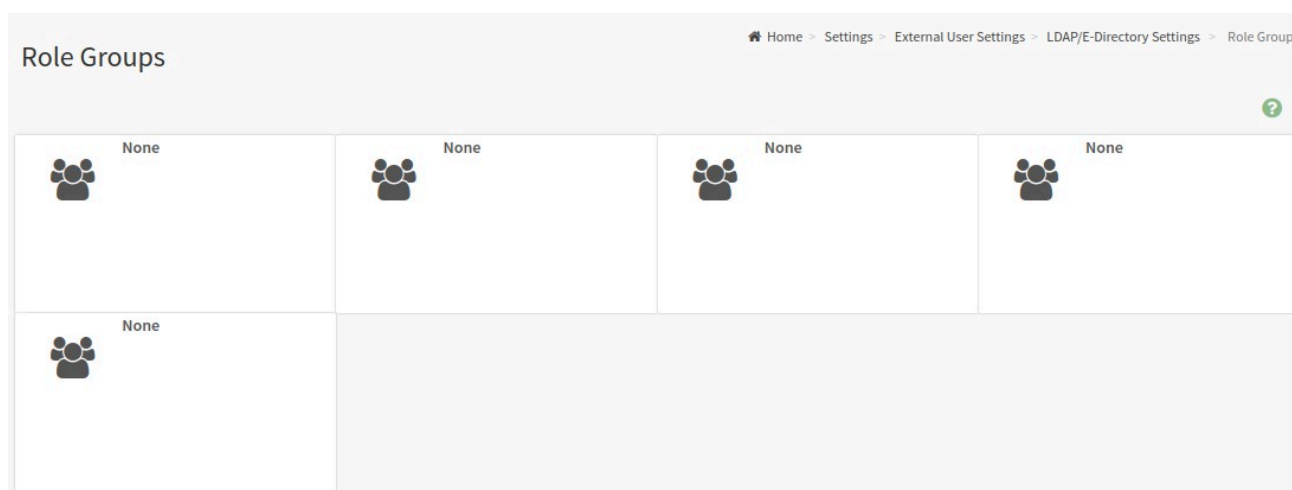
Notes:

- Search base is a string of 4 to 253 alpha-numeric characters.
- It must start with an alphabetical character.
- Special Symbols like dot(.), comma(,), hyphen(-), underscore(_), equal-to(=) are allowed.
- Example: ou=login,dc=domain,dc=com

10. In the Attribute of User Login drop down, select the attribute to be used to identify the user. Available attributes are: **cn** or **uid**.
11. If SSL or StartTLS is selected for Encryption Type:
 - a. For CA certificate file, browse to select the certificate of the trusted CA certs.
 - b. For Certificate File, browse to find the client certificate filename.
 - c. For Private Key, browse to find the client private key filename.
12. Click **Save** to save the settings.

To add a Role Group:

1. Verify the **Enable LDAP/E-Directory Authentication** check box is selected in the General LDAP Settings page.
2. In the LDAP/E-Directory Settings page, click **Role Groups**. The Role Groups page is displayed.



3. Double click on a blank row. The Add Role Group page is displayed.

Role Groups

[Home](#) > [Settings](#) > [External User Settings](#) > [LDAP/E-Directory Settings](#) > [Role Groups](#) > [Role Groups](#)

Group Name

Group Domain
eg., dc=domain

Group Privilege

☐ KVM Access

☐ VMedia Access

Save

- In the Group Name field, type the name that identifies the role group.

Notes:

- Maximum size of Role Group Name is 255 bytes.
- Special symbols hyphen and underscore are allowed.

- In the Group Domain field, type the Role Group Domain where the role group is located.

Notes:

- Maximum size of Group Domain Name is 255 bytes.
- It must start with an alphabetical character.
- Special Symbols like dot(.), comma(,), hyphen(-), underscore(_), equal-to(=) are allowed.
- Example: cn=manager,ou=login, dc=domain,dc=com

- From the Group Privilege drop down, select the level of privilege (User, Administrator, Operator, None) to assign to this role group.
- Select one or both of the following check boxes:
 - KVM Access
 - VMedia Access
- Click **Save** to save the new role group and return to the Role Group List.

To delete a Role Group:

- In the Role Groups Page, select the row that you wish to delete.

2. Click **Delete Role Group**.

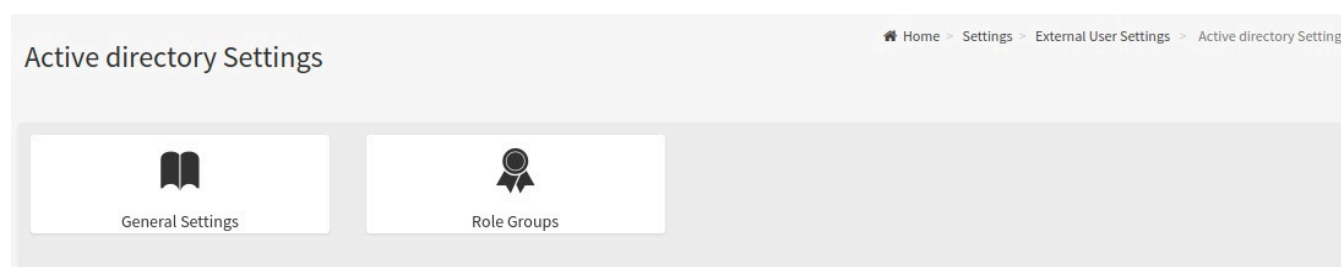
Active Directory Settings

An active directory (sometimes referred to as AD) is a directory structure used on Microsoft Windows based computers and servers to store information and data about networks and domains.

An active directory does a variety of functions including the ability to provide information on objects. It also helps to organize these objects for easy retrieval and access, allows access by end users and administrators and allows the administrator to set security up for the directory.

In IPMI WebGUI, Active Directory allows you to configure the Active Directory Server Settings. The displayed table shows any configured Role Groups and the available slots. You can modify, add or delete role groups from here. Group domain can be the AD domain or a trusted domain. Group Name should correspond to the name of an actual AD group.

To open the Active Directory Settings page, click **Settings > External User Settings > Active Directory**.



The available setting are:

- **General Settings:** This page is used to configure Active Directory General Settings. Options are Enable Active Directory Authentication, Secret User Name, Secret Password, User Domain name, and up to three Domain Controller Server Addresses.
- **Role Groups:** To add a new role group to the device. Alternatively, double click on a free slot to add a role group.

To configure General Active Directory Settings:

1. In the Active Directory Settings page, click **General Settings**. The General Active Directory Settings page is displayed.

General Active Directory Settings

[Home](#) > [Settings](#) > [External User Settings](#) > [Active directory Settings](#) > [General Active Directory Settings](#)



☒ Enable Active Directory Authentication

☒ SSL

Secret Username

Secret Password

User Domain Name

Domain Controller Server Address 1

Domain Controller Server Address 2

Domain Controller Server Address 3

 Save

2. Select the **Enable Active directory Authentication** check box to enable Active Directory Authentication.

Note: If you have enabled Active Directory Authentication, enter the required information to access the Active Directory server.

3. Select the **SSL** check box to enable SSL.
4. In the Secret Username and Secret Password fields, type the required credentials.

Notes:

- Secret username/password for AD is not mandatory. When secret username & password is empty, Authentication fails will be always treated as Invalid Password error. For Invalid Password error PAM will not try other Authentication Methods.
- User Name is a string of 1 to 64 alpha-numeric characters.
- It must start with an alphabetical character.
- It is case-sensitive.
- Special characters like comma, period, colon, semicolon, slash, backslash, square brackets, angle brackets, pipe, equal, plus, asterisk, question mark, ampersand, double quotes, space are not

allowed.

- Password must be at least 6 characters long and will not allow more than 127 characters.
- White space is not allowed.

5. In the User Domain Name field, type the name for the user. For example: MyDomain.com
6. In the Domain Controller Server Address 1, Domain Controller Server Address 2 and Domain Controller Server Address 3 fields, type the appropriate IP addresses.

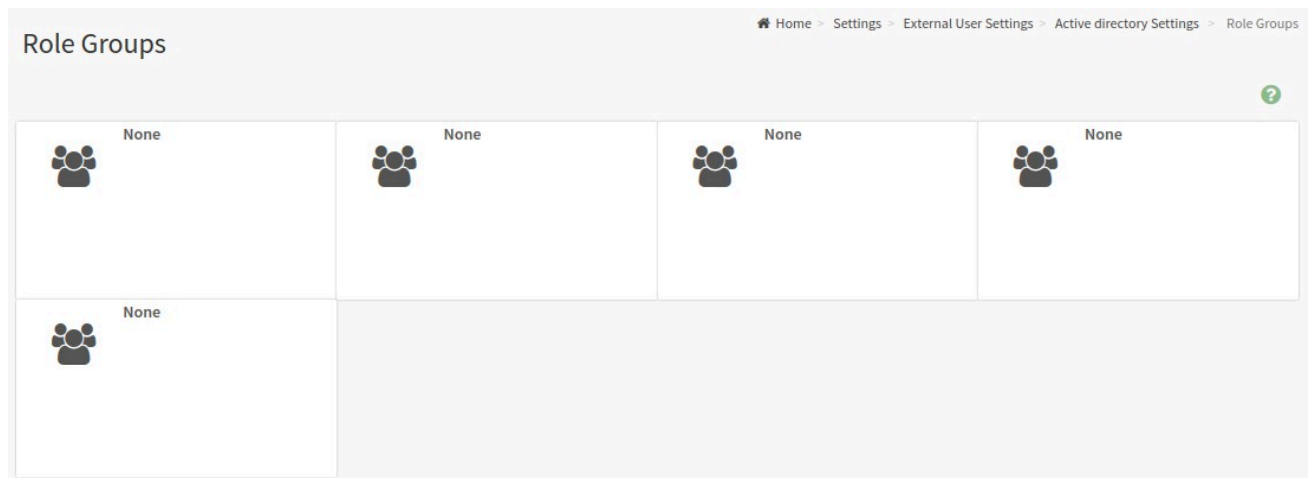
Notes:

- At least one Domain Controller Server Address must be configured.
- IP Address made of 4 numbers separated by dots as in "xxx.xxx.xxx.xxx".
- Each number ranges from 0 to 255.
- First number must not be 0.
- Domain Controller Server Addresses will supports IPv4 Address format and IPv6 Address format.

7. Click **Save** to save the entered settings and return to the Active Directory Settings page.

To add a Role Group:

1. Verify the **Enable Active directory Authentication** check box is selected in the General Active Directory Settings page.
2. In the Active Directory Settings page, click **Role Groups**. The Role Groups page is displayed.



3. Double click on a blank row. The Add Role Group page is displayed.

Role Groups

[Home](#) > [Settings](#) > [External User Settings](#) > [Active directory Settings](#) > [Role Groups](#) > [Role Groups](#)

?

Group Name

Group Domain

eg., MyDomain.com

Group Privilege

☐ KVM Access

☐ VMedia Access

Save

- In the Group Name field, type the name that identifies the role group.

Notes:

- Maximum size of Role Group Name is 255 bytes.
- Special symbols hyphen and underscore are allowed.

- In the Group Domain field, type the Role Group Domain where the role group is located.

Notes:

- Maximum size of Group Domain Name is 255 bytes.
- It must start with an alphabetical character.
- Special Symbols like dot(.), comma(.), hyphen(-), underscore(_), equal-to(=) are allowed.
- Example: cn=manager,ou=login, dc=domain,dc=com

- From the Group Privilege drop down, select the level of privilege (User, Administrator, Operator, None) to assign to this role group.
- Select one or both of the following check boxes:
 - KVM Access
 - VMedia Access
- Click **Save** to save the new role group and return to the Role Group List.

To delete a Role Group:

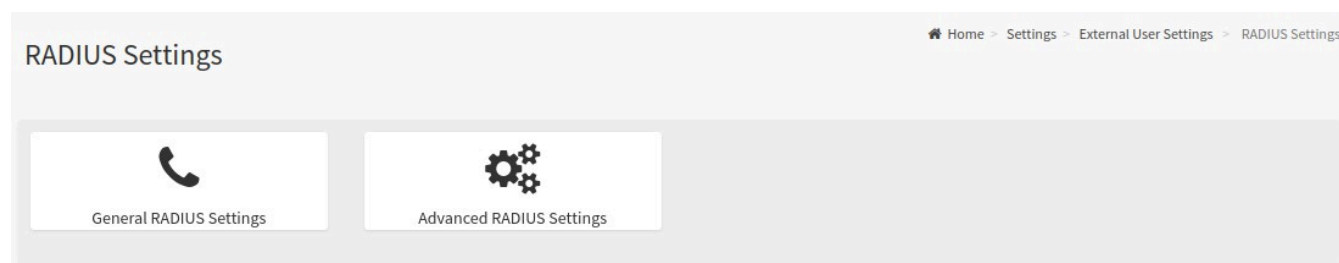
- In the Role Groups Page, select the row that you wish to delete.

2. Click **Delete Role Group**.

RADIUS Settings

RADIUS is a modular, high performance and feature-rich RADIUS suite including server, clients, development libraries and numerous additional RADIUS related utilities.

To open the RADIUS Settings page, click **Settings > External User Settings > RADIUS Settings**.

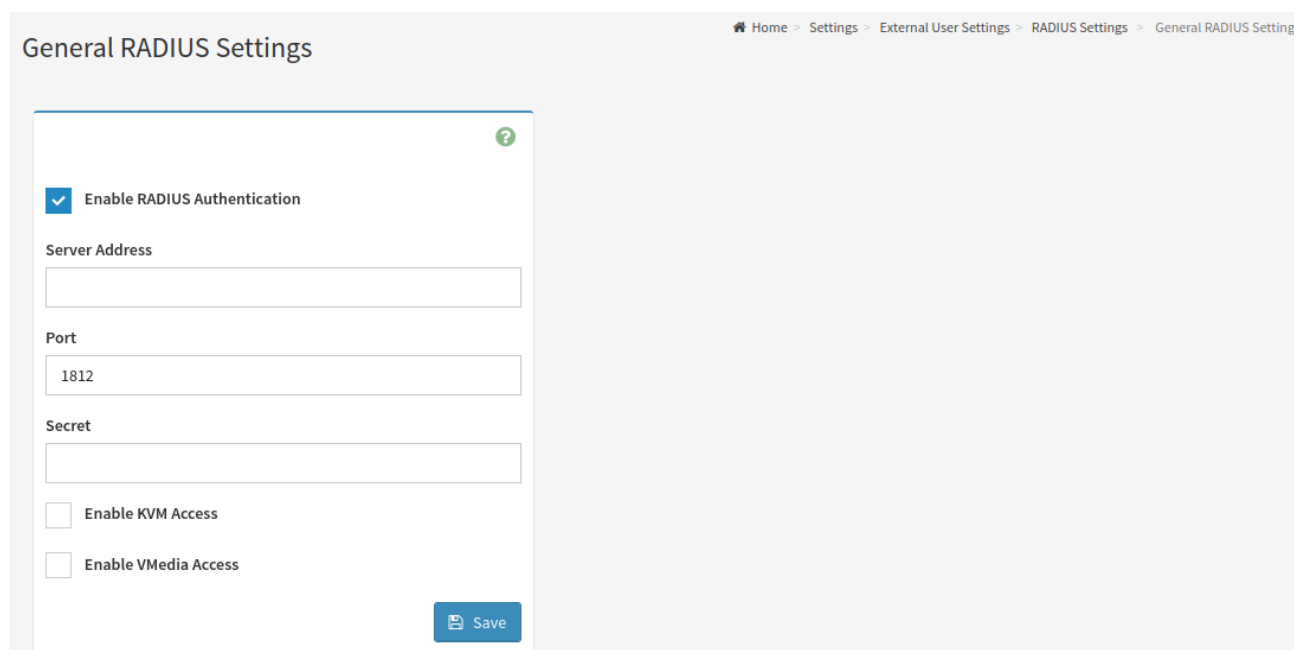


The available setting are:

- General RADIUS Settings
- Advanced RADIUS Settings

To configure General RADIUS Settings:

1. In the RADIUS Settings page, click **General RADIUS Settings**. The General RADIUS Settings page is displayed.



2. Select the **Enable RADIUS Authentication** check box to enable RADIUS Authentication.
3. In the Server Address field, type the IP address of the RADIUS serve.

Notes:

- For an IP Address, you can use IPv4 and IPv6 format.
- FQDN (Fully Qualified Domain Name) format is supported.

4. In the Port field, type the RADIUS Port number.

Notes:

- The default port is 1812.
- Port value ranges are from 1 to 65535.

5. In the Secret field, type the authentication secret for the RADIUS server.

Notes:

- This field does not allow more than 31 characters.
- Secret must be at least 4 characters long.
- White space is not allowed.

6. Select the **Enable KVM Access** check box to provide access to KVM for RADIUS authenticated users.
7. Select the **Enable VMedia Access**, to provide access to VMedia for RADIUS authenticated users.
8. Click **Save** to save the entered settings and return to the RADIUS Settings page.

To configure advanced RADIUS settings:

1. Verify the **Enable RADIUS Authentication** check box is selected in the General RADIUS Settings page.
2. In the RADIUS Settings page, click **Advanced RADIUS Settings**. The Advanced RADIUS Settings page is displayed.

Note: All fields on this page do not allow more than 127 characters. Special characters like ';' and '#' are also not allowed.

Advanced RADIUS Settings

Home > Settings > External User Settings > RADIUS Settings > Advanced RADIUS Settings

RADIUS Authorization ?

Administrator

H=4

Operator

H=3

User

H=2

OEM Proprietary

H=1

No Access

H=0

Save

- Configure the fields with Vendor Specific Attribute from the Server side.

Example 1:

```
testadmin Auth-Type :=PAP, Cleartext-Password:="admin"
```

```
Auth-Type :=PAP, Vendor-Specific="H=4"
```

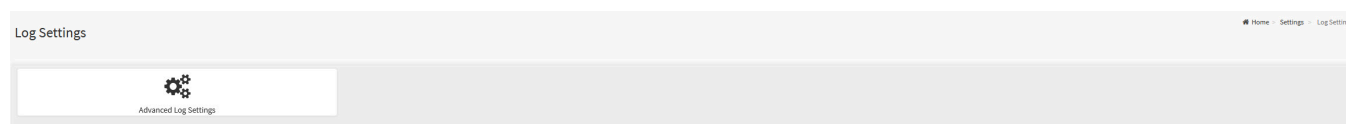
Example 2:

```
testoperator Auth-Type := PAP, Cleartext-Password := "operator" Auth-Type :=PAP, Vendor-Specific="H=3"
```

- If you change the Vendor Specific value on the server, you should change the same values in this page.
- Click **Save** to save the entered settings and return to the RADIUS Settings page.

Log Settings

The Log Settings page enables advanced configuration of the audit log. To open the Log Settings page, click **Settings > Log Settings**.



To configure advanced log settings:

- In the Log Settings page, click **Advanced Log Settings**. The Advanced Log Settings page is displayed.

Advanced Log Settings

[Home](#) > [Settings](#) > [Log Settings](#) > [Advanced Log Settings](#)

☒ Local Log

☒ Remote Log

Port Type
☐ UDP ☐ TCP

File Size

Rotate Count

Remote Log Server

Remote Server Port

☒ Enable Audit Log

 Save

2. Select the **Local Log** check box to save the logs locally.
3. Select the **Remote Log** check box to save the logs to a remote machine.

Notes:

- You can select either Local Log/Remote Log or both Logs as per the requirement.
- Either one of the Log selection is mandatory.
- Local file resides at /var/log/

4. For Port Type, select the type of port you are using **TCP** or **UDP**. Port Type selection is available only if the Remote Log check box is selected.
5. In the File Size field, type the size of the file in bytes. This field is configurable only if the Local Log check box is selected.

Note: The file size ranges from 3 to 65535. Log files are rotated when they grow larger than the size configured, and taking into account the last rotation time interval (1 minute).

6. In the Rotate Count field, type **0** or **1** to back up the log information in back up files. This field is configurable only if the Local Log check box is selected.

Note: When log information exceeds the file size, the old log information is automatically moved to back up files based on the rotate count value. If rotate count is zero, the old log information gets cleared permanently.

7. In the Remote Log Server field, type the Remote server address to log the system events. This field is configurable only if the Remote Log check box is selected.

Note: The Server address supports the following:

- IPv4 address format.
- FQDN (Fully qualified domain name) format.
- Maximum allowed size is 64 bytes.

8. In the Remote Server Port field, type the Remote Server port address to log the system events. This field is configurable only if the Remote Log check box is selected.
9. Select the **Enable Audit Log** check box to enable the audit log.
10. If **TCP** is selected for Port Type, for CA certificate file, browse to select the certificate of the trusted CA certs.

Note: CA certificate file should be of the type pem.

11. Click **Save** to save the changes.

Media Redirection Settings

This page is used to configure the media for redirection. To open the Media Redirection page, click **Settings > Media Redirection Settings**.

Media Redirection

🏠 Home > ⚙️ Settings > Media Redirection



General Settings



VMedia Instance Settings



Remote Session



Active Redirections

- [General Settings](#)
- [VMedia Instance Settings](#)

- Remote Session
- Active Redirections

General Settings

This page is used to configure General Media Settings. To open the General Media Settings section, click **Settings > Media Redirection Settings > General Settings**.

General Settings

Home > Settings > Media Redirection > General Settings

Click here to go to [Remote Media](#).

☐ Remote Media Support

Save

Remote Media Support: To enable or disable Remote Media support, select or clear the check box. If it is selected, then the following Remote Media types are displayed:

- Mount CD/DVD
- Mount Harddisk

To configure General Media Redirection Settings:

1. Select the **Mount CD/DVD** check box.

General Settings

Home > Settings > Media Redirection > General Settings

Click here to go to [Remote Media](#).

☒ Remote Media Support

☒ Mount CD/DVD

Server Address for CD/DVD Images

Server IP or Host name

Path in server

e.g., /opt/3mcdifs

Share Type for CD/DVD

☐ nfs ☐ cifs

Domain Name

Username

Password

☐ Same settings for Harddisk Images

☐ Mount Harddisk

Retry Interval

15

Retry Count

3

Save

2. In the Server Address for CD/DVD Images field, type the address of the server where the Remote media images are stored.

3. In the Path in server field, type the source path to the remote media images.

Note: Path must be alpha-numeric and the following special characters are only allowed, '/'(backward slash), '\'(forward slash), '_'(underscore), '.'(dot). This allows 256 characters or less.

4. For Share Type for CD/DVD, select the share type of the remote media server **NFS**, **Samba(CIFS)**, **HTTPS** or **SMBE (Encrypted CIFS)**.

Note: To display or hide the SMBE option in the Media Redirection General Settings page, enable/disable "CONFIG_SPX_FEATURE_RMEDIA_CIFS_ENC" respectively.

5. In the Domain Name, Username and Password fields, if share Type is **Samba(CIFS)** or **HTTPS** or **SMBE (Encrypted CIFS)**, type the user credentials to authenticate on the server.
6. Select the **Same settings for Harddisk Images** check box to select same media type data configurations for all the remote media types.
7. Select the **Mount Harddisk** check box.
8. Perform steps 2-6.
9. In the Retry Interval field, type the retry interval to reconnect the remote media.
10. In the Retry Count field, type the retry count to reconnect the remote media.

Note: Retry Count and interval are used only if there is any disruption in the active redirected image. In this case, the media will try to reconnect based on configured retry count and interval values.

11. Click **Save**.

VMedia Instance Settings

This page is used to configure Virtual Media device settings. To open the VMedia Instance Settings section, click **Settings > Media Redirection Settings > VMedia Instance Settings**.

VMedia Instance Settings

[Home](#) >
 [Settings](#) >
 [Media Redirection](#) >
 VMedia Instance Settings

?

CD/DVD device instances

4
▼

Hard disk instances

4
▼

Remote KVM CD/DVD device instances

4
▼

Remote KVM Hard disk instances

4
▼

☒ Power Save Mode

Save

To configure VMedia Instance Settings:

1. From the CD/DVD device instances dropdown, select the number of CD/DVD devices supported for Virtual Media redirection. Up to four devices can be added.
2. From the Harddisk instances dropdown, select the number of hard disk devices supported for Virtual Media redirection. Up to four devices can be added.
3. From the Remote KVM CD/DVD device instances dropdown, select the number of CD/DVD devices supported for KVM Virtual Media redirection.
4. From the Remote KVM Hard disk instances dropdown, select the number of Hard disk devices supported for KVM Virtual Media redirection.
5. Select the **Power Save Mode** check box to enable the virtual USB devices visibility in the host. If this option is enabled, Virtual media devices will be connected to the Host machine only at the instance launching KVM session. If this option is disabled, Virtual media devices will remain connected to the host machine all the time irrespective of KVM session status. If media is redirected from a kvm/media client and the client application is closed then the respective media session will be disconnected.
6. Click **Save**.

Note: Virtual Media configuration changes will restart all the media services. So configuration changes will be blocked when any active media redirection is present.

Remote Session

This page is used to configure Remote Session configuration settings. To open the VMedia Instance Settings section, click **Settings > Redirection Settings > Remote Session**.

Remote Session
Home > Settings > Media Redirection > Remote Session

?

Click here to go to [Remote Control](#).

☒ KVM Single Port Application

Keyboard Language

Auto Detect (AD)
▼

Retry Count

3
▼

Retry Time Interval(Seconds)

10
▼

☒ Server Monitor OFF Feature Status

☐ Automatically OFF Server Monitor, When KVM Launches

Save

To configure Remote Session:

1. The **KVM Single Port Application** check box is enabled by default. KVM session will use its dedicated port whereas both Web and KVM sessions will be established only via Web Port. Since singleport application performs better when compared to kvm/media tunnel encryption, singleport feature will always be enabled. You can't disable singleport support in runtime.
2. From the Keyboard Language dropdown, select the keyboard supported languages.
3. In the Retry Count field, type the number of attempts the KVM client will make to reconnect the KVM session. The retry count value ranges from 1 to 20.
4. In the Retry Time Interval (Seconds) field, type the time duration between two consecutive reconnect attempts. The KVM client will wait for a time interval equal to this value, after making a reconnect attempt, before trying to connect again. The retry interval value is mentioned in seconds and it ranges between 5 to 30 seconds.
5. Select the **Server Monitor OFF Feature Status** check box to Lock or Unlock the Local host monitor from the remote KVM window. If this option is disabled, you cannot Lock or Unlock the Local host monitor from the remote KVM window.
6. Select the **Automatically OFF Server Monitor, When KVM Launches** check box to automatically Lock the local monitor during viewer launch.
7. Click **Save**.

Active Redirections

This page is used to display a table showing information about the active redirected media. To open the Active Redirections section, click **Settings > Media Redirection Settings > Active Redirections**.

Active Redirections Home > Settings > Media Redirection > Active Redirections

Media Type	Media Instance	Client Type	Image Name	Redirection Status	Client IP
CD/DVD	0	RMedia	ubuntu-16.04.7-desktop-amd64.iso	Started with media boost	~
CD/DVD	1	RMedia	Windows7Ultimate_X64_TW.iso	Started	~
Hard disk	0	RMedia	Dos3.3-2.img	Started	~

The information displayed in the table is:

- **Media Type:** The type of media devices (CD/DVD) supported for Active Redirections.
- **Media instances:** The number of media devices supported for Active Redirections.
- **Client Type:** The Client type (JViewer/RMedia) used for active media redirection.
- **Image Name:** The name of the media devices supported image for Active Redirections.
- **Redirection Status:** The status media for Active Redirections.
- **Client IP:** The IP of the connected media devices (CD/DVD) supported for Active Redirections.

Note: Remote Media connection will use loopback socket for communication. So the '~' symbol will be displayed for loopback ip(127.0.0.1 (or) ::1) in the media session information page.

Network Settings

This page is used to configure the network settings for the available LAN channels. To open the Network Settings page, click **Settings > Network Settings**.



- Network IP Settings
- Network Link Configuration
- DNS Configuration

Network IP Settings

This page is used to configure Network IP Settings. To open the Network IP Settings section, click **Settings > Network Settings > Network IP Settings**.

Home > Settings > Network > Network IP Settings

Network IP Settings

?

☒ Enable LAN

LAN Interface

eth0

MAC Address

08:35:71:21:78:F9

☒ Enable IPv4

☐ Enable IPv4 DHCP

IPv4 Address

192.168.1.1

IPv4 Subnet

255.255.255.0

IPv4 Gateway

0.0.0.0

☒ Enable IPv6

☐ Enable IPv6 DHCP

IPv6 Index

0

IPv6 Address

Subnet Prefix Length

0

IPv6 Gateway

::

☐ Enable VLAN

VLAN ID

0

VLAN Priority

0

Save

To configure Network IP Settings:

1. Select the **Enable LAN** check box.

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2. From the LAN Interface dropdown, select the LAN interface. The MAC Address field displays the MAC Address of the device. This is a read only field.
3. Select the **Enable IPv4** check box. This option is to enable/disable the IPv4 settings in the device.
4. Select the **Enable IPv4 DHCP** check box. This option is to enable IPv4 DHCP support for the selected interface. When selecting this check box the IPv4 Address, IPv4 Subnet, and IPv4 Gateway fields are disabled.
5. In the IPv4 Address field, type the static IPv4 address of the device.
6. In the IPv4 Subnet field, type the Subnet Mask of the device.
7. In the IPv4 Gateway field, type the default gateway of the device.

Notes:

- IP Address made of 4 numbers separated by dots as in "xxx.xxx.xxx.xxx".
- Each Number ranges from 0 to 255.
- First Number must not be 0.

8. Select the **Enable IPv6** check box. This option is to enable/disable the IPv6 configuration settings.
9. Select the **Enable IPv6 DHCP** check box. This option is to enable/disable the IPv6 settings in the device. It dynamically configures IPv6 address using DHCP (Dynamic Host Configuration Protocol). When selecting this check box the IPv6 Index, IPv6 Address, Subnet Prefix Length and IPv6 Gateway fields are disabled.
10. From the IPv6 Index dropdown, select a static IPv6 index to be configured to the device (e.g.: 0).
11. In the IPv6 Address field, type a static IPv6 address to be configured to the device (e.g: 2004::2010).
12. In the Subnet Prefix length field, type the subnet prefix length for the IPv6 settings. Value ranges from 0 to 128.
13. In the IPv6 Gateway field, type a v6 default gateway for the IPv6 settings.
14. Select the **Enable VLAN** check box. This option is to enable/disable the VLAN support for selected interface.
15. In the VLAN ID field, type an ID for this VLAN configuration.

Notes:

- Value ranges from 2 to 4094
- VLAN ID cannot be changed without resetting the VLAN configuration.
- VLAN 0, 1 & 4095 are reserved VLAN ID's.

16. In the VLAN Priority field, type the priority for VLAN configuration.

Notes:

- Value ranges from 0 to 7.

- 7 is the highest priority for VLAN.

17. Click **Save**.

Network Link Configuration

This page is used to configure the network link configuration for available network interfaces. To the Network Link Configuration section, click **Settings > Network Settings > Network Link Configuration**.

Network Link Configuration

LAN Interface

eth0

▼

☒ Auto Negotiation

Link Speed

1000 Mbps

Duplex Mode

FULL Duplex

Save

To configure Network Link Configuration:

1. From the LAN Interface dropdown, select the required network interface from the list to which the Link speed and duplex mode are to be configured.
2. Select the **Auto Negotiation** check box. This option is to enable the device to perform automatic configuration to achieve the best possible mode of operation (speed and duplex) over a link. When this check box is selected, the Link Speed and Duplex Mode fields are disabled.
3. From the Link Speed dropdown, select the link speed from the supported capabilities of the network interface. It can be 10/100/1000 Mbps.
4. From the Duplex Mode selection field, Select **Half Duplex** or **Full Duplex**.
5. Click **Save**.

DNS Configuration

This page is used to manage the DNS settings of a device. The Domain Name System (DNS) is a distributed

hierarchical naming system for computers, services, or any resource connected to the Internet or a private network. It associates the information with domain names assigned to each of the participants. Most importantly, it translates domain names meaningful to humans into the numerical (binary) identifiers associated with networking equipment for the purpose of locating and addressing these devices worldwide.

To open the DNS Server Settings section, click **Settings > Network Settings > DNS Configuration**.

DNS Configuration



☒ DNS Enabled

☐ mDNS Enabled

Host Name Setting

☒ Automatic ☐ Manual

Host Name

AMI08357127C3BF

BMC Registration Settings

BMC Interface:

eth0

☒ Register BMC

Registration method:

☒ Nsupdate ☐ DHCP Client FQDN ☐ Hostname

Eth0 TSIG Configuration

☐ TSIG Authentication Enabled

Current TSIG Private File Info

Not Available

New TSIG Private File



Domain Setting

☒ Automatic ☐ Manual

Domain Interface

Domain Name Server Setting

☒ Automatic ☐ Manual

DNS Interface

eth0

IP Priority

☒ IPv4 ☐ IPv6

To configure DNS Configuration settings:

1. Select the **DNS Enabled** check box. This option is to enable/disable all the DNS Service Configurations.
2. Select the **mDNS Enable** check box. This option is to enable/disable the mDNS Support Configurations.
3. From the Host Name Setting selection field, select **Automatic** or **Manual**.

The Host Name field displays the host name of the device. If the Host setting is selected as Manual, then specify the host name of the device.

Notes:

- Value ranges from 1 to 64 alpha-numeric characters.
- Special characters '-' (hyphen) and '_' (underscore) are allowed.
- It must not start or end with a '-' (hyphen). IE browsers won't work correctly if any part of the host name contains underscore (_) character.

4. Select the **Register BMC** check box. This option is to enable/disable registration using a registration method.
5. In the Registration Method selection field, select the registration method, **NS Update**, **DHCP Client FQDN** or **Hostname**.
6. Select the **TSIG Authentication Enabled** check box. This option enables TSIG authentication while registering DNS via nsupdate. Separate TSIG files can be uploaded for each LAN interface.
7. In the Current TSIG Private File field, information of Current TSIG private file along with its uploaded date/time is displayed (read only).
8. In the New TSIG Private File field, browse and navigate to select the TSIG private file. The TSIG file should be of private type.
9. In the Domain Setting selection field, select:
 - **Automatic** - The domain name is configured automatically and the Domain Name field is disabled. The **Domain Interface** drop down is displayed where you can select the domain interface of the device.
 - **Manual** - The domain name is configured manually. The **Domain Name** field is displayed where you type the domain name of the device.
10. In the Domain Name Server Setting selection field, select:
 - **Automatic** - The domain name is configured automatically. The **Domain Interface** drop down is displayed where you can select the domain interface of the device.
 - **Manual** - Specify the DNS (Domain Name System) server address to be configured.
11. In the IP Priority selection field, select:
 - **IPv4** - You can use two IPv4 and one IPv6 DNS servers.
 - **IPv6** - You can use one IPv4 and two IPv6 DNS servers.

This selection field is not available when Domain Name Server Setting is set to **Manual**.

12. Click **Save**.

PAM Order Settings

This page is used to configure the PAM ordering for user authentication. To open the PAM Ordering section, click **Settings > PAM Order Settings**.

Notes:

- Do not to keep same username for different PAM modules.
- If Authentication fails, the reason could be invalid User or Invalid Password.
- If Radius Authentication fails, it is not possible to differentiate whether it is invalid user or invalid password. Therefore, it is always treated as Invalid username error and PAM tries different Authentication Methods.
- Radius must be the last in the PAM order for the same reason. If RADIUS is not the last in the PAM order, the User ID of the logged in users using other authentication services is shown as the RADIUS User ID.
- If the active directory contains a secret username and the password is empty, authentication fails will be always treated as Invalid Password error. For Invalid Password error PAM will not try other Authentication Methods. So it is recommended to keep AD before RADIUS if RADIUS Authentication is enabled or in the last location in PAM order if RADIUS Authentication is enabled.

Home > Settings > PAM Order

PAM Order

?

PAM Authentication Order

IPMI

LDAP

ACTIVE DIRECTORY

RADIUS

Save

To configure PAM order:

1. Select the required PAM module and click and drag the required PAM module. It can be moved UP or DOWN to change its arrangement order.
2. Click **Save**.

Notes Whenever the configuration is modified, the web server will be restarted automatically. Logged-in session will be logged out.

Services

This page displays the basic information about running services. Only an Administrator can modify the service. To open the Services section, click **Settings > Services**.

Services

Home > Settings > Services

Service	Status	Interfaces	Secure Port	Timeout	Maximum Sessions	
web	Active	both	443	1800	20	 
kvm	Active	both	443	1800	4	 
cd-media	Active	both	443	N/A	4	 
hd-media	Active	both	443	N/A	4	 

The Services table contains the following columns:

- **Service:** Displays the service name of the selected slot (read-only).
- **Status:** Displays the current status of the service, active or inactive.
- **Interfaces:** Displays the interface in which service is running.
- **Secure Port:** Displays the secure port number for the service.
 - The Web default port is 443
 - The KVM default port is 7582
 - The CD Media default port is 5124
 - The HD Media default port is 5127

Notes:

- If a single port feature is enabled, the KVM and Media ports cannot be edited.
- Port value ranges from 1 to 65535.

- **Timeout:** Displays the session timeout value of the service. For web and telnet service, user can configure the session timeout value.
 - Web timeout value ranges from 300 to 1800 seconds.
 - KVM timeout value ranges from 300 to 1800 seconds.
 - If KVM is launched, the web session timeout will not take effect.
- **Maximum Sessions:** Displays the maximum number of allowed sessions for the service.

To view or terminate the current active sessions for the service:

Note: All active sessions are terminated if the unit is rebooted.

1. Click  next to the desired service to view the details about the active sessions for the service. The Service Sessions page is displayed.

Service Sessions Home > Settings > Services > Service Sessions ?

Active Session - Web

Session ID	Session Type	User ID	User Name	Client IP	Privilege	
15	Web HTTPS	2	admin	192.168.122.131	Administrator	
16	Web HTTPS	2	admin	192.168.122.131	Administrator	

2. Click  to terminate the particular session of the service.

To modify an existing service:

1. In the Services page, click  next to the desired service. The Service configuration page is displayed.

Note: Whenever the configuration is modified, the service are restarted automatically. You need to close the existing opened session for the service if needed.

Service Configuration

[Home](#) > [Settings](#) > [Services](#) > [Service Configuration](#)

?

Service Name

web

☒ Active

Interface Name

both

Secure port

443

Timeout

1800

Maximum Sessions

20

Save

2. Select the **Active** check box to activate the current state.

Note: Interfaces, Secure port, Time out and Maximum Sessions are not active unless the current state is active.

3. From the Interface Name dropdown select any one of the available interfaces.
4. In the Secure port field, type the secure port number.
5. In the Timeout field, type the timeout value.
6. Click **Save**.

SMTP Settings

Simple Mail Transfer Protocol (SMTP) is an Internet standard for electronic mail (email) transmission across Internet Protocol (IP) networks. Using the IPMI Web GUI, you can configure the SMTP settings of the device. To open the SMTP Settings page, click **Settings > SMTP Settings**.

SMTP Settings



LAN Interface

eth0

▼

Sender Email ID

☒ Primary SMTP Support

Primary Server Name

Primary Server IP

Primary SMTP port

25

Primary Secure SMTP port

465

☐ Primary SMTP Authentication

Primary Username

Primary Password

☐ Primary SMTP SSLTLS Enable

☐ Primary SMTP STARTTLS Enable

☒ Secondary SMTP Support

Secondary Server Name

Secondary Server IP

Secondary SMTP port

25

Secondary SMTP Secure port

465

☐ Secondary SMTP Authentication

To configure SMTP settings:

1. From the LAN Interface dropdown, select LAN interface.
2. In the Sender Email ID field, type a valid 'Sender Address' to indicate whenever an email is sent.
3. Select the **Primary SMTP Support** check box to enable/disable SMTP Support.
4. In the Primary Server Name field, type the 'Machine Name' from where the email is sent.

Notes:

- Machine Name is a string of maximum 15 alpha-numeric characters.
- Space, special characters are not allowed.

5. In the Primary Server IP field, type the IP address of the SMTP Server (mandatory).
6. In the Primary SMTP Port field, type the SMTP Normal Port. Default Port is 25. The Port value ranges from 1 to 65535.
7. In the Primary Secure SMTP Port field, type the SMTP Secure Port. Default Port is 465. The Port value ranges from 1 to 65535.
8. In the Primary Server IP: The IP address of the SMTP Server. It is a mandatory field.

Notes:

- IP Address made of 4 numbers separated by dots as in "xxx.xxx. xxx.xxx".
- Each Number ranges from 0 to 255.
- First Number must not be 0.
- Supports IPv4 Address format and IPv6 Address format.

9. Select the **Primary SMTP Authentication** check box to enable/disable SMTP Authentication.

The supported SMTP Server Authentication types are:

- CRAM-MD5
- LOGIN
- PLAIN

If the SMTP server does not support any one of the above authentication types, the you get an error message stating, "Authentication type is not supported by SMTP Server".

10. In the Primary Username field, type the username to access SMTP Accounts.

Notes:

- User Name can be of length 4 to 64 alpha-numeric characters, dot(.), hyphen(-), at sign (@), and underscore(_).
- It must start with an alphabet.
- Other Special Characters are not allowed.

11. In the Primary Password field, type the password for the SMTP User Account.

Notes:

- Password must be at least 4 characters long.
- White space is not allowed.
- This field does not allow more than 64 characters.

12. Select the **Primary SMTP SSLTLS Enable** check box to enable/disable SSLTLS support.
13. In the Upload SMTP CA Certificate File field, browse to the file that contains the certificate of the trusted CA certs and upload it. CACERT key file should be of pem type.
14. In the Upload SMTP Certificate File field, browse to the client certificate file and upload it. CERT key file should be of pem type.
15. In the Upload SMTP Private Key field, browse to the client private key file and upload it. SMTP key file should be of pem type.
16. Select the **Primary SMTP STARTTLS Enable** check box to enable/disable STARTTLS support.
17. In the Upload SMTP CA Certificate File field, browse to the file that contains the certificate of the trusted CA certs and upload it. CACERT key file should be of pem type.
18. In the Upload SMTP Certificate File field, browse to the client certificate file and upload it. CERT key file should be of pem type.
19. In the Upload SMTP Private Key field, browse to the client private key file and upload it. SMTP key file should be of pem type.
20. Select the Secondary SMTP Support check box to enable/disable secondary SMTP Server configuration.
21. Repeat steps 4-19.
22. Click **Save**.

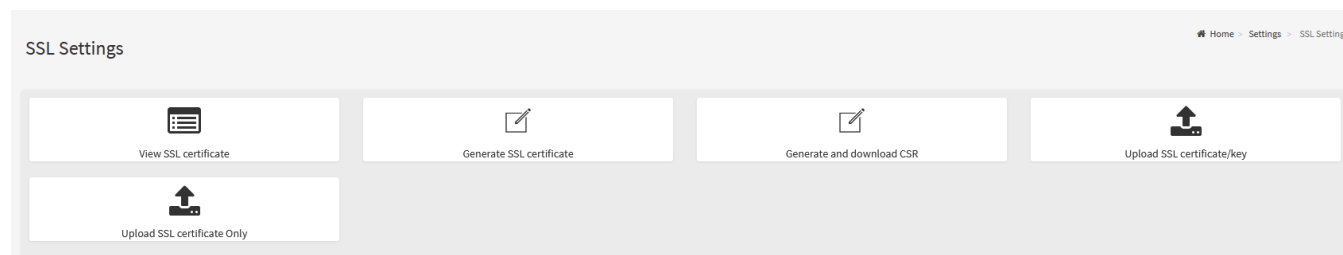
SSL Settings

The Secure Socket Layer protocol was created to ensure secure transactions between web servers and browsers. The protocol uses a third party, a Certificate Authority (CA), to identify one end or both end of the transactions.

In the NG7 appliances, the IPMI natively lacks PQC capabilities, and retrofitting it is not feasible. The highest

supported configuration for these appliances is TLS 1.3 using a standard Elliptic Curve (P-256) and AES_256_GCM.

Using IPMI Web GUI, configure SSL certificate. Using this, the device can be accessed in a secured mode. To open the SSL Certificate Configuration page, click **Settings > SSL Settings**.



- [View SSL Certificate](#)
- [Generate SSL Certificate](#)
- [Generate and Download CSR](#)
- [Upload SSL Certificate](#)
- [Upload SSL Certificate Only](#)

View SSL Certificate

In this page you can see basic information about the uploaded SSL certificate. It displays the following fields:

- Version Serial Number
- Serial number
- Signature Algorithm
- Public Key
- Issuer Common Name (CN)
- Issuer Organization (O)
- Issuer Organization Unit (OU)
- Issuer City or Locality (L)
- Issuer State or Province (ST)
- Issuer Country (C)
- Issuer Email Address
- Valid From
- Valid Till
- Issued to Common Name (CN)
- Issued to Organization (O)
- Issued to Organization Unit (OU)
- Issued to City or Locality (L)
- Issued to State or Province (ST)
- Issued to Country (C)
- Issued to Email Address

View SSL Certificate

Current Certificate Information



Certificate Version

1

Serial Number

66971B4E407454C3D84E273A04CDFFA0FF1D53DB

Signature Algorithm

sha256WithRSAEncryption

Public Key

(2048 bit)

Issuer Common Name (CN)

BMC

Issuer Organization (O)

SWRD

Issuer Organization Unit (OU)

CASWELL

Issuer City or Locality (L)

Taipei

Issuer State or Province (ST)

TP

Issuer Country (C)

TW

Issuer Email Address

info@cas-well.com

Valid From

Mar 3 10:56:23 2023 GMT

Valid Till

Feb 28 10:56:23 2033 GMT

Issued to Common Name (CN)

BMC

Issued to Organization (O)

SWRD

Issued to Organization Unit (OU)

CASWELL

Issued to City or Locality (L)

Taipei

Issued to State or Province (ST)

TP

TW

Generate SSL Certificate

In this page you can generate the SSL certificate based on configuration details.

Generate SSL Certificate

[Home](#) > [Settings](#) > [SSL Settings](#) > [Generate SSL Certificate](#)

?

Common Name (CN)

Organization (O)

Organization Unit (OU)

City or Locality (L)

State or Province (ST)

Country (C)

Email Address

Valid for

in days

Key Length

2048 bits

Save

To generate the SSL Certificate:

1. In the Common Name (CN) field, type the CN for which the certificate is to be generated.

Notes:

- Maximum length of 64 characters.
- It is a string of alpha-numeric characters.
- Special characters '#' and '\$' are not allowed.

2. In the Organization (O) field, type the organization name for which the certificate is to be generated.

Notes:

- Maximum length of 64 characters.
- It is a string of alpha-numeric characters.
- Special characters '#' and '\$' are not allowed.

3. In the Organization Unit (OU) field, type the over all organization section unit name for which certificate is to be generated.

Notes:

- Maximum length of 64 characters.
- It is a string of alpha-numeric characters.
- Special characters '#' and '\$' are not allowed.

4. In the City or Locality (L) field, type the city or locality of the organization (mandatory).

Notes:

- Maximum length of 128 characters.
- It is a string of alpha-numeric characters.
- Special characters '#' and '\$' are not allowed.

5. In the State or Province (ST) field, type the state or province of the organization (mandatory).

Notes:

- Maximum length of 64 characters.
- It is a string of alpha-numeric characters.
- Special characters '#' and '\$' are not allowed.

6. In the Country (C) field, type the country code of the organization (mandatory).

Notes:

- Only two characters are allowed.
- Special characters are not allowed.

7. In the Email Address field, type the email address of the organization (mandatory).
8. In the Valid for field, type the validity of the certificate. Value ranges from 1 to 3650 days.
9. From the Key Length dropdown, select the key length bit value of the certificate.
10. Click **Save**.

Generate and Download CSR

In this page you can generate and download the Certificate Signing Request (CSR).

The screenshot shows a web interface titled "Generate and download CSR". It features a form with the following fields: Common Name (CN), Organization (O), Organization Unit (OU), City or Locality (L), State or Province (ST), Country (C), Email Address, and Key Length (a dropdown menu currently showing "2048 bits"). A blue button labeled "Generate and download" is positioned at the bottom right of the form. The top right of the page has a navigation menu with links: Home, Settings, SSL Settings, and Generate and download CSR.

To generate/download the CSR:

1. In the Common Name (CN) field, type the CN for which the certificate is to be generated.

Notes:

- Maximum of 64 alpha-numeric characters.
- Special characters '#' and '\$' are not allowed.

2. In the Organization (O) field, type the name of the organization for which certificate is to be generated.

Notes:

- Maximum length of 64 alpha-numeric characters
- Special characters '#' and '\$' are not allowed.

3. In the Organization Unit (OU) field, type the section or unit of the organization for which certificate is to be generated.

Notes:

- Maximum length of 64 alpha-numeric characters.
- Special characters '#' and '\$' are not allowed.

4. In the City or Locality (L) field, type the city or locality.

Notes:

- Maximum length of 128 alpha-numeric characters.
- Special characters '#' and '\$' are not allowed.

5. In the State or Province (ST) field, type the state or province.

Notes:

- Maximum length of 128 alpha-numeric characters.
- Special characters '#' and '\$' are not allowed.

6. In the Country (C) field, type the country code.

Notes:

- Only two characters are allowed.
- Special characters are not allowed.

7. In the Email Address field, type the email address of the organization. The maximum allowed size for the email ID is 64 bytes, which includes username and domain name.
8. From the Key Length dropdown, select the key length bit value of the CSR.
9. Click **Generate and Download**.

Upload SSL Certificate

In this page, you can upload the certificate and private key file.

To upload the SSL Certificate:

1. In the Current Certificate field, you can see the current certificate and uploaded date/time (read-only).
2. In the New Certificate field, browse to the certificate file and select it. The file should be of pem type.
3. In the Current Private Key field, you can see the current private key information (read-only).
4. In the New Private Key field, browse to the private key file and select it. The file should be of pem type.
5. Click **Save**.

Notes:

- After successful upload, the HTTPs service restarts in order to use the newly uploaded SSL certificate.
- If you perform a Factory Restore Defaults, the SSL certificate at run-time moves to the SSL certificate provided at the built-time.

Upload SSL Certificate Only

In this page, you can upload the SSL certificate.

Upload SSL Certificate Only

Current Certificate

Sun Jan 1 01:08:37 2012

New Certificate

...

Save

To upload an SSL certificate:

1. In the Current Certificate field, you can see the information of the current certificate and the date/time of its upload (read-only).
2. In the New Certificate field, browse to the new certificate file and select it. The file should be of pem type.
3. Click **Save**.

System Firewall

The System Firewall page allows you to configure the firewall settings. The firewall rule can be set for an IP or range of IP addresses or port numbers. To view this page, you must at least be an operator. Only administrators can add or delete a firewall. To open the System Firewall page, click **Settings > System Firewall**.

System Firewall

Home > Settings > System Firewall



General Firewall Settings



IP Address Firewall Rules



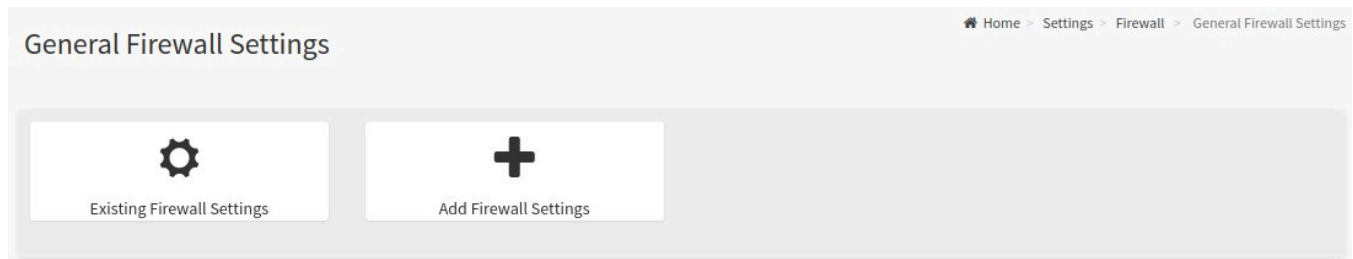
Port Firewall Rules

- [General Firewall Settings](#)
- [IP Address Firewall Rules](#)
- [Port Firewall Rules](#)

General Firewall Settings

In this page, you can view existing firewall setting or add firewall settings.

To open the General Firewall Setting page, click **Settings > System Firewall > General Firewall Settings**.

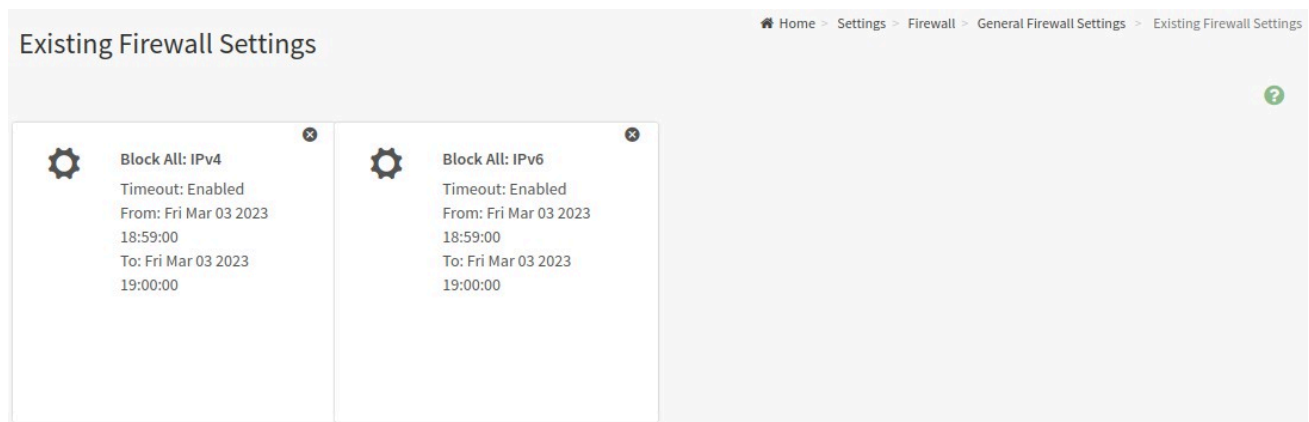


The available setting are:

- **Existing Firewall Settings:** View current firewall settings. A blank page opens if you did not add anything in "Add Firewall settings".
- **Add Firewall Settings:** Add firewall settings.

To view existing firewall settings:

1. In the General Firewall Setting page, click **Existing Firewall Settings**. The Existing Firewall Settings page is displayed showing a list of rules.



2. Click a rule from the list to see its details.

Existing Firewall Settings

[Home](#) > [Settings](#) > [Firewall](#) > [General Firewall Settings](#) > [Existing Firewall Settings](#) > [Existing Firewall Settings](#)

?

Block All

IPv4

☐ Flush All

☒ Timeout

Start Date&Time

Fri Mar 03 2023 18:59:00

End Date&Time

Fri Mar 03 2023 19:00:00

Delete

- Click **Delete** to delete the system firewall rule.

To add firewall settings:

- In the General Firewall Setting page, click **Add Firewall Settings**. The Add Firewall Settings page is displayed.

Add Firewall Settings

[Home](#) > [Settings](#) > [Firewall](#) > [General Firewall Settings](#) > [Add Firewall Settings](#)

?

Block All

IPv4

☐ Flush All

☐ Timeout

Start Date

YYYY/MM/DD

Start Time

End Date

YYYY/MM/DD

End Time

Save

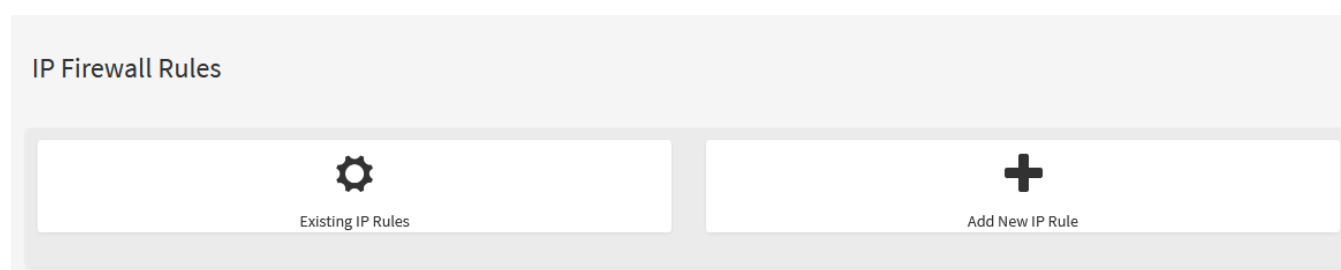
- From the Block All dropdown, select what incoming IPs and Ports are to be blocked.
- Select the **Flush All** check box to flush all the system firewall rules.
- Select the **Timeout** check box to enable or disable firewall rules with timeout.

5. In the Start Date field, type the date from which the respective firewall rule is effective from.
6. In the Start Time field, type time from which the respective firewall rule is effective from.
7. In the End Date field, type the date to which the respective firewall rule is effective to.
8. In the End Time field, type the time to which the respective firewall rule is effective to. The time should be in the dd-mm-yy:hh-mm format.
9. Click **Save**.

IP Address Firewall Rules

In this page, you can view existing firewall rules or add firewall rules.

To open the IP Firewall Rules page, click **Settings > System Firewall > IP Address Firewall Rules**.

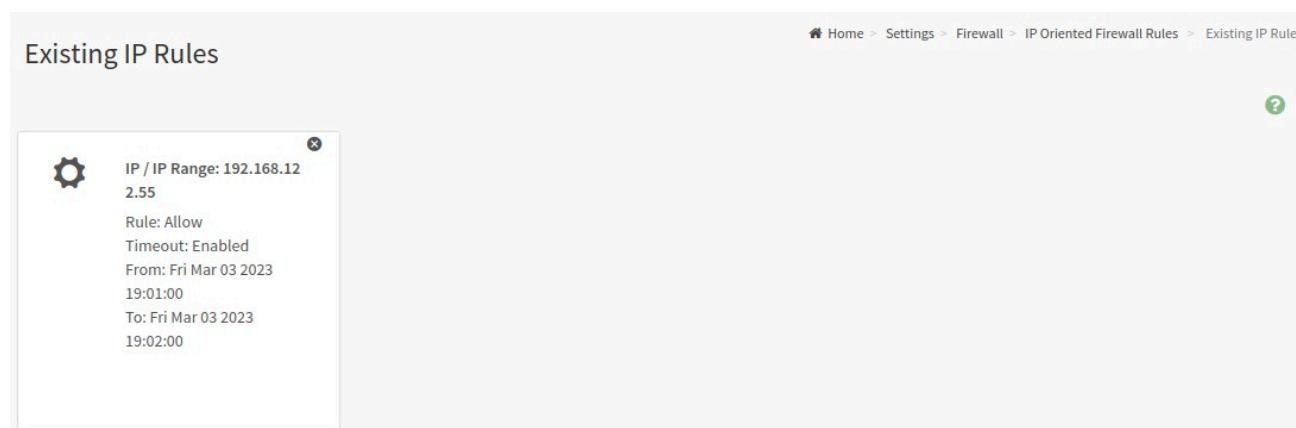


The available setting are:

- **Existing IP Rules:** View current IP rules. A blank page opens if you did not add anything in "Add New IP Rule".
- **Add New IP Rule:** Add a new IP rule.

To view existing IP rules:

1. In the IP Address Firewall Rules page, click **Existing IP Rules**. The Existing IP Rules page is displayed showing a list of rules.



2. Click a rule from the list to see its details.

Existing IP Rules

[Home](#) > [Settings](#) > [Firewall](#) > [IP Oriented Firewall Rules](#) > [Existing IP Rules](#) > Existing IP Rules



IP Single (or) Range Start

192.168.122.55

☒ **Enable Timeout**

Start Date&Time

Fri Mar 03 2023 19:01:00

End Date&Time

Fri Mar 03 2023 19:02:00

Rule

Allow

Delete

3. Click **Delete** to delete the IP rule.

To add an IP rule:

1. In the IP Address Firewall Rules page, click **Add New IP Rule**. The Add IP Rule page is displayed.

Add IP Rule

[Home](#) > [Settings](#) > [Firewall](#) > [IP Oriented Firewall Rules](#) > Add IP Rule



IP Single (or) Range Start

IP Range End

optional

☐ **Enable Timeout**

Start Date

YYYY/MM/DD 

Start Time



End Date

YYYY/MM/DD 

End Time



Rule

Allow 

Save

2. In the IP Single (or) IP Range Start field, type the IP address or a range of IP addresses.

Notes:

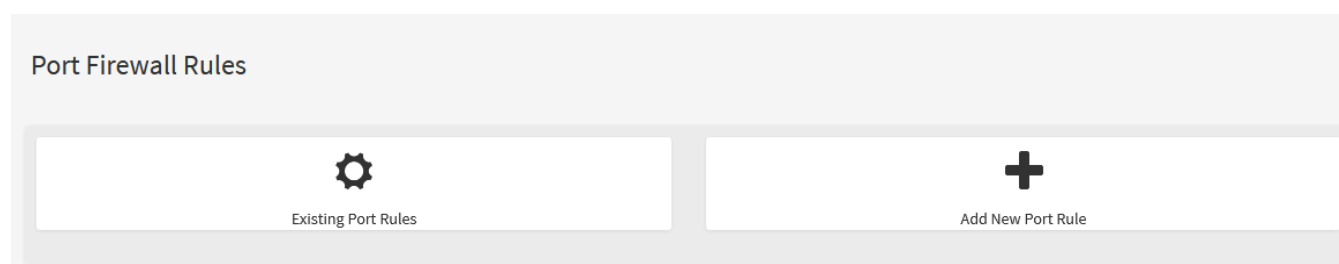
- IP Address support IPv4 Address format only
- IPv4 Address made of 4 numbers separated by dots as in xxx.xxx.xxx.xxx.
- Each number ranges from 0 to 255.
- First number must not be 0.
- IPv6 Address made of 8 groups of 4 Hexadecimal digits separated by colon as in
xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx.

3. In the IP Range End field, type the IP range end value.
4. Select the **Enable Timeout** check box to enable/disable firewall rules with timeout.
5. In the Start Date field, type the date from which the respective firewall rule is effective from.
6. In the Start Time field, type time from which the respective firewall rule is effective from.
7. In the End Date field, type the date to which the respective firewall rule is effective to.
8. In the End Time field, type the time to which the respective firewall rule is effective to. The time should be in the dd-mm-yy:hh-mm format.
9. From the Rule dropdown, select the action to be performed when the rule is broken.
10. Click **Save**.

Port Firewall Rules

In this page, you can view existing port firewall rules or add port firewall rules.

To open the Port Firewall Rules page, click **Settings > System Firewall > Port Firewall Rules**.



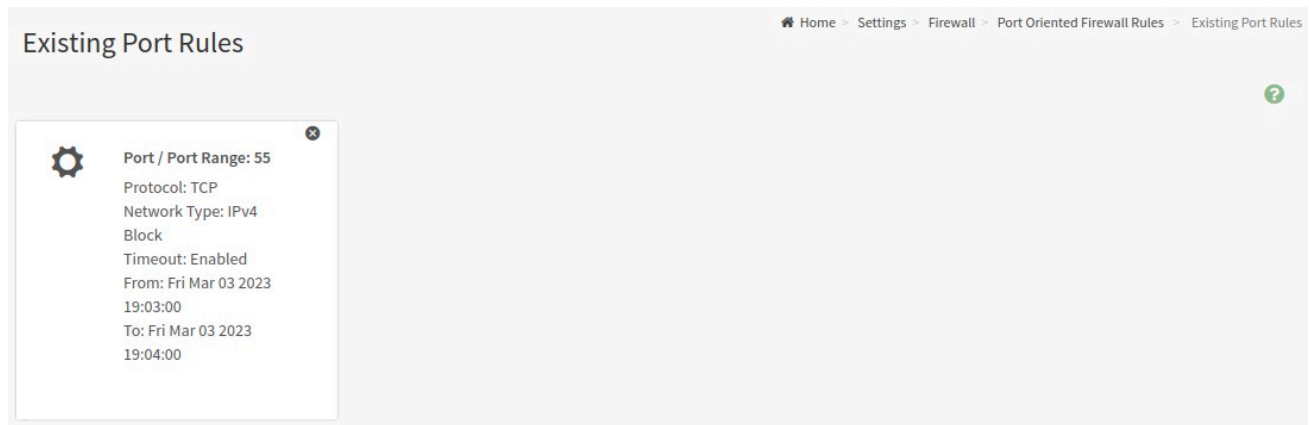
The available setting are:

- **Existing Port Rules:** View current port rules. A blank page opens if you did not add anything in "Add New Port Rule".
- **Add New Port Rule:** Add a new port rule.

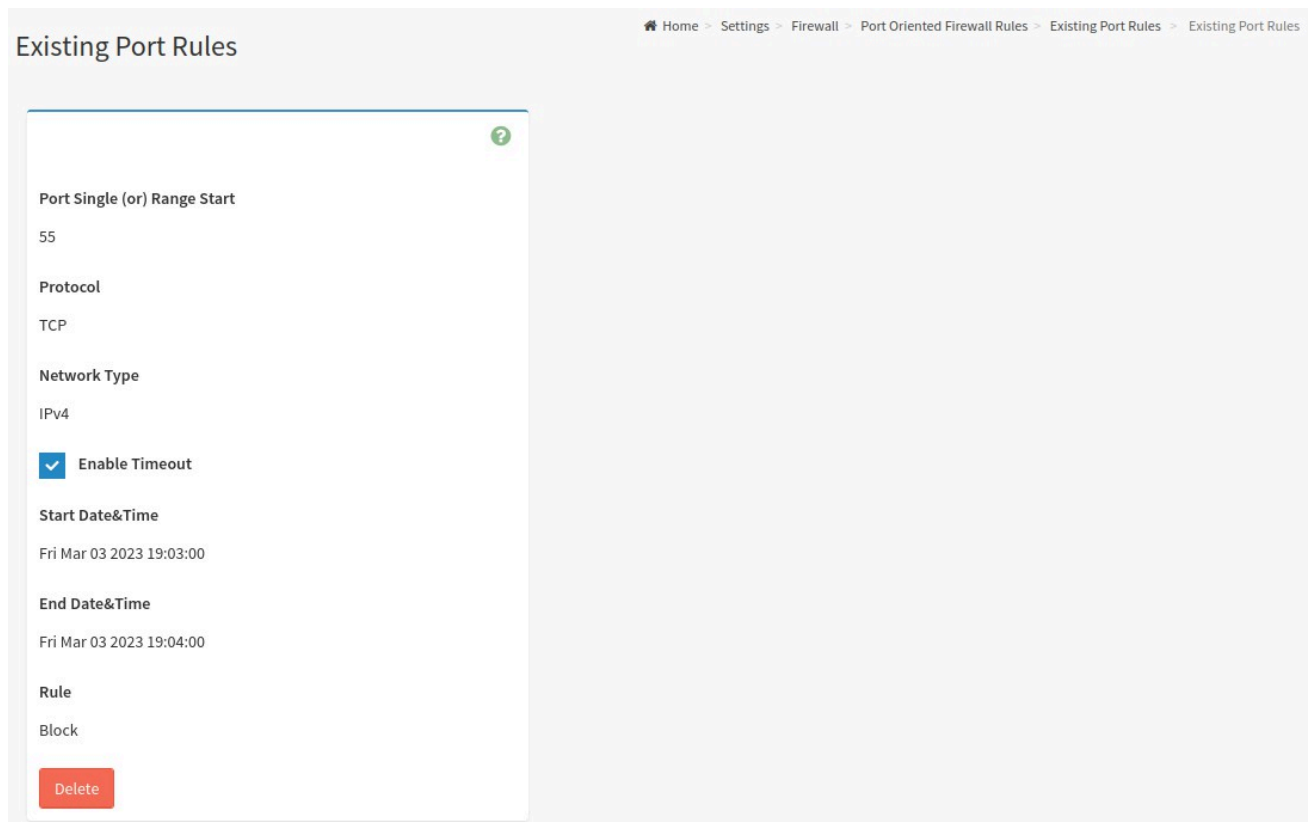
To view existing port rules:

1. In the Port Firewall Rules page, click **Existing Port Rules**. The Existing Port Rules page is displayed showing

a list of rules.



2. Click a rule from the list to see its details.



3. Click **Delete** to delete the port rule.

To add a port rule:

1. In the Port Firewall Rules page, click **Add New Port Rule**. The Add Port Rule page is displayed.

Add Port Rule

Port Single (or) Range Start

Port Range End

optional

Protocol

TCP

Network Type

IPv4

☐ Enable Timeout

Start Date

YYYY/MM/DD

Start Time

End Date

YYYY/MM/DD

End Time

Rule

Allow

Save

2. In the Port Single (or) IP Range Start field, type the port number or a range of port numbers. Port value ranges from 1 to 65535.
3. In the Port Range End field, type the port range end value.
4. In the Protocol dropdown, select **TCP**, **UDP** or **Both**.
5. In the Network Type dropdown, select **IPv4**, **IPv6** or **Both**.
6. Select the **Enable Timeout** check box to enable/disable firewall rules with timeout.
7. In the Start Date field, type the date from which the respective firewall rule is effective from.
8. In the Start Time field, type time from which the respective firewall rule is effective from.
9. In the End Date field, type the date to which the respective firewall rule is effective to.
10. In the End Time field, type the time to which the respective firewall rule is effective to. The time should be in the dd-mm-yy:hh-mm format.
11. From the Rule dropdown, select the action to be performed when the rule is broken.
12. Click **Save**.

User Management Configuration

[Home](#) > [Settings](#) > [User Management](#) > [User Management Configuration](#)



Username

Password Size

16 bytes

Password

Confirm Password

☐ **Enable User Access**

Enable Channel Access

☐ Channel 1

☐ Channel 2

☐ Channel 8

Privilege(Channel 1)

None

Privilege(Channel 2)

None

Privilege(Channel 8)

None

☐ **KVM Access**

☐ **VMedia Access**

☐ **SNMP Access**

SNMP Access level

SNMP Authentication Protocol

SNMP Privacy Protocol

Email Format

Email ID

Existing SSH Key

Not Available

Upload SSH Key

 ...

 Save

2. In the Username field, type the name of the user.

Notes:

- Name must be a string of 1 to 16 alpha-numeric characters.
- It must start with an alphabetical character.
- It is case-sensitive.
- Special characters '-'(hyphen), '_'(underscore), '@'(at sign) are allowed.
- For 20 Bytes password, LAN session will not be established.

3. From the Password Size dropdown, select the preferred size for this user's password.
4. In the Password and Confirm Password fields, type and confirm the password.

Notes:

- Password should have a minimum of 8 characters.
- Password should be the combination of alphabets, numbers, symbol and upper case characters.
- White space and below mentioned characters are not allowed.
- The combination of <> characters is not allowed, ex: "123<4567>".
- This field does not allow more than 16/20 characters based on Password size field value.
- The password should be a string, if you try to set password using "ipmitool user set password".

Characters not allowed:

Hex	Char	Hex	Char
00	NUL '\0'	0A	LF '\n' (new line)
01	SOH (start of heading)	0B	VT '\v' (vertical tab)
02	STX (start of text)	0C	FF '\f' (form feed)
03	ETX (end of text)	0D	CR '\r' (carriage ret)

04	EOT (end of transmission)	0E	SO (shift out)
05	ENQ (enquiry)	0F	SO (shift out)
06	ACK (acknowledge)	10	DLE (data link escape)
07	BEL 'a' (bell)	11	DC1 (device control 1)
08	BS 'b' (backspace)	12	DC2 (device control 2)
09	HT 't' (horizontal tab)	13	DC3 (device control 3)
14	DC4 (device control 4)	1B	ESC (escape)
15	NAK (negative ack.)	1C	FS (file separator)
16	SYN (synchronous idle)	1D	GS (group separator)
17	ETB (end of trans. blk)	1E	RS (record separator)
18	CAN (cancel)	1F	US (unit separator)
19	EM (end of medium)	20	SPACE
1A	SUB (substitute)	7F	DEL

5. Select the **Enable User Access** check box to enable the network access for the appropriate user.

Notes:

- Enabling Channel User Access will intern assign the IPMI messaging privilege to the specific Channel user.
- It is recommended that the IPMI messaging option should be enabled for the user to enable the User Access option, while creating User through IPMI.

6. In the Enable Channel Access field, select the channel/channels to enable the network access for the appropriate channels.
7. In the Privilege dropdown, select the privilege assigned to the user which could be Administrator, Operator, User, OEM or None. By default, the channel privileges will be displayed based on the channel availability.

Note: Callback privilege is displayed in Privilege field only if its assigned by other interfaces. By default ,Callback privilege is not available to set privilege as like other privilege options from Web UI.

8. Select the **KVM Access** check box to assign the KVM privilege for the user.
9. Select the **VMedia Access** check box to assign the VMedia privilege for the user.

Notes:

- The term VMedia represents H5Viewer and JViewer clients.
- It is recommended that the privileges support to KVM and VMedia should be provided only to the ADMIN user and shouldn't be provided to USER and OPERATOR privilege level users. The Admin user can provide the privilege support to USER and OPERATOR privilege level users at their own risk.
- While modifying the KVM and VMedia access by logged in User, it will prompt you with the alert message to log out the current session to reflect the changes.

10. Select the **SNMP Access** check box to enable SNMP access for the user. Password field is mandatory, if SNMP Status is enabled.
11. From the SNMP Access dropdown, select the level option for user. Either it can be **Read Only** or **Read Write**.
12. From the SNMP Authentication Protocol dropdown, select an SNMP Authentication Protocol for this user. Password field is mandatory, if Authentication protocol is changed.
13. From the SNMP Privacy Protocol drop down, select the encryption algorithm to be used for the SNMP settings.
14. From the Email Format dropdown, select the format for the email. This format is used, when sending emails. Two type of formats are available:
 - **AMI-Format:** The subject of this mail format is 'Alert from (your Host name)'. The mail content shows

sensor information, ex: Sensor type and Description.

- **Fixed-Subject Format:** This format displays the message according to user's setting. You must set the subject and message for email alert.

15. In the Email ID field, type the email ID of the user. If the user forgets the password, the new password is mailed to the configured email address.

In the Upload SSH Key field, browse and select the SSH key file. SSH key file should be of pub type.

16. Click **Save**.

To modify a user:

Note: If any changes made in User Access/Privilege or KVM/VMedia access or Change Password, it will be reflected in the next login only.

1. Click on the active user tab. This opens a User screen as shown in the screenshot below.

User Management Configuration

[Home](#) > [Settings](#) > [User Management](#) > [User Management Configuration](#)



Username

Logged-In Password

☐ **Change Password**

Password Size

16 bytes

Password

Confirm Password

☒ **Enable User Access**

Enable Channel Access

☒ Channel 1

☒ Channel 2

☐ Channel 8

Privilege(Channel 1)

User

Privilege(Channel 2)

User

Privilege(Channel 8)

User

☒ **KVM Access**

☒ **VMedia Access**

☐ **SNMP Access**

SNMP Access level

SNMP Authentication Protocol

SNMP Privacy Protocol

Email Format

AMI-Format

Email ID

Existing SSH Key

Upload SSH Key



Delete

 Save

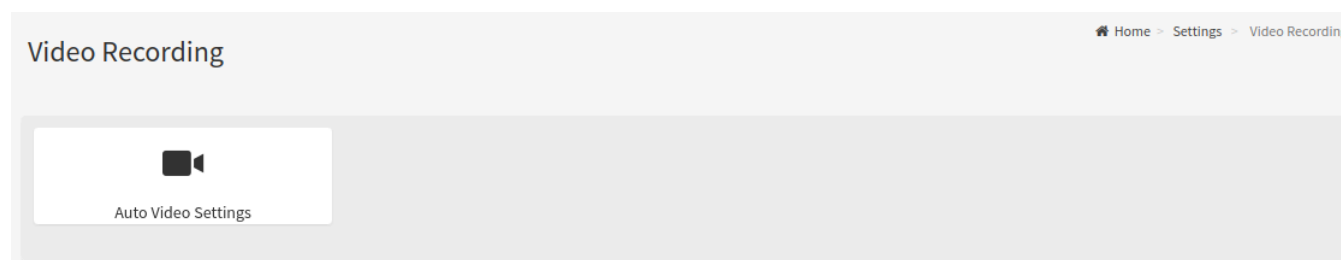
2. Replace the Username in the given field to modify the username.
3. In Logged-In Password field, type the password of the currently logged in user.
4. Select the **Change Password** check box if you wish to change the existing Password.

Note: If user login with admin and testuser in two different Web-Browsers and change the password for testuser account from admin account then the testuser account will automatically get logged-out from Web-Browser where the testuser is logged in.

5. Repeat steps 3 to 15 from the "Add new user" procedure above.
6. Click **Save**.
7. Click **Delete** to delete the user.

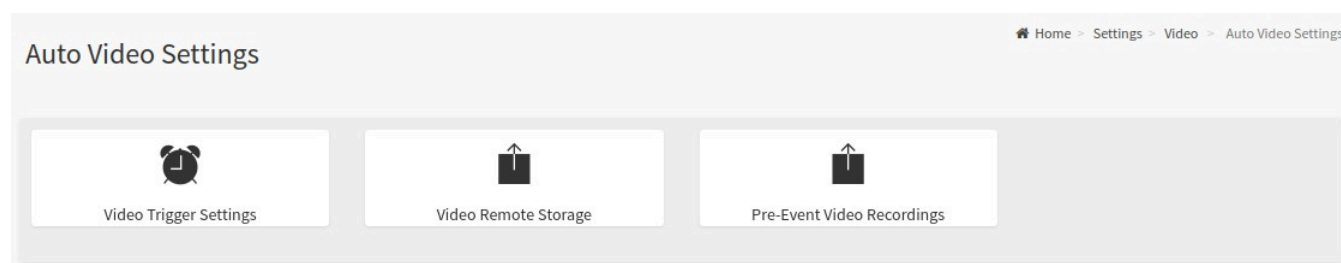
Video Recording

The Video Recording page is used to configure the events that trigger the auto video recording function of the KVM server. To open the Video Recording page, click **Settings > Video Recording**.



The available setting is Auto Video Settings, which consists of:

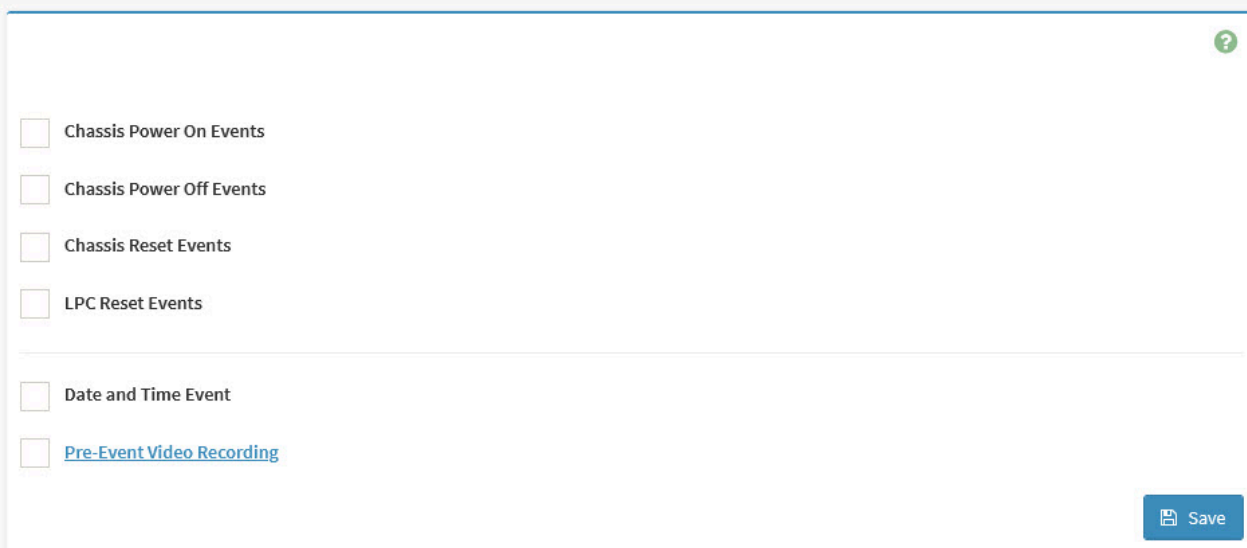
- Video Trigger Settings
- Video Remote Storage
- Pre-Event Video Recordings



To configure video trigger settings:

1. In the Auto Video Recording page, click **Video Trigger Settings**. The Video Trigger Settings page is displayed.

Video Trigger Settings



Video Trigger Settings

☐ Chassis Power On Events

☐ Chassis Power Off Events

☐ Chassis Reset Events

☐ LPC Reset Events

☐ Date and Time Event

☐ [Pre-Event Video Recording](#)

 Save

2. Select the events that trigger video recording.

Notes:

- KVM service should be enabled to perform auto-video recording.
- The date and time event should be in advance of the current system date and time.

3. Select the **Pre-Event Video Recording** check box and select for Crash Reset **Pre-crash** or **Pre-reset**.
4. Click **Save**.

To configure video remote storage:

1. In the Auto Video Recording page, click **Video Remote Storage**. The Video Remote Storage page is displayed.

Video Remote Storage

[Home](#) > [Settings](#) > [Video](#) > [Auto settings](#) > [Video Remote Storage](#)

☐ Record Video to Remote Server

Maximum Dumps

Maximum Duration (Sec)

Maximum Size (MB)

Server Address

Path in server

Share Type
☒ NFS ☐ CIFS

2. Select the **Record Video to Remote Server** check box to enable the Remote Video Support.
3. In the Maximum Dumps field, type a value from 1 to 100. The recorded video file should meet either the size constraint or duration constraint, according to the configured settings, depending on which constraint is met first.
4. In the Maximum Duration (Sec) field, type a value from 1 to 3600.
5. In the Maximum Size (MB) field, type a value from 1 to 500.
6. In the Server Address field, type the address of the server where remote videos are to be stored. The supported values are:
 - IP Address (Both IPv4 and IPv6 format).
 - FQDN (Fully qualified domain name) format.
7. In the Path in server field, type the source path. Path must be alpha-numeric and the following special characters are only allowed: '/'(backward slash), '\' (forward slash), '-'hyphen, '_'(underscore), '.'(dot) and ':'(colon). This field will not allow more than 256 characters.
8. In the Share Type field, select the share type of the remote video server. If the selected share type is CIFS, type the User Name, Password and Domain Name in the respective fields.
9. Click **Save**.

To configure pre-event video recordings settings:

1. In the Auto Video Recording page, click **Pre-Event Video Recordings**. The Pre-Event Video Recordings page is displayed.

Pre-Event Video Recordings

Home > Settings > Video > Auto settings > Pre-Event Video Recordings

?

This page is used to configure the Pre-Event video recording options. Pre-Event video recording is disabled by default. To enable the Pre-Event video recording, go to the [Triggers Configuration](#) page.

Note:
Disable/Enable pre-event recording selection for newly modified configuration to take effect.

Video Quality

Very Low

Compression Mode

High

Frames Per Second

1

Video Duration

10

Save

- From the Video Quality dropdown, select ranges (very low, low, high, average and normal).
- From the Compression Mode dropdown, select modes (high, normal, low, no).
- From the Frames Per Second dropdown, select frames/sec (1-4)
- From the Video Duration dropdown, select the duration of video, select second (10-60).
- Click **Save**.

Notes:

- Pre-Event video recording do not occur, while active KVM session or Post-event video recording is in progress.
- For supporting video playback using standard video players (e.g. VLC), the downloaded AVI video file is created with minimum of 3 FPS. Configured FPS value is not applicable here.

Pre-Event

Pre-Event video recording files are named as per event captured. For example - if any video is recorded for Crash Event, the recorded file is named as **pre_crash_video_x.dat**, where x is file count, similarly if it is recorded for reset event it is named as pre_reset_video_x.dat.

Post-Event

Post-Event video recording files will be named as shown below:

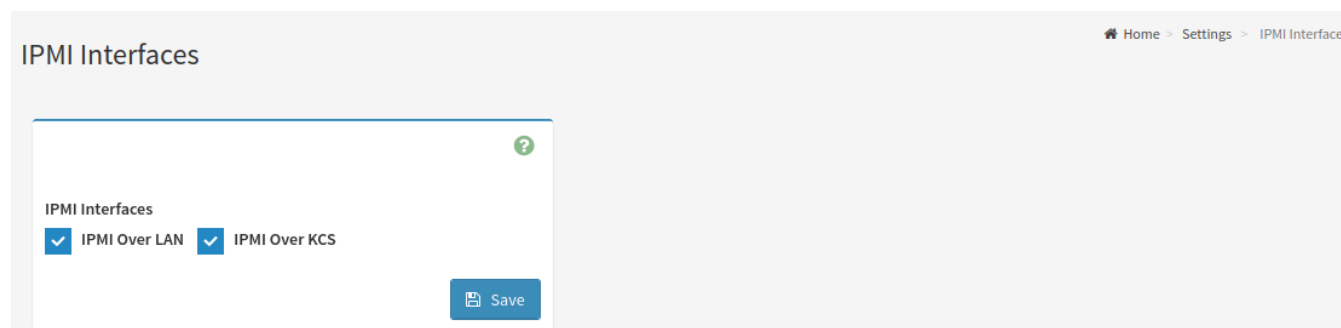
video_dump_<Hostname>_%Y%m%dT%H%M%S.dat

Notes:

- If video resolution change during auto-video recording then multiple video files are downloaded. That is, a video file is generated for each resolution change. So in case of resolution change, video file is downloaded with name video_dump %Y%m%dT%H%M-%S_partN.dat, where N is the number of files downloaded for a particular video recording.
- Multiple video files are generated only for HTML video download option. In case of Java video download application, single video file is downloaded even when resolution changes occurs. This behavior difference between HTML video download option and Java video download application is due to browser memory constrain.
- In some cases, the "power off " post-event cannot record the screen because of shutdown process, i.e., host boot into BIOS/DOS, is too short.
- When the video file cannot be recorded due to some reasons, the video file cannot be downloaded/played.

IPMI Interfaces

This page is used to configure the IPMI Interfaces. To open the IPMI interfaces page, click **Settings > IPMI Interfaces**.



To configure IPMI interfaces:

1. Select the **IPMI Over LAN** check box, which allows the user to perform IPMI communication over LAN.
2. Select the **IPMI Over KCS** check box, which allows the user to perform IPMI communication over KCS.
3. Click **Save**.

Remote Control

The Remote Control page enables you to configure remote viewers that give you remote access to the device. The options are: H5Viewer and JViewer. To open the Remote Control page, click **Remote Control**.

Remote Control

Remote KVM

Home > Remote Control

?

H5Viewer

Click here to go to Remote Session Settings.

Launch H5Viewer

JViewer

Launch JViewer

- H5Viewer
- JViewer

H5Viewer

The system and browser requirements for Remote Control are given below:

System Requirements

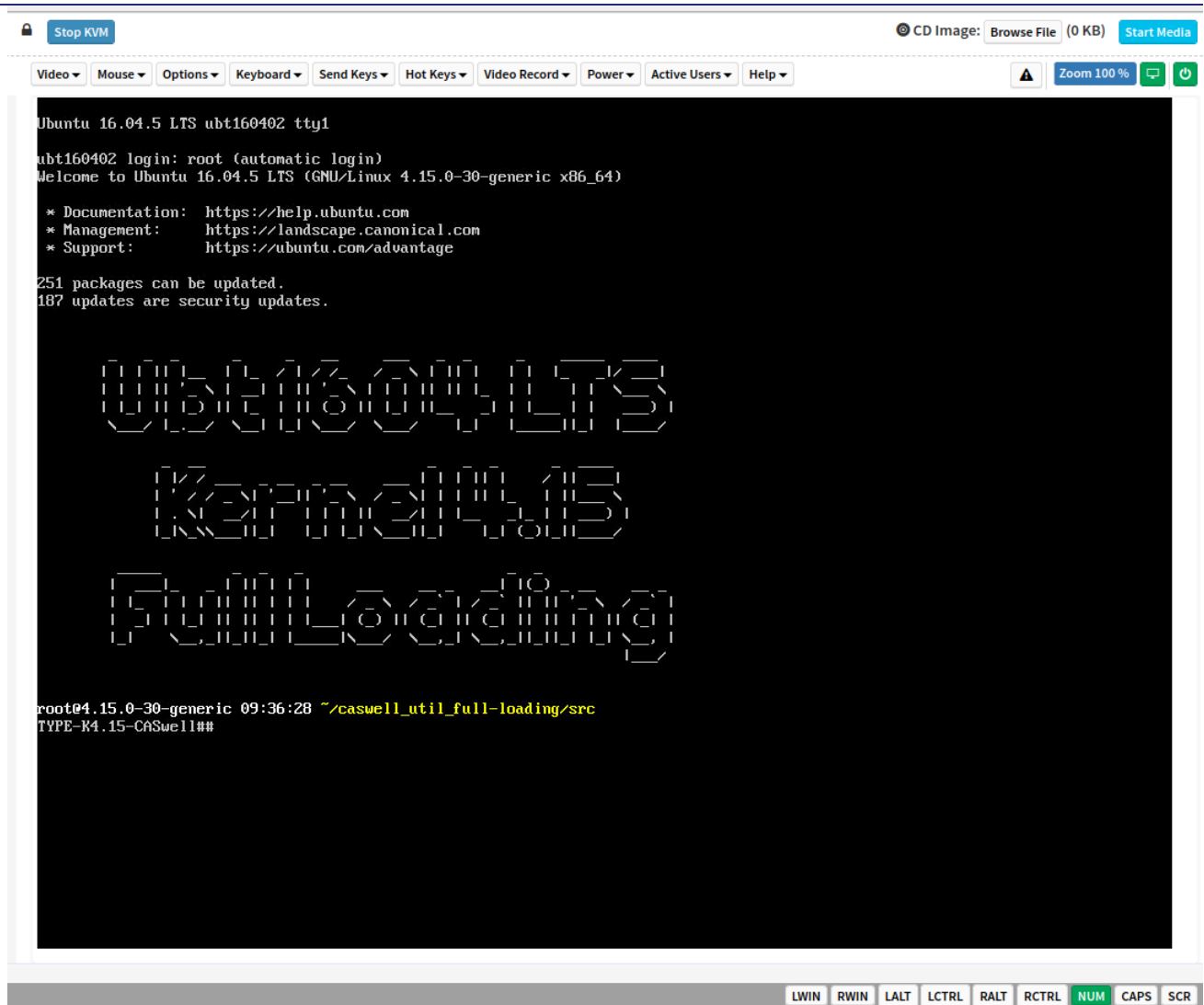
- Client machine with 8GB RAM
- If the client machine has 4GB RAM or lower, there will be lag in Video/Keyboard/ Mouse/Media redirection functionality

Supported Browsers

- Chrome latest version
- Firefox (with limited support)
- Microsoft Chromium-based Edge
- Safari (On Mac only)

To Start / Stop H5Viewer:

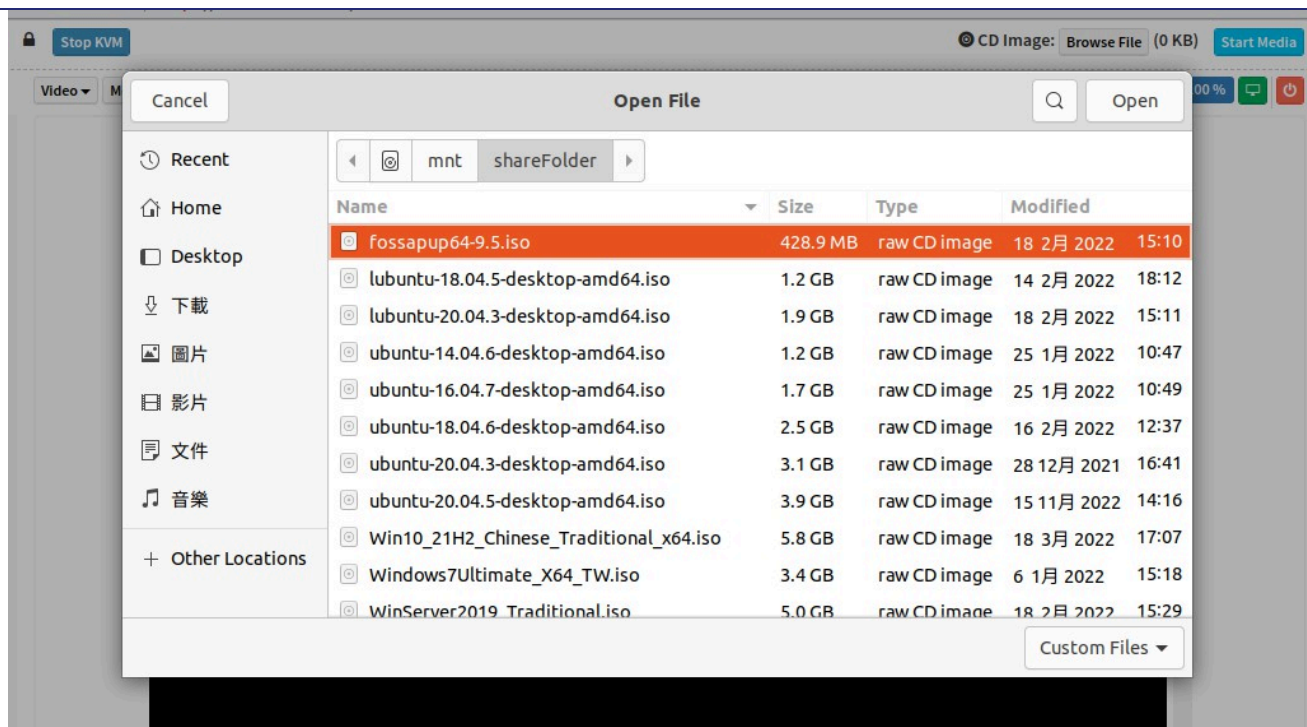
1. Click **Launch H5Viewer** to open the Remote Control KVM page.



2. To stop the H5Viewer video redirection, click Stop KVM.

To Start / Stop Media:

1. Click **Browse** to select CD Image. After selecting the image, **Select / Unselect** media boost option.



2. Click **Start Media** to redirect the selected CD image file to the Host.
3. To stop the CD Image redirection, click **Stop Media**.

Notes:

- If media boost mode is selected, the processes related to media redirection will have high priority than other processes. This improves media performance but other processes will have limited access to CPU cycle.
- If CD/DVD instance is started with media boost mode, the next CD/DVD instance is started without any pop-up message.

The Menu contains the following sub menu items.

▪ Video

Pause Video: This option is used for pausing Console Redirection.

Resume Video: This option is used to resume the Console Redirection when the session is paused. Refresh Video: This option can be used to update the display shown in the Console Redirection window.

Display on: This option is used to unlock the host display.

Display off: This option is used to lock the host display.

Capture Screen: This option helps to take the screenshot of the host screen and save it in the client's system.

▪ Mouse

Show Client Cursor: This menu item can be used to show or hide the local mouse cursor on the remote client system.

▪ Options

Zoom:

- **Normal:** By default this option is selected.
- **Zoom In:** For increasing the screen size. This zoom varies from 100% to 150% with an interval of 10%
- **Zoom Out:** For decreasing the screen size. This zoom varies from 100% to 50% with an interval of 10%
- **Block Privilege Request:** To enable or disable the access privilege of the user.

▪ Keyboard

Keyboard Layout: This feature is fully compatible when host and client have the same keyboard language layout. If the client and host language layouts differ, some special characters will not be compatible.

List of Host Physical Keyboard languages supported in H5Viewer:

- English U.S.
- German.
- Japanese.

Notes:

- Some keys like Windows, Print Screen have OS precedence and due to this H5Viewer might not be able to capture those OS precedence key and send it to the host.
- In some cases, after press OS precedence (Windows, Print Screen) keys press H5Viewer keyboard may not work as expected. This is due to browser focus loss. Since browser doesn't have control key press/release will not be done properly.
- It is highly recommended to use the Windows key option from Send Keys or using Hot keys options.

Soft Keyboard: This option allows you to select the keyboard layout. It will show the dialog similar to Windows On-screen keyboard. If the client and host languages are different, you can select the soft keyboard that corresponds to the host keyboard layout from the list shown in H5Viewer, and use it to avoid typo errors.

Different Linux systems follow different keyboard layouts. So the softkeyboard displayed uses standard windows keyboard layout irrespective of the host OS.

Soft keyboard is applicable only for H5Viewer Application not for other applications in the client system.

List of Soft Physical Keyboard languages supported in H5Viewer:

- English US
 - English UK
-

-
- Spanish
 - French
 - German
 - Italian
 - Korean
 - Chinese Simplified
 - Chinese Traditional
 - **Send Keys**

This option is used to key items. This menu contains the following sub menu items.

Hold Down

- **Right Ctrl Key:** This menu item can be used to act as the right-side <CTRL> key when in Console Redirection.
- **Right Alt Key:** This menu item can be used to act as the right-side <ALT> key when in Console Redirection.
- **Right Windows Key:** This menu item can be used to act as the right-side <WIN> key when in Console Redirection.
- **Left Ctrl Key:** This menu item can be used to act as the left-side <CTRL> key when in Console Redirection.
- **Left Alt Key:** This menu item can be used to act as the left-side <ALT> key when in Console Redirection.
- **Left Windows Key:** This menu item can be used to act as the left-side <WIN> key when in Console Redirection. You can also decide how the key should be pressed: Hold Down or Press and Release.

Press and Release

- **Ctrl+Alt+Del:** This menu item can be used to act as if you depressed the <CTRL>, <ALT> and keys down simultaneously on the server that you are redirecting.
- **Left Windows Key:** This menu item can be used to act as the left-side <WIN> key when in Console Redirection. You can also decide how the key should be pressed: Hold Down or Press and Release.
- **Right Windows Key:** This menu item can be used to act as the right-side <WIN> key when in Console Redirection.
- **Context Menu Key:** This menu item can be used to act as the context menu key, when in Console Redirection.
- **Print Screen Key:** This menu item can be used to act as the print screen key, when in Console Redirection.
- **Hot Keys**

This menu is used to add the user configurable shortcut keys to invoke in the host machine. The configured key events are saved in the BMC.

Add Hot Keys: This menu is used to enable macros. Click Add to macros.

- **Video Record**

Record Video: This option is to start recording the screen.

Stop Recording: This option is used to stop the recording.

Record Settings: This option is used to set video record duration and video compression value. Video record

duration value should be in the range of 1 to 1800 seconds. Video Compression value should be in the range of 0.1 (Low image quality) to 0.9. (High image quality). In H5Viewer video recording, the recorded video duration will not be the same as the configured duration. The recorded duration will be close to the configured duration.

Notes:

- The Maximum video file size allowed is around 40MB. If the video file size reaches its max size limit, the recorded file is downloaded and recording will be in progress until the configured video recording time is reached . The video file is saved as video_date-month-year_hr-min-sec_partno in client side video recording.
- Users have to take care of saving the video files in different browsers.
- When H5Viewer focus is lost and if video recording is in progress, the recording will be stopped with a notification message and the recorded video file will be discarded.
- Due to browser limitation, Set timeout/set interval will be delayed from specified time of interval when browser window loses focus, hence video server will not send the video packets to H5Viewer and so the video recording will be stopped.

▪ Power

The power options are to perform any power cycle operation.

Reset Server: To reboot the system without powering off (warm boot).

Immediate Shutdown: To perform Power OFF Immediately. **Orderly Shutdown:** To Power OFF the server in proper order. **Power ON Server:** To Power ON the server.

Power Cycle Serve: To first power off, and then reboot the system (cold boot).

▪ Active Users

Click this option to display the active users and their system IP address.

Active KVM Session can be terminated when there are multiple KVM Session from Primary [FULL Privilege KVM Session].

▪ Help

Click this option to get more information About H5Viewer. The KVM Remote Console utility version and plugin version will be displayed.

▪ Quick Buttons

The upper right of H5Viewer window displays all the quick buttons. These quick buttons allow you to perform the below functions by clicking them.

Quick Buttons	Explanation
	This quick button will show / hide notifications dropdown menu, which will contains the list of notifications displayed by H5Viewer.
	It shows the current zoom value in percentage.
	This quick button is used to display the current host monitor status. If icon is in green color then host monitor is unlocked. If the icon is in red color host monitor is locked. By clicking the button host monitor status can be toggled.
	This quick button is used to display the current server power status. If the icon is in green color, the server status is powered on. If the icon is in red color, the server status is powered off. Click the button to toggle immediate power off / power on the host.

▪ Status bar buttons



Num/Caps/Scroll lock buttons are LED status buttons that denotes the current status of Num/Caps/ Scroll lock in the host.

▪ Keyboard LED Sync

When the H5Viewer is launched, the keyboard locks status and LEDs denoting the lock status of the host machine, should be in sync with the client machine. That is, if the Num/Caps/Scroll lock is enabled/disabled in the client machine, the same should be updated in the host machine as well.

Notes:

- Client Side Limitations

Due to web browser related security concerns, this feature has the following limitations:

- Host LED status will be synced with client LED status, only if user presses any key in client keyboard when H5Viewer window is in focus.
- Client keyboard LED status cannot be updated from web browsers.

- **Host Side Limitations**

- In some Linux hosts, when the host is booted into text mode, CAPS LOCK LED status will not be updated properly. CAPS LOCK LED won't turn ON/OFF while changing the CAPS lock status in the host OS.
- In such cases, H5Viewer CAPS LOCK synchronization functionality will not work properly.
- Example: Typing letters in H5Viewer (after pressing CAPS LOCK) will toggle between lower to upper case inside host.

- **Control keys**

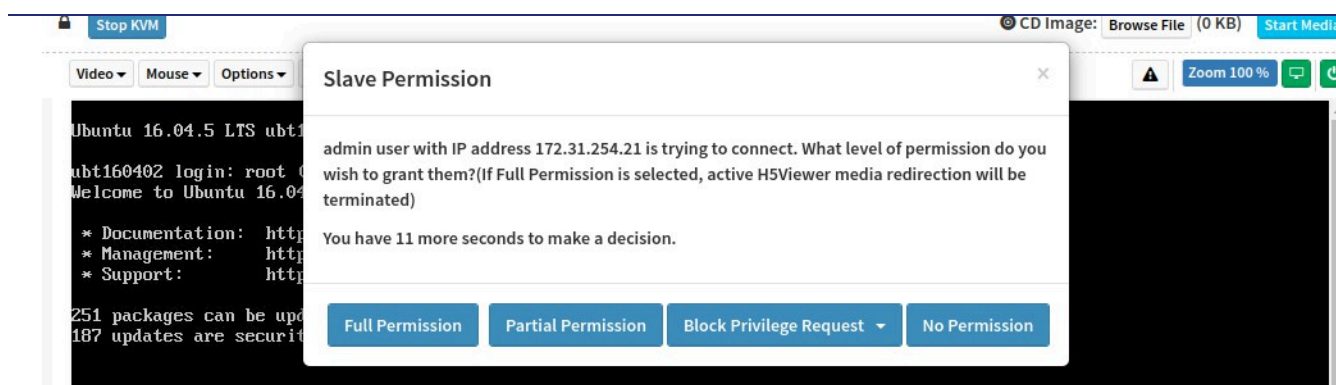
This option provides the same functionality of **Send Keys > Hold Down** menu item. Select any of the menu items, it will highlight the corresponding status bar button in green color. Similarly by clicking the buttons will toggle the selection status of the corresponding menu item.

- **KVM Sharing**

LOM stack supports "N" number of KVM Redirection sessions. Only one full permission JViewer / H5Viewer session at a time. With Full permission in JViewer / H5Viewer, the user can control the KVM redirection, and the other JViewer / H5Viewer users can only view the video redirected from the server without intervention.

When the First user launches JViewer / H5Viewer, the user will get full permission to control the host during KVM redirection. When another JViewer / H5Viewer session is launched, the Video server will send KVM sharing permission request packet to the current session, for the new Requesting session.

Once the requesting session is authenticated, a packet containing the information such as the client IP/hostname and user name of the newly authenticated or logged in user, will be send to the current session. The first client shows the dialog as a shown below:



Clicking the button in the dialog box will trigger specified action:

- **Full Permission:** When this button is clicked, the requesting session will receive full access permission, and the current (full permission) session will have a partial KVM access permission only. Partial Permission: When this button is clicked, the requesting session will receive partial permission and can only view server display (Video only).
- **Block Privilege Request > Partial Permission:** Once this option is selected, both newly requesting session and active partial privileged session will get partial permission as auto response and can only view server display. Further request will be served by auto response mechanism.
- **Block Privilege Request > No Permission:** Once this option is selected, both newly requesting session and active partial privileged session access will be denied as auto response. Further request will be served by auto response mechanism.
- **No Permission:** When this button is clicked, the requesting session access will be denied.

JViewer

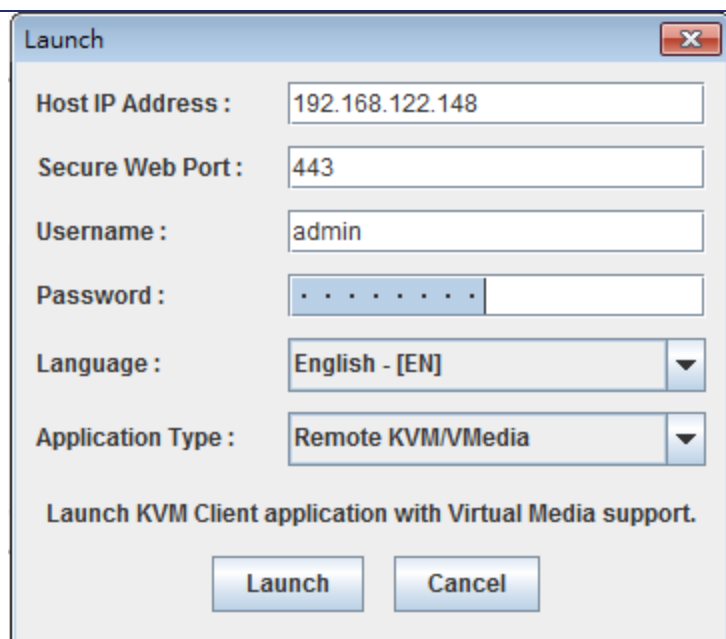
This is an OS independent plug-in, which can be used in Windows as well as Linux with the help of JRE. JRE should be installed in the client's system.

Notes:

- It is recommended to use JDK 8, openJDK 8 or higher but lower than (including) JDK 11.
- BMC will not be aware of NAT configuration settings. So launching JViewer from Web/Stand- Alone Application is not supported under NAT environment.

To Start / Stop JViewer:

1. Click **Launch JViewer** to download the jviewer.jar file.
2. Open the **jviewer.jar** file.
 - **On Windows:** Double click the jviewer.jar icon.
 - **On Linux:** Open a terminal, execute command `# sudo java -jar ./jviewer.jar`
3. Fill the required information then click **Launch**.



The image shows a 'Launch' dialog box with a title bar containing a close button. It contains several input fields and dropdown menus. The fields are: 'Host IP Address' with the value '192.168.122.148', 'Secure Web Port' with the value '443', 'Username' with the value 'admin', and 'Password' which is masked with dots. There are also two dropdown menus: 'Language' set to 'English - [EN]' and 'Application Type' set to 'Remote KVM/VMedia'. At the bottom, there is a message 'Launch KVM Client application with Virtual Media support.' and two buttons: 'Launch' and 'Cancel'.

Launch

Host IP Address : 192.168.122.148

Secure Web Port : 443

Username : admin

Password :

Language : English - [EN]

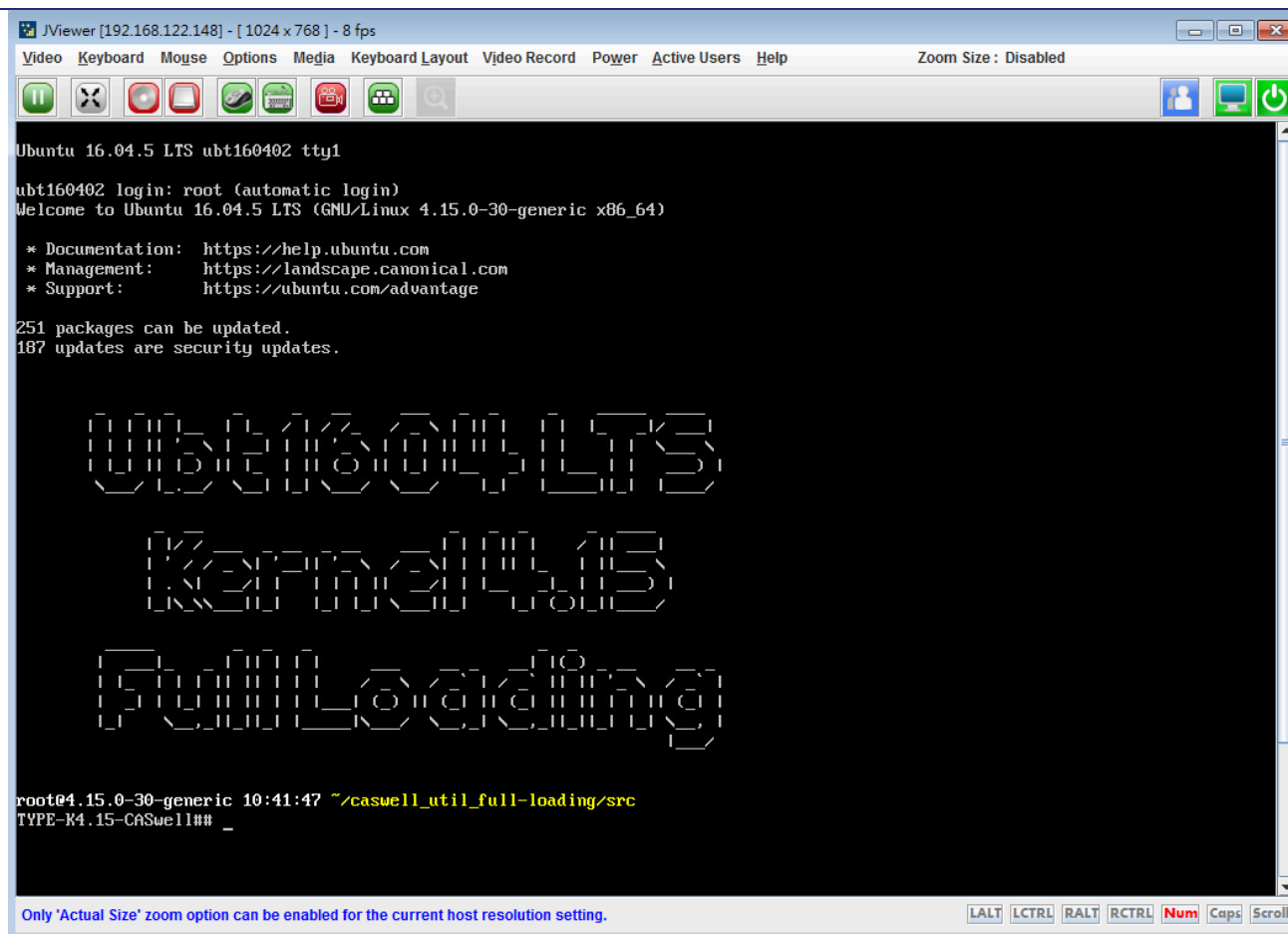
Application Type : Remote KVM/VMedia

Launch KVM Client application with Virtual Media support.

Launch Cancel

Notes:

- Host IP Address is the IP address of IPMI Web GUI.
- Secure Web Port is 443.
- The user should be with KVM Access enabled.
- Application Type should select Remote KVM/VMedia.



4. To stop the JViewer console redirection, click **Video > Exit**.

The Console Redirection menu bar consists of the following menu items.

- **Video**

Pause redirection: This option is used for pausing Console Redirection.

Resume Redirection: This option is used to resume the Console Redirection when the session is paused.

Refresh Video: This option can be used to update the display shown in the Console Redirection window.

Capture Screen: This option helps to take the screenshot of the host screen and save it in the client's system

Compression Mode: This option helps to compress the Video data transfer to the specific mode.

DTC Quantization Table: This option helps to choose the video quality.

Turn OFF Host Display/Host Video Output: If you enable this option, the server display will be blank but you can view the screen in Console Redirection. If you disable this option, the display will be back in the server screen.

Full Screen: This option is used to view the Console Redirection in full screen mode (Maximize). This menu is enabled only when both the client and host resolution are same.

Exit: This option is used to exit the console redirection screen.

▪ Keyboard

Hold Right Ctrl Key: This menu item can be used to act as the right-side <CTRL> key when in Console Redirection.

Hold Right Alt Key: This menu item can be used to act as the right-side <ALT> key when in Console Redirection.

Hold Left Ctrl Key: This menu item can be used to act as the left-side <CTRL> key when in Console Redirection.

Hold Left Alt Key: This menu item can be used to act as the left-side <ALT> key when in Console Redirection.

Left Windows Key: This menu item can be used to act as the left-side <WIN> key when in Console Redirection. You can also decide how the key should be pressed: Hold Down or Press and Release. **Right Windows Key:** This menu item can be used to act as the right-side <WIN> key when in Console Redirection. You can also decide how the key should be pressed: Hold Down or Press and Release.

Ctrl+Alt+Del: This menu item can be used to act as if you depressed the <CTRL>, <ALT> and keys down simultaneously on the server that you are redirecting.

Context menu: This menu item can be used to act as the context menu key, when in Console Redirection.

Hot Keys: This menu is used to add the user configurable shortcut keys to invoke in the host machine. The configured key events are saved in the BMC.

Full Keyboard Support: Enable this option to provide full keyboard support. This option is used to trigger the Ctrl and Alt key directly to host from the physical keyboard.

▪ Mouse

Show Cursor: This menu item can be used to show or hide the local mouse cursor on the remote client system.

Mouse Calibration: This menu item can be used only if the mouse mode is relative.

Notes:

- In this step, the mouse threshold settings on the remote server will be discovered.
- The local mouse cursor is displayed in RED color and the remote cursor is part of the remote video screen. Both the cursors will be synchronized in the beginning.
- Please use '+' or '-' keys to change the threshold settings until both the cursors go out of synch.
- Please detect the first reading on which cursors go out of synch.
- Once this is detected, use 'ALT-T' to save the threshold value.

▪ Options

Band width (Except RVAS video driver): The Bandwidth Usage option allows you to adjust the bandwidth. You can select one of the following:

- Auto Detect: This option is used to detect the network bandwidth usage of the BMC automatically.
- 256 Kbps
- 512 Kbps
- 1 Mbps
- 10 Mbps
- 100 Mbps

Zoom:

- **Zoom In:** For increasing the screen size. This zoom varies from 100% to 150% with an interval of 10%.
- **Zoom Out:** For decreasing the screen size. This zoom varies from 100% to 50% with an interval of 10%.
- **Actual Size:** By default this option is selected.
- **Fit to Client Resolution:** If the host screen resolution is greater than the client screen resolution, choose this option to fit the host screen to client screen. The host video will be scaled down and rendered in the KVM console. In this case, the host mouse cursor will appear smaller than the client mouse cursor. So the client and host mouse cursors might not be in perfect sync.
- **Fit to Host Resolution:** If the host screen resolution is lesser than the client screen resolution, choose this option to resize the JViewer frame to the host resolution.

Send IPMI Command: This option opens the IPMI Command dialog. Enter the raw IPMI command in Hexadecimal field as Hexadecimal value and click Send. The Response will be displayed as shown in the screenshot below.

The image shows a window titled "IPMI Command Dialog". It has two tabs at the top: "Hexadecimal" and "ASCII". Below the tabs is a large text area for input. At the bottom, there is a "Command :" label, followed by two input fields, one for "Hexadecimal" and one for "ASCII". To the right of these fields are two buttons: "Send" and "Clear".

GUI Languages: Choose the desired GUI language.

Block Privilege Request: Full privileged sessions can use this option to block incoming request from partial privileged sessions by setting an auto response as either "Allow only Video" or "Deny Access".

The image shows a menu item "Block Privilege Request" with a right-pointing arrow. To its right is a sub-menu with two options, each preceded by a checkbox: "Allow only Video" and "Deny Access".

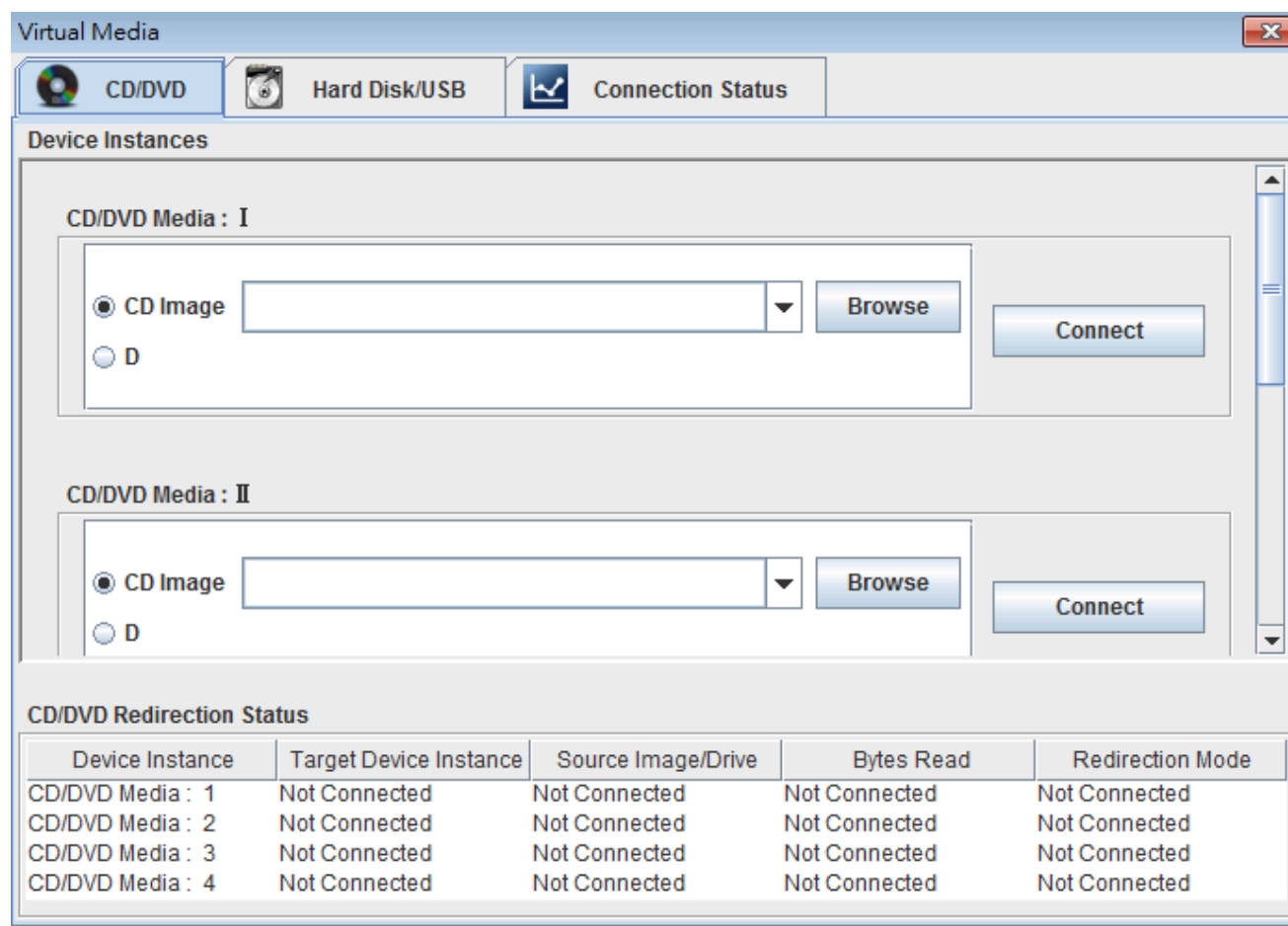
Notes:

- This menu option is available only for Full permission session and partially privileged sessions will not have this option in the menu. Either of the options can only be selected. Both options cannot be selected together. To disable "Block Privilege Request" none of the options should be selected in the menu.
- If "Allow only Video" is selected, then the secondary session will be notified as "KVM Primary Session blocked incoming request" and it will always receive "Video Only" (Partial Permission).
- If "Deny Access" is selected, then the secondary session will be notified as "KVM Primary Session blocked incoming request" and the incoming KVM session will be closed.

Request Full Permission: Partially Permitted sessions can use this option to request the Full permission from the existing full permitted session. This menu option is available only for partially privileged session and Full permission sessions will not have this option in the menu.

Media

Virtual Media Application: The virtual media application will allow you to redirect different media to the host system. The application supports CD/DVD, Hard Disk/USB devices as well as image files.



Notes:

- For more efficient performance, we recommend using the dedicated port interface to start the media.
- If there are two device panels for each device, and when you click the Connect button, then the redirected device panel will be disabled.
- Unmounting device will make the driver disconnect device when using Auto Attach. Hence, when unmounting one USB key, the other USB key will be disconnected and then reconnected.

The Virtual media application can be launched as a standalone application from the StandAlone connection dialog. It can also be launched from the JViewer, using the Virtual Media menu. When launched from JViewer, this application will work like a child dialog of the JViewer.

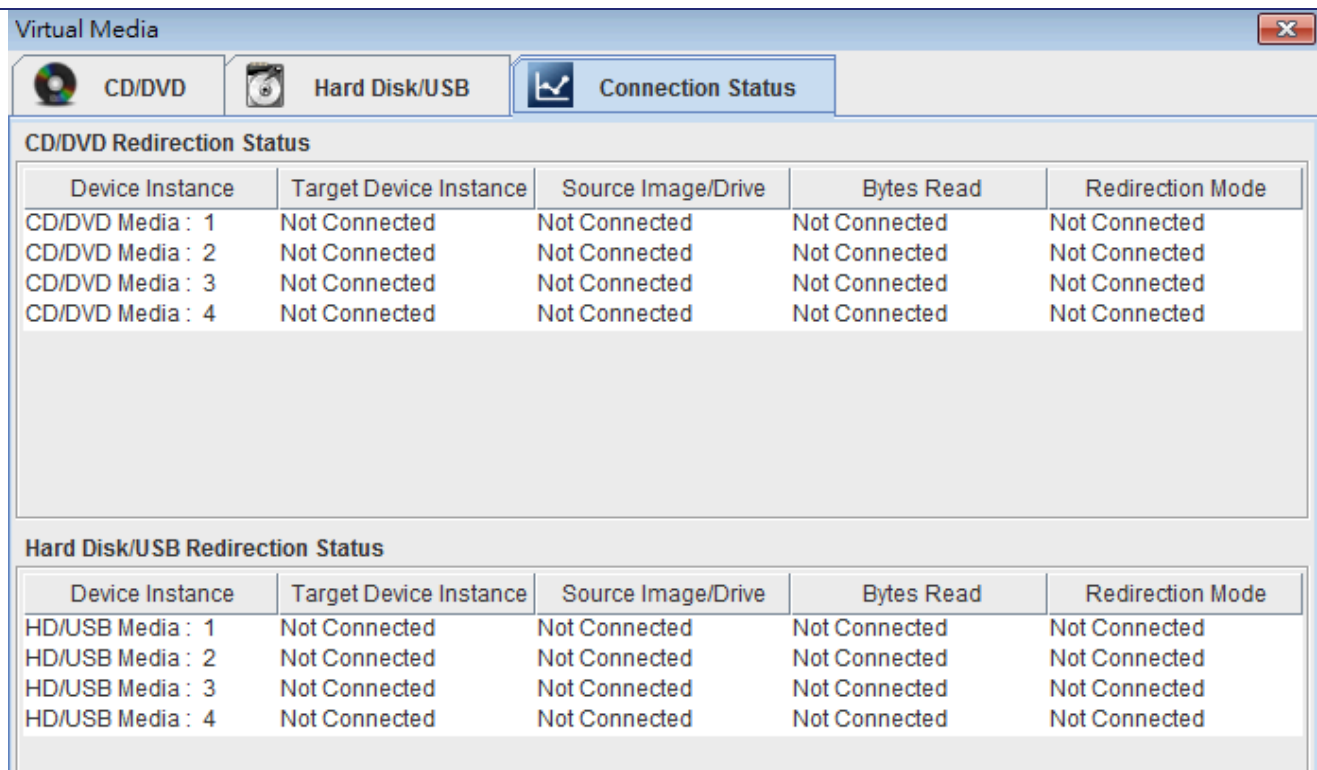
- CD/DVD Media:** This tab can be used to start or stop the redirection of a physical DVD/ CD-ROM drive and DVD/CD image file of ISO/NRG file format.

- **Hard disk/USB:** This tab can be used to start or stop the redirection of a Hard Disk/USB key image and USB key image such as img/ima.

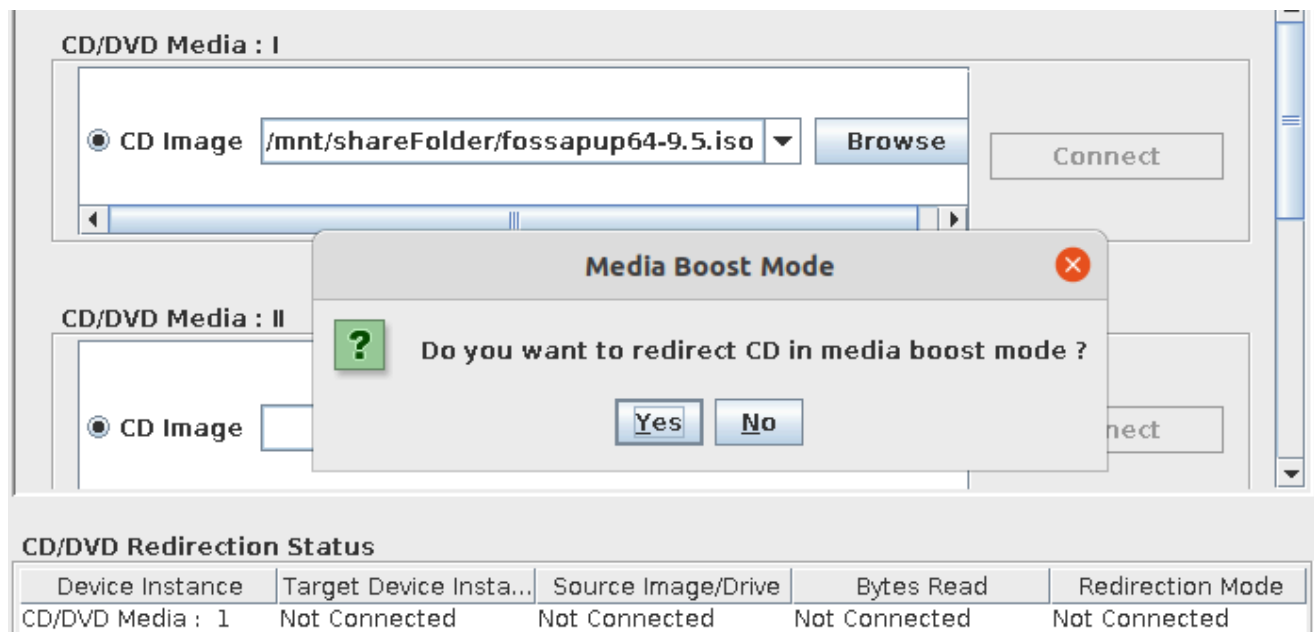
Notes:

- For redirecting Hard disk drives, you should have administrator privilege (root user in the case of Linux clients).
- For Windows 7 and above, the web browser from which the KVM redirection will be initiated, should be launched using "Run as Administrator" option. If there are multiple instances of the web browser open simultaneously, ensure that all the instances are launched using the "Run as Administrator" option.
- For Windows client, if the logical drive of the physical drive is dismounted then the logical device is redirected with Read/Write Permission else it is redirected with Read permission only. The USB/ Hard disk drive can be redirected as whole physical drive or individual logical drives.
- For MAC client, External USB Hard disk redirection is only supported. The External Hard disk Drives should be unmounted from the client before being redirected.
- For Linux client, fixed hard drive is redirected only as Read Mode. It does not support write mode. The USB/Hard disk drive will be redirected as whole physical drive.
- For Hard disk image redirection, only the file extension is validated. The Harddisk/USB key device/ image will be redirected to the host as it is. The BMC will not validate the harddisk medium, the host OS will take care of this. This is applicable for all the media redirection client applications.
- The internal hard disk drives in the client machine will not be listed. This information will be displayed in the status bar of the Virtual Media application.
- If files with hidden attribute are visible in the file open dialog, then the file can be opened and redirected.
- If the file is not visible in the file open dialog, the user shall mention the path of the image file in the file name field of the file open dialog and then open the image.
- Continuously clicking connect/disconnect buttons without giving any delay in-between may cause failure in media redirection, since the host may take few seconds to connect/disconnect the media device.

Connection Status: This tab provides a collective view of the redirection status of various virtual media devices. VMedia Privilege only restricts initiating/starting media redirection. If a device is already being redirected and attached to the host, then in host it will be visible as normal device. Hence it will be accessible to all the KVM sessions. Which includes 'KVM Privilege only' sessions as well.



Media Boost Mode: Media boost mode is applicable only for one VMedia instance. This support is available only for CD. On starting CD redirecting via JViewer/VMAApp, a pop up with an option to use media boost mode will open.



If option 'yes' is selected and no other vmedia instance is redirected in media boost mode, the redirection state will be updated as "Media Boost Mode".

CD/DVD Redirection Status

Device Instance	Target Device Insta...	Source Image/Drive	Bytes Read	Redirection Mode
CD/DVD Media : 1	Virtual CD/DVD : 0	/mnt/shareFolder/f...	589 KB	Media Boost Mode
CD/DVD Media : 2	Not Connected	Not Connected	Not Connected	Not Connected

Notes:

- If media boost mode is selected, the processes related to media redirection will have high priority than other processes. This will improve media performance but other processes will have limited access to CPU cycle.
- If CD/DVD instance is started with media boost mode, the next CD/DVD instance will be started without any pop-up message.

▪ Keyboard Layout

Auto Detect: This option is used to detect keyboard layout automatically. If the client and host keyboard layouts are the same, then for all the supported physical keyboard layouts, you must select this option to avoid typo errors. If the host and client languages differ, users can choose the host language layout in the menu and thereby can directly use the physical keyboard.

Physical Keyboard: This feature is fully compatible when host and client have the same keyboard language layout. If the client and host language layouts differ, some special characters will not be compatible.

- **Host Platform:** This feature contains two options Windows and Linux. When working with Windows host, Windows option should be selected. Similarly when working with Linux host, Linux option should be selected. This option should be selected properly for the Physical keyboard layout cross mapping to work properly. By default, Windows will be selected.

List of Host Physical Keyboard languages supported in JViewer:

- English – US
- English – UK
- French
- French (Belgium)
- German (Germany)
- German (Switzerland)
- Japanese
- Spanish
- Italian
- Danish
- Finnish
- Norwegian (Norway)
- Portuguese (Portugal)
- Swedish
- Dutch (Netherlands)

-
- Dutch (Belgium)
 - Turkish – F
 - Turkish – Q

Soft Keyboard: This option allows you to select the keyboard layout. It will show the dialog similar to Windows On-screen keyboard. If the client and host languages are different, you can select the soft keyboard that corresponds to the host keyboard layout from the list shown in JViewer, and use it to avoid typo errors. Different Linux systems follow different keyboard layouts. So the softkey board displayed uses standard windows keyboard layout irrespective of the host OS.

List of Soft Physical Keyboard languages supported in JViewer:

- English – US
- English – UK
- Spanish
- French
- German (Germany)
- Italian
- Danish
- Finnish
- German (Switzerland)
- Norwegian (Norway)
- Portuguese (Portugal)
- Swedish
- Hebrew
- French (Belgium)
- Dutch (Netherland)
- Dutch (Belgium)
- Russian (Russia)
- Japanese (QWERTY)
- Japanese (Hiragana)
- Japanese (Katakana)
- Turkish – F
- Turkish – Q

Soft keyboard is applicable only for JViewer Application not for other application in the client system.

▪ Video Record

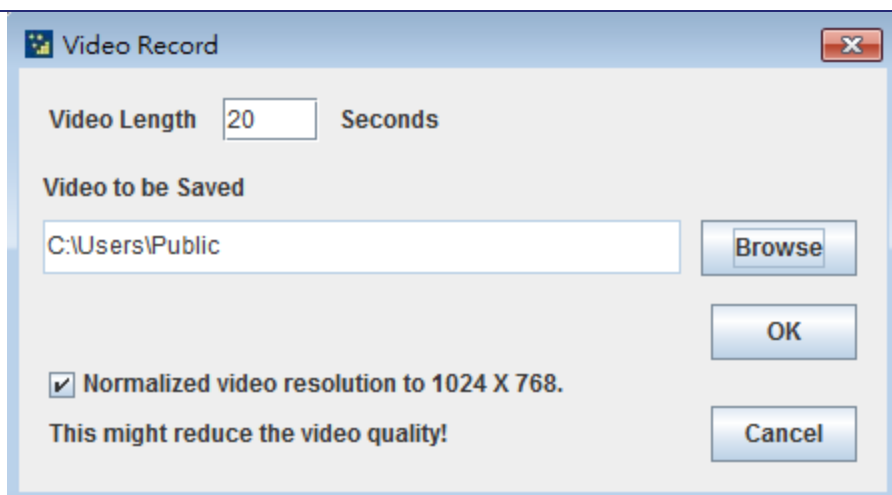
Start Record: This option is to start recording the screen.

Stop Record: This option is used to stop the recording.

Settings: To set the settings for video recording.

To Video Record:

- a. Click **Video Record > Settings** to open the settings page.
-



- b. Enter the **Video Length** in seconds.
- c. **Browse** and enter the location where you want the video to be saved.
- d. Enable the option Normalized video resolution to 1024X768.
- e. Click **OK** to save the entries and return to the Console Redirection screen.
- f. Click **Cancel** if you don't wish to save the entries.
- g. In the Console Redirection window, click **Video Record > Start Record**.
- h. Record the process.
- i. To stop the recording, click **Video Record > Stop Record**.

▪ Power

The power option is to perform any power cycle operation.

Reset Server: To reboot the system without powering off (warm boot).

Immediate Shutdown: To immediately power off the server.

Orderly Shutdown: To initiate operating system shutdown prior to the shutdown.

Power On Server: To power on the server.

Power Cycle Server: To first power off, and then reboot the system (cold boot).

▪ Active Users

Click this option to display the active users and their system ip address.

▪ Help

JViewer: Displays the copyright and version information.

▪ Quick Buttons

The lower right of Console Redirection windows displays all the quick buttons. These quick buttons help you to perform these functions by just clicking them. This option is available only when you launch the Java Console.

Quick Buttons	Explanation
	This key is used to play the Console redirection after being paused.
	This key can be used for pausing Console Redirection.
	This button is used to view the Console Redirection in full screen mode. Note: Set your client system resolution the same as the host system resolution so that you can view the server in full screen.
	This quick button is used to show or hide the soft keyboard.
	Drag this to zoom in or out.
	This quick button is used to record the video.
	This quick button is used to show or hide the mouse cursor on the remote client system.
	Active Users
	This quick button will work like a toggle button if the icon is in green color server status is power on by clicking the button immediate shutdown action will be triggered in host If the icon is in red color server status is power off. Click the button to power on the host.
	This quick button displays the available hotkeys.
	These quick buttons will pop up a virtual media where you can configure the media.

▪ Keyboard LED Sync

When the JViewer is launched from a client machine, the keyboard locks status and LEDs denoting the lock status, in the host machine should be in sync with that in the client machine. That is if the Num/Caps/Scroll lock is enabled in the host, the same should be enabled in the client machine as well.

The host keyboard LED status will be synced with the client keyboard, the lock indicators in the JViewer status bar, and the JViewer Softkeyboard.

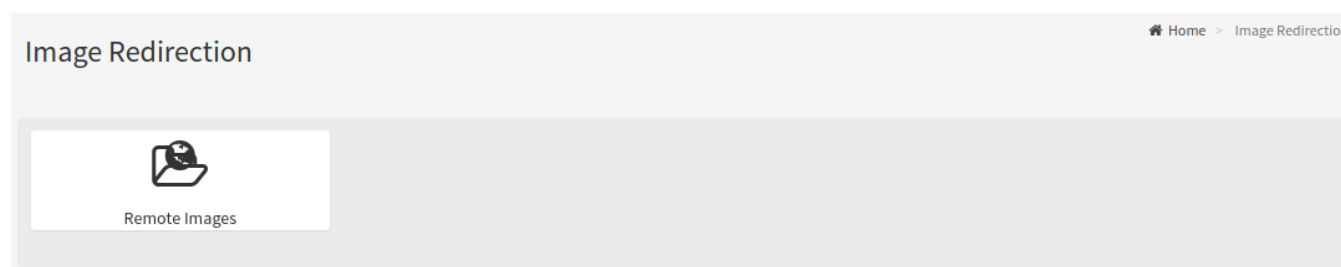
The client keyboard's LED status before launching JViewer, or before the JViewer gains focus, will be set back to the client when the focus is lost from the JViewer, or when the JViewer is closed.

Notes:

- For Macintosh OS X clients, the client keyboard LED sync will not work as the OS does not allow user applications to alter the keyboard LED status. However the keyboard lock indicators on the JViewer status bar, and the JViewer Softkeyboard lock status will sync with the host keyboard LED status.
- In the case of latest Linux distributions used as host, the keyboard LED sync will not work if the lock status is changed using the host physical keyboard directly. However the synch will work if the LED status is changed using the onscreen keyboard available in the host OS.
- Open a child dialog in JViewer will cause the focus shift out of JViewer. The client keyboard's LED status before launching JViewer, or the JViewer gains focus, will be set back to the client in this case.

Image Redirection

This page is used to configure the images into BMC for redirection. This can be done by mounting the image from the remote system, **Remote Media**. To open the Images Redirection page, click **Images Redirection**.



Notes:

- For more efficient performance, we recommend using the dedicated port interface to start the media.
- Media will always be redirected using the lowest available instance number, regardless of the instance slot opted in Web GUI. This is applicable for both local and remote media images.

To configure images for redirection:

1. In the Image Redirection page, click **Remote Images**. The Remote Media page is displayed.

The page shows a table showing the configured images on BMC. You can configure images of the remote media server. To configure the image, you need to enable Remote Media support under **Settings > Media Redirection > General Settings**.

Remote Media
Emulate CD/DVD/HDD images in the network to host as media through BMC

Home > Image Redirection > Remote Media

?

Click here to go to [Media General Settings](#) or .

Refresh Image List
Sync Image Status

Media Type	Media Instance	Image Name	Redirection Status	Connected Server Session Index	
CD/DVD	0	Windows7Ult	~	N/A	▶ ■ ▲
CD/DVD	1	Windows7Ult	~	N/A	▶ ■ ▲
CD/DVD	2	Windows7Ult	~	N/A	▶ ■ ▲
CD/DVD	3	Windows7Ult	~	N/A	▶ ■ ▲
Hard disk	0	Dos6.22.img	~	N/A	▶ ■ ▲
Hard disk	1	Dos6.22.img	~	N/A	▶ ■ ▲
Hard disk	2	Dos6.22.img	~	N/A	▶ ■ ▲
Hard disk	3	Dos6.22.img	~	N/A	▶ ■ ▲

Notes:

- More than one image can be configured for each image type. At maximum 4 images can be configurable.
- To start/stop redirection and to delete an image, you must have Administrator Privileges.
- Free slots are denoted by "~".
- Supported CD/DVD format: ISO9660, UDF(v1.02~v2.60).
- Supported CD/DVD media file type: (*.iso), (*.nrg).
- Supported HDD media file type: (*.img), (*.ima).

Media Type: Displays type of Media such as CD/DVD.

Media Instance: Displays total media instance count.

Image Name: Displays the default recovery image name on the server.

Status: Displays the status of the media.

Session Index: Displays Media Server Session Index. **Start/Stop Redirection:** To start or stop Media redirection. **Pause:** To Pause the Media redirection.

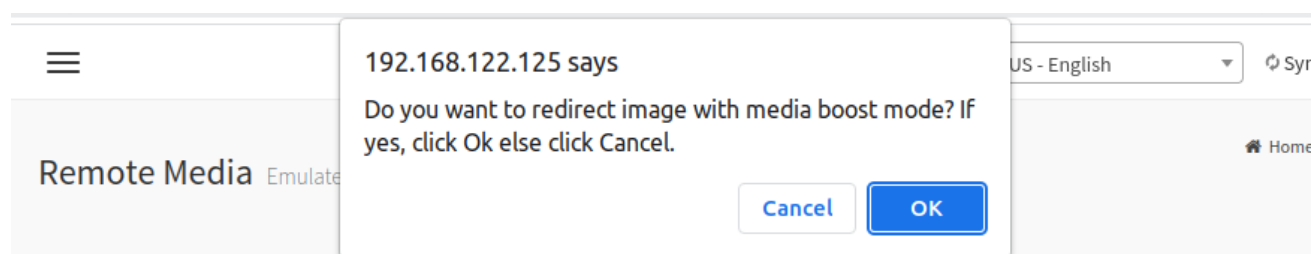
Refresh Image List: To get latest Image lists from the Remote Storage.

Sync Image Status: Click Sync Image Status to turn on/off the redirection status of images from the BMC.

2. Click  and make sure the Remote Media Support option is enabled.
3. Select a configured slot and click  to start the Remote media redirection. It is a toggle button, if the image is successfully redirected, then click  to stop the Remote media redirection. If you want to pause the Remote media Redirection, click . Redirection needs to be stopped to clear the image.
4. Perform CD Redirection with Media Boost Mode as follows:
 - a. Select CD media configured slot and click  to start the remote media redirection.
 - b. This action prompts you with the message and click **OK** to redirect the image with the media boost mode. Or else, click **Cancel** to stop this action. A sample screenshot is displayed below.

Notes:

- If media boost mode is selected, the processes related to media redirection will have high priority than other processes. This will improve media performance but other processes will have limited access to CPU cycle.
- If CD/DVD instance is started with media boost mode, the next CD/DVD instance will be started without any pop-up message.

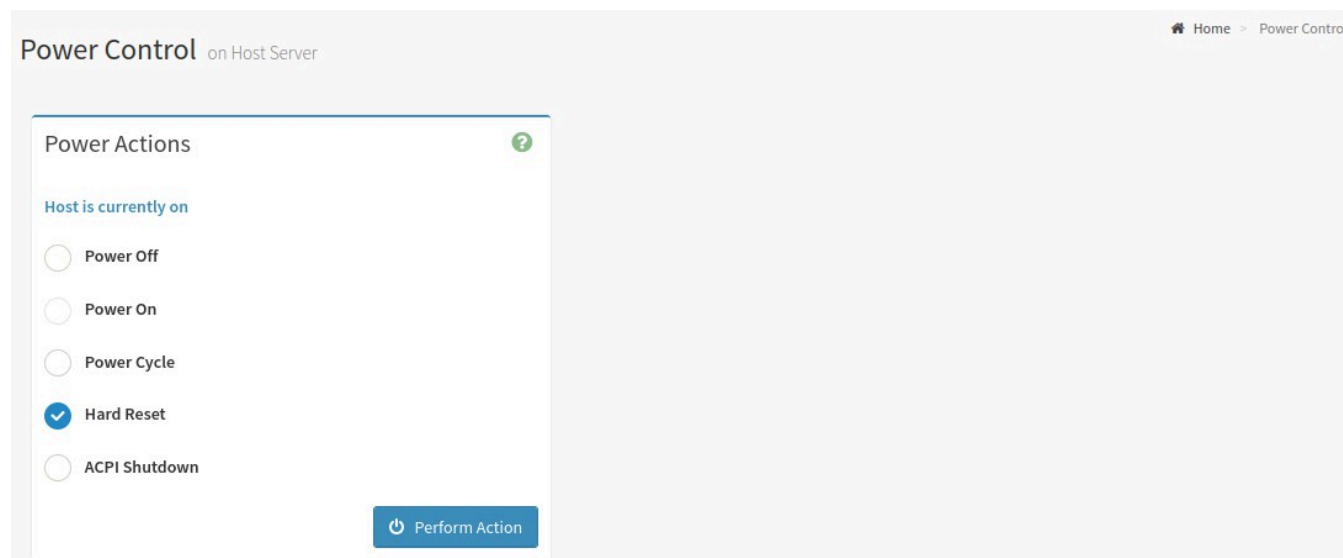


5. To clear an image status, select an image and click  to clear image status from the device.
6. Click **Refresh Image** list to get latest Image lists from the Remote Storage. The Latest Image Names list will be displayed in the Image Name dropdown list.

Power Control

This page allows you to view and control the power of your server. To open the Power Control page, click **Power**

Control.

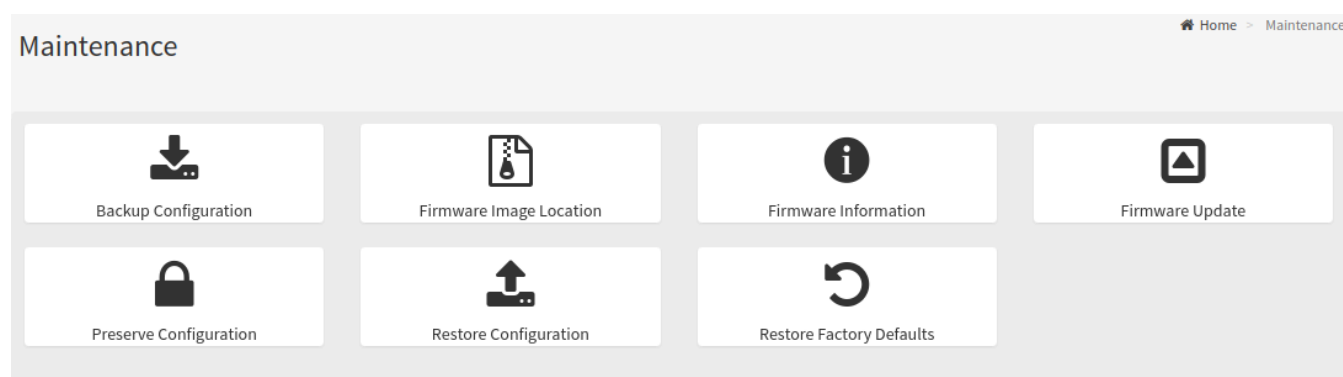


- **Power Off:** To immediately power off the server.
- **Power On:** To power on the server.
- **Power Cycle:** This option will first power off, and then reboot the system (cold boot).
- **Hard Reset:** This option will reboot the system without powering off (warm boot).
- **ACPI Shutdown:** This option to initiate operating system shutdown prior to the shutdown.
- **Perform Action:** Click this option to perform the selected operation.

Select an action and click **Perform Action**. During Execution you are asked to confirm your choice. Upon confirmation, you will be informed about the status after a few minutes.

Maintenance

This page allows you to do maintenance tasks on the device. To open the Maintenance page, click **Maintenance**.



- Backup Configuration
- Firmware Image Location
- Firmware Information
- Firmware Update

- Preserve Configuration
- Restore Configuration
- Restore Factory Defaults

Backup Configuration

This page allows you to select the specific configuration items to be backup in case of "Backup Configuration". To open the Backup Configuration page, click **Maintenance > Backup Configuration**.

- **Check All:** To select all the configuration list.
- **Download:** To download and save the configuration files backup from BMC to client system. During backup, because of security concern, the mechanism parses sensitive data to filter it out and not backup sensitive files. All IPMI users password be set to default password. User has to set password again after restoring configuration by using default user in case of login failure.

To backup the configuration:

1. Click **Check All** to backup all the configuration items or check the configuration that needs to be backup. The Backup Configuration page will appear as shown in the above screenshot. Network configurations are inter-related to IPMI, and hence by default IPMI configurations will be selected automatically when you select "Network and Services" to be backed up.
2. Click **Download** to save the backup file to the client system.
3. Click **OK** to perform the backup action. The Backup file will be saved in the client system.
4. Click **Cancel** to cancel the backup process.

Firmware Image Location

This page is used to configure TFTP location of BMC firmware image. To open the Firmware Image Location page, click **Maintenance > Firmware Image Location**.

Firmware Image Location Home > Maintenance > Firmware Image Location

Image Location Type

☒ Web Upload during flash

☐ TFTP Server

Save

To configure the firmware image location:

1. In the Image Location Type field, select **Web Upload during flash** or **TFTP Server**.
2. If you selected TFTP, in the TFTP Server Address field, type the address of the server where the firmware image is stored.

Notes:

- The Server supports both IPv4 and IPv6 addresses.
- IP Address made of 4 numbers separated by dots as in "xxx.xxx.xxx.xxx".
- Each number ranges from 0 to 255.
- First number must not be 0.
- IPv6 Address made of 8 groups of 4 Hexadecimal digits separated by colon as in "xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx".
- Hexadecimal digits are expressed as lower-case letters.

3. If you selected TFTP, in the TFTP Image Name field, type the full Source path with filename of the firmware image is stored on TFTP Server.
4. If you selected TFTP, in the TFTP Retry Count field, type the number of times to be retried in case a transfer failure occurs. Retry count ranges from 0 to 255.
5. Click **Save**.

Firmware Information

This page is used to configure the Firmware Information settings. To open the Firmware Information page, click **Maintenance > Firmware Information**.

Firmware Information

Home > Maintenance > Firmware Information

Active Firmware

Active Image ID

1

Build Date

Feb 22 2023

Build Time

11:00:56 CST

Firmware version

1.00.1003

- **Firmware Information Build Date:** Describes the Build Date of the active BMC image.
- **Build Time:** Describes the Build Time of the active BMC image.
- **Firmware version:** Describes the Firmware version of the active BMC image.

Firmware Update

The firmware update is performed with the assistance of Imperva support.

Important: DO NOT use the wizard under this tab to perform the firmware update.

Contact Imperva support to get the proper instructions for your IPMI version.

Preserve Configuration

This page allows the user to configure the preserve configuration items, which will be used by the Restore factory defaults to preserve the existing configuration without overwriting with defaults/ Firmware Upgrade configuration. To open the Preserve Configuration page, click **Maintenance > Preserve Configuration**.

Preserve Configuration Home > Maintenance > Preserve Configuration

Click here to go to [Firmware Update](#) or [Restore Factory Defaults](#)

☐ Check All

☐ SDR

☒ FRU

☐ SEL

☐ IPMI

☐ Network

☐ NTP

☐ SNMP

☐ SSH

☐ KVM

☐ Authentication

☐ Syslog

☐ Web

☐ Extlog

Save

To preserve the configuration:

1. Click **Firmware Update** or **Restore Configuration** to view Firmware Update or go to the Restore Configuration page accordingly.
2. Select the **Check All** check box to check the entire configuration list.

OR

Select the required items individually.

3. Click **Save**.

Restore Configuration

This page allows you to restore the configuration files from the client system to the BMC. To open the Restore Configuration page, click **Maintenance > Restore Configuration**.

Restore Configuration Home > Maintenance > Restore Configuration

?

Config File

 ...

 Save

To restore the configuration:

1. In the Config File field, click **Browse** to navigate and select the backed up configuration file.
2. Click **Save**. The Restore Configuration page is displayed.
3. Click **OK**.

Restore Factory Defaults

This option is used to restore the factory defaults of the device firmware. This section lists the configuration items that will be preserved during restore factory default configuration. To open the Restore Factory Defaults page, click **Maintenance > Restore Factory Defaults**.

Warning: After entering restore factory widgets, other web pages and services will not work. All open widgets will be closed automatically. The device will reset and reboot within a few minutes.

Restore Factory Defaults

Home > Maintenance > Restore Factory Defaults

?

The following checked configurations will be preserved through the restore operation. You can make changes to the list in the [preserve configuration](#) page.

☐

SDR

☒

FRU

☐

SEL

☐

IPMI

☐

Network

☐

NTP

☐

SNMP

☐

SSH

☐

KVM

☐

Authentication

☐

Syslog

☐

Web

☐

Extlog

Save

To restore factory defaults:

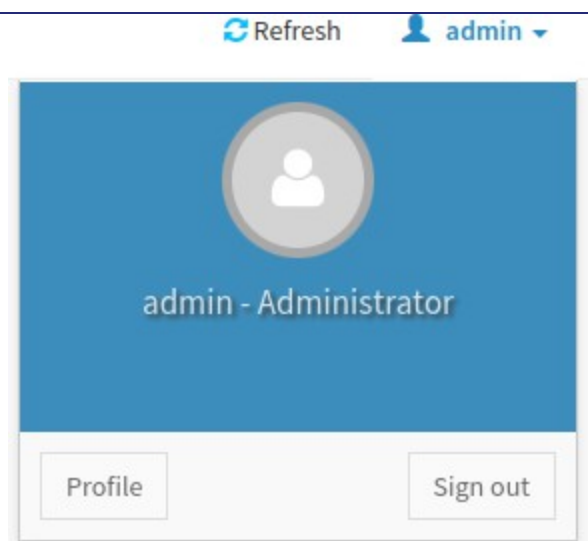
1. Click **Preserve Configuration** to redirect to the Preserve Configuration page, which is used to preserve the particular configuration not to be overwritten by the default configuration.
2. Click **Save** to restore the factory defaults of the device firmware.

When Restore Factory Defaults action is performed, there might be some log events present after performing restore operation. Those events might be newly generated which can be verified using its timestamp.

Sign Out

To log out from the IPMI Web GUI:

1. Click the user name on the top right corner of the screen.



2. Click Sign Out.

OR

1. On the Menu Bar, click **Sign Out**. A confirmation dialog box is displayed.
2. Click **OK**.

Venafi Encryption Director Integration

This appendix describes the SecureSphere - Venafi Encryption Director integration, and includes the following:

- [Overview of Integration of Venafi Encryption Director with SecureSphere](#)
- [Integrating the Venafi Encryption Director](#)

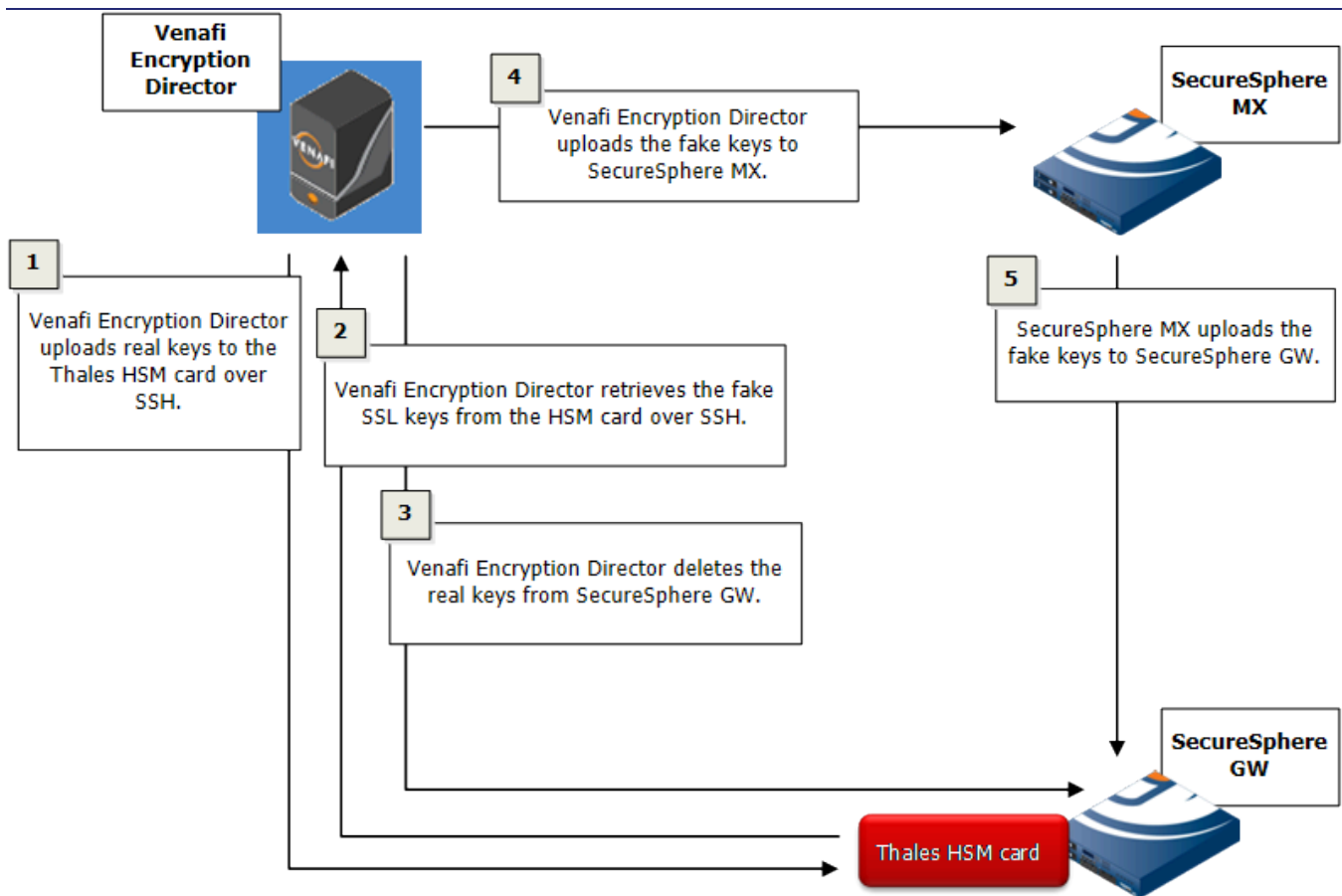
Overview of Integration of Venafi Encryption Director with SecureSphere

Venafi Encryption Director manages SSL keys across global computing infrastructures that stretch from the data center to the cloud and beyond. Patented technologies, including a centralized enrollment portal for all major certificate authorities (CAs), provide easy-to-deploy interoperability, scalability and orchestration across multiple encryption types, operating environments, CAs, HSMs, applications, directories and other enterprise systems.



Note: This topic provides an overview of how the integration with Imperva On-Premises works with Venafi. For instructions on configuring the integration, see [Integrating the Venafi Encryption Director](#).

The SecureSphere-Venafi Encryption Director integration enables downloading SSL keys to the SecureSphere Management Server (MX).



The steps involved in the integration (shown in the figure above) are as follows:

1. Venafi Encryption Director uploads the real SSL keys to the Thales HSM card installed on the SecureSphere Gateway, over SSH.

During this process, the real SSL keys are temporarily present in the SecureSphere Gateway's RAM. The keys will be deleted in step 5.

2. The HSM card creates fake SSL keys corresponding to the real SSL keys, and Venafi Encryption Director retrieves the fake SSL keys from the HSM card.
3. Venafi Encryption Director deletes the real SSL keys from the SecureSphere Gateway.

At this point, the real SSL keys are stored only on the HSM card.

4. Venafi Encryption Director uploads the fake SSL keys to the SecureSphere MX.
5. SecureSphere MX uploads the fake SSL keys to the SecureSphere Gateway.

SecureSphere GW needs the fake SSL keys to retrieve the real SSL keys from the HSM card as required.

Integrating the Venafi Encryption Director

When integrating Venafi, per Venafi's instructions, it is necessary to temporarily disable Cross Site Request

Forgery (CSRF) protection.

To temporarily disable Cross Site Request Forgery (CSRF) protection:

1. From the MX CLI, navigate to `/opt/SecureSphere/server/SecureSphere/jakarta-tomcat-secsph/webapps/SecureSphere/WEB-INF/`.
2. Open the `bootstrap.properties` file and add the following string:

```
client.include.test.cpt=false
```

3. Save your changes in the file, and then reboot the MX.

Integrating Venafi Encryption Director involves uploading the SSL keys using the On-Premises API.

For instructions on using the API to upload SSL keys, see [Uploading SSL Certificates](#).

For additional information regarding uploading the SSL keys, see [Notes on SSL Certificates](#).

Command Line Interface

This section is a reference to the SecureSphere command line tools and reviews the following topics:

- `impcfg`
- `impctl`
- `impcli`
- Miscellaneous Commands

impcfg

By default, there are two ways to run `impcfg`:

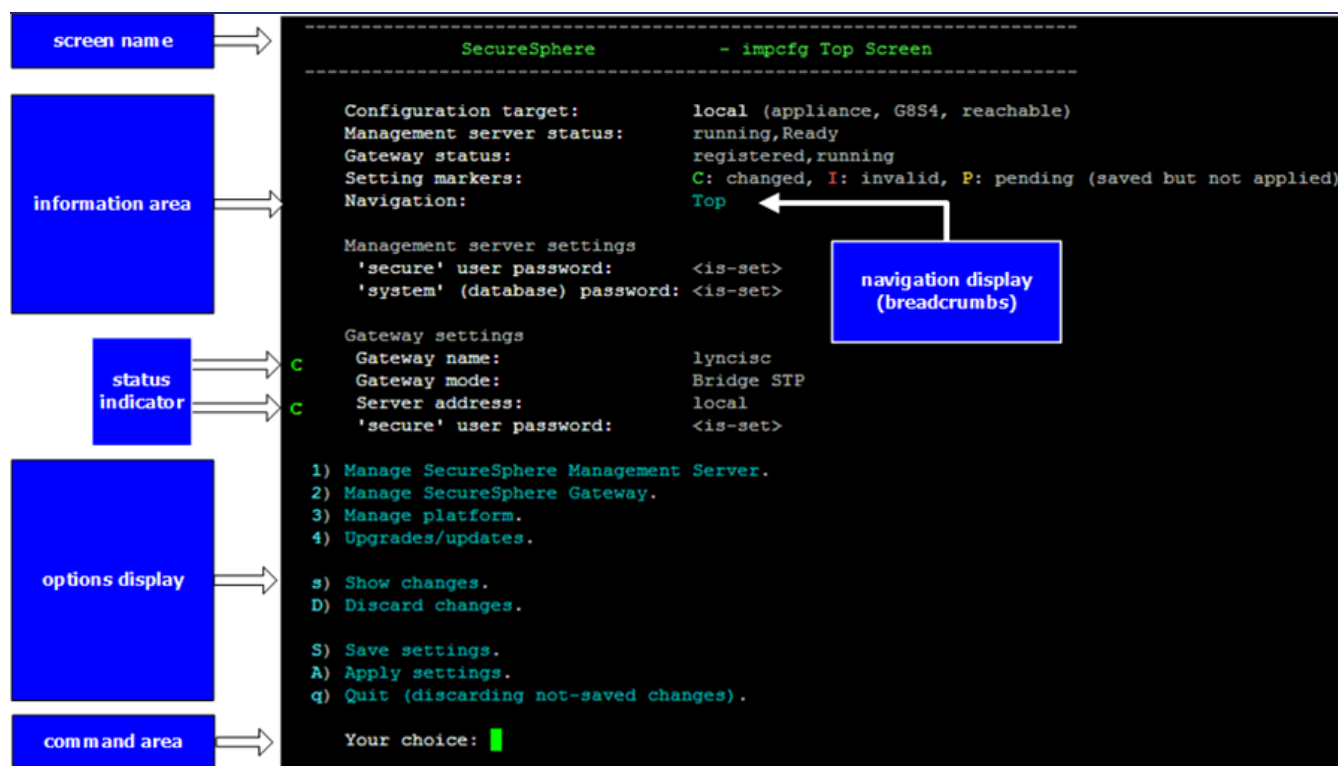
- Log in to the appliance remotely over SSH as another CLI user and run `impcfg`.
- Log in to the appliance remotely over SSH as another CLI user and use the `admin` command, and then run `impcfg`.



Note: Some terminal emulators correctly interpret the Backspace key to delete the previous character from the stream sent to the application as well as from the displayed text, while others send a control sequence, so that the stream the application sees is different from the displayed text. You should determine the behavior of your terminal emulator before using the Backspace key. In Hypterterminal, you can avoid this problem by using the DEL option.

Top Screen

The first screen displayed is the **Top** screen, shown in the following figure.



The following table lists the components of all `impcfg` screens.

impcfg Screen Components

Screen Component	color	Description
screen name	green	The name of the screen and the SecureSphere version number.
information area		Displays information about the Gateway or Management Server being configured.
status indicator		Entries in the information area may be shown with the following status indicators: - I (red I) = invalid setting - C (green C) = changed (unsaved setting) - P (yellow P) = pending (saved but not applied)
options display	blue	A list of command numbers with their meanings. In addition to the numbered commands, you can choose one of the following: - e = End this level - j = Jump up to a previous level

Screen Component	color	Description
		- t = Jump up to the top level - q = Quit (discarding non-saved changes) - S = Save settings - A = Apply settings - C = Confirm action
command area	green	The area in which the administrator enters commands and their arguments. Often, the default value [in square brackets] is displayed here.
navigation display	blue	"Breadcrumbs" indicating which screen is currently being displayed and the path followed to reach it. In the example shown in the figure impcfg Screen Layout above. We are in the Gateway Management screen, so the path is Top -> Gateway .

In the **Top** screen, you can configure SecureSphere by entering the number of one of the displayed options, as listed below.

Top Screen Configuration Options

Option	Name	Description
The numbers of these options change, depending on the appliance's configuration	Manage Secure Sphere Management Server	Configure and manage a SecureSphere Management Server. For more information, see Configuring a Management Server .
	Manage SecureSphere Gateway	Configure and manage a SecureSphere Gateway. For more information, see Configuring a Gateway .
	Manage platform	Configure and manage an appliance. For more information, see Configuring the Platform .

Option	Name	Description
s	Show changes	Display a list of changes made in this session.
D	Discard changes	Discard all the changes made in this session.
S	Save settings	Save all the configuration changes made in this session without applying them.
A	Apply settings	This option performs the following: ▪ Saves all the configuration changes made in this session. ▪ Applies all pending configuration changes (including changes made in previous sessions which were saved but not applied) to the Management Server and/or Gateways as appropriate. Except for those <code>impcfg</code> menu items which explicitly indicate that they are executed immediately, all changes take effect only when they are applied using this option.
q	Quit (discarding not-saved changes)	Quit the <code>impcfg</code> shell without saving any changes made and return to the command prompt.

impcfg Functionality Map

The table **impcfg Functionality Map** below shows the complete `impcfg` functionality.



Note: (I) means immediate activation, that is, the command is executed and, if relevant, the SecureSphere database is updated immediately.

impcfg Functionality Map

Top Screen Menu	Level 1 menu	Level 2 menu	Level 3 menu	For more information, see
Manage SecureSphere Server	Deactivate server			Configuring a Management Server
	Restart server (I)			
	Stop server (I)			
	Change management password			
	Change database password			
	Deactivate server			
Manage SecureSphere Gateway	Perform actions (start, stop, etc.)	Unregister gateway (I)		Configuring a Gateway
		Soft gateway restart (I)		
		Hard gateway restart (teardown/prepare resources) (I)		
		Stop gateway (I)		
		Refresh gateway		

Top Screen Menu	Level 1 menu	Level 2 menu	Level 3 menu	For more information, see
		status (I)		
Manage SecureSphere Gateway (continued)	Change gateway name			
	Change server address /password			
	Change operation mode	Sniffing		Configuring a Gateway as a Sniffing Gateway Note: The management server must be running when you execute this command.
		Bridge STP		Configuring a Gateway as a Bridge
		Bridge IMPVHA		
		Reverse Proxy		Configuring a Reverse Proxy
	Manage hardware security modules (HSM)	enable nCipher HSM		Hardware Security Modules (HSM)
		enable SafeNet HSM		
Manage SecureSphere Gateway (continued)	Manage interfaces and routes (for a bridge IMPVHA)	Create bridge		

Top Screen Menu	Level 1 menu	Level 2 menu	Level 3 menu	For more information, see
		Delete bridge		
		Set bridge MTU		
		Toggle High Availability on a Bridge		
	Manage interfaces and routes <i>(for a bridge STP)</i>	Create bridge		Configuring a Gateway as a Bridge
		Delete bridge		
		Set bridge MTU		
		Toggle High Availability on a Bridge		
	Manage interfaces and routes <i>(for a sniffing gateway)</i>	Set sniffing interface(s)		Configuring a Gateway as a Sniffing Gateway
		Set blocking interface		
		Add/modify blocking net		
		Delete blocking mask		
Manage platform	Manage network	Management interface	Change device	Management Interface

Top Screen Menu	Level 1 menu	Level 2 menu	Level 3 menu	For more information, see
			Change IP address	
			Change netmask	
			Locate device (I)	
		LAN interface	Change device	LAN Interface
			Change IP address	
			Change netmask	
			Locate device (I)	
		Default gateway		Default Gateway
		Static routes	Add route	Static Routes
			Delete route	
		Name Resolution (DNS client)	Enable DNS client	Name Resolution (DNS client)
			Set domain	
			Set search list	
			Set name	

Top Screen Menu	Level 1 menu	Level 2 menu	Level 3 menu	For more information, see
			servers	
		Locate network devices		Locate Network Devices
	Manage time	Add time server		Time Servers
		Delete time server		
	Manage users			Users Management
	Manage hostname			Hostname Management
	Manage bootloader password			Bootloader Password Management
	Reboot the appliance (I)			Configuring the Platform
	Shutdown the appliance (I)			Configuring the Platform
Upgrade / updates				This option is reserved for future use.

impctl

`impctl` is a lower level configuration tool that runs on the appliances. `impcfg` runs `impctl` on the appliance to deploy the configuration that the administrator defines in `impcfg`.



Warning: `impctl` is for highly experienced administrators only. Inexperienced users can seriously damage the appliance's configuration, requiring reinstallation from scratch, and they should therefore use `impcfg` instead.

- Safe `impctl` Commands
- Stopping and Starting the MX in an MX-HA Environment
- LDAP Authentication for SecureSphere CLI Users
- Configuring SNMP v3
- Miscellaneous `impctl` Commands

Safe `impctl` Commands

The table below lists the commonly used "safe" `impctl` commands.

`impctl` - Selected "Safe" Functionality

<code>impctl</code> command	Description
<code>impctl show commands --verbose</code>	List <code>impctl</code> commands and their arguments.
<code>impctl status</code>	Display the status of the appliance.
<code>impctl server start</code>	Start the MX server.
<code>impctl server stop</code>	Stop the MX server.
<code>impctl server restart</code>	Stop and then start the MX server.
<code>impctl gateway start</code>	Start the Gateway.
<code>impctl gateway stop</code>	Stop the Gateway.
<code>impctl gateway restart</code>	Stop and then start the Gateway.

impctl command	Description
<code>impctl gateway register</code>	Register the Gateway to its MX server.
<code>impctl gateway unregister</code>	Unregister the Gateway from its MX server.
<code>impctl gateway config --encryption=yes</code>	Encrypt database audit files stored on the Gateway.
<code>impctl hardening config --root-source-ip-exception=<IP address></code>	Specify an IP address from which user root is allowed to login over SSH.

Stopping and Starting the MX in an MX-HA Environment

The correct way start and stop the MX in an MX-HA environment is by using the following commands:

```
impctl server ha start
```

```
impctl server ha stop
```



Warning: Do not use `impcfg` or `impctl` to start and stop the MX as you would in a non-MX-HA environment (as described in [Configuring a Management Server](#)). Use only the above commands.

LDAP Authentication for SecureSphere CLI Users

Imperva SecureSphere now supports CLI user authentication using LDAP. Using this feature, you can manage SecureSphere MX, GW and SOM CLI users. This allows for a simplified and centralized user management and authentication process.

To use LDAP Authentication for SecureSphere CLI Users:

1. Join a new domain using the following command (by default, all users and groups are denied):

```
impctl platform ldap config --domain=imperva.platform --ldap-user=<ldap-user>
```

```
name>
```

where `<ldap-user name>` is a user with domain admin access rights.

2. Allow a user or a group using the following command:

```
impctl platform ldap allow --user=<user name>
```

-OR-

```
impctl platform ldap allow --group=<group name>
```

3. Deny a user or a group using the following command:

```
impctl platform ldap deny --user=<user name>
```

-OR-

```
impctl platform ldap deny --group=<group name>
```

4. Leave a domain authentication using the following command:

```
impctl platform ldap remove --domain=<domain name>
```

5. Show ldap domain authentication status using the following command:

```
impctl platform ldap show
```

Configuring SNMP v3

Imperva WAF gateways support monitoring of the On-Premises platform using SNMPv3, allowing for better security of messages by using authentication and encryption. The security features provided in SNMPv3 are as follows:

- **Message integrity:** Ensures that a packet has not been tampered with during transit.
- **Authentication:** Determines that the message is from a valid source.
- **Encryption:** Scrambles the content of a packet to prevent it from being learned by an unauthorized source.

SNMPv3 is a security model in which an authentication strategy is set up for a user and the group in which the user resides. The security level is the permitted level of security within a security model. A combination of a security model and a security level determines which security mechanism is used when handling an SNMP packet. The table below describes the combinations of SNMPv3 security models and levels.

Level	Authentication	Encryption	What Happens
noAuthNoPriv	Username	No	Uses a username match for authentication.

Level	Authentication	Encryption	What Happens
authNoPriv	Message-Digest Algorithm 5 (MD5) or Secure Hash Algorithm (SHA)	No	Provides authentication based on the Hashed Message Authentication Code (HMAC)-MD5 or HMAC-SHA algorithms.
authPriv	MD5 or SHA	DES or AES	Provides authentication based on the HMAC-MD5 or HMAC-SHA algorithms. In addition to authentication, provides DES 56-bit encryption based on the Cipher Block Chaining (CBC)-DES (DES-56) standard or AES.

To configure SNMPv3:

1. Login via SSH to the appliance you want to configure with SNMPv3 as root.
2. Verify you have the required authentication and encryption methods and the credentials from your SNMP administrator.
3. Run the command below:

```
impctl platform snmp config --version=3 --authentication-password=<your authentication password> --authentication-method=<your authentication method> --encryption-password=<your encryption password> --encryption-method=<your encryption method> --user-name=<your username>
```

To check the configuration of SNMPv3 users:

Run the command: `impctl platform snmp status`

An example output is:

```
[root@Test ~]# impctl platform snmp status

snmpd.service - Simple Network Management Protocol (SNMP) Daemon.

Loaded: loaded (/usr/lib/systemd/system/snmpd.service; enabled; vendor preset: disabled)

Active: active (running) since Tue 2020-06-09 15:16:42 IDT; 15s ago

Main PID: 1485 (snmpd)

CGroup: /system.slice/snmpd.service

1485 /usr/sbin/snmpd -LS0-6d -f

Jun 09 15:16:42 Test systemd[1]: Starting Simple Network Management Protocol (SNMP) Daemon....
```

```
Jun 09 15:16:42 Test snmpd[1485]: NET-SNMP version 5.7.2
```

```
Jun 09 15:16:42 Test systemd[1]: Started Simple Network Management Protocol (SNMP) Daemon..
```

```
Enabled SNMP versions:
```

```
SNMPv1
```

```
SNMPv2
```

```
SNMPv3
```

```
Configured Users:
```

```
imperva
```

Other **impctl** commands you can use:

- `impctl platform snmp remove --user-name=<username>`

Use this command for credential rotation, or to simply remove a user from SNMPv3

- `impctl platform snmp start`

Use this command to start the SNMP service

- `impctl platform snmp stop`

Use this command to stop the SNMP service

- `impctl platform snmp restart`

Use this command to restart the SNMP service

Miscellaneous **impctl** Commands

- [Disk Maintenance](#)
- [Configure Password Length](#)
- [Commands that Generate Events](#)
- [Platform System Events - Localization](#)
- [Configuring Security Banner](#)

Disk Maintenance

Use the following command on MX appliances with mirrored disks to determine if a disk has failed.

```
impctl platform storage raid adaptec arcoconf getconfig 1
```

The status of the appliance's disk drives is displayed.

Below is an example of a portion of the command's output. Note that in this example, "Device 0" has failed (its State is given as "Failed").

Physical Device information

```
Device #0
  Device is a Hard drive
  State                      : Failed
  Supported                  : Yes
  Transfer Speed             : Failed
  Reported Channel,Device    : 0,0
  Reported Location          : Connector 0, Device 0
  Vendor                     : WDC
  Model                      : WD1002FBYS-02A6B
  Firmware                   : 03.00C06
  Serial number              : WD-WMATV1847734
  Size                       : 0 MB
  Write Cache                : Unknown
  FRU                        : None
  S.M.A.R.T.                 : No

Device #1
  Device is a Hard drive
  State                      : Online
  Supported                  : Yes
  Transfer Speed             : SATA 3.0 Gb/s
  Reported Channel,Device    : 0,1
  Reported Location          : Connector 0, Device 1
  Vendor                     : WDC
  Model                      : WD1002FBYS-02A6B
  Firmware                   : 03.00C06
  Serial number              : WD-WMATV1847454
  Size                       : 953869 MB
  Write Cache                : Enabled (write-back)
  FRU                        : None
  S.M.A.R.T.                 : No
```

To determine which disk is "Device 1", use the following command:

```
impctl platform storage raid adaptec arcoconf IDENTIFY 1 DEVICE 0 1
```

To determine which disk is "Device 0", use the following command:

```
impctl platform storage raid adaptec arcoconf IDENTIFY 1 DEVICE 0 0
```

If this command is successful, the LED of the device lights up. If the device has failed, the command may not succeed, that is, the LED may not light up, depending on the nature of the failure.

Configure Password Length

Password length should contain 7 characters minimum and 30 characters maximum (default values are: min=7 and max=14).

The password length configuration affects the system users: imperva, root, secure, grub, db and newly created users.

This command can only be run by the 'root' user.

The commands are:

```
impctl security password-strength --min-length=<minimum number of characters>
```

```
impctl security password-strength --max-length=<maximum number of characters>
```

For example:

```
impctl security password-strength --min-length=7
impctl security password-strength --max-length=14
```

Commands that Generate Events

The following command generate a system event when run:

- `impctl platform time config`
- `impctl platform user config`
- `impctl security entropy-source`
- `impctl security password-strength`
- `impctl gateway start/stop`
- `patch`



Note: An unclear error argument `--key-number: expected one argument` can be displayed when you are using an unsupported character for the `impctl platform time config --key <KEY> password`. It means one of the following unsupported characters was used: `!@#$%^&*()_+{}|;:","<.>/?`~=-`

Platform System Events - Localization

A new tag called "Platform" is available in the event type (instead of "Regular") for identification.

Platform system events appear with the "Platform" tag and support translation to languages.

The commands that generate system events with the platform tag are:

- `impctl gateway start/stop`
- `impctl server start/stop`
- `patch`
- `impctl security banner config --text`
- `impctl security banner config --file`
- `impctl security banner config --display`
- `impctl platform time config --server`
- `impctl platform time config --now`
- `impctl security password-strength --min-length`
- `impctl security password-strength --max-length`
- `impctl platform user config --name`
- `impctl platform user create --name`
- `touch /dev/tpm0`
- `impctl security entropy-source --config`

Configuring Security Banner

You can configure a security banner that pops up in the MX GUI after the login page and when logging in via SSH.

To configure a security banner:

1. Run the command:

```
impctl security banner config
```

2. Enable the display to true by running the command:

```
impctl security banner config --display=true
```

3. Check the banner has been enabled by running the command:

```
impctl security banner show
```

4. Create a .txt file under the root directory.
5. Call the file by running the command:

```
impctl security banner config --file=/root/<name of file you created>.txt
```

6. Restart the server by running the command:

```
impctl server start
```

impcli

The Imperva Command Line Interface (impcli) is a proprietary shell that was developed in order to be a network appliance as our appliance should be and not as a generic operating systems. This shell provides a secure command line interface and contains a variety of commands and parameters that are most commonly used by customers.

The shell features the following use cases:

- Security. The commands that users can run the parameters that users can use are controlled, and all user inputs are logged.
- Usability. The user interface includes a commands list, help, parameters and auto completion.
- Profiling. The following shell profiled are supported:
 - Default - In this mode, all commands are available and user can run the 'admin' command to become root.
 - Sealed - In this mode, limited commands are available without the ability to switch to root user. For more information on sealed mode, see [Sealed Mode](#).

The following table indicates the available users in the shell and their usage.

User	Description
admin	Default user in the impcli shell environment. The default password for this user is admin. Users are required to change the password when logging in for the first time (not FTL).
root	Default Linux super user. Users cannot login with the 'root' user unless they change the default password in the FTL. The 'root' user needs to have its password changed every 90 days. By default, you cannot connect to the appliance as root or secure over SSH. To login as root, you must first connect as a CLI user and use the admin command. You can specify an IP address from which the user root is allowed to login over SSH with the following command: <pre>impctl hardening config --root-source-ip-exception=<IP address></pre>

User	Description
grub (bootloader)	The grub user is only for managing the grub bootloader. It is not a CLI user. You can change the bootloader password during FTL. The password can also be set using <code>impcfg</code> .



Note: In the latest versions of the shell (12 and up), the 'secure' user (`impcfg` user) is removed.

- Sealed Mode

Sealed Mode

Sealed mode enables you to lock the machine and permit only limited commands. When in sealed mode:

- The `impccli` profile is changed from 'default' mode to 'sealed box' mode
- The `admin` command that switches you to 'root' is not available
- All users that are logged into the sealed machine are not able to run the 'admin' command
- The Run a Shell Command followed action is not available
- You can enter an "unlock password" (defined during the seal process) in order to unseal the machine. This password is stored in a one-way encryption format in `/etc/shadow`
- There is no support for `scp/sftp`

To seal a SecureSphere machine:

1. In the command prompt, type **seal**.
2. Type **y** to continue.
3. Type a password that will be used to unseal the machine if necessary.
4. Retype the password to confirm.
5. Reconnect to SecureSphere to apply changes.

Note:



Make sure your password has the following characteristics:

- It must have no fewer than 10 characters and no more than 14 characters.
- It must have at least one number, one capital letter, and one special character from:

* + = # % ^ : / ~ . , [] _

- It cannot have more than two characters repeated in succession.



Note: You can permanently seal the machine by using the command **seal --strict**. However also note that issuing this command is irreversible. You will not be able to unseal it once executed.

To unseal a SecureSphere machine:

1. In the command prompt, type **unseal**.
2. Type the password that you defined during the seal process.
3. Reconnect to SecureSphere to apply changes.

The following table indicates the commands that are available in "default" and "sealed" modes.

Impcli Available Commands

Command	Description	Available in Sealed Mode
admin	Switch to admin mode	No
date	Show the system date and time	Yes
df	Show file system disk space usage	Yes
export-local	Export database to local file system	Yes
export-remote	Export database and upload it to URL	Yes
gti-remote	Upload get-tech-info to URL	Yes
gwlog	View gateway component log	Yes

Command	Description	Available in Sealed Mode
hades-show	Show contents of hades files	Yes
hades-watch	Monitor contents of hades files	Yes
help	List available commands	Yes
history	Show command history	Yes
hostname	Show the system hostname	Yes
id	Print current user and group information	Yes
ifconfig	Show network interface settings	Yes
impcfg	Imperva SecureSphere configuration menu utility	Different parameters in locked mode
impctl	Imperva SecureSphere command line control utility	Yes
import-local	Import database from local backup	Yes
import-remote	Import database from remote URL	Yes
lock	Change profile to locked mode	Yes
netstat	Show network statistics	Yes
patch-remote	Download and install SecureSphere patch from URL	Yes

Command	Description	Available in Sealed Mode
ping	Send ICMP ECHO_REQUEST to network hosts	Yes
quit	Exit SecureSphere Shell	Yes
rc	Show return code from last command	Yes
reboot	Reboot the machine	Yes
shutdown	Shutdown the machine	Yes
svlog	View server component log	Yes
syslog	View system log (/var/log/messages)	Yes
tcpdump	Dump traffic on a network	Yes
top	Monitor process activity	Yes
tracert	Print the route packets trace to network host	Yes
tset	Terminal reset	Yes
unlock	Change profile to unlocked mode	Yes
version	Show version information	Yes
wget	Test URL connectivity	Yes

Miscellaneous Commands

This section provides information about miscellaneous impcli commands and reviews the following:

- Powering Down/Up the Appliance
- Changing the Password for Management Server - Gateway Communication
- SSH Authorized Keys Management

Powering Down/Up the Appliance



Note: This section relates only to the NG (X series and M series) appliances.

Both of the following OS commands power down the appliance:

- `poweroff`
- `shutdown -P now`

To power ON the appliance:

- Push the front panel power button.

To power ON the appliance using the back power switch:

- Use either of the above commands to power down the appliance
- Move the power switch to OFF
- Remove the power cord and wait 20 seconds
- Re-insert the power cord
- Power up the appliance again.

For appliances with only one power supply, you can alternatively wait two minutes before powering up the appliance again.



Warning: These commands are the only correct, supported ways to power down the appliance.

Changing the Password for Management Server - Gateway Communication

Communication between a Management Server and its Gateways (including Gateway registration to the Management Server) is secured by a password.

To change the Password for Management Server - Gateway Communication:

On the Gateway, do one of the following:

1. Either run the following commands:

```
impctl service stop --transient gateway
```

```
impctl gateway unregister
```

```
impctl server config --password XXX (where XXX is your new password)
```

```
impctl service start --transient gateway
```

2. Or change the password via impcfg:

- a. Run `impcfg`.
- b. In the `impcfg` window, select **Manage SecureSphere Gateway -> Change Management Server address/password**.
- c. Following the instructions, enter the new password for the user "secure" and save it.

Note:

Make sure your password has the following characteristics:



- It must have no fewer than 10 characters and no more than 14 characters.
- It must have at least one number, one capital letter, and one special character from:
 * + = # % ^ : / ~ . , [] _
- It cannot have more than two characters repeated in succession.

For more information about `impcfg` functionality, refer to `impcfg` Functionality Map.

SSH Authorized Keys Management

SecureSphere Enables the creation of SSH Trust between two servers by adding the public key of one server into the "authorized_keys" of the second. This enable SSH connection without password, but does not create the the key. You can also manage the keys (check their validation and if they already exist).

To create an SSH Trust run the command: `impctl platform ssh authorized create -key=<> --user=<> --all`

To delete an SSH Trust run the command: `impctl platform ssh authorized delete -key=<> --user=<> --all`

To show an SSH Trust run the command: `impctl platform ssh authorized show -key=<> --user=<> --all`

FIPS 140 Compliance

Note: The following topic details historical information. It pertains to WAF gateway versions that are prior to v14.4. Imperva is working on updating it to today's product line versions.

This topic describes the WAF gateway FIPS (Federal Information Processing Standard) 140-2 compliance.

Encrypted Communications

WAF gateway uses FIPS-certified encryption modules to perform cryptographic operations within the cryptographic boundary. The cryptographic modules used by WAF gateway are compiled and operated in FIPS mode and perform the appropriate self tests at initialization.

MX Encrypted Communications

WAF gateway Management Server operates under Java JRE version 1.7. As part of its operation it performs cryptographic operations such as archiving, communications to clients, and communications with the Gateways it manages. All of these operations are performed by the RSA Crypto-J version 4.0 FIPS package which is certified at FIPS level 1. The NIST certificate number is 2058 (see [NIST Certificate #2058](#)).

FIPS-Approved Algorithms

The following algorithms are applied when using Crypto-J in FIPS mode:

- AES (Cert. #2249)
- DSA (Cert. #701)
- ECDSA (Cert. #357)
- HMAC (Cert. #1378)
- RNG (Cert. #1123)
- RSA (Cert. #1154)
- SHS (Cert. #1938)
- Triple-DES (Cert. #1408)
- PBKDF (vendor affirmed)
- CVL (Cert. #39)
- DRBG (Cert. #273)

FIPS-Non-Approved Algorithms

The following algorithms are **not** approved when using Crypto-J in FIPS mode:

- AES-GCM (non-compliant)
- DES
- Diffie-Hellman

-
- DESX
 - ECAES
 - EC Diffie-Hellman
 - ECIES
 - RNG (X9.31 non-compliant, MD5 and SHA1)
 - RC5
 - RSA OAEP (for key transport)
 - RSA Keypair Generation MultiPrime
 - RSA (key wrapping; key establishment methodology provides between 80 and 150 bits of encryption strength; non-compliant less than 80 bits of encryption strength);
 - HMAC-MD5
 - Raw RSA
 - RC4
 - RC2
 - ECDHC
 - MD2
 - MD5
 - PBE (SHA1 and Triple-DES)
 - RIPEMD 160

SecureSphere Gateway and Platform Communications

WAF gateway and platform communications are accomplished using OpenSSL version FIPS 2.0.1. This OpenSSL version is used for all incoming and outgoing connections related to the WAF gateway daemon. OpenSSL FIPS 2.0.1 is FIPS-certified with overall level 1 with certificate number 1747 (see [NIST Certificate #1747](#)).

FIPS-Approved Algorithms

The following algorithms are applied when using OpenSSL in FIPS mode:

- Triple-DES (Certs. #1223, #1346, #1398, #1465, #1492, #1522, #1695, #1742 and #1780)
- AES (Certs. #1884, #2116, #2234, #2342, #2394, #2484, #2824, #2929 and #3090)
- SHS (Certs. #1655, #1840, #1923, #2019, #2056, #2102, #2368, #2465 and #2553)
- HMAC (Certs. #1126, #1288, #1363, #1451, #1485, #1526, #1768, #1856 and #1937)
- RSA (Certs. #960, #1086, #1145, #1205, #1237, #1273, #1477, #1535 and #1581)
- RNG (Certs. #985, #1087, #1119, #1166, #1186, #1202, #1278, #1292 and #1314)
- DRBG (Certs. #157, #229, #264, #292, #316, #342, #485, #540 and #607)
- DSA (Certs. #589, #661, #693, #734, #748, #764, #853, #870 and #896)
- ECDSA (Certs. #264, #270, #315, #347, #378, #383, #394, #413, #496, #528 and #558)
- CVL (Certs. #10, #12, #24, #36, #49, #53, #71, #85, #260, #331 and #372)

FIPS-Non-Approved Algorithms

The following algorithms are **not** approved when using OpenSSL in FIPS mode:

- DES
-

-
- Diffie-Hellman (key agreement; key establishment methodology provides between 80 and 256 bits of encryption strength);
 - RSA (key wrapping; key establishment methodology provides between 80 and 150 bits of encryption strength);
 - DSA (Cert. #250; non-compliant)

HSM Solution

WAF gateway can be integrated with nCipher's netHSM which is FIPS level 2 (or 3, depending on configuration) certified, which provides FIPS-certified key storage solutions. For more information, see Imperva's support site for solution number 903.

WAF Gateway in FIPS Mode

In order to decrypt web traffic in FIPS mode, WAF gateway uses the FIPS-certified NSS cryptographic module. The module is both level 1 and 2 certified. SecureSphere uses it in level 1 with NIST certificate number 1747 (see [NIST Certificate #1747](#)).

Limitations

The following features are **not** supported in FIPS mode:

- Assessment – all tests that are categorized as "OS Level"
- Assessment – all tests that are categorized as "Track Changes"
- Action Sets – Archive to SCP Location Action Interface
- Web Reverse Proxy when the Reverse Proxy encrypts/decrypts data
- Web TRP (Transparent Reverse Proxy) when the TRP encrypts/decrypts data
- RSA keys shorter than 1024 bits

Activating FIPS Mode - MX

To activate FIPS mode on the MX:

1. In the **Admin** workspace, click **System Definitions**.
2. In the **System Definitions** pane, select **FIPS Settings**.
3. Enable **FIPS Mode**.
4. Click **Save**.

If you are in delayed activation mode, you need to activate this settings. For more information, see [Activating Settings](#).

Management Server Disaster Recovery (MX-DR)

SecureSphere Management Server High Availability (automatic fail over) is not available over the WAN. Traditional backup and restore is available over the WAN, but is very time consuming, and is an undesirable feature during a Disaster Recovery (DR) scenario.

To decrease the amount of manual intervention and time required to recover, a warm-standby Management Server is configured on the DR site. This hardware is only used in the event of a DR scenario. Typically transferring the backup file is the most time consuming phase of recovery. To reduce the amount of time consumed in recovery, an automated transfer of an additional copy of the backup file to the Management Server at the DR site, is configured. This process does require some manual operator intervention.

Critical functionalities that are available when configuring MX-DR:

- The last two backups are stored on the DR Management Server
- The DR Management Server enables dual licensing
- Command line Re-registration of active gateways
- All SSL keys are automatically transferred

This section provides information on how to set up two Imperva Management Servers in an MX-DR configuration.



Note: The process discussed in this section is a relatively new process and has many sub components. If the MX-DR scripting process fails, you still have the option to fall back to a standard backup and recovery process, which is effective, but more time consuming and intensive.

The process entails:

Management Server DR Task Overview

	Task Overview	Description	For more information, see
1	Prerequisites	Step to be taken prior to configuring MX-DR.	Prerequisites
2	Architectural Procedures	Procedures to configure items created in advance, available capabilities of the SecureSphere system during various phases of failover and interactive tasks required by the administrator during restore.	Architectural Procedures

	Task Overview	Description	For more information, see
3	Backup Procedure	Creation and implementation of the MX-DR process on the Imperva SecureSphere Management Servers.	Backup Procedure
4	Recovery Procedure	Steps to bring the DR Management Server online and make the gateways functional.	Recovery Procedure

- Prerequisites
- Architectural Procedures
- Backup Procedure
- Recovery Procedure

Prerequisites

Prior to performing the MX-DR configuration process, make sure you have the following items ready:

- Two Management Server licenses. Each must have the same products and information with them
- Root level password for initial configuration
- Putty or some other form of SSH terminal software
- The DR Management Server **MUST** be on the same version and patch level as the primary Management Server
- Knowledge of the secure and system user passwords
- If an MX-DR deployment already exists, you need to remove the existing scripts and SecureSphere configurations.

Architectural Procedures

The below tables give architecture and process overview of MX-DR configuration.



Note: Since the DR Management Server is being brought online using a backup from a snapshot taken at a point in time, there will be some data loss on the Management Server. Any changes made to the policies and/or Management Server settings **AFTER** the system export is created will be lost, this cannot be prevented with MX-DR.

Configuration Items Created in Advance

Task	Description
Followed Action	Created on Primary MX, configured to run on system export
OS Script	Called by followed action, moves the backup file to the DR Management Server.

Capabilities of the SecureSphere System Available During Various Phases of Failover

Phase	Elapsed Time	Agent Impact	Gateway Impact	Description
MX1 up	NA	NA	NA	Business as usual
MX1 down	0	Acting as usual	Gateways queue data locally	No ability to generate reports or change policies
Backup of MX Reloading	1.5 - 2 hours	Acting as usual	Gateways queue data locally	No ability to generate reports or change policies
Standby MX Up	2 - 2.5 hours	Acting as usual	Gateways queue data locally	No ability to generate reports or change policies
Gateways forced to new MX	Few milliseconds	Acting as usual	Small packet loss while gateway registers gateway queue begins to process	Ability to generate reports or change policies returns
System Running on Standby MX	2-3 hours	Acting as usual	gateways processing locally queued data	- All abilities restored - Gap in alerts data from the time of the last backup to the time of the failure - Fast view data will have a gap until the next nightly run

Interactive Tasks Required by Administrator During Restore

Task	Time	Tool	Description
Standby MX Up and Running	NA	NA	Start state
Stop Server process on MX	5 min	CLI	Must have MX GUI stopped for system restore
Reload backup from local disk	0.5 to 2 hours	CLI	Need to know system passwords and backup encryption password
Restart MX GUI	20 to 30 min	CLI	Typical startup time for java process which runs the GUI
Force Gateways to register to new MX	Few milliseconds	CLI	Gateways will stop and start
Verify all gateways and agents are working properly	15 min	GUI	It make take a little bit of time and several refreshes of the various parts of the MX GUI, before everything appears correctly

Backup Procedure

The MX-DR backup procedure steps enable the creation and implementation of the MX-DR process on the Imperva SecureSphere Management Servers as follows:

Backup Task Overview

	Task Overview	Description	For more information, see
1	Creating SSH Trust	Create a trusted connection to allow the configuration to be loaded onto the second management server without the need for a password.	Creating SSH Trust
2	Loading Licenses	Load the license for the	Loading Licenses

	Task Overview	Description	For more information, see
		primary and secondary Management servers into the Primary.	
3	Creating Followed Action	Create followed action for archiving.	Creating Followed Action
4	Modifying System Archive	Modify system archive to use the new followed action.	Modifying System Archive

Creating SSH Trust

To load the configuration of the primary server onto the secondary server without having to enter a password each time, you need to create a trusted SSH connection between the two servers. This section provides the procedure for the AWS environment and for other environments.

AWS environment

To create a trusted SSH connection between the primary and secondary MX servers (on an AWS environment):

1. Open an SSH connection to the secondary MX server.
2. Login using root credentials.
3. Manually configure a trusted SSH root connection between the MXs.
4. Open an SSH connection to the primary MX server.
5. Login using root credentials.
6. Run the following command to set up the folder structure and allow the mxserver user to connect to the MX-DR:

```
impctl server dr create --ip=<secondary server IP address>
```

A confirmation message stating the connection was successfully set is displayed.

7. Perform the procedure described in [Loading Licenses](#).

Other environments

To create a trusted SSH connection between the primary and secondary MX servers (on a non-AWS environment):

1. Open an SSH connection to the secondary MX server.

2. Login using root credentials.
3. Run the following command to enable the primary MX server to connect as root to the secondary MX server:

```
impctl hardening config --root-source-ip-exception=<primary server IP address>
```

4. Open an SSH connection to the primary MX server.
5. Login using root credentials.
6. Run the following command to set up the folder structure and allow the mxserver user to connect to the MX-DR:

```
impctl server dr create --ip=<secondary server IP address>
```

7. Type the password of the secondary MX server and hit Enter.
8. A confirmation message stating the connection was successfully set is displayed.
9. Perform the procedure described in [Loading Licenses](#).

Loading Licenses

After creating a trusted connection between the two servers, you need to load the license (you prepared in the Prerequisites stage) for the primary and secondary MX servers into the primary MX server. This enables the configuration to contain the licenses and SecureSphere challenges for both management servers and allows the configuration to be loaded onto the secondary MX server without the need for relicensing. The following procedure describes how this is done.

To load the license for the primary and secondary MX servers into the primary MX server:

1. Log into the SecureSphere UI on the primary MX server with administrator credentials.
2. Select the **Admin** workspace.
3. Click on the **Licensing** tab.
4. Click **Action > Upload License File**.
5. In the **Upload License File** window, click **Browse** and navigate to the license file.
6. Select the **Allow license upload for secondary MX server** check box.
7. Click **Upload**.



Note: After loading the license, you might see an invalid status indicated. This is OK and can be ignored.

Creating a Followed Action

After loading the license, you need to create a followed action for archiving. The following procedure describes how this is done.

To create a followed action for archiving:

1. Select the **Main** workspace.
2. Click **Policies > Action Sets**.
3. In the **Select** pane, click  to create a new action set.
4. In the **Action Set** window, type a name for the action set, for example: **Copy System Configuration To Secondary MX**.
5. From the **Apply to event type** dropdown select **Archiving**.
6. Click **Create**.
7. In the **Select** pane, select the action set you just created.
8. In the **Action Set** pane, under **Available Action Interfaces**, click  next to the **OS Command > Run A Shell Command** action interface. The action interface moves under **Selected Actions**.



Note: If the machine is on AWS, verify that the machine is unsealed.

9. Expand the **OS Command > Run A Shell Command** action interface and fill the fields as follow:

Name: Copy System Export To Secondary MX

Command: transfer

Arguments: --ip=<secondary server IP address> --days=<number of days between backups> --export=\${Job.file}

Working Dir: /opt/SecureSphere/etc/impctl/bin/server/dr

10. Click **Save**.
11. Perform the procedure described in [Modifying System Archive](#).

Modifying System Archive

After creating the followed action, the last step is to modify the system archive to use the new followed action. The following procedure describes how this is done.

To modify the system archive to use the new followed action:

1. Select the **Admin** workspace.
2. Click on the **Maintenance** tab.
3. In the **Maintenance** pane select **Export System**.
4. In the **Export Settings** pane, under **Archiving Action**, select the **Copy System Export To Secondary MX** followed action.
5. Click **Save**.

6. Click **Export Now**.
7. The **Export** window is displayed showing the progress. Click **OK** when completed.

Recovery Procedure

When you need to recover from a disaster, you need to bring the secondary MX server online and register the gateways to it. The procedure entails:

Recovery Task Overview

	Task Overview	Description	For more information, see
1	Stopping the Secondary MX Server	Stop the Server on the Secondary MX Server.	Stopping Secondary MX Server
2	Restoring Configuration and Restarting the Secondary MX	Restore the exported system configuration to the secondary MX and restarting it.	Restoring Configuration
3	Registering the Gateways to the Secondary MX Server	Use CLI command and force the gateways to register to the restored secondary MX.	Registering Gateways to the Secondary MX Server

Stopping Secondary MX Server

Before you begin the recovery process, you need to stop the secondary MX server. The following procedure describes how this is done.

To stop the secondary MX server:

1. Open an SSH connection to the secondary MX server.
2. Login using root credentials.
3. Run the command `impctl server stop` to stop the server.
4. Run the command `impctl server status` to confirm that the server stopped. The confirmation message `not-running` is displayed.
5. Perform the procedure described in [Restoring Configuration](#).

Restoring Configuration

After stopping the secondary MX server, you need to import the exported system configuration to the secondary MX server. The following procedure describes how this is done.



Note: When storing the system export file, the filename is denoted as follows: `exportdb-<hostname>-<year>-<month>-<day>-<time>`. For example, `exportdb-ZR1_Gateway_RC-2016-11-23-11-56-59.tgz` means the export file is from the ZR1_Gateway_RC appliance and was created on October 08, 2014 at 11:56am.

To help minimize the loss of data, always use the latest export.

To import the exported system configuration to the secondary MX server

1. Run the command `ll /var/MXDR/importdb/` and locate the latest system export file (tgz file).
2. Run the command `cd /var/tmp`
3. Run the command

```
cp /var/MXDR/importdb/exportdb-<hostname>-<year>-<month>-<day>.tgz ./
```

to copy the latest system backup to the `/var/tmp` directory.

For example,

```
cp /var/MXDR/importdb/exportdb-SS950MX-2014-10-08.tgz ./
```

4. Run the command `full_expimp.sh` to start the import script.
5. From the menu select **2** (Import).
6. Type the **System** user password for the secondary MX server.
7. From the menu, select **1** (Drop target schemas (if exists)).
8. From the menu, select **1** (Copy configuration files during the import).
9. Type the **Secure** user password.
10. Type the directory path and filename including extension to the configuration export file that you want to import.



Note: Since you are in the `/var/tmp` directory, there is no need for a path. However, if you ran `full_expimp.sh` from another directory you would have to include the path to the import file.

11. Type the password used to encrypt the configuration export.



Note: This password can be different for each file and is set at the time of export. If no password is set in the System Export screen of the Maintenance tab, the MX server automatically uses the SYSTEM user password. This guarantees that the system export is encrypted even if the user does not supply a password.

12. Type **Y** to begin the import of the configuration file.

You can monitor the import by tailing the import log file in another SSH session using `tail -f /var/tmp/exportdb-<hostname>-<year>-<month>-<day>.log`



Note: Depending on the size of this import, the import process may take up to a few hours. Average time is roughly 20-30 minutes, again depending on size of import.

13. When the import process is completed, type **y** to start the server.
14. Run the command `impctl server status` to verify the server started. The confirmation message: `running, ready` is displayed.
15. Perform the procedure described in [Registering Gateways to the Secondary MX Server](#).

Registering Gateways to the Secondary MX Server

After the system configuration was imported into the secondary MX server and it up and running, the gateways are still pointing to the primary MX server. They now need to be pointing to the secondary MX server. The following procedure describes how this is done.

To register the gateways to the secondary MX server:

1. If you are working on an AWS environment, create a new gateway stack with the secondary MX server's IP address and delete the previous gateway stack. In the secondary MX's UI, remove the old gateways from the DR MX.

For a non-AWS environment, from the secondary MX server, run the command `impctl force-remote-registration --mx-ip=<secondary MX server IP address> --gateway-password=<Imperva user password>` to force the gateways to change their IPs to the secondary MX server.



Notes: This process may take several minutes depending on the number of gateways being registered.

2. Log into the secondary MX server UI and go to **Setup > Gateways** to check that the gateways are in running status.
3. Perform checks to ensure that data is being seen in the secondary MX server.
4. Check the agents to ensure they are connected.

PCI Compliance

This appendix describes PCI compliance, and includes:

- [PCI Data Security Standard](#)
- [PCI Compliance](#)

PCI Data Security Standard

PCI DSS requirements apply to all system components that are included in or connected to the cardholder data environment. The cardholder data environment is that part of the network that possesses cardholder data or sensitive authentication data, including network components, servers and applications.

- Network components may include but are not limited to firewalls, switches, routers, wireless access points, network appliances, and other security appliances.
- Server types may include but are not limited to the following: web, database, authentication, mail, proxy, network time protocol (NTP), and domain name server (DNS).
- Applications may include but not limited to all purchased and custom applications, including internal and external (Internet) applications.

SecureSphere and PCI DSS

SecureSphere can assist administrators in achieving PCI compliance for their systems by tracking, monitoring and, if necessary, blocking access to cardholder data. At the same time, SecureSphere Management Servers, Gateways and SecureSphere Agents are considered network components in the cardholder data environment, and are thus themselves subject to the PCI DSS.

The requirements relevant to SecureSphere fall into the following area:

- Protecting the SecureSphere appliance itself, for example, through the use of firewalls. For more information, see [Protecting the SecureSphere Appliance](#).

PCI Compliance

This section reviews the following topics:

- [Protecting \(Hardening\) the SecureSphere Appliance](#)
- [Configuring the Gateway for PCI Compliance](#)

Protecting (Hardening) the SecureSphere Appliance

The default SecureSphere configuration provides protection (hardening) for the SecureSphere appliance as follows:

- Root user access (see [SecureSphere Users](#)) to the appliance OS is available only locally, and not over SSH. SSH access to the appliance OS is available only for users other than `root` and `secure` (see [CLI Users](#)), and these users can execute only a limited set of OS commands.
- Passwords for all SecureSphere administrators and users must be at least 7 characters long, and contain at least 1 alphabetic and at least 1 numeric character, in conformance with PCI requirements.
- When you log in as the SecureSphere administrator for the first time, you are required to change your password.
- All SecureSphere administrators and users must change their passwords at least once every 90 days (configurable in **Admin > System Definitions > Security and Authentication > Password Settings**). The new password cannot be the same as any of the previous four passwords.
- All SecureSphere administrators and users are locked out after 3 consecutive failed authentication attempts.
- GUI sessions are terminated automatically after 30 minutes of inactivity. CLI sessions are terminated automatically after 15 minutes of inactivity.
- SecureSphere log files are accessible only by authorized administrators.
- SecureSphere logs all administrator access (logins) to SecureSphere appliances and to the SecureSphere GUI.
- SecureSphere uses SSH to secure administrative access to the SecureSphere Management Server, and SSL/TLS to secure connections between the SecureSphere Management Server and SecureSphere Gateways.

In addition, administrators should restrict (harden) the physical access to the SecureSphere appliance to authorized personnel only.

Configuring the Gateway for PCI Compliance

SecureSphere enables protecting your data when working with credit card payments and when you need to host your data securely with a PCI compliant hosting provider. This is possible by turning on or off ciphers in the `bootstrap.xml` file.

To turn ciphers on or off:

1. On the gateway, open the file `/opt/SecureSphere/etc/bootstrap.xml`.
2. Find the section `<argusSSL>`.

Under the `<ciphers>` section, the supported ciphers are all set to **true** (turned on).

3. Set the ciphers you want to turn off to **false**.
4. Under the `<gateway-is-server>` section, set the TLS versions you want to turn off to **false**.
5. Under the `<gateway-is-client>` section, set the maximum and minimum TLS versions to comply with PCI.



Note: In SecureSphere versions 12.0 and above, `<max_tls_version>` must be set to **TLS_1_2**.

6. Save the file.
7. Restart the gateway.

If for any reason the `<ciphers>`, `<gateway-is-server>` or `<gateway-is-client>` sections are missing in the `bootstrap.xml` file, copy the below section, paste it in the `bootstrap.xml` file under the line `</highLevelCipherSuites>` and make the necessary changes as described in the above procedure.

```
<ciphers> <-- list of ciphers GW supports for all connections except agent data
channel connection>

<TLS_DHE_RSA_WITH_AES_128_CBC_SHA>true</TLS_DHE_RSA_WITH_AES_128_CBC_SHA>
<TLS_DHE_RSA_WITH_AES_128_CBC_SHA256>true</TLS_DHE_RSA_WITH_AES_128_CBC_SHA256>
<TLS_DHE_RSA_WITH_AES_128_GCM_SHA256>true</TLS_DHE_RSA_WITH_AES_128_GCM_SHA256>
<TLS_DHE_RSA_WITH_AES_256_CBC_SHA>true</TLS_DHE_RSA_WITH_AES_256_CBC_SHA>
<TLS_DHE_RSA_WITH_AES_256_CBC_SHA256>true</TLS_DHE_RSA_WITH_AES_256_CBC_SHA256>
<TLS_DHE_RSA_WITH_AES_256_GCM_SHA384>true</TLS_DHE_RSA_WITH_AES_256_GCM_SHA384>
<TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA>true</TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA>
<TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256>true</TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256>
<TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA>true</TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA>
<TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384>true</TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384>
<TLS_RSA_WITH_AES_128_CBC_SHA>true</TLS_RSA_WITH_AES_128_CBC_SHA>
<TLS_RSA_WITH_AES_128_CBC_SHA256>true</TLS_RSA_WITH_AES_128_CBC_SHA256>
<TLS_RSA_WITH_AES_128_GCM_SHA256>true</TLS_RSA_WITH_AES_128_GCM_SHA256>
<TLS_RSA_WITH_AES_256_CBC_SHA>true</TLS_RSA_WITH_AES_256_CBC_SHA>
<TLS_RSA_WITH_AES_256_CBC_SHA256>true</TLS_RSA_WITH_AES_256_CBC_SHA256>
<TLS_RSA_WITH_AES_256_GCM_SHA384>true</TLS_RSA_WITH_AES_256_GCM_SHA384>

</ciphers>

<versions>

<gateway-is-server> <!-- list of TLS/SSL version for GW which acts as server
(for example, for GW GUI connection from client application)-->
```

```
<TLS_1_0>true</TLS_1_0>

<TLS_1_1>true</TLS_1_1>

<TLS_1_2>true</TLS_1_2>

</gateway-is-server>

<gateway-is-client> <!-- list of TLS/SSL version for GW which acts as client
(for example, towards MX)-->

<min_tls_version>TLS_1_0</min_tls_version>

<max_tls_version>TLS_1_2</max_tls_version>

</gateway-is-client>

</versions>

<tls-supported-versions>

<server> <!-- list of TLS/SSL version for GW which acts as server on agent data
channel connection-->

<kssl> <!-- list of ciphers GW supports for agent data channel connection-->
```

ABR Migration Tool

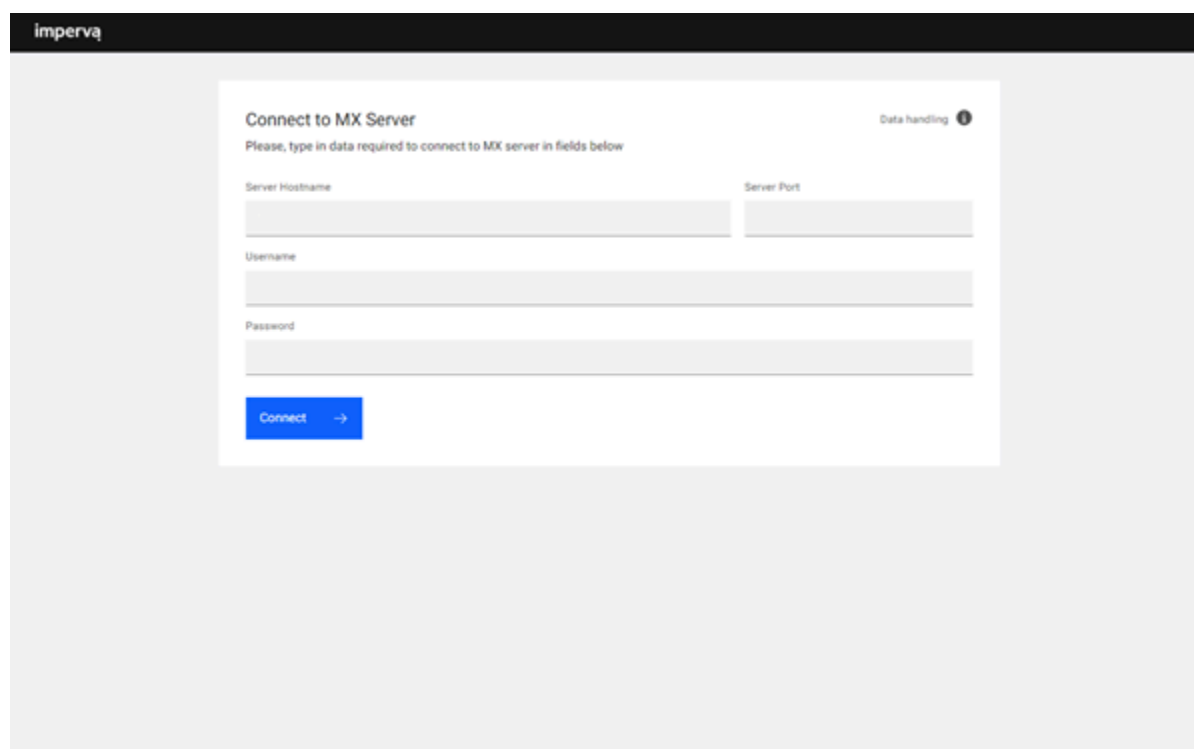
If you are intending on using the ABR feature of the Imperva WAF Gateway, Imperva developed an external migration tool that helps in migrating your existing bridge rules to Advanced Bridge in an easy manner.

- [How To Use](#)
- [ABR Migration Tool Clarifications](#)

How To Use

To use the ABR migration tool:

1. Go to the [Imperva Customer Portal](#), click **Downloads**, then navigate to **/Downloads/Tools/ABR Migration Tool**.
2. Download the tool package from the folder appropriate to your operating system (Linux or Windows) to the machine that has connectivity with the MX.
3. Unpack the tool package and double click on the application executable, **AbrMigTool-win32-x64.exe** for Windows or **AbrMigTool** for Linux. The connect to MX Server screen is displayed.



4. Fill the fields as follows:
 - **Server Hostname:** The IP address where your MX is deployed
 - **Server Port:** The port on which your MX is deployed. This port is set to **8083** and cannot be changed.
 - **Username:** The user name you want to connect to the MX Server with. By default this field is set to **admin**.

- **Password:** The password of the user you are trying to connect with
- Click **Connect**. The tool connects to the MX Server and a list of bridge mode rules and TRP rules that need to be created is displayed.

Bridge mode rules

How to use ⓘ

Search

⌵

☐ Site	Server Group	Service	Hostname	Port	Certificate
☐ Fresh Site	Server Group One	Service One	10.100.52.101	443	Cert2
☐ Fresh Site	Server Group One	Service One	10.100.52.101	444	Cert1
☐ Fresh Site	Server Group One	Service One	10.100.52.102	443	Cert2
☐ Fresh Site	Server Group One	Service One	10.100.52.102	443	Cert1
☐ Fresh Site	Server Group One	Service One	10.100.52.102	444	Cert2
☐ Fresh Site	Server Group One	Service One	10.100.52.102	444	Cert1

TRP rules to create

⌵

Search

⌵

☐ Site	Server Group	Service	Hostname	Port	Certificate
☐ Fresh Site	Server Group One	ivan HTTP	10.100.52.102	808	⊗
☐ Fresh Site	Server Group One	ivan HTTP	10.100.52.101	808	⊗
☐ Fresh Site	Server Group One	Service One	10.100.52.102	81	⊗
☐ Fresh Site	Server Group One	Service One	10.100.52.102	80	⊗
☐ Fresh Site	Server Group One	Service One	10.100.52.101	444	Cert2
☐ Fresh Site	Server Group One	Service One	10.100.52.101	443	Cert1
☐ Fresh Site	Server Group One	Service One	10.100.52.101	81	⊗
☐ Fresh Site	Server Group One	Service One	10.100.52.101	80	⊗
☐ Fresh Site	SG 2	Some Service	10.100.52.203	80	⊗
☐ Default Site	SG	IvanService	10.100.97.203	80	⊗

Submit rules migration

- Select the check boxes for the bridge rules you want to migrate to TRP rules.
- Select the check boxes for the TRP rules you want to create.
- Click on the **Submit rules migration** button.
- When the migration completes, download the revert file.
- Connect to the gateway via SSH and run the command **impcfg**. The **impcfg Top Screen** is displayed.
- Type **2**. The **Gateway Management Screen** is displayed.
- Type **4**. A list of gateway operation modes is displayed.
- Type **2** or **3** according to your needs. The question **Would you like to enable Advanced Bridge?** is displayed.

14. Type **1**. The **Gateway Management Screen** is displayed.
15. Type **t**. The **impcfg Top Screen** is displayed.
16. Type **S**.
17. Type **A**. The gateway restarts and ABR mode is enabled.
18. Login to the Management Server UI.
19. Go to **Setup > Sites** and for every service in the service groups that were selected for ABR migration, in the **Reverse Proxy** tab, under the **Transparent Reverse Proxy** section, select the **Enable Transparent Reverse Proxy** check box.
20. Click **Save**.

To restore the MX to its original state:

1. Double click on the application executable, **AbrMigTool-win32-x64.exe** for Windows or **AbrMigTool** for Linux. The connect to MX Server screen is displayed.

2. Fill the fields as follows:
 - **Server Hostname:** The IP address where your MX is deployed
 - **Server Port:** The port on which your MX is deployed. This port is set to **8083** and cannot be changed.
 - **Username:** The user name you want to connect to the MX Server with. By default this field is set to **admin**.
 - **Password:** The password of the user you are trying to connect with
3. Click **Connect**.
4. Click on the **Revert from file** tab.
5. Click the link to browse to the Revert file you downloaded after the migration process or drag and drop it in the shown box. A list of bridge rules to restore and a list of TRP rules to delete is displayed.
6. Click **Submit revert migration**.

ABR Migration Tool Clarifications

The following are further clarifications regarding the ABR migration tool.

- For revert to work, the site tree must remain unchanged
- Imperva does not guarantee the integrity of the migration if something is changed on the MX during migration
- The tool is available for Linux and Windows machines only.

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PATENT INFORMATION

The software described by this document may be covered by one or more of the following patents:

US Patent Nos. 7,640,235, 7,743,420, 7,752,662, 8,024,804, 8,051,484, 8,056,141, 8,135,948, 8,181,246, 8,392,963, 8,448,233, 8,453,255, 8,713,682, 8,752,208, 8,869,279 and 8,904,558, 8,973,142, 8,984,630, 8,997,232, 9,009,832, 9,027,136, 9,027,137, 9,128,941, 9,148,440, 9,148,446, 9,401,927, and 11, 579, 859..

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v14.7-waf-administration-guide

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